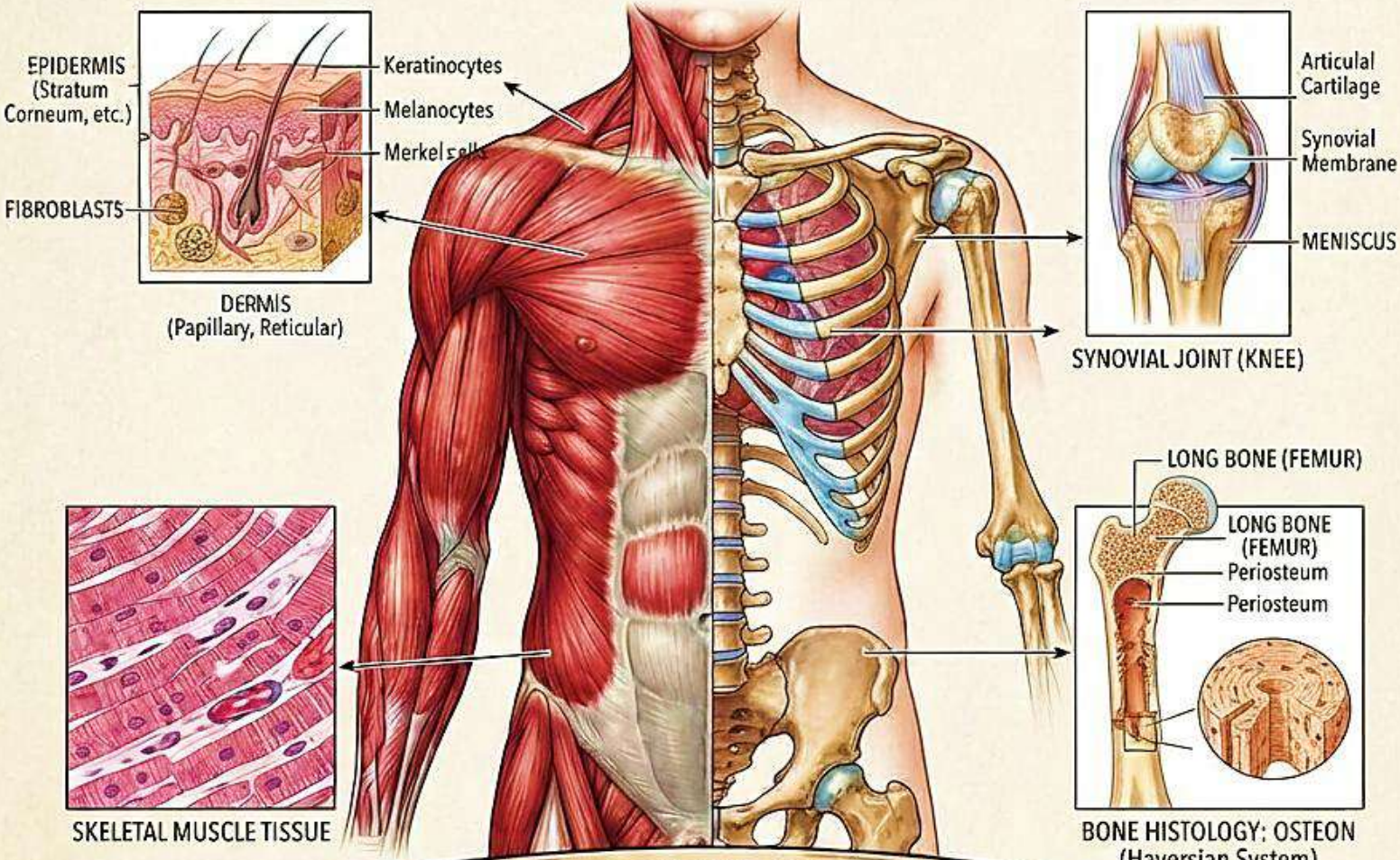
MUE
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PRINCIPLES OF MACROSCOPIC AND MICROSCOPIC STRUCTURE

PMS - 102

FOR FIRST YEAR MEDICAL STUDENTS



EDITED BY
THE FACULTY OF MEDICINE
MERIT UNIVERSITY

ACADEMIC YEAR 2026-2027

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Principles of Microscopic and Macroscopic Structures

PMS-102

Human Anatomy and Embryology

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Lectures 1 & 2:

OVERVIEW AND INTRODUCTION TO ANATOMY, ANATOMICAL TERMS

ILOs:

1. Describe the anatomical terms, position and movement.
2. Describe the normal structure of the body and its major organ systems and explain their functions.

ANATOMY

Definition of Anatomy: The science of the structure and function of the body.

Clinical Anatomy: Study of the macroscopic structure and function of the body as it relates to the practice of medicine and other health sciences.

Basic Anatomy: Study of the minimal amount of Anatomy consistent with the understanding of the overall structure and function of the body.

Anatomy is a Greek word first used by *Aristotle*.

Ana means through

Tom means to cut

Y suffix forming noun

So, Anatomy means 'to cut through' or 'dissect'

Human body consists of:

Systems: Each performs a special function in th body. Each system is formed of a group of organs**.**

Organ: Formed of different tissues grouped together forming a distinct structure. Each organ is formed of a group of tissues.

Tissue: Formed of cells similar in structure and function

Cell: The basic structural unit of the body. So, the cell is the simplest structural form of life.

Approaches to Studying Anatomy:

1. **Regional (Topographical) Anatomy:**

   It considers the organization of the human body as segments or major parts based on form and mass. So, the human body is divided into the following regions:

   - **Head**
   - **Neck**

- **Trunk:** Which is further divided into:
    - Chest
    - Abdomen
    - Pelvis
- **Limbs:** which are:
    - Upper limbs (right and left)
    - Lower limbs (right and left)

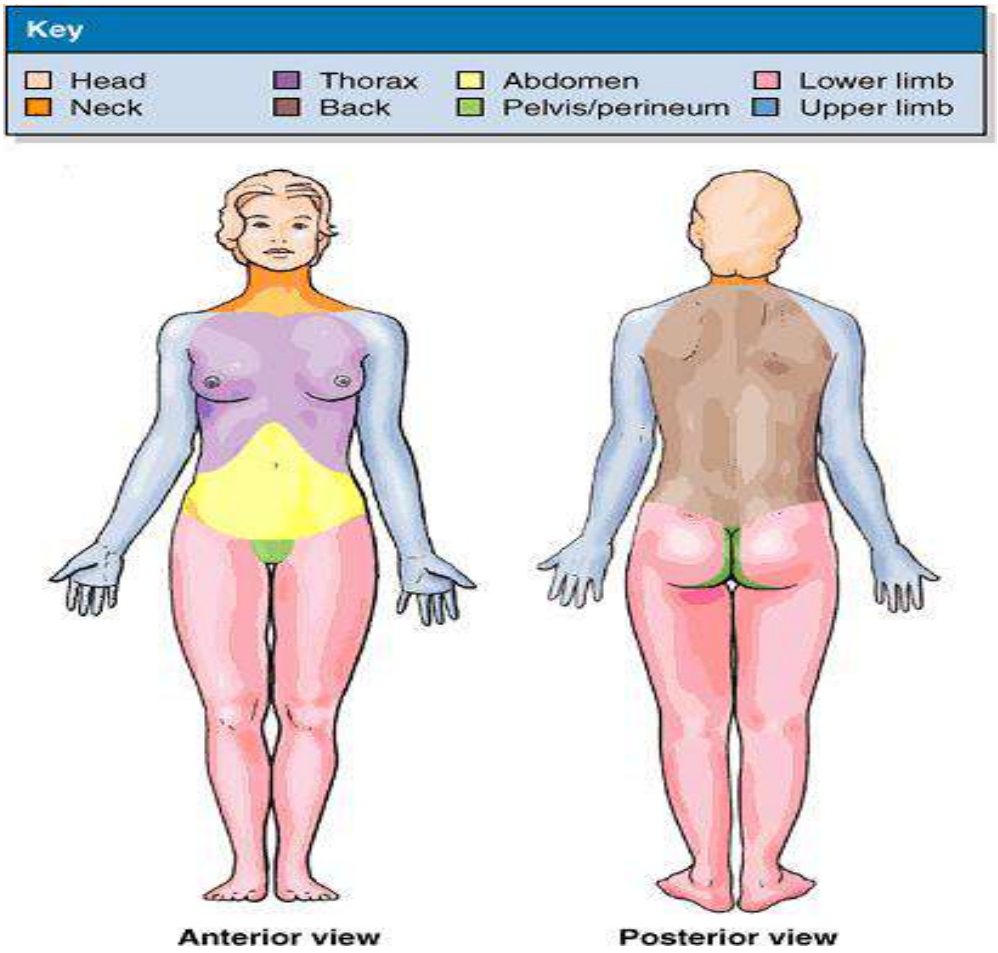


Figure 1: Major parts of the body as studied by regional anatomy (Moore et al, 2023).

II. Systemic Anatomy:

It recognizes the organization of the body's organs into systems that work together to carry out complex functions:

- **Integumentary System (Dermatology):** Consists of the skin and its appendages
- **Skeletal System (Osteology):** Consists of the bones and cartilage
- **Articular System (Arthrology):** Consists of joints and their associated ligaments
- **Muscular System (Myology):** Consists of muscles to move or position parts of the body
- **Nervous System (Neurology):** Consists of central nervous system (brain and spinal cord) and Peripheral nervous system (nerves, ganglia, motor and sensory endings)
- **Circulatory System (Angiology):** Consists of cardiovascular system (heart and blood vessels) and lymphatic system (lymph vessels and lymph nodes)
- **Digestive or Alimentary System: (Gastroenterology):** Consists of Organs and glands associated with digestion
- **Respiratory System (Pulmonology):** Consists of air passages and lungs
- **Urinary System (Urology):** Consists of kidneys, ureters, urinary bladder and urethra
- **Reproductive or Genital System (Gynecology for females and Andrology for males):** Consists of gonads (ovaries and testes), ducts and genitalia.
- **Endocrine System (Endocrinology):** Consists of ductless glands e.g. thyroid gland.

Descriptive Anatomic Terms

Terms Related to Position

All descriptions of the human body are based on that the person is in the anatomic position.

Anatomic Position (Figure 1):

- The person is standing erect.
- The upper limbs by the sides.
- The face and palms of the hands directed forward.

Various parts of the body are described in relation to certain imaginary planes (Figure 2).

Median Sagittal Plane: A vertical plane passing through the centre of the body, dividing it into equal right and left halves.

Paramedian Planes: Planes are situated to one or the other side of the median plane and parallel to the median plane.

Medial: A structure situated nearer to the median plane of the body than another.

Lateral: A structure that lies farther away from the median plane than another.

Coronal Planes: Planes are imaginary vertical planes at right angles to the median plane.

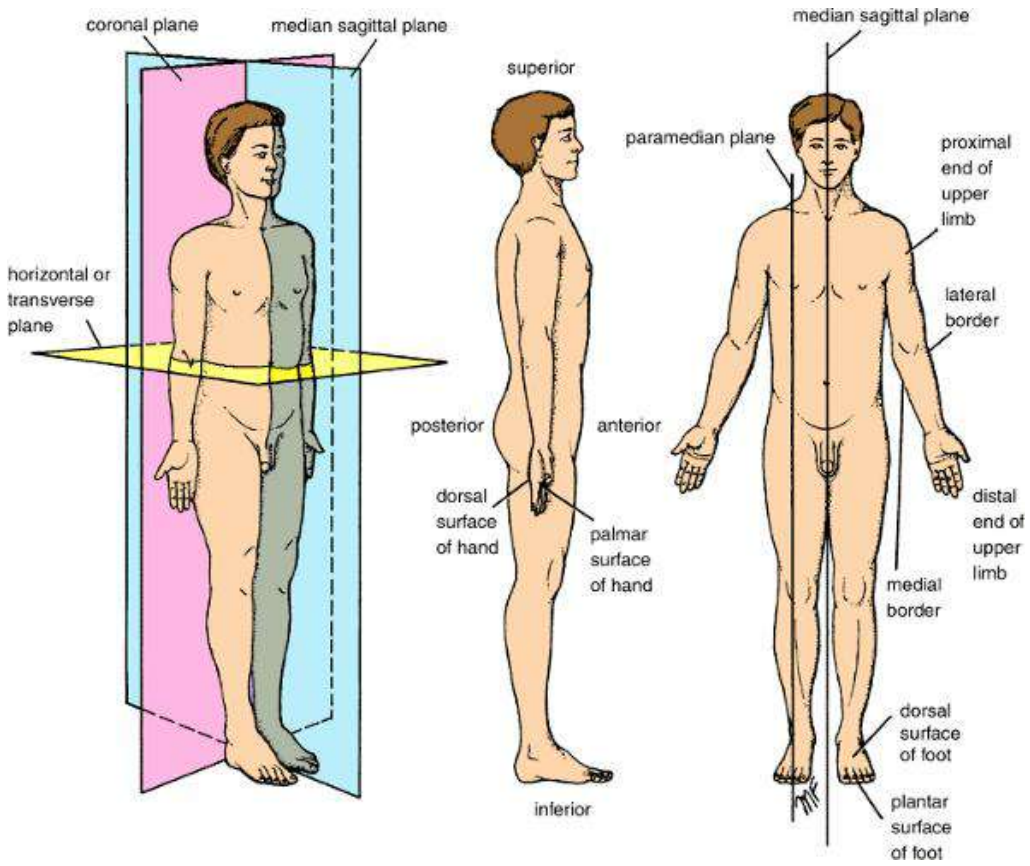


Figure 2: Anatomic position and Anatomic terms used in relation to position (Snell, 2019).

Horizontal, (Transverse) Plane: Planes are at right angles to both the median and the coronal planes.

Anterior: Front of the body.

Posterior: Back of the body.

- In describing the hand, the terms palmar and dorsal surfaces are used in place of anterior and posterior, and in describing the foot, the terms plantar and dorsal surfaces are used instead of lower and upper surfaces.

Proximal: Near to the roots of the limbs

Distal: Away from the roots of the limbs

- For example, the arm is proximal to the forearm, and the hand is distal to the forearm.
- The terms superficial and deep denote the relative distances of structures from the surface of the body, and the terms superior and inferior denote levels relatively high or low with reference to the upper and lower ends of the body.

Internal: Inside cavity.

External: Outside cavity.

- For example, the internal carotid artery is found inside the cranial cavity and the external carotid artery is found outside the cranial cavity.

Ipsilateral: The same side of the body.

- For example, the left hand and left foot are ipsilateral.

Contralateral: Refers to opposite sides of the body.

- For example, the left biceps brachii muscle and the right rectus femoris muscle are contralateral.

Supine position: The body is lying on the back.

Prone position: The body is lying face downward.

Terms Related to Movement (Figures 3 & 4):

Flexion: A movement that takes place in a sagittal plane.

- For example, flexion of the elbow joint approximates the anterior surface of the forearm to the anterior surface of the arm. It is usually an anterior movement, but it is occasionally posterior, as in the case of the knee joint.

Extension: Straightening the joint and usually takes place in a posterior direction.

- Lateral flexion is a movement of the trunk in the coronal plane.

Abduction: A movement of a limb away from the midline of the body in the coronal plane.

Adduction: A movement of a limb toward the body in the coronal plane.

- In the fingers and toes, abduction is applied to the spreading of these structures and adduction is applied to the drawing together of these structures.
- Movements of the thumb is different as it rotated at right angle of the other fingers.

Rotation: Movement of a part of the body around its long axis.

Medial rotation: Movement that results in the anterior surface of the part facing medially.

Lateral rotation: Movement that results in the anterior surface of the part facing laterally.

Pronation of the forearm: Medial rotation of the forearm; the palm of hand faces posteriorly.

Supination of the forearm: Lateral rotation of the forearm; the palm of hand faces anteriorly.

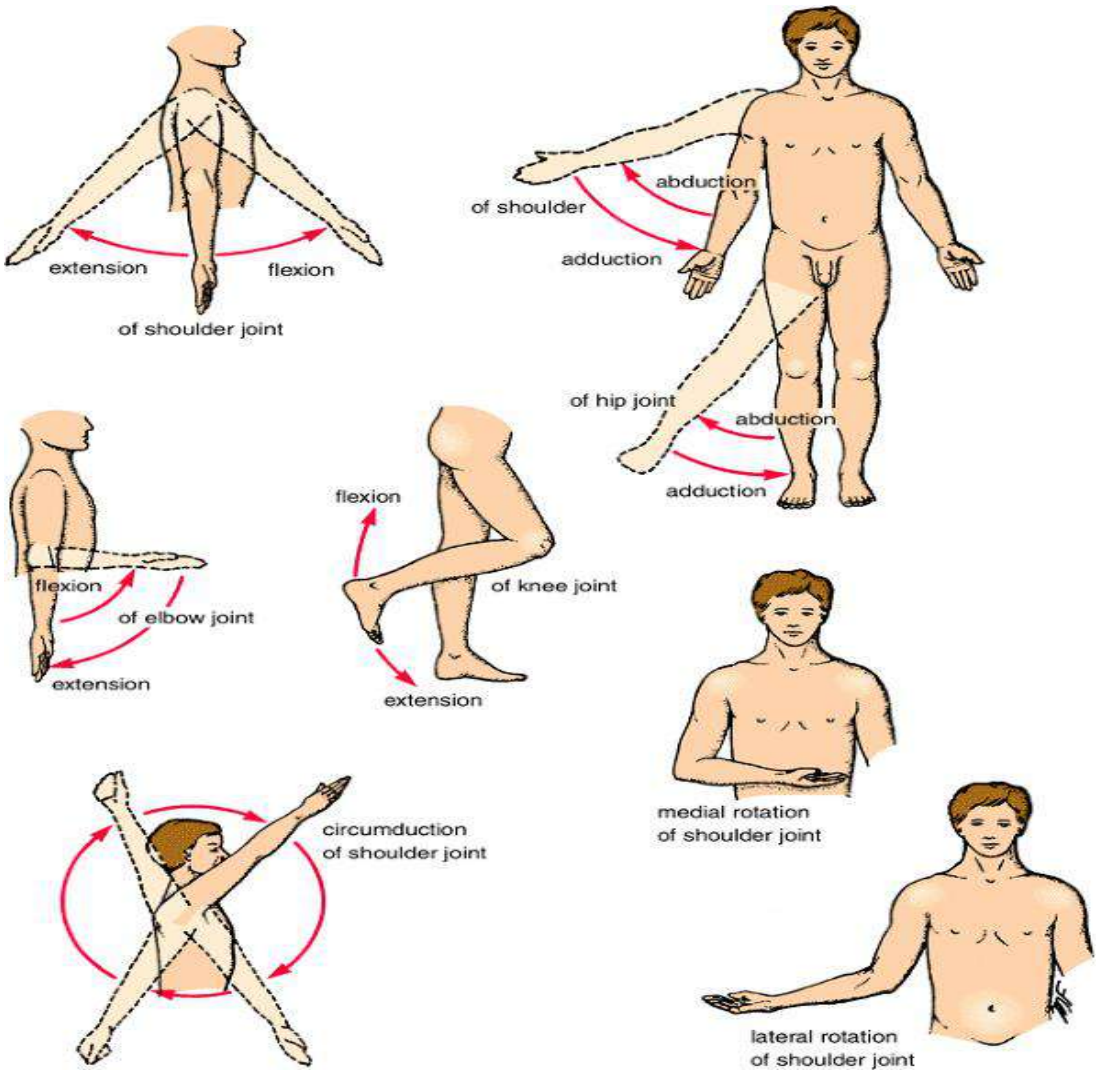


Figure 3: Some anatomic terms used in relation to movement (Snell, 2019).

Circumduction: Combination in sequence of the movements of flexion, extension, abduction, and adduction.

Protraction: To move forward.

Retraction: To move backward

- Describe the forward and backward movement of the jaw at temporomandibular joints.

Inversion: is the movement of the foot so that the sole faces in a medial direction.

Eversion: is the opposite movement of the foot so that the sole faces in a lateral direction.

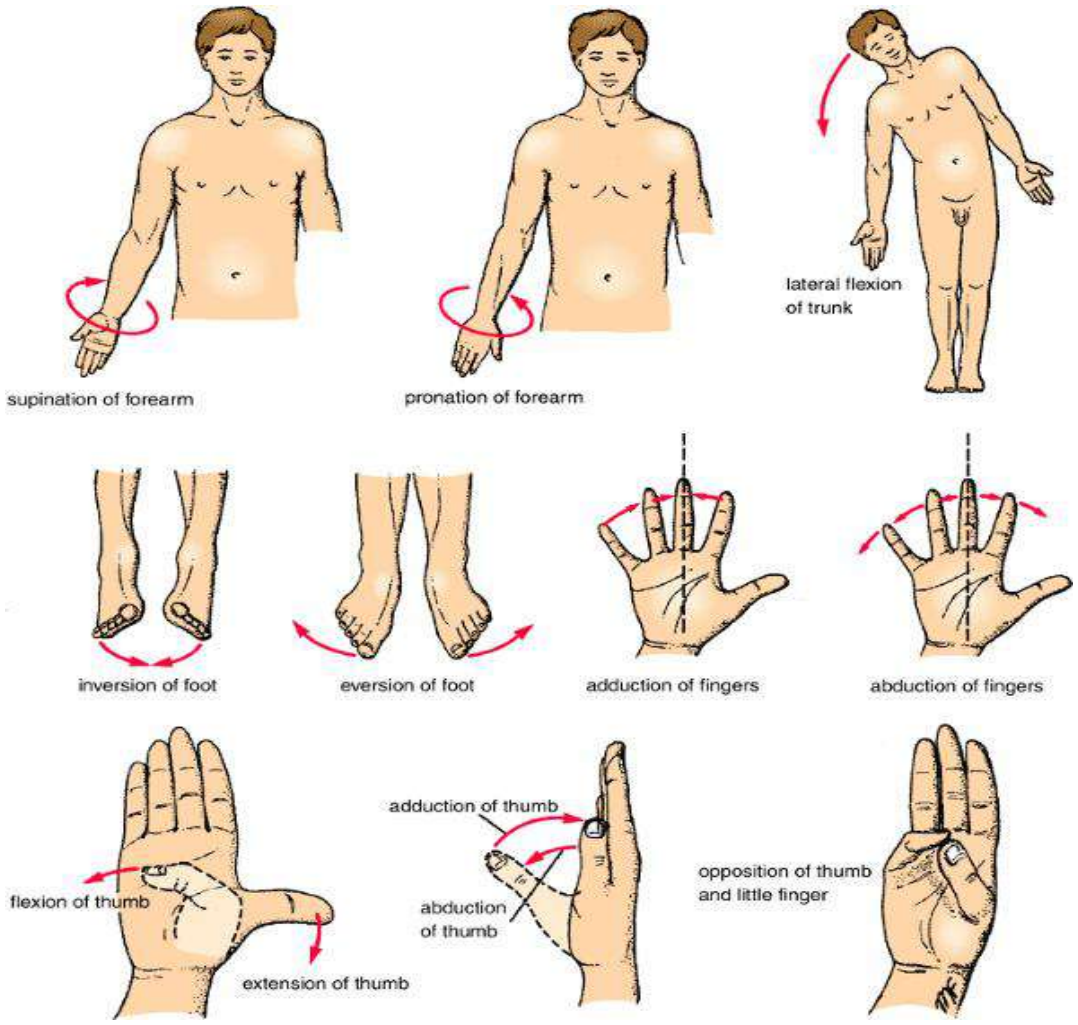


Figure 4: Anatomic terms used in relation to movement (Snell, 2019).

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1. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins. Pages 1-4.
2. Moore, Keith L.; Dalley and Arthur F. (2006): Clinically Oriented Anatomy, 5th Edition, Lippincott Williams & Wilkins. Pages: 12-18.

Lecture 3:

SKIN AND FASCIA

ILOs:

1. Define the skin
2. What are the divisions of the skin?
3. What are the functions of the skin?
4. What are the appendages of the skin?
5. Enumerate the skin having no sweat glands.
6. Enumerate the sites of superficial fascia devoid of adipose tissue?
7. What are the functions of superficial fascia?

Skin (Figure: 1):

- The largest organ in the body
- The outer layer covering the body
- Divided into two parts: the superficial part, the epidermis, and the deep part, the dermis.

Epidermis:

- Stratified epithelium.
- On the palms of the hands and the soles of the feet, the epidermis is extremely thick, to withstand the wear and tear that occurs in these regions.
- In other areas of the body, on the anterior surface of the arm and forearm, it is thin.

Dermis:

- Composed of dense connective tissue
- Contains many blood vessels, lymphatic vessels, and nerves.
- Shows variation in thickness in different parts of the body, thinner on the anterior than on the posterior surface.
- Thinner in women than in men.
- Connected to the underlying deep fascia or bones by the superficial fascia.

Skin Creases:

- Skin folds in the same place, over the joints.
- At these sites, the skin is thinner than elsewhere
- Firmly tethered to underlying structures by strong bands of fibrous tissue.

Appendages of the skin:

1. Nails.
2. Hair follicles.
3. Sebaceous glands.
4. Sweat glands.

Nails:

- Keratinized plates on the dorsal surfaces of the tips of the fingers and toes.
- The proximal edge of the plate is the root of the nail.
- Except for the distal edge of the plate, the nail is surrounded by nail folds.
- The surface of skin covered by the nail is the nail bed.

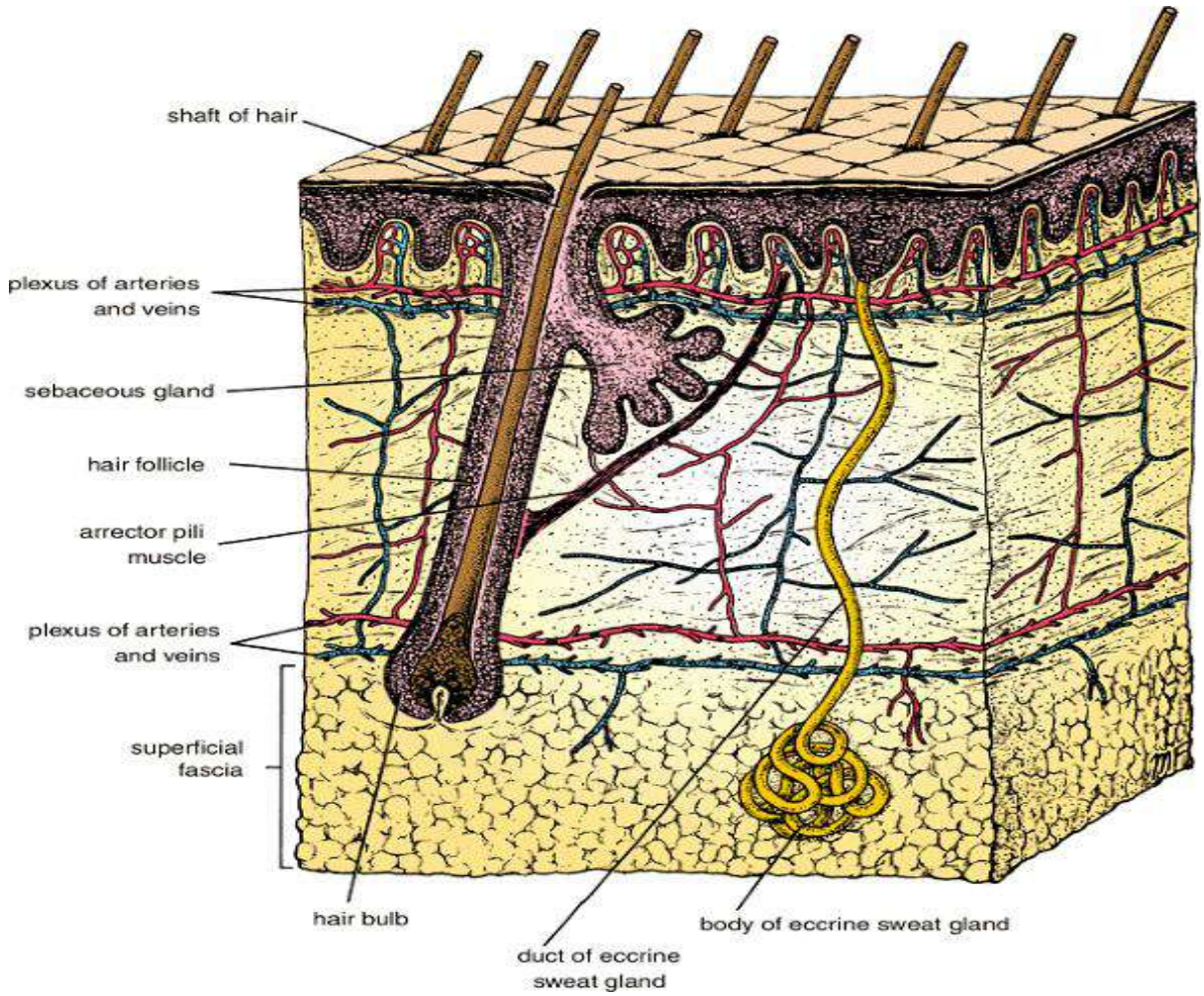


Figure 1: General structure of the skin and its relationship to the superficial fascia. (Snell, 2019).

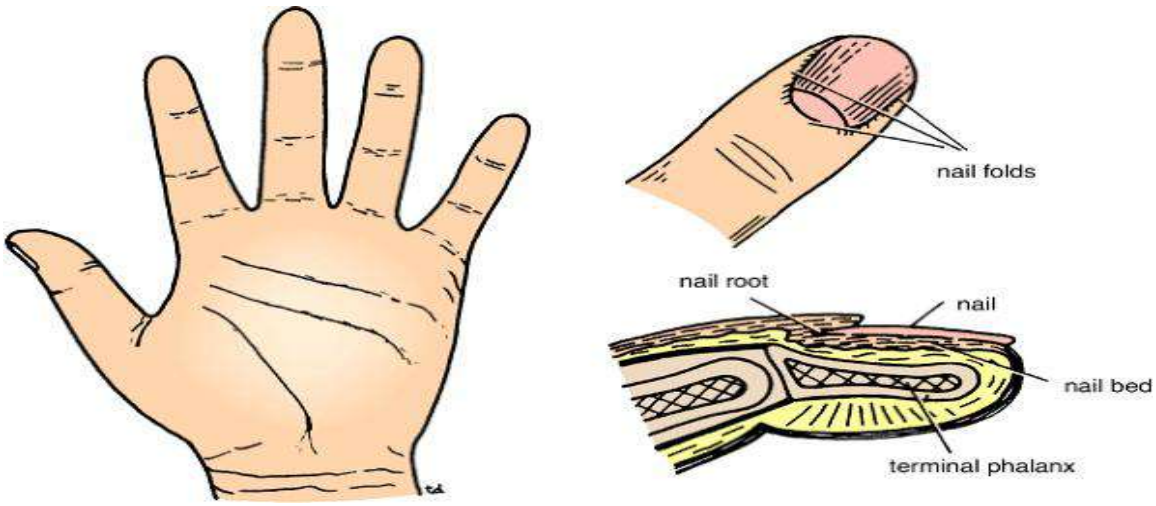


Figure 2: Various skin creases on the palmar surface of the hand and the anterior surface of the wrist joint. Also, relationship of the nail to other structures of the finger. (Snell, 2019).

Hairs:

- Grow out of follicles, which are invaginations of the epidermis into the dermis.
- The follicles have hair bulbs to penetrate to the deeper part of the dermis.
- Each hair bulb is concave at its end, occupied by vascular connective tissue, hair papilla.
- A smooth muscle, arrector pili, connects the undersurface of follicle to the dermis.
- The muscle is innervated by sympathetic nerve fibres.

Sites of skin having no hairs:

1. Lips.
2. Palms of the hands.
3. Sides of the fingers.
4. Glans penis.
5. Clitoris.
6. Labia minora.
7. Internal surface of the labia majora.
8. Soles and sides of the feet.
9. Sides of the toes.

Sebaceous Glands:

- Pour the sebum, onto the shafts of hairs as they pass up through the necks of the follicles.
- Situated on the sloping undersurface of the follicles and lie within the dermis.

- Sebum is an oily material that helps preserve the flexibility of the emerging hair.
- It also oils the surface epidermis around the mouth of follicle.

Sweat Glands:

- Extend through the full thickness of the dermis, and their extremities may lie in the superficial fascia.
- The most deeply penetrating structures of all the epidermal appendages.
- Long, spiral, tubular glands distributed over the surface of the body.

Sites of Skin Having no Sweat Glands:

1. Red margins of the lips.
2. Nail beds.
3. Glans penis.
4. Clitoris.

Function of the Skin:

1. Protection: of the body from environmental effects such as abrasions, fluid loss, ultraviolet radiations, micro-organisms
2. Sensation (pain, temperature, touch) by way of superficial nerves and their sensory endings.
3. Heat-regulation: through evaporation of sweat and/or dilation or constriction of superficial blood vessels.
4. Synthesis and storage of vitamin D.
5. Containment for the body's structures (tissues and organs) and vital substances (extra-cellular fluids) preventing dehydration.

FASCIAE

- Can be divided into two superficial and deep.
- Lie between the skin and the underlying muscles and bones.

I. Superficial Fascia (Subcutaneous Tissue) (Figures 2 & 3):

- Mixture of loose areolar and adipose tissue.

Sites of superficial fascia having many bundles of collagen fibres:

1. Scalp.
2. Back of neck.
3. Palms of hands.
4. Soles of feet.

Sites of superficial fascia devoid of adipose tissue:

1. Eyelids.
2. Auricle of the ear.
3. Penis.
4. Scrotum.
5. Clitoris.
6. Nipple and areola of the breast

Functions of Superficial Fascia:

1. Distributing layer in which blood vessels, lymphatics and nerves can travel.
2. Allows mobility of the skin on underlying structures, particularly over the joints.
3. Provides a depot for food storage.
4. Forms a useful insulating layer over body.

II. Deep Fascia:

- Membranous layer of connective tissue that invests the muscles and other deep structures.
- In the neck, it forms well-defined layers.
- In the thorax and abdomen, it is a thin film of areolar tissue covering the muscles and aponeuroses.
- In the limbs, it forms a definite sheath around the muscles and other structures.
- Fibrous septa extend from the deep surface of the membrane, between the groups of muscles, and in many places divide the interior of the limbs into compartments.
- In the region of joints, it may be thickened to form restraining bands, retinacula.

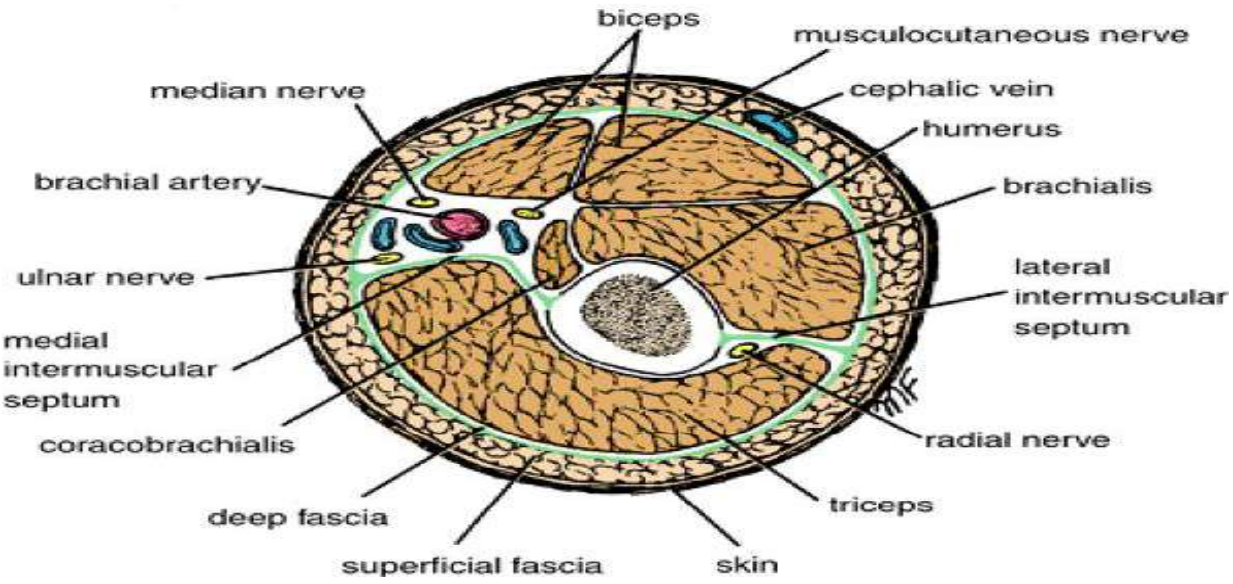


Figure 2: Section in the middle of right arm showing superficial and deep fasciae (Snell, 2019).

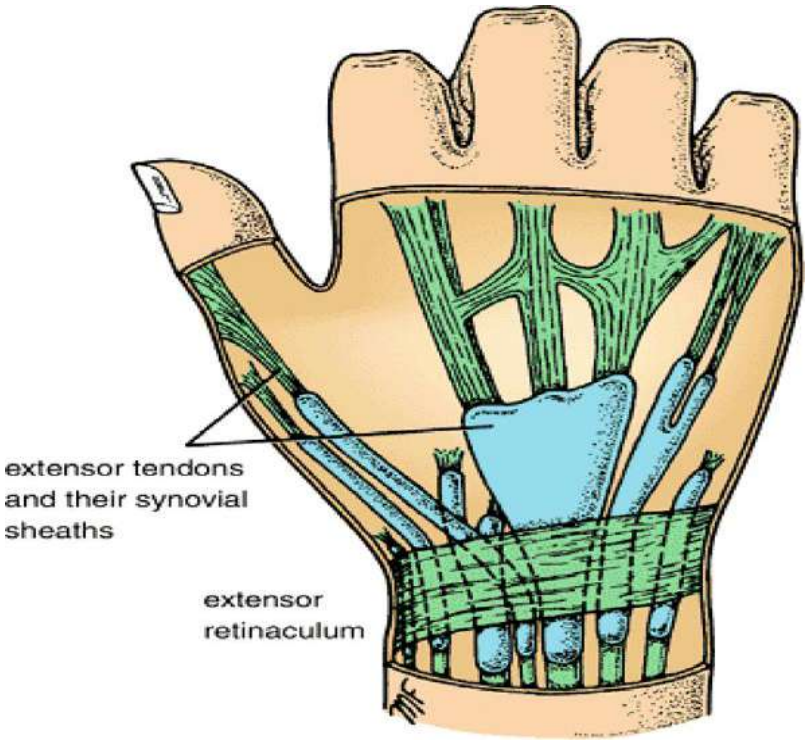


Figure 3: Extensor retinaculum on the posterior surface of the wrist (Snell, 2019).

References:

1. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.
2. Moore, Keith L.; Dalley and Arthur F. (2006): Clinically Oriented Anatomy, 5th Edition, Lippincott Williams & Wilkins. Pages 2-12.

Lecture 4:

MUSCLE

ILOs:

1. Define the muscle
2. What are the types of muscle?
3. Enumerate the types of skeletal muscles according to the shape of their fibres.
4. Describe the muscle actions.
5. What are the features of smooth muscles?
6. What are the features of the cardiac muscle?

Muscle:

Definition: Muscles represent fleshy tissue of the body which have a power of contraction.

Types of the muscles:

- The three types of muscle are skeletal, smooth, and cardiac.

I. Skeletal Muscle:

- Produces the movements of the skeleton.
- Sometimes called voluntary muscles.
- Made up of striped muscle fibres.
- Has two or more attachments.
- The attachment that moves the least is the origin.
- The attachment that moves the most is the insertion.
- Degree of mobility of the attachments may be reversed; therefore, the terms origin and insertion are interchangeable.
- Fleshy part of the muscle is referred to as its belly.
- Ends of a muscle are attached to bones, cartilage, or ligaments by cords of fibrous tissue, tendons.
- Flattened muscles are attached by a thin sheet of fibrous tissue, aponeurosis.
- Raphe is an interdigitation of the tendinous ends of fibres of flat muscles.
- Represent more than 40% of the whole-body weight.
- Their total number is 620 muscles all over the body.

Forms of Voluntary Muscles:

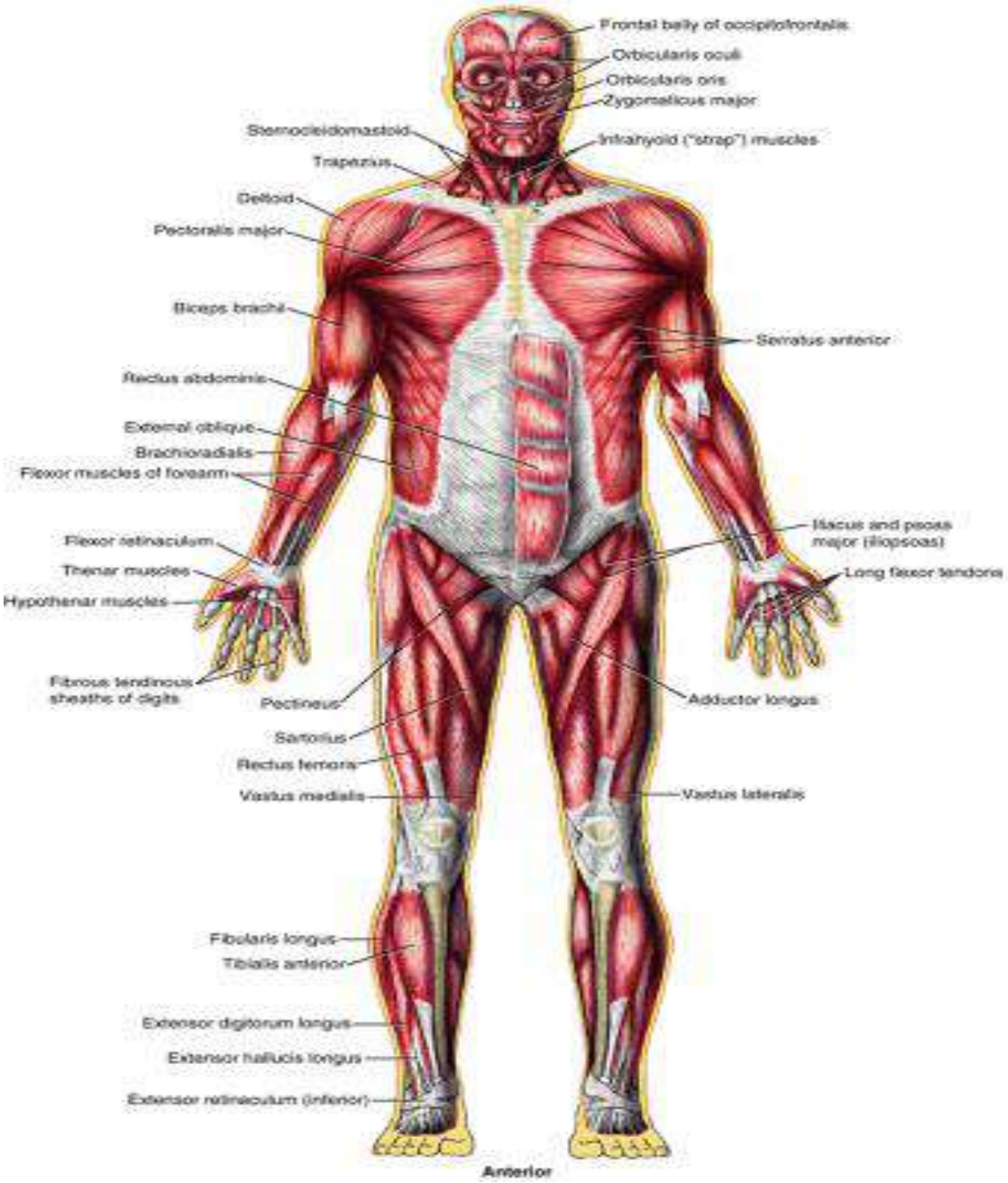
(According to shape and arrangement of fibres):-

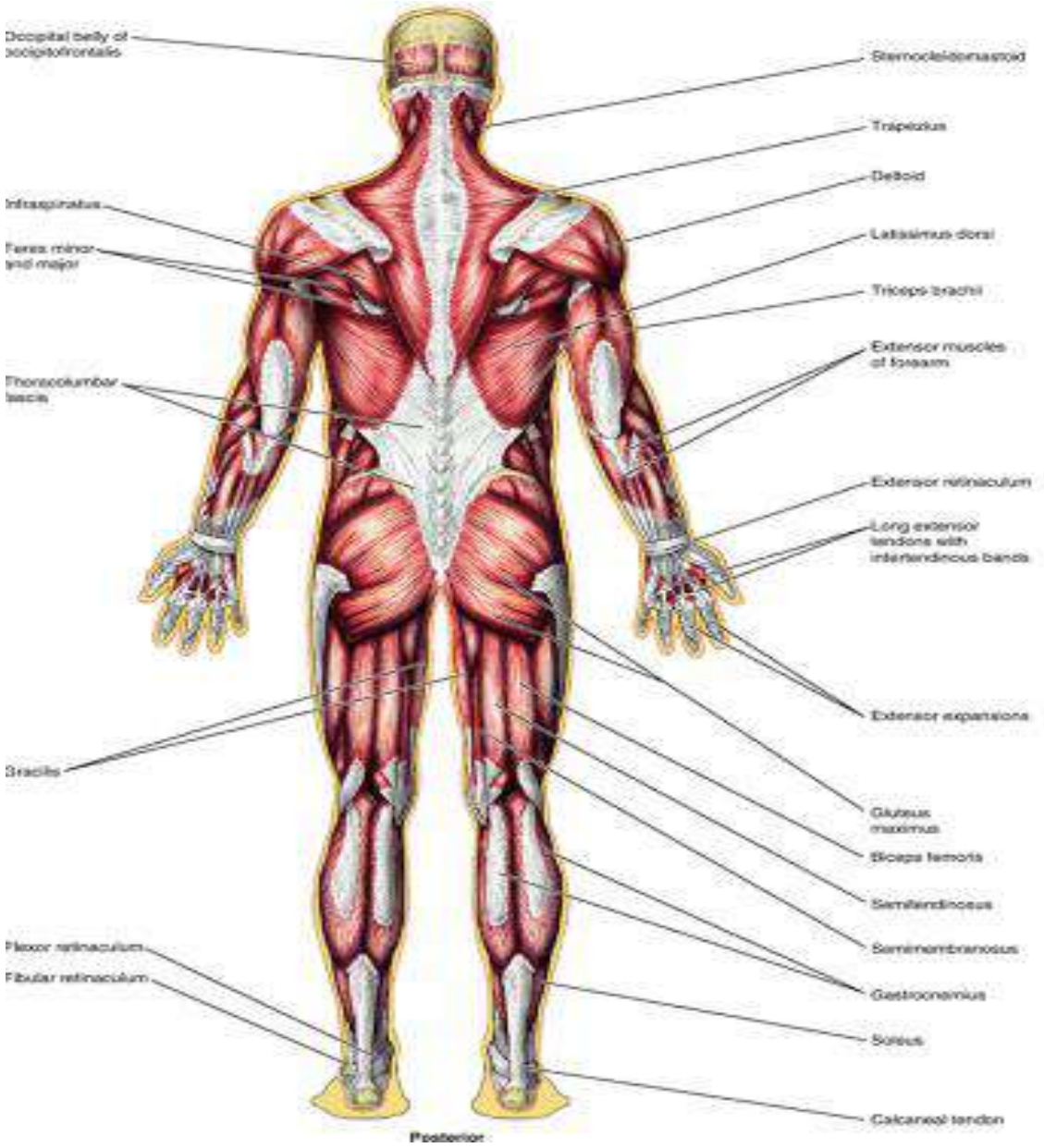
1- Strap or parallel muscles: Parallel fibres e.g. sartorius

2- Quadrate muscle: Quadrangular in shape e.g. pronator quadratus

3- Triangular muscle: Triangular in shape e.g. temporalis

4- Fusiform muscle: Spindle-shaped e.g. biceps, flexor carpi radialis.





Figures 1 & 2: Skeletal muscles (anterior and posterior aspects). (Moore et al, 2023)

5- Pennate muscle: The fibers are in oblique direction, feathery in shape; many forms:-

a- Uni-pennate: Its fibers come from one side of a tendon e.g. flexor pollicis longus

b- Bi-pennate: Its fibers come from both sides of a tendon e.g. rectus femoris

c- Multi-pennate: Series of bi-pennate fibers lie side by side e.g. deltoid muscle

d- Circum-pennate: Fibers come out from central tendon in different directions e.g. tibialis anterior.

Muscle Movement:

All movements are the result of coordinated action of many muscles

Muscles may work in (Muscle Action):-

1. Prime Mover:

- The chief muscle or a member of a chief group of muscles responsible for particular action e.g. quadriceps femoris is a prime mover in extension of the knee.

2. Antagonist:

- Any muscle opposes action of prime mover e.g. biceps femoris oppose action of quadriceps femoris.

3. Fixator:

- A muscle contracts isometrically to stabilize action of prime mover.
- Muscles attaching to shoulder girdle to trunk contract as fixators to allow deltoid to act on shoulder

4. Synergist:

- Prevent unwanted movements in intermediate joints when the prime mover crosses number of joints before reaching the joint where its main action takes place.
- Flexor and extensor muscles of carpus contract to fix wrist joint to allow flexor and extensor muscles of fingers to work efficiently.

Nerve Supply of Skeletal Muscle

- The nerve trunk to a muscle is a mixed nerve, about 60% is motor and 40% is sensory, and it also contains some sympathetic autonomic fibres.

II. Smooth Muscle:

Less common than voluntary

Characters of Smooth Muscle:

1. Not controlled by well
2. Under control of autonomic nervous system
3. Not attached to bones
4. Their fibres are smooth, plane, not stripped
5. They contract and relax in ring fashion (peristalsis)

Examples: Smooth muscles in wall of gastrointestinal tract, respiratory tract, urinary and genital systems and wall of the blood vessels

III- Cardiac Muscle:

- Single and the most important muscle in the body

Characters of cardiac muscle:

1. Not voluntary.
2. Not smooth.
3. Single.
4. Under control of autonomic nervous system.
4. Contracts in rhythmic fashion (pumping).

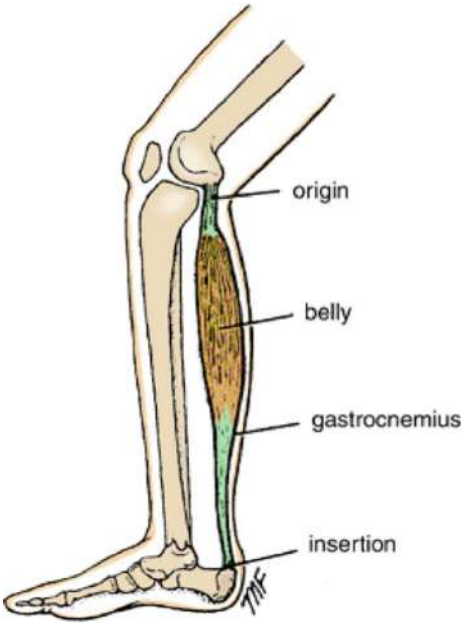


Figure 3: Origin, insertion, and belly of the gastrocnemius muscle (Snell, 2019).

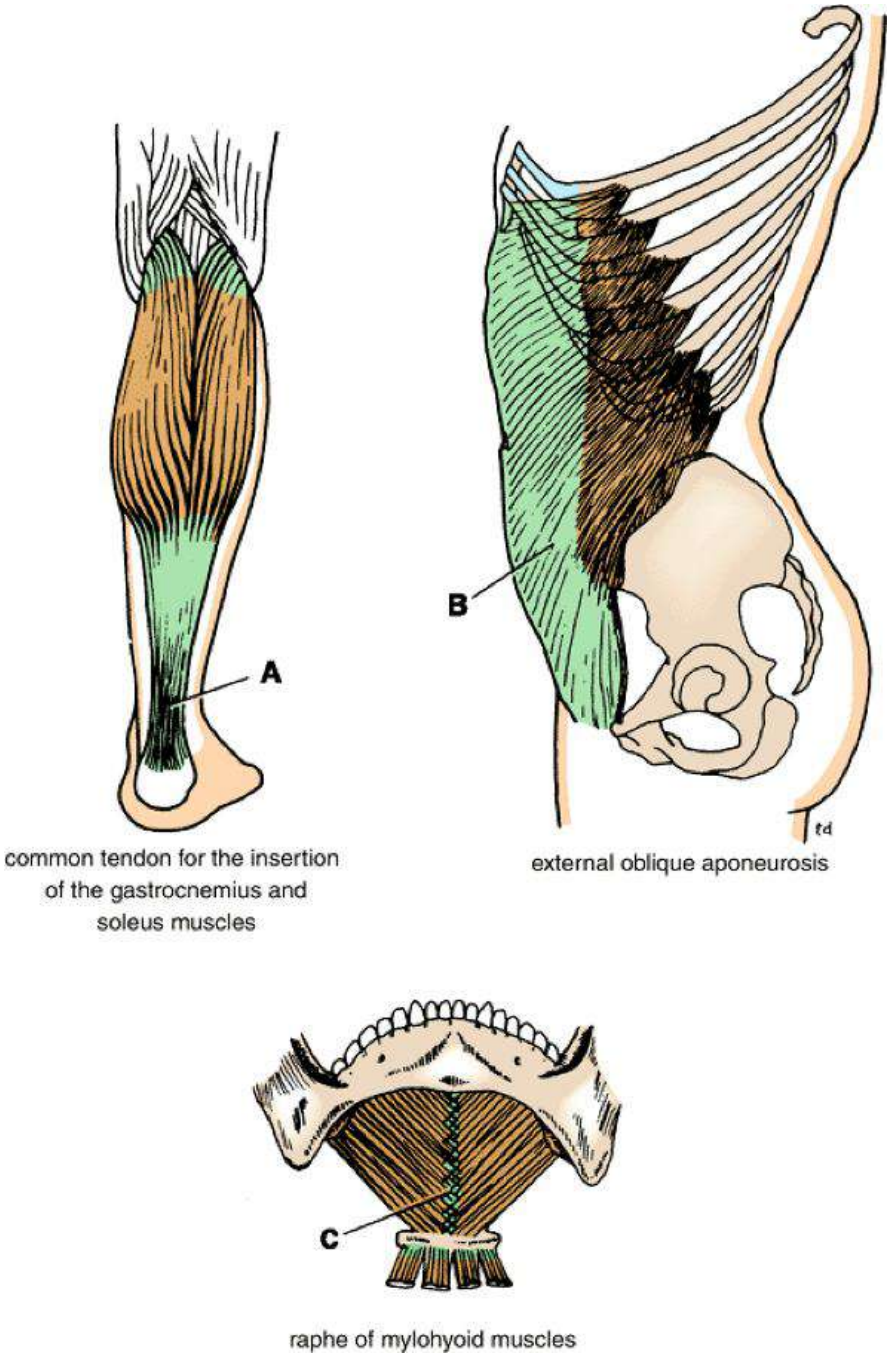


Figure 4: Examples of (A) a tendon, (B) an aponeurosis, and (C) a raphe (Snell, 2019)

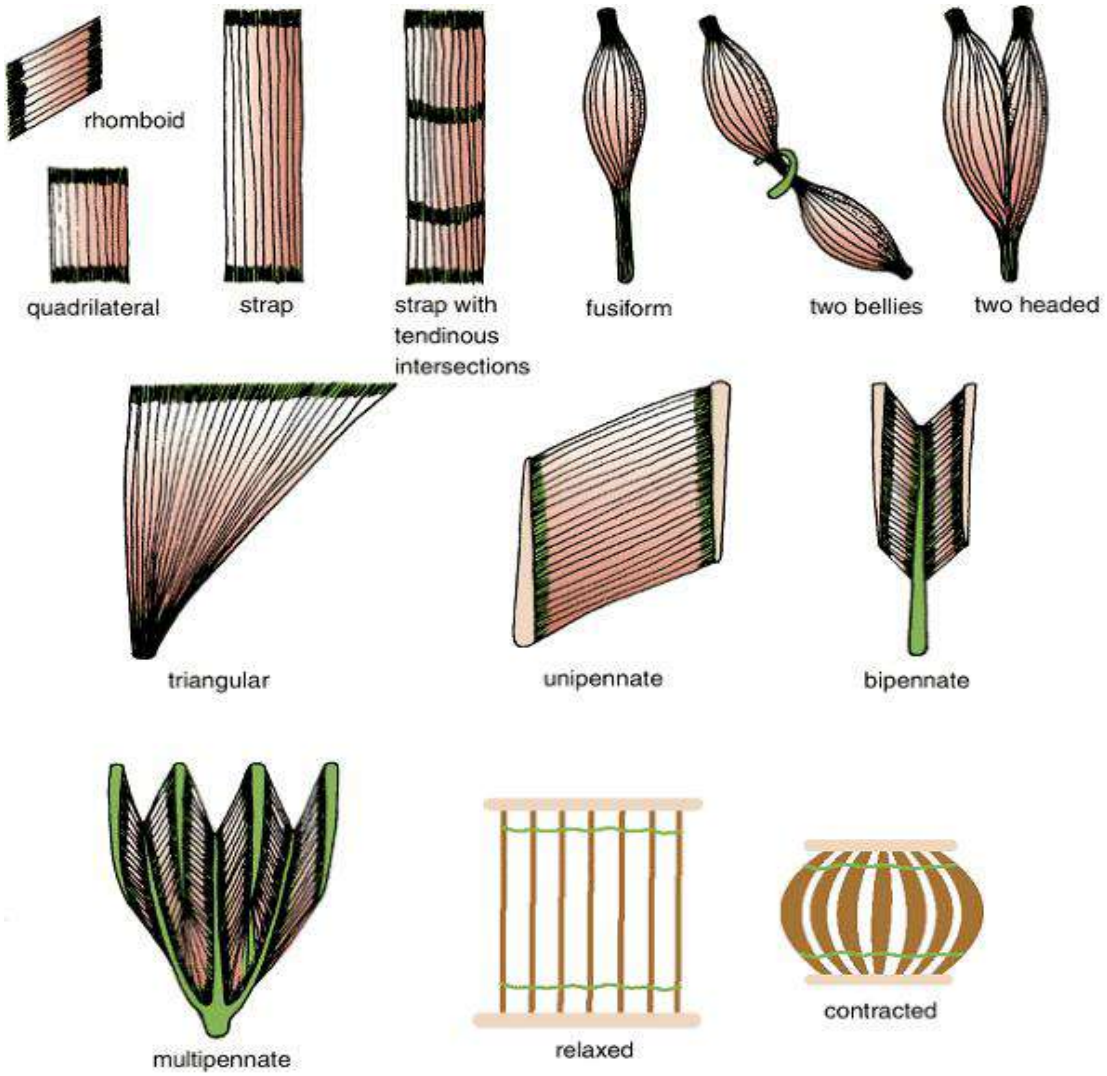


Figure 5: Different forms and action of the internal structure of skeletal muscle (Snell, 2019).

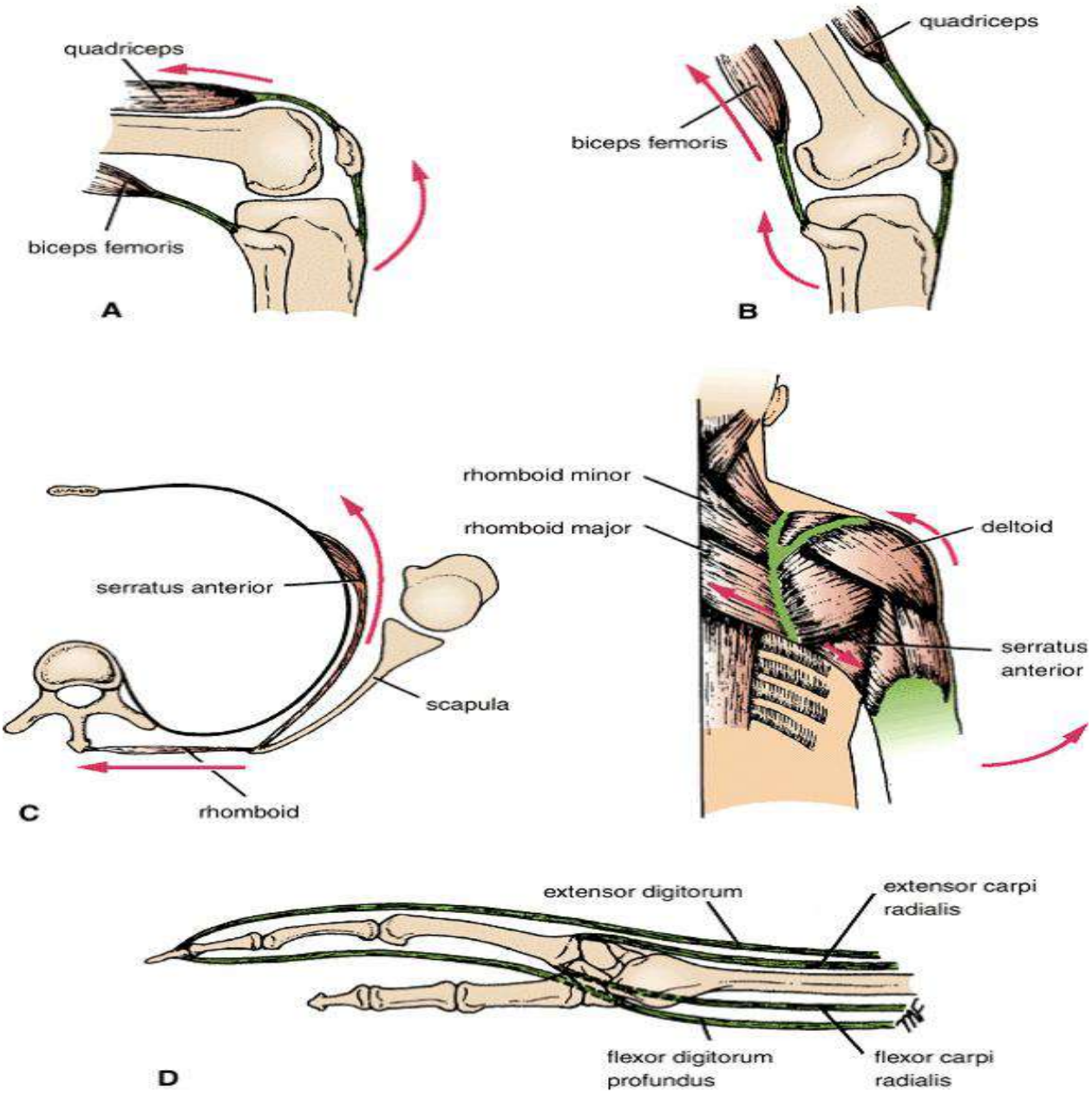


Figure 6: Different types of muscle action (Snell, 2019).

References:

1. Snell, R. S. (2007): Clinical Anatomy by Regions (Anatomy) 8th Edition. Pages 9-12.
2. Moore, Keith L.; Dalley and Arthur F. (2006): Clinically Oriented Anatomy, 5th Edition, Lippincott Williams & Wilkins. Pages 30-38.

Lectures 5 & 6:

SKELATAL SYSTEM

(BONES AND CARTILAGE)

ILOs:

1. Describe anatomical aspects of bones.
2. What are the types of bones with examples?
3. What are the functions of the bones?
4. Describe anatomical aspects of cartilages.
5. What are the types of cartilages with examples?

Bone:

- Living tissue.
- Hard because of the calcification of its extracellular matrix.
- Possesses a degree of elasticity because of the presence of organic fibres.

Function of Bone:

- Has a protective function; the skull and vertebral column; protect the brain and spinal cord from injury; the sternum and ribs protect the thoracic and upper abdominal viscera.
- Serves as a lever, as seen in the long bones of the limbs, and as an important storage area for calcium salts.
- Houses and protects within its cavities the delicate blood-forming bone marrow.

Forms of Bone: 2 forms: compact and cancellous.

1. **Compact bone:** Solid mass
2. **Cancellous bone:** Branching network of trabeculae.

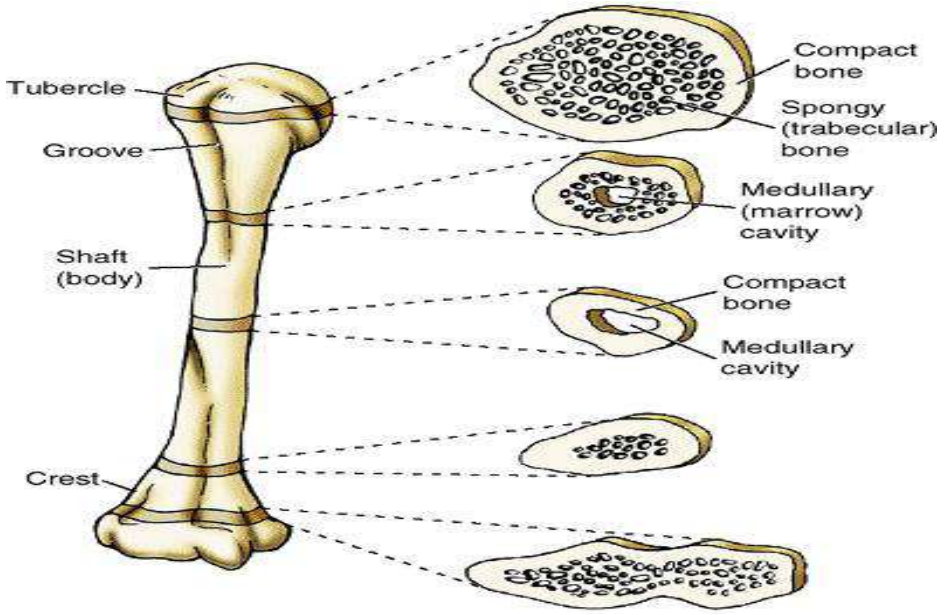


Figure I: Transverse sections of the humerus. Shaft of a living bone is a tube of compact bone that surrounds a medullary cavity containing red or yellow bone marrow or both. (More et al., 2007).

Classification of Bones: According to their general shape:

1. Long Bones:

Long bones are: Humerus, ulna, radius, metacarpals, femur, tibia, fibula, metacarpal, metatarsal and phalanges.

- All present in the limbs.
- Have a shaft, diaphysis, and usually an epiphysis at each end.
- During growing, diaphysis is separated from epiphysis by an epiphyseal cartilage.
- Part of diaphysis adjacent to the epiphyseal cartilage is called the metaphysis.
- The shaft has a central marrow cavity containing bone marrow.
- Outer part of the shaft is composed of compact bone that is covered by a connective tissue sheath, the periosteum.
- The ends of long bones are composed of cancellous bone surrounded by a thin layer of compact bone.

The articular surfaces of the ends of the bones are covered by hyaline cartilage

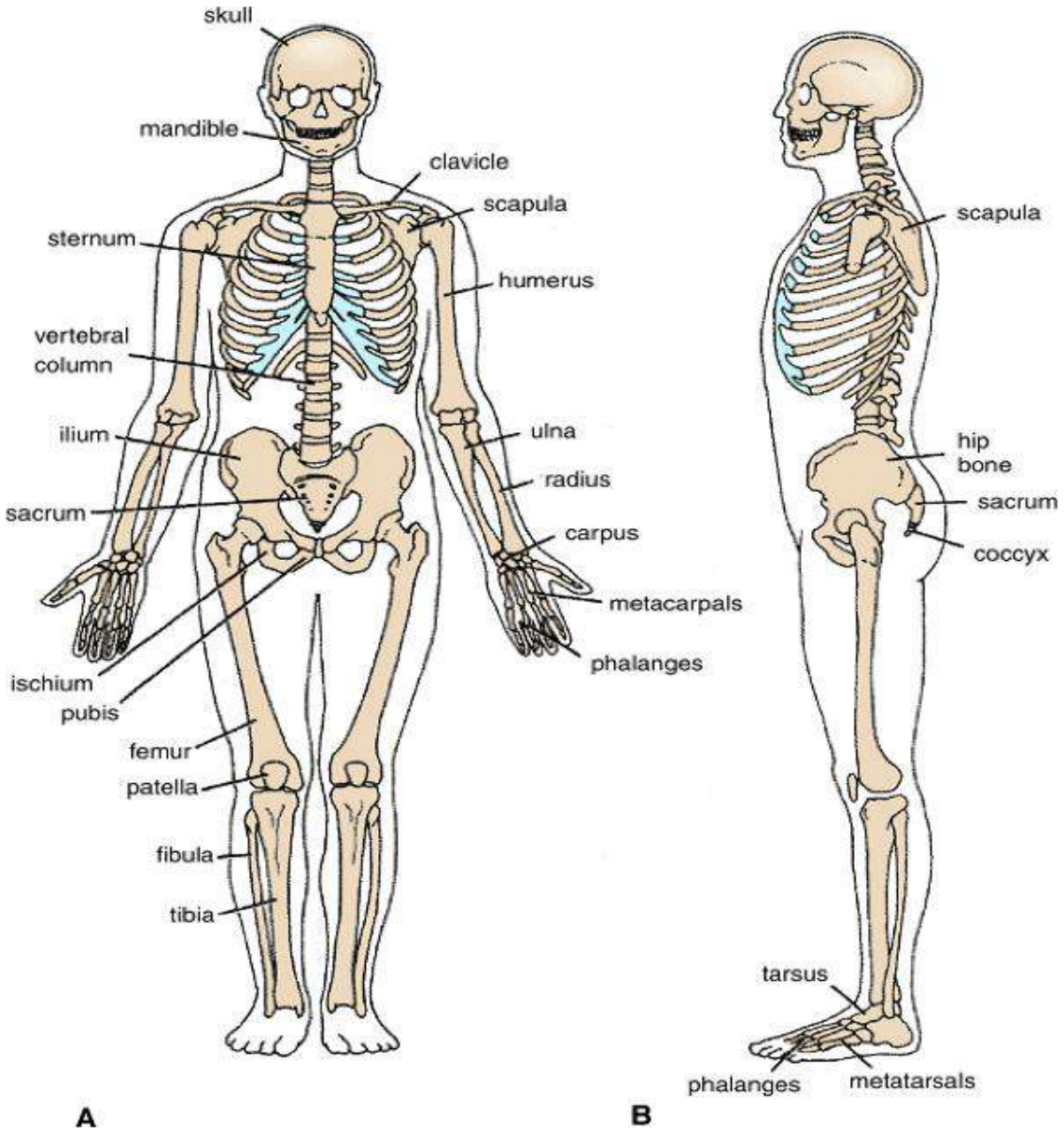


Figure 2: The skeleton. A. Anterior view. B. Lateral view. (Snell, 2019).

2. Short Bones:

Site of short bones: Found in the hand and foot (carpal and tarsal bones).

- **Carpal bones:** scaphoid, lunate, triquetral, pisiform, trapezium, trapezoid, capitate and hamate.
- **Tarsal bones:** calcaneus, talus, navicular, medial cuneiform, intermediate cuneiform, lateral cuneiform and cuboid.

- Composed of cancellous bone surrounded by a layer of compact bone.
- Covered with periosteum.
- The articular surfaces are covered by hyaline cartilage.

3. Flat Bones:

Examples of the flat bones:

1. Vault of the skull (e.g., frontal and parietal bones).
2. Scapulae, although irregular, it is flat.

- Composed of inner and outer layers of compact bone, tables, separated by a layer of cancellous bone, diplo.

4. Irregular Bones:

Examples of irregular bones:

1. Bones of skull.
2. Vertebrae.
3. Hip bones.

- Ill-defined shape due to presence of bony processes and extensions diverge in different directions.
- Thin shell of compact bone with an interior made up of cancellous bone.

5. Sesamoid Bones:

Example of sesamoid bones: Patella (the largest sesamoid bone in the body).

- Small nodules of bone found in certain tendons where they rub over bony surfaces.
- Reduce friction on the tendon; it can also alter the direction of pull of a tendon.

6. Pneumatic Bones:

Pneumatic bones are: Paranasal sinuses (maxillary, frontal, sphenoid and ethmoid air sinuses).

- Found in some bones of the face around the nose (contain small air cavities inside them).

Regional Classification of Bones

Region of Skeleton	Number of Bones
1. Axial skeleton	
Skull	
Cranium	8
Face	14
Auditory ossicles	6
Hyoid	1
Vertebrae (including sacrum and coccyx)	26
Sternum	1
Ribs	24
2. Appendicular skeleton	
Shoulder girdles	
Clavicle	2
Scapula	2
Upper extremities	
Humerus	2
Radius	2
Ulna	2
Carpals	16
Metacarpals	10
Phalanges	28
Pelvic girdle	
Hip bone	2
Lower extremities	
Femur	2
Patella	2
Fibula	2
Tibia	2
Tarsals	14
Metatarsals	10
Phalanges	28
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Blood Supply of Bones:

1. Nutrient artery: Bone marrow, branches in metaphysis (its branches are end arteries).

2. Epiphyseal artery: Epiphysis

3. Metaphyseal artery: Metaphysis

4. Periosteal artery: Periosteum.

Note: Cartilaginous epiphysis hyaline cartilage has NO blood supply.

- As ossification begins, blood vessels penetrate epiphysis ossification centres

Surface Markings of Bones

Bone Marking	Example
Linear elevation	
Line	Superior nuchal line of the occipital bone.
Ridge	Medial and lateral supracondylar ridges of the humerus.
Crest	Iliac crest of the hip bone.
Rounded elevation	
Tubercle	Pubic tubercle.
Protuberance	External occipital protuberance.
Tuberosity	Greater and lesser tuberosities of the humerus.
Malleolus	Medial malleolus of the tibia, lateral malleolus of the fibula.
Trochanter	Greater and lesser trochanters of the femur.
Sharp elevation	
Spine or spinous process	Ischial spine, spine of vertebra.
Styloid process	Styloid process of temporal bone.
Expanded ends for articulation	
Head	Head of humerus, head of femur.
Condyle (knucklelike process)	Medial and lateral condyles of femur.
Epicondyle (a prominence situated above condyle)	Medial and lateral epicondyles of femur.
Small flat area for articulation	
Facet	Facet on head of rib for articulation with vertebral body.
Depressions	
Notch	Greater sciatic notch of hip bone.
Groove or sulcus	Bicipital groove of humerus.
Fossa	Olecranon fossa of humerus, acetabular fossa of hip bone.
Openings	
Fissure	Superior orbital fissure.
Foramen	Infraorbital foramen of the maxilla.
Canal	Carotid canal of temporal bone.
Meatus	External acoustic meatus of temporal bone.

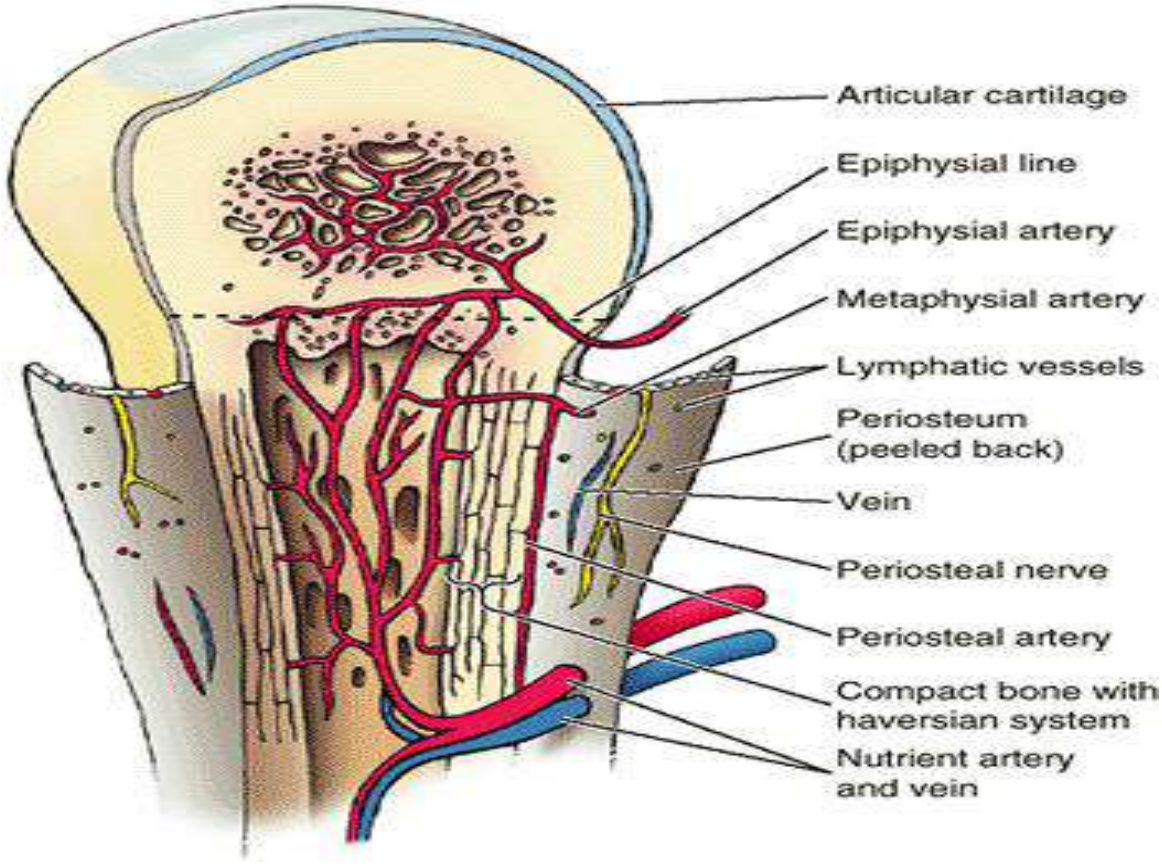


Figure 3: Vasculature and innervation of a long bone. (Moore et al., 2023).

Applied Anatomy of Bones:

Avascular Necrosis:

- Loss of the arterial supply to an epiphysis or other parts of a bone results in the death of bone tissue.

Bottom Line:

Bones grow through the processes of:

1. **Intramembraneous ossification:**

- Mesenchymal bone models are formed during the embryonic and prenatal periods.

2. **Endochondral ossification:**

- Cartilage models are formed during the fetal period, with bone replacing most of the cartilage after birth.

CARTILAGE

Cartilage:

- Connective tissue in which the cells and fibres are embedded in a gel-like matrix, responsible for its firmness and resilience.
- No blood vessels or lymph or nerves (insensitive).

There are three types of cartilage:

1. Hyaline Cartilage:

- Throughout childhood and adolescence, it plays an important part in the growth in length of long bones (epiphyseal plates).
- Covers the articular surfaces of synovial joints.
- Incapable of repairing when fractured.

2. Fibrocartilage:

- Found in the discs within joints (e.g., the temporomandibular joint, sternoclavicular joint, and knee joint) and on the articular surfaces of the clavicle and mandible.

3. Elastic cartilage:

- Possesses large numbers of elastic fibers embedded in matrix.
- Flexible
- Found in the auricle of ear, external auditory meatus, auditory tube, and epiglottis.

References:

1. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins. Pages 30-40.
2. Moore, Keith L.; Dalley and Arthur F. (2008): Clinically Oriented Anatomy, 5th Edition, Lippincott Williams & Wilkins. Pages 19-26.

Lecture 7:

JOINTS

ILOs:

- Outline types of body joints with examples.
- Outline the general joint structures.

Joints:

Definition of Joints:

- Site where two or more bones come together.
- Union or junction between two or more bones or rigid parts of the skeleton

Classification of Joints: According to the tissues that lie between the bones (Figure 1):

- Fibrous joints, cartilaginous joints, and synovial joints.

II. Fibrous Joints:

- Articulating surfaces of the bones are joined by fibrous tissue.
- Very little movement is possible.

Examples of fibrous joints:

1. Sutures of the vault of the skull.
2. Inferior tibiofibular joints.
3. Gomphosis (dentoalveolar syndesmosis).

II. Cartilaginous Joints: Divided into 2 types:

1. Primary cartilaginous joint:

- The bones are united by a plate of hyaline cartilage.
- No movement is possible.

Examples of primary cartilaginous joint:

1. Union between the epiphysis and the diaphysis of a growing bone.
2. Between the first rib and the manubrium sterni.
3. Gomphosis, between the teeth and their sockets.

2. Secondary cartilaginous joint:

- The bones are united by a plate of fibrocartilage.
- Articular surfaces of the bones are covered by a layer of hyaline cartilage.
- Small amount of movement is possible.

Examples of secondary cartilaginous joint:

1. Joints between the vertebral bodies.
2. Symphysis pubis.
3. Manubriosternal joint.

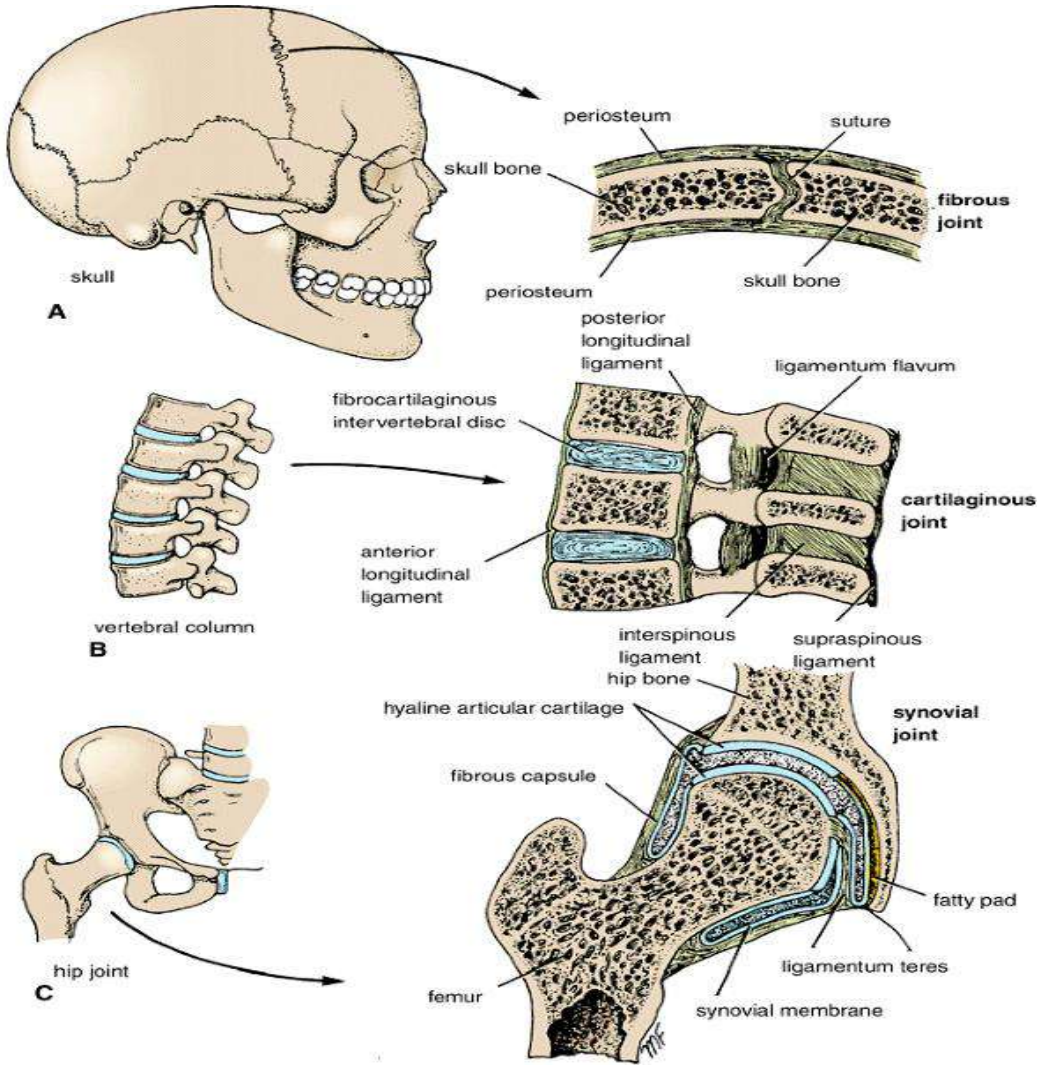


Figure 1: Types of joints. A. Fibrous joint (suture of skull). B. Secondary cartilaginous joint (joint between two vertebral bodies). C. Synovial joint (hip joint). (Snell, 2019).

3. Synovial Joints (Figure 2):

- Permits a great degree of freedom of movement.
- Articular surfaces of the bones are covered by a layer of hyaline cartilage.
- Articular surfaces are separated by a joint cavity.

- Cavity of the joint is lined by a synovial membrane.
- Capsule of the joint, fibrous membrane protects the synovial membrane.
- Synovial fluid, viscous fluid lubricating the articular surfaces.
- Synovial membrane produces the synovial fluid.

Classification of synovial joints (According to shape of articulating surfaces):

1. Plane synovial joints:

- Apposed articular surfaces are flat.
- The bones slide on each other.

Examples of plane joints:

1. Sternoclavicular joint.
2. Acromioclavicular joint.

2. Hinge joints:

- Resemble the hinge on a door.
- Flexion and extension movements are possible.

Examples of hinge joints:

1. Elbow joint.
2. Knee joint
3. Ankle joint.

3. Pivot synovial joints:

- Central bony pivot is surrounded by a bony ligamentous ring.
- Rotation is the only movement possible.

Examples of pivot synovial joints:

1. Atlantoaxial joint.
2. Superior radioulnar joint.

4. Condyloid synovial joints:

- Two distinct convex surfaces that articulate with two concave surfaces.
- Movements are flexion, extension, abduction, and adduction.
- Rotation is possible.

Example of condyloid synovial joints:

- Metacarpophalangeal joints or knuckle joints.

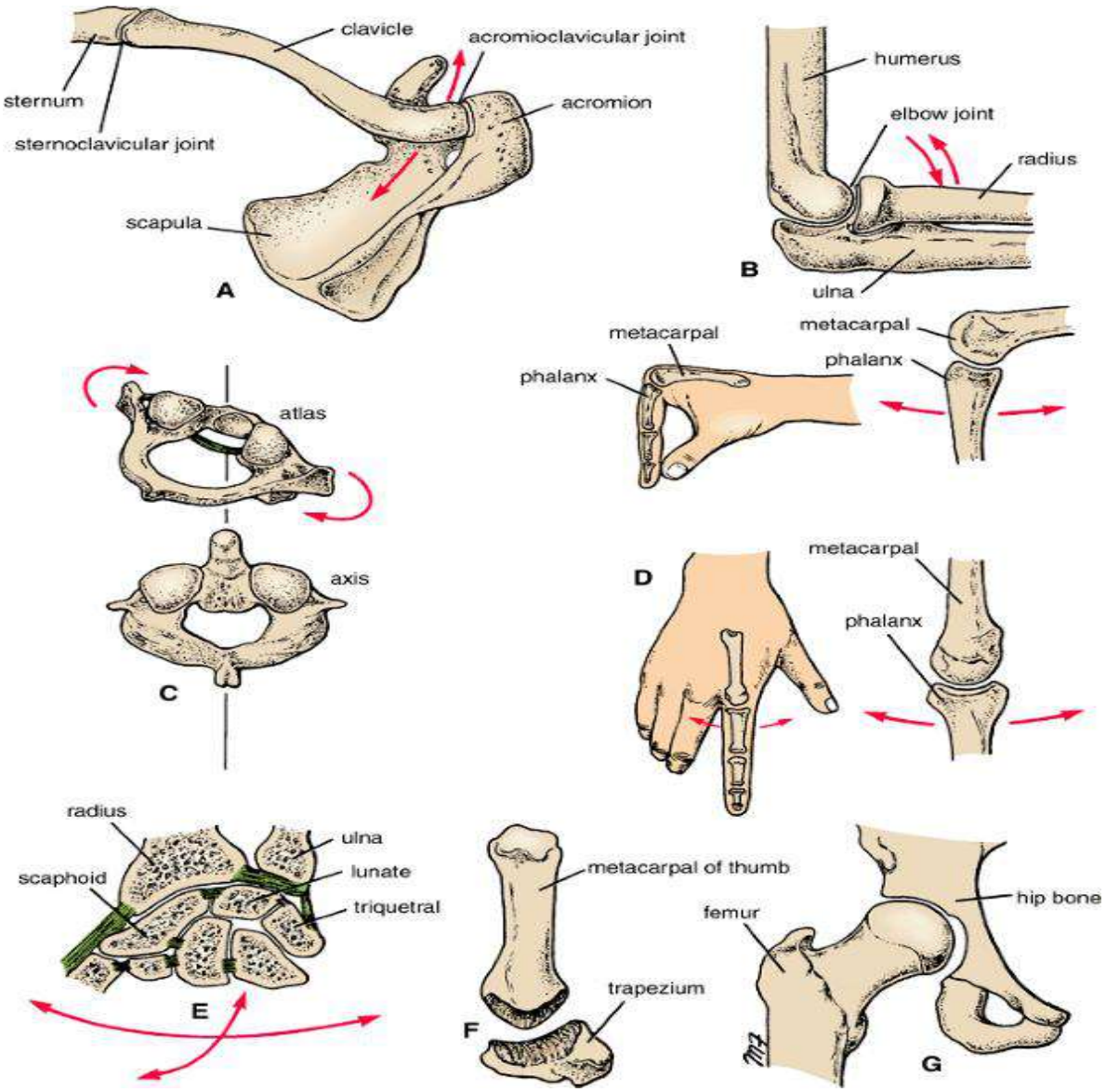


Figure 2: Types of synovial joints with examples. A. Plane joints (sternoclavicular joint). B. Hinge joint (elbow joint). C. Pivot joint (atlantoaxial joint). D. Condyloid joint (metacarpophalangeal joint). E. Ellipsoid joint (wrist joint). F. Saddle joint (carpometacarpal joint of the thumb). G. Ball-and-socket joint (hip joint). (Snell, 2019).

5. Ellipsoid joints:

- Elliptical convex articular surface fits into an elliptical concave articular surface.
- Movements are flexion, extension, abduction, and adduction.

- No rotation.

Examples of ellipsoid synovial joints: Wrist joint.

6. Saddle synovial joints:

- Articular surfaces are reciprocally concavoconvex.
- Resemble a saddle on a horse's back.
- Movements are flexion, extension, abduction, adduction, and rotation.

Example of saddle synovial joints: Carpometacarpal joint of the thumb.

7. Ball-and-socket joints:

- Ball-shaped head of one bone fits into a socket like concavity of another bone.
- Free movements are flexion, extension, abduction, adduction, medial rotation, lateral rotation, and circumduction.

Examples of ball and socket synovial joints:

1. Shoulder joint.
2. Hip joints.

Types of Synovial Joints According to Axis of Movements:

Uni-axial synovial joints:

- Move around single axis, either transverse or longitudinal: *Elbow, ankle joints*.
- Movements occur at these joints are; flexion and extension.

Bi-axial synovial joins:

- Move around two axes*: Wrist and metacarpo-phalangeal joints.*
- Movements occur are flexion & extension and adduction & abduction.

Poly-axial synovial joints:

- Move in three axes: *Shoulder and hip joints*.
- Movements at these joints are flexion and extension & adduction and abduction & medial and lateral rotations.

Arterial Supply of the Joints:

Articular Arteries:

- Arise from vessels around the joint.
- The arteries often anastomose to form networks (periarticular arterial anastomoses), which ensure a continuous blood supply to a joint throughout its range of movement.
- **Articular veins:** Communicating veins that accompany the arteries (venae comitantes), located in the joint capsule, mostly in the synovial membrane.

Nerve Supply of Joints:

- The capsule and ligaments receive many sensory nerve supply.
- Sensory nerve supplying a joint also supplies the muscles moving the joint and the skin overlying the insertions of these muscles (Hilton's law).

Stability of the Joints (Figure 3):

- Depends on 3 main factors: shape, size, and arrangement of the articular surfaces; ligaments; and tone of the muscles around the joint.

1. Articular Surfaces

- Ball-and-socket arrangement of the hip joint and mortise arrangement of the ankle joint are good examples of how bone shape plays an important role in joint stability.

2. Ligaments:

- **Fibrous ligaments**: Prevent excessive movement in a joint, ligaments of arches of foot.
- **Elastic ligaments:** Return to their original length after stretching, ligaments of auditory ossicles.

3. Muscle Tone:

- Resting tension in the muscles that resists passive stretching and maintains posture.
- In most joints, muscle tone is the main factor controlling joint stability.
- Muscle tone of the short muscles around the shoulder joint keeps head of the humerus in the glenoid cavity of the scapula.
- Without the action of these muscles, little force can dislocate this joint.

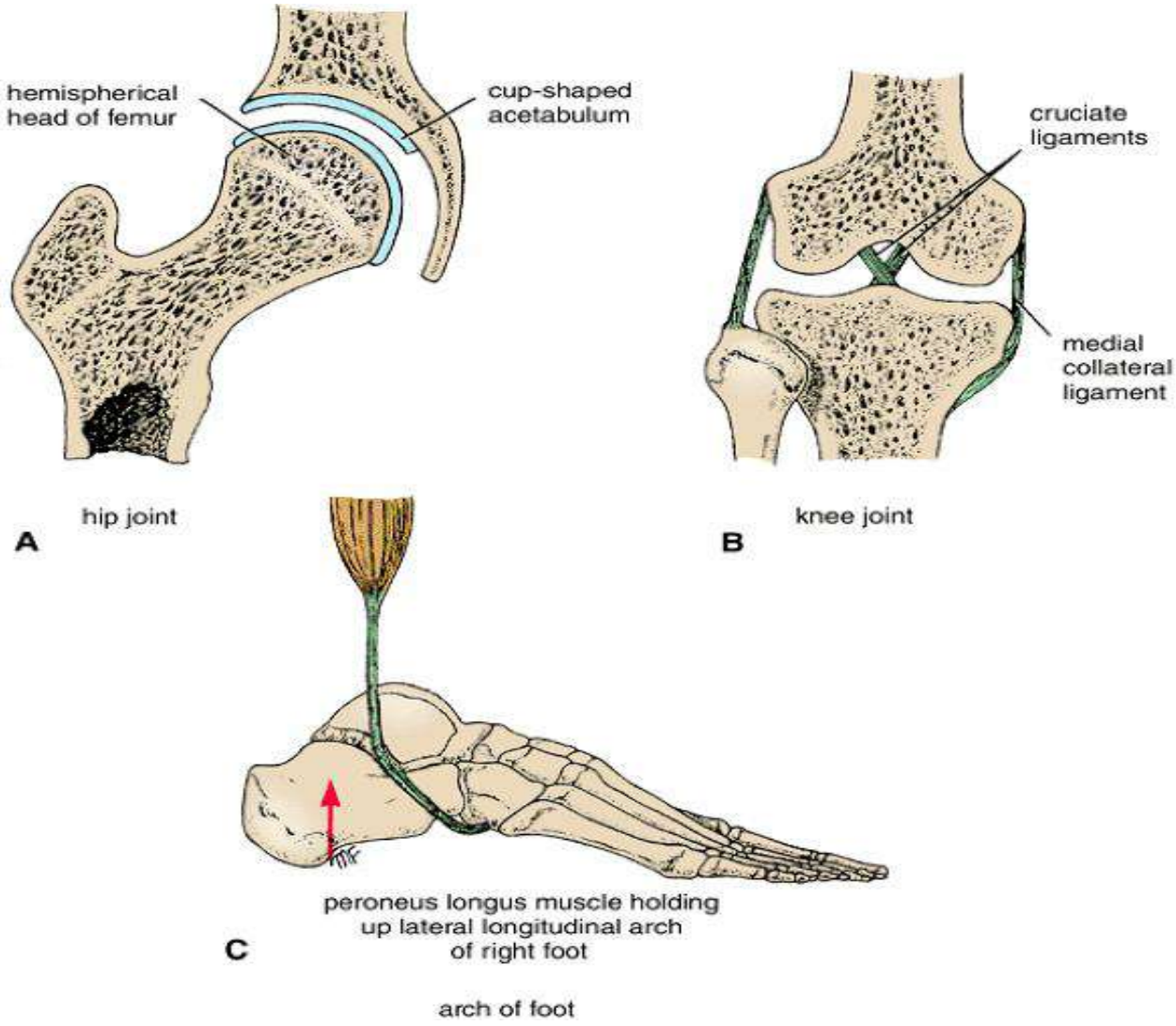


Figure 3: The three main factors responsible for stabilizing a joint. A. Shape of articular surfaces. B. Ligaments. C. Muscle tone. (Snell, 2019).

Bursae (Figure 4):

- Lubricating device consisting of a closed fibrous sac lined with a smooth membrane.
- Its walls are separated by a film of viscous fluid.
- Found wherever tendons rub against bones, ligaments, or other tendons.

Example of closed bursae: Prepatellar bursa.

- Occasionally, the cavity of a bursa communicates with the cavity of a synovial joint.

Example of communicating bursae:

1. Suprapatellar bursa communicates with the knee joint
2. Subscapularis bursa communicates with the shoulder joint.

Synovial Sheath (Figure 4):

- Tubular bursa that surrounds a tendon.
- The tendon invaginates the bursa from one side so that the tendon becomes suspended within the bursa by a mesotendon.
- The mesotendon enables blood vessels to enter the tendon along its course.
- When the range of movement is extensive, the mesotendon may remain in the form of narrow threads, vincula (long flexor tendons of the fingers and toes).
- Synovial sheaths occur where tendons pass under ligaments and retinacula and through osseofibrous tunnels.
- Reduces friction between the tendon and its surrounding structures.

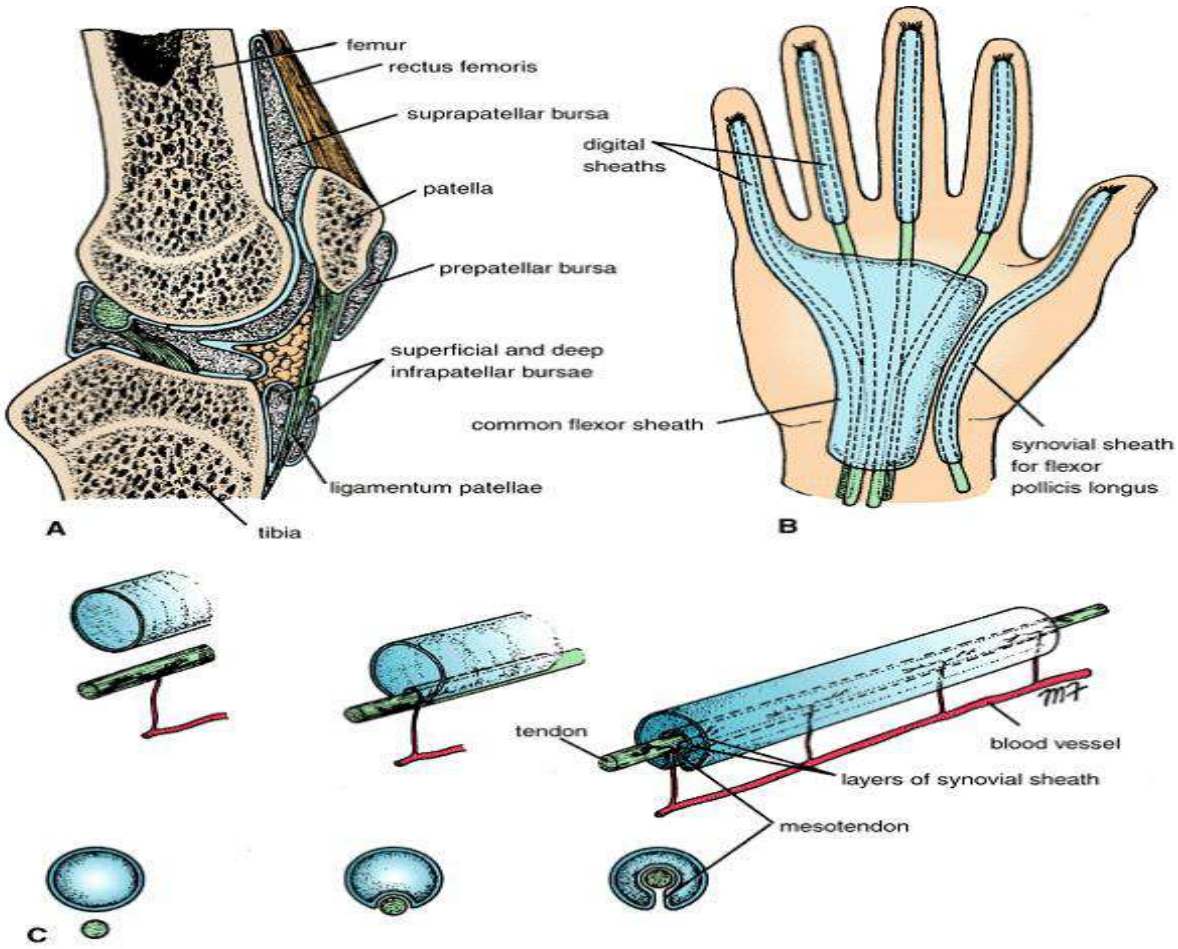


Figure 4: A. Four bursae related to the front of the knee joint. Note that the suprapatellar bursa communicates with the cavity of the joint. B. Synovial sheaths around the long tendons of the fingers. C. How tendon indents synovial sheath during development, and how blood vessels reach the tendon through the mesotendon. (Snell, 2019).

The Bottom Line

- A joint is a union between two or more bones or rigid parts of the skeleton.
- Tere are 3 types of joints: fibrous, cartilaginous, and synovial.
- Synovial joints: The most common type, can be classified into plane, hinge, saddle, condyloid, ball and socket, and pivot.
- Joints receive their blood supply from articular arteries that often form networks.
- Articular veins are usually located in the synovial membrane.
- Joints are richly innervated and transmit the sensation of proprioception, an awareness of movement and position of parts of the body.

References:

3. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins. Pages 12-18.
4. Moore, Keith L.; Dalley and Arthur F. (2008): Clinically Oriented Anatomy, 5th Edition, Lippincott Williams & Wilkins. Pages 26-30.

Lecture 8 & 9:

DIGESTIVE SYSTEM

ILOs:

- Describe general anatomical structures of the digestive system?
- Define the digestive system?
- Enumerate the organs forming the digestive tube?
- What are the salivary glands and mention their function?
- Describe anatomy of the mouth and its structure.
- Describe the pharynx and mention its parts?
- What is the beginning, end and length of the oesophagus?
- Describe anatomy of the stomach and name the structures forming the stomach bed?
- Describe the anatomy of the small intestine?
- Describe anatomy of the large intestine?
- What are differences between the small and large intestine?
- Describe anatomy of the liver, gall bladder and pancreas?

Digestive System (Figure 1):

- It consists of all the organs which participate in the process of digestion that starts with entry of food into the mouth to exit as stools from the anus.

Organs of the digestive System are:

1. Organs of digestive tube: Mouth, pharynx, esophagus, stomach, intestine, anus.

2. Salivary glands: Parotid, submandibular, sublingual.

3. Liver, gall bladder, pancreas.

Mouth (Figure 2): Divided into vestibule and mouth proper.

Lips and cheeks: Composed chiefly of muscles and fat covered with skin and lined with mucous membrane.

Gums: Dense fibrous tissues covered with mucous membrane.

Teeth:

A) Permanent teeth (Figure 3): Full set of adult teeth consists of 32, 8 in each half jaw counting from before backwards are: 2 incisors, 1 canine, 2 premolars, and 3 molars.

B) Deciduous teeth (Primary or milk teeth):

- 20 in number. 5 in each half jaw, counting from before backwards are 2 incisors, 1 canine and 2 molars.
- Eruption of deciduous teeth starts at 6 months of age and is completed at 2 years.
- Eruption of permeant teeth begins at 6 years of age and ends at 12 years.
- The third permanent molar erupts around 18 years (Wisdom tooth).

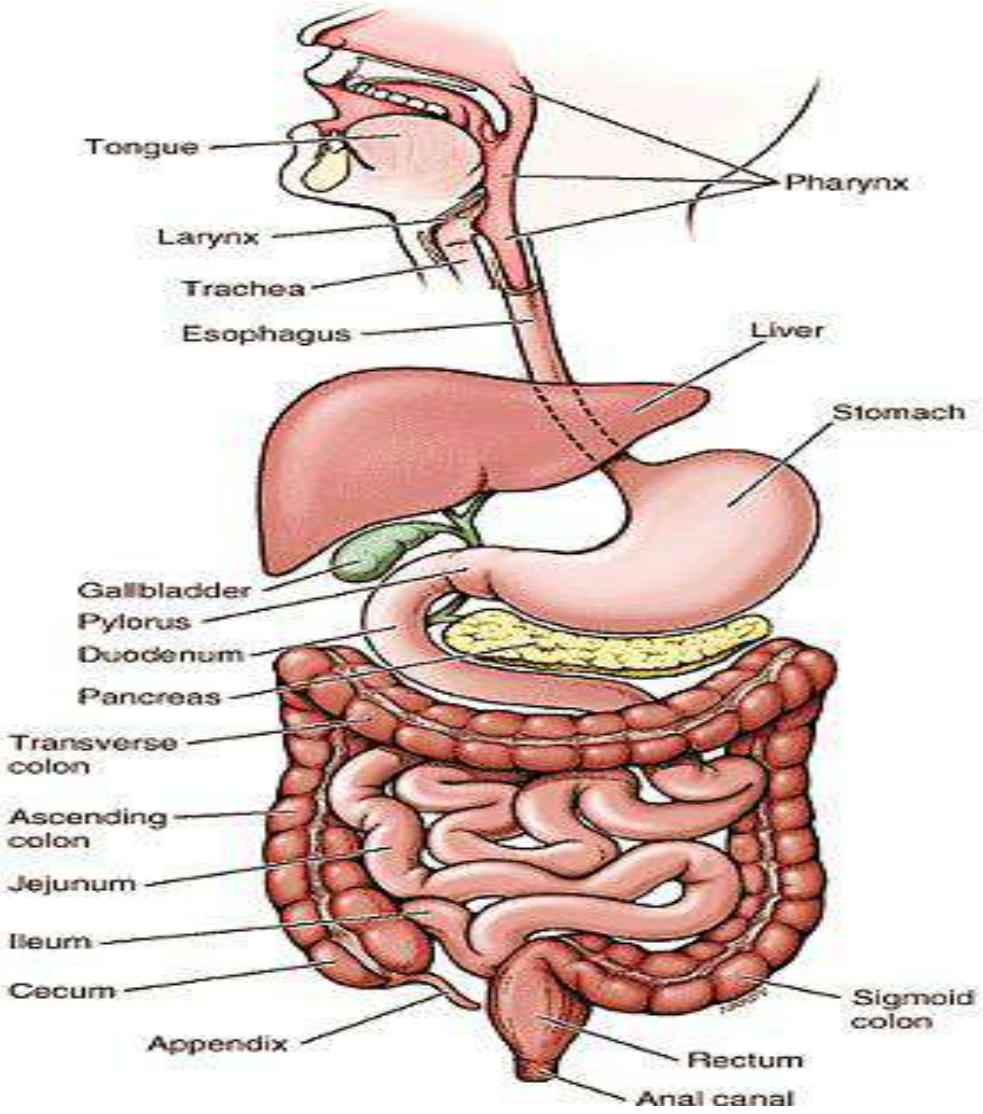


Figure 1: Anatomy of the digestive system. (Drake et al., 2007).

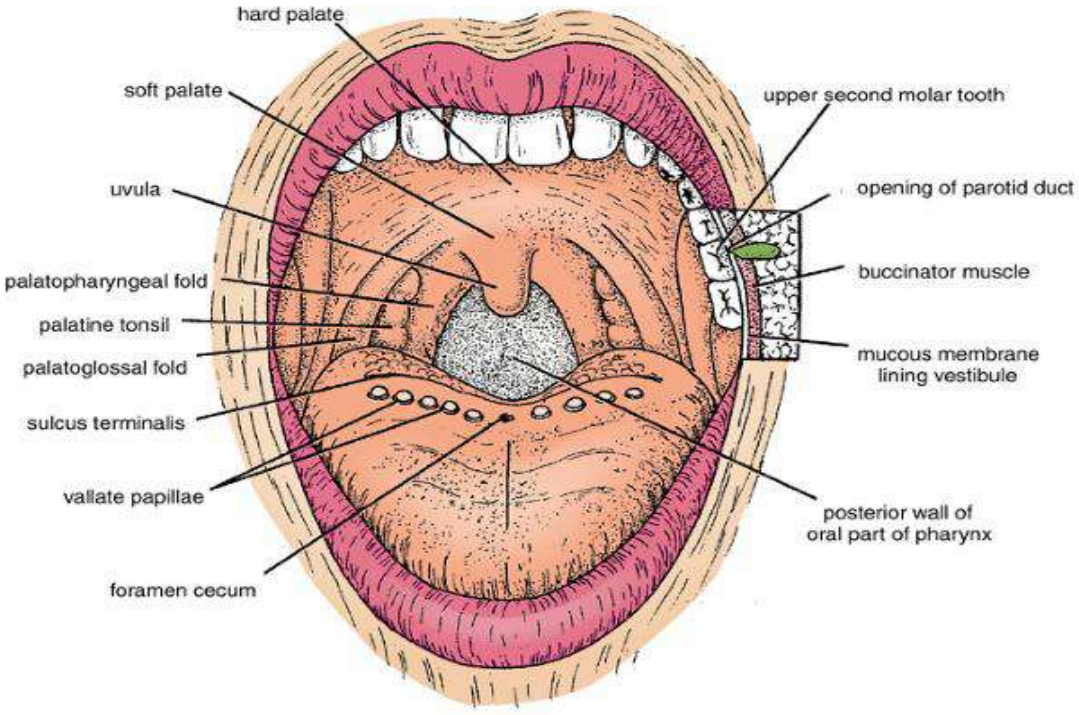


Figure 2: Cavity of the mouth. (Snell, 2019).

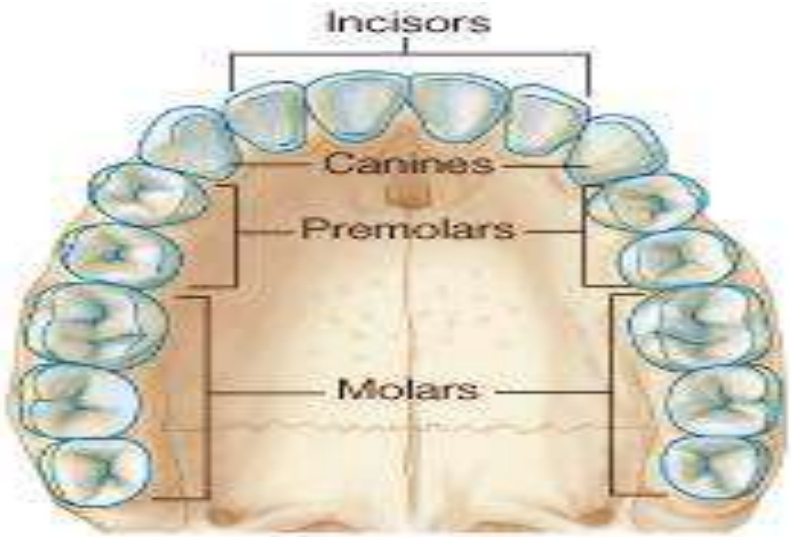


Figure 3: Permanent teeth. (Drake et al., 2007).

Tongue (Figure 4):

- Mobile muscular organ that bulges upwards from floor of the mouth.
- Concerned with mastication, deglutition, speech and taste.
- Mucous membrane of the anterior 2/3 is roughened by presence of papillae that is responsible for taste sensation.

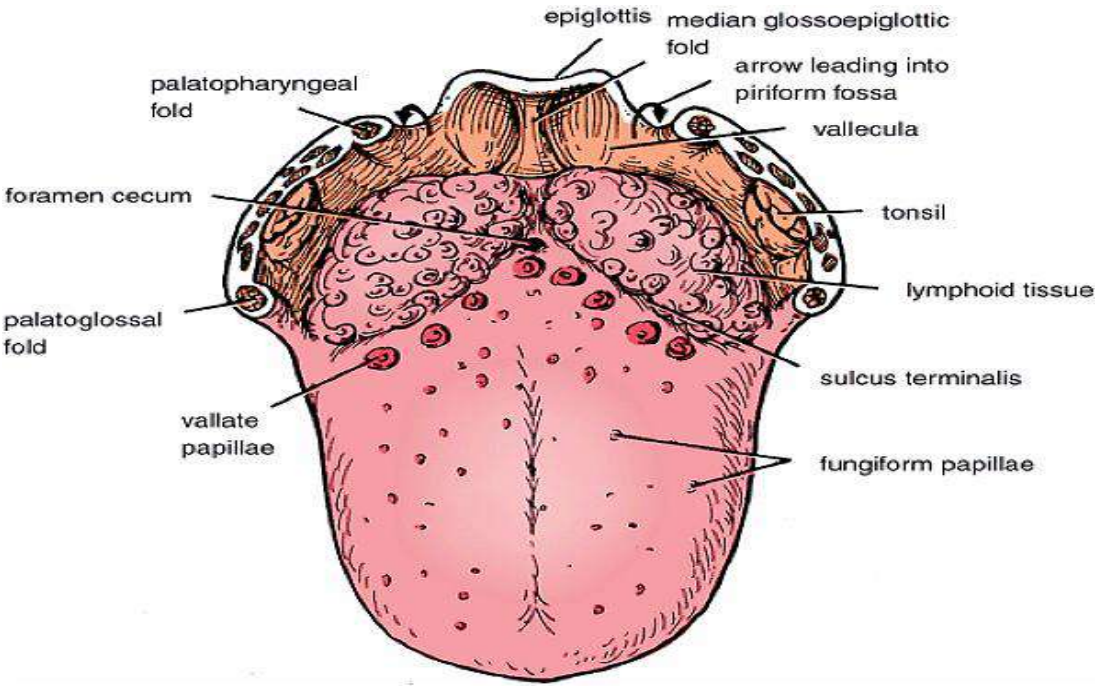


Figure 4: Dorsal surface of the tongue showing the valleculae, the epiglottis, and the entrance into the piriform fossa on each side (arrows). (Snell, 2019)

Salivary Glands:

Salivary glands are: 1. Parotid 2. Submandibular 3. Sublingual

Function of salivary glands: Secretion of saliva which helps in digestion of the food.

PHARYNX:

Divided into (Figure 5):

1. Nasopharynx
2. Oropharynx
3. Laryngo-pharynx

Length: 12-14 cm

Beginning of pharynx: Base of skull

End of pharynx: Continues as esophagus at level of the six cervical vertebra.

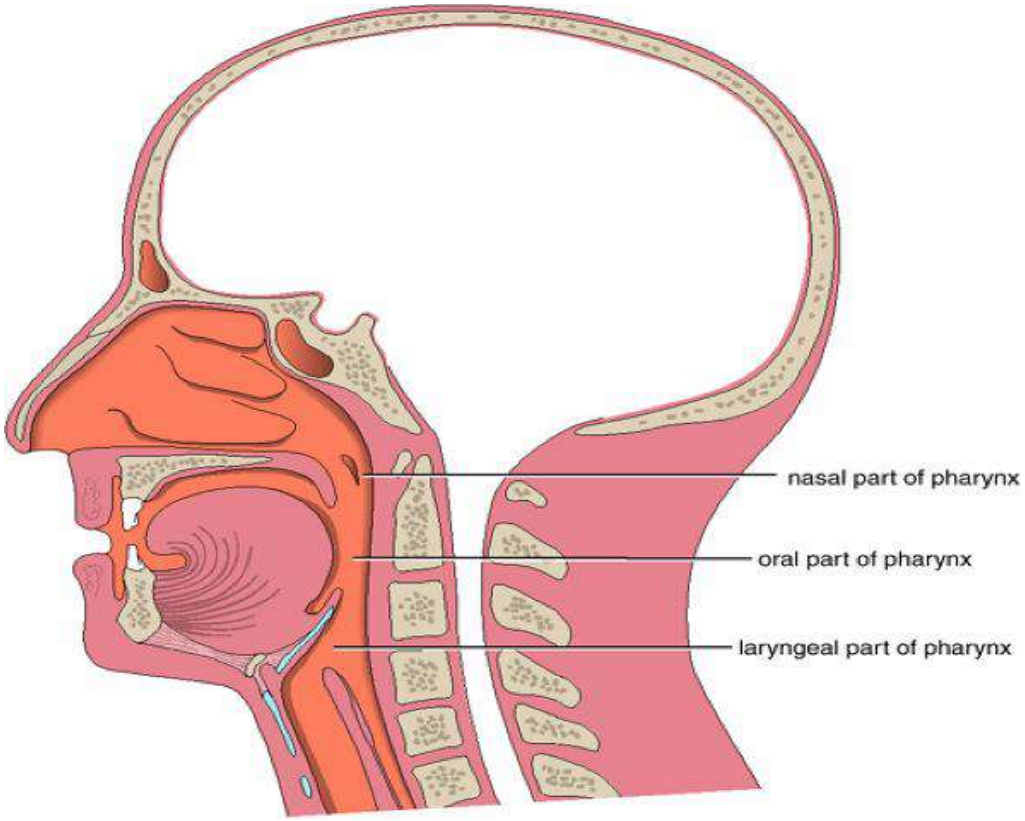


Figure 5: Sagittal section in the head and neck. Parts of the pharynx. (Snell, 2019).

ESOPHAGUS (Figure 6):

Length: 25 cm

Beginning: Continuation of pharynx at level of the six cervical vertebra.

Ends: Cardiac opening of the stomach.

- At level of the eleventh thoracic vertebra.

STOMACH:

Stomach has (Figure 7):

- **2 Openings:** Cardiac and pyloric orifices.
- **2 curvatures:** Greater and lesser curvatures.
- **2 surfaces:** Anterior and posterior surfaces

Parts of the stomach:

- **Fundus:** Dome-shaped, projects upward to left of cardiac orifice, usually fill with air.
- **Body:** Extends from level of cardiac orifice to level of incisura angularis.
- **Pyloric antrum:** From incisura angularis to pylorus.

- **Pylorus:** Tubular part of the stomach that has a thick muscular wall (pyloric sphincter) and its cavity is called pyloric canal.

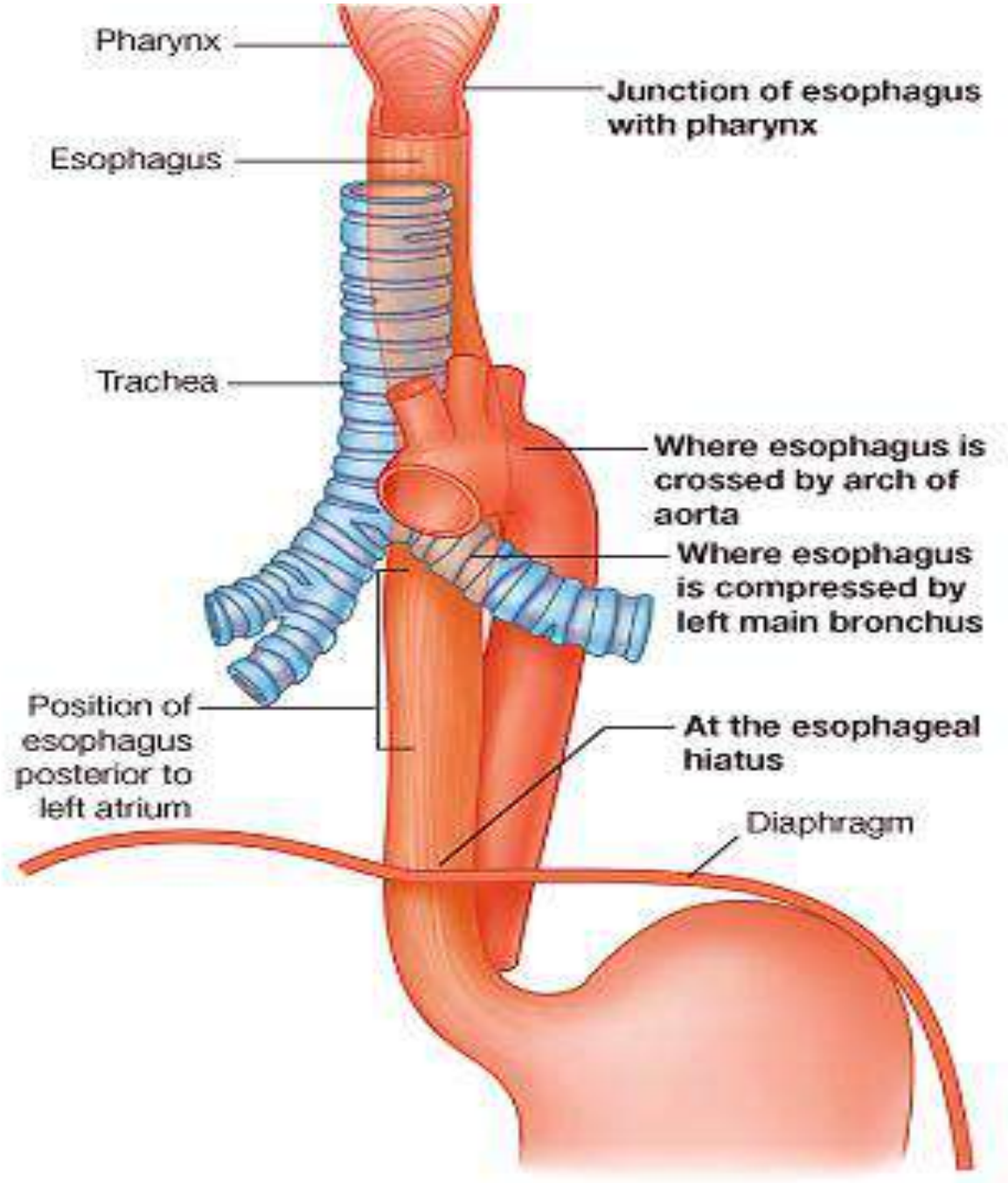


Figure 6: Beginning and end of esophagus and main relations.

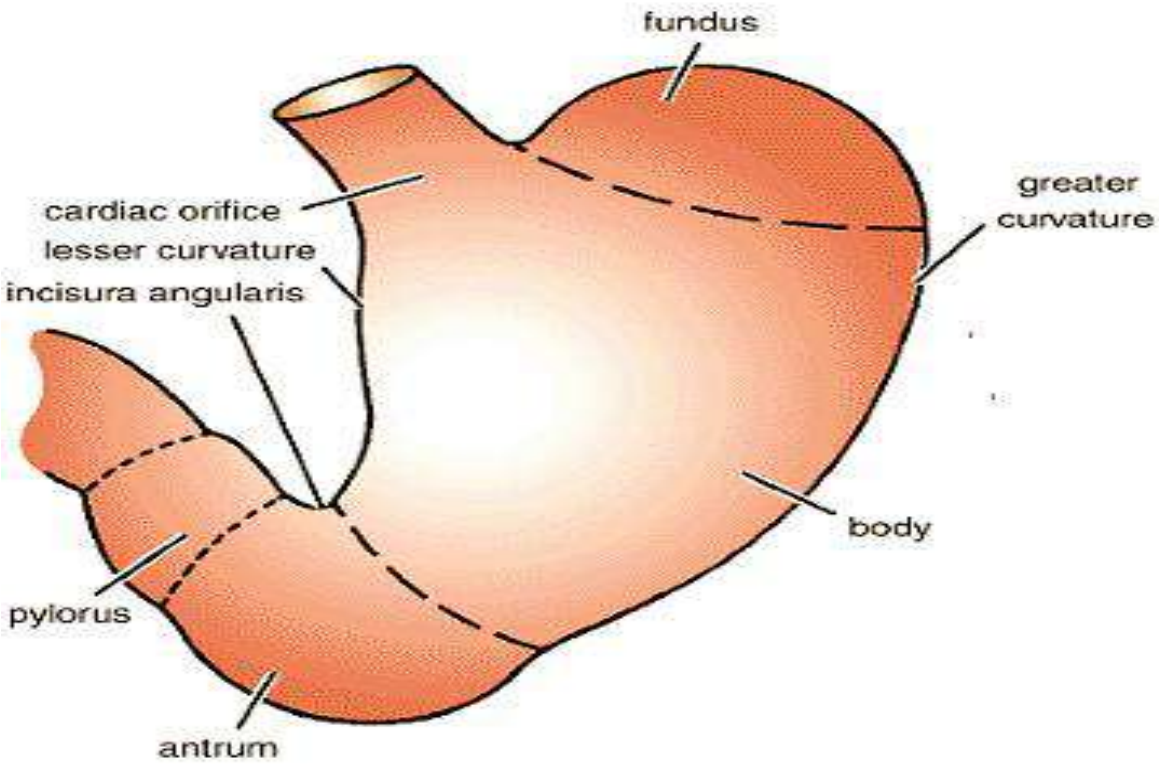


Figure 7: Parts of the stomach. (Snell, 2019).

Posterior relations of the stomach (stomach bed) (Figure 8):

1. Lesser sac of peritoneum.
2. Diaphragm.
3. Spleen.
4. Left suprarenal gland.
5. Left kidney.
6. Splenic artery.
7. Pancreas.
8. Transverse mesocolon.
9. Transverse colon.
10. Greater sac of peritoneum.

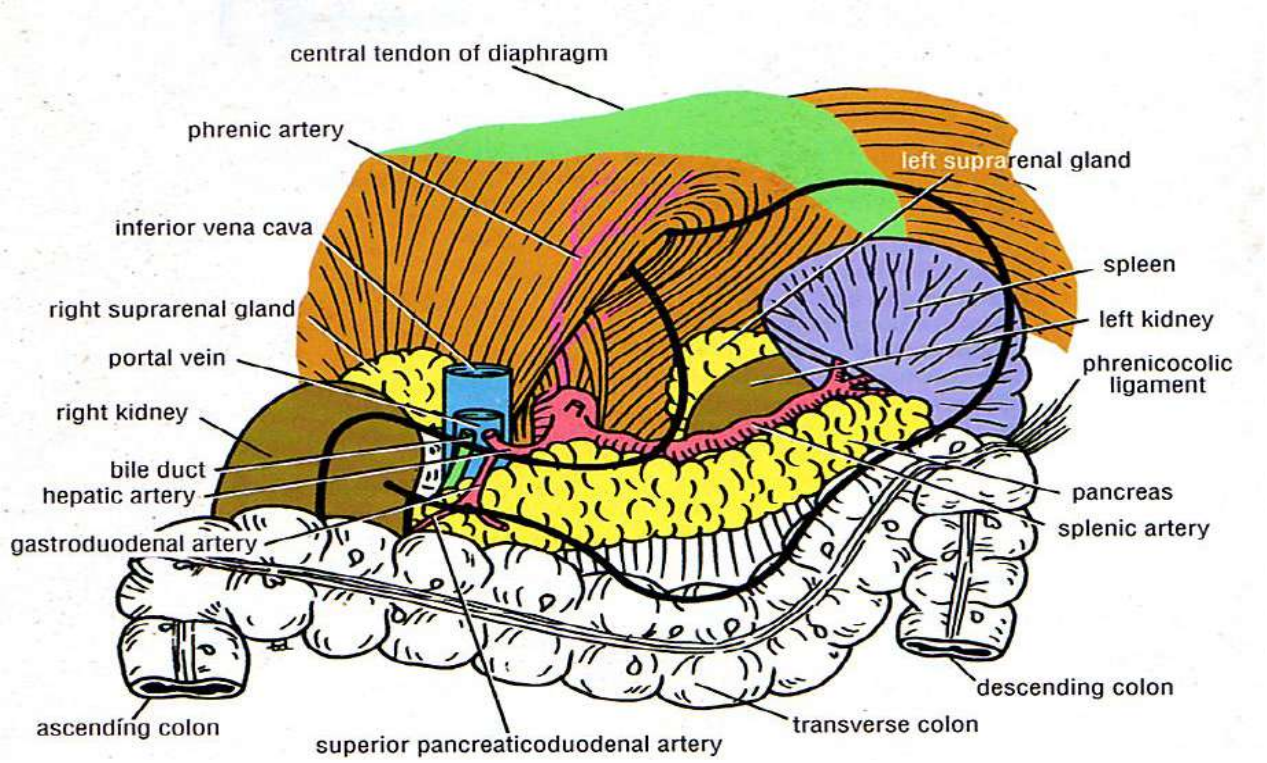


Figure 8: Posterior relations of the stomach (stomach bed). (Snell, 2019).

SMALL INTESTINE:

- Concerned with completion of digestion and absorption of digested food.

Formed of: 1. Duodenum 2. Jejunum 3. Ileum

Length: About 7 meters long.

DUODENUM

- C-shaped tube about 10 inches (25 Cm) long.
- Joins stomach to jejunum.
- Situated in epigastric and umbilical regions.
- Curves around the head of the pancreas
- **Formed of 4 parts:** First, second, third and fourth parts.
- Main pancreatic duct and common bile open into second part in major duodenal papilla.
- Accessory pancreatic duct into second part of duodenum in minor duodenal papilla.

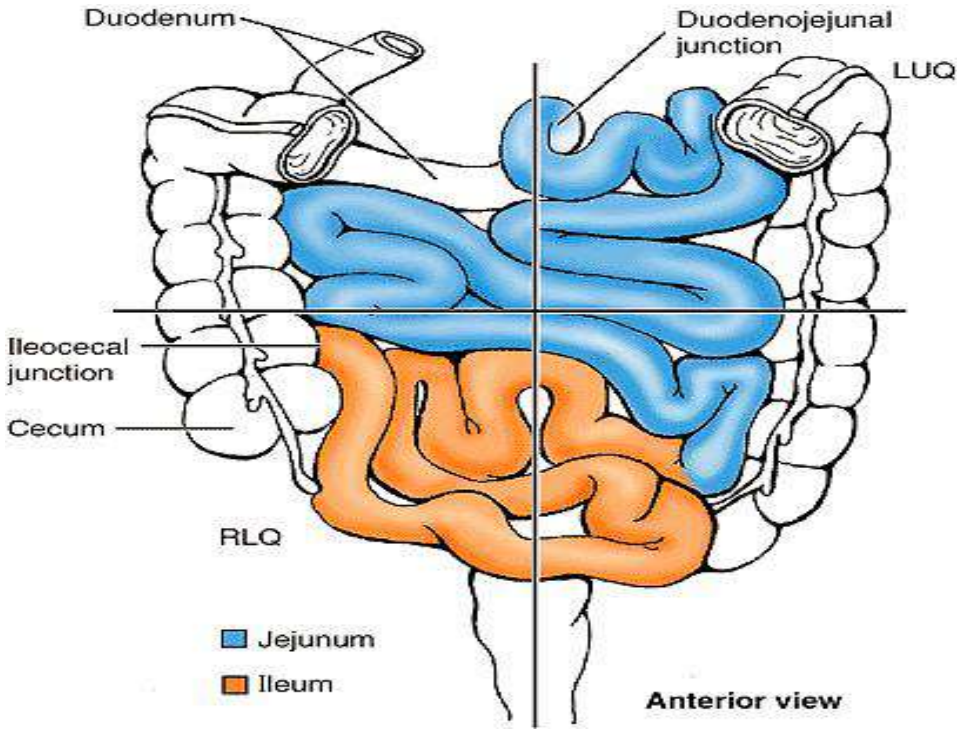


Figure 9: Small intestine; duodenum, jejunum and ilium. (Moore et al, 2023).

LARGE INTESTINE (Figure 10):

- **About:** 1.5 meters.
- **Beginning:** Ileocecal junction.
- **Ends:** Anal orifice.

Divided into:

1. Caecum.
2. Vermiform appendix.
3. Ascending colon.
4. Transverse colon.
5. Descending colon,
6. Sigmoid colon (pelvic colon).
7. Rectum.
8. Anal canal.

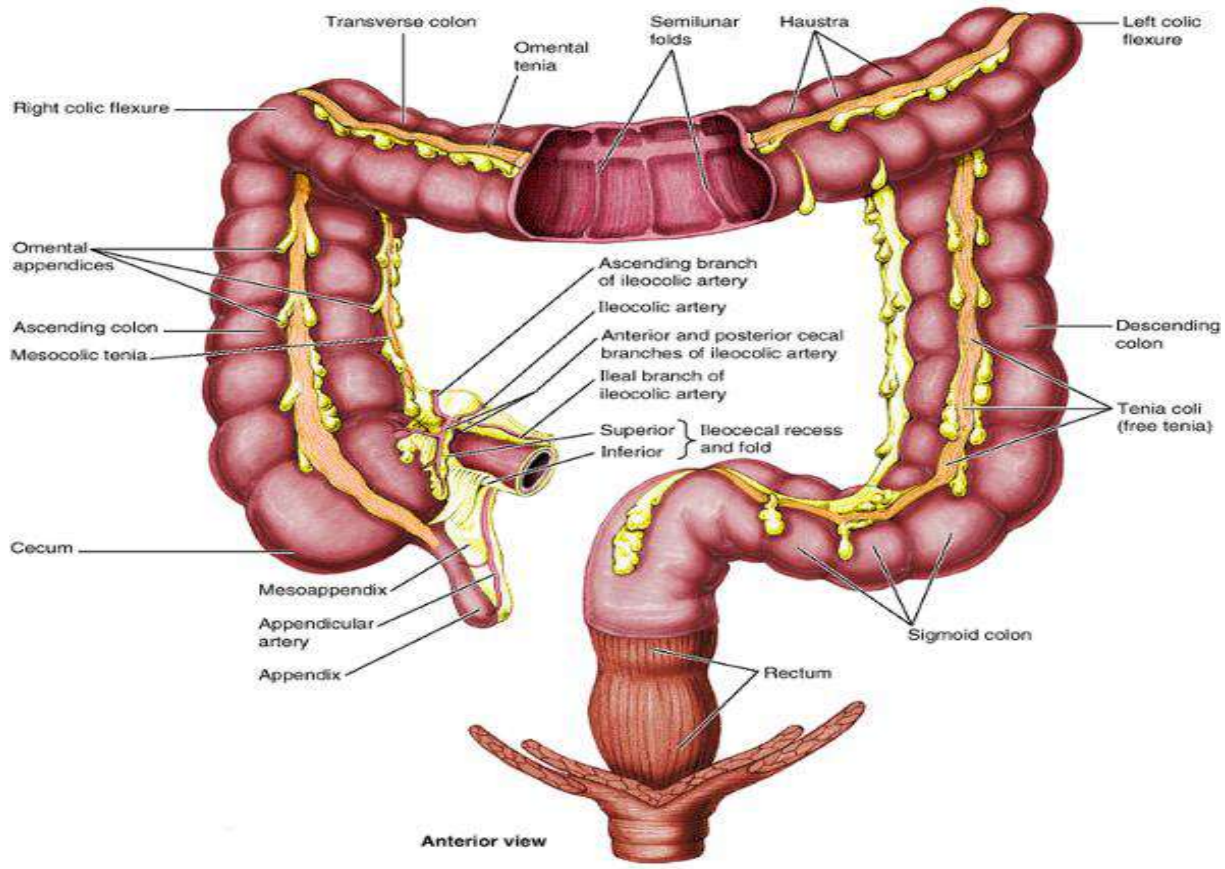


Figure 10: Large intestine. (Moore et al, 2023).

LIVER:

- The largest gland in the body.

Function of the liver:

1. Concerned with metabolism of the absorbed food.
2. Detoxification of chemicals
3. Secretion of bile salts which emerge from the liver through right and left hepatic ducts.

Site of the liver: Lies under the right copula of the diaphragm.

Formed of (Figure 11): 2 large lobes right and left, and 2 small lobes caudate and quadrate lobes which are present on under surface of liver.

- Caudate and quadrate lobes are separated by (porta hepatis) hilum of the liver.

Differences between the small and large intestine

	Small intestine	Large intestine
Length	About long 6-7 meters	About long 1.5 meters
Caliber	Smaller	Larger
Mobility	Freely mobile	Less freely mobile
Teniae coli (3 bands of longitudinal muscles)	Absent	Present
Sacculations	Absent	Present
Appendices epiplica (fat)	Absent	Present over free surface of colon except caecum, appendix and rectum
Mucous membrane	Has permanent folds, called plicae circulares	Absent
Villi in mucosa	Present	Absent
Peyer's patches (lymphatic aggregations)	Present	Absent

Porta Hepatis (Figure 12):

- Hilum of the liver.
- Found on postero-inferior surface of liver.
- Lies between caudate and quadrate lobes.

Contents of porta hepatis:

1. Hepatic bile ducts.
2. Hepatic artery.
3. Portal vein.
4. Sympathetic and parasympathetic nerve fibres.
5. Hepatic lymph nodes.

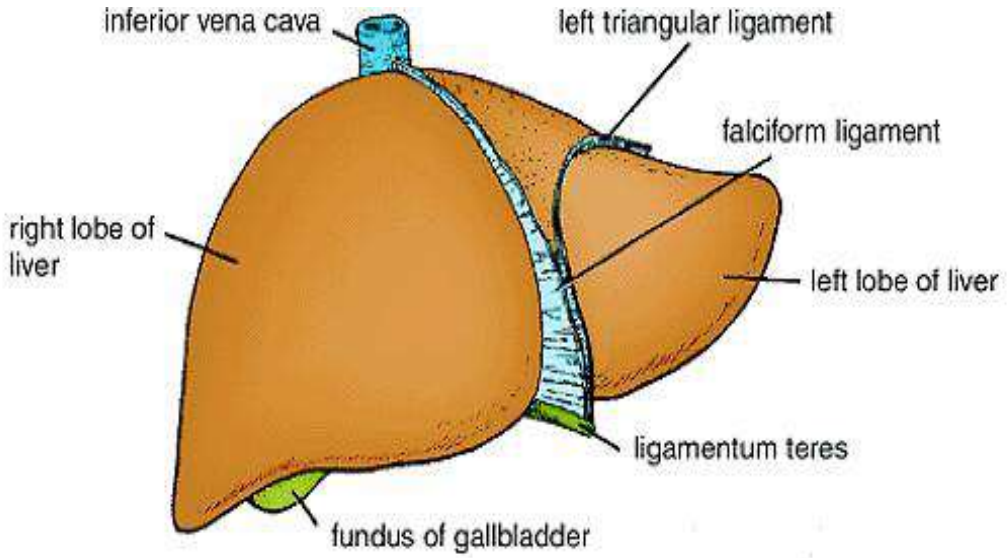


Figure 11: Liver as seen from in front (2 large lobes). (Snell, 2019)

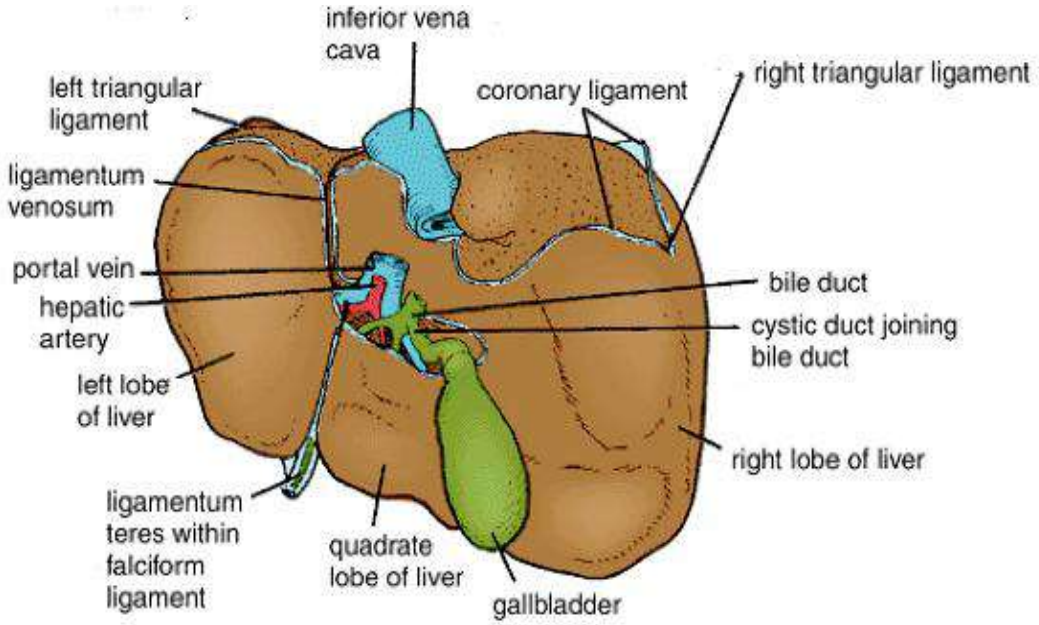


Figure 12: Liver as seen from behind (4 lobes; 2 large and 2 small). (Snell, 2019).

EXTRA HEPATIC BILIARY PASSAGES

Consists of: 1. Right hepatic duct 2. Left hepatic duct 3. Common hepatic 4. Cystic duct 5. Bile duct.

Note:

- Right and left hepatic ducts unite to form common hepatic duct.
- Common hepatic duct unites with cystic duct to form common bile duct.
- Common bile duct opens into the second part of duodenum with main pancreatic ducr.

GALL BLADDER (Figure 13):

- ❖ Pear-shaped sac

Site of gall bladder: Under surface of the liver

Capacity of gall bladder:

- ❖ 30-50 ml
- ❖ Stores bile
- ❖ Bile concentrates by absorbing water

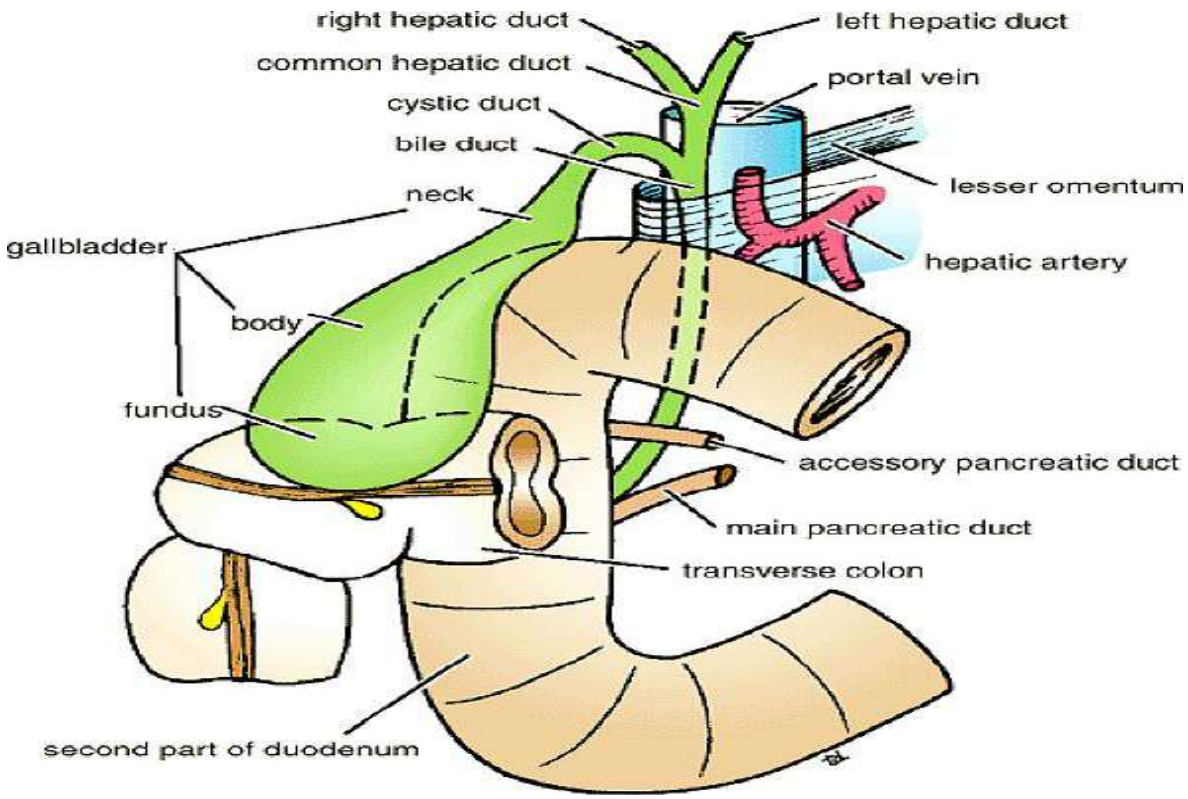


Figure 13: Extrahepatic biliary passages with gall bladder. (Snell, 2019).

PANCREAS (Figure 14):

- Both an exocrine and an endocrine gland.
- Exocrine portion produces a secretion that contains enzymes capable of hydrolyzing proteins, fats, and carbohydrates.
- Endocrine portion, pancreatic islets (islets of Langerhans), produces the hormones insulin and glucagon, which play a key role in carbohydrate metabolism.
- Soft and lobulated and situated on posterior abdominal wall behind the peritoneum.
- Divided into a head, neck, body, and tail.
- Has a duct opens into the second part of duodenum.

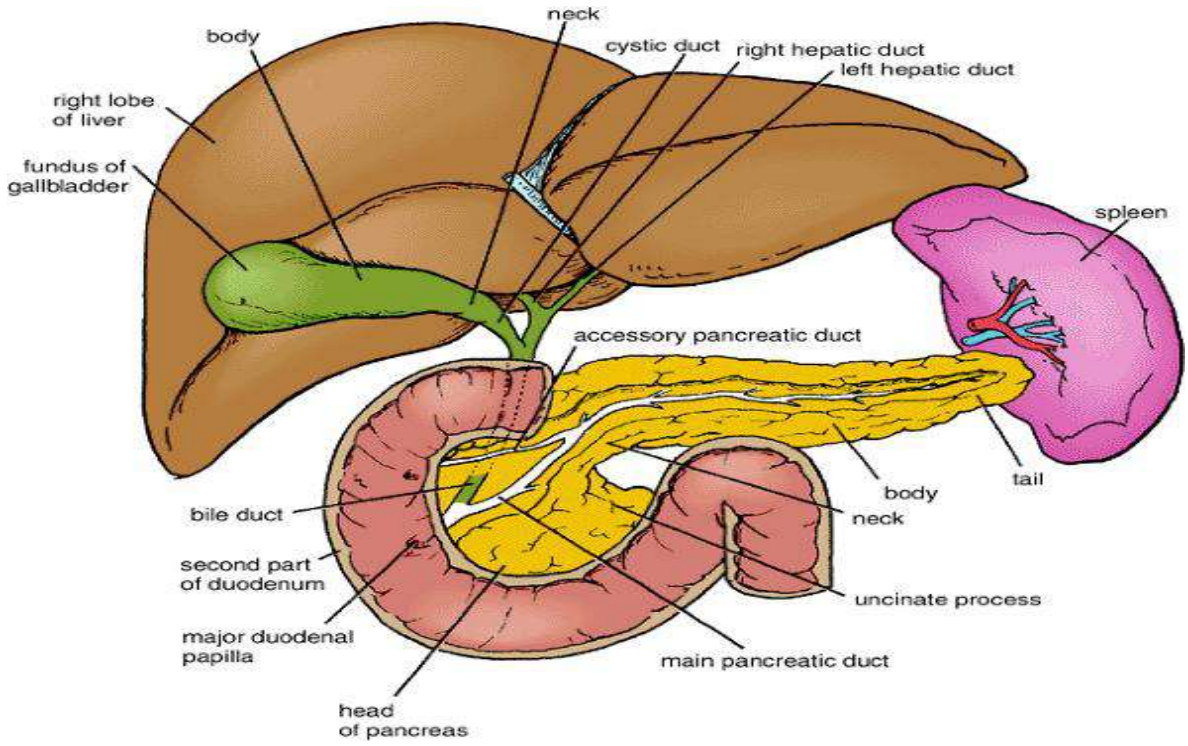


Figure 14: Pancreas. (Snell, 2019).

References:

1. Snell, R. S. (2007): Clinical Anatomy by Regions (Anatomy) 8th Edition.
2. Moore, Keith L. and Dalley A. F. (2008): Clinically Oriented Anatomy, 5th Edition, Lippincott Williams & Wilkins.
3. Drake L. R., Vogl W. and Mitchell A. W. M. (2007): Gray's Anatomy for Students.
4. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021).

Lecture 10:

RESPIRATORY SYSTEM

ILOs:

1. Describe the normal anatomy of the nasal cavity.
2. Describe the normal anatomy of the pharynx
3. Describe the normal anatomy of the larynx.
4. Describe the normal anatomy of the trachea
5. Describe the normal anatomy and differences between the right and left bronchi
6. Describe the normal anatomy and differences between the right and left lungs.

RESPIRATORY SYSTEM:

Definition of the respiratory system:

- A network of organs and tissues responsible for gas exchange in the body.

The respiratory system has two portions:

I. Conducting

II. Respiratory

I. Conducting portion (air passages) comprises:

1. Nasal cavities.
2. Pharynx.
3. Larynx.
4. Trachea.
5. Bronchi.
6. Bronchioles

II. Respiratory portion comprises: The lung which contains:

1. Respiratory bronchioles
2. Alveolar ductules
3. Alveoli.

NASAL CAVITIES:

- Right and a left are separated from each other by a median partition, nasal septum.

Function of the nose plays an important role in:

1. Sense of smell
2. Moistens inhaled air
3. Warms inhaled air
4. Prevents particulate matter from entering the airways.

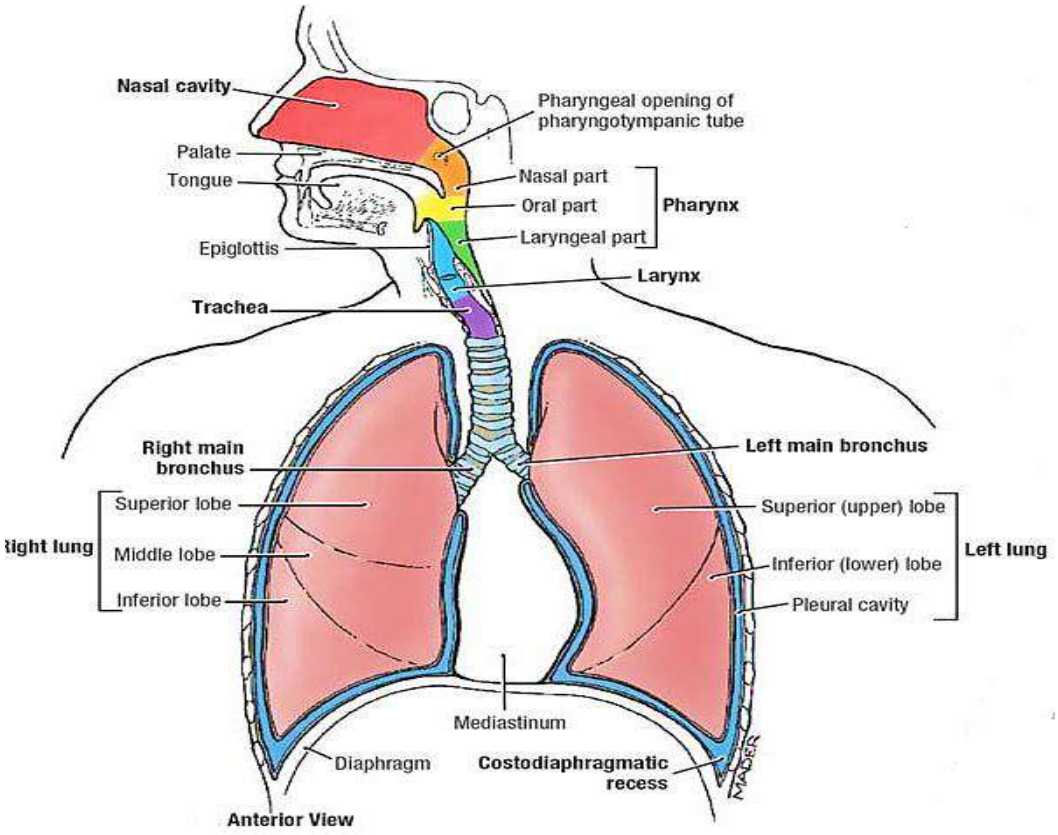


Figure 1: Structure of the respiratory system. (Agur and Dalley, 2009).

PHARYNX:

- Fibromuscular chamber 12-14 Cm long.
- Attached above to the base of the skull.
- Continuous below with the oesophagus at level of C 6.
- Communicating with it from front are the nasal (nasopharynx), oral (oropharynx), and laryngeal (laryngopharynx) cavities.

LARYNX (VOICE BOX):

Beginning: Opens at the lowest part of the pharynx at level of C3.

End: Continuous below with the trachea at level of C 6.

- Kept rigid by 9 cartilages which are united by membranes.
- The vocal cords lie inside it.

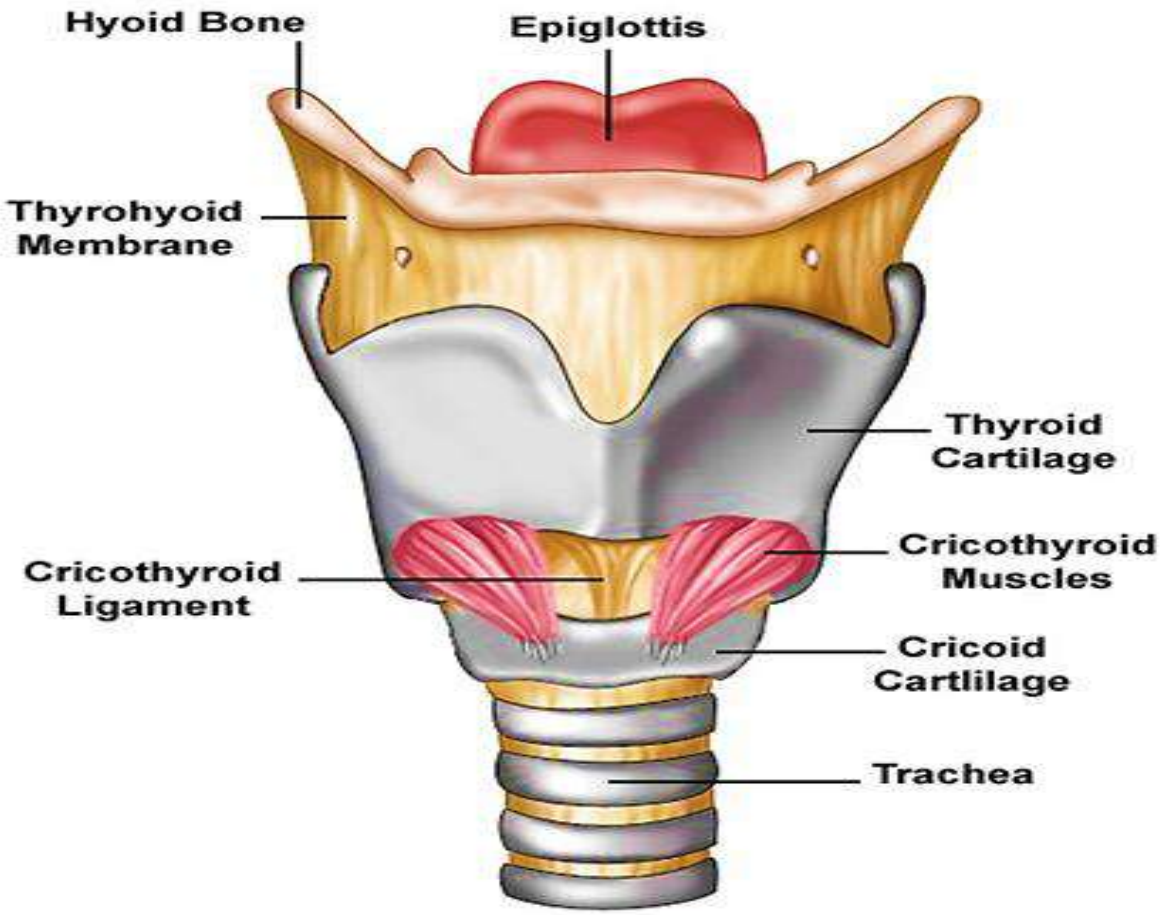


Figure 2: Structure of the larynx.

TRACHEA (WINDPIPE):

- Elastic tube.
- About 10 cm long.
- Kept patent by about 20 U-shaped rings of hyaline cartilage.
- Opens posteriorly.

Beginning of Trachea: Continuation of the larynx at the level of C 6.

End of Trachea: At level of the sternal angle (T 4).

- Bifurcates into the right and left bronchi.

BRONCHI:

- Right and left principal bronchi
- Have the same structure as the trachea.
- After an oblique course, each bronchus enters the respective lung at the hilum.
- Give off branches which in turn branch and rebranch like a tree.
- Right bronchus is shorter, wider and more vertical than the left bronchus.
- Foreign bodies normally enter the right bronchus.

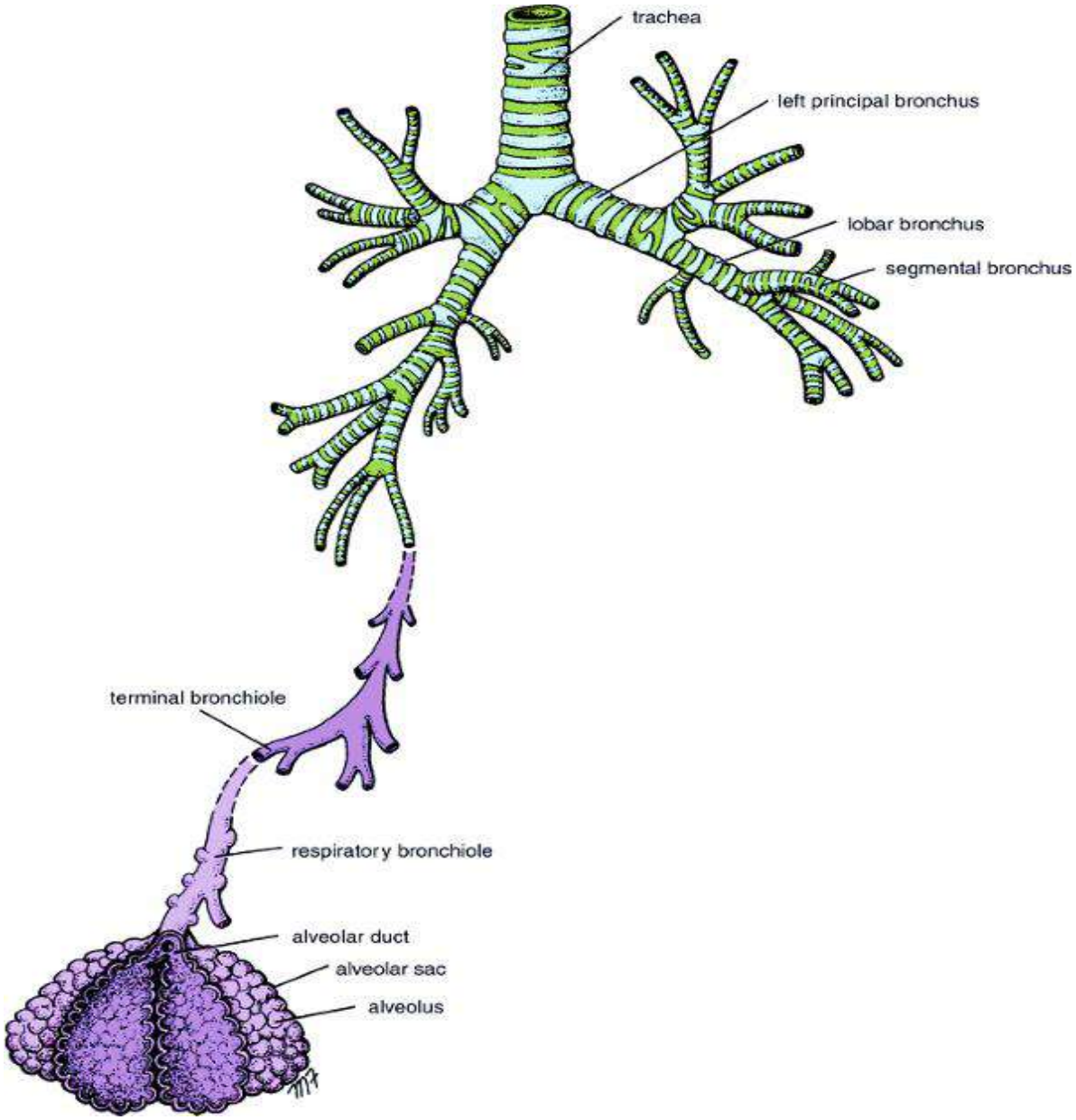


Figure 3: Trachea, bronchi and alveoli. (Snell, 2019).

LUNGS (Figure 4):

- The essential organs of respiration.
- Responsible for the uptake of oxygen into the blood and the removal of carbon dioxide.
- There are 2 lungs; right and left.
- Conical in shape.
- Each lung has an apex and base, costal and medial surfaces, anterior, posterior and inferior borders and a hilum.

Hilum of the lung: On the medial surface and serves for passage of structures in the root of the lung: the pulmonary vessels, primary bronchi, nerves, and lymphatics.

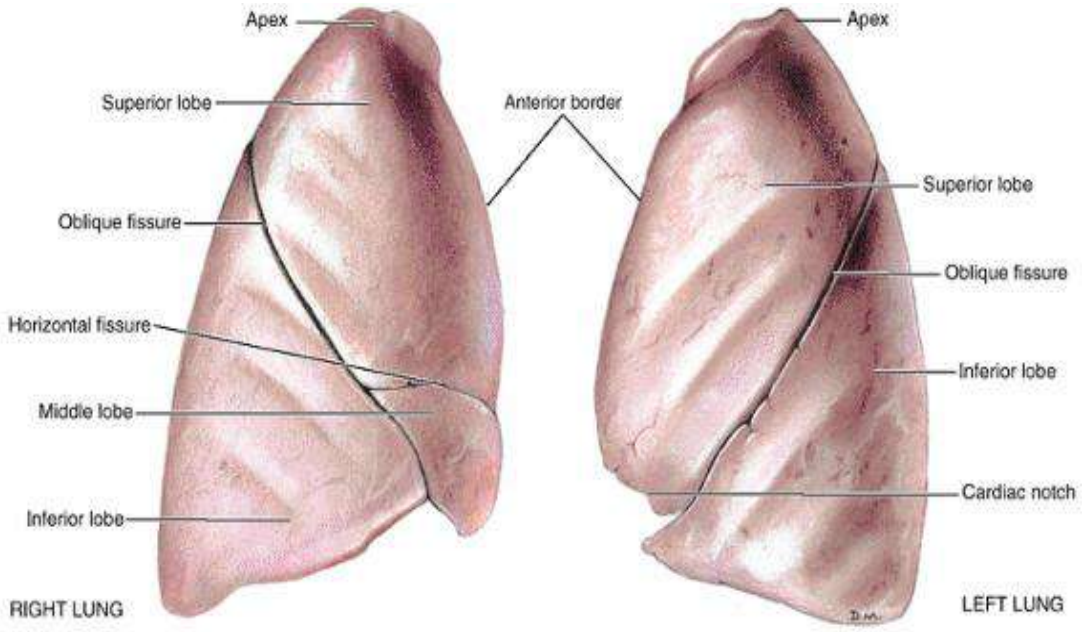


Figure 4: Right and left lungs. (Moore et al, 2023).

Differences between the right and left lungs

Serial	Right Lung	Left Lung
1. Length	Shorter than the left (Right lobe of the liver)	Longer than the right
2. Width	Larger	Thinner (Heart, cardiac notch)
3. Weight	Heavier	Lighter
4. Fissures	Two (transverse and oblique)	One (oblique)
5. Lobes	Three (superior, middle, inferior)	Two (superior and inferior)
6. Bronchopulmonary segments	10 (3 upper lobe, 2 middle lobe and 5 lower lobe)	8-9 (? upper lobe and ? Lower lobe)
7. Relation with the heart	Venous side of the heart (right atrium and right ventricle)	Arterial side of the heart (Left atrium and left ventricle)

References:

1. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.
2. Moore, Keith L. and Dalley A. F. (2006): Clinically Oriented Anatomy, 5th Edition, Lippincott Williams & Wilkins.
3. Drake L. R., Vogl W. and Mitchell A. W. M. (2007): Gray's Anatomy for Students.
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5. Agur, Anne M.R.; Dalley, Arthur F. (2009): Grant's Atlas of Anatomy, 12th Edition. Lippincott Williams & Wilkins.

Lectures 11 & 12:

NERVOUS SYSTEM

ILOs:

1. Mention the organs forming the central nervous system.
2. Mention the structures forming the peripheral nervous system.
3. Mention the types of autonomic nervous system and their location.
4. Describe anatomy of the cells in the central nervous system, their location and function.
5. Name the cavities of the brain and their sites.
6. Mention the lobes of the cerebral hemispheres and their function.
7. Describe the structure of the grey and white matters of the brain.
8. How many spinal nerves are there in the body and their distribution.
9. Enumerate the cranial nerves.

Nervous System:

Structurally divided into: 2 main parts:

1. Central nervous system (Figure 1): Consists of the brain and spinal cord.

2. Peripheral nervous system: Consists of 12 pairs of cranial nerves and 31 pairs of spinal nerves and their associated ganglia.

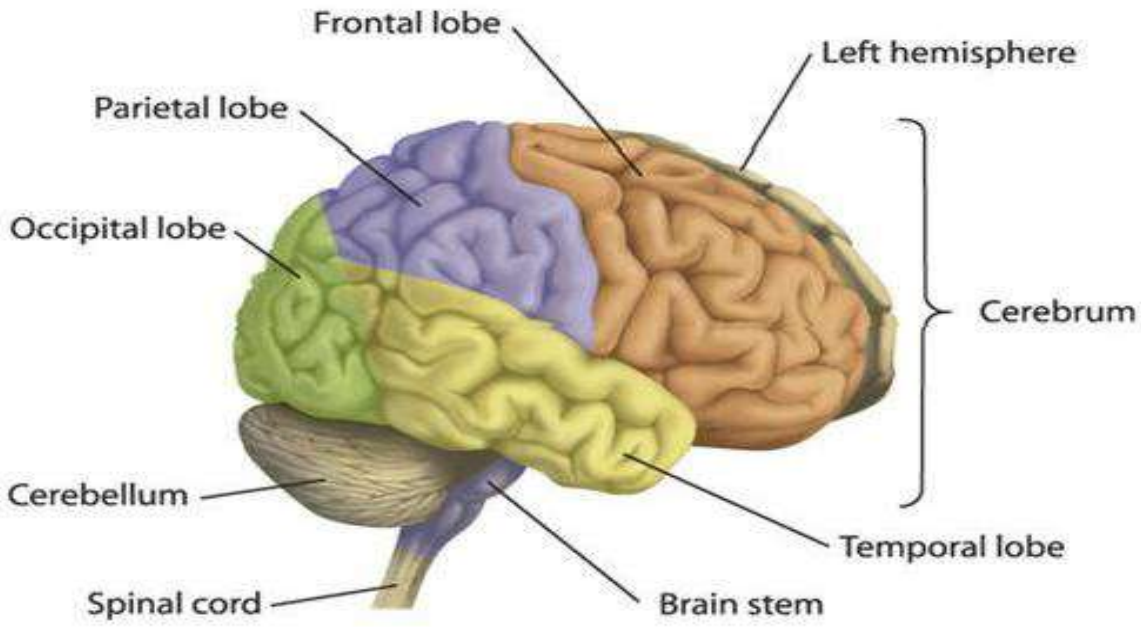


Figure 1: Central nervous system.

Brain: Consists of:

a. Cerebrum
b. Cerebellum
c. Brain stem: Consists of midbrain, pons and medulla oblongata

Cerebrum (Figure 1):

- 2 cerebral hemispheres connected by corpus callosum.
- Formed of 4 lobes; frontal, parietal, occipital and temporal.
- Frontal lobe: Motor.
- Parietal lobe: Somatosensory.
- Temporal lobe: Auditory.
- Occipital lobe: Vision.

Spinal cord:

Beginning of the spinal cord: Continuation of medulla oblongata at foramen magnum.

End of adult spinal cord: At the level of L 1- L2 vertebrae**.**

Functionally:

- Nervous system can be further divided into:

1. Somatic nervous system: Controls voluntary activities.
2. Autonomic nervous system: Controls involuntary activities.

- The nervous system, together with the endocrine system, controls and integrates the activities of the different parts of the body.

Central Nervous System:

- Composed of large numbers of nerve cells and their processes, supported by specialized tissue called neuroglia.

Neuron (Figure 2): The nerve cell and all its processes.

- The nerve cell has two types of processes, called dendrites and an axon.

Dendrites: The short processes of the cell body.

Axon: The longest process of the cell body.

Interior of the central nervous system: Organized into grey and white matter.

Gray matter: Consists of nerve cells embedded in neuroglia.

White matter: Consists of nerve fibres (axons) embedded in neuroglia.

Glial cells: Consists of (Figure 3):

- **Astrocytes:** protoplasmic and fibrous give neurons both structural and nutritional support.
- **Oligodendrocytes:** Formation of myelin sheath inside CNS and
- **Microglia:** Phagocytes.

Peripheral Nervous System:

Cranial Nerves:

- 12 pairs of cranial nerves
- Leave the brain and pass through foramina in the skull.
- All the nerves are distributed in the head and neck except the Xth (vagus), which also supplies structures in the thorax and abdomen.

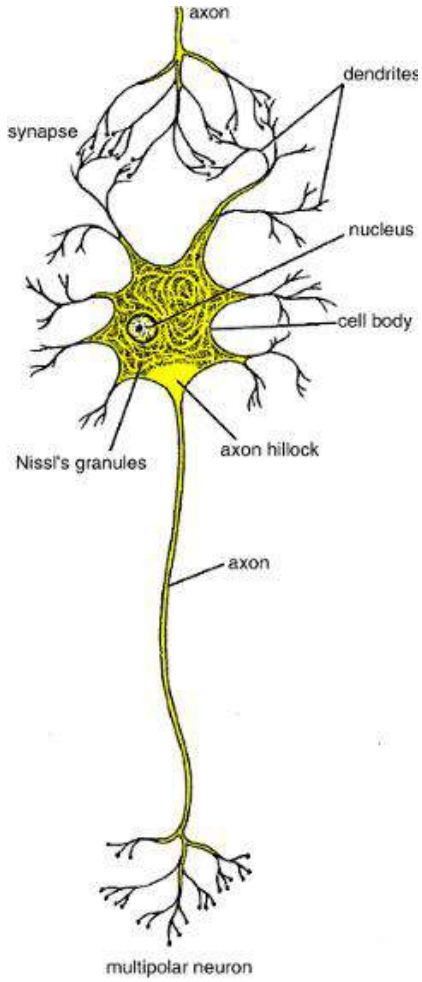


Figure 2: Structure of the neuron. (Snell, 2019).

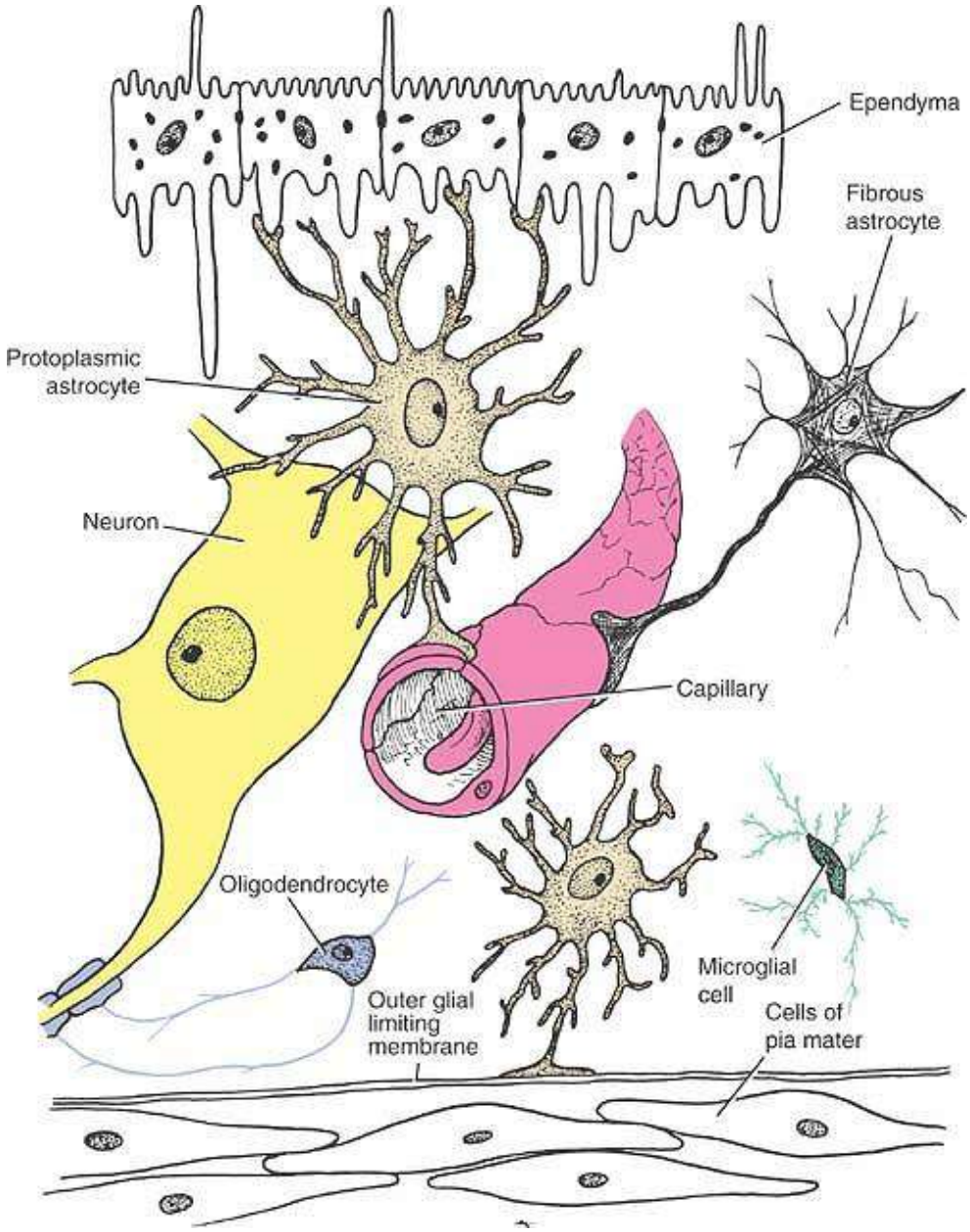


Figure 3: Arrangement of different types of neuroglial cells.

Basal Ganglia:

- Number of nuclear masses
- Buried within the white matter

The most are:

1. Caudate nucleus
2. Putamen
3. Globus pallidus

4. Claustrum
5. Amygdaloid body

Diencephalon:

- Continuous with rostral part of midbrain.
- **Lies:** Between the brain stem and cerebral hemispheres.

Diencephalon comprises:

1. Epithalamus.
2. Thalamus.
3. Subthalamus.
4. Hypothalamus.
5. Metathalamus.

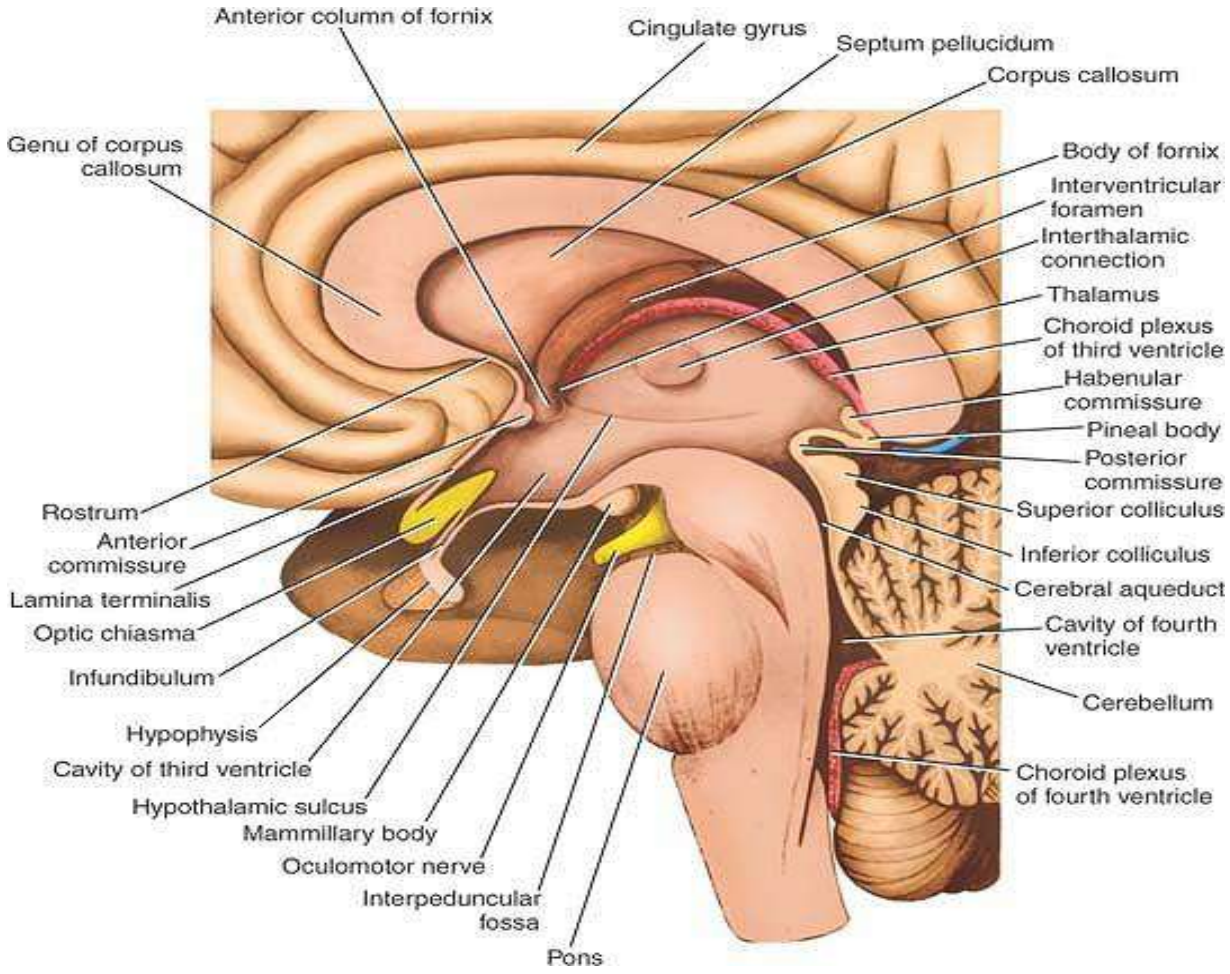


Figure 4: Diencephalon. (Snell, 2018).

Cranial nerves are (Figure 5):

1. Olfactory nerve
2. Optic nerve
3. Oculomotor nerve
4. Trochlear nerve
5. Trigeminal nerve
6. Abducent nerve
7. Facial nerve
8. Vestibulo-cochlear nerve
9. Glosso-pharyngeal nerve
10. Vagus nerve
11. Accessory nerve
12. Hypoglossal nerve

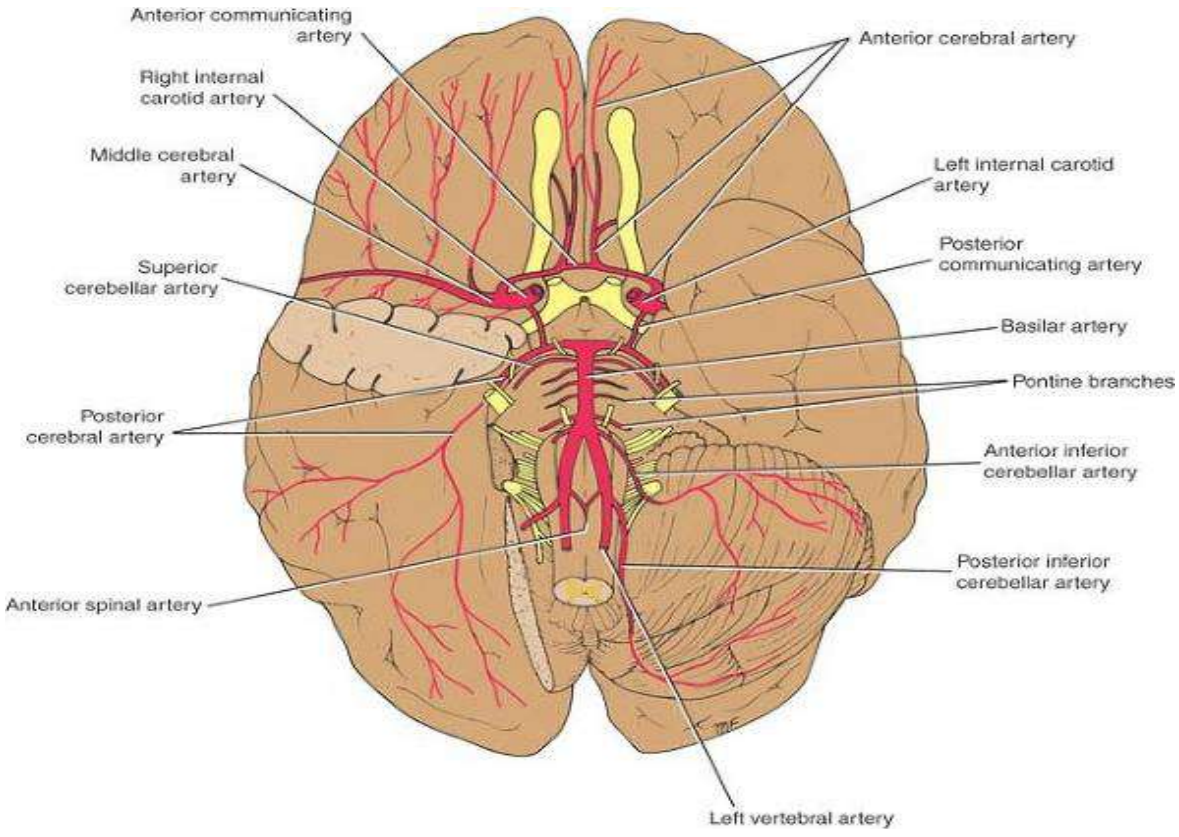


Figure 5: Cranial nerves. (Snell, 2018).

Spinal Nerves (Figure 6):

- 31 pairs of spinal nerves.
- Leave the spinal cord.
- Pass through intervertebral foramina in the vertebral column.
- Named according to the region of vertebral column with which they are associated: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, and 1 coccygeal.
- Note: 8 cervical nerves and only 7 cervical vertebrae and that there is one coccygeal nerve and four coccygeal vertebrae.
- **Each spinal nerve is mixed and formed of:**
    a. Anterior root (efferent, motor).
    b. Posterior root (afferent, sensory).
    c. Trunk: Union of anterior and posterior roots (mixed).
    d. Anterior ramus (mixed): Supplies all the body except the back.
    e. Posterior ramus (mixed); Supplies the back.

Nerve Plexuses: Union of anterior rami in some places; cervical, brachial, lumbar, sacral.

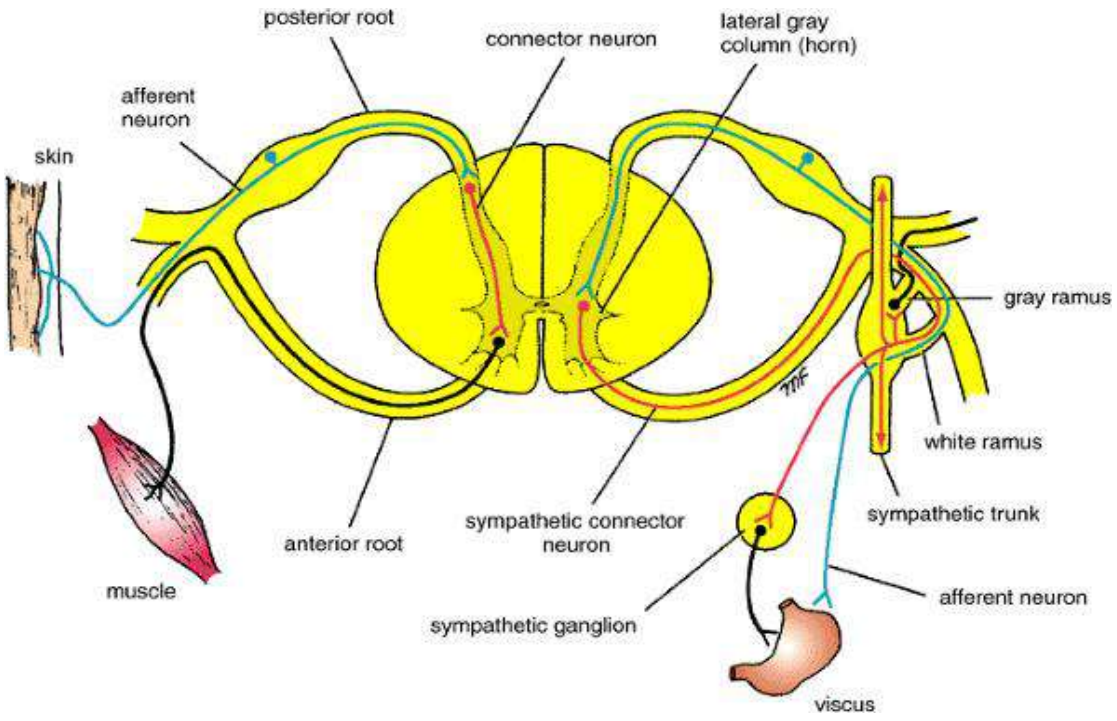


Figure 6: Spinal nerves. (Snell, 2019).

B. Autonomic Nervous System:

1. Sympathetic nervous system: Thoraco-lumbar (T1 – L 2).

2. Parasympathetic nervous system: Cranio-sacral (Cr III, VII, IX, X and S 2,3,4).

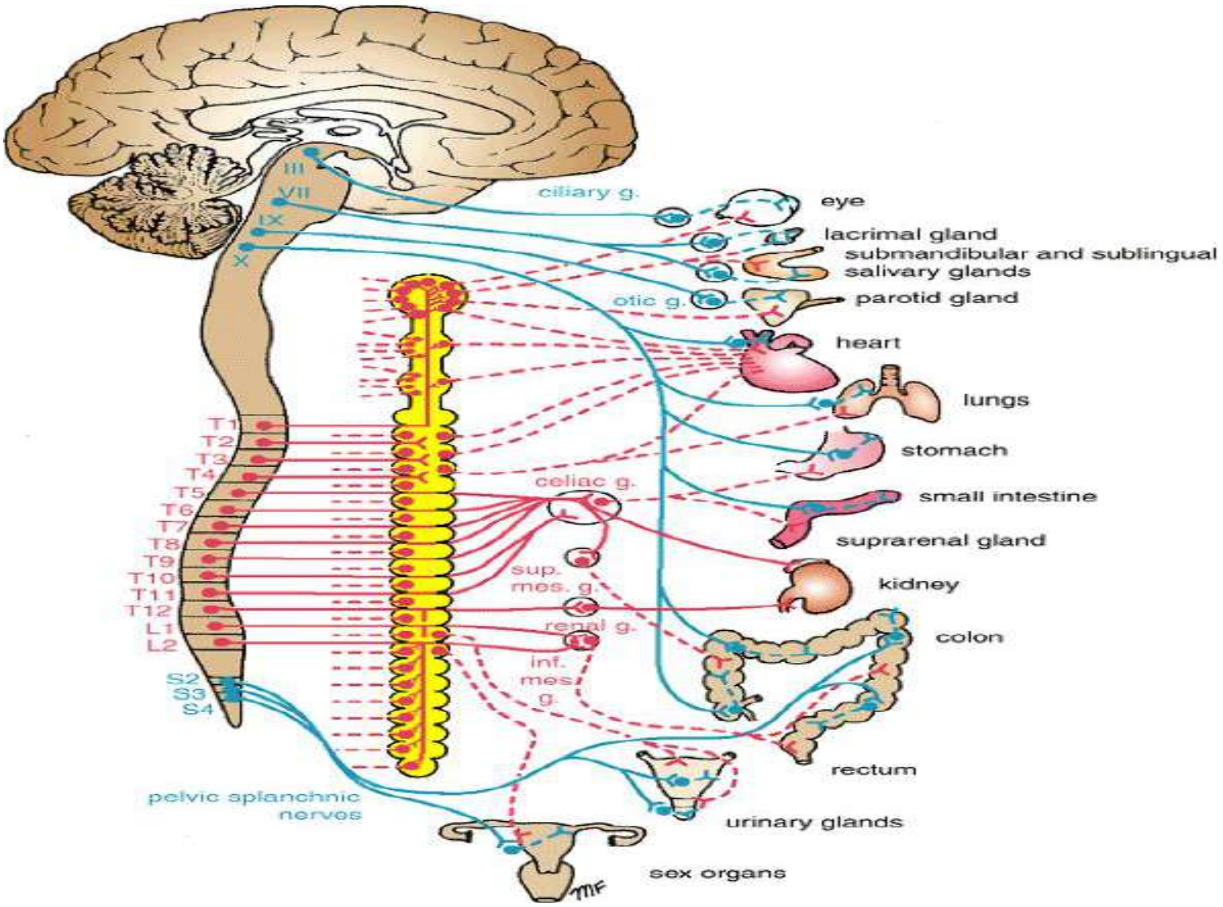


Figure 7: Autonomic nervous system. (Snell, 2019).

References:

1. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.
2. Moore, Keith L. and Dalley A. F. (2023): Clinically Oriented Anatomy, 9th Edition, Lippincott Williams & Wilkins.
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Lecture: 13:

CIRCULATORY SYSTEM
(CARDIOVASCULAR AND LYMPHATIC SYSTEMS)

ILOs:

1. Describe the normal anatomy of cardiovascular system?
2. Describe the anatomy of the arteries and its varieties?
3. Define with examples arterial anastomosis, end arteries?
4. Define with examples tortious arteries?
5. Define arterial pulsation and enumerate some palpable pulsating arteries?
6. Define with examples venae comitantes?
7. Describe the anatomy of the veins and its varieties?

CIRCULATORY SYSTEM:

- Transports fluids throughout the body.
- Consists of the cardiovascular and lymphatic systems.
- The heart and blood vessels make up the blood transportation network, cardiovascular system.

BLOOD VESSELS (Figures 1, 2 & 3):

- 3 types: arteries, veins, and capillaries.

ARTERIES:

- Transport blood from the heart and distribute it to the various tissues of the body by means of their branches.
- Arteries do not have valves.

Arterioles:

- The smallest arteries, <0.1 mm in diameter.

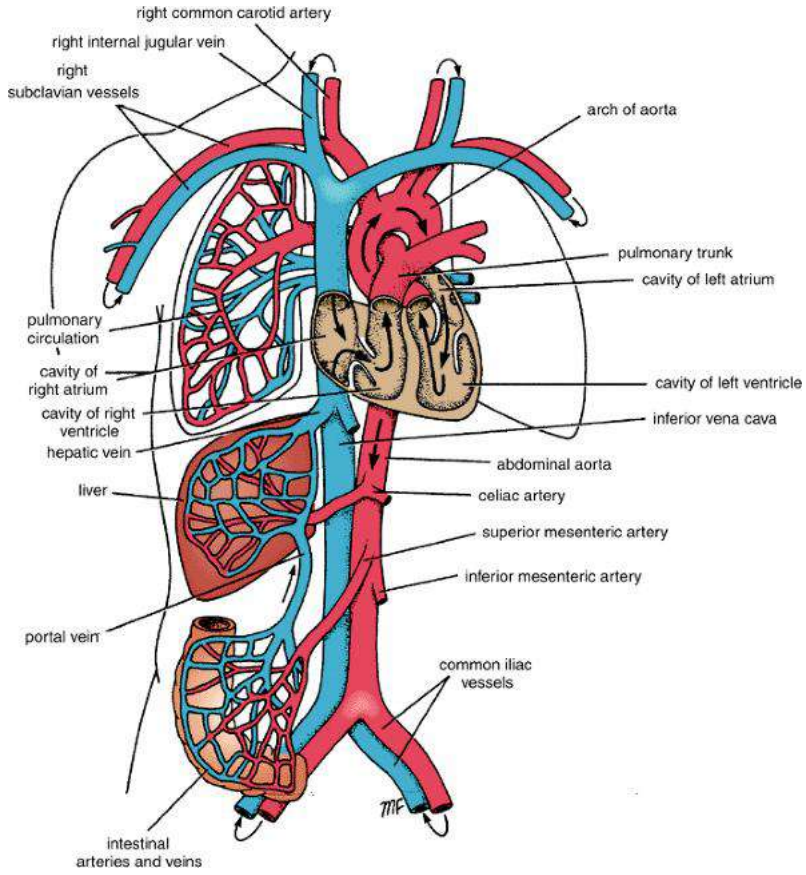


Figure 1: Cardiovascular system. (Snell, 2019).

Anastomosis of Arteries:

- Joining of branches of arteries.
- Example: Gastric arteries, genicular arteries

Anatomic end arteries:

- Vessels whose terminal branches do not anastomose with branches of arteries supplying adjacent areas.
- Example: Central artery of the retina.

Functional end arteries:

- Vessels whose terminal branches do anastomose with those of adjacent arteries, but the calibre of the anastomosis is insufficient to keep the tissue alive should one of the arteries become blocked.
- Example: Coronary arteries.

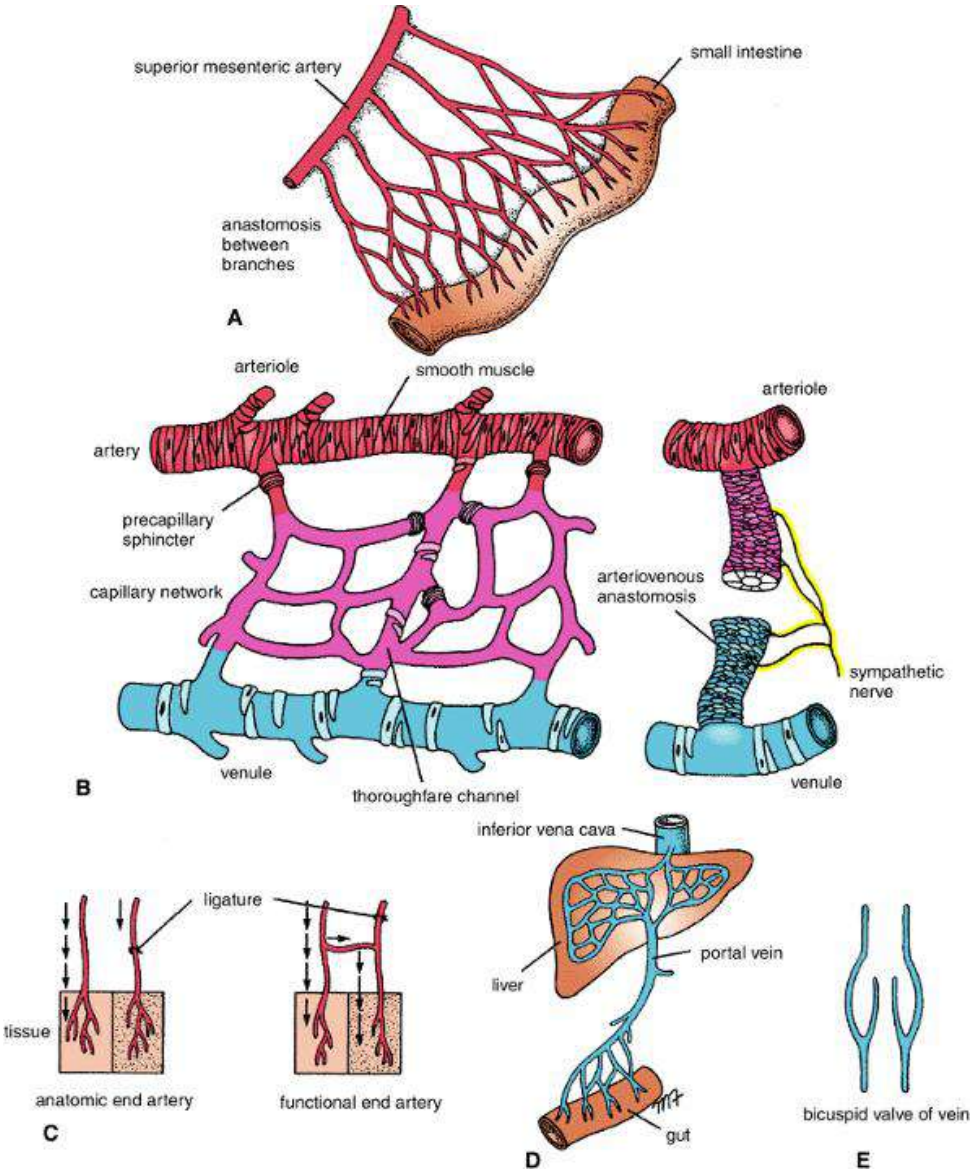


Figure 2: A. Anastomosis between the branches of the superior mesenteric artery. B. Capillary network and an arteriovenous anastomosis. C. Anatomic end artery and functional end artery. D. Portal system. E. Structure of the bicuspid valve in a vein. (Snell, 2019).

Capillaries:

- Microscopic vessels in the form of a network connecting the arterioles to the venules.

Sinusoids:

- Resemble capillaries in that they are thin-walled blood vessels.
- Have an irregular cross diameter.

- Wider than capillaries.
- **Found in:** Bone marrow, the spleen, the liver, and some endocrine glands.

Arteriovenous Anastomoses:

- Direct connections occur between the arteries and veins without the intervention of capillaries.
- **Found in:** Tips of the fingers and toes.

Arterial Pulsation:

- Wall of the arteries pulsates with each cardiac beat.
- Can be felt in arteries where they are superficial and pass on a rigid substance as bone.

Examples of palpable pulsating arteries:

1. Superficial temporal artery
2. Facial artery
3. Common carotid artery
4. Axillary artery
5. Brachial artery
6. Radial artery (the most commonly used)
7. Femoral artery
8. Popliteal artery
9. Dorsalis pedis artery

VEINS:

- Vessels that transport blood back to the heart.
- Many veins have valves.

Venules: The smallest veins.

- The smaller veins, or tributaries, unite to form larger veins,
- Commonly join with one another to form venous plexuses.

Venae comitantes (Figure 4): Two or more veins, accompany medium-size deep arteries.

- Serves as a counter current heat exchanger, the warm arterial blood warming the cooler venous blood as it returns to the heart from a cold extremity.

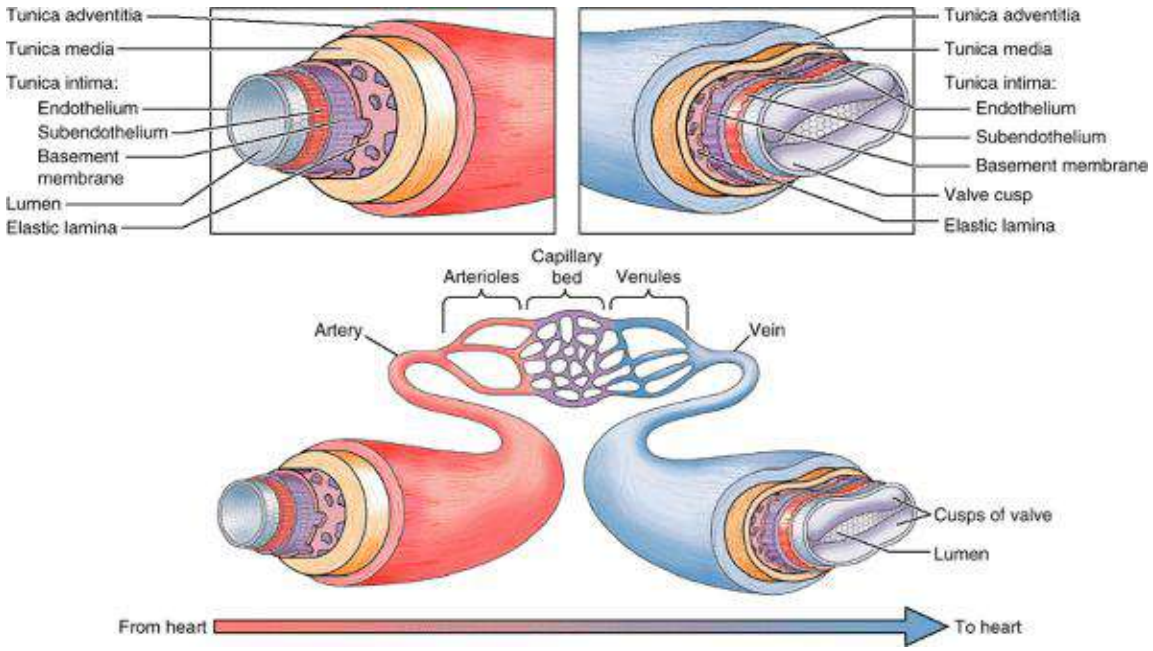


Figure 3: Blood vessels. (Moore and Dalley, 2023).

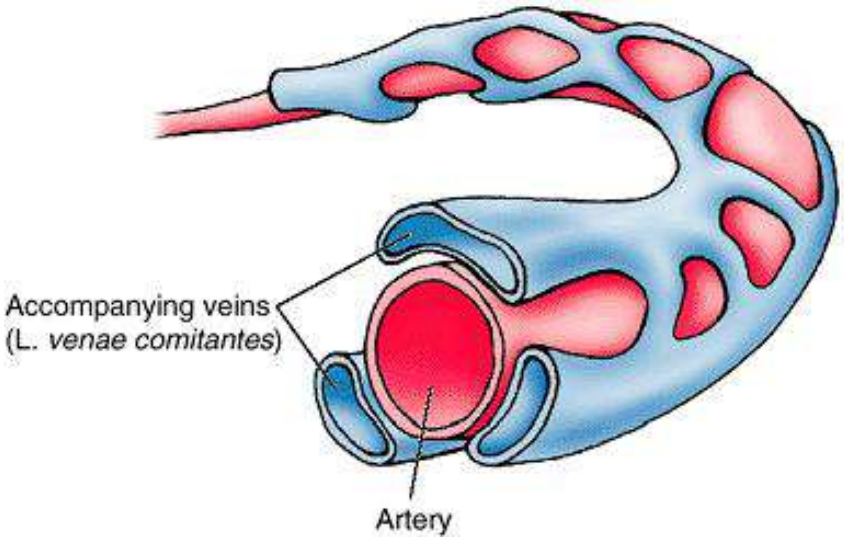


Figure 4: Accompanying veins (venae comitantes). (Moore and Dalley, 2023).

Portal Vein:

- Veins leaving the gastrointestinal tract do not go directly to the heart but converge on the this vein
- Enters the liver and breaks up again into veins of diminishing size, which ultimately join the sinusoids, in the liver.

Portal system: System of vessels interposed between two capillary beds.

LYMPHATIC SYSTEM

ILOs:

. Define the lymphatic system and its function

. Define the lymphoid tissue

. Mention the sites that produce lymphocytes

. Define the lymphatic vessels and their types

. What are the sites which do not have lymphatic vessels?

. Define the lymph

. Define the lymph capillaries

. Describe anatomy of the lymph nodes

. Where is the end of the right lymphatic trunk?

. Where is the end of the thoracic duct?

Lymphatic System (Figure 3):

- Consists of lymphatic tissues and lymphatic vessels.
- Essentially a drainage system.
- There is no circulation.

Lymphatic Tissues: Type of connective tissue that contains large numbers of lymphocytes.

Lymphatic Tissues Organs:

1. Thymus.
2. Lymph nodes.
3. Spleen.
4. Lymphatic nodules.

Function of the Lymphatic Tissue:

- Essential for the immunologic defences of the body against bacteria and viruses.

Lymphatic Vessels:

- Tubes that assist the cardiovascular system in the removal of tissue fluid from the tissue spaces of the body; the vessels then return the fluid to the blood.

- Have a beaded appearance because of presence of numerous valves along their course.

Tissues and Organs of the Body Have NO Lymphatic Vessels:

(All tissues and organs of the body have lymphatic vessels except):

1. Central nervous system.
2. Eyeball.
3. Internal ear.
4. Epidermis of the skin.
5. Cartilage.
6. Bone.

Lymph:

- Tissue fluid once it has entered a lymphatic vessel.

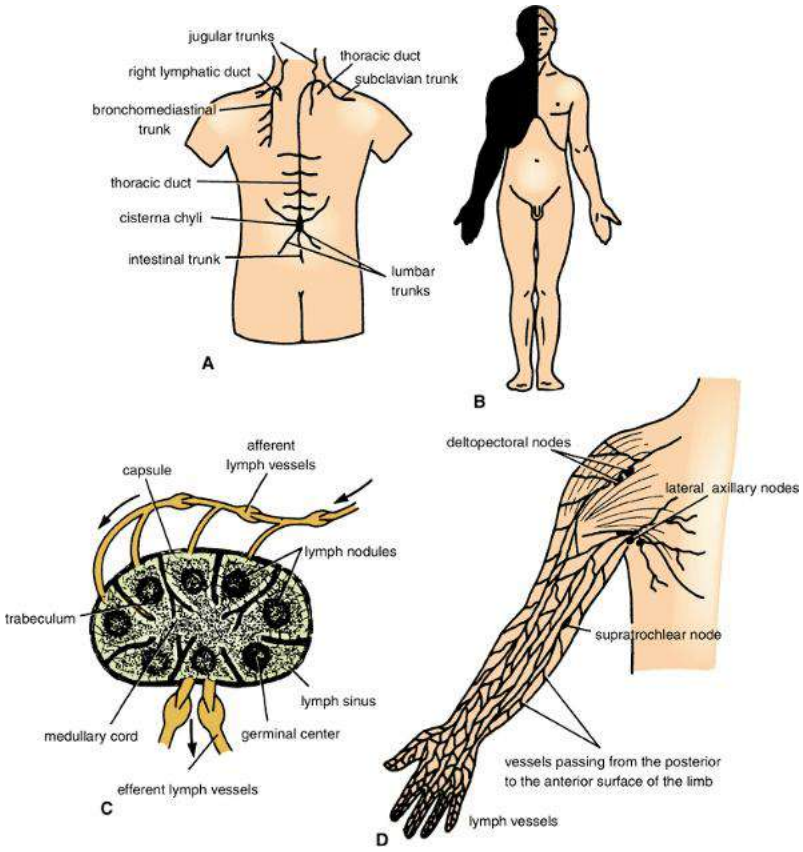
Lymph capillaries:

- Network of fine vessels that drain lymph from the tissues.
- The capillaries are in turn drained by small lymph vessels, which unite to form large lymph vessels.
- Before lymph is returned to the bloodstream, it passes through at least one lymph node and often through several.

Afferent Lymph Vessels: Lymph vessels that carry lymph to a lymph node.

Efferent Lymph Vessels: Lymph vessels that transport it away from a node

End of the lymph: Reaches the venous blood at the root of neck by the right lymphatic duct and the thoracic duct.



References:

5. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.
6. Moore, Keith L. and Dalley A. F. (2023): Clinically Oriented Anatomy, 9th Edition, Lippincott Williams & Wilkins.
7. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021).

Lecture 14:

URINARY SYSTEM

ILOs:

- Mention the organs forming the urinary system?
- Describe anatomy of the kidney and its relations?
- Describe anatomy of the ureter and its constrictions?
- Describe anatomy of the urinary bladder?
- Describe anatomy of the urethra in male and its parts?
- Describe the urethra in female?
- What are the differences between the female and male urethra?

URINARY SYSTEM: Consists of:

- 2 kidneys
- 2 ureters
- 1 Urinary bladder
- 1 Urethra

KIDNEYS:

- One on each side of the vertebral column.
- Extend from level of the twelfth thoracic vertebra to the third lumbar vertebra.
- Right kidney is lower than left one.
- 12 cm length, 6 cm width and 3 cm thickness

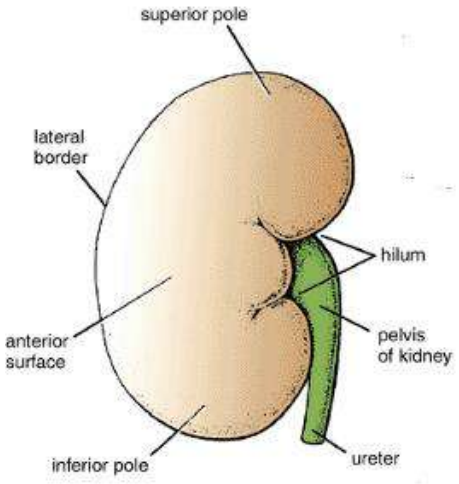


Figure 1: Right kidney. (Snell, 2019).

- Has 2 surfaces (anterior and posterior), 2 borders (medial and lateral, and 2 ends (upper and lower).
- Lateral border is convex, medial border is convex at both ends and concave at its middle
- Middle of the medial border represents the hilum.

The hilum of the kidney gives passage to:

1. Renal vein
2. Renal artery
3. Parasympathetic nerves and
4. Pelvis of kidney (dilated sac extends outside the hilum to be continuous with the ureter).
5. Lymph vessels

Coverings of the kidney:

1. Fibrous capsule.
2. Perirenal fat.
3. Renal fascia.
4. Pararenal fat.

Structure of the kidney:

- Tissue of the kidney is differentiated into outer cortex and inner medulla.

RELATIONS OF THE KIDNEYS:

Anterior Relations of the Right Kidney:

1. Right suprarenal gland.
2. Duodenum.
3. Liver.
4. Colon.
5. Small intestine.

Anterior Relations of the Left Kidney:

1. Left suprarenal gland.
2. Spleen.

3. Pancreas.
4. Stomach.
5. Colon.
6. Small intestine.

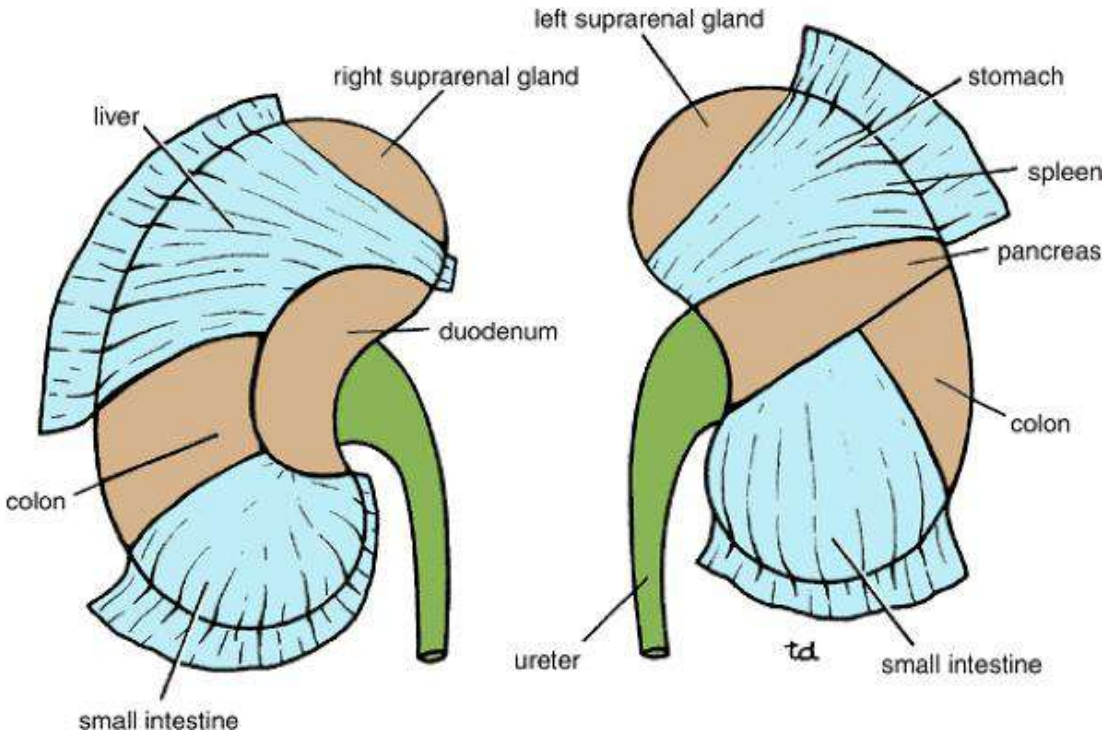


Figure 2: Anterior relations of the kidneys. (Snell, 2019).

Posterior Relations of the Kidneys (Figure 4):

1. Diaphragm
2. Muscles of posterior abdominal wall:

   A. Psoas major

   B. Quadratus lumborum

   C. Transversus abdominis

Ureters:

- Two muscular tubes one on each side.
- Convey urine from the kidneys to the urinary bladder.
- Each is 25 to 30 cm in length

- Half of its course is in abdomen, while another half in the pelvis
- Continuation of pelvis of the kidney
- Opens on postero-superior angle of the urinary bladder

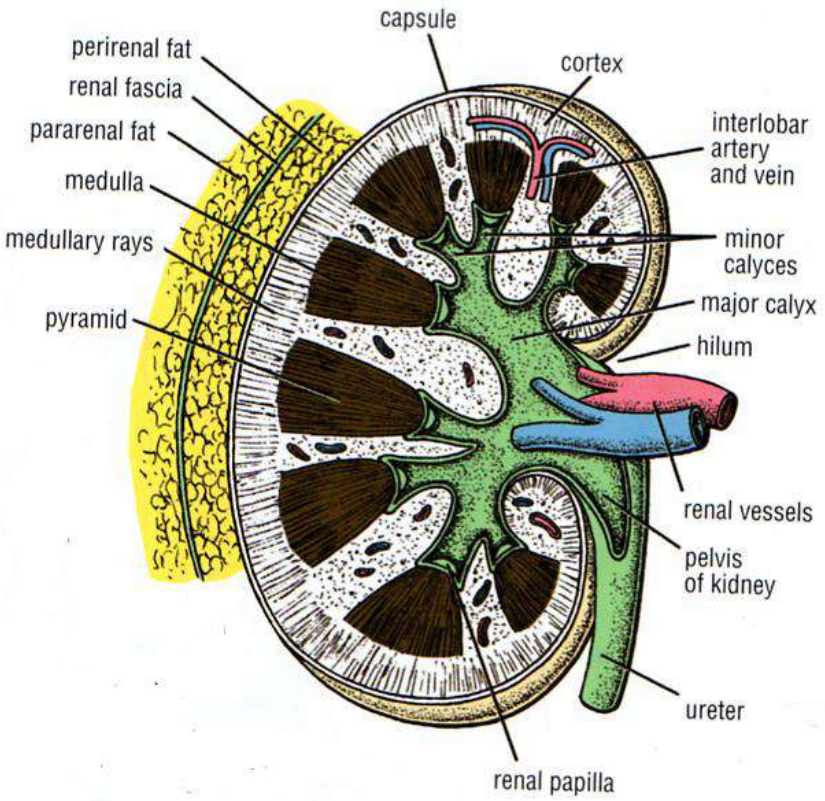


Figure 3: Coverings and structure of the kidney. (Snell, 2019).

During course of ureter, it has 3 sites of constriction (Figure 4):

1. At pelvi-ureteric junction
2. Where ureter crosses common iliac artery
3. At uretero-vesical junction

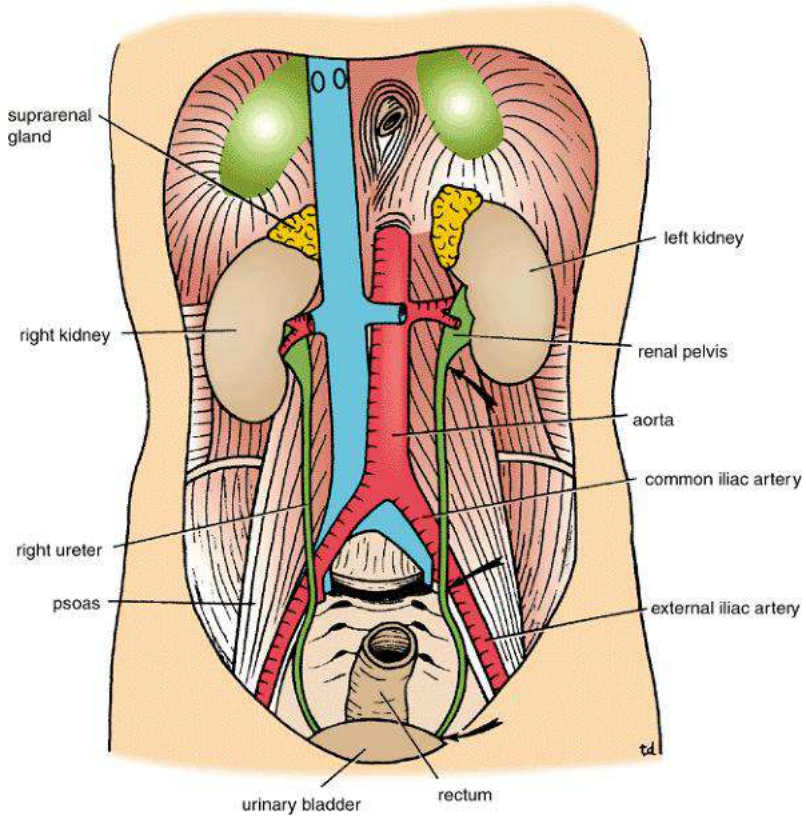


Figure 4: Posterior abdominal wall, Kidneys, Constrictions of the ureters. (Snell, 2019).

URINARY BLADDER:

- Hollow viscus with strong muscular wall which serves as a *reservoir for urine*.
- Lies in pelvic cavity directly behind pubic bones.
- Its shape is a three-sided pyramid placed on one of its angles, its capacity 220 cc.
- Can accommodate up to 500 cc without discomfort.
- Apex; behind upper part of symphysis pubis, continuous with *median umbilical ligament*
- Base is directed posteriorly.
- Has 3 surfaces 1 superior and 2 inferolateral.
- Each surface and base are triangular, related to seminal vesicles and ampulla of vas deferens on each side (in male) and cervix of uterus and vagina (in female).
- Ureter enters supero-lateral angle of base.
- Inferior angle (neck) gives rise to urethra.
- In male, the neck rests on the prostate and continuous with prostatic part of urethra.
- In female, it is continuous with the urethra.

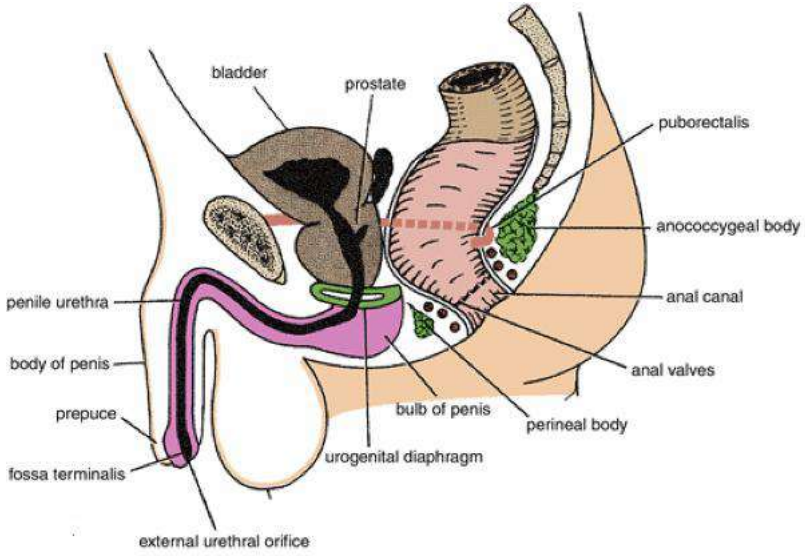


Figure 5: Sagittal section in male pelvis, urinary bladder and urethra. (Snell, 2019).

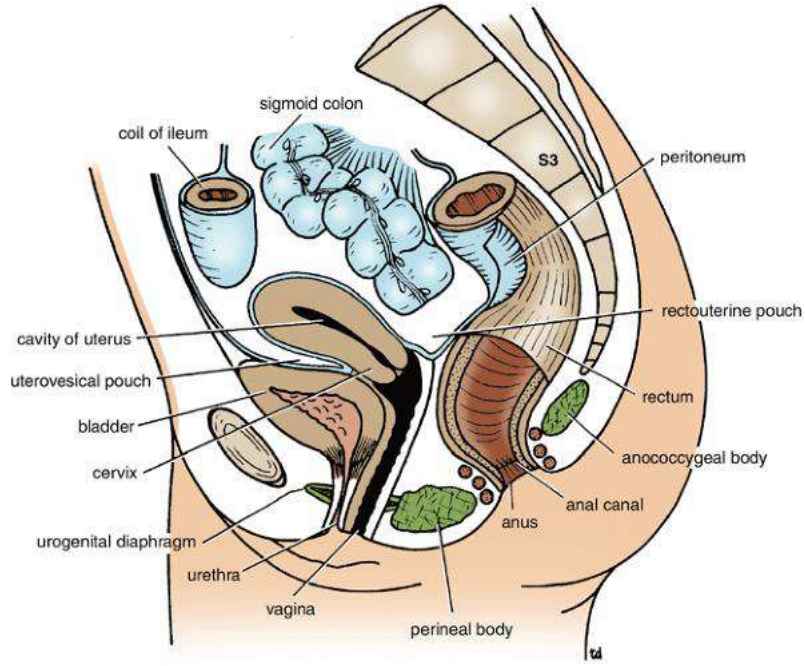


Figure 6: Sagittal section in female pelvis, urinary bladder and urethra. (Snell, 2019).

URETHRA

MALE URETHRA

- 20 cm long,

Divided into 3 parts:

1. **Prostatic part:** 3 cm in long, widest part, begins at neck of urinary bladder, surrounded by prostate. Ejaculatory duct opens into prostatic urethra.
2. **Membranous urethra:** 1 cm in long, narrowest, shortest. Surrounded by sphincter urethra.
3. **Spongy part:** 16 cm, longest, traverse penis, ended by external urethral orifice.

FEMALE URETHRA

- Very short 4 cm in long.
- Begins at neck of the bladder.
- Its upper part is surrounded by *sphincter urethra*.
- Shorter, wider and more dilatable than male urethra.

References:

6. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.
7. Moore, Keith L. and Dalley A. F. (2023): Clinically Oriented Anatomy, 9th Edition, Lippincott Williams & Wilkins.
8. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021).

Lecture 15:

GENITAL SYSTEMS

ILOs:

1. Mention the organs forming the male genital system?
2. Describe anatomy of the testes?
3. Describe anatomy of the epididymis and its parts?
4. Describe anatomy of the vas deferens?
5. Describe anatomy of the spermatic and its contents?
6. Describe anatomy of the seminal vesicle?
7. Describe anatomy of the prostate?
8. Describe anatomy of the external genitalia in the male?
9. Mention the organs forming the female genital system?
10. Describe anatomy of the ovary?
11. Describe anatomy of the uterine tube and its parts?
12. Describe anatomy of the uterus, parts and normal position?
13. Describe anatomy of the vagina?
14. Describe anatomy of the vulva?

MALE REPRODUCTIVE ORGANS

I. Internal Male Reproductive Organs (Figure 1):

1. Testis
2. Epididymis
3. Ductus deferens
4. Seminal vesicles
5. Ejaculatory ducts
6. Prostate
7. Bulbourethral glands

Ii. External Genital Organs:

1. Penis
2. Scrotum

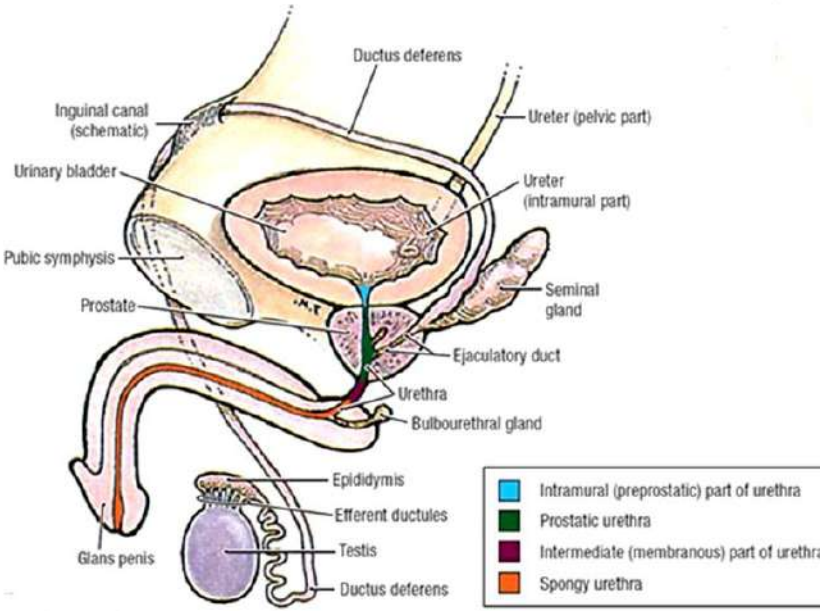


Figure 1: Male genital system.

TESTES:

- Primary male sex organ.
- Lies in the scrotum where temperature is less than that of the body which is suitable for process of spermatogenesis.
- 5 cm in length, 2.5 cm in breadth and 3 cm in antero-superior diameter.
- Suspended by the spermatic cord.
- Ovoid in shape having 2 borders (anterior and posterior), 2 ends (upper and lower) and 2 surfaces (lateral and medial).
- Covered by thick fibrous capsule (tunica albuginea)
- Composed of numerous convoluted, seminiferous tubules.
- Each tubule is 2 feet in length.
- From upper part of testis about 6-12 *efferent ductules* arise.
- Open into head of the epididymis.

Function of the Testes:

1. Production of the sperms.
2. Secretion of testosterone.

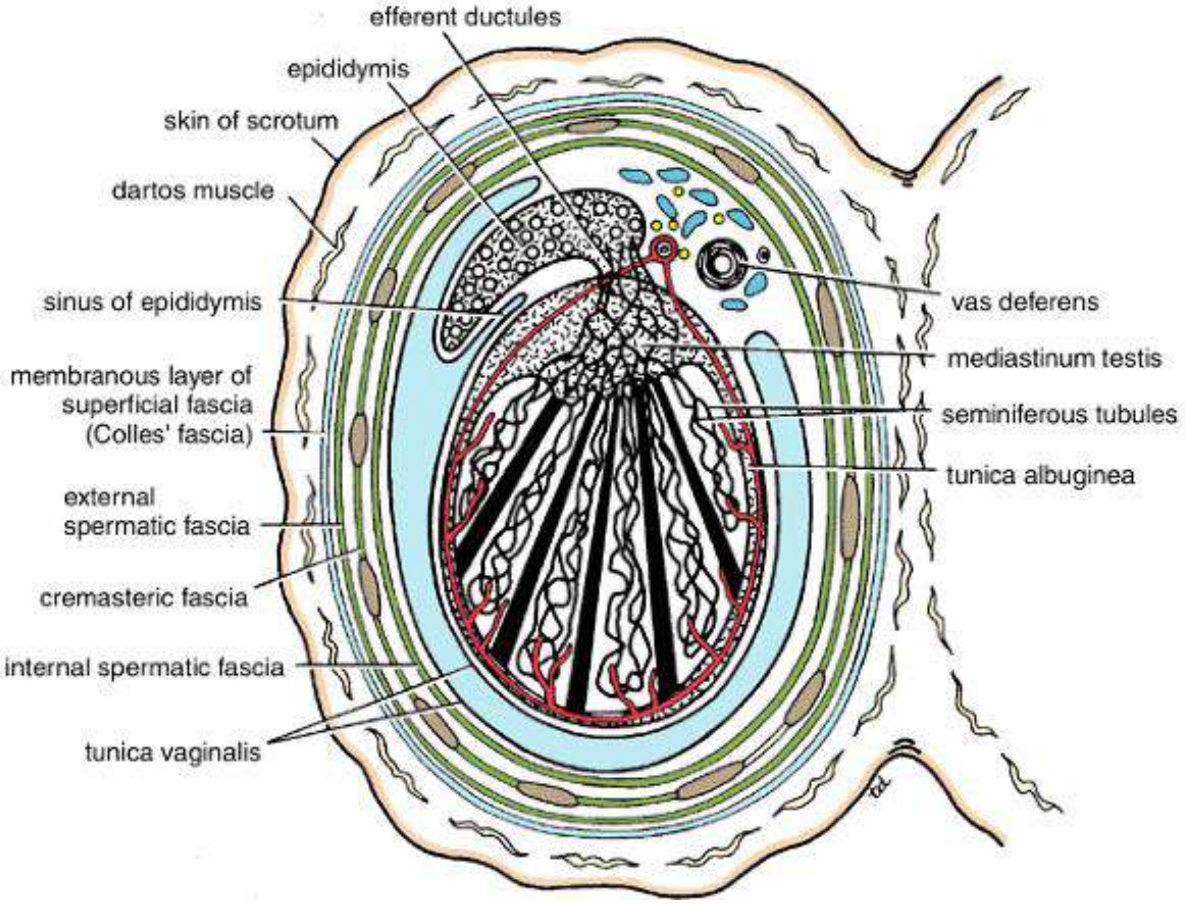


Figure 2: Structure of the testis. (Snell, 2019).

VAS DEFERENS

- Duct which transfers sperms from testis.
- Has thick muscular wall and narrow lumen.
- Each vas deferens connects duct of epididymis to urethra.
- 45 cm long.
- Ascends through spermatic cord to abdominal wall.
- Passes into pelvis from deep inguinal ring.
- On pelvic wall, lies behind peritoneum to reach base of urinary bladder.
- Unites with duct of seminal vesicle to form ejaculatory duct.

SPERMATIC CORD

Position of the Spermatic Cord:

- Lies in the inguinal canal and the scrotum.
- Extends from lower end of testis (below) to deep inguinal ring (above) where ductus deferens enters the pelvis.
- The left cord is longer than the right one.

Contents of the Spermatic Cord:

1. Vas deferens.
2. Artery of ductus deferens.
3. Vessels and nerves of testis:
   a. Testicular artery: Arises from abdominal aorta.
   b. Sympathetic fibres: Surround testicular artery forming testicular plexus.
   c. Pampiniform plexus of veins: Formed by several veins which emerge from back of testis.
   d. Lymph vessels: Drain testes and epididymis and accompany testicular blood vessels up to lateral (para-aortic) lymph nodes
4. Vestigue of processus vaginalis: Fibrous strand of obliterated remains of processus vaginalis

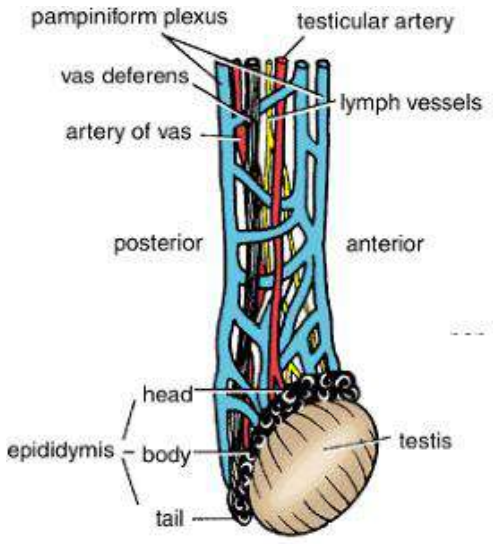


Figure 3: Structure of the spermatic cord. (Snell, 2019).

EJACULATORY DUCTS:

- Each is 2 cm long and pierces prostate.
- Open into prostatic urethra close to its fellow.

SEMINAL VESICLES:

- 2 pear shaped lobulated pouches on each side of base of the urinary bladder
- Each 5 cm long.
- Secret alkaline secretion of seminal fluid.
- Secret fructose for nutrition of the sperms.
- Their 2 ducts unite with 2 ducti deferens to form 2 ejaculatory ducts.

PROSTATE:

- Fibromuscular accessory gland.
- Encircle neck of the urinary bladder.
- Conical in shape.
- Has a base, an apex, anterior, posterior, and two inferolateral surfaces.
- Base directed upward continuous with neck of the urinary bladder.
- Apex is directed downwards.

Function of the Prostate:

- Produce secretion rich in enzymes which liquefy the semen to allow the sperms to travel into the female genitalia.

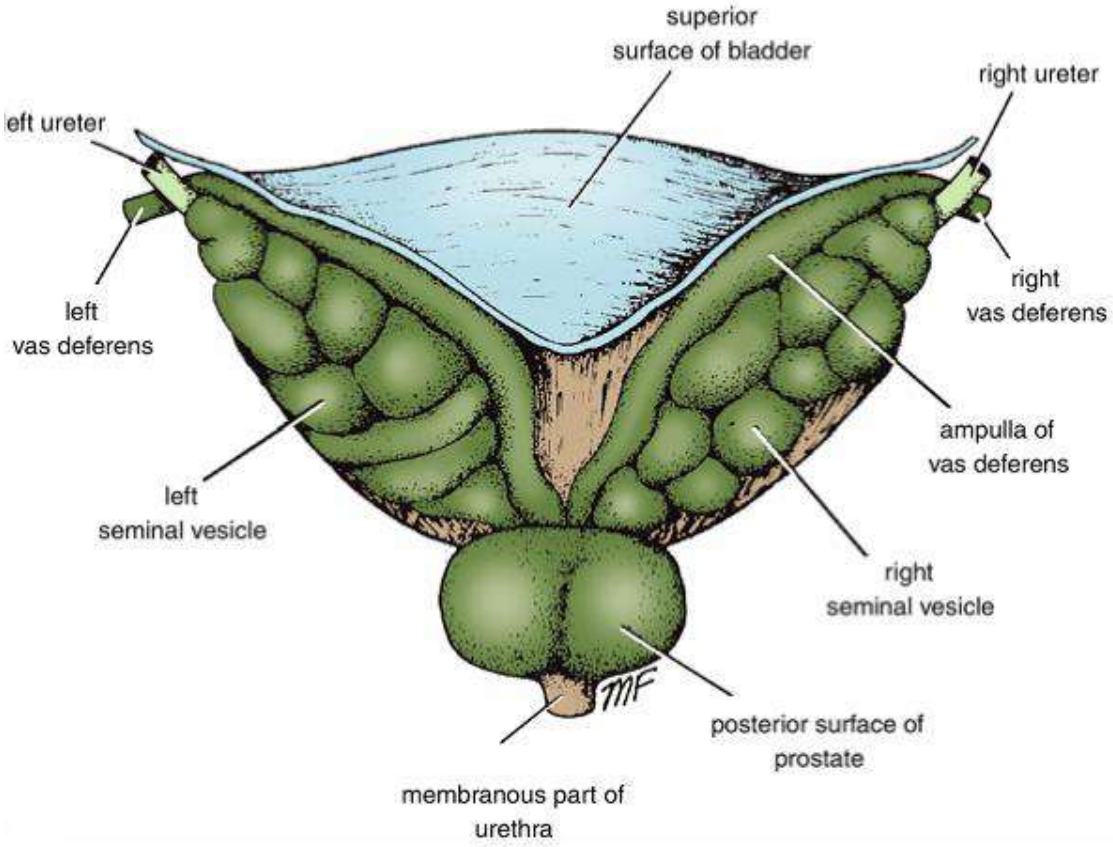


Figure 4: Seminal vesicles, vas deferens and prostate. (Snell, 2019).

BULBOURETHERAL GLANDS:

- Pair of pea-sized yellowish glands.
- Lie within fibers of sphincter urethra.
- Ducts of bulbourethral ducts open into spongy part of urethra.

PENIS:

- Formed of 2 main parts: Root and body
- Its anterior end is enlarged to form *glans penis*.

Body: Formed of 3 elongated erectile bodies; one ventral (corpus spongiosum and 2 dorsal (corpora cavernosa).

Root: Bulb (posterior expanded part of corpus spongiosum) and 2 crura (posterior parts of corpora cavernosa).

SCROTUM:

- Sac of skin and subcutaneous tissue.

Contains: 2 testes and lower parts of spermatic cords.

FEMALE REPRODUCTIVE ORGANS

I. Internal Genitalia:

1. 2 Ovaries.
2. 2 Uterine tubes.
3. 1 Uterus.
4. 1 Vagina.

II. Internal Genitalia:

1. Clitoris.
2. Mons pubis.
3. Labia majora.
4. Labia minora.
5. Greater vestibular glands.
6. Vaginal orifice.
7. Vestibule.
8. Pudendal cleft.

OVARIES

- Present in the pelvis in ovarian fossa.
- One on each side of uterus, on back of broad ligament.
- Each ovary is almond-shaped, 5 cm long, 1.5 cm wide and I cm thickness.

Function of the Ovary:

1. Production of the ova.
2. Secretion of estrogen and progesterone.

UTERINE TUBES (FALLOPIAN TUBES):

- 2 tubes, one on each of the uterus.
- 10 cm long each tube.
- Lies in free anterior border of broad ligament
- Lateral or abdominal end opens into peritoneal cavity close to the ovary, and medial end open into uterine cavity.
- Abdominal opening is funnel-shaped (infundibulum).
- Uterine tubes receive the ovum from the ovary and propels it into the uterine cavity.

Parts of Uterine Tube: (From Medial to Lateral):

1. Uterine (intramural).
2. Isthmus.
3. Ampulla.
4. Infundibulum.

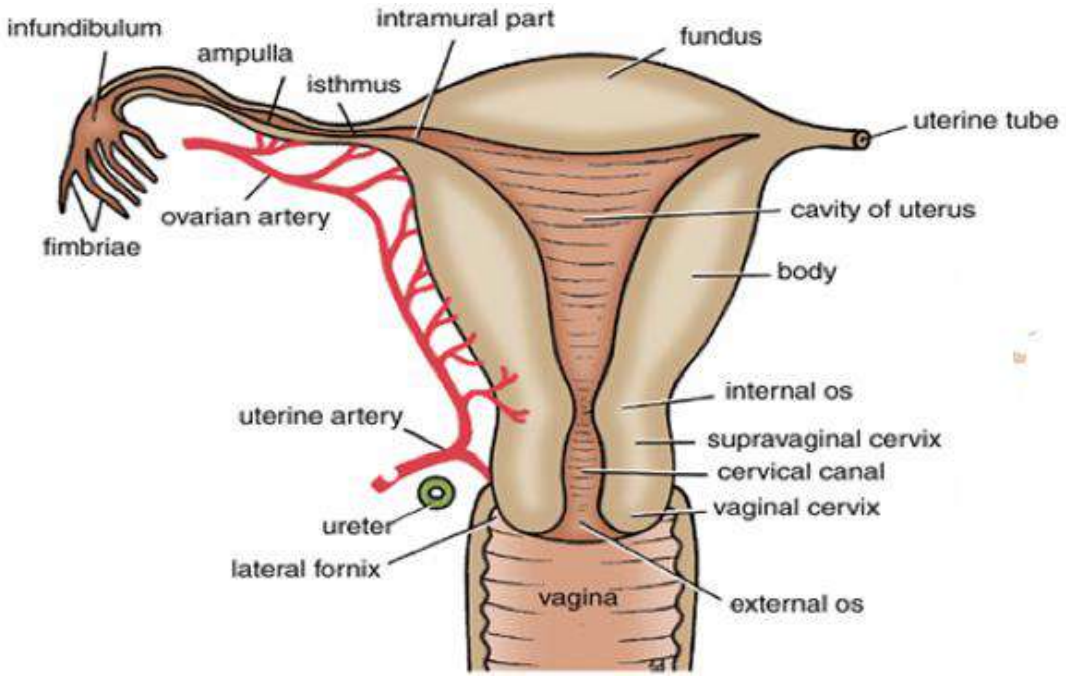


Figure 4: Uterine tubes and uterus. (Snell, 2019).

UTERUS:

- Hollow thick-muscular pear-shape organ situated in lesser pelvis.
- Lies between urinary bladder ant. and rectum post.
- Measures 3 inches in length, 2 inches width and 1 inch thickness.

Openings: Above receives openings of uterine tubes, below opens into vagina.

1. Fundus: Part above entrance of the tubes, convex in all directions.

2. Body: 2 inches long, narrows from fundus to cervix.

3. Cervix: 1 inch long and 1 inch thick, projects through the vagina (upper ½ above vagina, supra-vaginal and lower ½ inside vagina; vaginal).

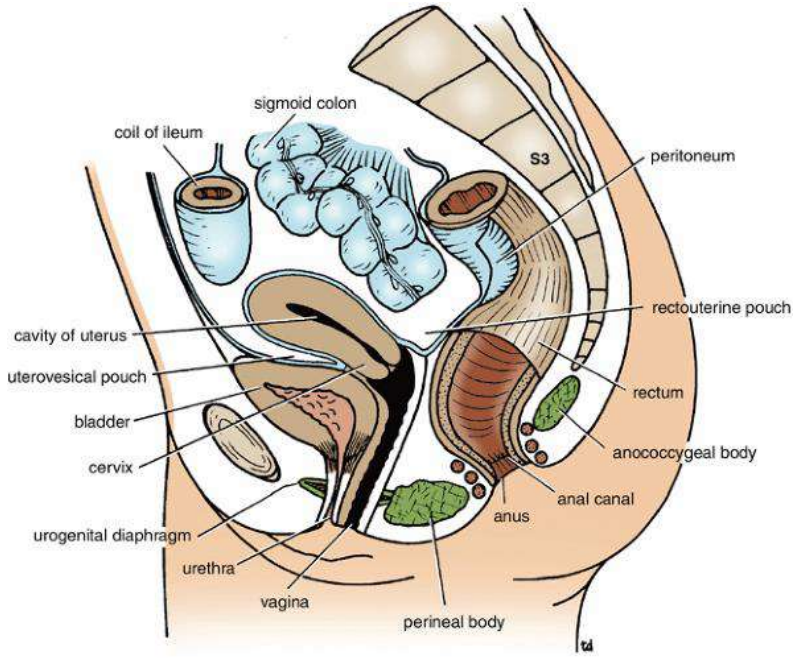


Figure 5: Uterus and vagina. (Snell, 2019).

Position of uterus: Anteflexed anteverted:

- **Anteverted:** Long axis of uterus (body and cervix) makes right angle with long axis of vagina.
- **Anteflexed:** Body of uterus bent forwards on itself at level of internal os forming angle of 170.
- In some women, uterus acquires abnormal position of retroversion and retroflexion.

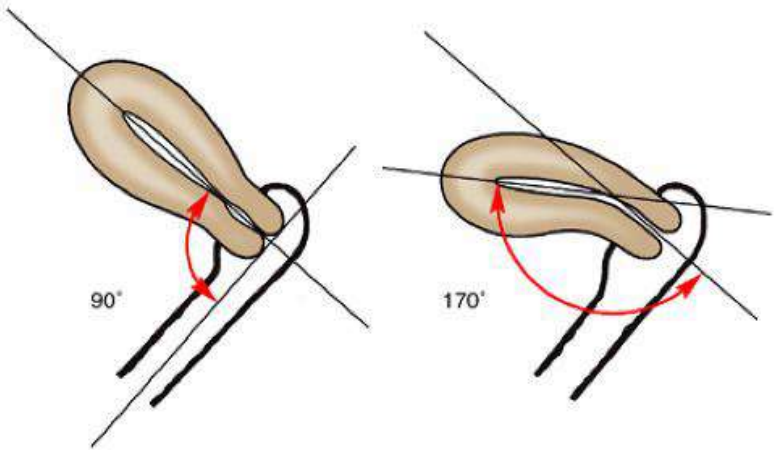


Figure 6: Normal position of uterus. (Snell, 2019).

VAGINA:

- Connects the uterus with the vestibule.
- Lies between bladder and urethra anterior and rectum and anal canal posterior.
- Anterior wall 6 cm and posterior wall 9 cm; anterior wall is pierced by the cervix.
- Forms gutter around the cervix; vaginal fornices.

VULVA (External genital organ of female):

Mons Pubis: Hairy fatty elevation of skin in front of pubic bones.

Clitoris: Highly sensitive organ of 2-3 cm and located anteriorly in vulva.

Labia majora: 2 large hairy folds of skin, provide protection for urethral and vaginal orifices.

Labia Minora: 2 small hairless folds of skin that lie between labia majora.

Vestibule of Vagina:

- Triangular area bounded by labia minora and anteriorly by clitoris.
- Urethral orifice opens in front of vaginal orifice, presents 2 cm behind clitoris.
- In virgin, the vaginal orifice is protected by hymen (mucous fold having central opening).

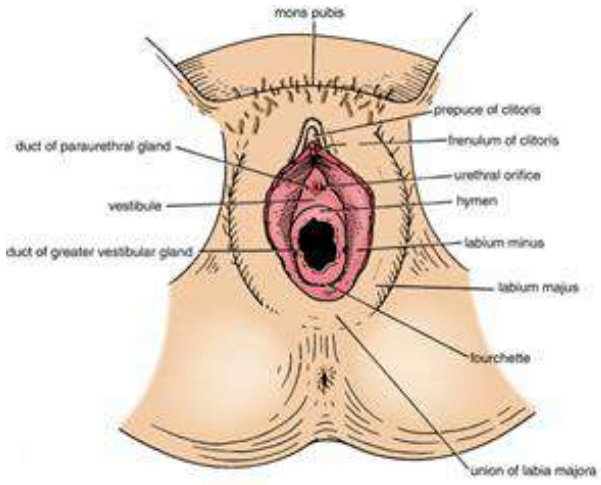


Figure 7: Normal position of uterus. (Snell, 2019).

References:

1. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.
2. Moore, Keith L. and Dalley A. F. (2023): Clinically Oriented Anatomy, 9th Edition, Lippincott Williams & Wilkins.
3. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021).

Lecture 16:

CHROMOSOME THEORY OF INHERITANCE

Objectives of the lecture:

By end of the lecture the student should be able to:

1. Define terms of embryology; conceptus, embryo, fetus, prenatal, postnatal….
2. Define terms of genetic constituents, DNA, gene, chromosomes….
3. Define and describe steps and results of cell divisions, mitosis, meiosis.

Embryology: Study of the developing human being during intrauterine life

Consists of three phases:

1. **Conceptus:** From fertilization to end of 2nd week
2. **Embryo:** From the 3rd week to 8th week of developing
3. **Foetus:** From the 3rd month till birth

Human Development:

-A continuous process that begins when an oocyte (ovum) from a female is fertilized by a sperm (spermatozoon) from a male to form the zygote.

Unicellular Zygote:

-Divides many times and becomes progressively transformed into a multicellular human being through cell division, migration, growth, differentiation, rotation, ...

Prenatal period: The main developmental changes occurring before birth.

Postnatal period: The changes occurring after birth.

Infancy:

- The earliest period of extrauterine life; roughly the first year after birth.
- An infant aged 1 month or less is called a newborn or neonate.

Childhood:

- The period from about 13months until puberty (sexual maturity).
- Primary (deciduous) teeth continue to appear and are later replaced by secondary (permanent) teeth.

Puberty:

- The period, usually between the ages of 12 to 15 years in girls and 13 to 16 years in boys.
- During which secondary sexual characteristics develop.

Adolescence: The period from about 11to 19 years of age.

- Characterized by rapid physical and sexual maturation.

Adulthood: Between the age of 18 and 21 years.

- Ossification and growth are completed during early adulthood.

Adult:

1. Early Adult: From 20-40 years

2. Late Adult: From 40-60 years

Aging: From 60 years till death

DEOXYRIBONUCLEIC ACID (DNA)

- Contains genetic information used in development and function of all living organisms.
- Works as a long-term storage (blueprint) of information needed to construct components of cells (proteins, RNA)

GENE: Segment of DNA, involved in regulation of the use of such genetic information

Genetic code: Formed of a sequence of 3 nucleotides.

Chromosome Components

1. **Chromatid:** One of the identical copies (one half of a replicated chromosome)
2. **Centromere:**

- Region near middle of chromosome where two sister chromatids come in contact
- Its position can be metacentric, sub-centric, areocentric, telocentric.

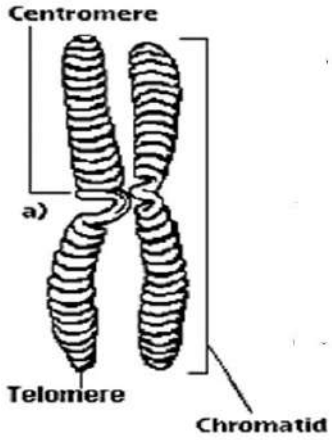


Figure 1: Components of Chromosome.

Types Of Chromosomes:

1. Autosomes: Chromosomes not involved in sex determination, 22 pairs.

2. Sex chromosomes: Responsible for sex determination, 1 pair.

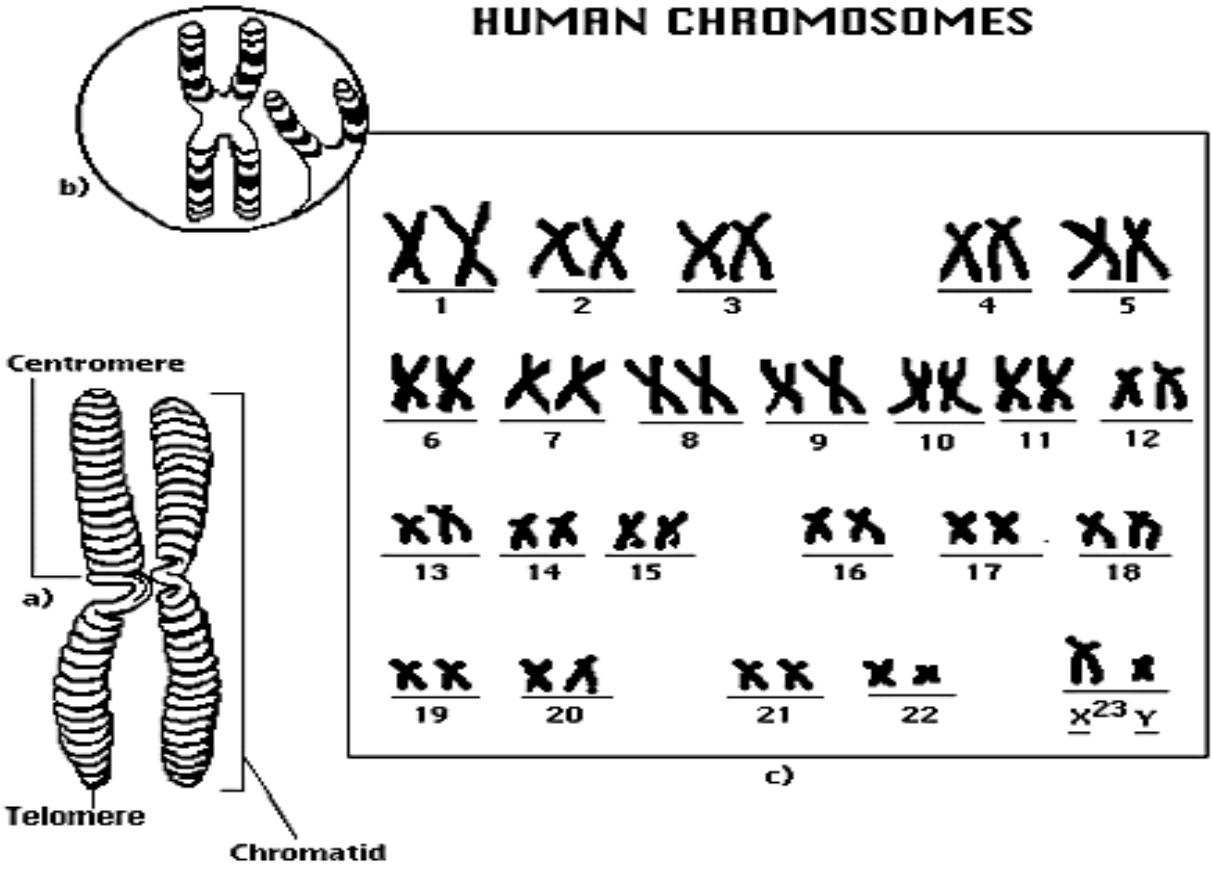


Figure 2: Human Chromosomes.

Notes:

- Each pair of chromosomes: one from the father and one from the mother.
- Member of each pair comes in close contact only during meiotic division.

Genes:

- Region of DNA that controls a hereditary characteristic (trait).
- Humans have approximately 35,000 genes on 46 chromosomes.

Trait:

- Distinct character of an organism that may be inherited.
- Some characters are visible such eye colors.
- Some are not visible such as blood grouping, risk for certain diseases

Types of Traits:

1. Dominant trait: A trait that is expressed.

2. Recessive trait: A trait that is masked.

Genome: All DNA in a cell**.**

CELL DIVISIONS

Types of Cell Divisions:

1. Mitotic cell division: Occurs in somatic cells.

2. Meiotic cell Division: Occurs in male and female germ cells**.**

MITOTIC CELL DIVISION

Definition of Mitotic Cell Division:

- Process by which one cell divides and gives rise to two daughter cells, identical to the parent cell.

Stages of mitotic cell division:

Replication phase: Somatic cell replicates its DNA before entering mitotic division.

- Chromosomes are long and spread through the nucleus and cannot be seen by light microscope.

1. Prophase: Chromosomes contract and condense, so become shorter and thicker.

2. Prometaphase: Chromatids become visible as individual units.

3. Metaphase: Chromosomes line up in equatorial plane

- Each chromosome is attached by mitotic spindle (microtubules extending from centromeres to centrioles).

4. Anaphase: Centromere divides**.**

- Each chromatid migrates to opposite poles of the spindle.

5. Telophase: Chromosomes lengthen

- Nuclear envelope reforms with division of cytoplasm

Result of mitotic cell division:

- Two daughter cells each receives one half of all the doubled chromosome material and thus maintains the same number of chromosomes as the mother cell.

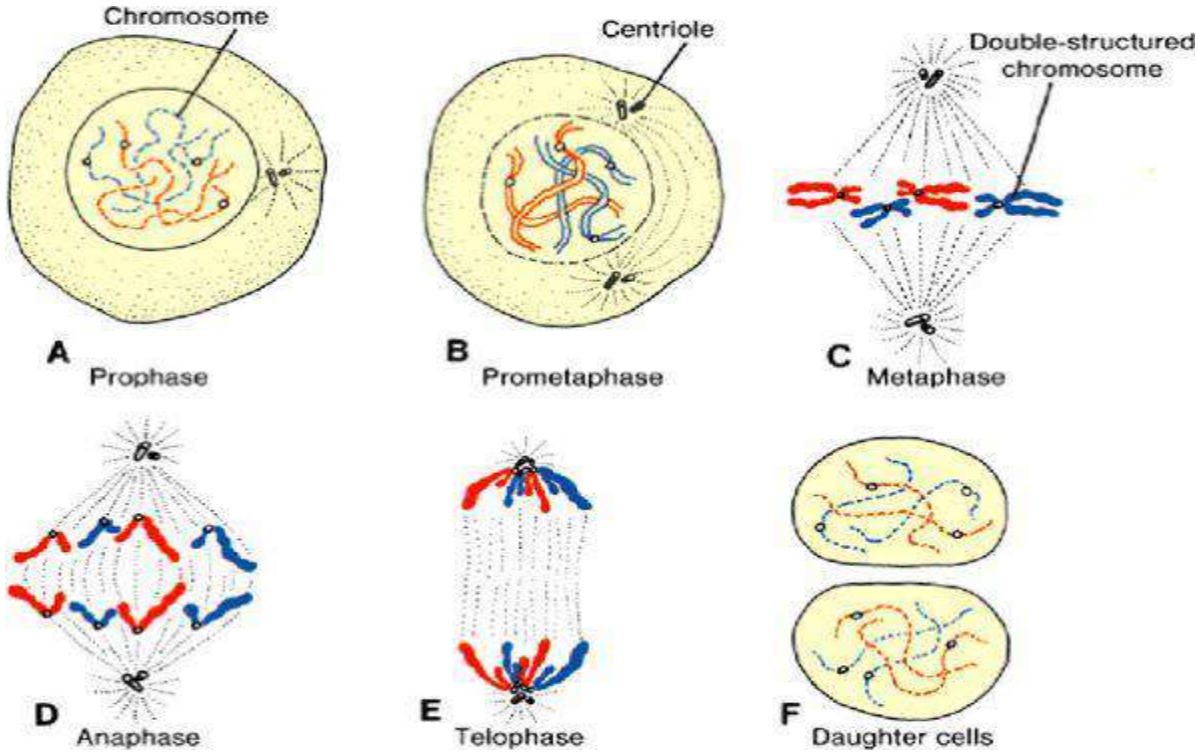


Figure 3: Various stages of mitosis. Blue, paternal chromosomes; red, maternal chromosomes.

MEIOTIC CELL DIVISION

- Occurs in male and female germ cells.
- Male and female primitive germ cells replicate their DNA just before entering the division.
- So, the germ cells contain double the normal amount of DNA (46 chromosomes).

Notees:

1. Male primitive germ cell=primary spermatocyte.
2. Female primitive germ cell= primary oocyte.

FIRST MEIOTIC DIVISION

Features of First Meiotic Division:

1. Pairing of homologous chromosomes: (Bivalents)

- Point for point
- Centromere regions do not pair
- Such pair contains 4 chromatids

2. Cross over: Interchange of chromatid segments.

3. Chiasma stage: Before homologous chromosomes separation, take the x appearance, and blocks of genes are exchanged.

4. Separation: Each chromosome migrates to opposite pole of the cell.

Result of First Meiotic Division: 2 cells each contains 23 double structured chromosomes, so it contains the same amount of DNA as in the somatic cell.

Second Meiotic Division:

- Begins shortly after first meiotic division
- No DNA synthesis occurs in advance of this division.
- Double-structured chromosomes in each cell split at the centromere.

Results of Second Meiotic Division:

- 4 daughter cells.
- Each contains 23 single chromosomes, and each are different.

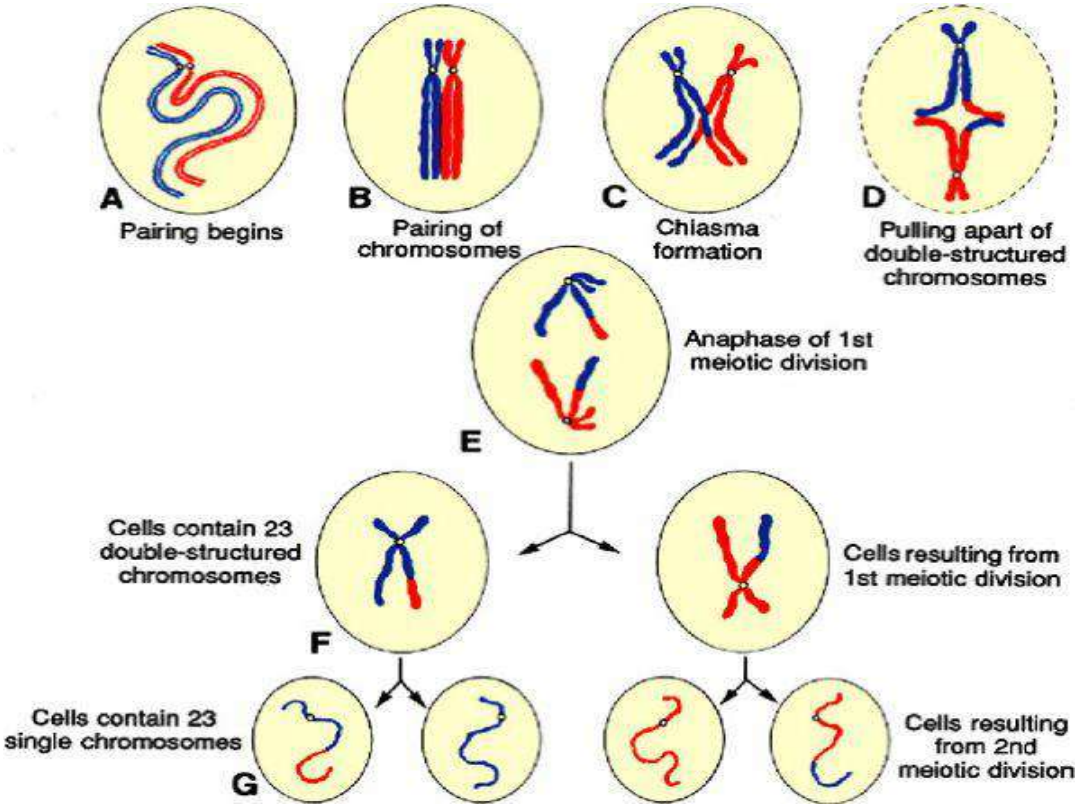


Figure 4: First and second meiotic divisions. (Sadler, 2018).

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.

2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.

Lecture 17:

GAMETOGENESIS

Objectives of the lecture:

By end of the lecture the student should be able to:

1. Define gametogenesis.
2. Describe steps of spermatogenesis.
3. What are the anomalies of the sperms.
4. Describe steps of oogenesis.
6. Describe Down, Klinefelter and Turner syndromes
5. Describe the anomalies of the ova.

Gametogenesis:

- Conversion of germ cells into male and female gametes (eggs and spermatozoa)
- Increase in the number of germ cells by mitosis
- Reduction in chromosomal number by meiosis
- Structural and functional maturation of the eggs and spermatozoa
- The first phase of gametogenesis is identical in both males and females, whereas distinct differences exist between the male and female patterns in the last phases

Primordial Germ Cells:

- Earliest precursors of gametes.
- Arise outside the gonads in the walls of the yolk sac.
- Migrate into the gonads during early embryonic development.

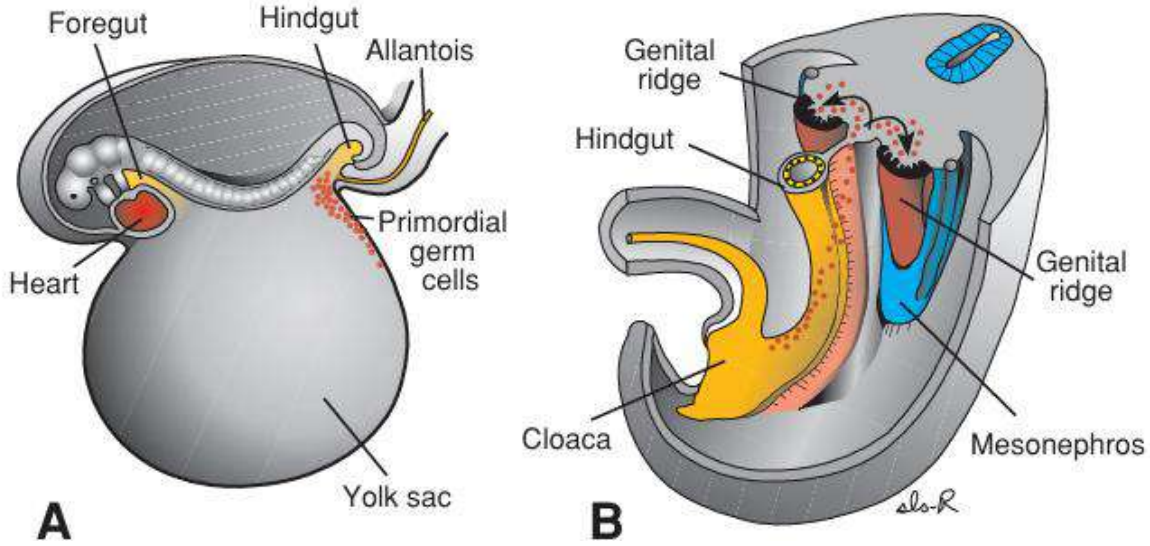


Figure 1: A. Primordial germ cells in wall of yolk sac and their migration B. (Sadler, 2018).

Meiosis (Maturation Division) in Primary Spermatocyte

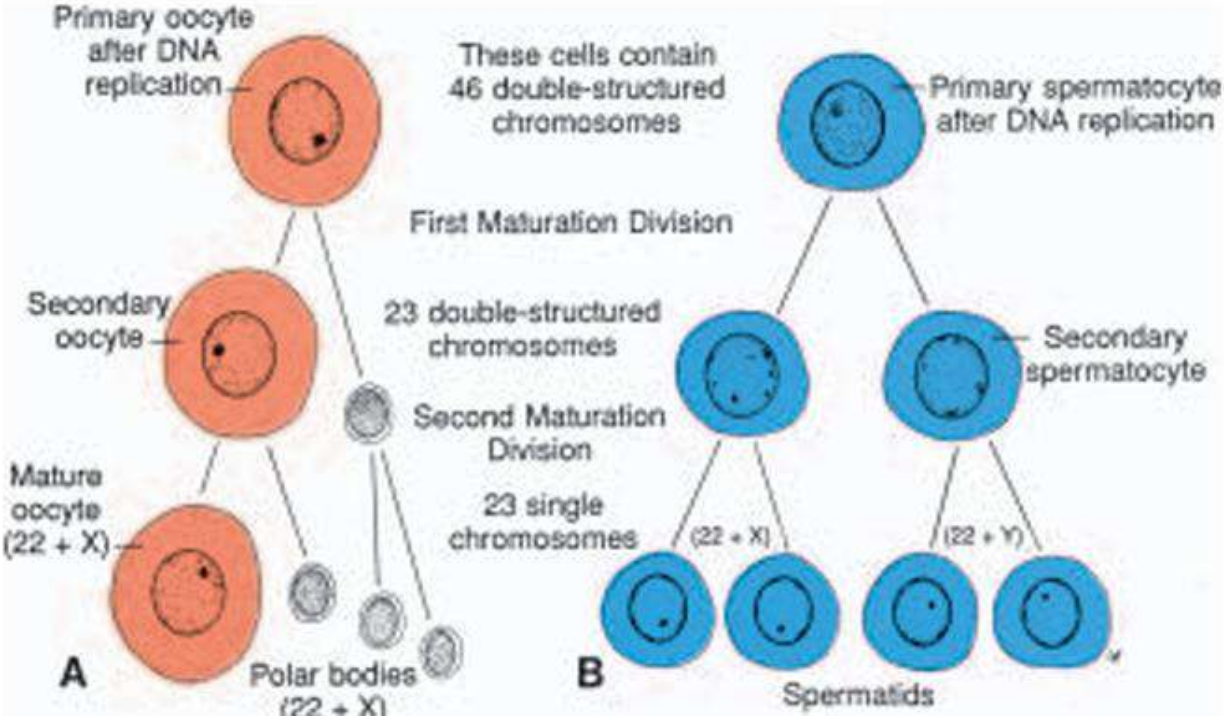


Figure 2: Events occurring during the first and second maturation divisions. A. Primary oocyte produces only one mature oocyte. Primary spermatocyte produces four spermatids, all develop into spermatozoa.

Results of Meiotic Division:

- Primary spermatocyte gives rise to 4 daughter cells, 2 with 22 + 1 X-chromosome, and 2 with 22 + 1 Y-chromosomes, (4 spermatids).
- One primary oocyte, results into one mature oocyte and three polar bodies (which degenerate later), each has 22+1 X chromosome.

Nondisjunction:

- Both members of a chromosome pair fail to separate and then move into one cell, so it receives 24 chromosomes and the other only 22, when fertilization occurs, one zygote will have 47 chromosomes (trisomy) or 45 chromosomes (monosomy).

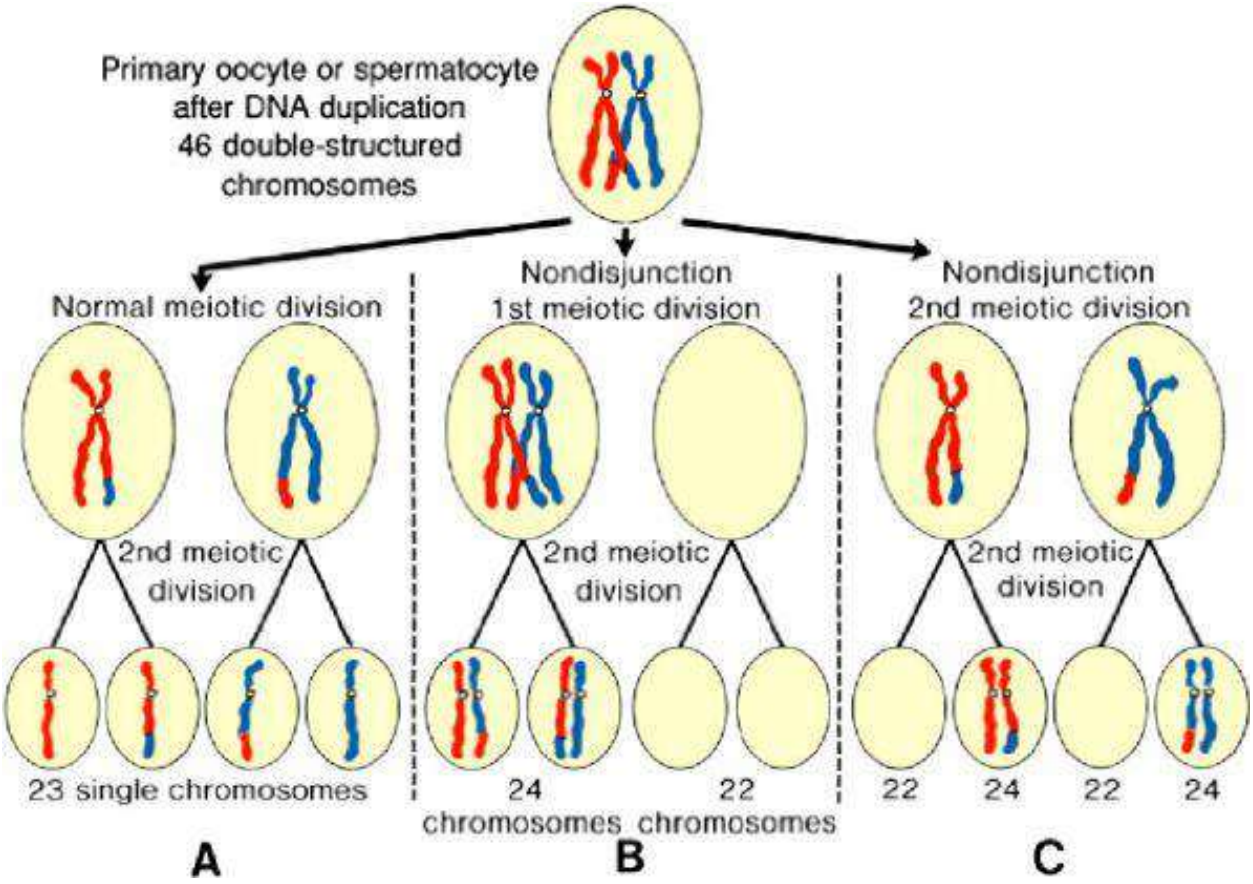


Figure 3: A. Normal maturation divisions. B. Nondisjunction in the first meiotic division. C. Nondisjunction in the second meiotic division.

Trisomy:

- One extra chromosome abbreviated as 2N+1
- Nondisjunction occurs most commonly during meiosis 1

They are generally lethal except in:

1. Trisomy 21 (Down syndrome)
2. Xxy Klinefelter syndrome

Trisomy 21 (Down Syndrome):

Cause: Maternal age

Incidence: If mother age below 25 it is 1: 2000, by age 40 its 1: 100

Features (clinical picture):

1. Flat facial profile
2. Small ears
3. Large tongue
4. Upward slanting eyes
5. Single crease in the hand
6. Loose joints
7. Low muscle tone



Figure 4: A&B and C. Children with Down syndrome:
A&B. Flat, broad face, oblique palpebral fissures, epicanthus, and furrowed lower lip.
C. Broad hand with single transverse or simian crease.

Klinefelter Syndrome:

- Found only in males
- The cells have 47 chromosomes XXY
- Detected at puberty
- Sterility
- Hyalinization of seminiferous tubules
- Testicular atrophy
- Normal phallus development but gynecomastia (enlarged breasts).

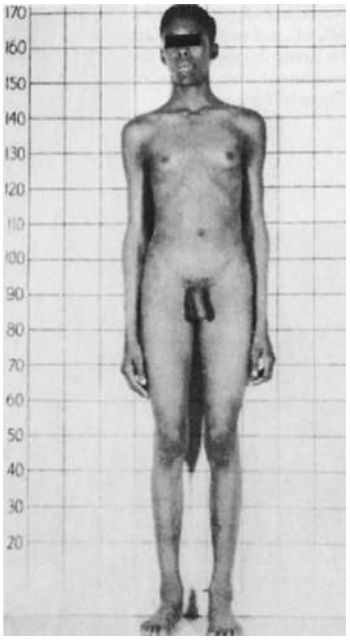


Figure 4: Klinefelter syndrome:

Turner Syndrome:

- Only monosomy compatible with life.
- 45 X0 : Girl missing one chromosome.

Clinical picture:

- Poor develop ovaries, Short, no monthly cycles.
- Infertile, webbed neck, dropping eyelids, low set ears.
- Increase carrying angle (cubitus vulgas), edema.
- Blood pressure, Heart, Kidney problems.

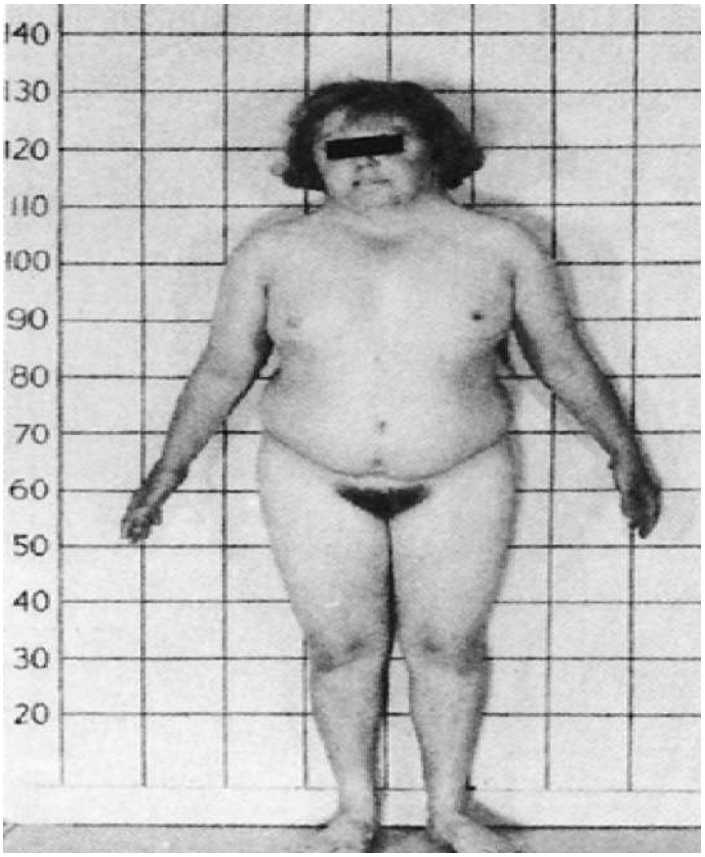


Figure 5: Patient with Turner syndrome. Main characteristics are webbed neck, short stature, broad chest, and absence of sexual maturation.

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.

2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-16

Lecture 18:

OOGENESIS

Objectives of the lecture:

By end of the lecture the student should be able to:

1. Define oogenesis.
2. Describe the steps of the oogenesis.
3. What are the anomalies of the ova?

Oogenesis

Definition: Differentiation of primordial germ cells into oocytes.

Stages of Oogenesis:

1. Prenatal maturation:

- Once a primordial germ cell reaches the ovary, it differentiates into oogonia.
- By a few mitotic divisions, the oogonia increase their number and form clusters of cells.
- Each cluster derived from a single primordial germ cell surrounded by a layer of flat epithelial cells (follicular cells) to form primordial follicle.
- **By 4th month of development:** Total number of germ cells in ovary is about 7 million.
- **By 7th month:** Starts to degenerate (only 400,000 reach puberty, and only 500 ovulated).

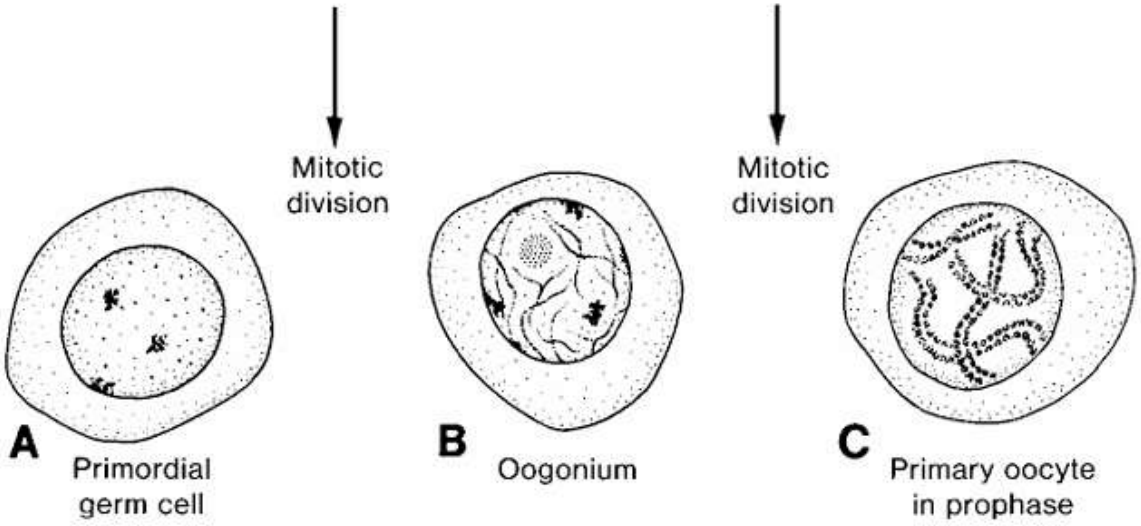


Figure 1: Differentiation of primordial germ cells into oogonia begins after their arrival in the ovary. By the third month of development, some oogonia give rise to primary oocytes that enter prophase of the first meiotic division. This prophase may last 40 years and finishes only when the cell begins its final maturation. During this period, it carries 46 chromosomes. (Sadler, 2018)

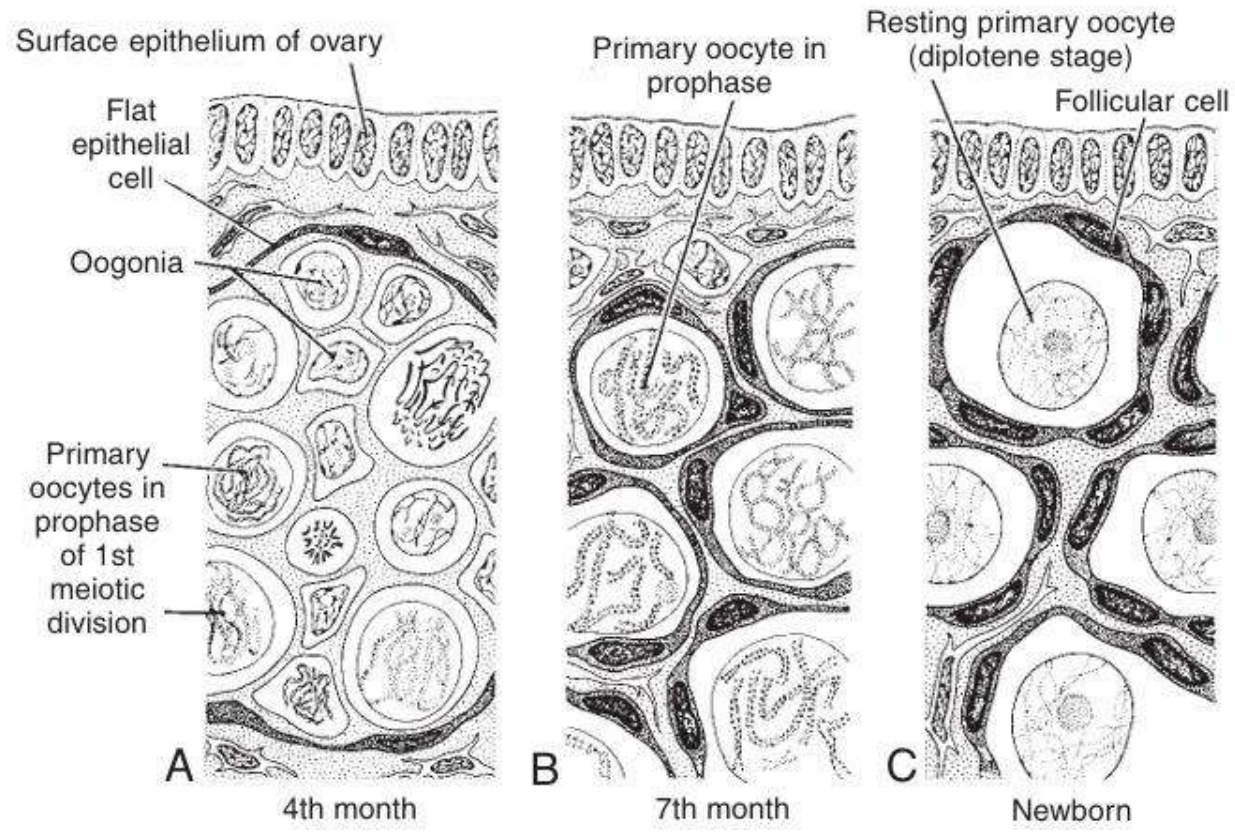


Figure 2: Segment of the ovary at different stages of development. A. Oogonia are grouped in clusters in the cortical part of the ovary. Some show mitosis; others differentiated into primary oocytes and entered prophase of the first meiotic division. B. Almost all oogonia are transformed into primary oocytes in prophase of the first meiotic division. C. Primordial follicle.

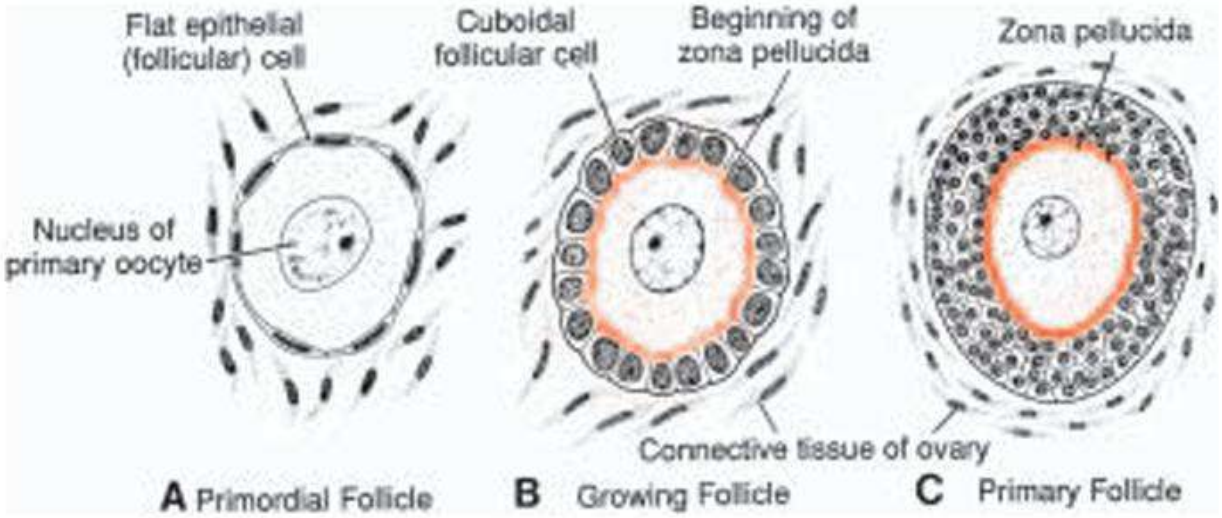


Figure 3: A. Primordial follicle. B. Early Preantral stage. C. Mature primary (preantral) follicle. (Sadler, 2018).

2. Dormant Stage:

- **At birth:** All primary oocytes, enter resting stage in prophase (diplotene stage), and do not finish their first meiotic division till puberty, due to action of oocyte maturation inhibition (OMI), secreted by follicular cells.
- So, some oocytes may remain dormant in diplotene stage of first meiotic division for 40 years or more.

3. Postnatal Maturation:

With onset of puberty:

- Each month, 15 to 20 follicles selected from this pool begin to mature, passing through three stages:

1. Primary or preantral.
2) Secondary or antral (vesicular or Graafian).
3) Preovulatory.

- The antral stage is the longest, whereas the preovulatory stage is about 37 hours before ovulation.
- As the primary oocyte begins to grow, surrounding follicular cells change from flat to cuboidal and proliferate to produce a stratified epithelium of granulosa cells, and the unit is called a primary follicle.
- Granulosa cells rest on a basement membrane separating them from surrounding stromal cells that form the theca folliculi.
- Also, granulosa cells and the oocyte secrete a layer of glycoproteins on the surface of the oocyte, forming the zona pellucida.
- As follicles continue to grow, cells of the theca folliculi organize into an inner layer of secretory cells, the theca interna, and an outer fibrous capsule, the theca externa.
- Also, small, finger-like processes of the follicular cells extend across the zona pellucida and interdigitate with microvilli of the plasma membrane of the oocyte.
- These processes transport materials from follicular cells to the oocyte.
- As development continues, fluid-filled spaces appear between granulosa cells.
- Union of these spaces forms the antrum, and the follicle is termed a secondary (vesicular, Graafian) follicle.
- Initially, the antrum is crescent shaped, but with time, it enlarges.
- Granulosa cells surrounding the oocyte remain intact and form the cumulus oophorus.
- At maturity, Graffian follicle may be 25 mm or more in diameter.
- It is surrounded by the theca interna (composed of cells having characteristics of steroid secretion, rich in blood vessels), and the theca externa (merges with ovarian stroma).
- With each ovarian cycle, a few follicles begin to develop, but usually only one reaches full maturity. The others degenerate and become atretic.

- When the secondary follicle is mature, a surge in luteinizing hormone (LH) induces the preovulatory growth phase.
- Meiosis I is completed, resulting in formation of two unequal daughter cells, each with 23 double structured chromosomes.
- One cell, the secondary oocyte, receives most of the cytoplasm; the other, the first polar body, receives none.
- The first polar body lies between the zona pellucida and the cell membrane of the secondary oocyte in the perivitelline space.
- The cell then enters meiosis II but arrests in metaphase approximately 3 hours before ovulation.
- Meiosis II is completed only if the oocyte is fertilized; otherwise, the cell degenerates approximately 24 hours after ovulation.
- The first polar body also undergoes a second division.

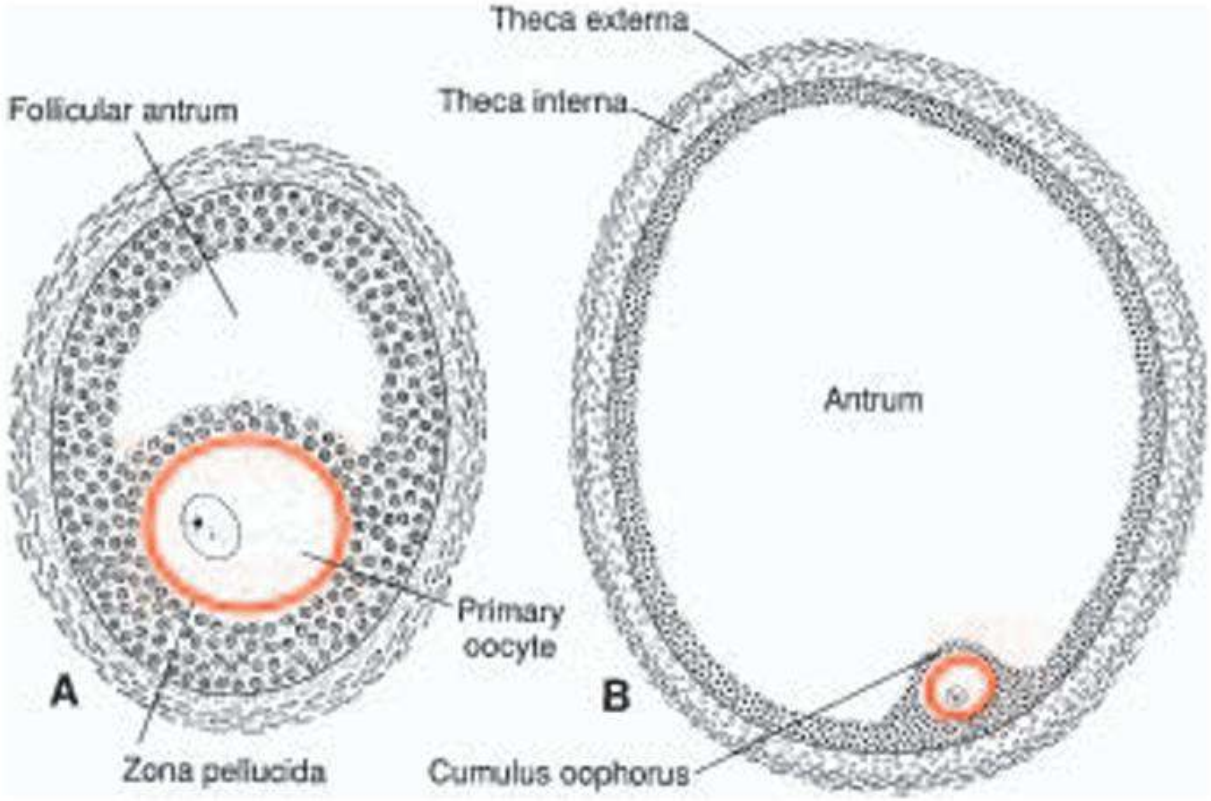


Figure 4: A. Secondary (antral) stage follicle. The oocyte, surrounded by the zona pellucida, is off-centre; the antrum has developed by fluid accumulation between intercellular spaces. B. Mature secondary (graafian) follicle. The antrum has enlarged, is filled with follicular fluid, and is surrounded by a stratified layer of granulosa cells. The oocyte is embedded in some granulosa cells, the cumulus oophorus. (Sadler, 2018).

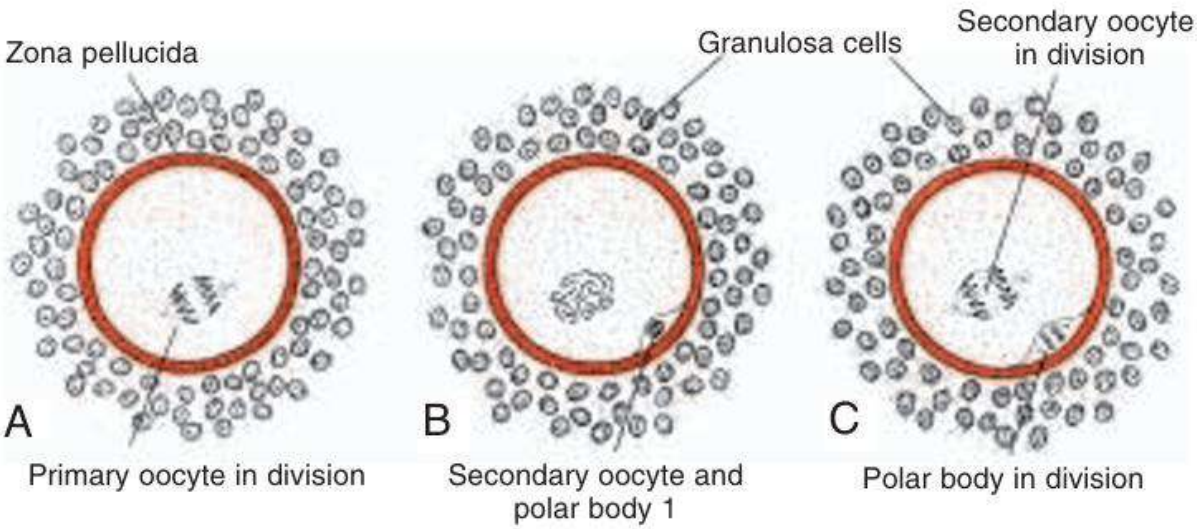


Figure 5: Maturation of the oocyte. A. Primary oocyte showing spindle of the first meiotic division. B. Secondary oocyte and first polar body. C. Secondary oocyte showing spindle of the second meiotic division. The first polar body is also dividing.

Abnormal Female Germ Cells:

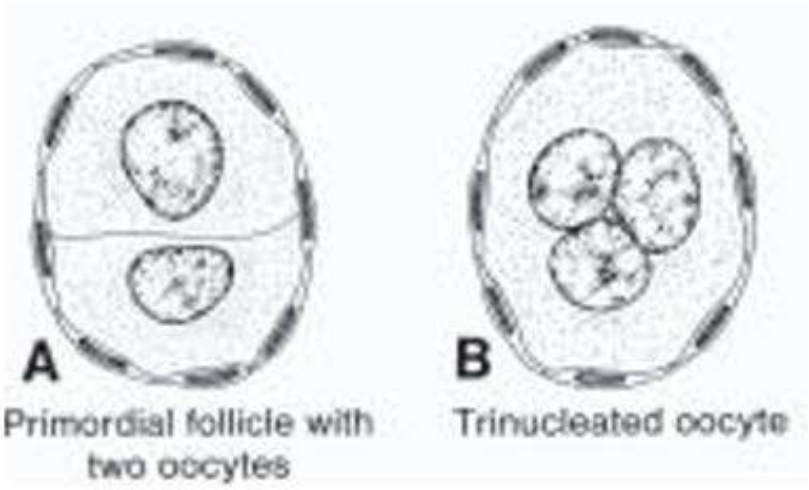


Figure 6: Primordial follicle with two oocytes (A) and trinucleated oocyte (B). (Sadler, 2018).

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.

2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.

Lecture 19:

SPERMATOGENESIS

Objectives of the lecture:

By end of the lecture the student should be able to:

1. Define spermatogenesis.
2. Describe the steps of the spermatogenesis.
3. Define spermiogenesis.
4. What are the anomalies of the sperms?

Spermatogenesis:

Definition: Transformation of spermatogonia into spermatozoa in the testes.

At Birth: Germ cells are in sex cords of the testis (large pale, surrounded by supporting cells).

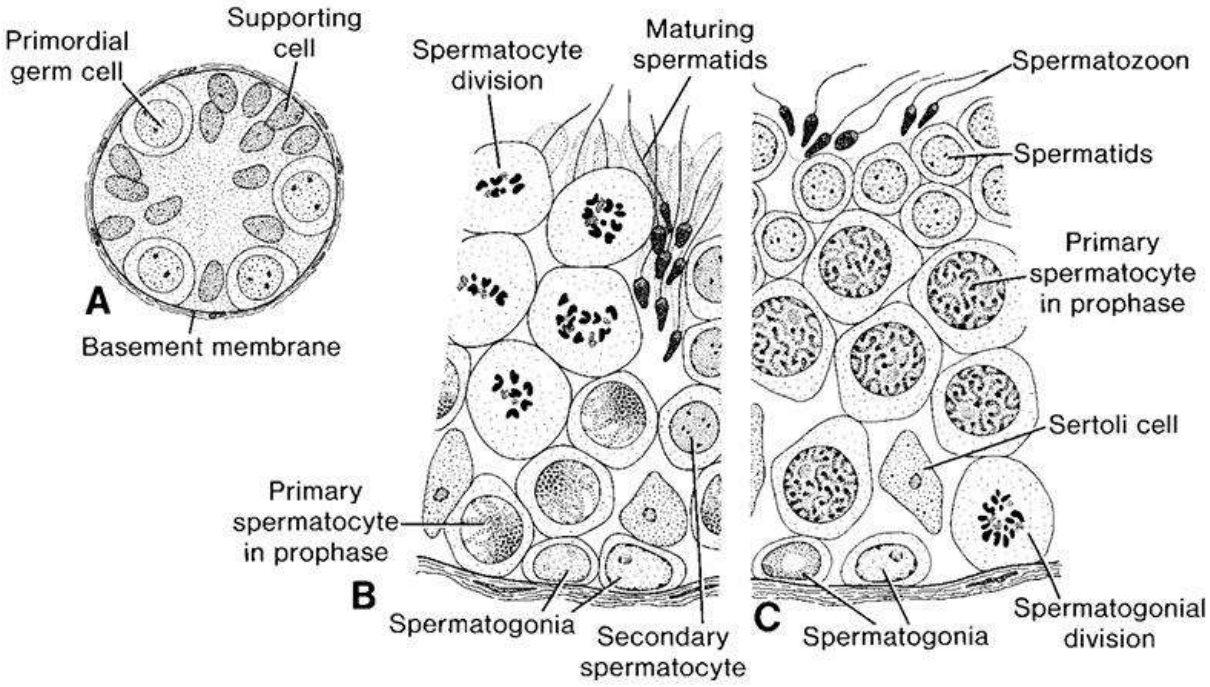


Figure 1: A. Cross section through primitive sex cords of a newborn boy showing primordial germ cells and supporting cells. B and C. Two segments of a seminiferous tubule in transverse section. Note the different stages of spermatogenesis. (Sadler, 2018).

Shortly Before Puberty: Sex cords acquire a lumen and transformed into seminiferous tubules.

PGCs differentiated into:

Spermatogonial stem cells: Will give type A spermatogonia

Type A spermatogonia:

- Works as a continuous reserve of stem cells by mitotic division.
- Its last cell division will give rise to type B spermatogonia.

Type B spermatogonia: Will differentiate into primary spermatocytes.

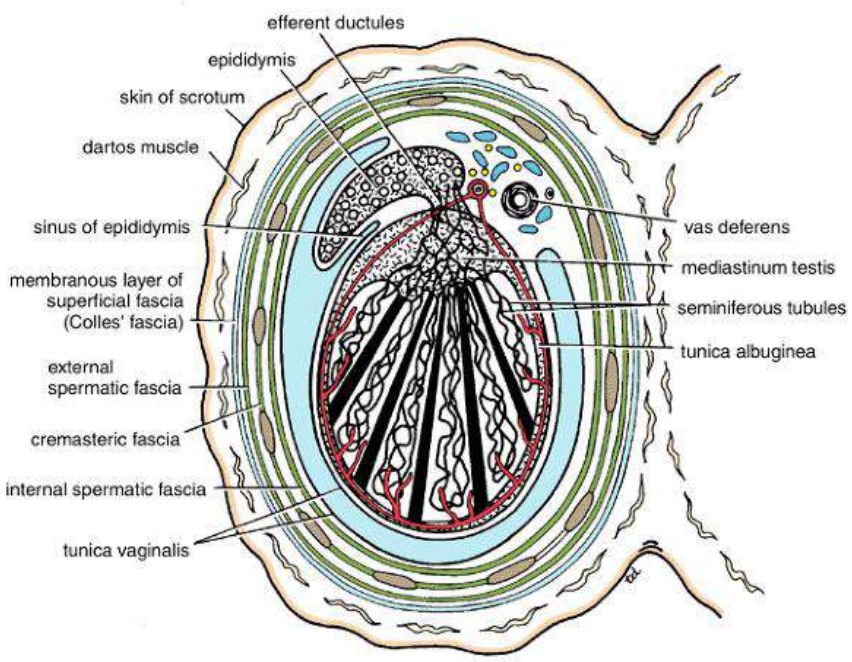


Figure 2: Section in the testes to show seminiferous tubules. (Snell, 2019).

Secondary spermatocytes: Enter meiosis 2 to form spermatids (contain haploid number).

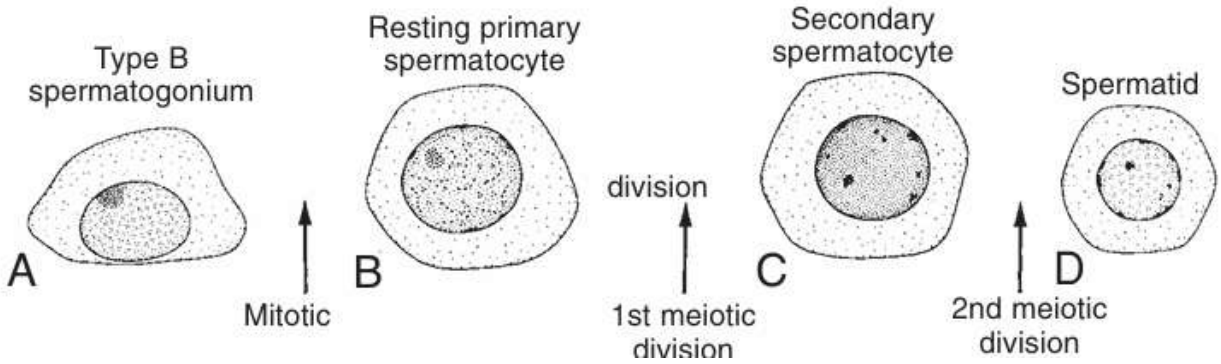


Figure 3: The products of meiosis during spermatogenesis in humans. (Sadler, 2018).

- **Once type A cells:** Leave the stem cells population cytoplasmic bridges join the cells in each succeeding division, so progeny of a single A spermatogonium form a cluster, that maintains contact throughout differentiation and gives a clone of germ cells.

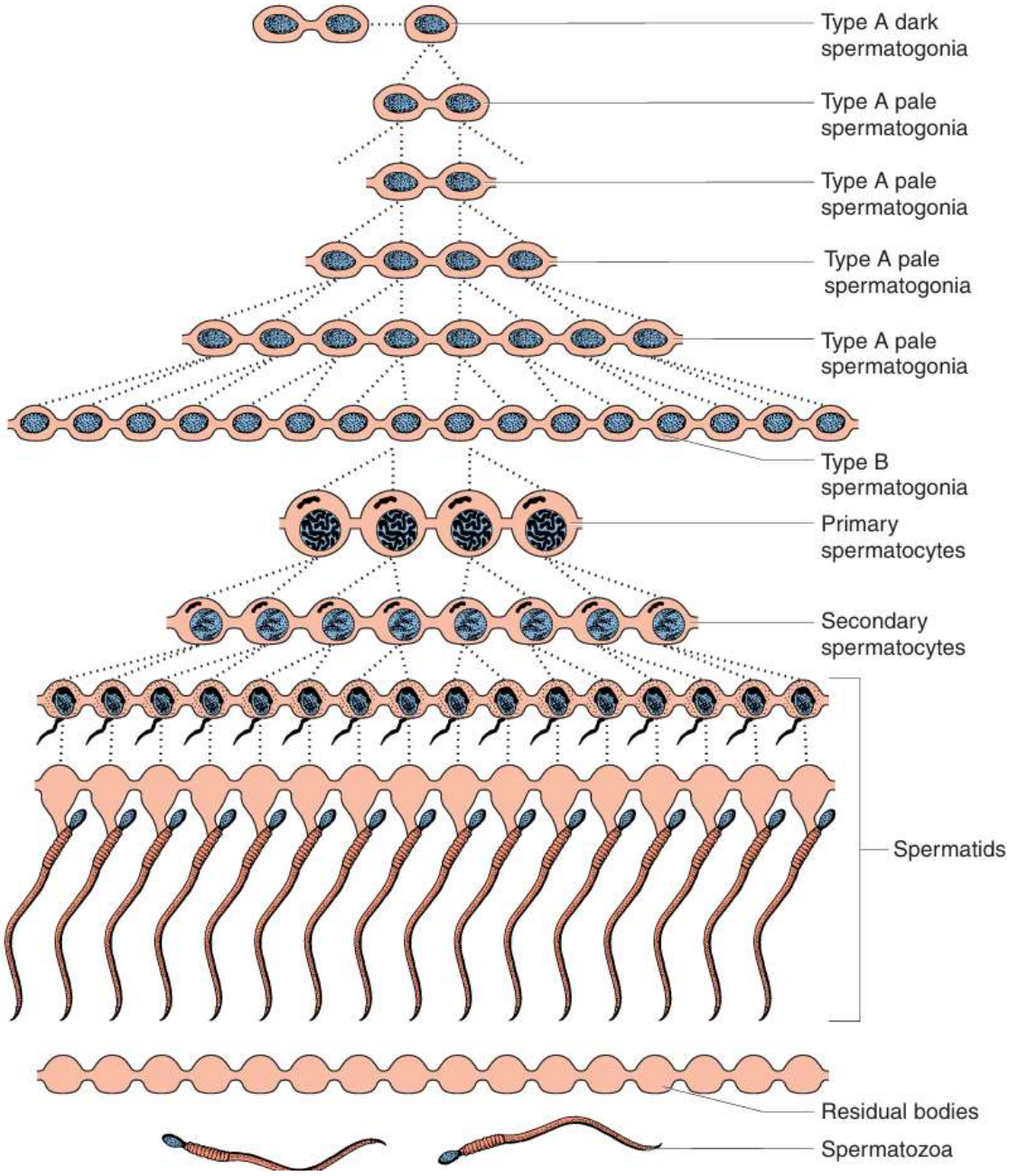


Figure 4: Type A spermatogonia, derived from the spermatogonial stem cell population, represent the first cells in the process of spermatogenesis. Clones of cells are established, and cytoplasmic bridges join cells in each succeeding division until individual sperm are separated from residual bodies. (Sadler, 2019).

Spermatogonia and Spermatids: Embedded in deep recesses of Sertoli cells which provide support, protection, nutrition and assist in release of mature spermatozoa.

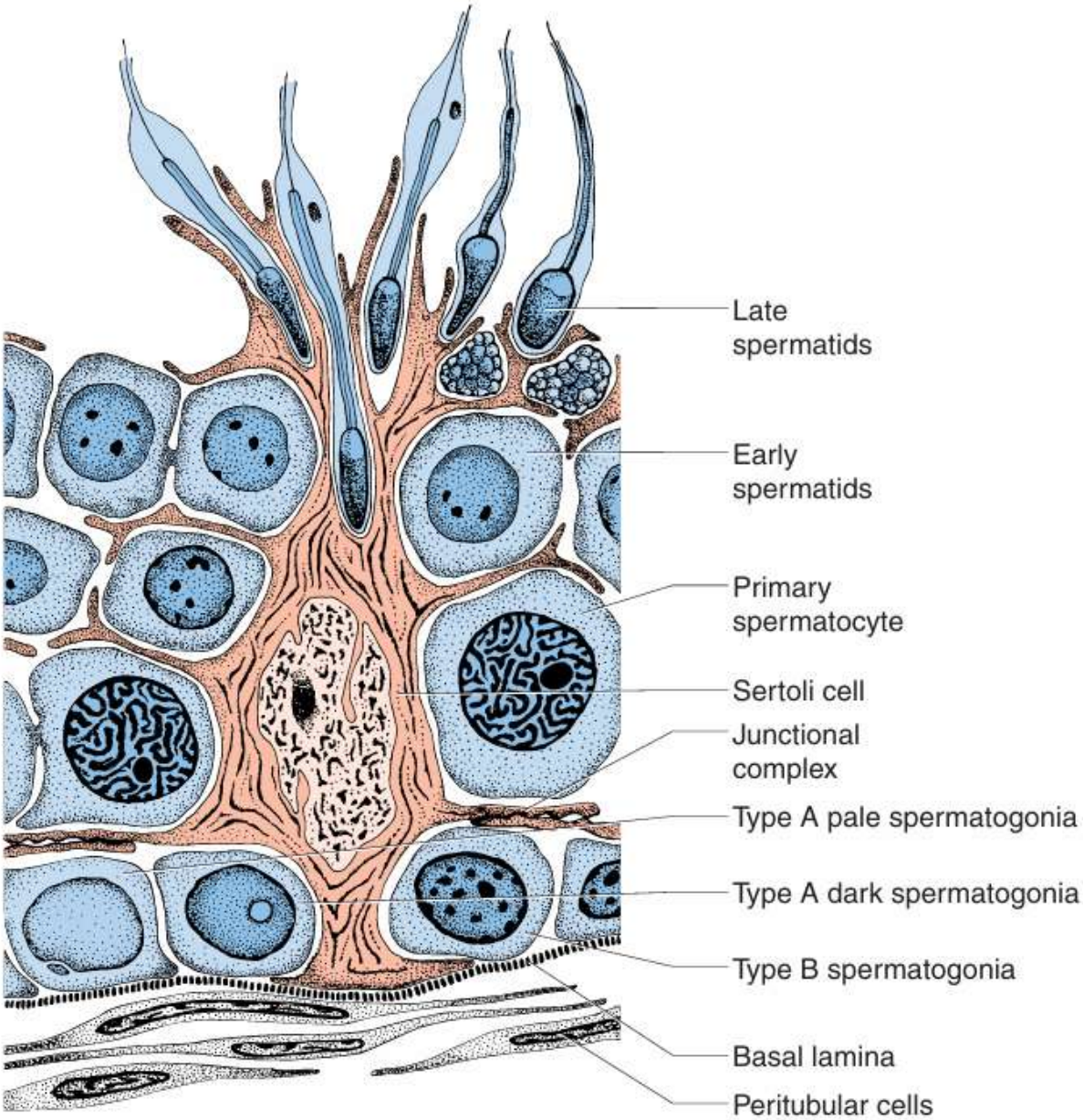


Figure 5: Sertoli cells and maturing spermatocytes. Spermatogonia, spermatocytes, and early spermatids occupy depressions in basal aspects of the cell; late spermatids are in deep recesses near the apex. (Sadler, 2019).

Spermatogenesis is regulated by:

- Luteinizing hormone (LH) production by the pituitary gland.
- LH binds to receptors on Leydig cells and stimulates testosterone production, which in turn binds to Sertoli cells to promote spermatogenesis.
- Follicle stimulating hormone (FSH) is also essential because its binding to Sertoli cells stimulates testicular fluid production and synthesis of intracellular androgen receptor proteins.

SPERMIOGENESIS

Definition: Transformation of spermatids into spermatozoa by:

1. Acrosome formation: Covers half the nuclear surface and contains enzymes.
2. Condensation of the nucleus.
3. Formation of neck, middle piece and tail.
4. Shedding most of the cytoplasm.

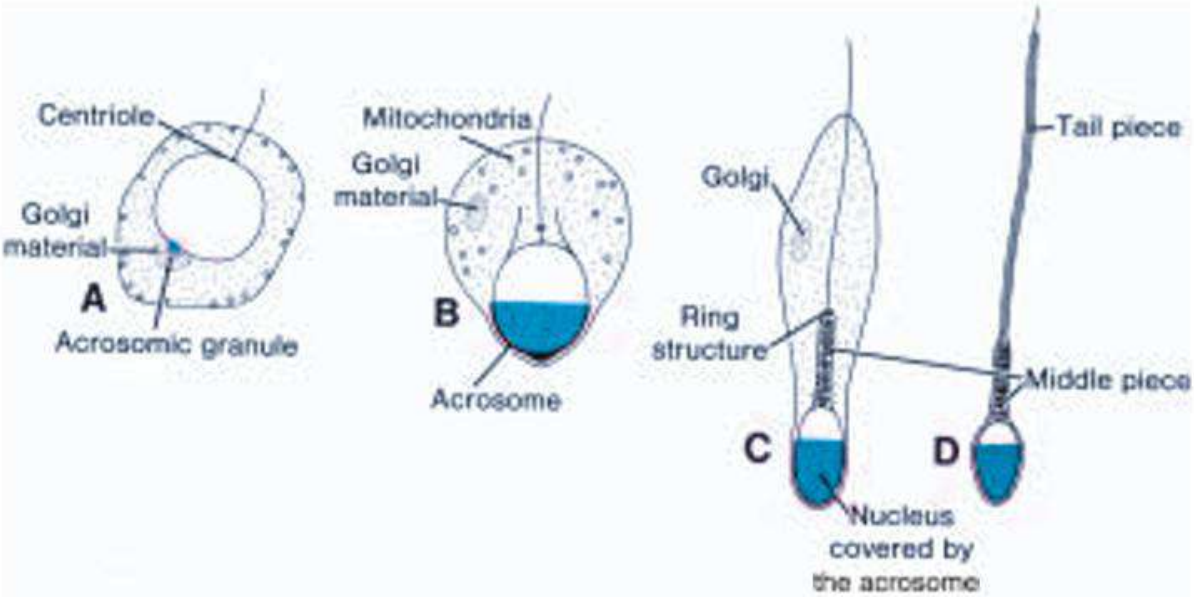


Figure 6: Stages in transformation of the human spermatid into the spermatozoon.

Note:

- In humans, the time required for a spermatogonium to develop into a mature spermatozoon is about 64 days.
- When fully formed, spermatozoa enter the lumen of seminiferous tubules.
- From there, they are pushed toward the epididymis by contractile elements in the wall of the seminiferous tubules.
- Initially spermatozoa only slightly motile.
- Spermatozoa obtain full motility in the epididymis.

Abnormal Male Germ Cells (Figure 7).

1. Abnormal head.
2. Abnormal tail.
3. Abnormal size: giant or very small (dwarfs).
4. Joined.

- Abnormal spermatozoa are seen frequently.
- Up to 10% of all spermatozoa have observable defects.
- If more than 25% are abnormal this may affect fertility.
- Sperm with morphologic abnormalities lack normal motility, probably do not fertilize oocytes.

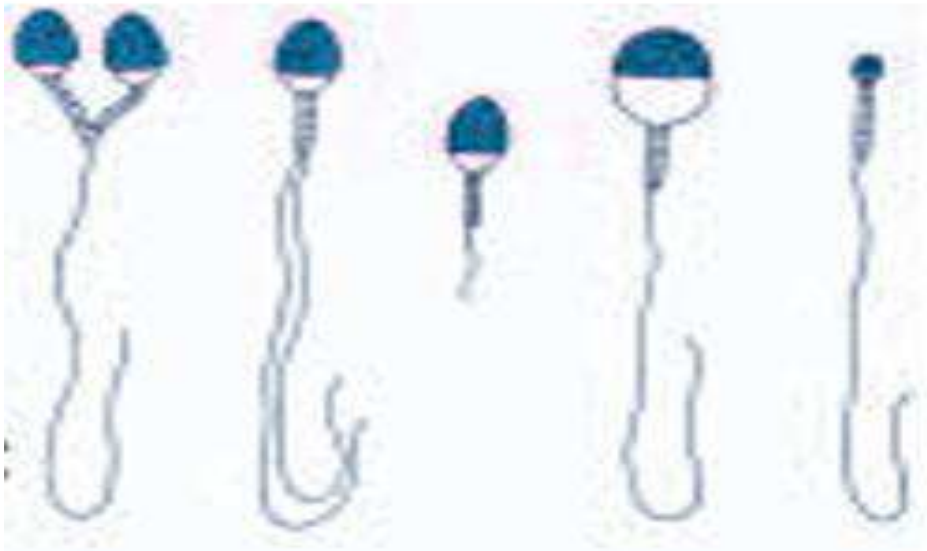


Figure 7: Various types of abnormal spermatozoa.

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.

2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.

3. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.

Lecture: 20

OVARIAN CYCLE OVULATION

ILOs:

1. When the monthly cycles begin in the female?
2. Enumerate the hormones that control the ovarian cycle.
3. What are the functions of oestrogen?
4. What are the functions of LH?
5. Describe the mechanism of ovulation.
6. Identify time of Ovulation.

OVARIAN CYCLE

At puberty:

- The female begins to undergo regular monthly cycles.
- These sexual cycles are controlled by the hypothalamus.
- Gonadotropin-releasing hormone (GnRH) produced by the hypothalamus acts on cells of the anterior pituitary gland, which in turn secrete gonadotropins.
- These hormones, follicle-stimulating hormone (FSH) and luteinizing hormone (LH), stimulate and control cyclic changes in the ovary.

At the beginning of each ovarian cycle:

- 15 to 20 primary stage follicles are stimulated to grow under the influence of FSH.
- Under normal conditions, only one of these follicles reaches full maturity, and only one oocyte is discharged; the others degenerate and become atretic.
- Most follicles degenerate without ever reaching full maturity.
- When a follicle becomes atretic, the oocyte and surrounding follicular cells degenerate and are replaced by connective tissue, forming a corpus atreticum.
- FSH also stimulates maturation of follicular (granulosa) cells surrounding the oocyte.
- Granulosa and thecal cells produce oestrogens.

Function of Oestrogen:

1. Cause the uterine endometrium to enter the follicular or proliferative phase.
2. Cause thinning of the cervical mucus to allow passage of sperm.
3. Stimulate the pituitary gland to secrete LH.

At mid-cycle, there is an LH surge:

Function of LH:

1. Elevates concentrations of maturation-promoting factor, causing oocytes to complete meiosis I and initiate meiosis II.
2. Stimulates production of progesterone by follicular stromal cells (luteinization).
3. Causes follicular rupture and ovulation.

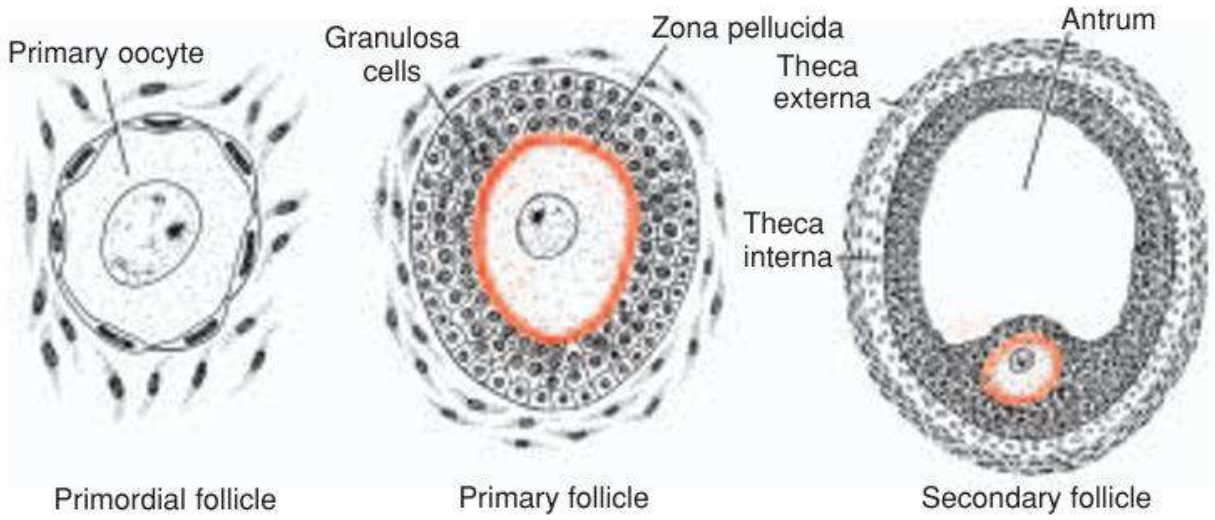


Figure 1: From the pool of primordial follicles, every day some begin to grow and develop into primary (preantral) follicles. As the cycle progresses, FSH secretion recruits primary follicles to begin development into secondary (antral, Graafian) follicles. (Sadler, 2018).

OVULATION:

- In the days immediately preceding ovulation, under the influence of FSH and LH, the secondary follicle grows rapidly to a diameter of 25 mm.
- Coincident with final development of the secondary follicle, there is an abrupt increase in LH that causes the primary oocyte to complete meiosis I and the follicle to enter the preovulatory stage.
- Meiosis II is also initiated, but the oocyte is arrested in metaphase 3 hours before ovulation.
- **In meantime:** Surface of the ovary begins to bulge locally, and at the apex, an avascular spot, the stigma, appears.
- High concentration of LH increases collagenase activity, resulting in digestion of collagen fibres surrounding the follicle.
- Prostaglandin levels also increase in response to the LH surge and cause local muscular con tractions in the ovarian wall.
- Those contractions extrude the oocyte, which together with its surrounding granulosa cells from region of the cumulus oophorus, breaks free (ovulation) and floats out of the ovary.
- Some of the cumulus oophorus cells then rearrange themselves around the zona pellucida to form the corona radiata.

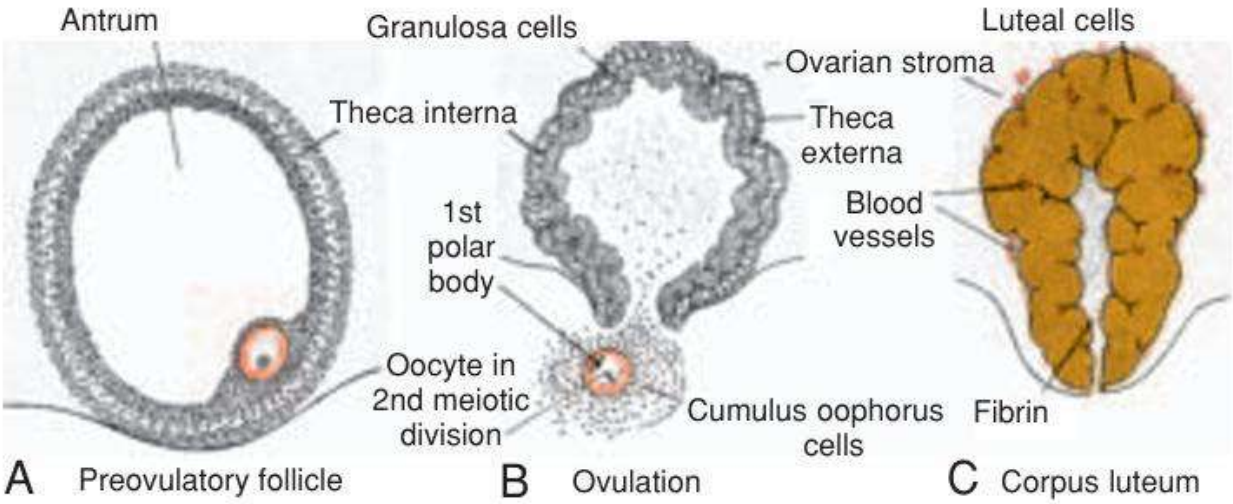


Figure 2: A. Preovulatory follicle bulging at ovarian surface. **B.** Ovulation. Oocyte, in metaphase of meiosis II, is discharged from the ovary together with many cumulus oophorus cells. Follicular cells remaining inside collapsed follicle differentiate into lutean cells. **C.** Corpus luteum. Large size of corpus luteum, by hypertrophy and accumulation of lipid in granulosa and theca interna cells. Remaining cavity of the follicle is filled with fibrin. (Sadler, 2018).

Note on Ovulation:

- During ovulation, some women feel a slight pain, middle pain because it normally occurs near the middle of the menstrual cycle.
- Ovulation is generally accompanied by a rise in basal temperature, which can be monitored to aid in determining when release of the oocyte occurs.
- Some women fail to ovulate because of a low concentration of gonadotropins.
- Administration of an agent to stimulate gonadotropin release and hence ovulation can be employed.
- Such drugs often produce multiple ovulations, so that the risk of multiple pregnancies is 10 times higher in these women than in the general population

Corpus Luteum:

- After ovulation, granulosa cells remaining in the wall of ruptured follicle, together with cells from the theca interna, are vascularized by surrounding vessels.
- Under the influence of LH, these cells develop a yellowish pigment and change into lutean cells, which form the corpus luteum and secrete the hormone progesterone.
- Progesterone, together with estrogenic hormones, causes the uterine mucosa to enter the progestational or secretory stage in preparation for implantation of the embryo.

Oocyte Transport:

- Shortly before ovulation, fimbriae of the uterine tube begin to sweep over the surface of the ovary, and the tube itself begins to contract rhythmically.

- The oocyte surrounded by some granulosa cells is carried into the tube by these sweeping movements of the fimbriae and by motion of cilia on the epithelial lining.
- In the tube, cumulus cells withdraw their cytoplasmic processes from the zona pellucida and lose contact with the oocyte.
- The oocyte in the uterine tube is propelled by the cilia.
- In humans, the fertilized oocyte reaches the uterine lumen in approximately 3 to 4 days

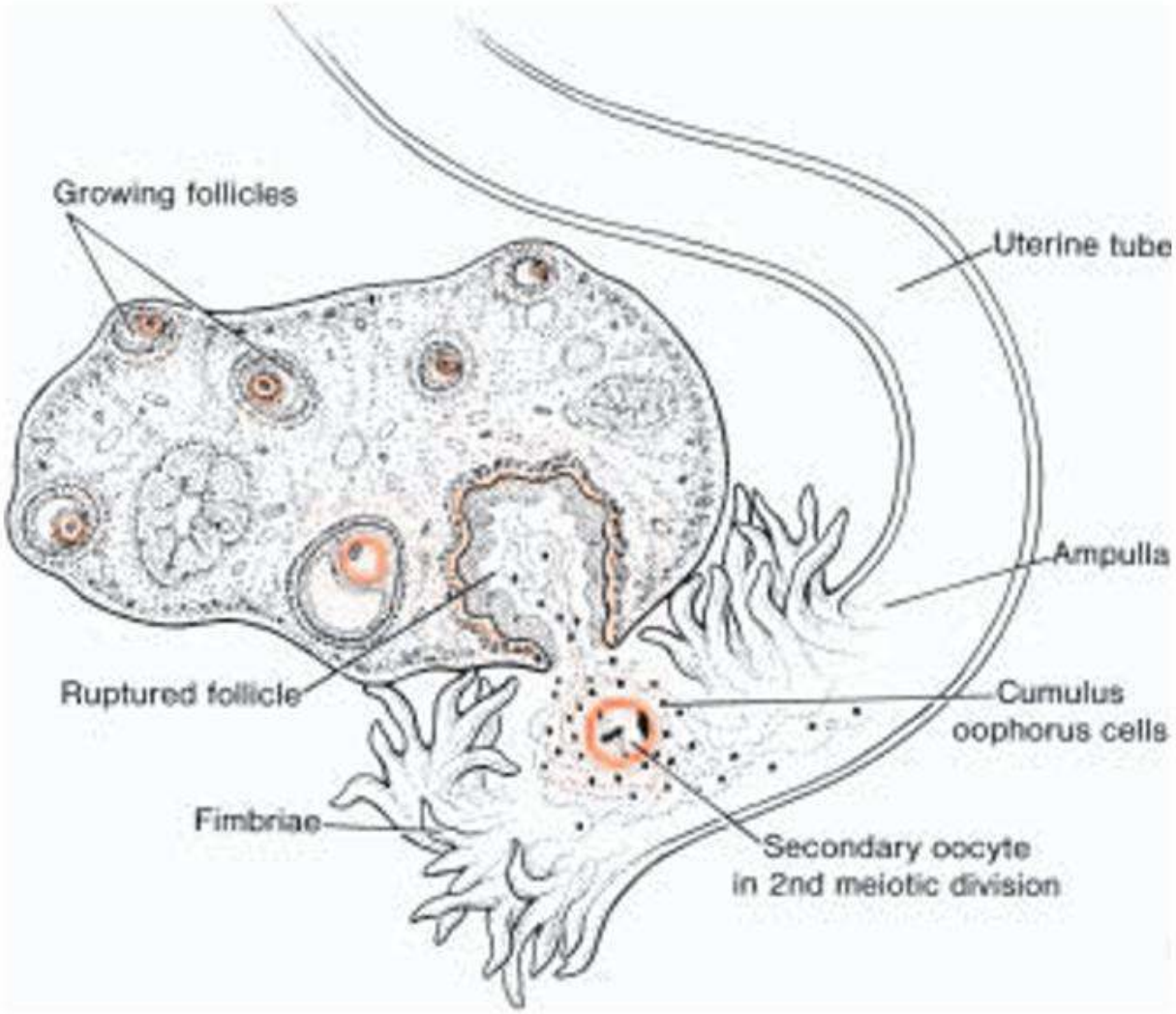


Figure 3: Relation of fimbriae and ovary. Fimbriae collect the oocyte and sweep it into the uterine tube. (Sadler, 2018).

Corpus Albicans:

If fertilization does not occur:

- The corpus luteum reaches maximum development approximately 9 days after ovulation.
- The corpus luteum shrinks because of degeneration of lutean cells and forms a mass of fibrotic scar tissue, corpus albicans.
- Progesterone production decreases, precipitating menstrual bleeding.

If the oocyte is fertilized:

- Degeneration of the corpus luteum is prevented by human chorionic gonadotropin (hCG) hormone secreted by syncytiotrophoblast of developing embryo.
- The corpus luteum continues to grow and forms the corpus luteum of pregnancy.

By the end of the third month:

- Corpus luteum of pregnancy may be one-third to one-half of the total size of the ovary.
- Yellowish luteal cells continue to secrete progesterone until end of the fourth month; then, they regress slowly as secretion of progesterone by trophoblastic component of the placenta becomes adequate for maintenance of pregnancy.

References:

- 1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
- 2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
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Lecture: 21

FERTILIZATION

ILOs:

1. Define fertilization.
2. Summarize stages of fertilization and zygot formation.
3. Enumerate results of fertilization.
4. Recognize infertility (male and female factors).

Fertilization:

Definition of fertilization:

- The process by which male and female gametes fuse.

Site of fertilization:

- Ampullary region of the uterine tube, widest part of the tube and close to the ovary.

Notes:

- Spermatozoa may remain viable in the female reproductive tract for several days.
- Only 1% of sperms deposited in the vagina enter the cervix, where they may survive for many hours.
- Movement of sperm from the cervix to the oviduct is accomplished by their own propulsion, assisted by movements of fluids created by uterine cilia.
- The trip from cervix to oviduct requires a minimum of 2 to 7 hours.
- After reaching the isthmus, sperm become less motile and cease their migration.
- At ovulation, sperm again become motile, perhaps because of chemo-attractants produced by cumulus cells surrounding the egg, and swim to the ampulla where fertilization usually occurs.

Spermatozoa Preparation for Fertilization:

1. **Capacitation:**
2. **Acrosome reaction.**

Capacitation:

- Period of conditioning in female reproductive tract that in the human lasts about 7 hours.
- During this time a glycoprotein coat and seminal plasma proteins are removed from the plasma membrane that overlies the acrosomal region of the spermatozoa.
- Only capacitated sperm can pass through the corona cells and undergo acrosome reaction.

Acrosome reaction:

- Occurs after binding of the sperm to the zona pellucida.
- Induced by zona proteins.

- This reaction culminates in the release of enzymes needed to penetrate the zona pellucida, including acrosin and trypsin-like substances.

Phases of Fertilization: 3 Phases:

1. Penetration of the corona radiata.
2. Penetration of the zona pellucida.
3. Fusion of the oocyte and sperm cell membranes.

Phase 1: Penetration of the Corona Radiata:

- Of the 200 to 300 million spermatozoa deposited in the female genital tract, only 300 to 500 reach the site of fertilization.
- Only one of these fertilizes the egg.
- Other sperms aid fertilizing sperm in penetrating the barriers protecting female gamete.
- Capacitated sperm pass freely through corona cells.

Phase 2: Penetration of the Zona Pellucida:

- The zona is a glycoprotein shell surrounding the egg that facilitates and maintains sperm binding and induces the acrosome reaction.
- Release of acrosomal enzymes (acrosin) allows sperm to penetrate the zona, meeting the plasma membrane of the oocyte.
- Permeability of the zona pellucida changes when the head of sperm meets the oocyte surface.
- This contact results in release of lysosomal enzymes from cortical granules lining the plasma membrane of the oocyte.
- These enzymes alter properties of zona pellucida to prevent sperm penetration and inactivate species-specific receptor sites for spermatozoa on the zona surface.
- Only one sperm able to penetrate the oocyte.

Phase 3: Fusion of the Oocyte and Sperm Cell Membranes

- After adhesion, the plasma membranes of the sperm and egg fuse.
- Because the plasma membrane covering the acrosomal head cap disappears during the acrosome reaction, actual fusion is accomplished between the oocyte membrane and the membrane that covers the posterior region of sperm head.
- In the human, both the head and tail of the spermatozoon enter the cytoplasm of oocyte, but the plasma membrane is left behind on the oocyte surface.

Egg Response to Penetration of the Sperm:

1. Cortical and zona reactions:

- As a result of release of cortical oocyte granules, which contain lysosomal enzymes:

a. Oocyte membrane becomes impenetrable to other spermatozoa.

b. Zona pellucida alters its structure and composition to prevent sperm penetration.

- These reactions prevent polyspermy.

2. Resumption of the Second Meiotic Division.

- Oocyte finishes its second meiotic division immediately after entry of the spermatozoon. One of the daughter cells, which receives hardly any cytoplasm, second polar body; other daughter cell is definitive oocyte.
- Its chromosomes (22+X) arrange themselves in a vesicular nucleus; female pronucleus.

3. Metabolic activation of the egg.

- The activating factor is carried by the spermatozoon.
- Post fusion activation initiates cellular and molecular events associated with early embryogenesis.

Zygote Formation:

- The sperm moves forward until it lies close to the female pronucleus.
- Its nucleus becomes swollen and forms male pronucleus.
- The tail detaches and degenerates.
- Male and female pronuclei come into close contact and lose their nuclear envelopes.
- During growth of male and female pronuclei (both haploid), each pronucleus replicates its DNA.
- If it does not, each cell of the two-cell zygote has only half of normal amount of DNA.
- Immediately after DNA synthesis, chromosomes organize on the spindle in preparation for a normal mitotic division.
- The 23 maternal and 23 paternal (double) chromosomes split longitudinally at the centromere, and sister chromatids move to opposite poles, providing each cell of the zygote with normal diploid number of chromosomes and DNA.
- As sister chromatids move to opposite poles, a deep furrow appears on surface of the cell, gradually dividing the cytoplasm into two parts.

Results of Fertilization:

1. Restoration of diploid number of chromosomes, half from father and half from mother.

- The zygote contains a new combination of chromosomes different from both parents.

2. Determination of the sex of the new individual.

- An X-carrying sperm produces a female (XX) embryo, and a Y-carrying sperm produces a male (XY) embryo. Chromosomal sex of the embryo is determined at fertilization.

3. Initiation of cleavage.

- Without fertilization, the oocyte usually degenerates 24 hours after ovulation.

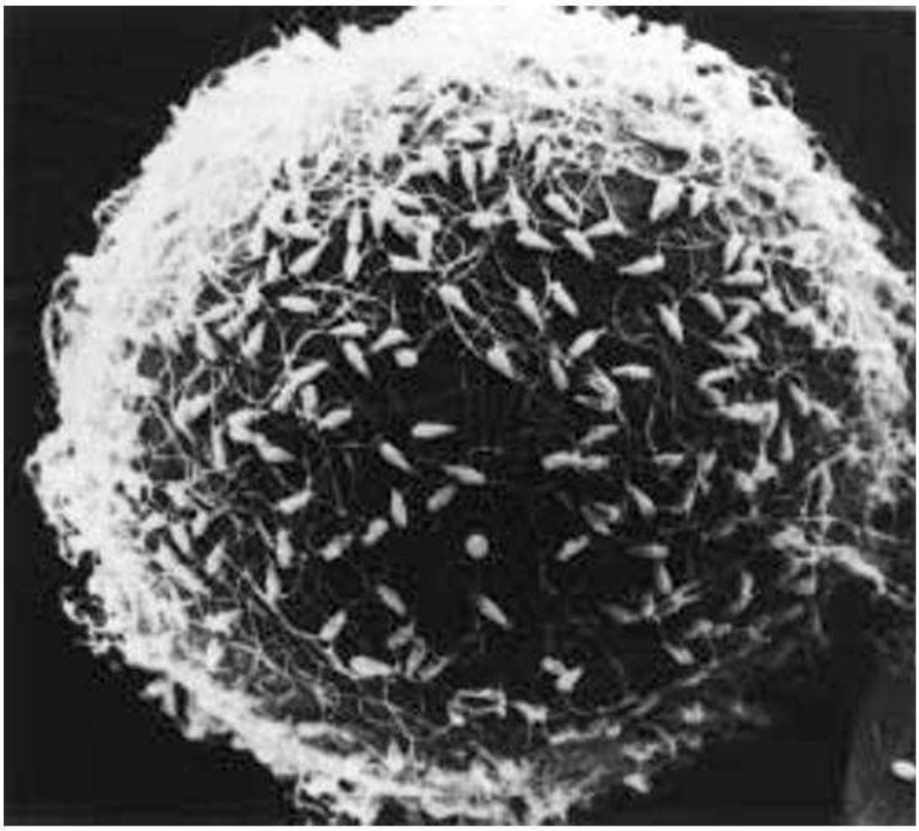


Figure 1: Scanning electron micrograph of sperm binding to the zona pellucida. (Sadler, 2018).

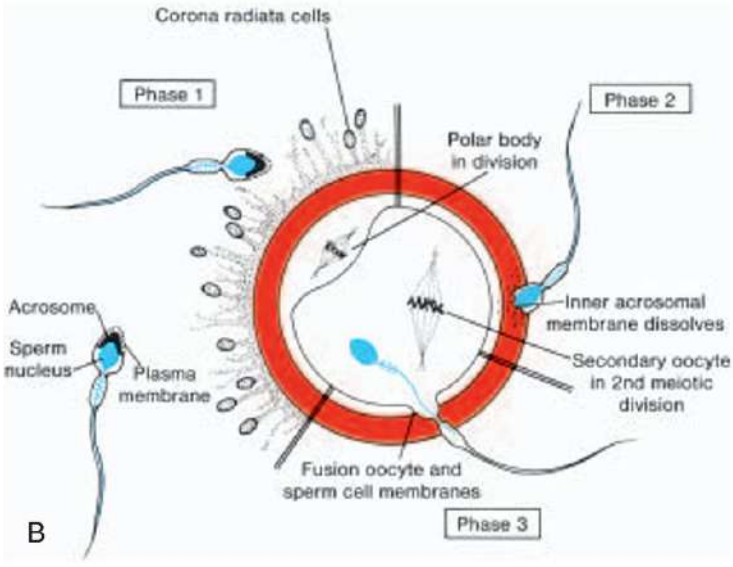


Figure 2: The three phases of oocyte penetration. Phase 1: Spermatozoa pass through the corona radiata barrier. Phase 2: One or more spermatozoa penetrate the zona pellucida. Phase 3: one spermatozoon penetrates the oocyte membrane while losing its own plasma membrane. Inset. Normal spermatocyte with acrosomal head cap. (Sadler, 2018).

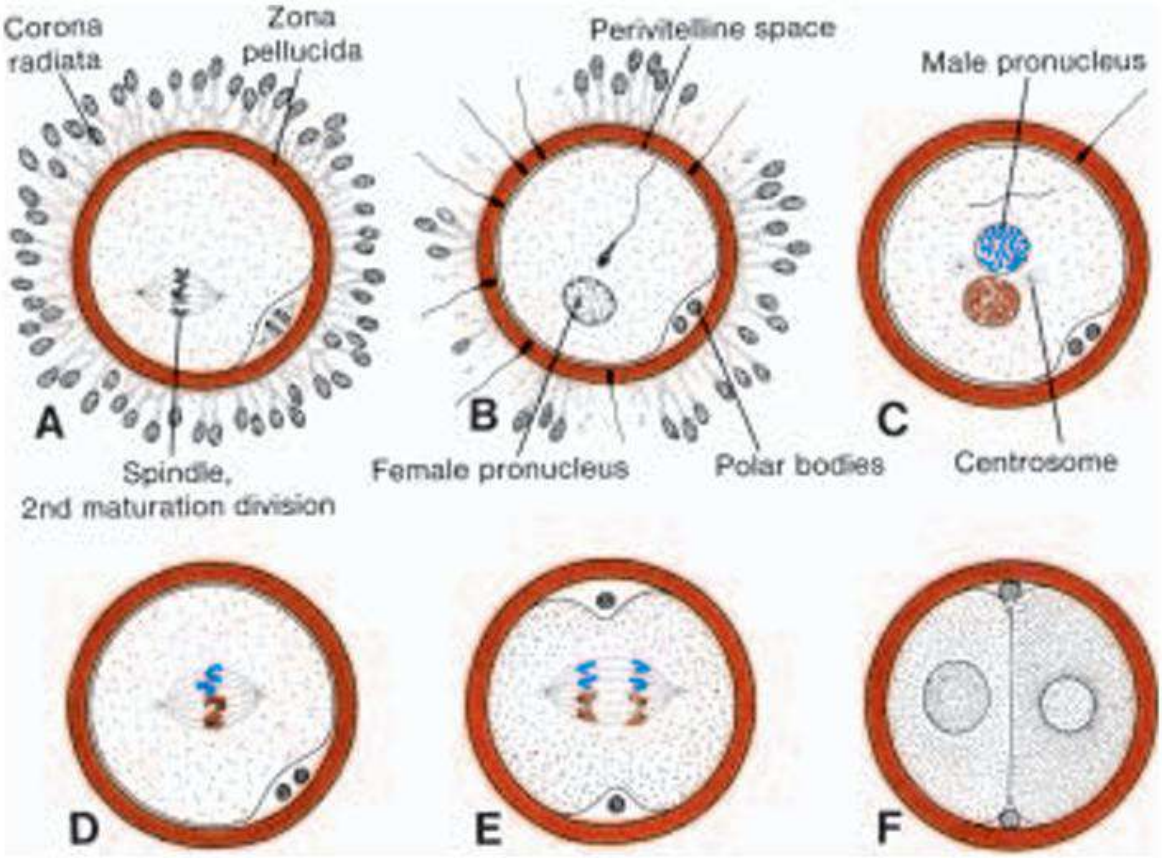


Figure 3: A. Oocyte immediately after ovulation, showing the spindle of second meiotic division. B. Spermatozoon penetrated the oocyte, which finished its second meiotic division. Chromosomes of the oocyte are arranged in a female pronucleus. Heads of several sperm are stuck in zona pellucida. C. Male and female pronuclei. D and E. Chromosomes arranged on the spindle, split longitudinally, and move to opposite poles. F. Two-cell stage. (Sadler, 2018).

Contraceptive Methods:

1. **Barrier techniques of contraception:**

- **Male condom:** Made of latex containing chemical spermicides, fits on the penis.
- **Female condom:** Made of polyurethane, lines the vagina.
- **Other barriers placed in vagina:** Diaphragm, cervical cap, and contraceptive sponge.

2. **Contraceptive pill:** Combination of oestrogen and progesterone analogue progestin, together inhibit ovulation but permit menstruation.
3. **Depo-Provera:** Progestin compound implanted subdermal or injected intramuscular to prevent ovulation for up to 5 years or 23 months, respectively.
4. **Male pill:** Synthetic androgen prevents both LH and FSH secretion and either stops sperm production (70–90% of men) or reduces it to a level of infertility.
5. **Intrauterine device (IUD):** Placed in the uterine cavity, preventing pregnancy by direct effects on sperm and oocytes or inhibition of preimplantation stages of development.

6. **Drug RU-486 (mifepristone):** Causes abortion if it is administered within 8 weeks of the previous menses, initiates menstruation, antiprogesterone agent.
7. **Tubal ligation:** Effective in female contraception.
8. **Vasectomy:** Effective in male contraception.

INFERTILITY:

- Problem for 15% to 30% of couples.

Causes of Infertility:

Male Infertility:

- Insufficient numbers of sperm and/or poor motility.
- The ejaculate has a volume of 3 to 4 ml, with about 100 million sperm per ml.
- Males with 20 million sperm per ml or 50 million sperm per total ejaculate are fertile.

Female Infertility:

1. Occluded oviducts (pelvic inflammatory disease).
2. Hostile cervical mucus.
3. Immunity to spermatozoa.
4. Absence of ovulation.

Treatment of Infertility:

1. In vitro fertilization (IVF) of human ova and embryo transfer:

- Follicle growth in the ovary is stimulated by administration of gonadotropins.
- Oocytes taken by laparoscopy from ovarian follicles just before ovulation when the oocyte is in the late stages of the first meiotic division.
- The egg and sperm are placed in a culture medium.
- Fertilized eggs monitered to the eight-cell stage and placed in uterus to develop to term.
- **Disadvantage of IVF:** Low success rate; 20% of fertilized ova develop to term.

2. Gamete intrafallopian transfer (GIFT):

- Oocytes and sperm into the ampulla of the fallopian tube, where fertilization takes place.

3. Zygote intrafallopian transfer (ZIFT): Fertilized oocytes are placed in ampullary region.

4. Intracytoplasmic sperm injection (ICSI):

- Oligozoospermia or even azoospermia), overcomes using a single sperm, obtained from any point in male reproductive tract.
- Injected into cytoplasm of the egg to cause fertilization.

References:

- 1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
- 2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
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Lecture 22:

CLEAVAGE AND BLASTOCYTE FORMATION

Objectives of the lecture:

By end of the lecture the student should be able to:

1. Define cleavage.
2. Describe the cleavage.
3. Describe the blastocyte formation.

Cleavage:

- Once zygote reached the two-cell stage, undergoes a series of mitotic divisions, increasing the numbers of cells.
- These cells, which become smaller with each cleavage division, blastomeres.
- Until the eight-cell stage, they form a loosely arranged clump.
- After the third cleavage, blastomeres maximize their contact with each other, forming a compact ball of cells held together by tight junctions.
- This process, compaction, segregates inner cells, which communicate by gap junctions, from outer cells.
- About 3 days after fertilization, cells of the compacted embryo divide again to form a 16-cell morula (mulberry).
- Inner cells of morula constitute inner cell mass, and surrounding cells compose outer cell mass.
- Inner cell mass gives rise to tissues of the embryo proper.
- Outer cell mass forms trophoblast, which contributes to the placenta.

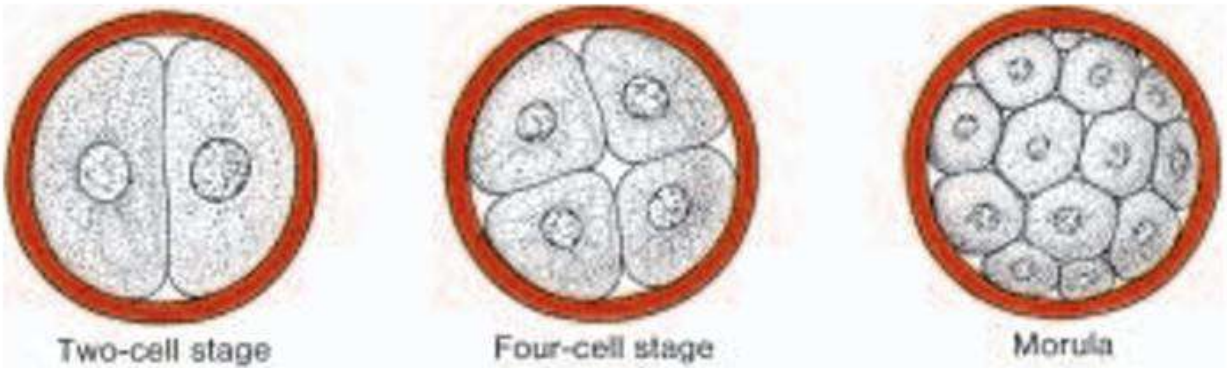


Figure 1: Development of zygote from two-cell stage to late morula stage. Two-cell stage is reached about 30 hours after fertilization; four-cell stage, at approximately 40 hours; 12- to 16-cell stage, at about 3 days; and late morula stage, at about 4 days. During this period, blastomeres surrounded by zona pellucida, disappears at end of the fourth day. (Sadler, 2018).

BLASTOCYST FORMATION

- About the time the morula enters the uterine cavity, fluid begins to penetrate through the zona pellucida into the intercellular spaces of the inner cell mass.
- Gradually intercellular spaces become confluent, and finally a single cavity, blastocele, forms.
- So, the embryo is a blastocyst.
- Cells of the inner cell mass; embryoblast, are at one pole, and those of outer cell mass, or trophoblast, flatten and form epithelial wall of the blastocyst.
- Zona pellucida has disappeared, allowing implantation to begin.
- In the human, trophoblastic cells over the embryoblast pole begin to penetrate between epithelial cells of the uterine mucosa about the sixth day.
- By end of the first week of development, human zygote passed through morula and blastocyst stages and has begun implantation in the uterine mucosa.

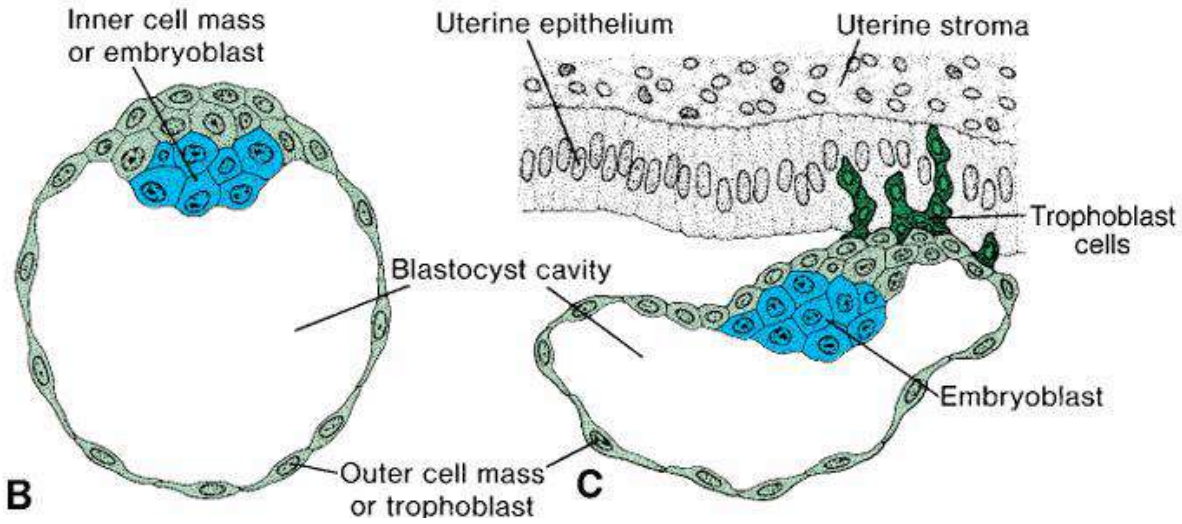


Figure 2: B. Human blastocyst recovered from the uterine cavity at approximately 4.5 days. Blue, inner cell mass or embryoblast; green, trophoblast. C. Blastocyst at the ninth day of development showing trophoblast cells at embryonic pole of the blastocyst penetrating uterine mucosa. (Sadler, 2018).

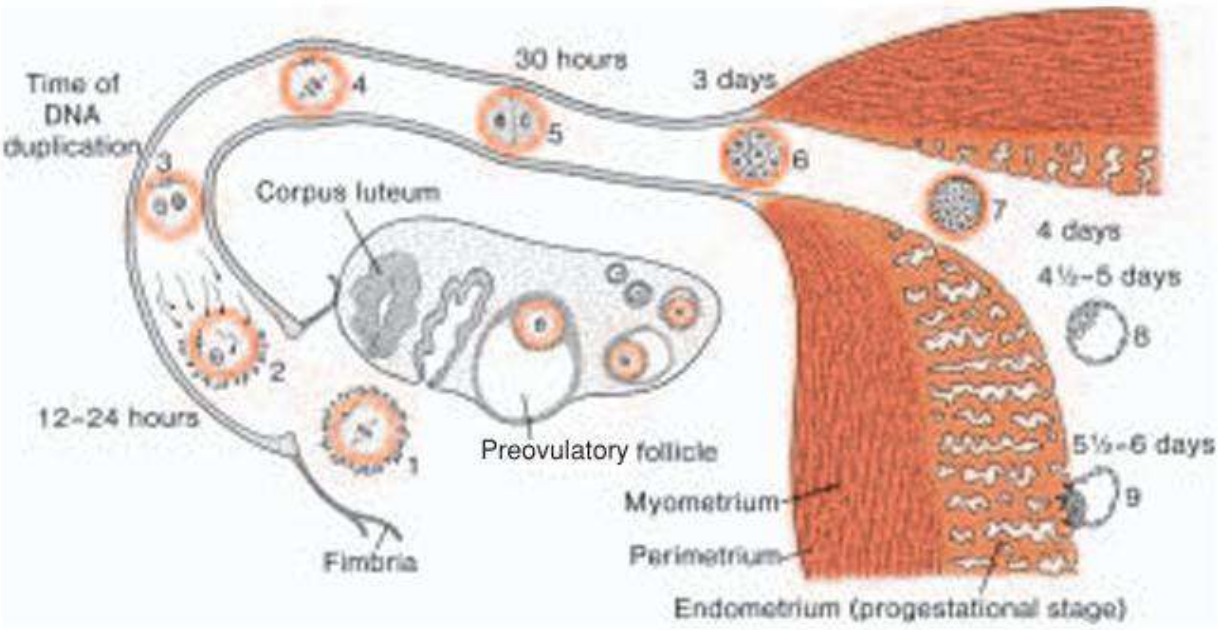


Figure 3: Events during the first week of human development. 1. Oocyte immediately after ovulation. 2. Fertilization 12 to 24 hours after ovulation. 3. Stage of the male and female pronuclei. 4. Spindle of the first mitotic division. 5. Two cell stage (30 hours of age). 6. Morula containing 12 to 16 blastomeres (3 days of age). 7. Advanced morula stage reaching the uterine lumen (4 days of age). 8, Early blastocyst stage (4.5 days of age). Zona pellucida has disappeared. 9. Early phase of implantation (blastocyst 6 days of age). The ovary shows stages of transformation between a primary follicle and a preovulatory follicle as well as a corpus luteum. The uterine endometrium is in progestational stage. (Sadler, 2018).

Abnormal Zygotes:

- Exact number of abnormal zygotes formed is unknown because they are usually lost within 2 to 3 weeks of fertilization, before the woman realizes she is pregnant.
- 50% of pregnancies end in spontaneous abortion and that half of these losses are result of chromosomal abnormalities.
- Single blastomeres from early-stage embryos can be removed and their DNA amplified for analysis.

References:

- 1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
- 2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
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Lecture 23:

UTERUS AT TIME OF IMPLANTATION

Objectives of the lecture:

By end of the lecture the student should be able to:

1. Define menstrual cycle.
2. Describe the uterine cycle.
3. Define implantation.
4. What is the normal site of implantation
5. Describe the implantation.
6. What are the sites of abnormal plantation?

Uterus at Time of Implantation:

Wall of the uterus consists of 3 layers:

1. **Endometrium:** Mucosa lining the inside wall.
2. **Myometrium:** Thick layer of smooth muscle.
3. **Perimetrium:** Peritoneal covering lining the outside wall.

From puberty (11–13 years) until menopause (45–50 years):

- Endometrium undergoes changes in a cycle of about 28 days under hormonal control by the ovary.

Menstrual cycle:

- Uterine endometrium passes through 3 stages:

1. **Follicular or proliferative phase.**
2. **Secretory or progestational phase.**
3. **Menstrual phase.**

1. **Proliferative phase:**

- Begins at the end of the menstrual phase.
- Under the influence of estrogen.
- Parallels growth of the ovarian follicles.

2. **Secretory phase:**

- Begins approximately 2 to 3 days after ovulation in response to progesterone produced by the corpus luteum.
- **If fertilization does not occur:** Shedding of the endometrium (compact and spongy layers) marks the beginning of the menstrual phase.

3. **Menstrual Phase:**

- If the oocyte is not fertilized, venules and sinusoidal spaces gradually become packed with blood cells, and an extensive diapedesis of blood into the tissue is seen.

- During the following 3 or 4 days, the compact and spongy layers are expelled from the uterus, and basal layer is the only part of endometrium that is retained.
- This layer, which is supplied by its own arteries, the basal arteries, functions as regenerative layer in rebuilding of glands and arteries in proliferative phase.
- Blood escapes from superficial arteries, and pieces of stroma and glands break away.

IMPLANTATION

- **At the time of implantation:** Mucosa of the uterus is in the secretory phase, during which time uterine glands and arteries become coiled and the tissue becomes succulent.

- As a result, 3 distinct layers can be recognized in the endometrium: a superficial compact layer, an intermediate spongy layer, and a thin basal layer.

Site of Implantation: Normally, human blastocyst implants in the endometrium along anterior or posterior wall of body of uterus, where it becomes embedded between openings of the glands.

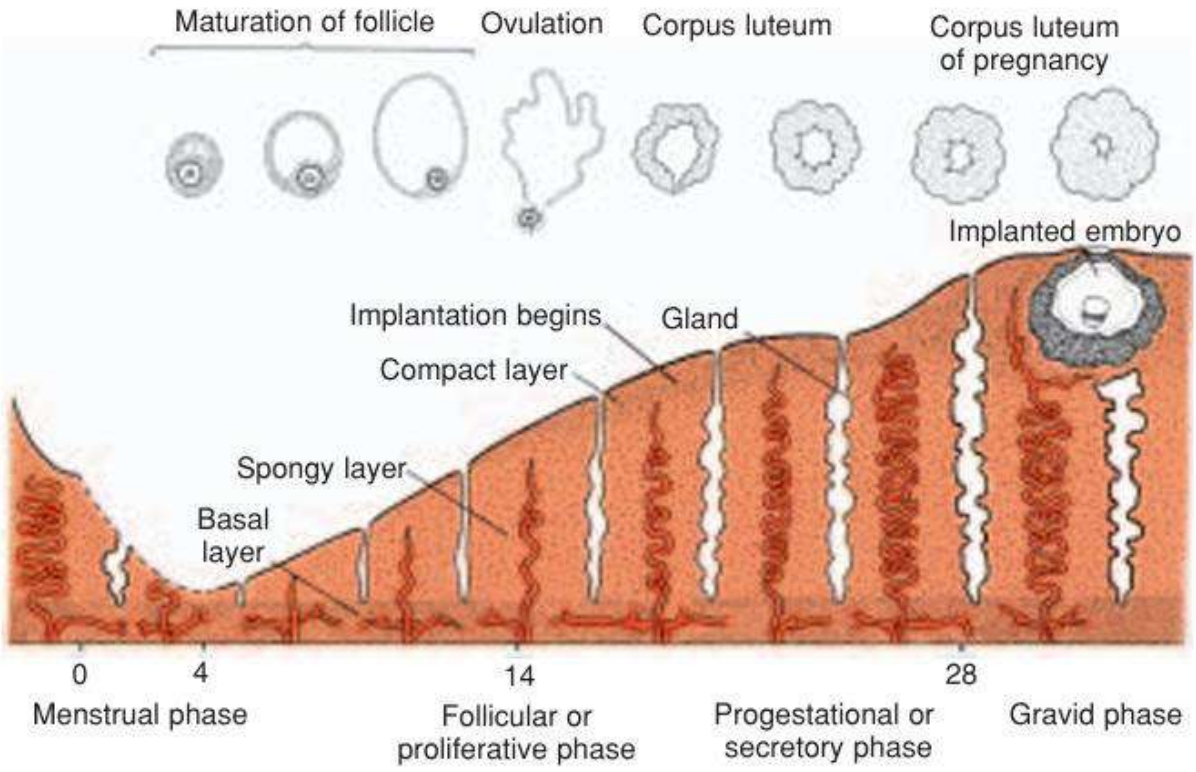


Figure 1: Changes in the uterine mucosa correlated with those in the ovary. Implantation of the blastocyst has caused development of a large corpus luteum of pregnancy. Secretory activity of the endometrium increases gradually because of large amounts of progesterone produced by the corpus luteum of pregnancy. (Sadler, 2018).

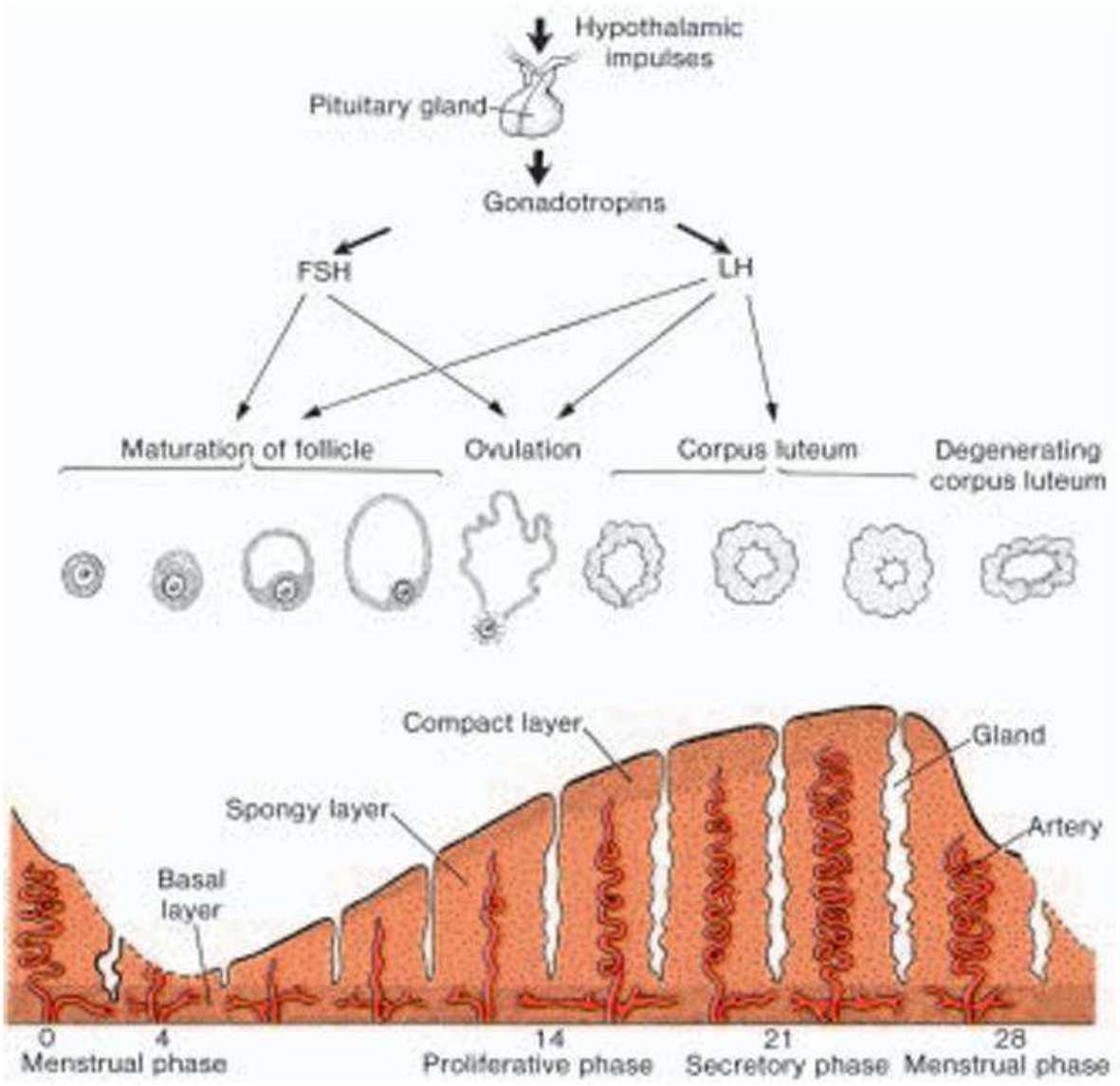


Figure 2: Changes in the uterine mucosa (endometrium) and corresponding changes in the ovary during a regular menstrual cycle without fertilization. (Sadler, 2018).

Abnormal implantation sites of the blastocyst:

1. Implantation in the abdominal cavity. The ovum most frequently implants in the rectouterine cavity (Douglas' pouch) but may implant at any place covered by peritoneum.
2. Implantation in the ampullary region of the tube.
3. Tubal implantation.
4. Interstitial implantation, that is, in the narrow portion of the uterine tube.
5. implantation in the region of the internal os, frequently resulting in placenta previa.
6. Ovarian implantation.

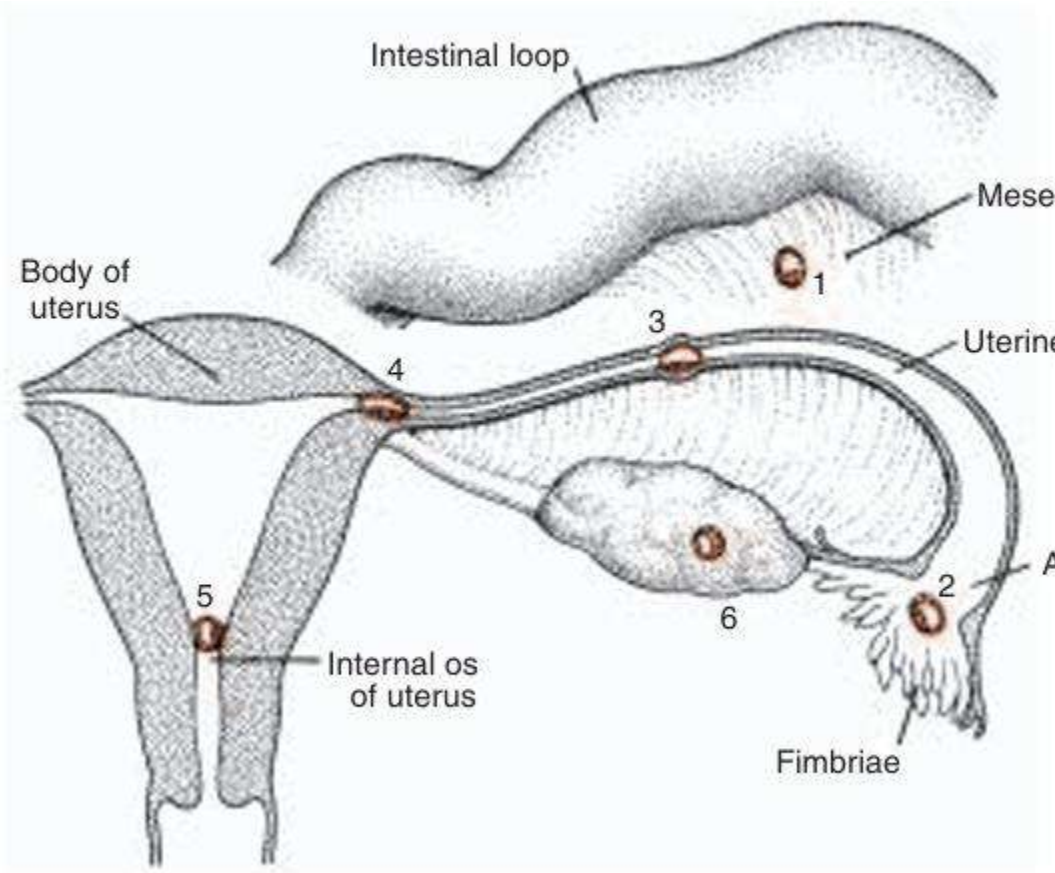


Figure 3: Abnormal implantation sites of the blastocyst. 1. Implantation in the abdominal cavity. The ovum most frequently implants in the rectouterine cavity (Douglas' pouch) but may implant at any place covered by peritoneum. 2. Implantation in the ampullary region of the tube. 3. Tubal implantation. 4. Interstitial implantation, that is, in the narrow portion of the uterine tube. 5. implantation in the region of the internal os, resulting in placenta previa. 6. Ovarian implantation. (Sadler, 2018).

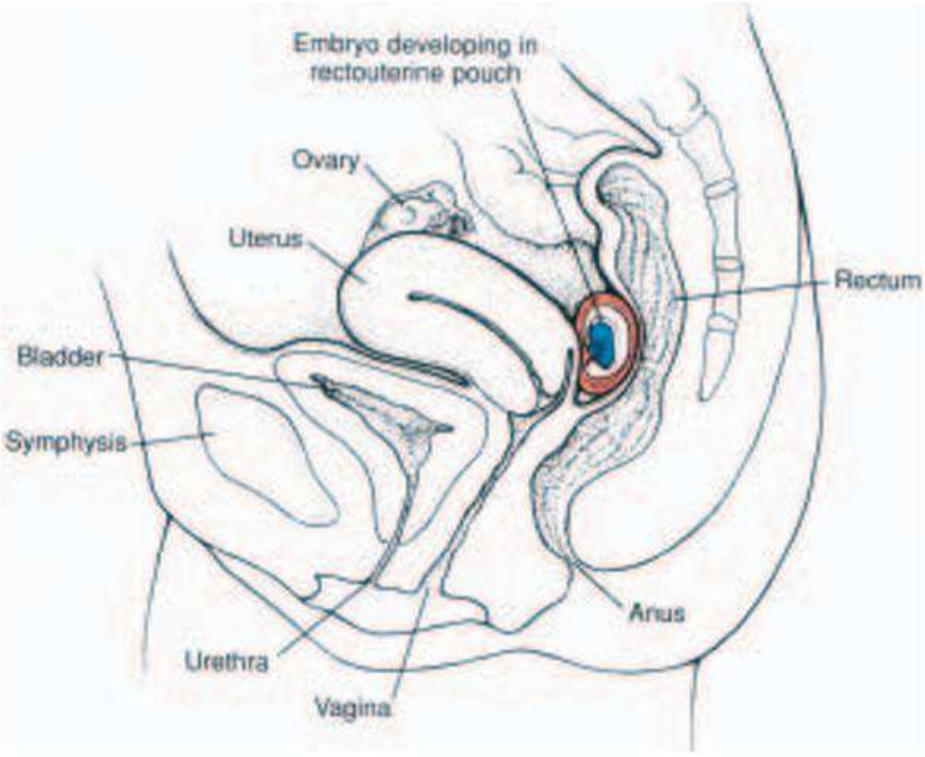


Figure 4: Midline section of bladder, uterus, and rectum to show an abdominal pregnancy in the rectouterine (Douglas') pouch. (Sadler, 2018).

References:

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Lecture 24:

2ND WEEK OF DEVELOPMENT

BILAMINAR GERM DISC

Day 8-9 and Day 11-13

Objectives of the lecture:

By end of the lecture the student should be able to:

1. Enumerate the changes occurring during the second week.
2. Describe the layers of trophoblast.
3. Describe the parts of the bilaminar germ disc
4. What are the sites of amniotic and yolk sac cavities?

Day 8 Human Blastocyst: (Figure 1):

- The blastocyst is partially embedded in the endometrial stroma.

In area over the embryoblast, trophoblast differentiated into 2 layers:

a. Cytotrophoblast: Inner layer of mononucleated cells.

b. Syncytiotrophoblast: Outer multinucleated zone without distinct cell boundaries.

- Mitotic figures are found in cytotrophoblast but not in syncytiotrophoblast.
- Cells in cytotrophoblast divide and migrate into syncytiotrophoblast, where they fuse and lose their cell membranes.

Cells of the embryoblast differentiate into 2 layers:

a. Hypoblast Layer: Small cuboidal cells adjacent to the blastocyst cavity.

b. Epiblast Layer: High columnar cells adjacent to the amniotic cavity.

- Together, the layers form a flat disc.

Amniotic cavity: Small cavity appears within the epiblast.

- This cavity enlarges to become lined by epiblast cells adjacent to cytotrophoblast; amnioblasts; together with rest of the epiblast.

- Endometrial stroma adjacent to the implantation site is oedematous and highly vascular. The large, tortuous glands secrete abundant glycogen and mucus.

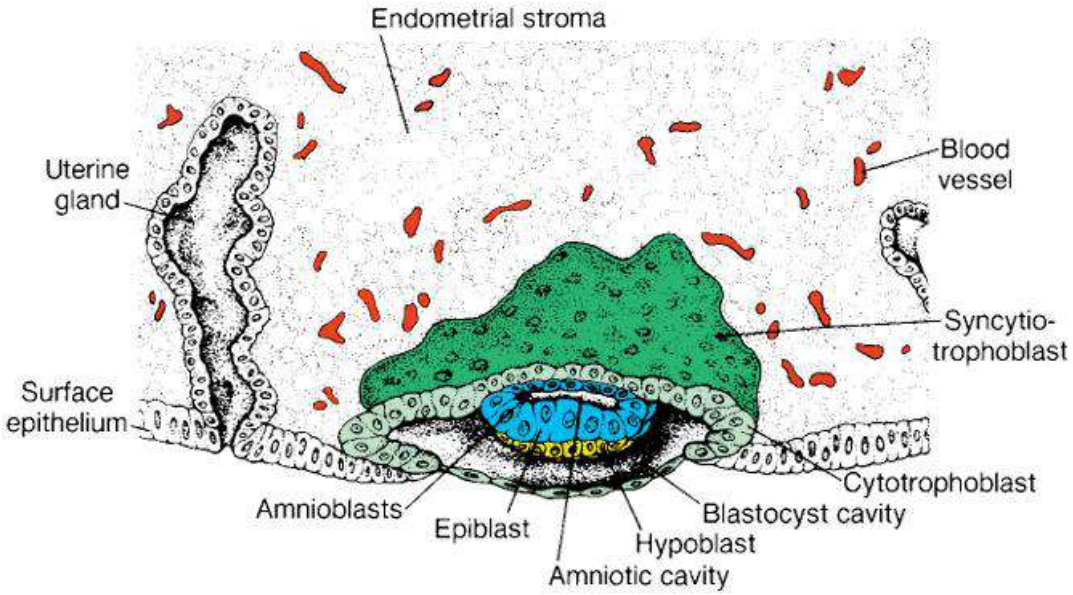


Figure 1: 8-day human blastocyst, partially embedded in the endometrial stroma. Trophoblast consists of inner layer; cytotrophoblast, and an outer layer; syncytiotrophoblast. Embryoblast formed by epiblast and hypoblast layers. Amniotic cavity appears as a small cleft. (Sadler, 2018).

Day 9 Human Blastocyst: (Figure 2):

- **Blastocyst:** More deeply embedded in the endometrium.
- **Penetration defect in surface epithelium:** Closed by a fibrin coagulum.
- The trophoblast shows considerable progress in development, particularly at the embryonic pole, where vacuoles appear in the syncytium.
- When these vacuoles fuse, they form large lacunae, and this phase of trophoblast development is thus known as the lacunar stage.
- At abembryonic pole, meanwhile, flattened cells probably originating from the hypoblast form a thin membrane, exocoelomic (Heuser's) membrane, that lines the inner surface of cytotrophoblast.
- Heuser's membrane, together with hypoblast, forms lining of exocoelomic cavity, or primitive yolk sac.

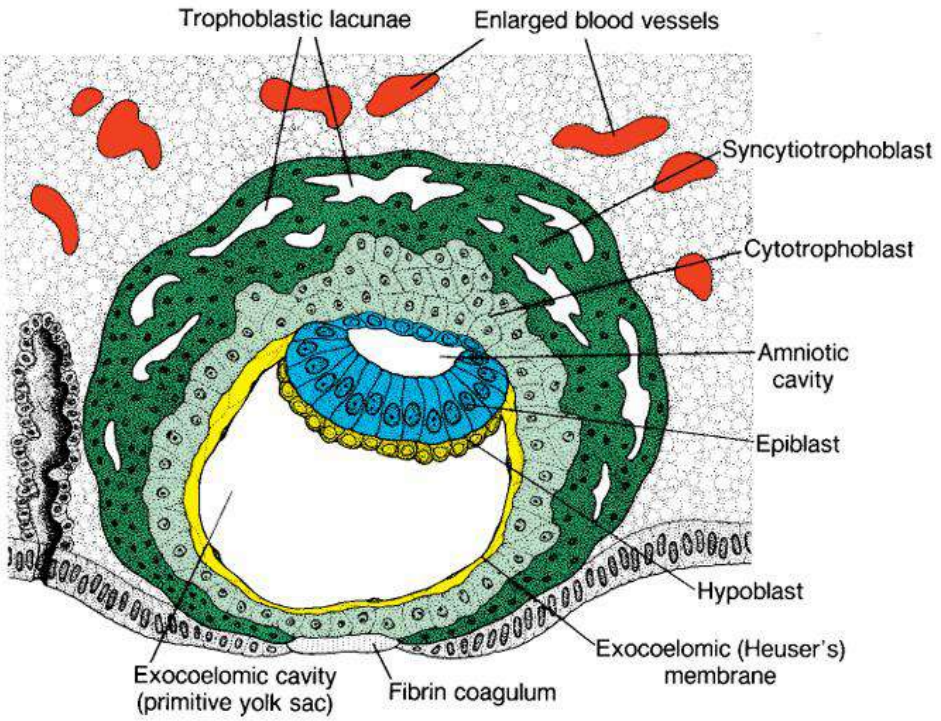


Figure 2: 9-day human blastocyst. Syncytiotrophoblast shows lacunae. Flat cells form exocoelomic membrane. Bilaminar disc consists of a layer of epiblast cells and a layer of hypoblast cells. Original surface defect closed by a fibrin coagulum. (Sadler, 2018).

Days 11 and 12 Human Blastocyst: (Figure 3):

- The blastocyst is completely embedded in the endometrial stroma.
- Surface epithelium covers the original defect in the uterine wall.
- Trophoblast is characterized by lacunar spaces in syncytium that form an intercommunicating network, evident at embryonic pole.
- Cells of syncytiotrophoblast penetrate deeper into the stroma and erode the endothelial lining of maternal capillaries.
- Maternal capillaries are congested and dilated, called sinusoids.
- Syncytial lacunae become continuous with the sinusoids and the maternal blood enters lacunar system.
- As the trophoblast continues to erode more sinusoids, maternal blood begins to flow through the trophoblastic system, establishing uteroplacental circulation.
- New population of cells appears between inner surface of the cytotrophoblast and outer surface of exocoelomic cavity.
- These cells, derived from yolk sac cells, form a fine, loose connective tissue, extraembryonic mesoderm, which eventually fills all the space between trophoblast externally and amnion and exocoelomic membrane internally.

- Large cavities develop in extraembryonic mesoderm, form, extraembryonic coelom, or chorionic cavity.
- This space surrounds the primitive yolk sac and amniotic cavity except where the germ disc is connected to the trophoblast by connecting stalk.
- Extraembryonic mesoderm lining the cytotrophoblast and amnion; extraembryonic somatopleuric mesoderm; lining covering yolk sac; extraembryonic splanchnopleuric mesoderm.

Decidua reaction:

- Cells of endometrium, become polyhedral and loaded with glycogen and lipids.
- Intercellular spaces are filled with extravasate.
- The tissue is oedematous area surrounding the implantation site and throughout the endometrium.

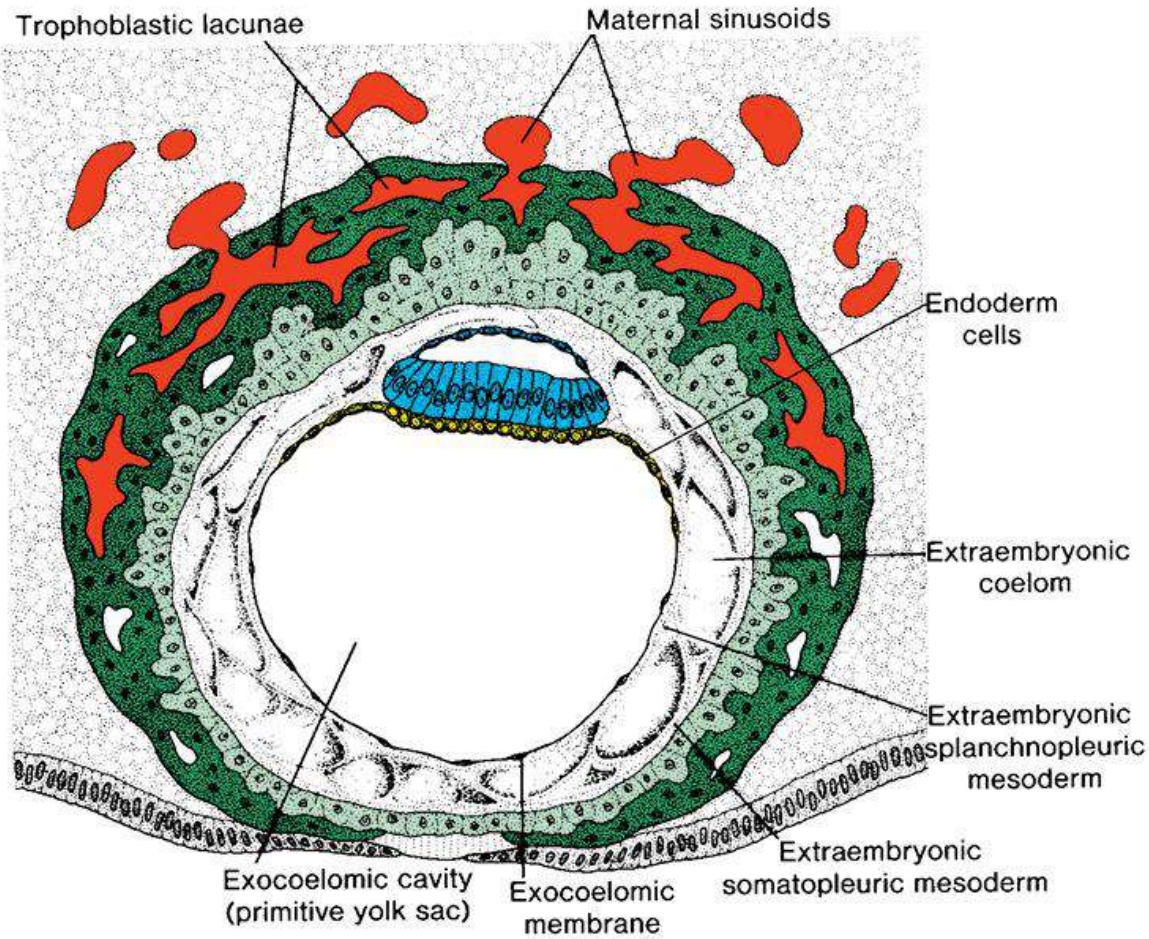


Figure 3: Human blastocyst of 12 days. Trophoblastic lacunae at embryonic pole are in open connection with maternal sinusoids in endometrial stroma. Extraembryonic mesoderm proliferates and fills the space between exocoelomic membrane and inner aspect of trophoblast. (Sadler, 2018).

Day 13 Human blastocyst: (Figure 4):

- Surface defect in the endometrium has usually healed.
- Bleeding occurs at implantation site because of increased blood flow into lacunar spaces.
- As the bleeding occurs at the 28th day of the menstrual cycle, it may be confused with normal menstrual bleeding and, cause inaccuracy in determining expected delivery date.
- Trophoblast is characterized by villous structures.
- Cells of cytotrophoblast proliferate locally and penetrate syncytiotrophoblast, forming cellular columns surrounded by syncytium.
- Cellular columns with the syncytial covering are known as primary villi.
- Hypoblast produces additional cells that migrate inside of exocoelomic membrane.
- These cells proliferate and form a new cavity within exocoelomic cavity, secondary yolk sac or definitive yolk sac.
- Yolk sac is much smaller than original exocoelomic cavity, or primitive yolk sac.
- During its formation, portions of exocoelomic cavity are pinched off; exocoelomic cysts, found in extraembryonic coelom or chorionic cavity.
- Extraembryonic coelom expands and forms the chorionic cavity.
- Extraembryonic mesoderm lining inside of cytotrophoblast; called chorionic plate.
- The only place where extraembryonic mesoderm traverses chorionic cavity is in connecting stalk.
- With development of blood vessels, the stalk becomes umbilical cord.

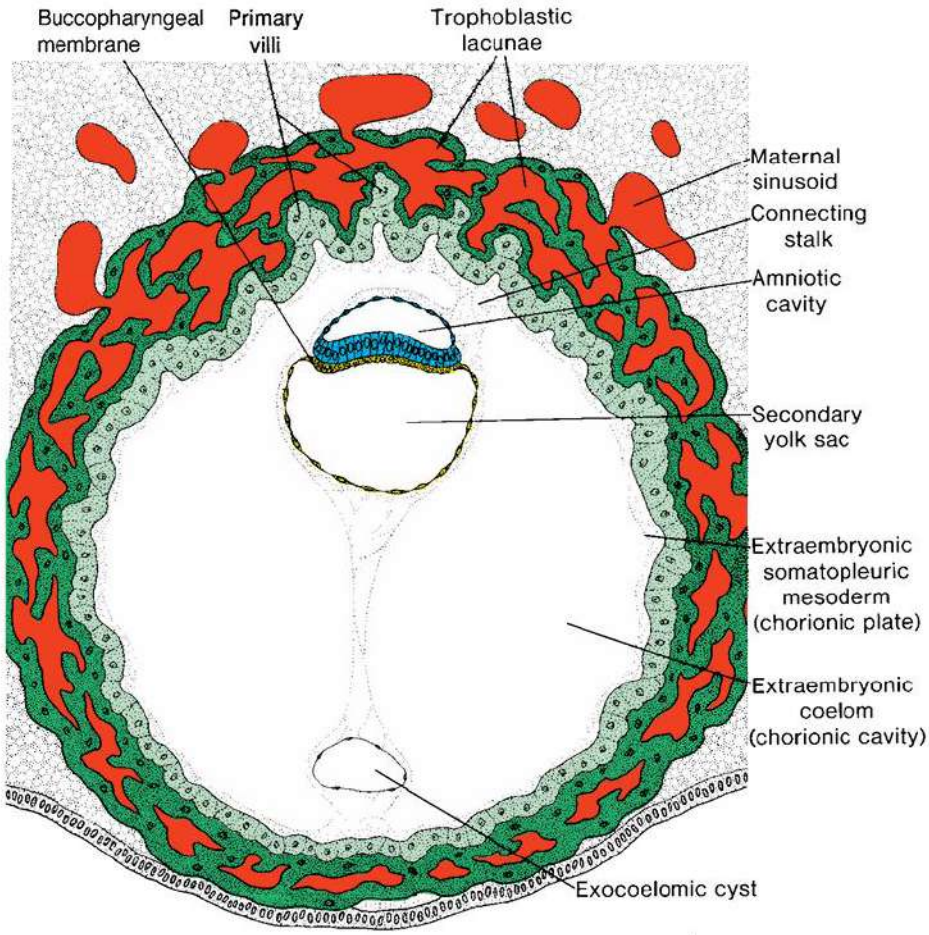


Figure 4: 13-day human blastocyst. Trophoblastic lacunae are present at embryonic and abembryonic pole, and uteroplacental circulation has begun. Primary villi and extraembryonic coelom or chorionic cavity seen. Secondary yolk sac lined with endoderm. (Sadler, 2018).

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
3. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.

Lecture 25:

THIRD WEEK OF DEVELOPMENT: TRILAMINAR GERM DISC

Objectives of the lecture:

By end of the lecture the student should be able to:

1. Enumerate the changes occurring during the third week.
2. Define gastrulation.
3. Describe the parts of the trilaminar germ disc
4. Describe formation of the notochord?
5. What is the fate of notochord?

GASTRULATION:

Definition: Formation of Embryonic Mesoderm and Endoderm.

Beginning of Gastrulation: With formation of the primitive streak on the surface of epiblast.

- The streak in a 15- to 16-day embryo, is visible as a narrow groove with bulging regions on either side.
- Cephalic end of the streak, the primitive node, slightly elevated area surrounding the small primitive pit.
- Cells of the epiblast migrate toward the primitive streak.
- Upon arrival in the region of the streak, they become flask-shaped, detach from the epiblast, and slip beneath it.
- This inward movement is known as invagination.
- Once the cells have invaginated, some displace the hypoblast, creating the embryonic endoderm, and others come to lie between the epiblast and newly created endoderm to form mesoderm.
- Cells remaining in the epiblast then form ectoderm.
- Thus, the epiblast, through process of gastrulation, is the source of all the germ layers, and cells in these layers will give rise to all the tissues and organs in the embryo.
- As more cells move between the epiblast and hypoblast layers, they begin to spread laterally and cephalad.
- They migrate beyond margin of the disc and establish contact with extraembryonic mesoderm covering the yolk sac and amnion.
- In the cephalic direction, they pass on each side of the prechordal plate.
- Prechordal plate forms between tip of the notochord and buccopharyngeal membrane and derived from some of the first cells that migrate through the node in a cephalic direction.
- Prechordal plate will be important for induction of the forebrain.
- Buccopharyngeal membrane at cranial end of the disc consists of a region of adherent ectoderm and endoderm cells that represents future opening of the oral cavity.

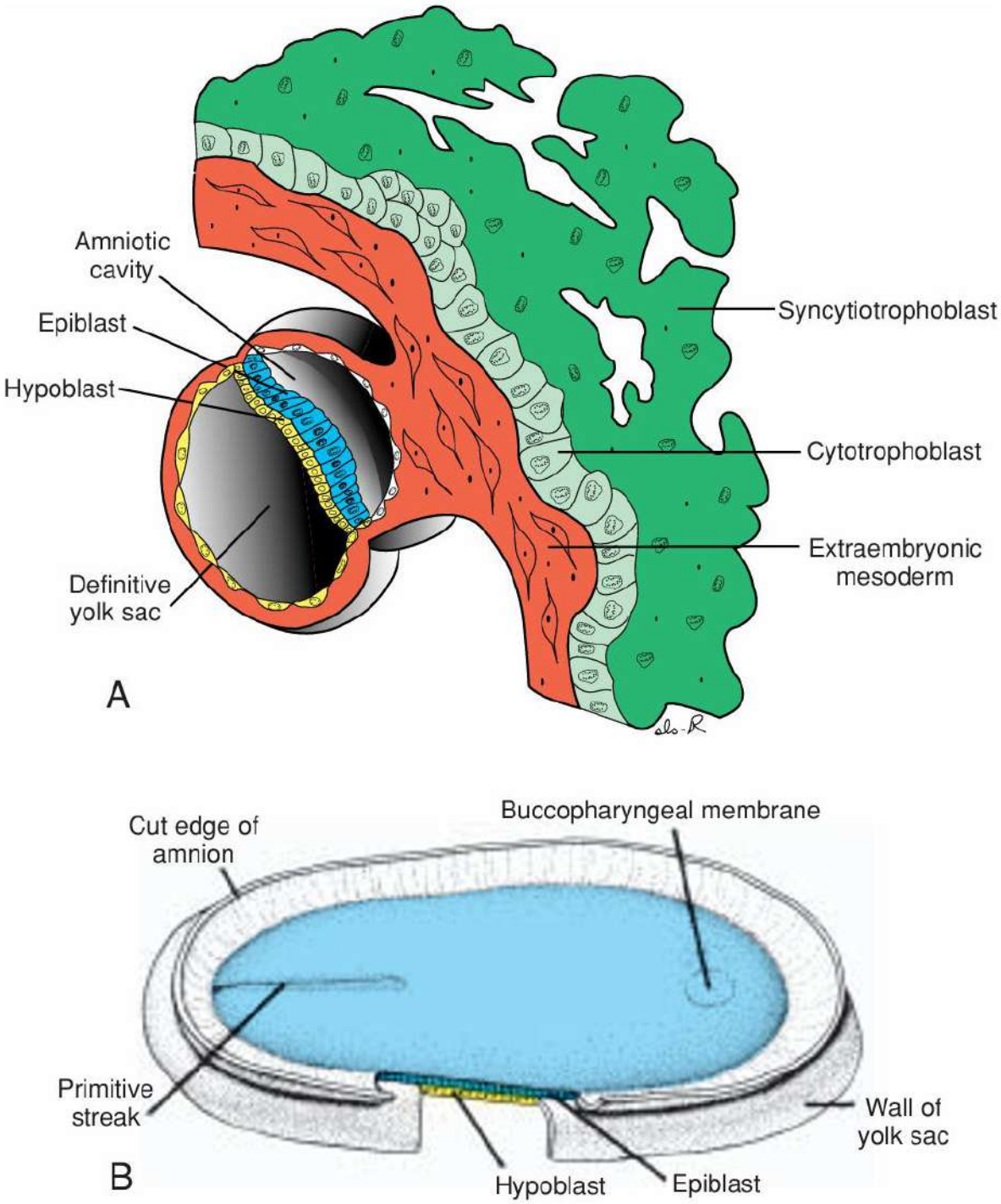


Figure 1: A. Implantation site at end of the second week. B. View of the germ disc at end of the second week of development. Amniotic cavity opened to permit a view on the dorsal side of the epiblast. Hypoblast and epiblast are in contact with each other and the primitive streak forms a groove in caudal region of the embryo. (Sadler, 2018).

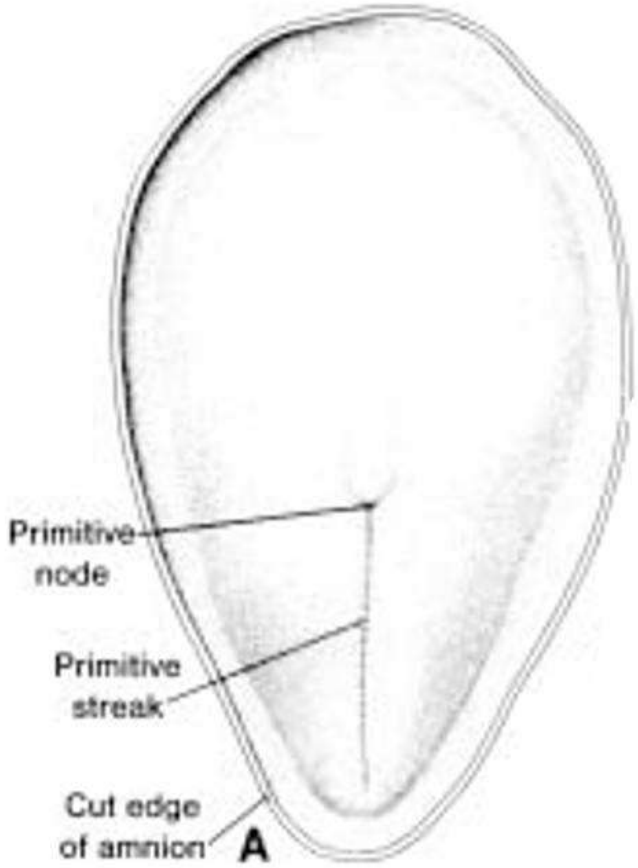


Figure 2: A. Dorsal aspect of an 18-day embryo. The embryo has a pear-shaped appearance and shows primitive streak and node at its caudal end. Primitive node and the notochord extending cranially. (Sadler, 2018).

FORMATION OF THE NOTOCHORD

- Prenotochordal cells invaginating in the primitive pit move forward cephalad until they reach prechordal plate.
- Prenotochordal cells become intercalated in hypoblast so that, the midline of embryo consists of two cell layers that form notochordal plate.
- The hypoblast is replaced by endoderm cells moving in at the streak, cells of notochordal plate proliferate and detach from endoderm.
- They form a solid cord of cells, definitive notochord, underlies the neural tube and serves as basis for the axial skeleton.
- Because elongation of the notochord is a dynamic process, the cranial end forms first, and caudal regions are added as primitive streak assumes a more caudal position.
- Notochord and prenotochordal cells extend cranially to prechordal plate (an area caudal to buccopharyngeal membrane) and caudally to primitive pit.
- At the point where the pit forms an indentation in the epiblast, neurenteric canal temporarily connects amniotic and yolk sac cavities.

- Cloacal membrane formed at caudal end of embryonic disc, similar to buccopharyngeal membrane, consists of ectoderm and endoderm cells with no intervening mesoderm.
- When cloacal membrane appears, posterior wall of the yolk sac forms a diverticulum that extends into connecting stalk.
- Allantois remains rudimentary and involved in abnormalities of bladder development.

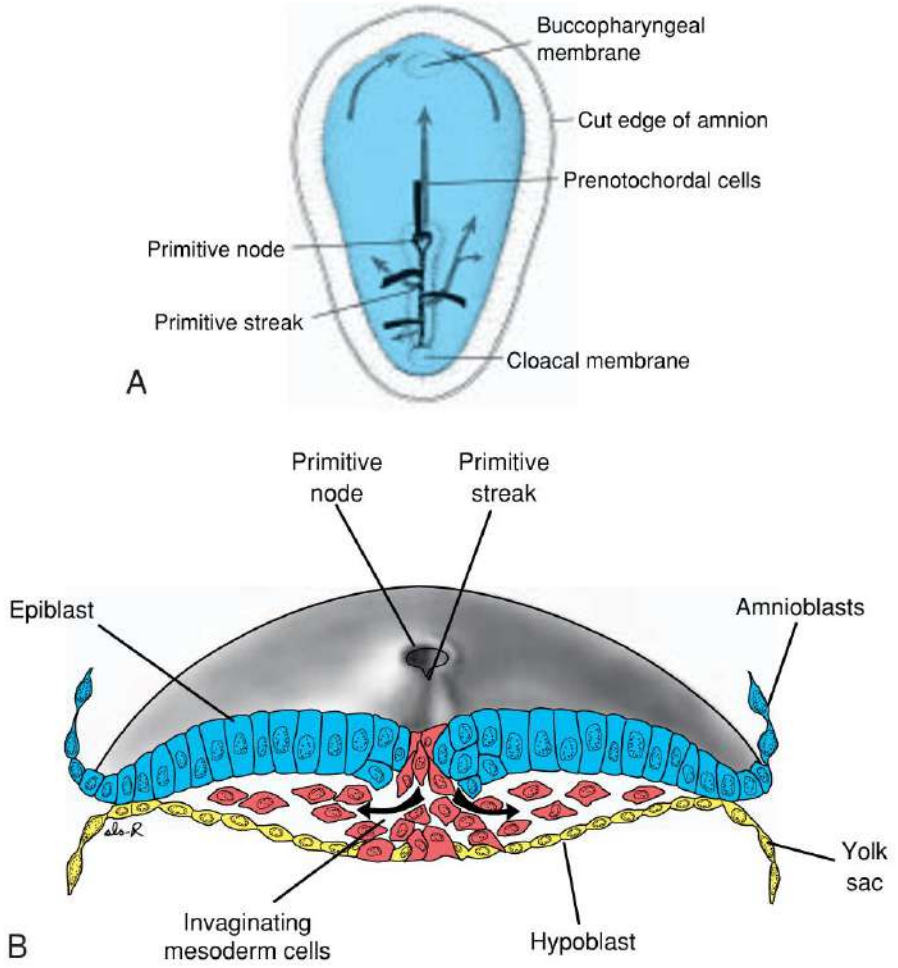


Figure 3: A. Dorsal side of germ disc from a 16-day embryo indicating movement of surface epiblast cells (solid black lines) through primitive streak and node and migration of cells between hypoblast and epiblast (broken lines). B. Cross section through cranial region of streak at 15 days showing invagination of epiblast cells. The first cells to move inward displace hypoblast to create definitive endoderm. Once definitive endoderm is established, inwardly moving epiblast forms mesoderm. (Sadler, 2018).

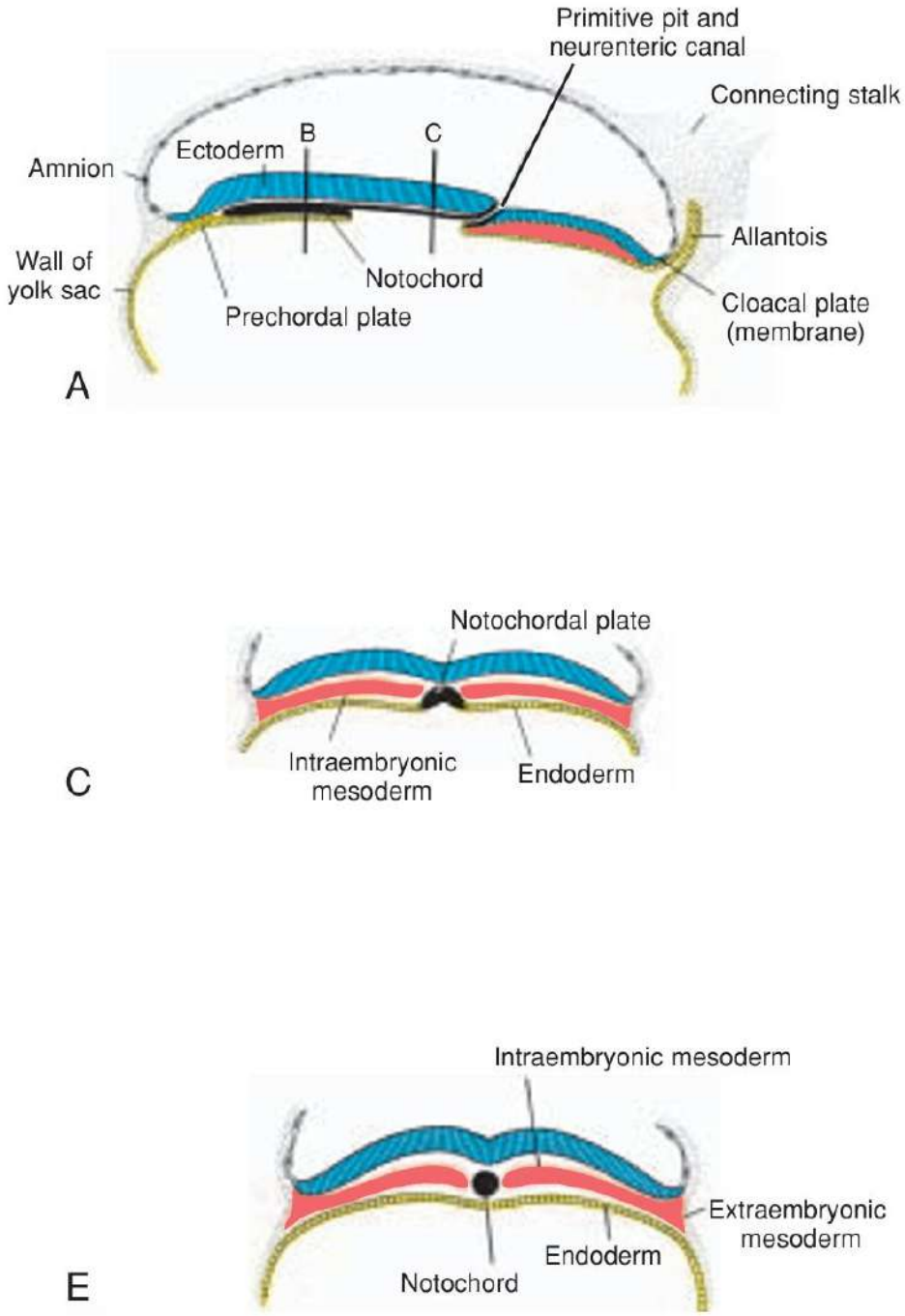


Figure 4: Formation of notochord. Prenotochordal cells migrate through primitive streak, intercalated in endoderm to form notochordal plate, and finally detach from endoderm to form definitive notochord. Because these events occur in a cranial-to-caudal sequence, portions of definitive notochord are established in head region first. A. Sagittal section through a 17-day embryo. The most cranial portion of definitive notochord formed, while prenotochordal cells caudal to this region are intercalated into endoderm as notochordal plate. C. Cross section through region of notochordal plate. Notochordal plate detach from endoderm to form definitive notochord. E. Definitive notochord. (Sadler, 2018).

Fate of the Notochord:

1. Formation of nucleus pulposus of the intervertebral disc.
2. Apical ligament.
3. Degeneration inside the bodies of the vertebrae

Fate Map Established During Gastrulation:

- Regions of the epiblast that migrate and ingress through the primitive streak have been mapped and their ultimate fates determined.
- Cells that ingress through cranial region of the node become notochord.
- Cells migrating at lateral edges of the node and from cranial end of the streak become paraxial mesoderm.
- Cells migrating through mid-streak region become intermediate mesoderm.
- Cells migrating through the more caudal part of streak form lateral plate mesoderm
- Cells migrating through the caudal-most part of streak contribute to extraembryonic mesoderm (other source of this tissue is primitive yolk sac, hypoblast).

Growth of the Embryonic Disc:

- The embryonic disc, initially flat and almost round, gradually becomes elongated, with a broad cephalic and a narrow caudal end.
- Expansion of the embryonic disc occurs mainly in cephalic region.
- Region of the primitive streak remains the same size.
- Growth and elongation of cephalic part of the disc are caused by a continuous migration of cells from the primitive streak region in a cephalic direction.
- Invagination of surface cells in primitive streak and their migration forward and laterally continues until end of the fourth week.
- Primitive streak shows regressive changes, rapidly shrinks, and disappears.
- Primitive streak at caudal end of the disc continues to supply new cells until end of the fourth week is important in development of the embryo.
- In cephalic part, germ layers begin their specific differentiation by middle of the third week, whereas in the caudal part, differentiation begins by the end of the fourth week.
- Gastrulation, (formation of germ layers) continues in caudal segments while cranial structures are differentiating, causing embryo to develop cephalocaudally.

TERATOGENESIS ASSOCIATED WITH GASTRULATION

Beginning of the third week of development:

- When gastrulation is initiated, is a highly sensitive stage for teratogenic insult.
- Fate maps can be made for various organ systems, eyes and brain, and these cell populations may be damaged by teratogens.

- High doses of alcohol at this stage kill cells in the anterior midline of the germ disc, producing a deficiency of the midline in craniofacial structures and resulting in holoprosencephaly.
- Gastrulation may be disrupted by genetic abnormalities and toxic insults.
- In caudal dysgenesis (sirenomelia), insufficient mesoderm is formed in caudal-most region of the embryo.
- Affected individuals exhibit hypoplasia and fusion of the lower limbs, vertebral abnormalities, renal agenesis, imperforate anus, and anomalies of the genital organs.



Figure 5: Sirenomelia (caudal dysgenesis). Loss of mesoderm in the lumbosacral region has resulted in fusion of the limb buds and other defects. (Sadler, 2018).

Tumors Associated with Gastrulation

- Sometimes, remnants of primitive streak persist in the sacrococcygeal region.
- These clusters of pluripotent cells proliferate and form tumours, sacrococcygeal teratomas, that commonly contain tissues derived from all three germ layers.
- The most common tumour in newborns, occurring with a frequency of one in 37,000.
- These tumours may also arise from primordial germ cells (PGCs) that fail to migrate to the gonadal ridge.

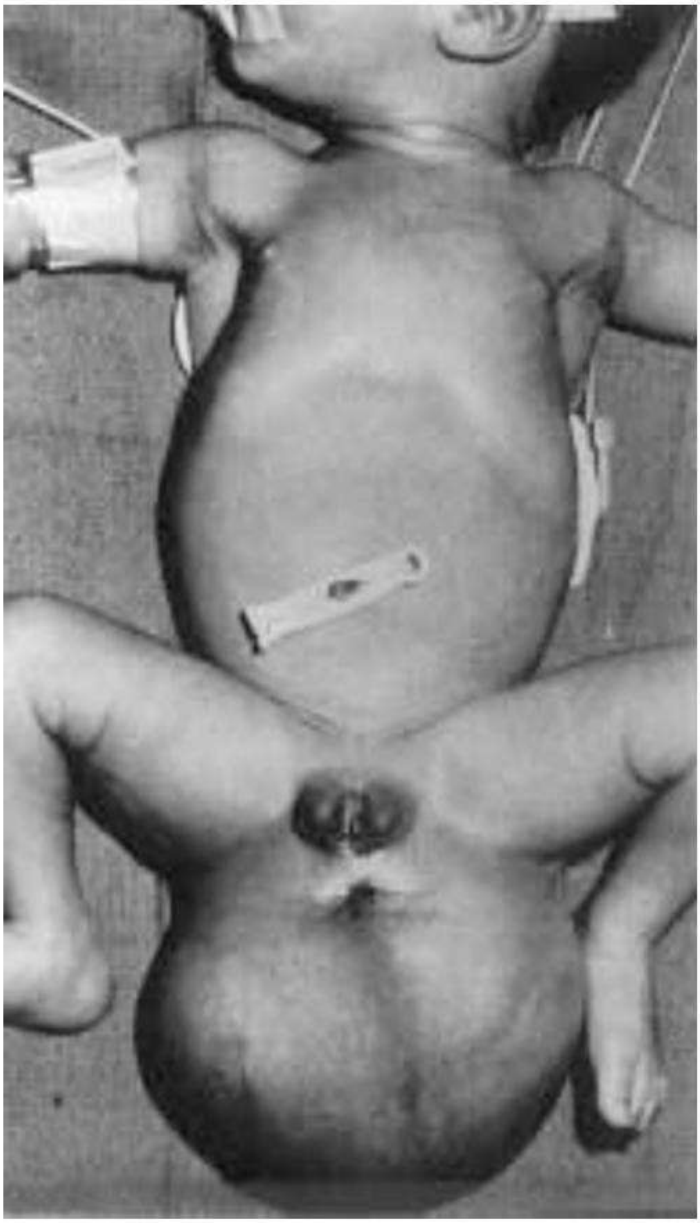


Figure 6: Sacrococcygeal teratoma resulting from remnants of the primitive streak. The teratoma may become malignant and are most common in females. (Sadler, 2018).

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
3. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.

Lecture 26:

THIRD WEEK OF DEVELOPMENT
FURTHER DEVELOPMENT OF THE TROPHOBLAST

Objectives of the lecture:

By end of the lecture the student should be able to:

1. What are the changes occurring in the trophoblast?
2. Describe structure of the primary chorionic villi.
3. Describe structure of the secondary chorionic villi.
4. Describe structure of the tertiary chorionic villi.

Further Development of the Trophoblast:

By the beginning of the third week:

- Trophoblast is characterized by primary villi, consist of a cytotrophoblastic core covered by a syncytial layer.
- Mesodermal cells penetrate the core of primary villi and grow toward the decidua, secondary villus.

By end of the third week:

- Mesodermal cells in the core of villus differentiate into blood cells and blood vessels, forming villous capillary system, tertiary villus or definitive placental villus.
- Capillaries in tertiary villi contact with capillaries developing in mesoderm of chorionic plate and in connecting stalk.
- The vessels contact with intraembryonic circulatory system, connecting placenta and embryo.

When the heart begins to beat in the fourth week of development:

Villous System: Supply the embryo proper with essential nutrients and oxygen.

- Cytotrophoblast cells in the villi penetrate syncytium to reach maternal endometrium, contact with extensions of neighbouring villous stems, forming cytotrophoblast shell.
- This shell gradually surrounds trophoblast and attaches chorionic sac to maternal endometrial tissue.
- **Stem or Anchoring Villi:** Villi that extend from chorionic plate to decidua basalis (decidual plate: part of endometrium where placenta will form).
- Branches from the sides of stem villi are free (terminal) villi, through which exchange of nutrients and other factors will occur.

Chorionic cavity:

- Becomes larger, embryo attached to its trophoblastic shell by connecting stalk.

Connecting stalk: Develops into umbilical cord, forms connection between placenta and embryo.

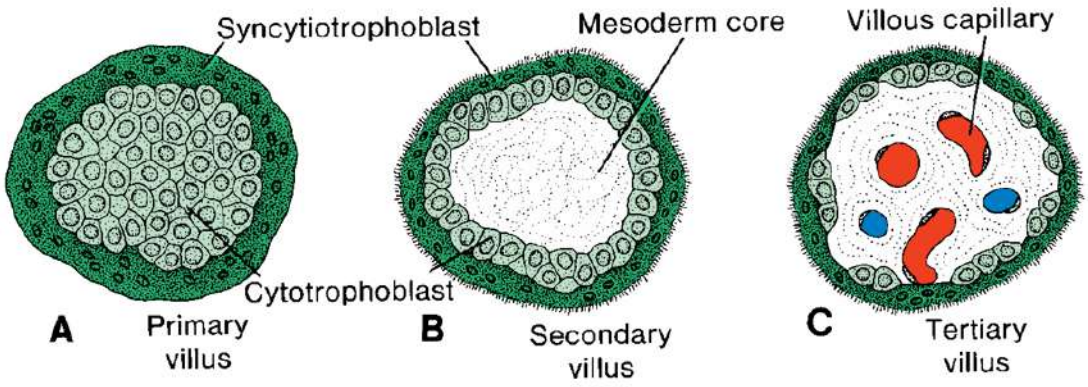


Figure 1: Development of a villus. A. Primary villus with a core of cytotrophoblastic cells covered by syncytium. B. Secondary villus with a core of mesoderm covered by cytotrophoblastic cells and syncytium. C. Mesoderm of the villus showing capillaries and venules. (sadler, 2018).

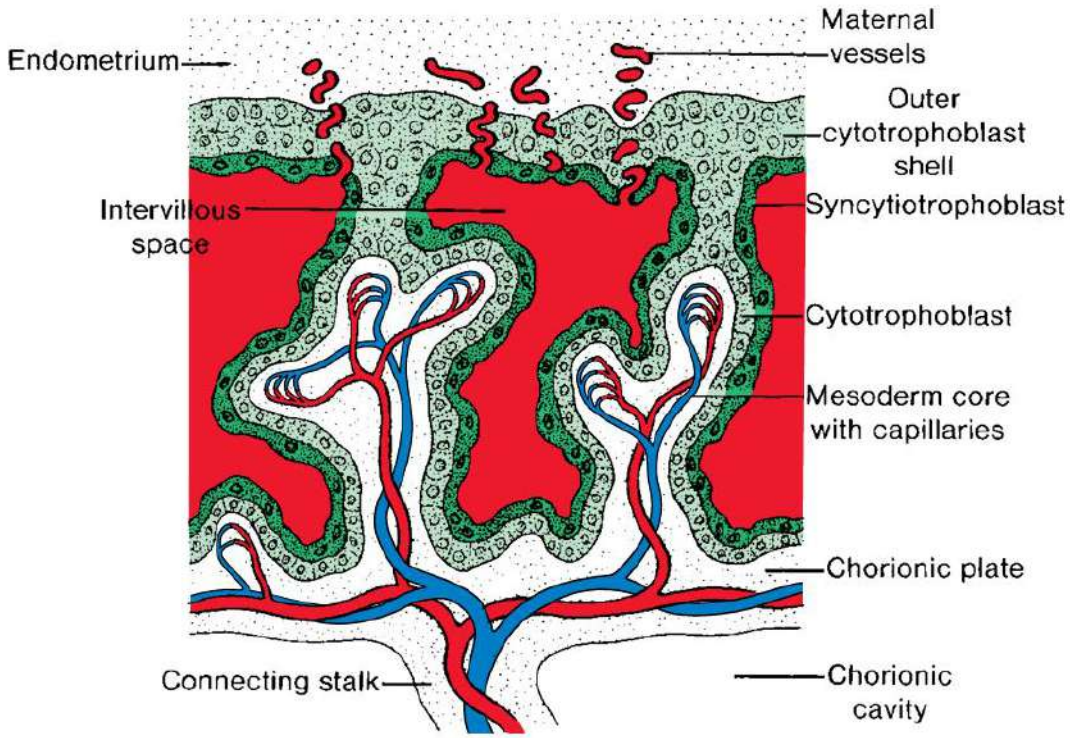


Figure 2: Villus at end of the third week of development. Maternal vessels penetrate cytotrophoblastic shell to enter intervillous spaces, which surround the villi. Capillaries in the villi are in contact with vessels in chorionic plate and in connecting stalk and connected to intraembryonic vessels. (Sadler, 2018).

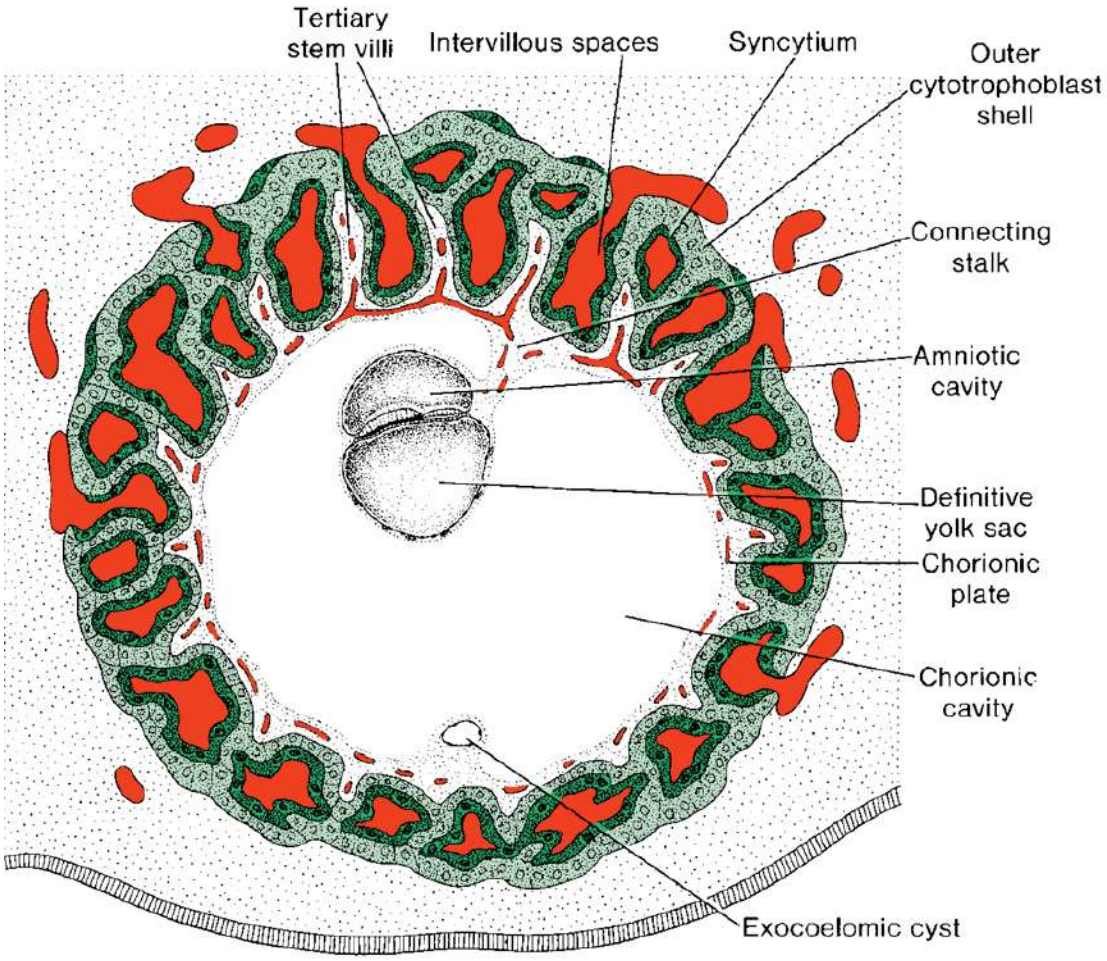


Figure 3: Presomite embryo and the trophoblast at the end of the third week. Tertiary and secondary stem villi give trophoblast a characteristic radial appearance. Inter villous spaces, lined with syncytium. Cytotrophoblastic cells surround trophoblast and contact endometrium. Embryo suspended in chorionic cavity by connecting stalk. (Sadler, 2018).

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
3. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.

Lecture 27:

THIRD TO EIGHTH WEEK: THE EMBRYONIC PERIOD

Objectives of the lecture:

By end of the lecture the student should be able to:

1. Describe the neurulation.
2. Enumerate the derivatives of the ectoderm germ layer.
3. Enumerate the structures derived from the ectoderm germ layer.

Embryonic Period or Period of Organogenesis:

- Occurs from the third to the eighth weeks of development.
- The time when each of the three germ layers, ectoderm, mesoderm, and endoderm, gives rise to a number of specific tissues and organs.

By end of the embryonic period:

- Main organ systems established.
- major features of the external body form recognizable by end of the second month.

DERIVATIVES OF THE ECTODERMAL GERM LAYER

At Beginning of the Third Week of Development:

- Ectodermal germ layer has shape of a disc that is broader in the cephalic than caudal region.
- Appearance of the notochord and prechordal mesoderm induces overlying ectoderm to thicken and form neural plate.
- Cells of the plate make up neuroectoderm and their induction represents initial event in the process of neurulation.

NEURULATION

- Elongated, slipper-shaped neural plate gradually expands toward primitive streak.

By end of the third week:

- Lateral edges of the neural plate become more elevated to form neural folds.
- Depressed mid-region forms the neural groove.
- Neural folds approach each other in the midline, where they fuse.
- Fusion begins in cervical region (fifth somite) and proceeds cranially and caudally.
- Neural tube is formed.
- Until fusion is complete, cephalic and caudal ends of the neural tube communicate with amniotic cavity by cranial and caudal neuropores, respectively.
- Closure of cranial neuropore occurs at approximately day 25, whereas the posterior neuropore closes at day 27.

- Neurulation is complete, and central nervous system is represented by a closed tubular structure with a narrow caudal portion, spinal cord, and a broader cephalic portion characterized by a number of dilations, brain vesicles.
- As the neural folds elevate and fuse, cells at lateral border or crest of neuroectoderm dissociate from their neighbours, *neural crest cells.*

Derivatives of the Ectodermal Germ Layer:

- Gives rise to organs and structures that maintain contact with the outside world:

1. Central nervous system.
2. Peripheral nervous system.
3. Sensory epithelium of the ear
4. Nose.
5. Eye.
6. Epidermis.
7. Hair.
8. Nails.
9. Subcutaneous glands.
10. Mammary glands.
11. Pituitary gland.
12. Enamel of the teeth.

Derivatives of Neural Crest Cells:

1. Connective tissue and bones of the face and skull Cranial nerve ganglia.
2. C cells of the thyroid gland.
3. Conotruncal septum in the heart.
4. Odontoblasts.
5. Dermis in face and neck.
6. Spinal (dorsal root) ganglia.
7. Sympathetic chain and preaortic ganglia.
8. Parasympathetic ganglia of the gastrointestinal tract
9. Adrenal medulla.
10. Schwann cells Glial cells.
11. Arachnoid and pia mater (leptomeninges).
12. Melanocytes.

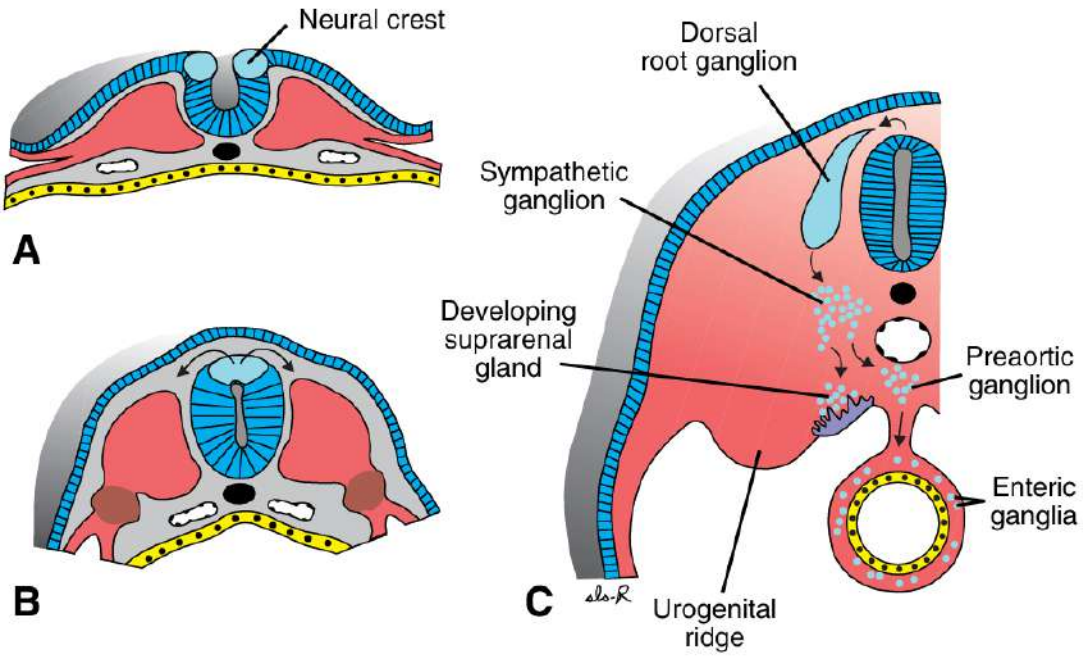


Figure: Formation and migration of neural crest cells in spinal cord. A and B. Crest cells form at the tips of neural folds. C. After migration, crest cells contribute to structures, including dorsal root ganglia, sympathetic chain ganglia, adrenal medulla, and other tissues. (Sadler, 2018).

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
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Lecture 28:

DERIVATIVES OF THE MESODERMAL GERM LAYER

Objectives of the lecture:

By end of the lecture the student should be able to:

1. Enumerate the derivatives of the paraxial mesoderm germ layer.
2. Enumerate the derivatives of the intermediate mesoderm germ layer.
3. Enumerate the derivatives of the lateral plate mesoderm germ layer.
4. Describe mechanism of development of the blood vessels.

Derivatives of the Mesodermal Germ Layer:

- Initially, cells of the mesodermal germ layer form a thin sheet of loosely woven tissue on each side of the midline.

By the 17th Day:

- Cells close to the midline proliferate and form, paraxial mesoderm.
- More laterally, mesoderm layer remains thin, lateral plate mesoderm.
- **With appearance and coalescence of intercellular cavities in lateral plate mesoderm, divided into 2 layers:**

1. Layer continuous with mesoderm covering amnion, somatic, parietal mesoderm layer.
2. Layer continuous with mesoderm covering yolk sac, splanchnic, visceral mesoderm layer.

- Together, these layers line a newly formed cavity, intraembryonic cavity, continuous with extraembryonic cavity on each side of the embryo.
- Intermediate mesoderm connects paraxial and lateral plate mesoderm.

PARAXIAL MESODERM

By the Beginning of the Third Week:

- Paraxial mesoderm is organized into segments.
- These segments, somitomeres, first appear in cephalic region of embryo, and their formation proceeds cephalocaudally.
- Each somitomere consists of mesodermal cells arranged in concentric whorls around center of the unit.
- In the head region, somitomeres form in association with segmentation of the neural plate into neuromeres and contribute to mesenchyme in the head.
- From occipital region caudally, somitomeres organize into somites.
- First pair of somites arises in occipital region of embryo at the 20th day of development.
- New somites appear in craniocaudal sequence at a rate of three pairs per day until, at end of the fifth week, 42 to 44 pairs are present.
- 4 occipital, 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, and 8 to 10 coccygeal pairs.
- First occipital and the last 5 to 7 coccygeal somites later disappear, while remaining somites form the axial skeleton.

By Beginning of the Fourth Week:

- **Cells forming Ventral and Medial Walls of Somite:** Lose their compact organization, become polymorphous, and shift their position to surround the notochord.
- Sclerotome, form a loosely woven tissue, mesenchyme.
- **Sclerotome Surround Spinal Cord and Notochord:** Form vertebral column.
- **Cells at Dorsolateral Portion of Somite:** Form limb and body wall musculature.
- **Cells at Dorsomedial Portion of Somite:** Proliferate and migrate down the ventral side of the remaining dorsal epithelium of the somite to form a new layer, the myotome.
- **Remaining Dorsal Epithelium**: Forms dermatome.
- **Myotome and Dermatome layers:** Constitute the dermomyotome.
- Each segmentally arranged myotome contributes to muscles of the back, while dermatomes disperse to form dermis and subcutaneous tissue of the skin.

Notes:

- Each myotome and dermatome retains its innervation from its segment of origin, no matter where the cells migrate.
- Each somite forms its own sclerotome (cartilage and bone component), its own myotome (segmental muscle component), and its own dermatome, segmental skin component.
- Each myotome and dermatome has its own segmental nerve component.

INTERMEDIATE MESODERM

- Intermediate mesoderm, connects paraxial mesoderm with lateral plate, differentiates into urogenital structures.
- **In cervical and upper thoracic regions:** Forms segmental cell clusters (nephrotomes), and an unsegmented mass of tissue, nephrogenic cord.
- **Excretory units of urinary system and gonads:** Develop from partly segmented and partly unsegmented intermediate mesoderm.

LATERAL PLATE MESODERM

- Lateral plate mesoderm splits into parietal and visceral layers, which line intraembryonic cavity and surround organs, respectively.
- **Mesoderm from parietal layer, together with overlying ectoderm:** Form lateral and ventral body wall.
- **Visceral layer and Embryonic Endoderm:** Form wall of the gut.
- **Mesoderm cells of parietal layer surrounding intraembryonic cavity:** Form thin membranes, mesothelial membranes, or serous membranes, line peritoneal, pleural, and pericardial cavities and secrete serous fluid.
- **Mesoderm cells of visceral layer:** Form a thin serous membrane around each organ.

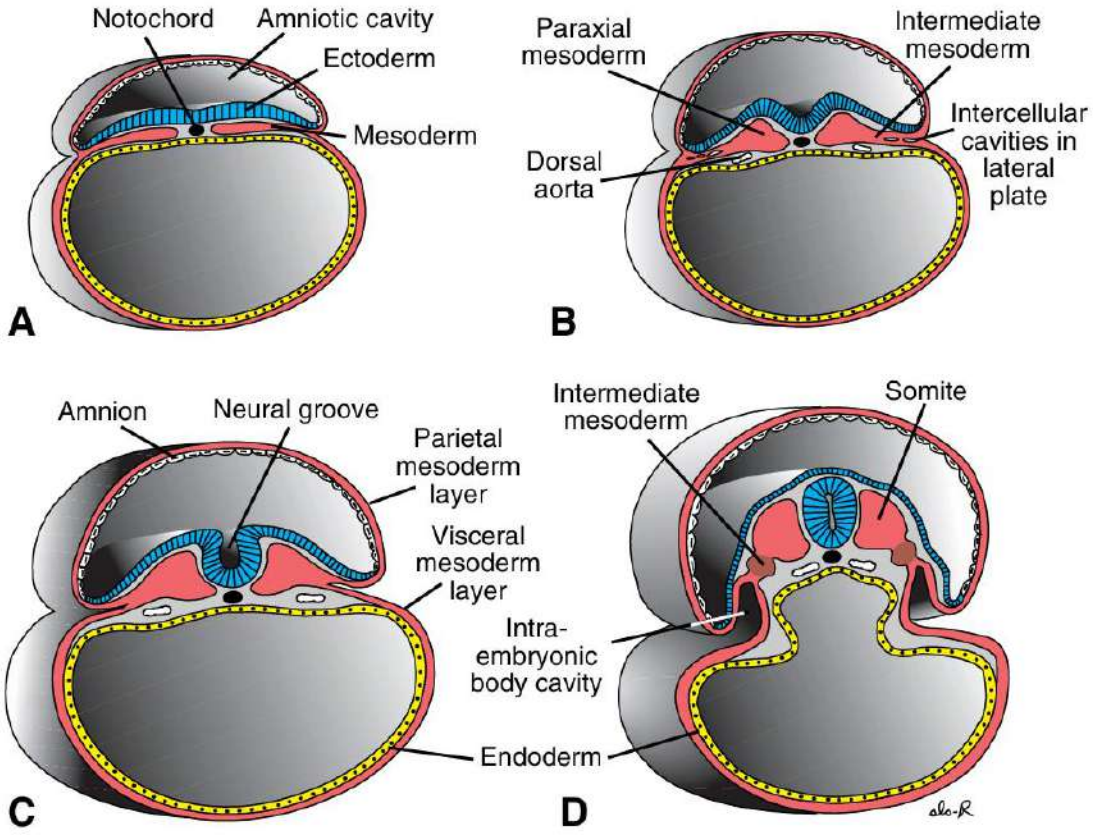


Figure 1: Sections showing development of mesodermal germ layer. A. Day 17. B. Day 19. C. Day 20. D. Day 21. The mesodermal sheet gives rise to paraxial mesoderm (future somites), intermediate mesoderm (future excretory units), and lateral plate, which is split into parietal and visceral mesoderm layers lining intraembryonic cavity. (Sadler, 2018).

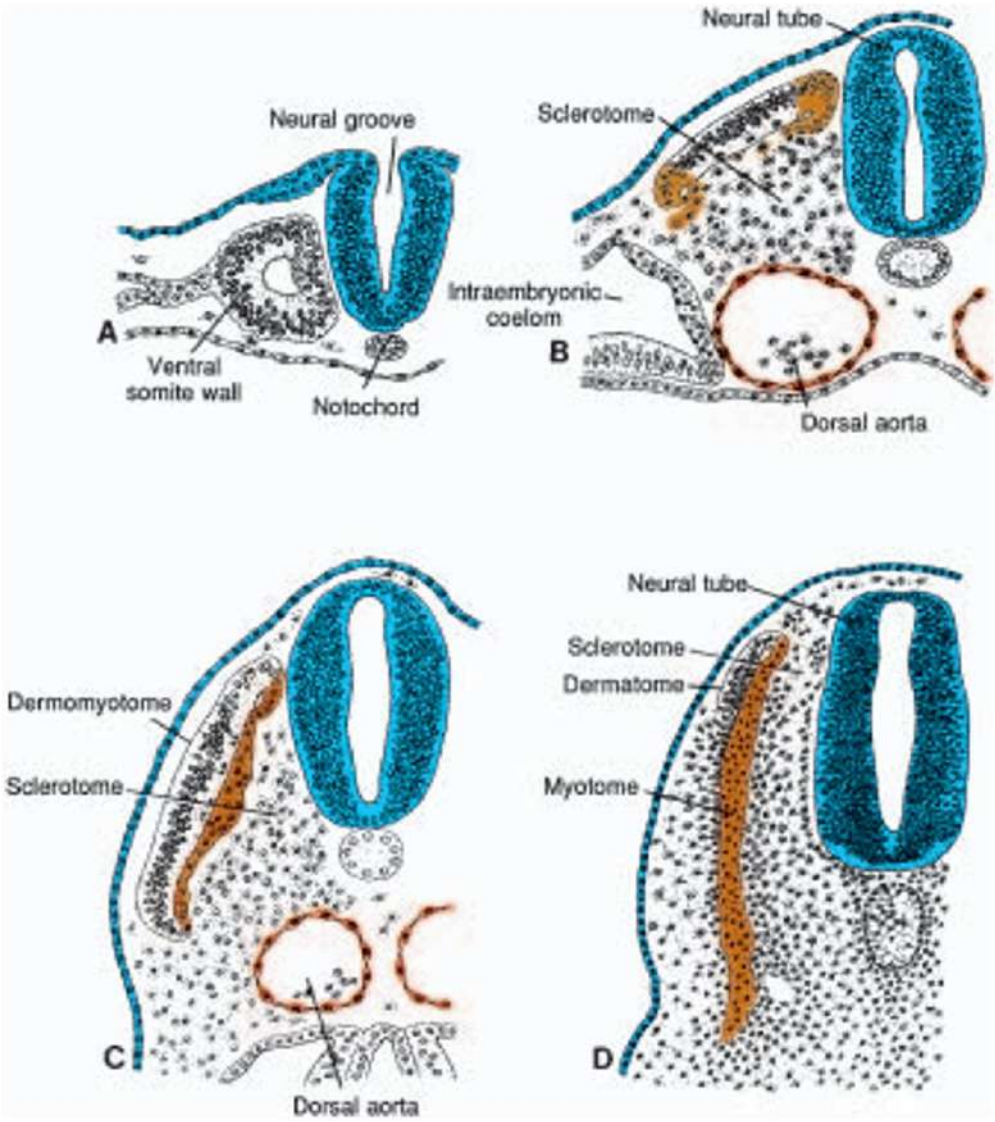


Figure 2: Development of a somite. A. Mesoderm cells arranged around a small cavity. B. Cells of ventral and medial walls of somite migrate in direction of notochord. Cells at dorsolateral portion of somite migrate as precursors to limb and body wall musculature. Dorsomedial cells migrate to form the myotome. C. Cells forming the myotome continue to extend beneath the dorsal epithelium. D. Dermatome cells lose their epithelial configuration and spread out under the overlying ectoderm to form dermis. (sadler, 2018).

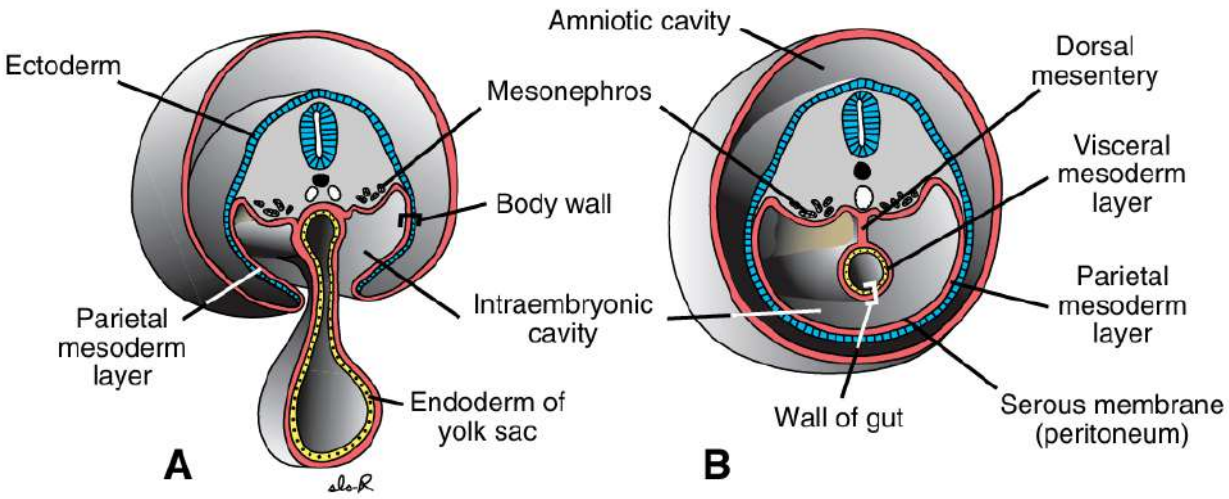


Figure 3: A. Section through a 21-day embryo in region of mesonephros showing parietal and visceral mesoderm layers. Intraembryonic cavities communicate with extraembryonic cavity (chorionic cavity). B. Section at end of the fourth week. Parietal mesoderm and ectoderm form ventral and lateral body wall. (Sadler, 2018).

BLOOD AND BLOOD VESSELS

Blood Vessels Form in Two Ways:

1. **Vasculogenesis:** Vessels arise from blood islands.
2. **Angiogenesis:** Sprouting from existing vessels.

- First blood islands appear in mesoderm surrounding the wall of yolk sac at 3 weeks of development and later in lateral plate mesoderm and other regions.
- Definitive hematopoietic stem cells arise from mesoderm surrounding the aorta in a site called the aorta-gonad-mesonephros region (AGM).
- These cells will colonize the liver, which becomes major hematopoietic organ of foetus.
- Stem cells from the liver will colonize the bone marrow, definitive blood-forming tissue.

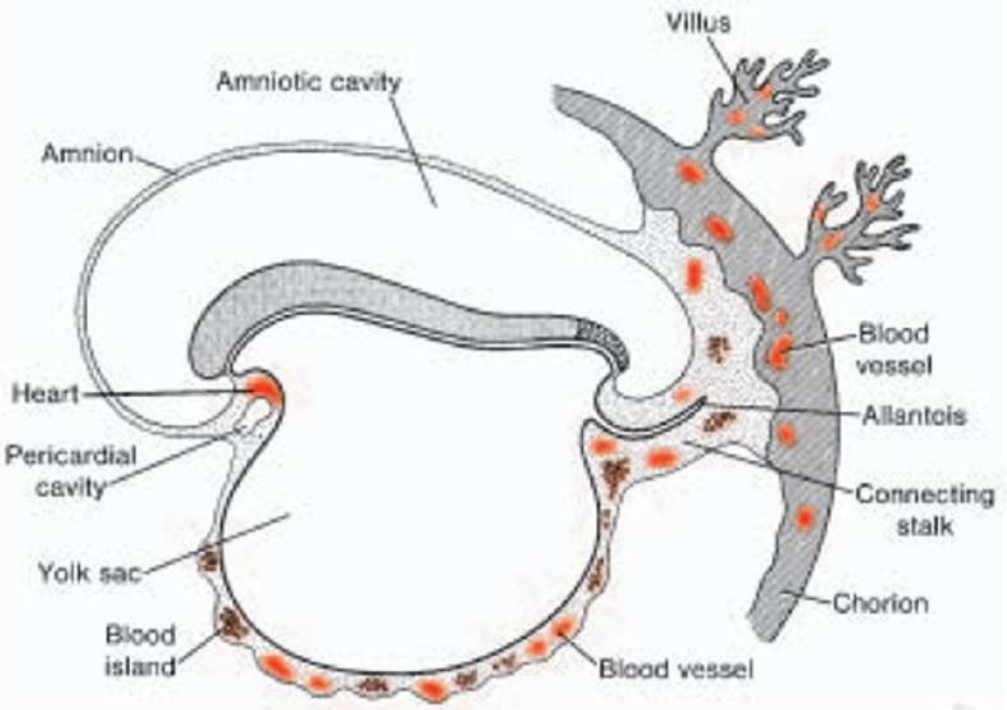


Figure 4: Extraembryonic blood vessel formation in the villi, chorion, connecting stalk, and wall of the yolk sac in a presomite embryo of 19 days. (Sadler, 2018).

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
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Lecture 29:

DERIVATIVES OF THE ENDODERMAL GERM LAYER

Objectives of the lecture:

By end of the lecture the student should be able to:

1. Enumerate the derivatives of the endoderm germ layer.
2. Enumerate results of folding.
3. What are the functions of the yolk sac?

Folding and Gastrointestinal Tract:

- Main organ system derived from the endodermal germ layer.
- Endoderm germ layer covers ventral surface of embryo and forms roof of the yolk sac.
- With development of the brain vesicles, embryonic disc bulge into amniotic cavity and fold cephalocaudally.
- Folding is in regions of the head and tail, where head fold and tail fold are formed.
- Portion of endoderm-lined cavity is incorporated into the body of embryo proper.
- **In Anterior Part:** Endoderm forms foregut.
- **In Tail Region:** Endoderm forms hindgut.
- Part between foregut and hindgut is midgut.
- **Midgut:** Temporarily communicates with yolk sac by way of a stalk, vitelline duct.
- **Vitelline duct:** Wide initially, with growth of embryo, becomes narrow and longer.
- At its cephalic end, the foregut is temporarily bounded by an ectodermal endodermal membrane called the buccopharyngeal membrane.

In the Fourth Week: Buccopharyngeal membrane ruptures, establishing an open connection between amniotic cavity and primitive gut.

- **Hindgut:** Terminates temporarily at cloacal membrane, which breaks down in seventh week to create the opening for anus.
- As result of rapid growth of somites, initial flat embryonic disc folds laterally, and embryo obtains a round appearance.
- Ventral body wall of the embryo is established except for a part in ventral abdominal region where yolk sac duct and connecting stalk are attached.
- When vitelline duct is obliterated, midgut lose its connection with original endoderm-lined cavity and obtain its free position in abdominal cavity.
- Partial incorporation of allantois into the body of embryo: Forms the cloaca.
- **Distal Portion of the Allantois**: Remains in connecting stalk.

By the Fifth Week:

- Yolk sac duct, allantois, and umbilical vessels are restricted to region of umbilical ring.
- In humans, yolk sac is vestigial and, has a nutritive role only in early stages of development.

Derivatives of Endoderm Germ Layer:

1. Epithelial lining of the primitive gut.
2. Intraembryonic portion of allantois
3. Intraembryonic portion of vitelline duct.
4. Epithelial lining of the respiratory tract.
5. Parenchyma of thyroid gland
6. Parenchyma of parathyroid liver, and pancreas.
7. Reticular stroma of the tonsil glands
8. Parenchyma of thymus gland.
9. Epithelial lining of the urinary bladder
10. Epithelial lining of the urethra.
11. Epithelial lining of tympanic cavity.
12. Epithelial lining of auditory tube.

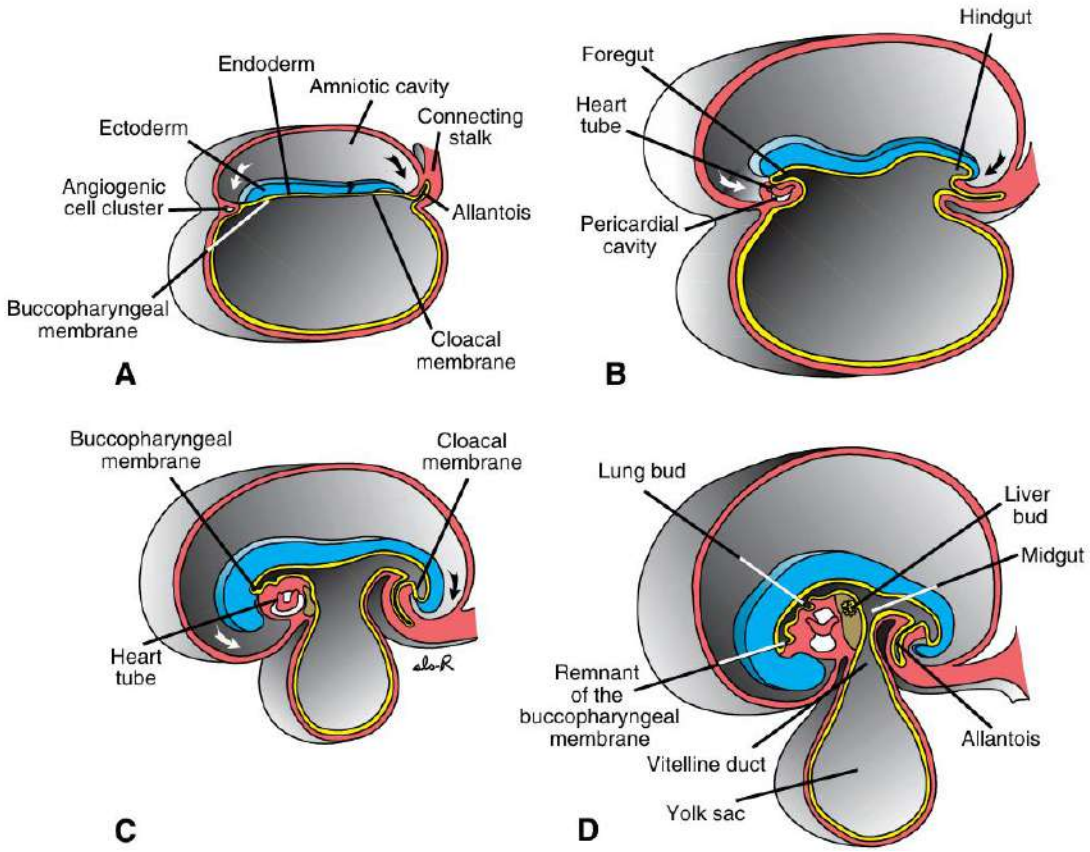


Figure 1: Sagittal midline sections of embryos at various stages of development to demonstrate cephalocaudal folding and its effect on position of endoderm-lined cavity. A. Presomite embryo. B. Embryo with 7 somites. C. Embryo with 14 somites. D. End of the first month. Angiogenic cell clusters in relation to buccopharyngeal membrane. (Sadler, 2018).

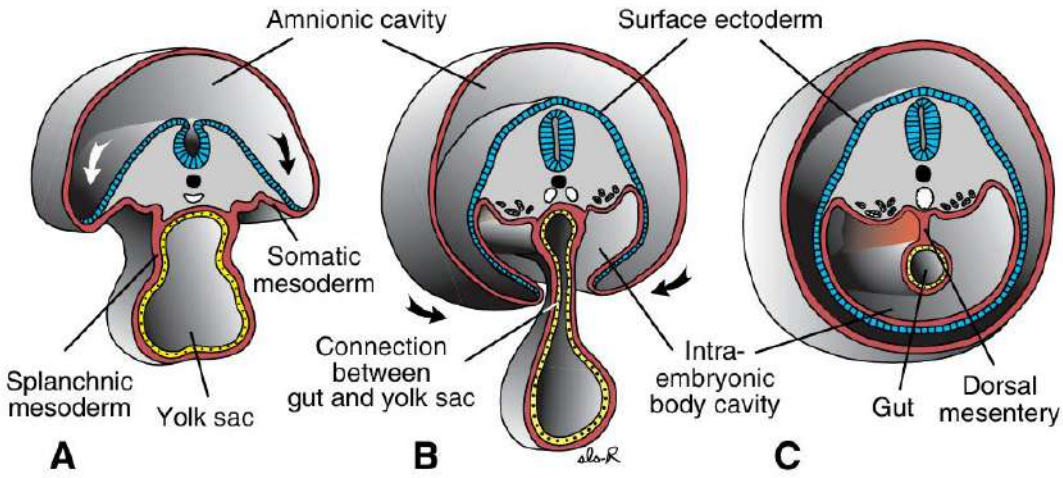


Figure 2: Sections through embryos at various stages of development to show effect of lateral folding on endoderm-lined cavity. A. Folding is initiated. B. Section through midgut to show connection between gut and yolk sac. C. Section below midgut to show closed ventral abdominal wall and gut suspended from dorsal abdominal wall by its mesentery. (Sadler, 2018).

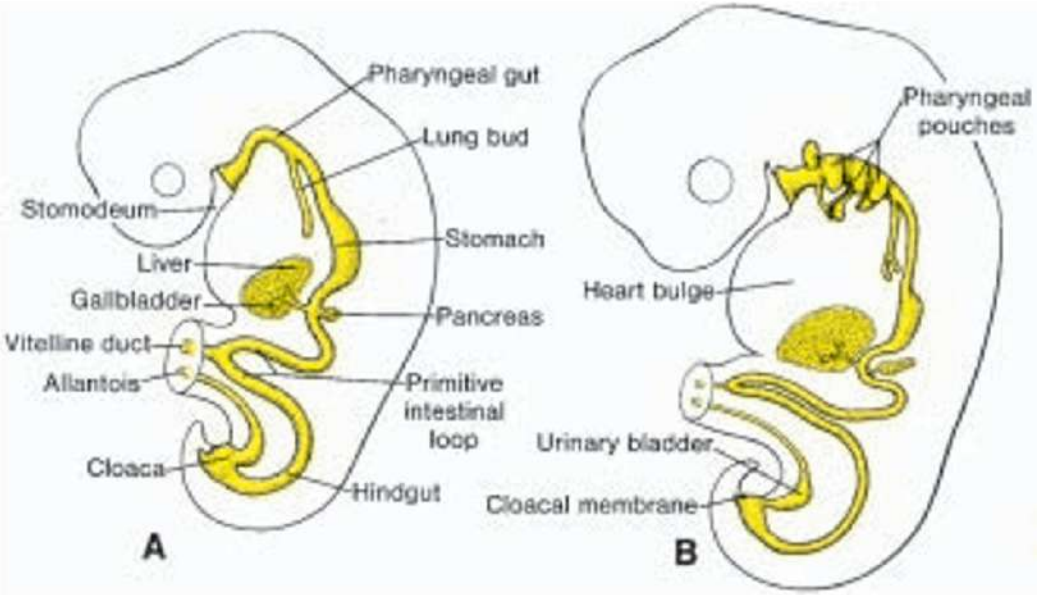


Figure 3: Embryos showing derivatives of endodermal germ layer. A. Pharyngeal pouches, epithelial lining of lung buds and trachea, liver, gallbladder, and pancreas. B. Urinary bladder is derived from cloaca and, at this age, is in open connection with allantois. (Sadler, 2018).

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
3. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.

Lecture 30:

EXTERNAL APPEARANCE DURING THE SECOND MONTH

Objectives of the lecture:

By end of the lecture the student should be able to:

- Know external appearance and changes in the second month of human development.

External Appearance During the Second Month:

At the End of the Fourth Week:

- When embryo has 28 somites, main external features are somites and pharyngeal arches.
- **Age of the embryo:** Usually expressed in somites.
- Because counting somites becomes difficult during second month of development, age of the embryo is indicated as crown-rump length (CRL) and expressed in millimeters.
- **CRL:** From vertex of skull to midpoint between apices of the buttocks.

Table 1: CRL Correlated to Approximate Age in Weeks

CRL (mm)	Approximate Age (week)
5–8	5
10–14	6
17–22	7
28–30	8

During the Second Month:

- Increase in head size.
- Formation of the limbs
- Formation of the face.
- Formation of the ears.
- Formation of the nose.
- Formation of the eyes.

By Beginning of the fifth Week:

- Forelimbs and hindlimbs appear as paddle-shaped buds.
- Forelimbs are located dorsal to pericardial swelling at level of the fourth cervical to first thoracic somites, innervated by brachial plexus.
- Hindlimb buds appear later caudal to attachment of umbilical stalk at level of lumbar and upper sacral somites.
- Terminal portions of the buds flatten, and a circular constriction separates them from proximal, more cylindrical segment.
- Four radial grooves separating five thicker areas appear on distal portion of the buds, foreshadowing formation of digits.

- The grooves, rays, appear in the hand region first and afterward in the foot, as upper limb is more advanced in development than lower limb.
- While fingers and toes are being formed, a second constriction divides proximal portion of the buds into two segments, and 3 parts characteristic of adult extremities can be recognized.

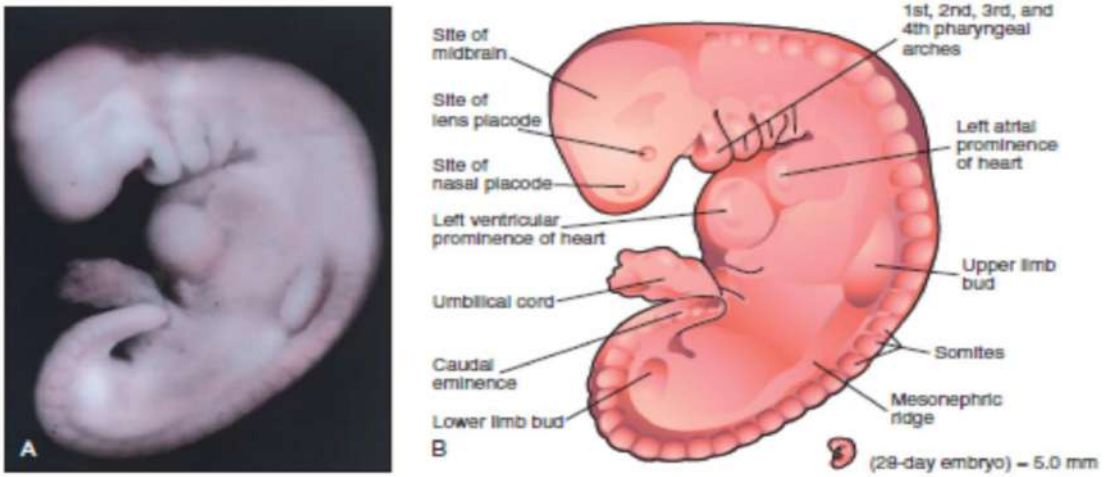


Figure 1: 28-day Embryo showing pharyngeal arches, upper and lower limb buds. (Sadler, 2018)

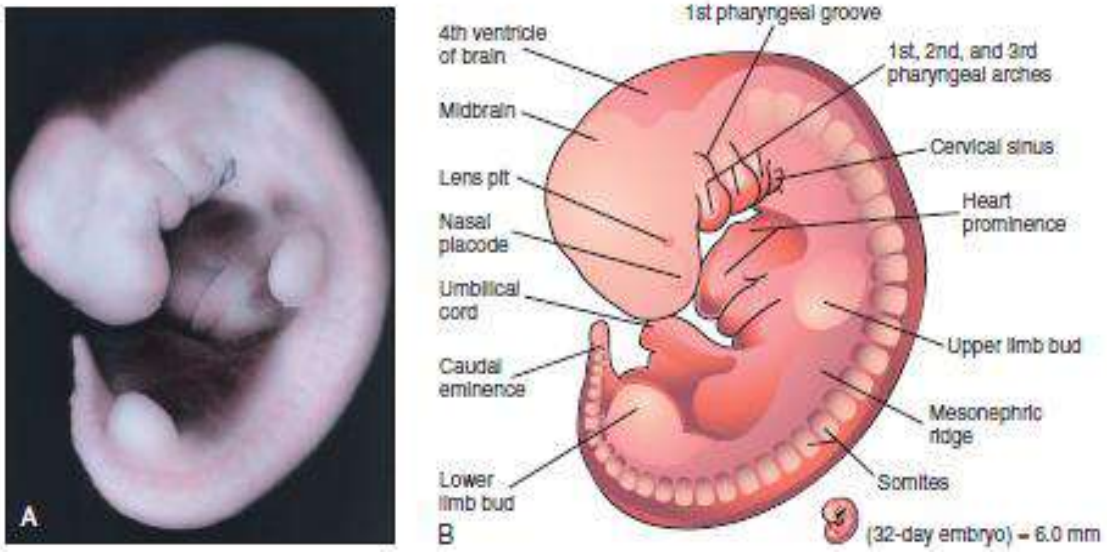


Figure 2: 32-day Embryo showing ear, eye, heart, upper and lower limbs buds buds.

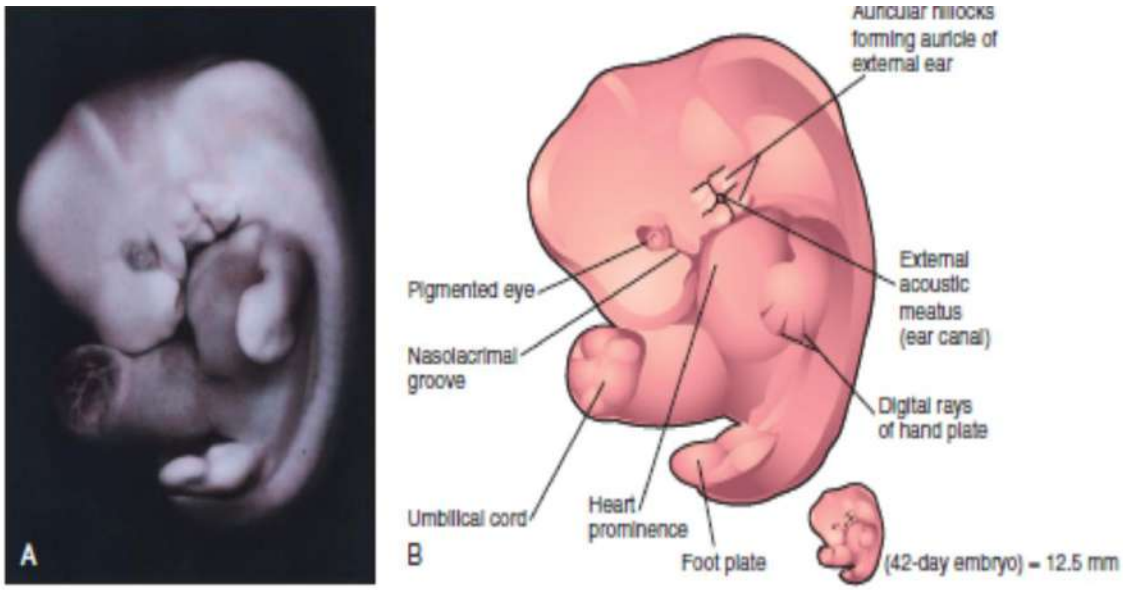


Figure 3: 42-day Embryo showing ear, eye, nasolacrimal groove, heart prominence, upper and lower limbs digital rays. (Sadler, 2018).

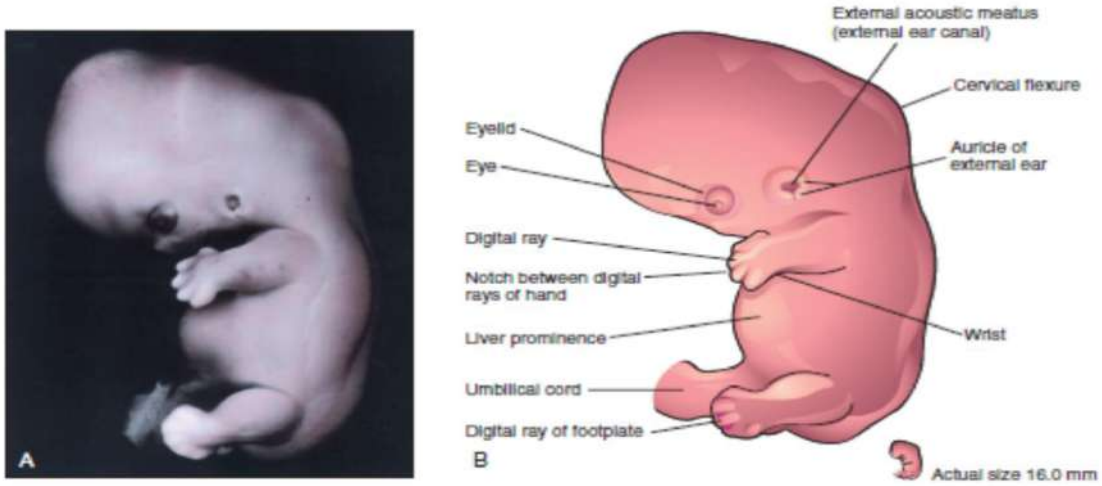


Figure 4: 48-day Embryo showing external acoustic meatus, eyelids and eye, heart, upper and lower limbs digital rays. (Sadler, 2018).

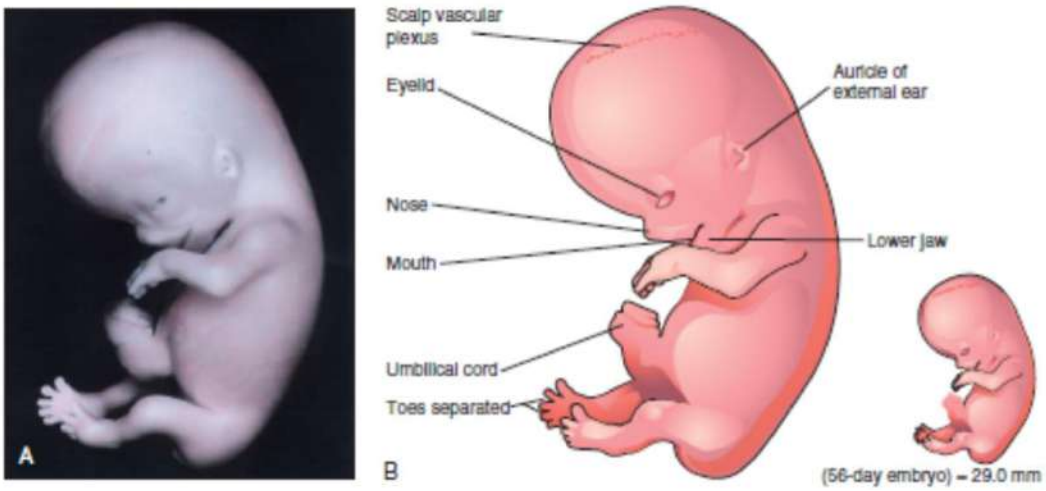


Figure 5: 56-day Embryo (end of embryonic period). The embryo has a distinct human appearance. (Sadler, 2018).

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.

2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.

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Lecture 31:

THIRD MONTH TO BIRTH: THE FETUS AND PLACENTA

Objectives of the lecture:

By end of the lecture the student should be able to:

- Know external appearance and changes in the third month to birth.

DEVELOPMENT OF THE FETUS:

Fetal Period:

- Period from beginning of the ninth week to birth.
- Maturation of tissues and organs and rapid growth of the body.

Length of the Fetus:

- **Crown-rump length (CRL):** (sitting height).
- **Crown-heel length (CHL):** Measurement from vertex of skull to heel (standing height).
- Expressed in centimetres.
- Correlated with the age of the fetus in weeks or months.
- Growth in length is during the third, fourth, and fifth months.
- Increase in weight is during the last 2 months of gestation.

Length of Pregnancy:

- Considered to be 280 days, or 40 weeks after onset of the last normal menstrual period (LNMP) or more accurately, 266 days or 38 weeks after fertilization.

Monthly Changes:

- Relative slowdown in growth of the head compared with the rest of the body.
- At the beginning of the third month the head constitutes approximately half of the CRL.

By beginning of the fifth month:

- Size of the head is one-third of CHL, and at birth it is one-fourth of the CHL.
- Over time, growth of the body accelerates but that of the head slows down.

During the Third Month:

- **Face:** Becomes more human looking.
- **Eyes:** Initially directed laterally, move to ventral aspect of the face.
- **Ears:** Come to lie close to their definitive position at the side of the head.
- **Limbs:** Reach their relative length in comparison with the rest of the body, although lower limbs are a little shorter and less well developed than upper extremities.
- **Primary ossification centres:** Present in long bones and skull by the 12th week.
- **External genitalia:** Develop to a degree that sex of the fetus can be determined by external examination (ultrasound) by the 12 week.

- **Intestinal Loops:** During the 6th week, cause a large swelling (herniation) in the umbilical cord, but by the 12th week the loops withdraw into the abdominal cavity.
- **Reflex activity:** Can be evoked in aborted fetuses, indicating muscular activity.

During the Fourth and Fifth Months:

- Fetus lengthens rapidly, and at end of the first half of intrauterine life its CRL is 15 cm, about half the total length of newborn.
- **Weight of the fetus:** Increases little during this period and by end of the fifth month is less than 500 g.
- Fetus is covered with fine hair, lanugo hair; eyebrows and head hair are also visible.
- **Movements of the fetus:** Can be felt by the mother by the fifth month.
- **During the second half of intrauterine life:** Weight increases, particularly during last 2.5 months, when 50% of full-term weight (3200 g) is added.

During the Sixth Month:

- **Skin of the fetus:** Reddish and has a wrinkled appearance because of lack of underlying connective tissue.
- Fetus born early in the sixth month has great difficulty surviving.
- Respiratory system and the central nervous system have not differentiated sufficiently, and coordination between the two systems is not yet well established.
- **By 6.5 to 7 months:** Fetus has a length of about 25 cm and weighs 1100 g.
- If born at this time, the infant has a 90% chance of surviving.

During the last 2 months:

- **Fetus:** Obtains well-rounded contours as result of deposition of subcutaneous fat.

By end of intrauterine life:

- **Skin**: Covered by a whitish, fatty substance (vernix caseosa) composed of secretory products from sebaceous glands.

At the end of the ninth month:

- **Skull:** Has the largest circumference of all parts of body, an important fact with regard to its passage through the birth canal.

At the time of birth:

- Weight of a normal fetus is 3000 to 3400 g; its CRL is 36 cm; and its CHL is 50 cm.
- Sexual characteristics are pronounced, and the testes should be in the scrotum.

Time of Birth:

- 266 days, or 38 weeks, after fertilization.
- **Oocyte:** Usually fertilized within 12 hours of ovulation.
- **Sperm:** deposited in reproductive tract up to 6 days prior to ovulation can survive to fertilize oocytes.
- Most pregnancies occur when sexual intercourse occurs within a 6-day period that ends on the day of ovulation.
- **Date of birth:** 280 days or 40 weeks.

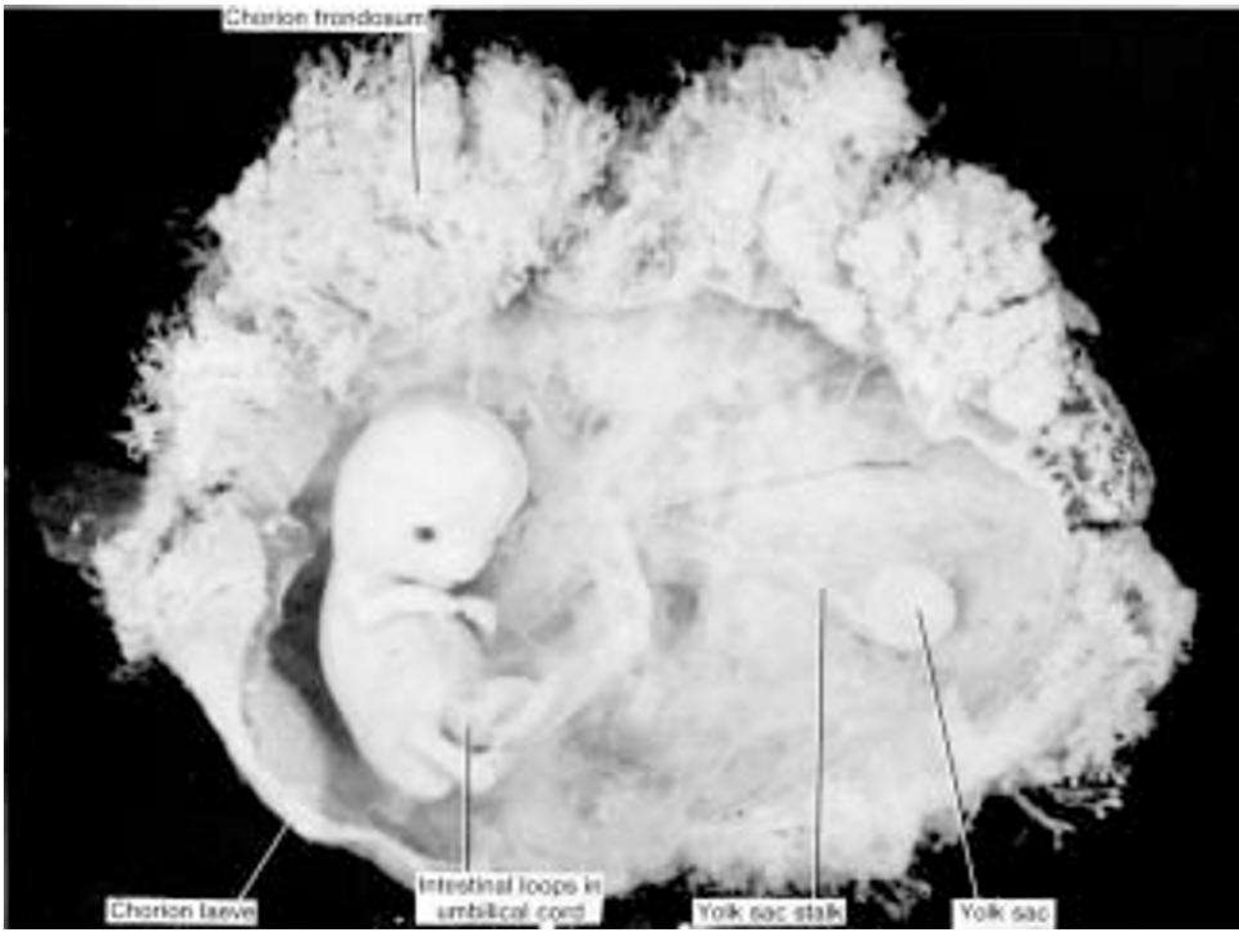


Figure 1: A 9-week fetus. Large size of head compared with that of the rest of body. Yolk sac and long vitelline duct are visible in the chorionic cavity. Note umbilical cord and herniation of intestinal loops. One side of chorion has many villi (chorion frondosum), while the other side is smooth (chorion laeve). (Sadler, 2018).

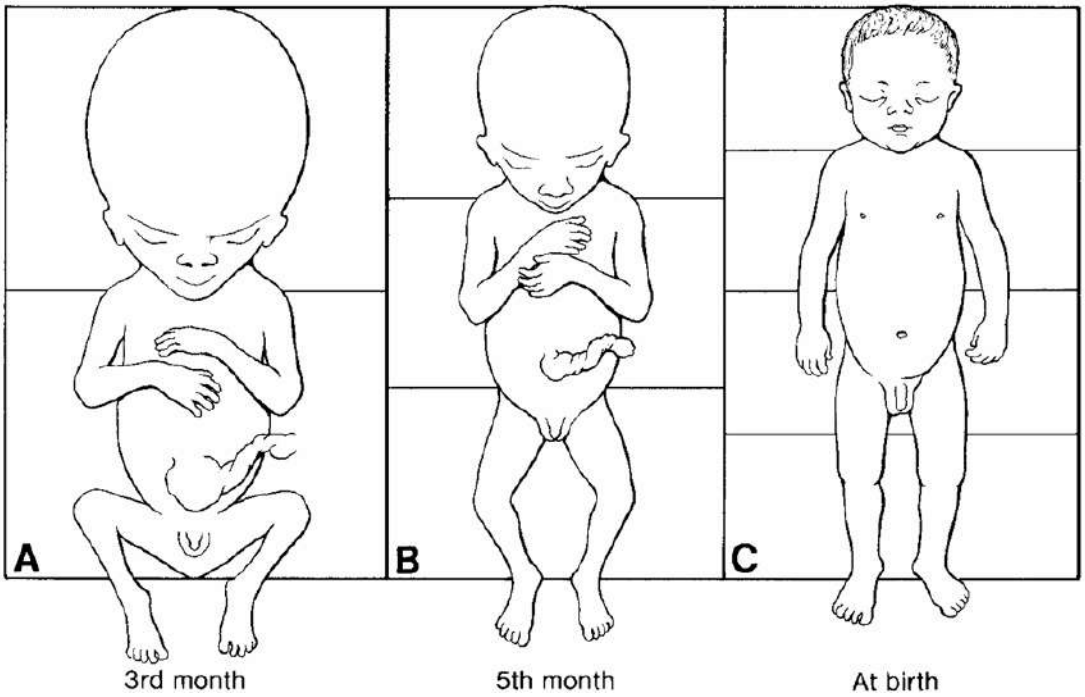


Figure 2: Size of head in relation to rest of body at stages of development. (Sadler, 2018).



Figure 3: 11-week fetus. Umbilical cord shows a swelling at its base, caused by herniated intestinal loops. Toes are developed, and sex of the fetus can be seen. Skull of this fetus lacks normal smooth contours. (Sadler, 2018).

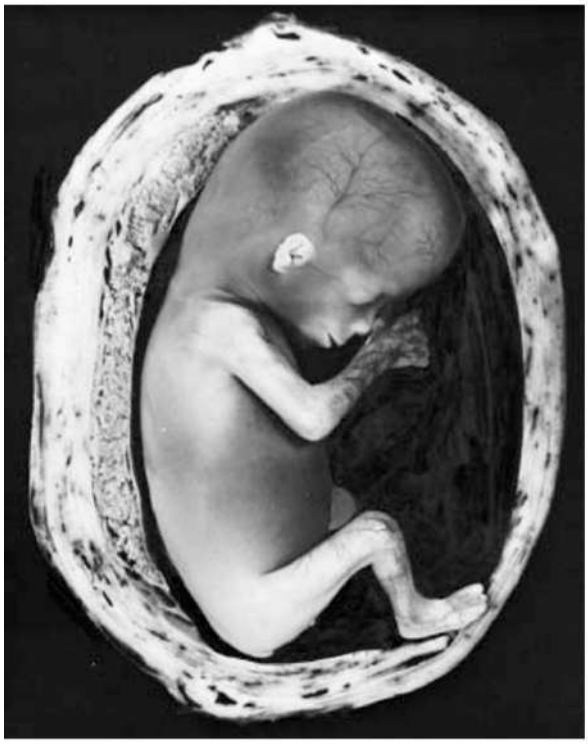


Figure 4: A12-week fetus in utero. Extremely thin skin and underlying blood vessels. Face has all of human characteristics, but ears are still primitive. (Sadler, 2018).



Figure 5:18-week-old fetus connected to placenta by its umbilical cord. Skin of fetus is thin because of lack of subcutaneous fat. Placenta with its cotyledons and amnion. (Sadler, 2018).



Figure 6: A7-month-oldfetus. This fetus would be able to survive. Well-rounded contours because of deposition of subcutaneous fat. Twisting of the umbilical cord. (Sadler, 2018).

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
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Lecture 32:

PLACENTA

Objectives of the lecture:

By end of the lecture the student should be able to:

- Enumerate fetal membranes.
- Describe the formation of the placenta.
- Enumerate functions of the placenta.
- Enumerate congenital of the placenta.

Fetal Membranes:

1. Amnion
2. Yolk sac
3. Allantois
4. Connecting stalk
5. Umbilical cord
6. Chorion
7. Placenta

Placenta:

- One of fetal membranes that carries out respiration, nutrition and excretion of embryo.
- Fully formed during the fourth month of pregnancy.

Formation of the Placenta: Fetal part (chorion frondosum) and maternal part (decidua basalis)

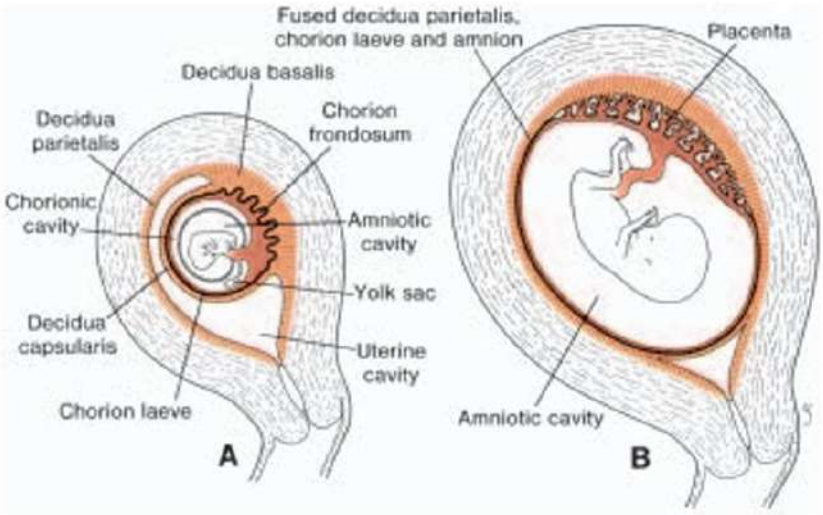


Figure 1: Formation of placenta from decidua basalis and chorion frondosum. A. End of the second month. At abembryonic pole, villi have disappeared (chorion laeve). B. End of the third month. Chorion and amnion have fused, and uterine cavity is obliterated by fusion of chorion laeve and decidua parietalis. (Sadler, 2018).

Structure of the Placenta:

By Beginning of Fourth Month: Has 2 sides

1. Fetal Portion: Formed of chorionic frondosum and bordered by chorionic plate.

2. Maternal portion: Formed of decidua basalis, of which decidual plate is the most incorporated into placenta.

- In-between the 2 portions intervillous spaces filled with maternal blood and villous trees grow into these blood lakes.

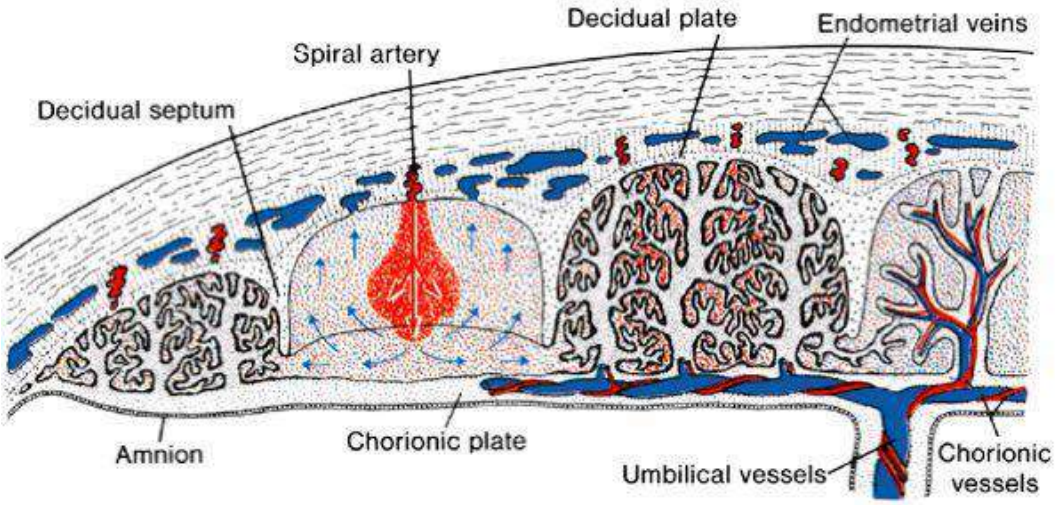


Figure 2: Placenta in the second half of pregnancy. Cotyledons are partially separated by decidual (maternal) septa. Intervillous blood returns to maternal circulation by way of endometrial veins. Small portions enter neighboring cotyledons. (Sadler, 2018)

Definition of Decidua: Endometrium of the pregnant uterus

- Functional layer of endometrium (shed during parturition).

Parts of Decidua:

1. **Decidua basalis:** Immediately deep to implanted ovum.
2. **Decidua capsularis:** Covering developing embryo.
3. **Decidua parietalis:** Lining rest of uterine cavity (decidua vera).

By 4-5 Months:

- **Decidual Septa:**
- Number of septa which project into intervillous spaces from decidua.
- These septa do not reach chorionic plate.
- Result in division of the placenta into 15-25 compartments (cotyledons).
- Formed of core of maternal tissue and covered by a layer of syncytial cells.

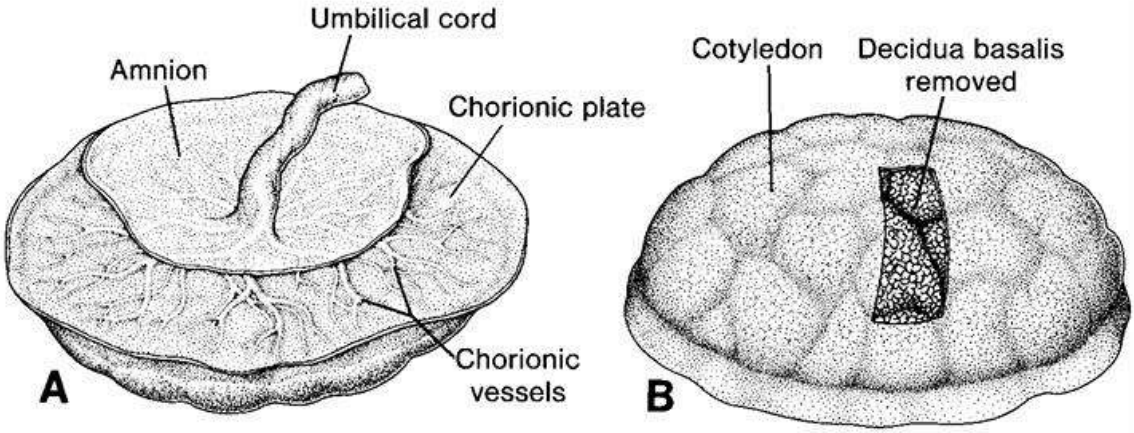


Figure 3: Full-term placenta. A. Fetal side. Chorionic plate and umbilical cord are covered by amnion. B. Maternal side showing the cotyledons. One area decidua has been removed. Maternal side of placenta is carefully inspected at birth, and one or more cotyledons with a whitish appearance are present because of excessive fibrinoid formation and infarction of a group of intervillous lakes. (Sadler, 2018).

Full Term Placenta:

- Covered 25 % of internal surface of uterus at any time
- Increase in thickness because of arborization of villi.
- Discoid in shape 20 X 3 Cm and weight about 500 Gm.

Placenta after birth: Comes out about 30 minutes after childbirth.

Placental examination after birth:

Placenta From Maternal Side:

- 15-20 cotyledons, covered by a thin layer decidua basalis.
- Decidual septa are seen as grooves between cotyledons.

Placenta From Fetal Side:

- Entire surface is covered by chorionic plate, with large number of arteries and veins (chorionic vessels), converge toward umbilical cord.
- Umbilical cord usually attached eccentric or marginal, rarely into the chorion outside the placenta = Velamentous insertion.
- Cotyledons receive their blood though 80-100 spiral arteries, that pierce decidual plate and enter intervillous spaces.
- Blood pressure in intervillous spaces is high, this pressure forces blood deep in these spaces to path numerous villi (around 10 m square).

- As the pressure decreases the blood flow back from the chorion plate toward the decidua, where it enters the endometrial veins, and back to maternal circulation.
- Intervillous spaces contain 150 ml of blood which changes 3-4 times per minute.
- Only villi where fetal vessels are in intimate contact with covering syncytial membrane, gas exchange occur.

Placental Membrane (Barrier)

- Not true barrier as many substances can cross through it freely.

Initially formed of:

1. Endothelial lining of fetal vessels
2. Connective tissue in villous core
3. Cytotrophoblastic layer
4. Syncytium

BY 4th months: 1 and 4 comes in intimate contact, this increase greatly rate of gas exchange.

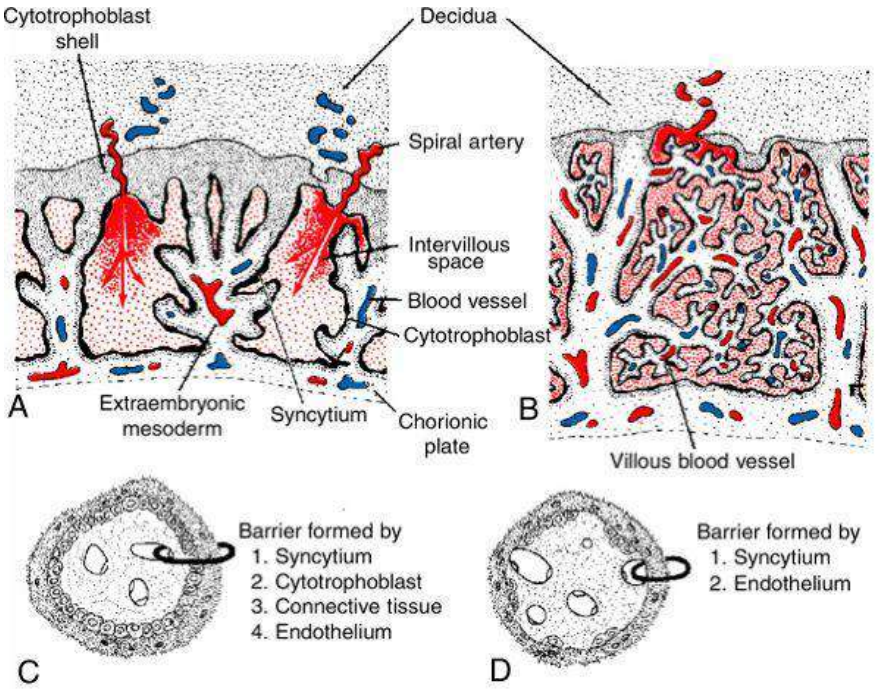


Figure 4: Structure of villi at various stages of development. A. During the fourth week. Extraembryonic mesoderm penetrates the stem villi in direction of decidual plate. B. During the fourth month. In many villi walls of the capillaries is in direct contact with syncytium. C and D. Enlargement of the villus as shown in A and B, respectively. (Sadler, 2018).

Functions of placenta

1. Respiration: Oxygen supplied by the mother diffuses across placental membrane into fetal blood. Fetal hemoglobin has greater affinity for oxygen than maternal hemoglobin.
Carbon dioxide passes readily in opposite direction. Placenta acts as a lung.

2. Nutrition: Water, salts, carbohydrates, fats, proteins and vitamins can pass from maternal to fetal blood.

3. Excretion: Products of fetal metabolism cross placentas into maternal circulation. Placenta acts as a kidney.

4. Protection: Foreign matters such as most bacteria are unable to cross placental barrier.

5. Endocrine: Placenta secretes estrogen, progesterone, chorionic gonadotropins and somatomammotropin.

6. Passive immunity: Transmission of maternal antibodies; IgG to give fetus.

Anomalies of the Placenta:

1. Diffuse Placenta (Placenta Membranacea): Chorion leave persists.

2. Placenta Bipartita or Tripartita: Placenta with two or three lobes.

3. Placenta Duplex, Triplex or Multiplex: Placenta has two or more separate lobes.

4. Placenta Succenturiata: Placenta has a small accessory lobe.

5. Placenta Fenestrate: Failure of development of placenta over a small area, the defect has appearance of a window.

6. Placenta Circumvallata: Placenta with central depression.
7. Battledore placenta: Marginal attachment of the cord.
8. Placenta Velamentosa: Cord is attached distance from margin of placenta. Blood vessels pass via membranes.
9. Small Placenta: With maternal hypertension.
10. Excessive Large Placenta: With fetal hydrops.
11. Placenta Previa:

a. Placenta previa lateralis
b. Placenta previa marginalis.
c. Placenta previa centralis

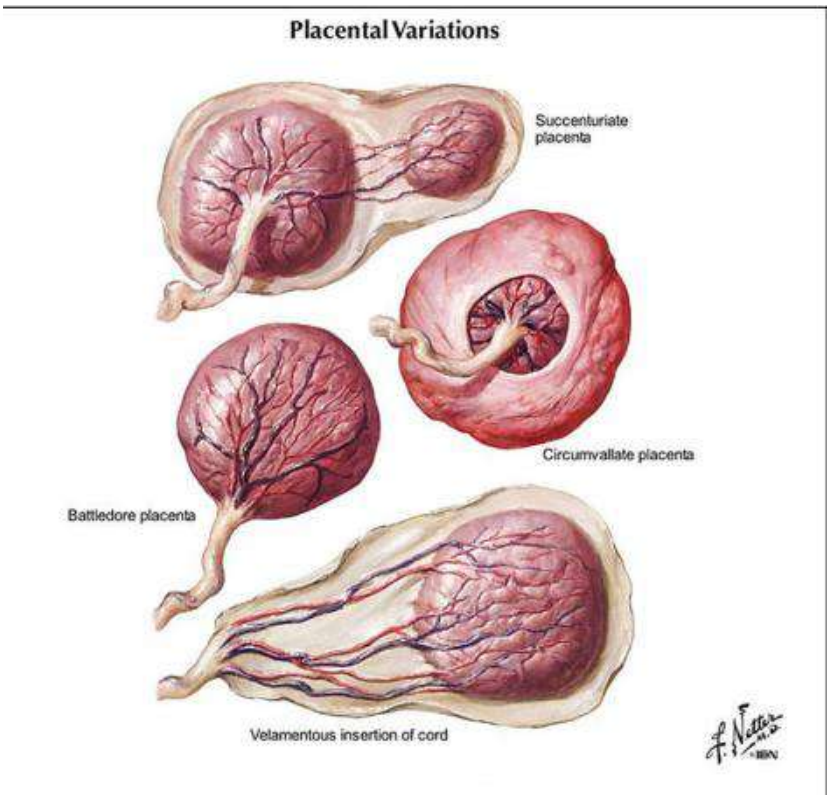


Figure 5: Placental Variations.

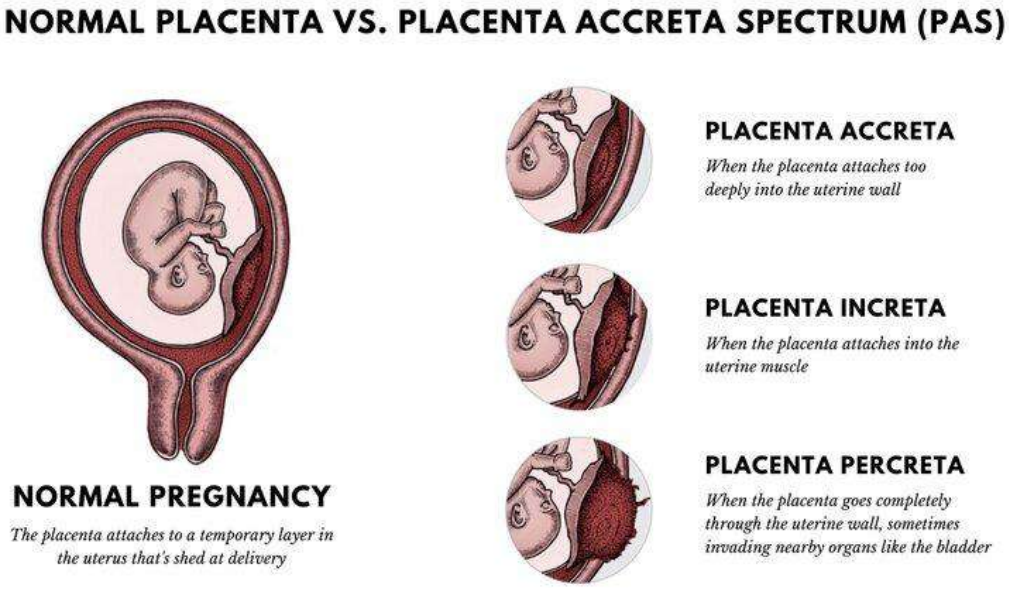


Figure 6: Placenta accreta, increta and percreta.

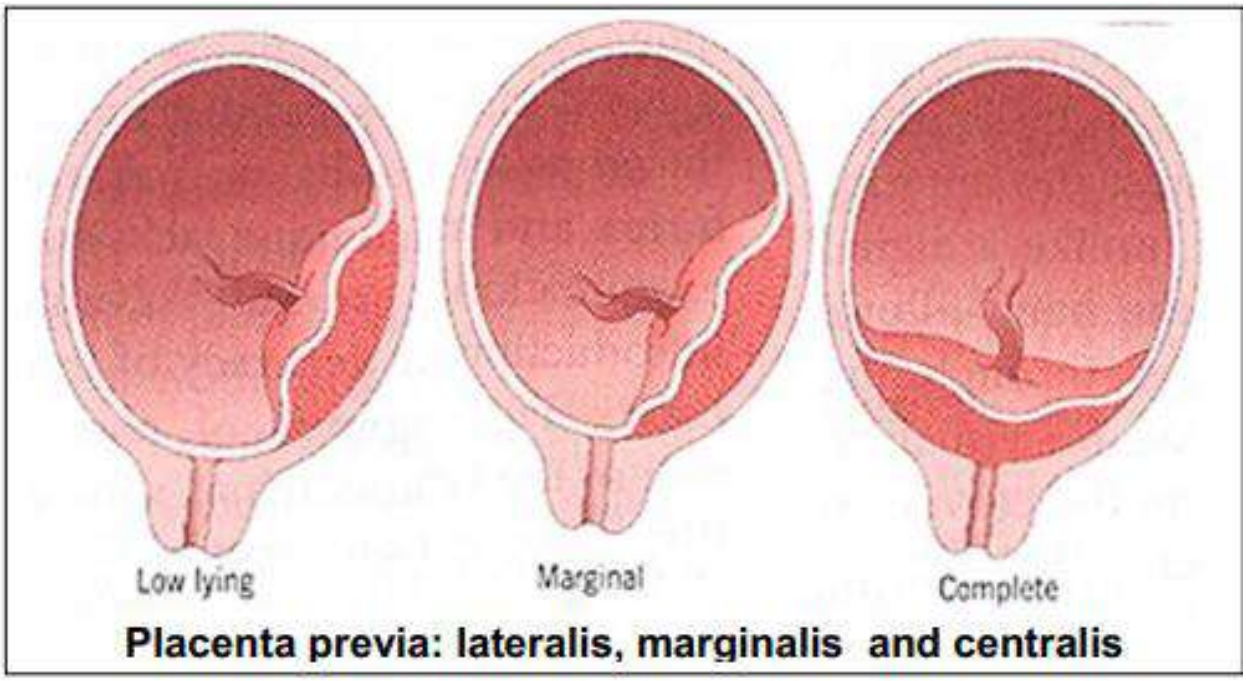


Figure 7: Types of placenta previa.

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
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Lecture 33:

UMBILICAL CORD

Objectives of the lecture:

By end of the lecture the student should be able to:

- Describe the formation of the umbilical cord.
- Enumerate structure of initially formed umbilical cord.
- Enumerate structure of full-term umbilical cord.
- Enumerate congenital of the umbilical cord.

Umbilical Cord:

Definition of Umbilical Cord:

- Tortuous structure that contains blood vessels which connect the placenta with the embryo.

Gross Appearance of Umbilical Cord:

- At end of pregnancy, about 50 Cm long and 2 Cm in diameter
- Mesodermal core of the cord becomes loose connective tissue called Warton's jelly.

Contents of Umbilical Cord:

1. **Wharton's Jelly:** Mucous connective tissue, invests other structures of the cord.
2. **Umbilical Vessels:**
   - a. **Two Umbilical Arteries:** Carry deoxygenated blood from the fetus to placenta.
   - b. **One Umbilical Vein**: At first two veins; the right vein disappears, and the left remains.
     - Left umbilical vein carries oxygenated blood from the placenta to the fetus**.**
3. **Vitelline Duct:** Connects midgut with definitive yolk sac**.**
4. **Allantois**
5. **Definitive Yolk Sac**
6. **Physiological Umbilical Hernia:** Contains loop of intestine.
7. **Remains of Extraembryonic Celom:** Accommodates the herniating gut.

Note: At middle of pregnancy, contents from 3-7 disappear

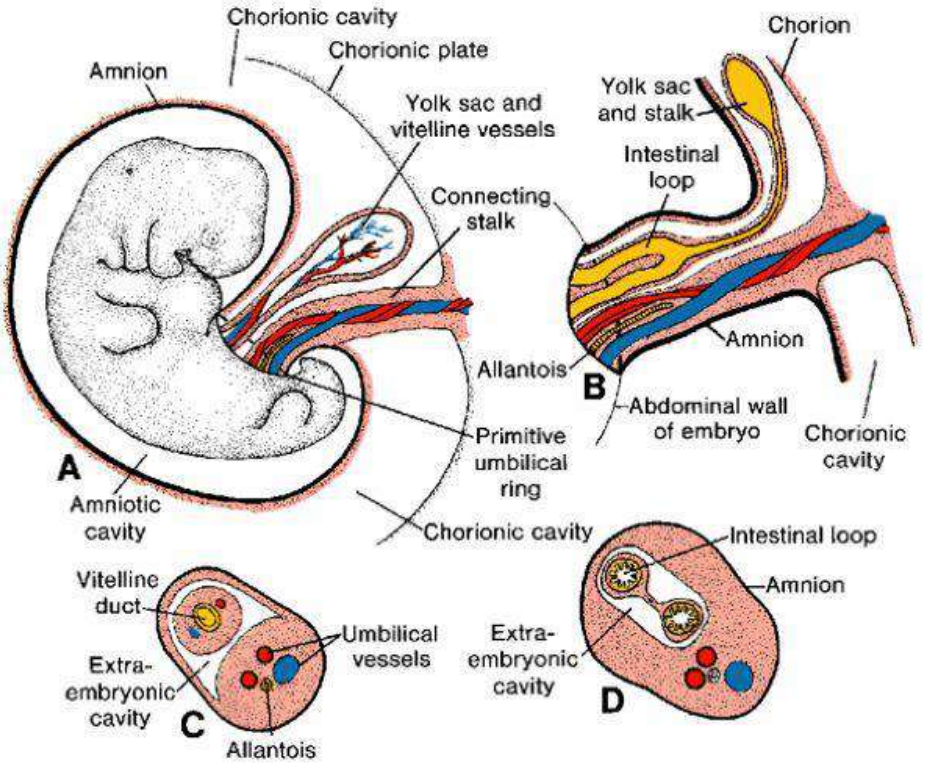


Figure 1: A. A 5-week embryo showing structures passing through primitive umbilical ring. B. Primitive umbilical cord of a 10-week embryo. C. Transverse section through the structures at the level of umbilical ring. D. Transverse section through primitive umbilical cord showing intestinal loops protruding in the cord. (Sadler, 2018).

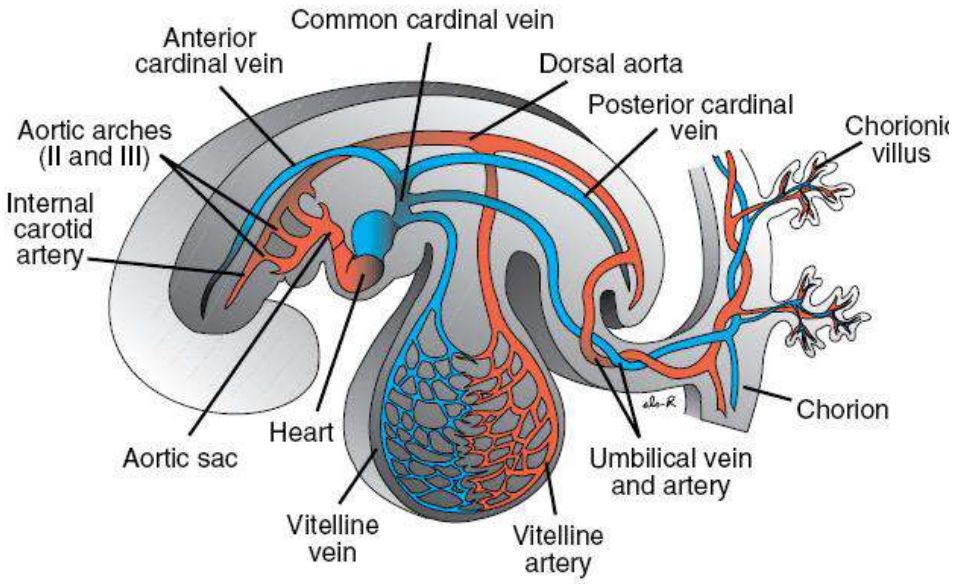


Figure 2: Main components of the umbilical cord in a 4-mm embryo (end of the fourth week).

Physiological Umbilical Hernia:

- Results due to rapid development of intestinal loops, and because of small abdominal cavity at that stage.
- Intestinal loops is pushed into umbilical cord, but it returns into abdominal cavity by end of the third month.

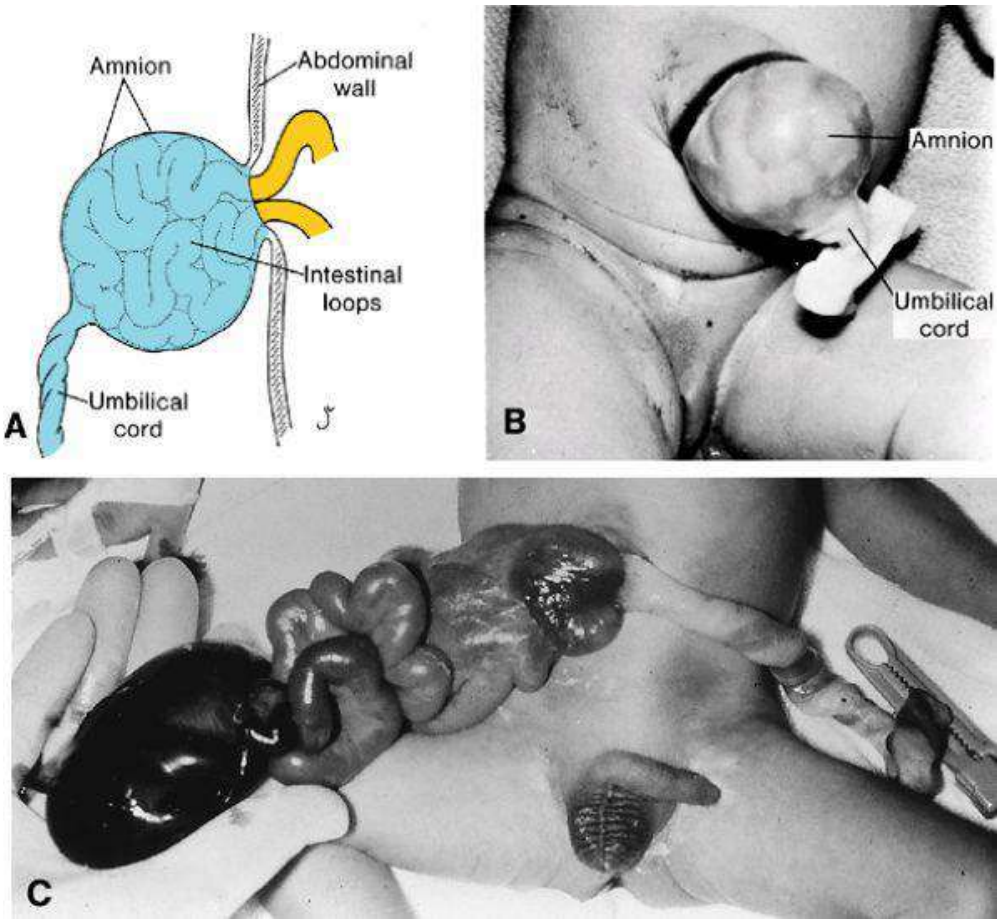


Figure 3: A. Omphalocele with failure of intestinal loops to return to the body cavity following physiological herniation. Herniated loops are covered by amnion. B. Omphalocele in a newborn. C. Newborn with gastroschisis. Loops of bowel return to the body cavity but herniate again through body wall, usually to the right of the umbilicus in the region of the regressing right umbilical vein. Unlike omphalocele, the defect is not covered by amnion. (Sadler, 2018).

Postnatal changes of umbilical cord:

At birth:

- The cord is tied off close to umbilicus
- It is advisable to leave 5 Cm of the cord between umbilicus and ligature, since a piece of intestine may be present at an umbilical hernia in the remains of extraembryonic celom.

Following ligature: Umbilical vessels constrict and thrombose.

Later: Stump of the cord sloughs off, and the umbilical scar tissue becomes restricted and slowly assumes shape of the umbilicus.
Umbilical arteries: Become medial umbilical ligaments
Left umbilical vein: Becomes ligamentum teres (round ligament) of the liver.

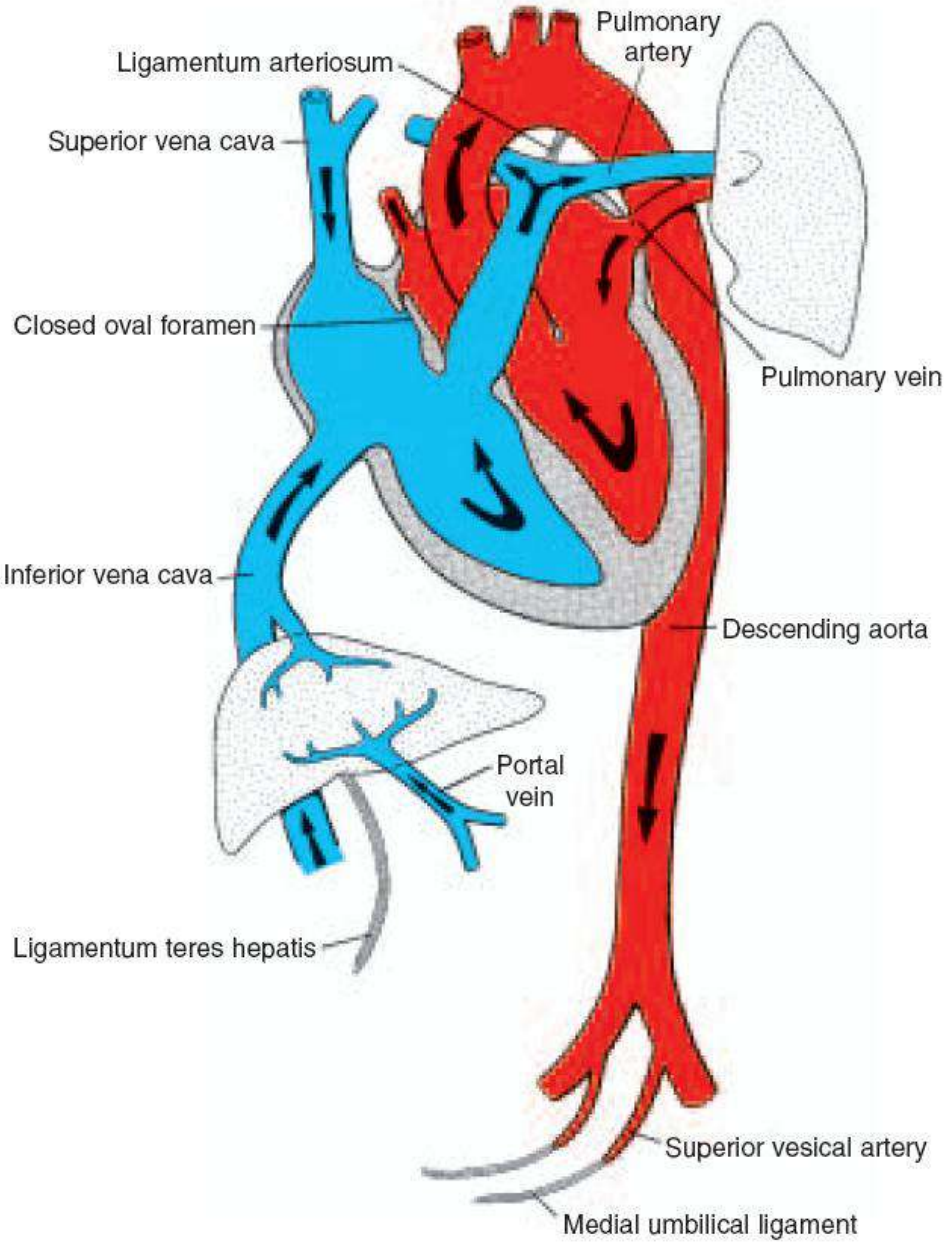


Figure 4: Postnatal changes of the umbilical vessels. The changes occur because of the beginning of respiration and interruption of placental blood flow. Arrows, direction of blood flow. (Sadler, 2018).

Anomalies of Umbilical Cord:

1. **Abnormally long cord:** Reach up to 1.5 meter. The long cord may be complicated by:
   - **a. Encircling neck or limb of the fetus**
   - **b. Forming knots:** which deprives the fetus from its blood supply
   - **c. Prolapsing of the cord:** Before delivery of the child
2. **Abnormally short cord:** 10 Cm or less, may cause difficult labor
3. **Abnormal insertion of the cord in the placenta:**
   - **a. Battledore Placenta:** Peripheral attachment.
   - **b. Placenta Velamentosa:** Attachment to the membranes distance from placenta.
4. **Umbilical Hernia.**

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
3. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.

Lecture 34:

AMNION, AMNIOTIC FLUID

Objectives of the lecture:

By end of the lecture the student should be able to:

- Describe the formation of the umbilical cord.
- Enumerate structure of initially formed umbilical cord.
- Enumerate structure of full-term umbilical cord.
- Enumerate congenital of the umbilical cord.

Amnion:

Definition of Amnion:

- Membranous sac that surrounds the embryo.
- Thin, tough, transparent and non-vascular membrane.
- The membrane forms wall of amniotic cavity.
- Amniotic cavity contains pale watery fluid (liquor amnii).

Origin of Amnion:

- **Amniotic Cavity:** Cavity appears within epiblast.
- Formed from ectodermal cells which enclose the amniotic cavity.
- Cells of roof of the cavity are called amniotic ectoderm or amnioblast.

Development of Amnion:

- First attached to trophoblast.
- Then, separate from trophoblast.
- Increases and expands on expense of extraembryonic celom and yolk sac.
- First attached to margins of embryonic disc, but after folding its attachment becomes limited to umbilical opening.

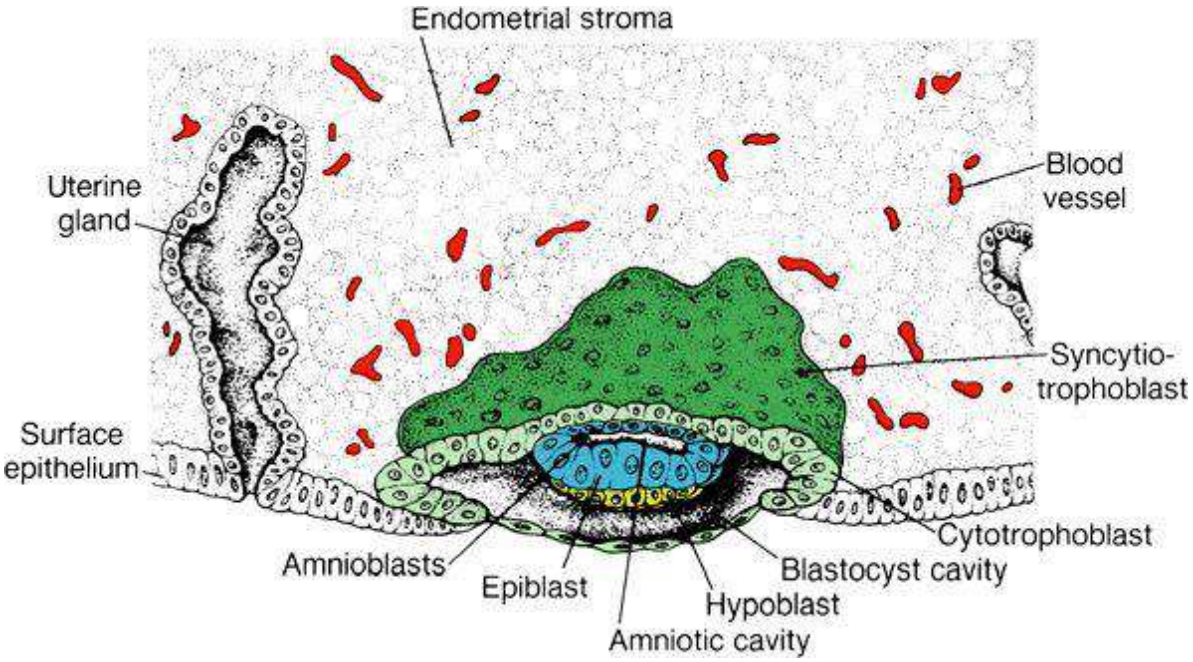


Figure 1: 7.5-day human blastocyst, partially embedded in endometrial stroma. Embryoblast, formed by epiblast and hypoblast layers. Amniotic cavities appear as a small cleft inside embryoblast. (Sadler, 2018).

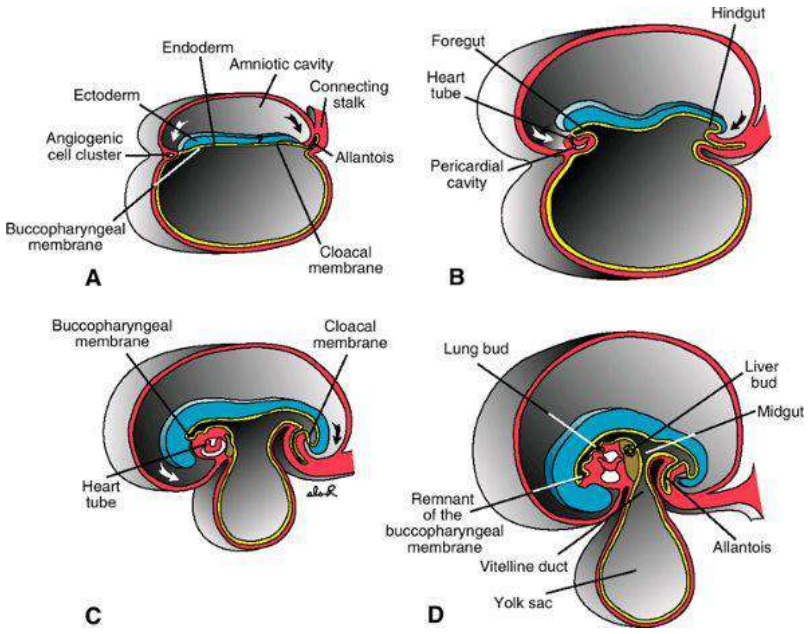


Figure 2: Sagittal midline sections of embryos at various stages of development to demonstrate cephalocaudal folding and its effect on amniotic cavity. A. Presomite embryo. B. Embryo with 7 somites. C. Embryo with 14 somites. D. End of the first month. (Sadler, 2018).

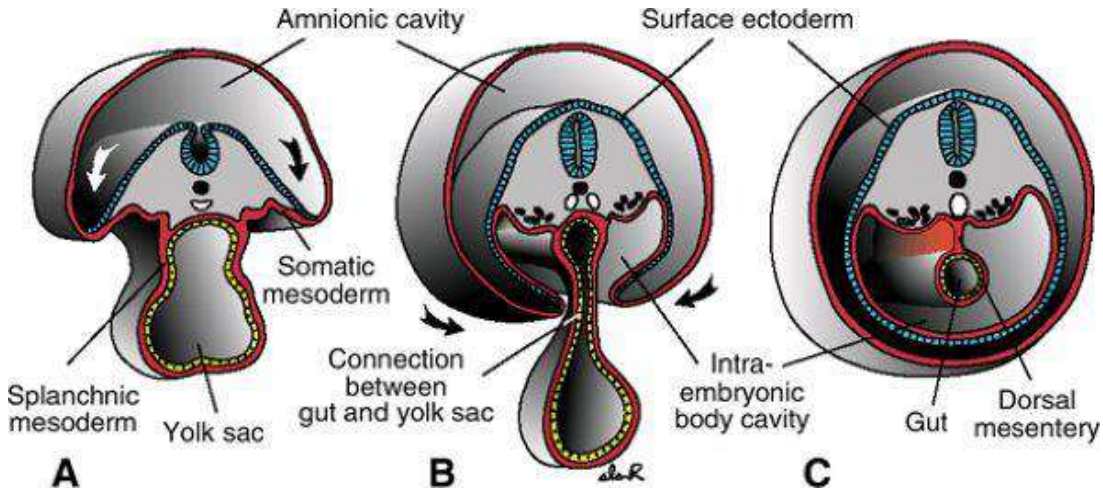


Figure 3: Transverse sections through embryos at various stages of development to show the effect of lateral folding on amniotic cavity. A. Folding is initiated. B. Transverse section through the midgut. C. Section just below the midgut. (Sadler, 2018).

Amniotic fluid:
Amount: One litter by end of pregnancy
Composition: 98% water and remaining 2% are solids; glucose, urea, amino acids and salts.
Source:

1. Amnioblasts
2. Kidneys of embryo
3. Transudation of fluids from placental area

Circulation of Amniotic Fluid:

- Once buccopharyngeal membrane ruptures, liquor enters digestive tract of embryo, absorbed by intestine into blood, passes by way of placenta into mother's blood.
- The liquor circulates and does not stagnate.
- Later in pregnancy the turnover of water in the liquor is at a rate of 1/3 of the volume each hour.

Fetal Functions:

1. Protection of embryos from external trauma and from uterine contractions during labor.
2. Keeps temperature of embryo constant.
3. Allows fetal movements.
4. Allows the fetus to drink fluids.

5. Liquor could be considered as a source of nutrition to fetus.

6. Amniotic fluid prevents adhesions between different parts of the embryo.

II. Maternal functions (during labor):

1. As labor starts pressure of uterus pushes down amnion into cervical canal.
2. With each succeeding uterine contraction (bag of waters) is forced more in cervical canal helping in dilatation of the cervix.
3. As the membranes stretch and descend, they push out the mucous which closes the cervical canal, and this together with slight bleeding is referred to as show.
4. Later, the membranes rupture and liquor escapes, acts as natural antiseptic of birth canal.
5. Escape of water facilitates process of delivery of the child.

Anomalies of Amnion:

1. Hydramnios or Polyhydramnios:

- Excess amniotic fluid (1500–2000 ml)
- If as much as 6 liters, present in amniotic cavity.
- **Causes:** twins, anencephaly, esophageal atresia.
- Maternal diabetes is the most common cause.

2. Oligohydramnios: Decreased amount of amniotic fluid (less than 400 ml).

- **Causes:** Absent renal function in the fetus is most common cause and premature rupture of membranes with prolonged leakage of amniotic fluid.

3. Amniotic adhesions: Adhesions between skin of fetus and amnion.

4. Unruptured membranes: Fetus is born inside amniotic sac.

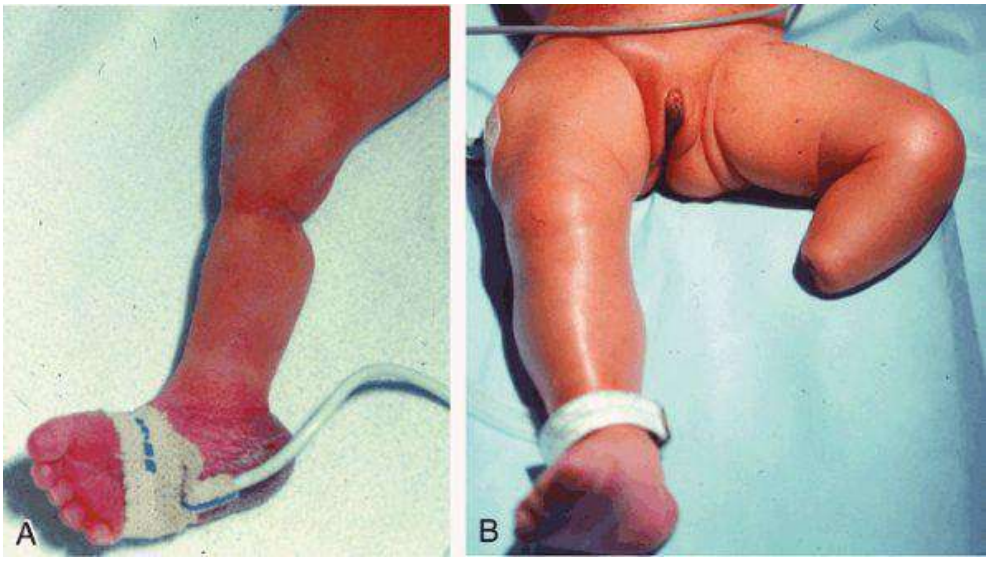


Figure 4: Limb abnormalities caused by amniotic bands. A. Limb constriction ring. B. Limb amputation. (Sadler, 2018).

Amniocentesis:

- Sample of amniotic fluid could be taken, centrifuged and analysed.
- To detect chromosal abnormalities of the fetus; Down syndrome, klinefelter;….

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
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Lecture 35:

MULTILPE PREGNANCY (TWINS)

Objectives of the lecture:

By end of the lecture the student should be able to:

- Define twins.
- What are the types of twins?
- Define dizygotic twins.
- Describe development of dizygotic twins.
- Define monozygotic twins.
- What are the characters of monozygotic twins.
- What are the relations of foetal membranes in monozygotic twins?
- Define joined twins and mention some varieties.

Multilpe Pregnancy (Twins):

Definition of Twins:

- Pregnancy in more than one embryo at the same time

- Common number is two embryos

Twining: Nearly simultaneous birth of two babies

Types of Twins

1. Dizygotic (Binovular) Twins:

- Constitutes about three quarters of multiple pregnancies.
- Two ova are ovulated nearly at the same time.
- Each will be fertilized by a separate sperm.
- The two ova may come from the same Graafian follicles or from different follicles.
- The two fertilized ova implanted separately, and each will have its own placenta and fetal membranes.
- The two placentas may fuse together, but each will have a separate circulation.
- The two babies have different genetic constitutions and may be of the same or opposite sex.
- Ovulation of more than one ovum at the same time is a hereditary character.

Incidence of dizygotic twins: Differs from one population to another, in Japan 1.3 per 1000 while in Nigeria 40 per 1000.

Note: FSH is used for treatment of unovular infertility. This usually results from ovulation of more than one ovum leading to multiple pregnancy.

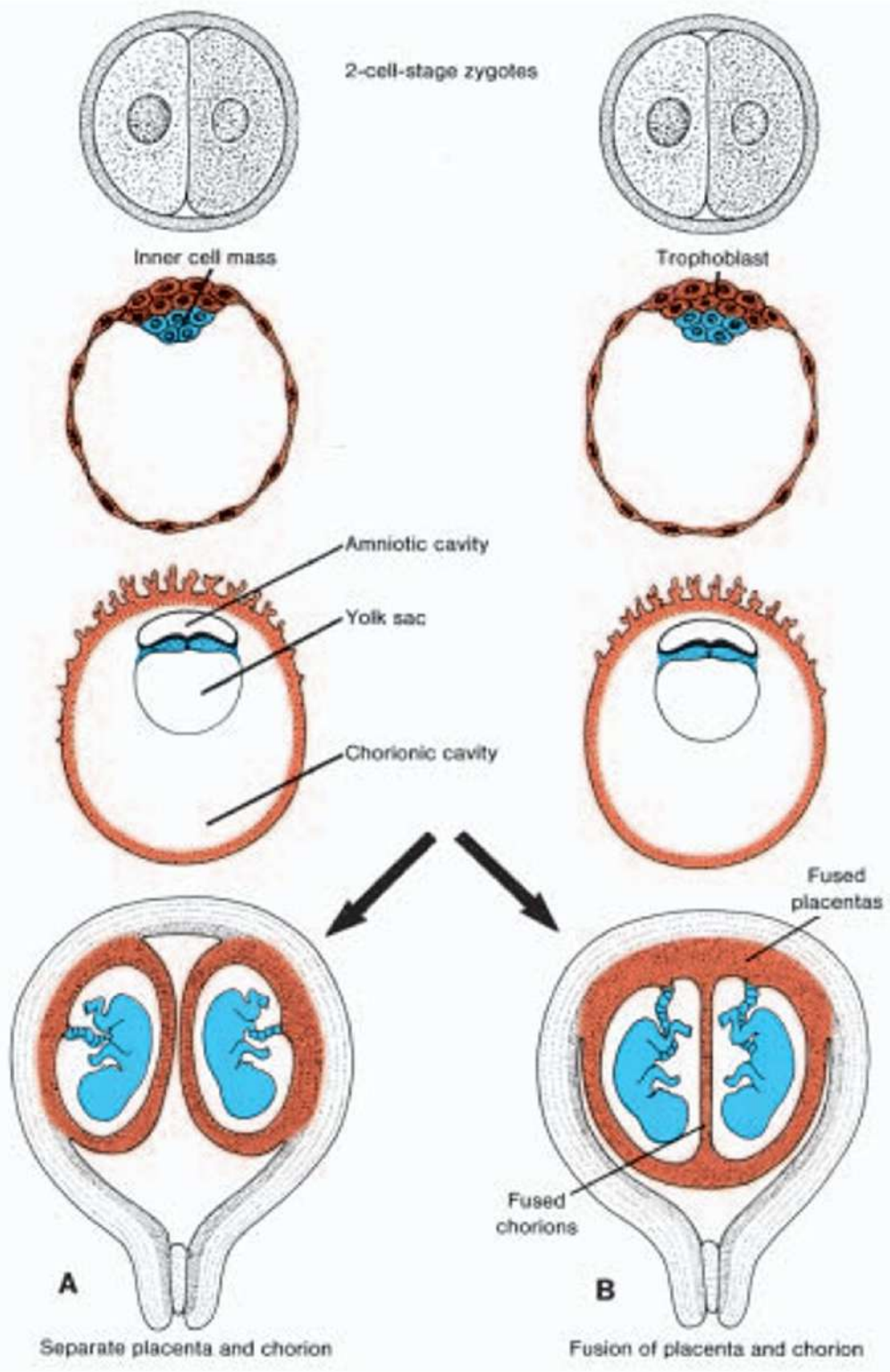


Figure 1: Development of dizygotic twins. A. Each embryo has its own amnion, chorion, and placenta. B. The placentas are fused. Each embryo receives appropriate amount of blood, but large anastomoses shunt more blood to one of the partners than to the other. (Sadler, 2018).

Monozygotic Twins:

- Occurs when one fertilized ovum splits into two or more embryos.

Two conditions could be recognized:

1. Splitting at two cell stage:

- Two embryos will have completely separated chorionic sac and placenta

2. If splitting occurs later:

- Two embryos may share something in common.
- If splitting occurs in stage of blastocyst with division of inner cell mass into two or more masses, the two resulting embryos will share same chorionic sac or amnion.

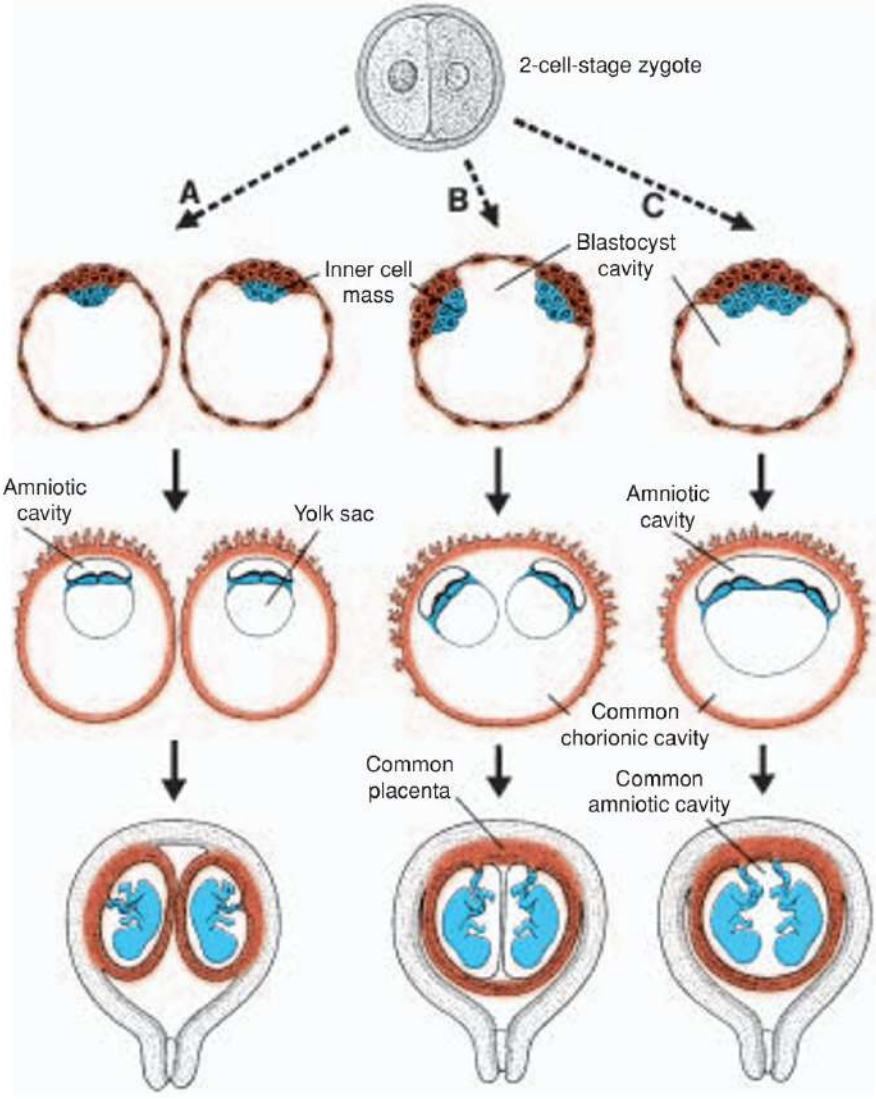


Figure 2: Relations of fetal membranes in monozygotic twins. A. Splitting occurs at two-cell stage, and each embryo has its own placenta, amniotic cavity, and chorionic cavity. B. Splitting of inner cell mass into 2 separated groups. Two embryos have a common placenta and chorionic sac but separate amniotic cavities. C. Splitting of inner cell mass at a late stage of development. Embryos have common placenta, amniotic cavity, and chorionic cavity. (Sadler, 2018).

Note:

- In rare conditions the splitting of the inner cell mass may be incomplete.
- This may result in incomplete splitting of primitive streak, resulting in joint twins which share some tissues.
- Shared tissue may be just the skin.

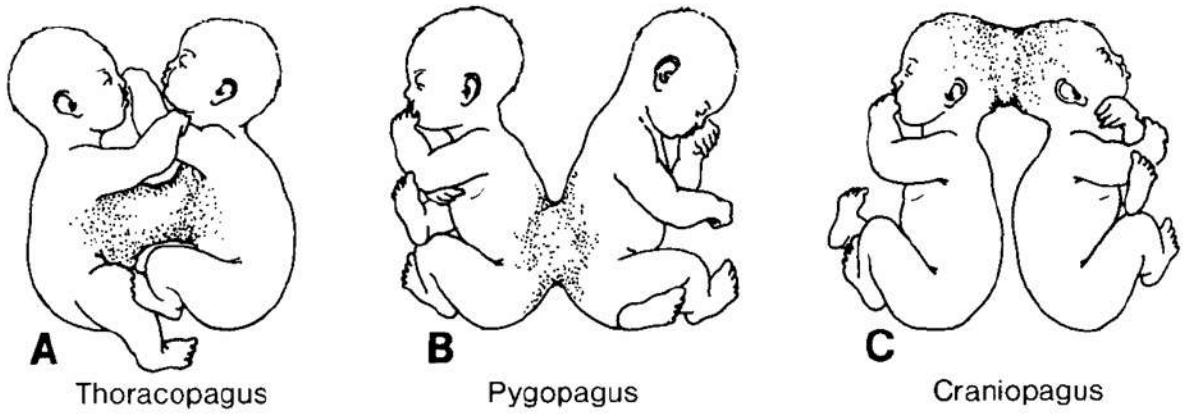


Figure 3: Thoracopagus, pygopagus, and craniopagus twins. Conjoined twins can be separated only if they have no vital parts in common. (Sadler, 2018).

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
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Lecture 36:

BIRTH DEFECTS AND PRENATAL DIAGNOSIS (CONGENITAL MALFORMATIONS)

Objectives of the lecture:

By end of the lecture the student should be able to:

- Define birth defects.
- Define teratology.
- Enumerate teratogens associated with human malformations.
- Enumerate congenital anomalies due to infectious agents.
- Enumerate congenital anomalies due to physical agents.
- Enumerate congenital anomalies due to chemical agents?
- Define syndrome.

Birth Defects (Congenital Malformations):

Definition:

- Malformation, birth defect and congenital anomaly are synonymous terms and used to describe structural, behavioral, functional and metabolic disorders present at birth.

Teratology: Science that studies congenital malformations.

Teratogens Associated with Human Malformations:

1. **Infectious Agents:** Rubella virus, Cytomegalovirus, Herpes simplex virus, Varicella virus, HIV, Toxoplasmosis, Syphilis.
2. **Physical Agents:** X-rays, Hyperthermia.
3. **Chemical Agents:** Thalidomide, Aminopterin, Diphenylhydantoin (phenytoin), Valproic acid, Trimethadione, Lithium, Amphetamines, Warfarin, ACE inhibitors, Cocaine, Alcohol, Isotretinoin (vitamin A), Industrial solvents, Organic mercury, Lead.
4. **Hormones:** Androgenic agents (ethisterone, norethisterone), Diethylstilbestrol (DES).
5. **Maternal Diabetes.**

Incidence of congenital malformations:

- 2-3% for major malformation - 15% for minor malformation

Importance of congenital malformations: It causes 21% of infant death.

Teratogen	Congenital Malformations
Infectious agents	
Rubella virus	Cataracts, glaucoma, heart defects, deafness, teeth
Cytomegalovirus	Microcephaly, blindness, mental retardation, fetal death
Herpes simplex virus	Microphthalmia, microcephaly, retinal dysplasia
Varicella virus	Limb hypoplasia, mental retardation, muscle atrophy
HIV	Microcephaly, growth retardation
Toxoplasmosis	Hydrocephalus, cerebral calcifications, microphthalmia
Syphilis	Mental retardation, deafness
Physical agents	
X-rays	Microcephaly, spina bifida, cleft palate, limb defects
Hyperthermia	Anencephaly, spina bifida, mental retardation, facial defects
Chemical agents	
Thalidomide	Limb defects, heart malformations
Aminopterin	Anencephaly, hydrocephaly, cleft lip and palate
Diphenylhydantoin (phenytoin)	Fetal hydantoin syndrome: facial defects, mental retardation
Valproic acid	Neural tube defects, heart, craniofacial, and limb anomalies
Trimethadione	Cleft palate, heart defects, urogenital and skeletal abnormalities
Lithium	Heart malformations
Amphetamines	Cleft lip and palate, heart defects
Warfarin	Chondrodysplasia, microcephaly
ACE inhibitors	Growth retardation, fetal death
Cocaine	Growth retardation, microcephaly, behavioral abnormalities, gastroschisis
Alcohol	Fetal alcohol syndrome, short palpebral fissures, maxillary hypoplasia, heart defects, mental retardation
Isotretinoin (vitamin A)	Vitamin A embryopathy: small, abnormally shaped ears, mandibular hypoplasia, cleft palate, heart defects
Industrial solvents	Low birth weight, craniofacial and neural tube defects
Organic mercury	Neurological symptoms similar to those of cerebral palsy
Lead	Growth retardation, neurological disorders
Hormones	
Androgenic agents (ethisterone, norethisterone)	Masculinization of female genitalia: fused labia, clitoral hypertrophy
Diethylstilbestrol (DES)	Malformation of the uterus, uterine tubes, and upper vagina; vaginal cancer; malformed testes
Maternal diabetes	Variety of malformations; heart and neural tube defects most common

Table: Teratogens associated with human malformations. (Sadler, 2018).



Figure 1: Child with microcephaly. Poor growth of the brain. Frequently associated with mental retardation. (Sadler, 2018).



Figure 2: Child with hydrocephalus. Enlarged head, thinning bones of skull and cerebral cortex. (Sadler, 2018).

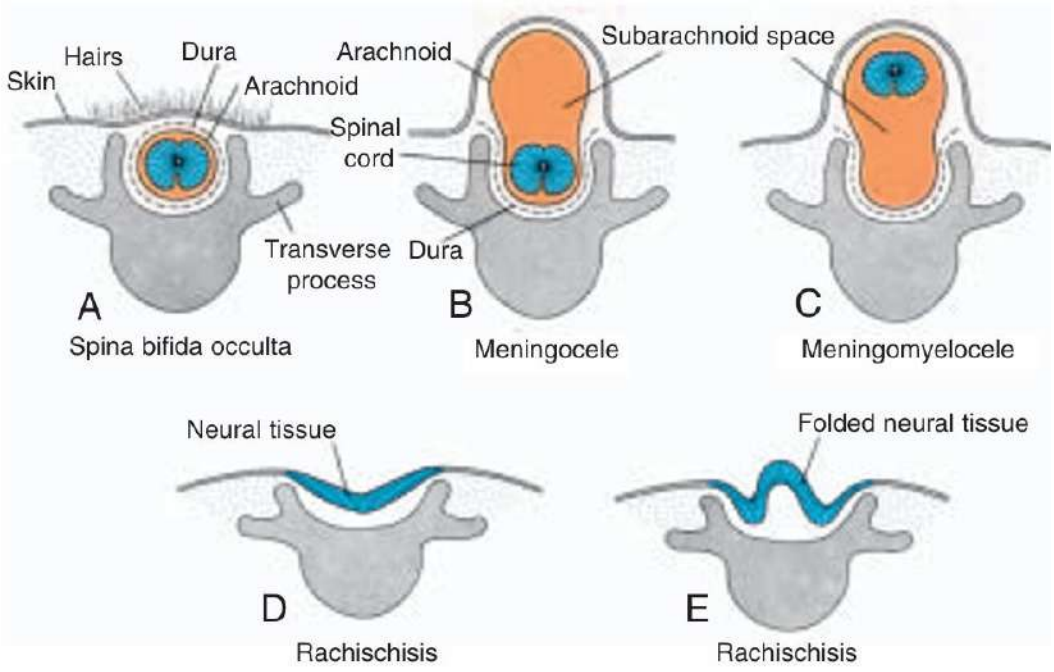


Figure 3: Various types of spina bifida.

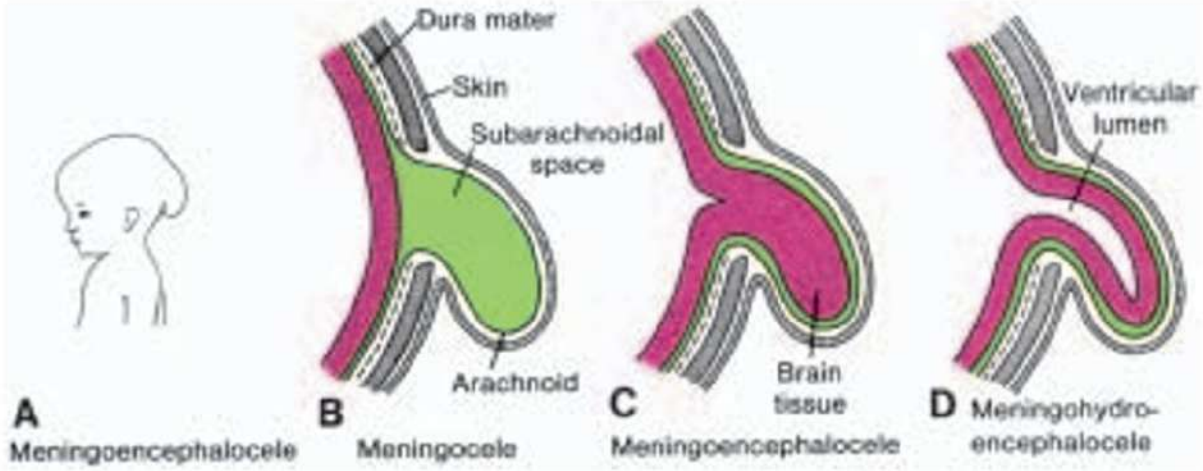


Figure 4: Various types of brain herniation due to abnormal ossification of the skull

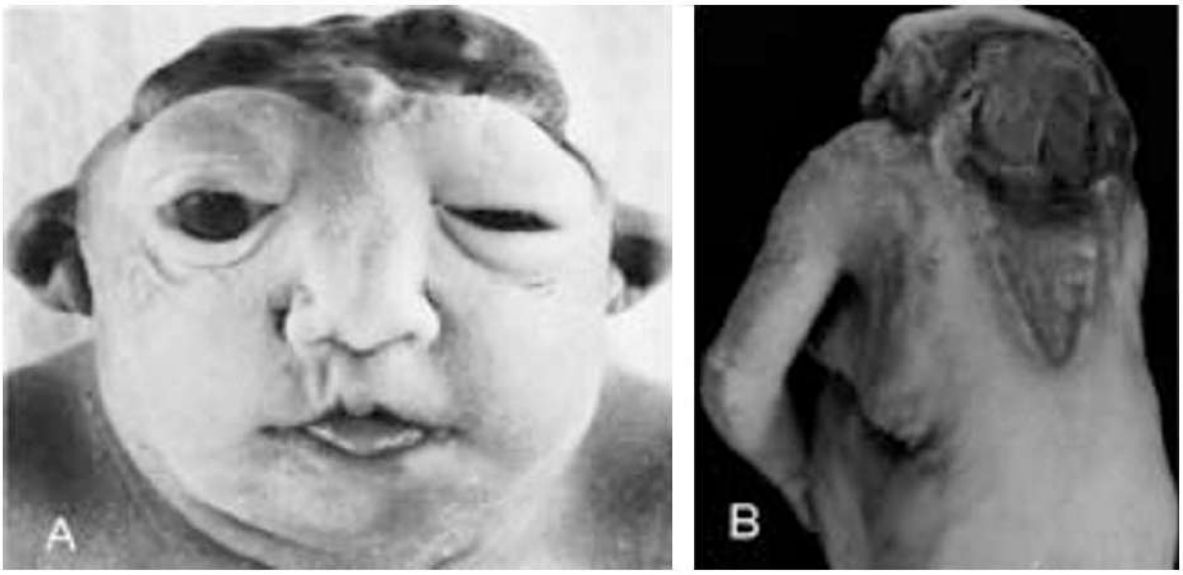


Figure 5: A. Anencephalic child, ventral view. This abnormality occurs frequently The child dies a few days after birth. B. Anencephalic child with spina bifida in the cervical and thoracic segments, dorsal view. (Sadler, 2018).

SYNDROME

Definition of Syndrome:

- A group of anomalies occurring together that have a specific common cause, so the risk is known and diagnosis is made.

Examples of Syndromes:

1. Down syndrome
2. Klinefelter's syndrome
3. Turner syndrome

ASSOCIATION

Definition of association:

- Nonrandomized appearance of two or more anomalies that occur together, more frequently than by the chance, but the cause has not been determined.

Why are they important?

- Because if you found one you have to search for the others.

Examples of Associations:

1. CHARGE Association:

C. Coloboma
H. Heart
A. Artesia
R. Retard growth
G. Genital
E. Ear anomalies

2. VACTERL Association:

V: Vertebral

A: Anal,

C: Cardiac

T: Tracheo-esophageal

R: Renal

L: Limb

Prevention of Malformations

1. Increase level of awareness: of risk to the mothers and physicians.

2. Early diagnosis: Using a variety of techniques.

a. Ultrasound: To determine fetal growth and any malformation.

b. Maternal serum screening: Good example is alpha fetoprotein for early diagnosis for neural tube defect.

c. Amniocentesis: Taken sample from amniotic fluid using needle, using it for cell culture and gene analysis.

d. Chorionic villous sample: Tissue sample is taken from placenta for gene analysis.

Amniocentesis and chorionic villous sample:

- Carry some risk and used only when indicated.
- Age above 35, history of neural tube defect in family or maternal history of chromosome anomalies x-liked disorders).

Treatment of congenital malformations:

1. **Blood transfusion:** As in cases of haemolytic blood disorders.
2. **Medication:** As in case foetal arrhythmias, either through Mather or directly in the foetus veins or muscles.
3. **Fetal surgery:** As in placing shunts into the urinary balder in case of obstructive urinary disease.
4. **Ex-utero surgery:** Uterus opened, and fetus operated directly as in case of repairing congenital diaphragmatic hernia or removing cysts from lung.
5. **Stem cells transplant:** Before 18 weeks no risk of rejection, as in case hematopoietic stem cells.
6. **Gene therapy:** As in case of inherited metabolic disorders cystic fibrosis.

References:

1. KAPLAN Medical USMLE Step 1 Lecture Notes Anatomy (2021). Pages 1-12.
2. Sadler, T. W.: Langman's Medical Embryology, 14th Edition by T.W. Sadler PhD (2018). Pages 1-168.
3. Snell, R. S. (2019): Clinical Anatomy by Regions (Anatomy) 9th Edition. Lippincott Williams & Wilkins.

Principles of Microscopic and Macroscopic Structures

PMS-102

histology and cell biology

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Level: 1

Academic year 2026-202

Lecture 1:

Micro techniques and Microscopy

Learning Objectives

After this lecture, students should be able to:

- Define histology.
- Identify the methods of studying cells and tissues.
- Recognize the light and electron microscopes.
- Know the cell and tissue culture.

Introduction to Histology

Histology is the study of the microscopic structure of cells, tissues and organs of the body, with the focus on how cells' structure and arrangement optimize functions specific to each organ.

- Tissues have two interacting components: cells and extracellular matrix (ECM).
- Cytology is the science which deals with the structure, function and molecular construction of the cell with the use of microscopes and various techniques.

Why do we study Histology?

- Study microscopic Structure of cells & Organs
- Correlate Structure to Function
- Understand the pathology of diseases

Microtechnique

Microtechnique is the art of preparing objects for examination under the microscope (light and electron microscopes) and of preserving objects so prepared.

Preparation of histological sections

A) For light microscope (LM):

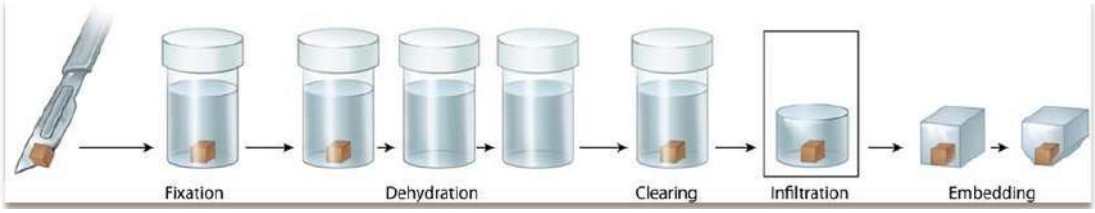
1. Paraffin technique.
2. Celloidin method.

3. Freezing technique.

B) For electron microscope (EM)

1) Paraffin technique (routine histological section)

1. Selection and obtaining tissue.
2. Fixation.
3. Washing.
4. Dehydration.
5. Clearing.
6. Infiltration and Embedding.
7. Microtomy and Sectioning.
8. Mounting.
9. Deparaffinization.
10. Hydration.
11. Staining.
12. Dehydration.
13. Clearing.
14. Mounting.
15. Cleaning and Labeling.



1. Selection and obtaining tissue

- Samples must be obtained from living tissue (biopsy) or very soon and as rapidly as possible after death (autopsy) using very sharp razors to prevent distortion of tissue during cutting.
- Samples should be small size to allow the penetration of fluid through it up to the most interior part.

2. Fixation

- The tissue is put immediately in a suitable fixative.
- The most commonly used fixative is formalin (buffered 10%).
- Purpose of fixation:
    1. Prevent autolysis as it destroys lysosomal enzymes.
    2. Prevent putrefaction by killing bacteria.
    3. Coagulate the proteins and harden the tissue.
    4. Some fixatives have **mordating** effect (increase the affinity of tissue for stains).

3. Washing

Washing out excess fixative after the tissue is fixed for the proper length of time (to removes substances in the fixative which might interfere with the following steps).

4. Dehydration

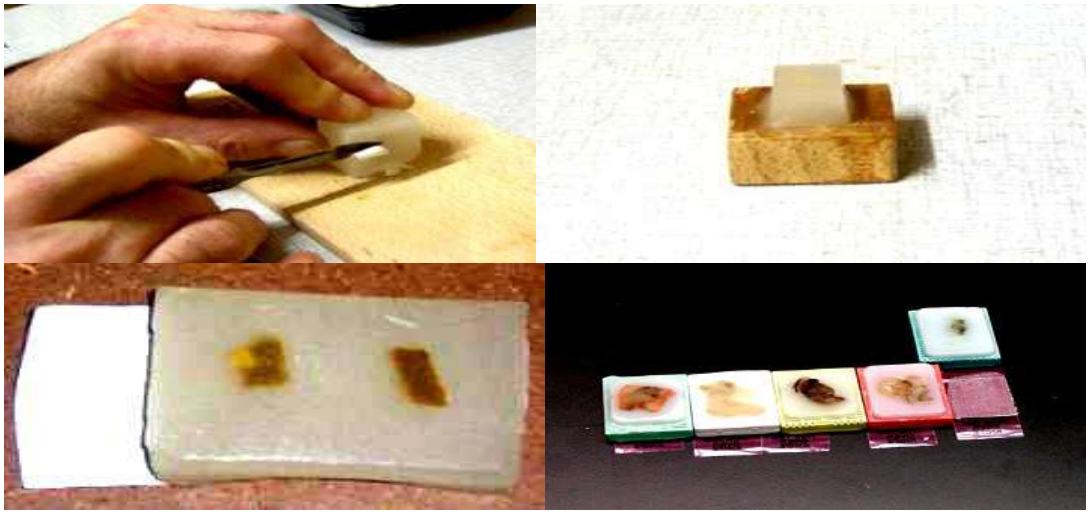
- Removal of water is a necessity to embed the tissue in a wax or resin material to be ready for sectioning.
- Dehydration is usually done by ascending grades of ethyl alcohol (50, 70, 90, 100%) to prevent sudden withdrawal of water (shrinkage) and damage of the tissue.

5. Clearing

Paraffin is not soluble in alcohol. So, alcohol is replaced by a substance in which paraffin is soluble, such as xylene, toluene, or benzene.

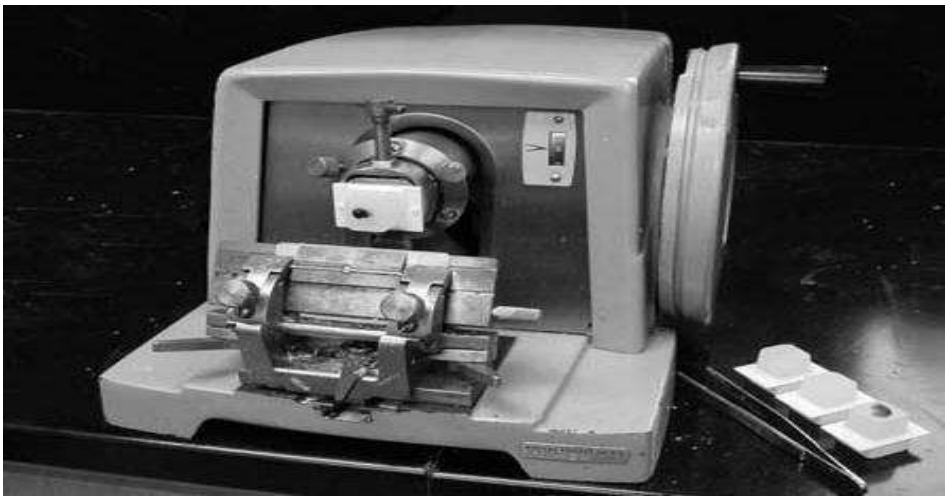
6. Infiltration and Embedding

- Wax must be surrounding (embedding) and passing into the inside of the tissue (infiltration).
- Infiltration is done first soft paraffin.
- Soft paraffin, thus, replaces the clearing agent and hard paraffin is used in the following step to conform the block around the tissue.



7. Microtomy and Sectioning

- Sectioning is done using the microtome.
- Microtomes in common use are usually the Rotatory microtome.
- Sliding microtome usually used in sectioning of dense tissue (bone and skin).
- Section thickness usually suitable for routine histological study is 4-7 um.





8. Mounting

Sections are mounted onto a glass slide and left to dry.

9. Deparaffinization

- Stains are usually in the form of aqueous solutions.
- Sections on the slide is surrounded and infiltrated with paraffin which immiscible with water.
- Paraffin is removed by xylol.

10. Hydration

- Hydration of the tissue is a necessity to enable the stain reach and react with the tissue components.
- Rehydration of the tissue is achieved by descending grades of alcohol (100, 90, 70, 50%), distilled water is then used to replace the alcohol.

11. Staining

- The nucleus is stained blue first by haemotoxylin then, the use of acidic dyes (e.g. eosin) to stain the cytoplasm red.
- Washing with water is done between the two steps.

12. Dehydration

by ascending grades of ethyl alcohol (50, 70, 90, 100%).

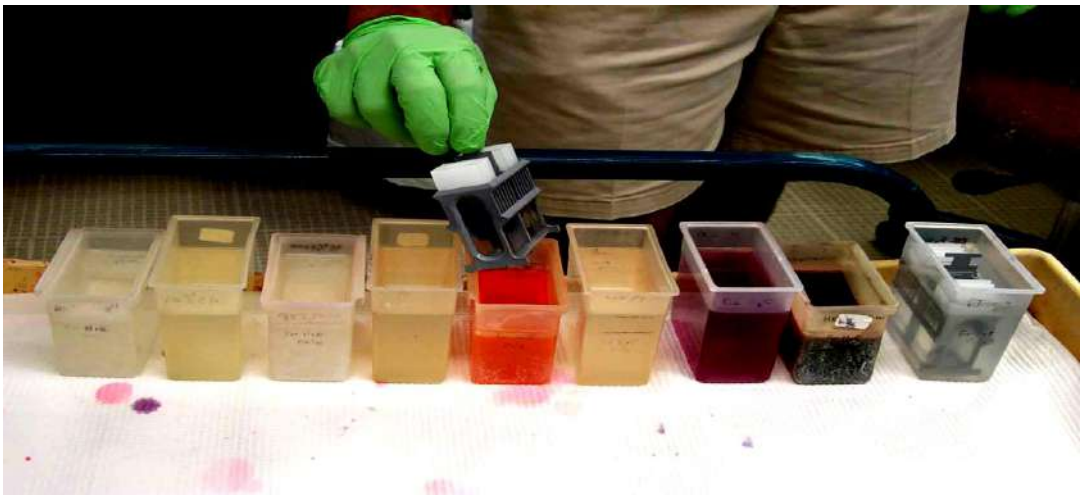
13. Clearing

by xylene, it makes the stained tissue sections transparent.

14. Mounting

the section is mounted (covered) with a very thin cover slip using a sticky mount (Canda balsam or De Pe Xe).

15. Cleaning and Labeling







Advantages of paraffin technique

- It takes a short time.
- It gives very thin serial sections (ribbon) that can be easily stained.

Disadvantages of paraffin technique

- The fixatives and heat used may damage the tissues.
- The fat contents are dissolved during preparation.
- It can not used for large pieces of tissues with large cavities.

2) Celloidin method

- The tissue is embedded in celloidin block.
- The sections are cut with a sliding microtome.

- **Advantages of Celloidin method:**

1. It preserves the tissues (because no heating is used).
2. It can be used for cutting large organs with plenty of folds and lumina.

Disadvantages of Celloidin method:

- Time-consuming.
- No serial sections.

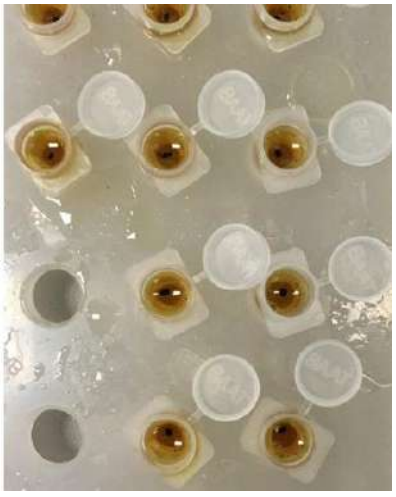
3) Frozen sections

- It is a much more rapid processing method for L.M. The tissue is rapidly frozen in liquid nitrogen (-195°C), preserving cell structures and hardening the tissue.
- Frozen sections are effective in histochemical studies of very sensitive enzymes or lipids because freezing preserves cell enzymes and lipids.

4) Plastic sections

It is used to prepare very thin plastic- embedded tissue sections for electron microscopic examination.





Staining

Most cells and ECM components are colorless and to be studied microscopically, tissue sections must be stained.

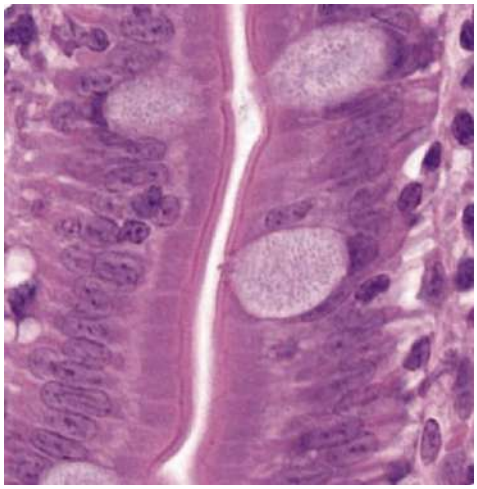
Staining methods make various tissue components from one another when examined by the microscope.

The types of staining methods:

1. General histological methods:

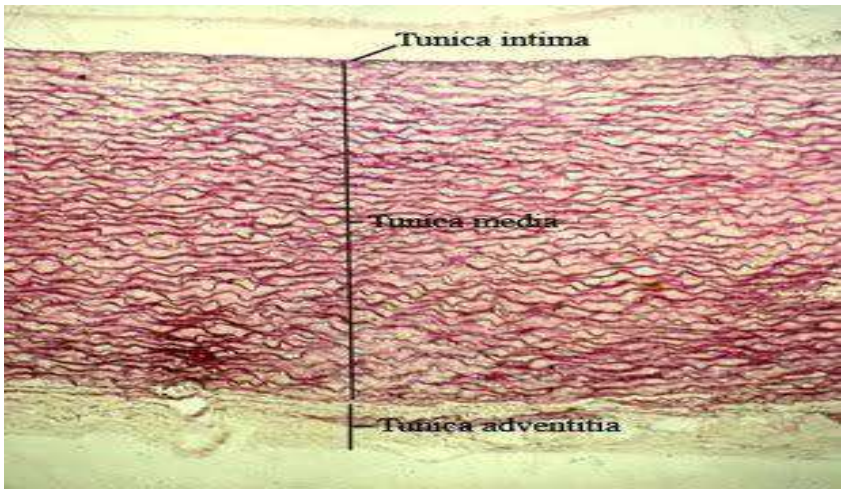
one dye may stain various structures; e. g. haematoxylin stains the nucleus and other structures in the cytoplasm.

- Hematoxylin is a basic dye which stains acidic cell components such as nucleic acids (DNA & RNA). The cell components which have an affinity for basic dyes are termed basophilic.
- Eosin is an acidic dye which stains basic cell components such as most cytoplasm proteins (e.g. mitochondria). The cell components which have an affinity for acidic dyes are termed acidophilic.



2. Specific (special) histological methods:

one dye stains only one structure or substance e.g., orcein stains only elastic fibers.



3. Vital staining methods:

are those which enter cells either by diffusion or by phagocytosis and stain cellular components without harming the cell. This makes it possible to study living cells.

Examples:

- **Janus Green B stain:** Used for superviatal staining of mitochondria in living cells.
- **Vital staining of Golgi complex:** The Golgi complex can be demonstrated under the phase contrast microscope or after vital staining.

4. Histochemical and cytochemical methods:

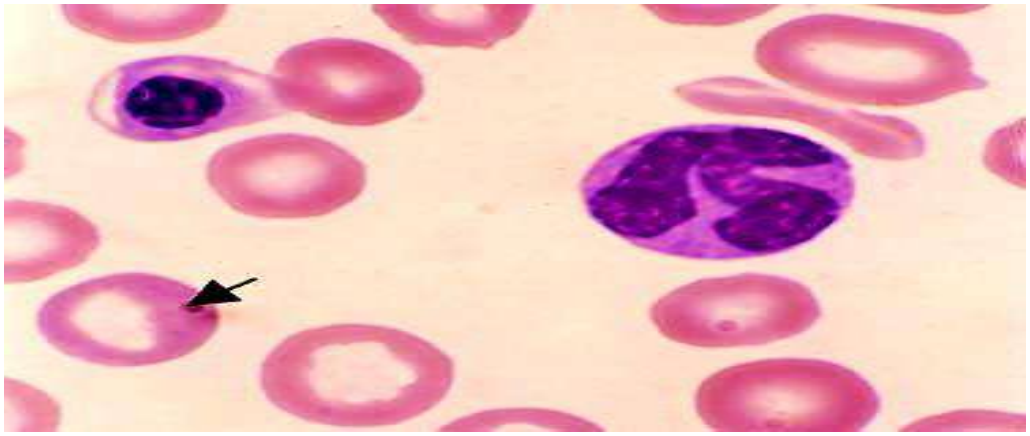
a definite substance such as fat or enzyme is identified and localized in cells or tissues by using either physical or chemical procedures on histological sections (usually frozen sections)

- a) Chemical methods e.g. identification of glycogen by using the periodic acid schiff (P.A.S)
- b) Physical methods due to certain physical properties of these substances e.g., fluorescent substances (vitamin A) can be seen directly.

5. Others:

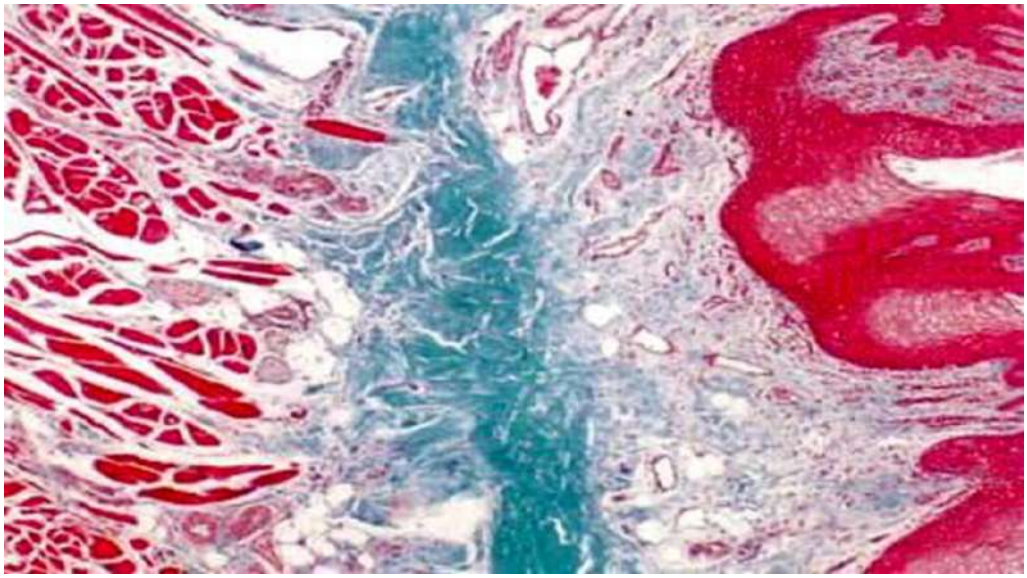
Neutral stains

It is a mixture of acidic and basic stains. It used to stain the blood elements in a blood film, e.g. Leishman's stain.



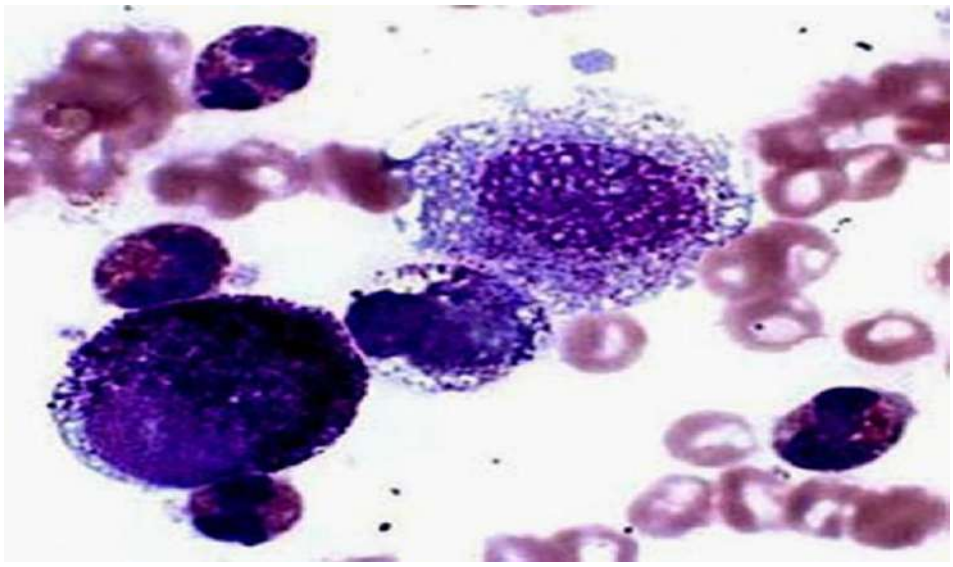
Trichrome stains

Three stains are used in combinations to give 3 colors to different tissue components. In addition to staining the nuclei and cytoplasm can differentiate collagen from smooth muscle as Masson's stain: it stains collagen green, nuclei blue and cytoplasm red.



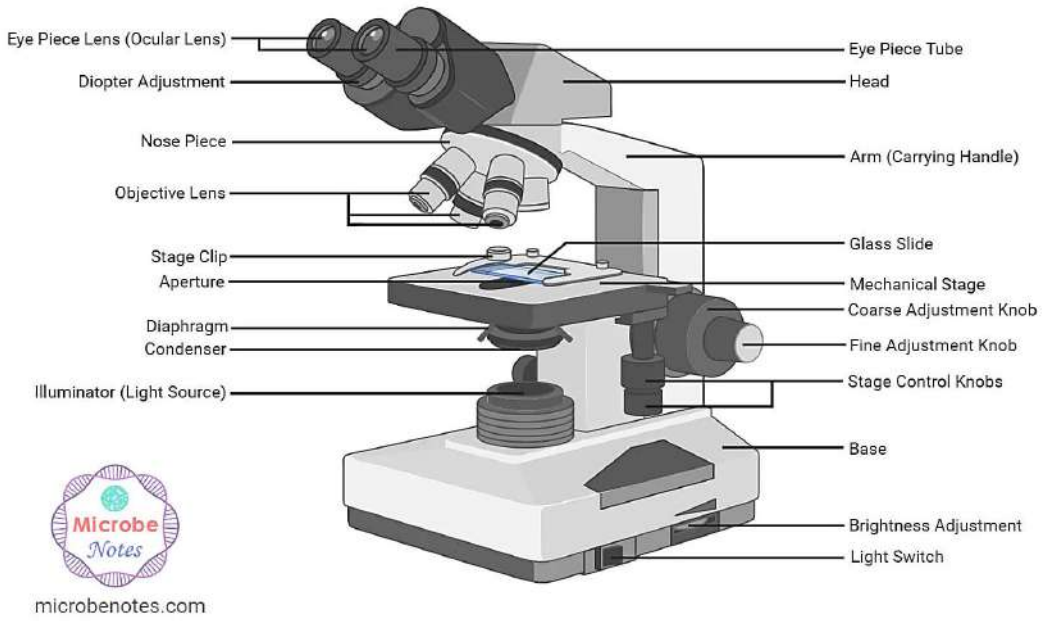
Metachromatic staining

It is the staining of certain cell components with a color which is different from that of the dye used. The phenomenon of altering the color is called metachromasia. It is due to the interaction between the dye and the cell component, producing a different compound with a different color. e.g. the staining of mucopolysaccharide granules of mast cells by toluidine blue which changes to the red or violet colors.



Microscopes

The microscope is an instrument having magnifying lenses for inspecting objects too small to be seen by the human eyes. Any microscope serves two main functions, namely resolution and magnification.



TYPES OF MICROSCOPES

A) Depend on ordinary visible light

1. Conventional light microscope.

Polarizing microscope.

2. Phase contrast microscope.
3. Dark-field microscope.

B) Depend on other than visible light

1. Fluorescent microscope.
2. X-ray microscope.
3. Laser microscopes.
4. Electron microscope.

Conventional light microscope

It performs its function through 2 systems:

The illumination system

Light, from the illumination source (electrical lamp) is directed and focused onto the specimen (mounted on a glass slide) through the aperture in the stage. This light then passes through the specimen and into the objective lens.

The optical system

- Condenser: collects and focuses light, producing a cone of light that illuminates the object to be observed.
- Objective lenses: enlarge and project the illuminated image of the object in the direction of the eyepiece.
- Eyepiece lens: further magnifies this image.

NB: **The total magnification** is obtained by multiplying the magnifying power of the objective and eyepiece lenses.

Resolution

- It is the power of the microscope to distinguish fine details.
- The limit of resolution (or the resolving power)
- It is defined as the smallest diameter of an object below which it cannot be seen at all with the microscope.

Resolution is defined as the ability to distinguish two very small and closely-spaced objects as separate entities.
Resolution is best when the distance separating the two tiny objects is small.

The limit of resolution of:

- human eye is 200 um=0.2 mm
- ordinary light microscope is 0.24 um=200 nm .
- electron microscopes is 0.24 nm.

Magnification

It is the increase in the linear dimensions of an object without resulting in any optical defects.

- Ordinary L.M. is based on the passage of light through stained tissue sections.
- The maximal resolving power of L.M. is approximately 0.2 µm which allows clear images magnified 1000 times.

Electron microscopes (E.M.)

are based on the passage of beams of electrons through the tissue to study fine tissue structures.
Transmission electron microscope (a type of E.M.) has resolving power as little as 1 nm which allows clear images magnified as much as 100,000 times.

Differences between light microscope (LM) and electron microscope (EM).

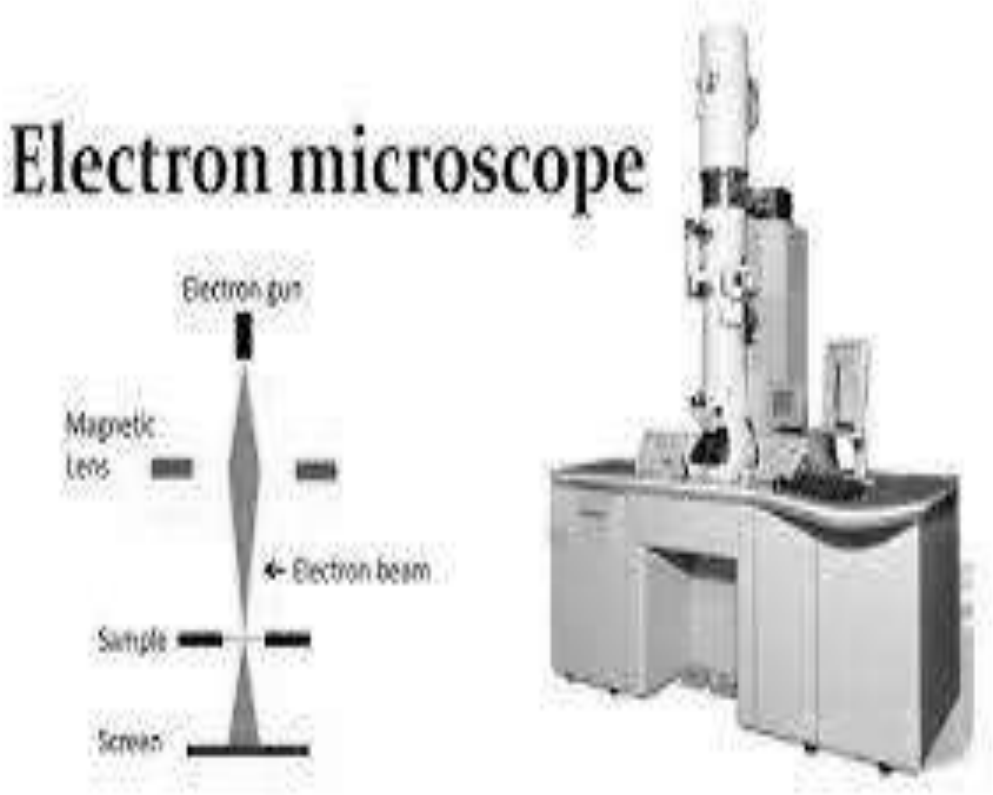
Feature	Light microscope (LM)	Electron microscope (EM)
Body tube	non-vacuumed tube	vacuumed tube
Illumination system	visible beam of light (day or electric)	invisible beam of electrons
Optical system	glass lenses	electro-magnetic coils
Magnification	up to 1500 times	up to 400.000 times or more
Resolution	around 0.2 µm	around 0.2 nm
Examination	• by human eyes • to see many cells with colored photo-micro-graphs (according to stains)	• by fluorescent screen • to see 1 or 2 cells with non-colored photo-micro-graphs (only black and white)

Feature	Light microscope (LM)	Electron microscope (EM)
Examples	- Ordinary light microscope - Phase contrast microscope	- Transmission EM → 2 dimensions

Types :

1. Ordinary transmission electron microscope (T E M)

- High voltage E. M. (HVEM) is a subtype of transmission electron microscope with the following advantages:
  a) It is useful in examining thick sections (5 microns). In TEM much thinner sections 50 nanometer (nm) must be used.
  b) Studying living cells and objects.



3. **Scanning electron microscope (SEM):**
   it shows a three dimensional image. It is useful in studying details of surfaces (as cilia).

Cell Culture

Definition: is the maintenance and study of live cells and tissues outside the body in culture (in vitro).

Uses:

1. Cell culture allows the direct observation of living cells.
2. Experiments which are technically impossible to perform in the body (invivo) can be accomplished in culture (in vitro).
3. In research on cell structure and function.
4. For genetic or chromosomal analyses.
5. To study molecular changes that occur in cancer.

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4- Wheater, P. R., Burkitt, H. G., & Daniels, V. G. (2006). *Wheater's functional histology: A text and colour atlas* (5th ed.). Churchill Livingstone/Elsevier.

Lecture 2 :

membranous Organoids

Learning Objectives

By the end of these lectures the student should be able to:

- Describe general structure of the cell.
- Enumerate members of membranous organoids.
- Demonstrate sites and structures (L/M & EM) of each organoid.
- Discuss histophysiology (functions) of each member of the membranous organoids.

Cell components

Cells are the structural and functional units of all living organisms.

The cell is composed of protoplasm and is surrounded by cell membrane. Protoplasm is composed of:

1- Cytoplasm.

2- Nucleus.

CYTOPLASM

It is a specific kind of protoplasm consisting of matrix.

MATRIX (CYTOSOL, CELL SAP)

It is colorless watery fluid or semi fluid substance in between organelles and inclusions.

- **LM:** it appears as homogenous or finely granular ground substance, commonly acidophilic in reaction, but may be basophilic or neutrophilic.
- **EM:** appears as amorphous substance with a variety of electron densities.
- **Function:** Provides a suitable medium for reactions occurring in the cytoplasm.

CYTOPLASMIC ORGANELLES

Organelles are metabolically active structures performing distinctive functions.

Divided into two parts:

Membranous organelles include:

1- Cell membrane

2- Endoplasmic reticulum (rough& smooth).

3- Golgi apparatus.

4- Lysosomes.

5- Peroxisomes.

6- Mitochondria.

7- Annulate lamellae

8- Secretory vesicles

9- Coated vesicles.

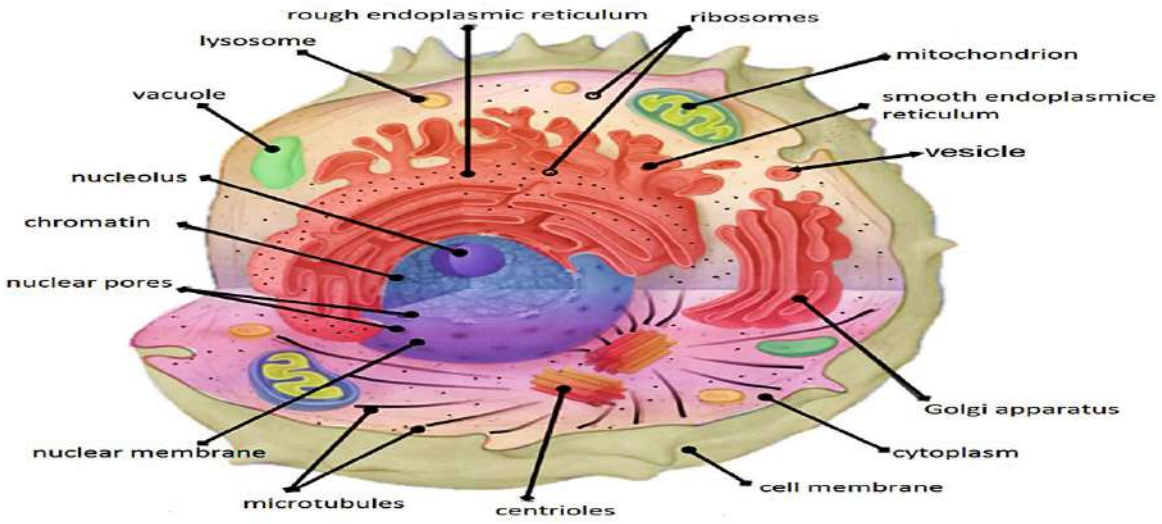
Non-membranous organelles include:

1- Ribosomes.

2- Microtubules.

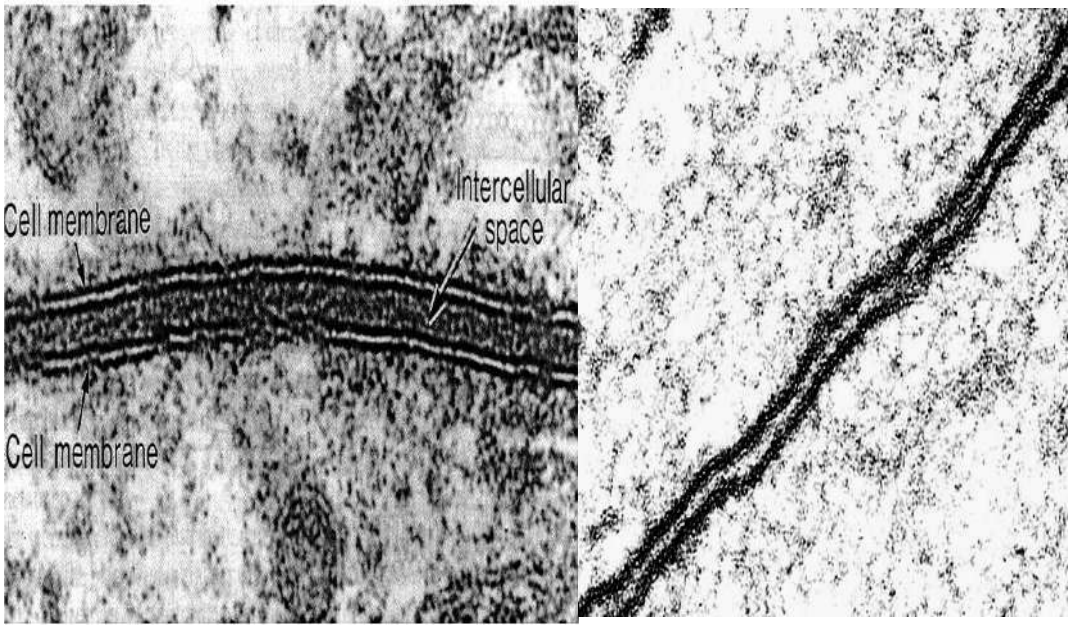
3- Centrioles.

4- Cell filaments.



Plasma Membrane

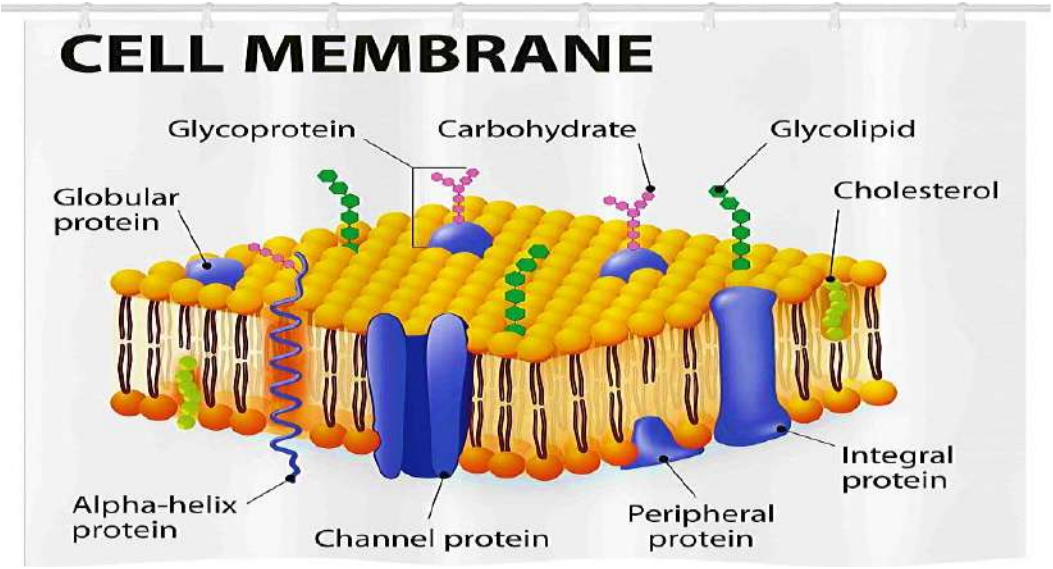
- The plasma membrane (also called plasmalemma) envelops cell & defines its outer limit.
- **L/M:** appears as a very thin limiting border line.
- **E/M:** appears as a 3- layered membrane (triallellar or unit membrane).



- usually about 7.5-10 nm. in thickness.
- Cell membrane appears as a trilaminar structure (2 thin, dense lines with intervening light area) after fixation in osmium tetroxide due to the deposition of reduced osmium on the hydrophilic groups present on each side of the lipid bilayer. (unit membrane)
- The inner (cytoplasmic) dense line is its inner leaflet, whereas the outer dense line is its outer leaflet.
- Plasma membrane consists of lipids, proteins and carbohydrates.

What is the cell's glycocalyx (cell coat)?

- It is a delicate cell surface coating.
- It is formed of oligosaccharide chains attached to both lipid and protein components of plasma membrane.



What is the fluid mosaic model of membrane structure?

- The plasma membrane is formed of alternating protein and lipid regions giving it a mosaic appearance.
- At body temperature, the lipid bilayer is fluid allowing free movement of some membrane proteins within the lipid bilayer.
- Thus, membrane proteins comprise a moveable mosaic within the fluid lipid bilayer (fluid mosaic model).

Functions of Plasma Membrane

1. **Microtransfer (permeability)**
    - It is the exchange of materials between the cell and its surroundings
    - It occurs by one of the following processes:

      A) Passive permeability

      B) Active permeability

      C) Selective permeability
2. **Macrotransfer (bulk transport)**
    - It is a specific, selective process utilizing energy and comprises endocytosis and exocytosis.
3. **Functions related to the cell coat**
    - Recognition of other cells.
    - Holding adjacent cells together.

- Intercellular communication.
- Acting as a filter.
- Forming a protective layer.

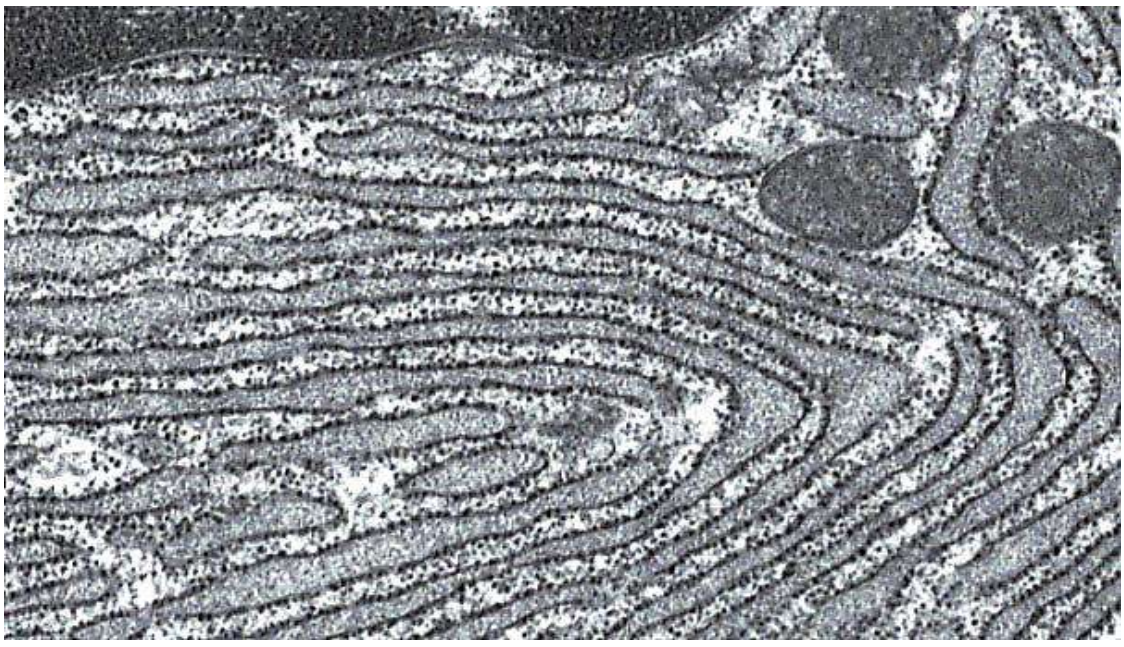
4. **Participation in the formation of other structure**
    - As the junctional complexes of the epithelial tissues, the basal lamina of epithelium, the myelin sheath of the neurolemma.

Endoplasmic Reticulum

- It is a network of channels present allover the cytoplasm, connected intermittently with the cell and nuclear membranes.
- It exists in 2 forms:

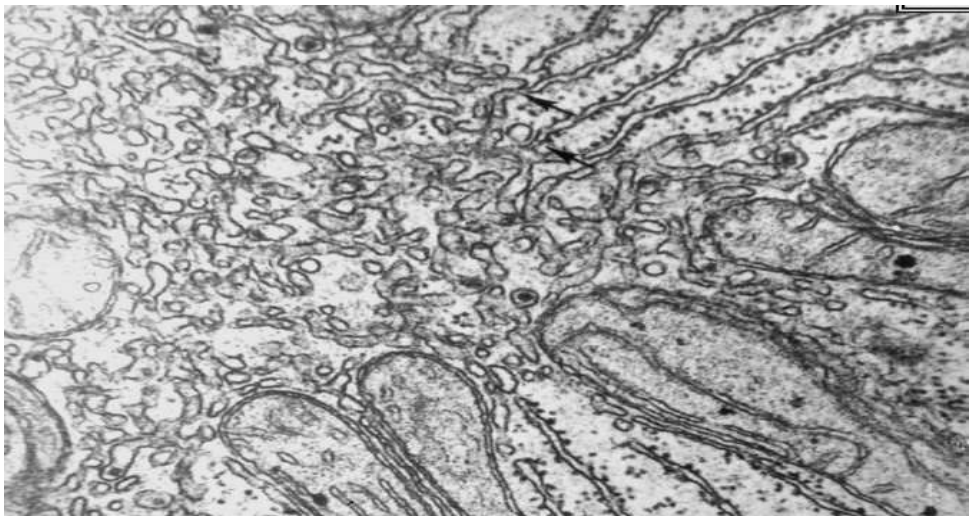
I. Rough Endoplasmic Reticulum (rER)

- **L/M:** Appears as a basophilia of the cytoplasm.
- **E/M:** Present in the form of tubules, vesicles or cisternae. Ribosomes attached to its cytoplasmic surface.
- Well developed in the cells which synthesize proteins for internal use as fibroblasts and for export as glandular secretory cells.
- **Functions Related to rER:**
    - Protein synthesis on the ribosomes.



II. Smooth endoplasmic reticulum (sER)

- **L/M:** Appears as a acidophilia of the cytoplasm.
- **E/M:** Present in the form of tortuous tubules, concentric or spiral arrays of highly fenestrated cisternae. Absence of ribosomes from its outer walls.
- Well developed in the cells of the adrenal cortex and the interstitial cells of the testis.
- **Functions:**
    - Lipid synthesis e.g. steroid hormones in adrenal cortex.
    - Calcium concentration: in muscular tissue for contraction.
    - Neutralization and detoxification: in liver cells where hormones and drugs are broken down.



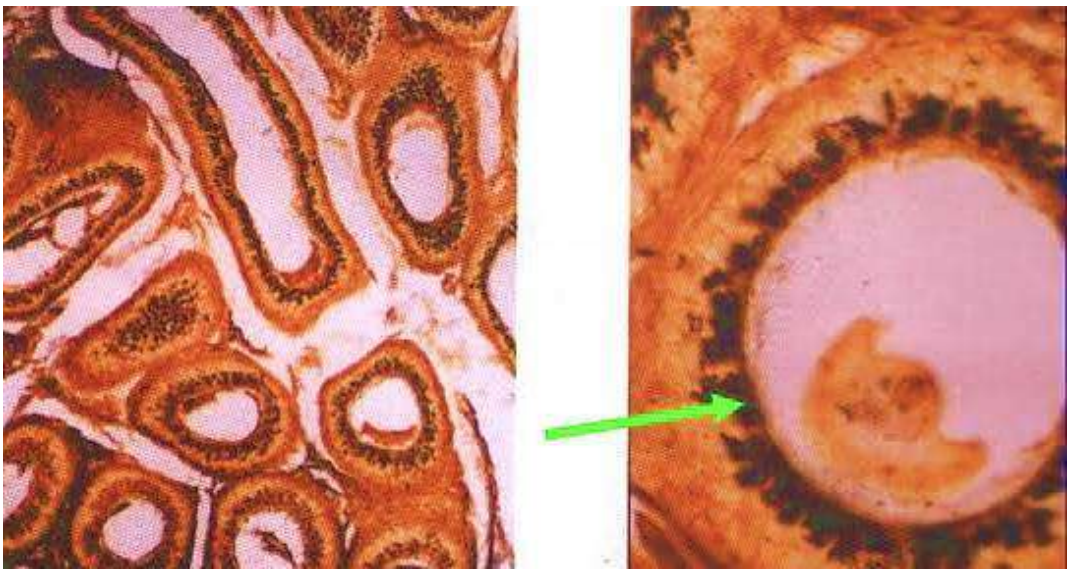
Golgi complex (apparatus)

It is one of the membranous organoids.

- It is present in nearly all cells and of variable position depending upon the cell activity.
- In glandular cells, it is present at one side of the nucleus (basal or apical) but in nerve cells, it is present surrounding the nucleus.
- It is also variable in size and shape.

L/M:

- under the phase contrast microscope or after vital staining.
- after osmotic acid fixation or silver impregnation.
- Appears as a fibrillar or granular network (positive Golgi image).
- In H&E stained sections, it appears as lightly stained areas (negative Golgi image) as in plasma cells.



E/M:

As consisting of 3 components:

A) Flattened vesicles or cisternae

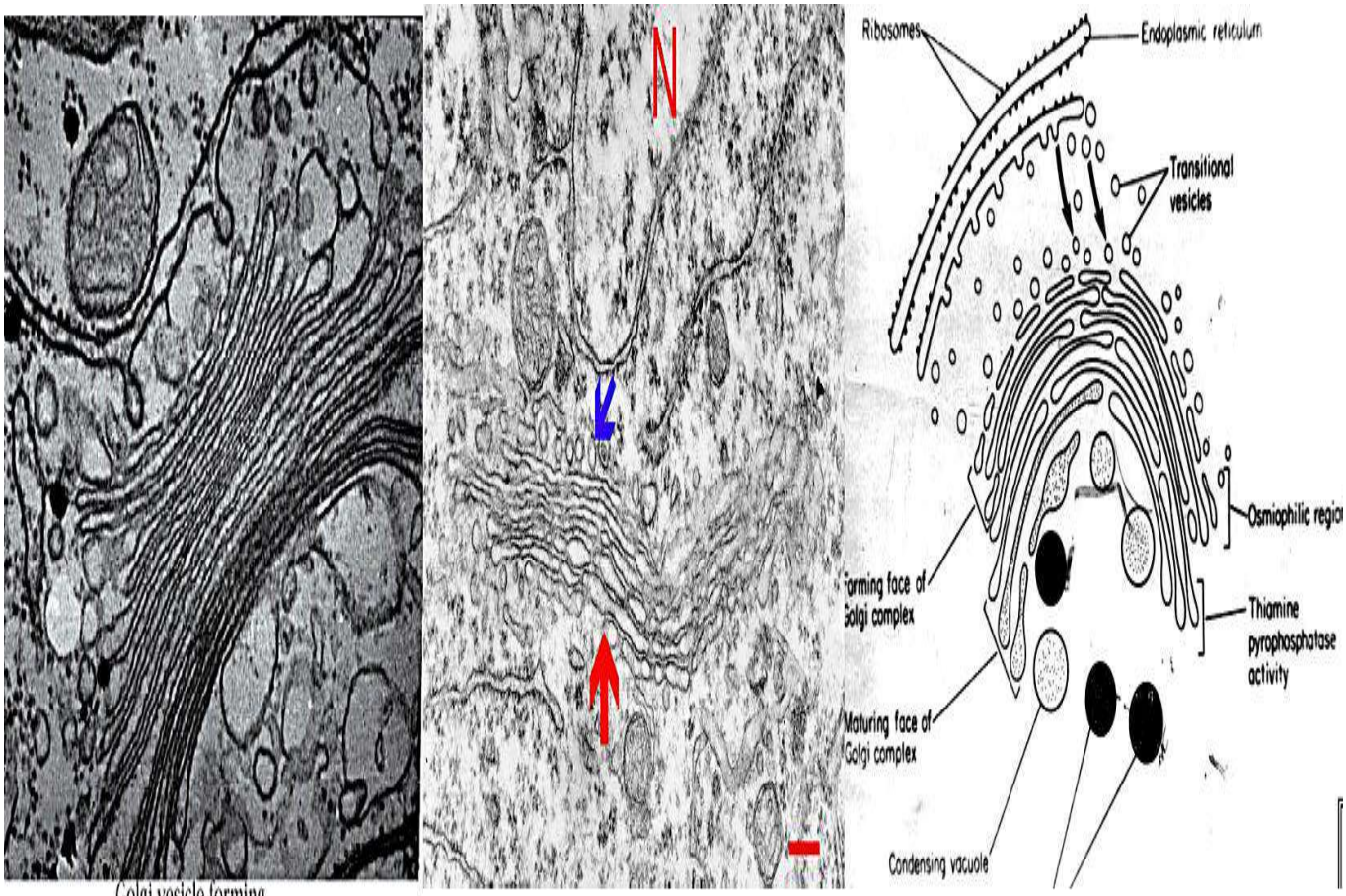
- These are a series (3-12 in number) of parallel, elongated, smooth-surfaced, double-walled membranous sacs forming stacks.
- They have a convex face (immature or Cis face) and a concave face (mature or Trans face).

B) Microvesicles or transfer vesicles

- These are smooth-surfaced small spherical vesicles arising by budding from the rER and seen around the periphery of the stack.
- They are migrating through the cytoplasm to fuse with the immature face where they discharge their contents thus causing expansion of the saccules.

C) Macrovessels or secretory vacuoles

- These are smooth- surfaced large spherical vesicles, usually present at one pole of the Golgi complex (at the mature face) and contain formed substances.
- They either migrate to fuse with the cell membrane and discharge their contents through it to the outside forming secretory granules or remain in the cytoplasm as lysosomes.



Golgi vesicle forming

Functions

- Processing, concentration and packaging of secretory products.
- Converting the inactive proinsulin, by enzymatic cleavage, into active insulin.
- Addition of carbohydrate fractions to proteins coming from rER to form glycoprotein.

Lysosomes

- They are membranous vesicles containing hydrolytic enzymes which are active at an acid PH e.g. lipases, nucleases, sulphatases,...
- They are present in all cells, especially reticuloepithelial system.
- They are variable in number, size, shape and origin in different cells.
- Can be demonstrated after especial cytochemical methods for their enzymes.
- Appears as homogenous (without internal structure) membrane bound vesicles, with unit membrane.

Types of lysosomes:

According to their contents (by histochemical studies) and their appearance, the following types can be distinguished:

1. Prelysosomes:

They contain only substrates and they may be:

a) **Heterophagosomes:** containing exogenous substrates as bacteria, derived by phagocytosis or pinocytosis.

b) **Autophagosomes:** containing endogenous substrates as cytoplasmic organelles.

2. Primary lysosomes:

They are small, spherical and their contents are homogenous or finely granular. They contain only hydrolytic enzymes.

3. Secondary lysosomes:

- They are larger in size and their contents are pleomorphic.
- They are formed by the fusion of the primary lysosomes with prelysosomes and they may be:

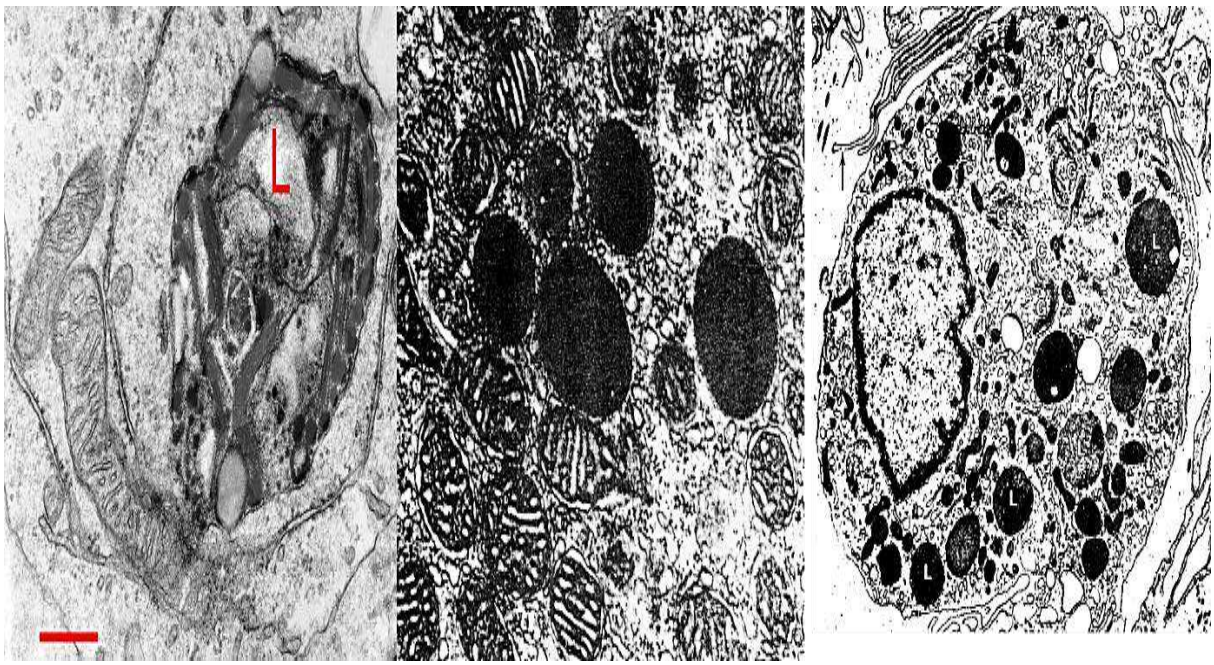
  a) **Heterolysosomes:** formed by the fusion of primary lysosomes with heterophagosomes.

  b) **Autolysosomes:** formed by the fusion of primary lysosomes with autophagosomes.

  c) **Multivesicular bodies:** formed by the fusion of primary lysosomes with pinocytic vesicles.

4. Post lysosomes (residual bodies):

- They are of large size and contain electron-dense pigments, granules and lipid-like inclusions. They have no enzymatic activity.



Functions of lysosomes

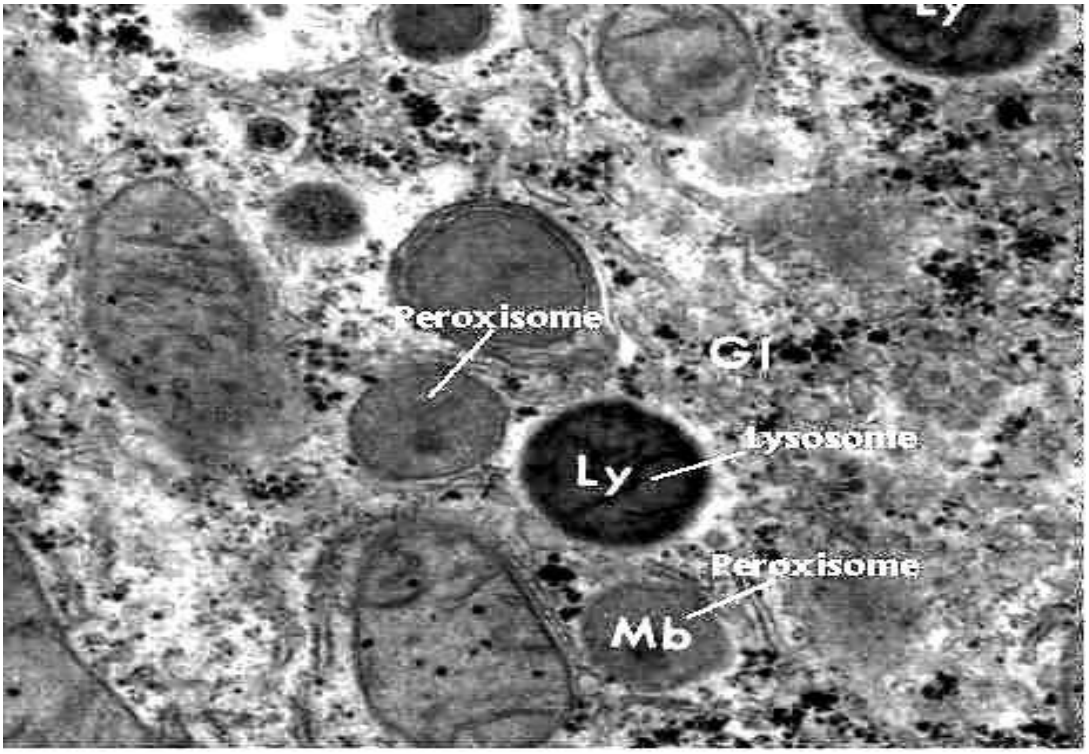
1. Defensive mechanism against bacteria, viruses etc.
2. Remodeling tissues and removing of the dead cells as in bone and cartilage.

Peroxisomes (Microbodies)

These are spherical membrane-limited organelles similar to lysosomes, but containing oxidative enzymes as oxidases and catalases. They are present in many cells especially hepatic cells and those of the proximal convoluted tubules of the kidney.

Functions

- Degradation of purines (by the oxidases) yielding hydrogen peroxide, then destroying it (by catalase) thus protecting the cell against it.

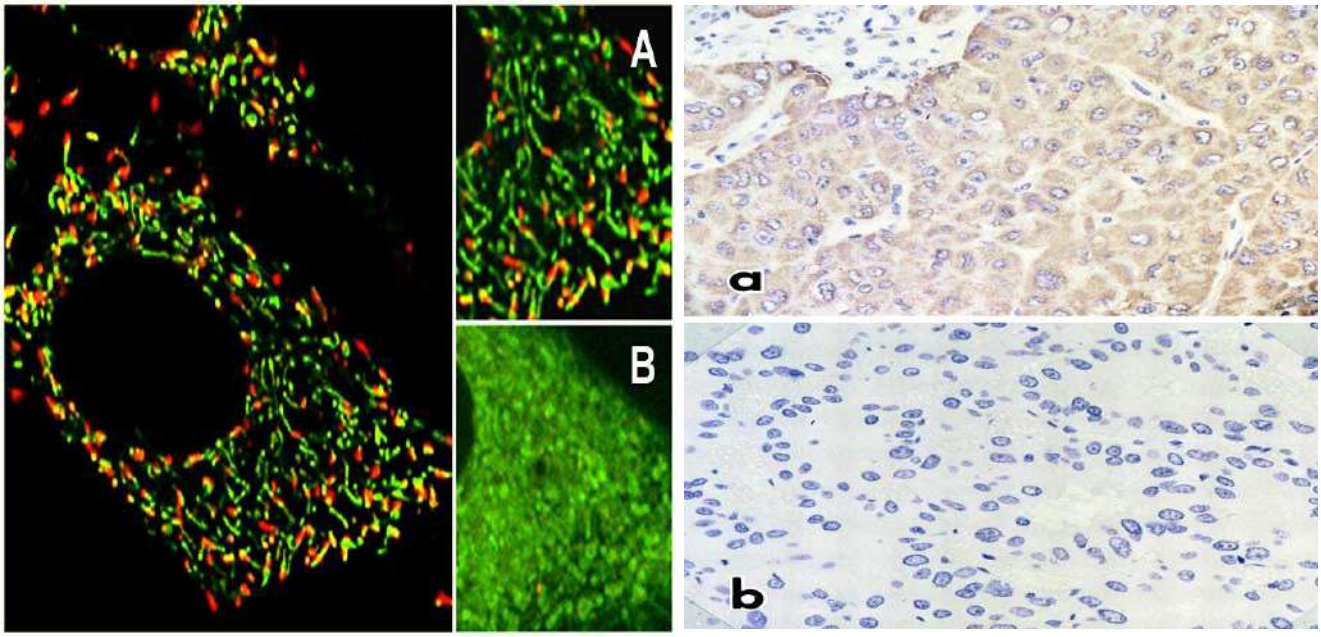


Mitochondria

These are membrane-bounded organoids distributed allover the cytoplasm in nearly all cells except RBCs and the terminal keratinocytes in the skin. They tend to accumulate in parts of intense metabolic activity as the apical ends of ciliated cells, middle piece of spermatozoa and base of ion transferring cells.

L/M

- These are membrane- bounded organoids distributed allover the cytoplasm in nearly all cells except RBCs and the terminal keratinocytes in the skin.
- They tend to accumulate in parts of intense metabolic activity as the apical ends of ciliated cells, middle piece of spermatozoa and base of ion transferring cells.
- **In living cells:** as cultured cells, under the phase contrast microscope or after superviatal staining (Janus green B stain).
- **In fixed cells:** after Fe hematoxylin stain.
- They appear as granules, rods or filaments of variable number, length and size. They are capable of movement, change shape, division and fusion.
-



E/M

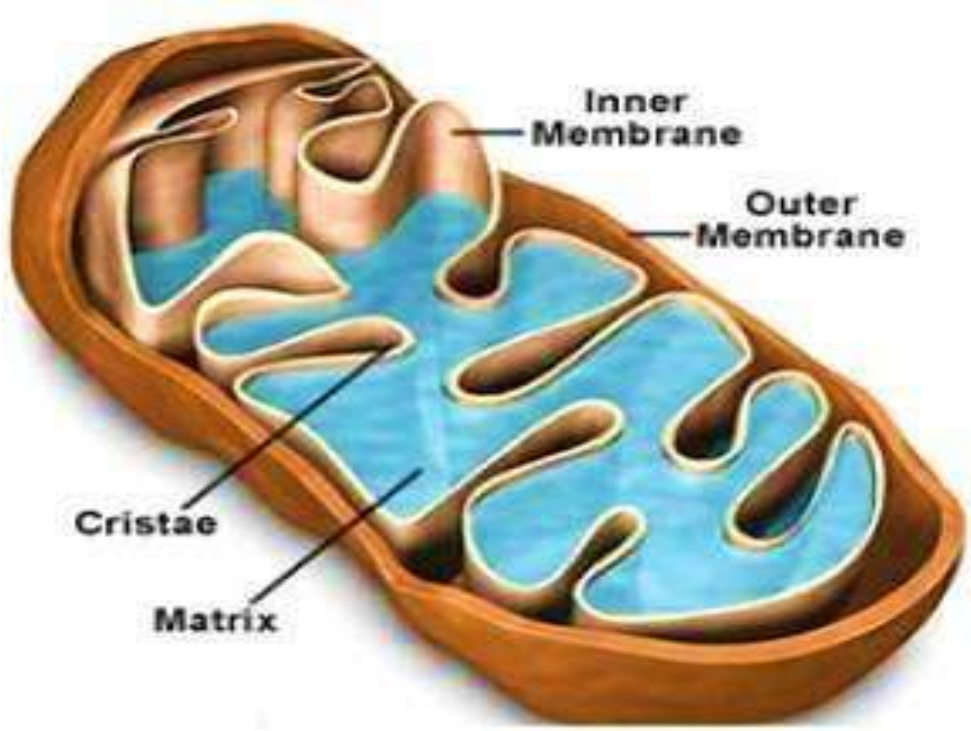
Elongated, spherical, rounded or filamentous organoid which is formed of:

A) **Outer membrane:** It is formed of the unit membrane and is smooth.

B) **Inner membrane:** It is formed of a special type of the unit membrane.

The inner membrane is thrown into shelf- like projections known as cristae of which the following types can be observed:

1. **Septate (typical):**
    - Present in most cells.
    - Situated transversely, longitudinally or obliquely.
2. **Tibular (villous):**
    - Present in the mitochondria of steroid secreting cell.



Spaces

1. **Inter membrane space:** which lies in between the 2 membranes and contains a low-density fluid.
2. **Inter cristae (matrix) space:** is enclosed by the inner membrane and filled with a fine granular material of variable electron density.



Mitochondrial matrix

- Contains a dense fluid with spherical or ovoid dense granules rich in Ca and Mg.
- The matrix rich in protein and containing some DNA and the 3 types of RNA.
- Enzymes for the citric acid (Kreps) cycle

Functions

- Mitochondria are known as the powerhouses of the cell; being the site of Krebs cycle and oxidative phosphorylation thus providing the ATP which is an energy rich compound.

Annulate Lamellae

- These are membrane- bounded cisternae, frequently near and resembling the nuclear membrane.
- They are found in many cells especially oocytes, spermatocytes and rapidly growing cells.

L/M:

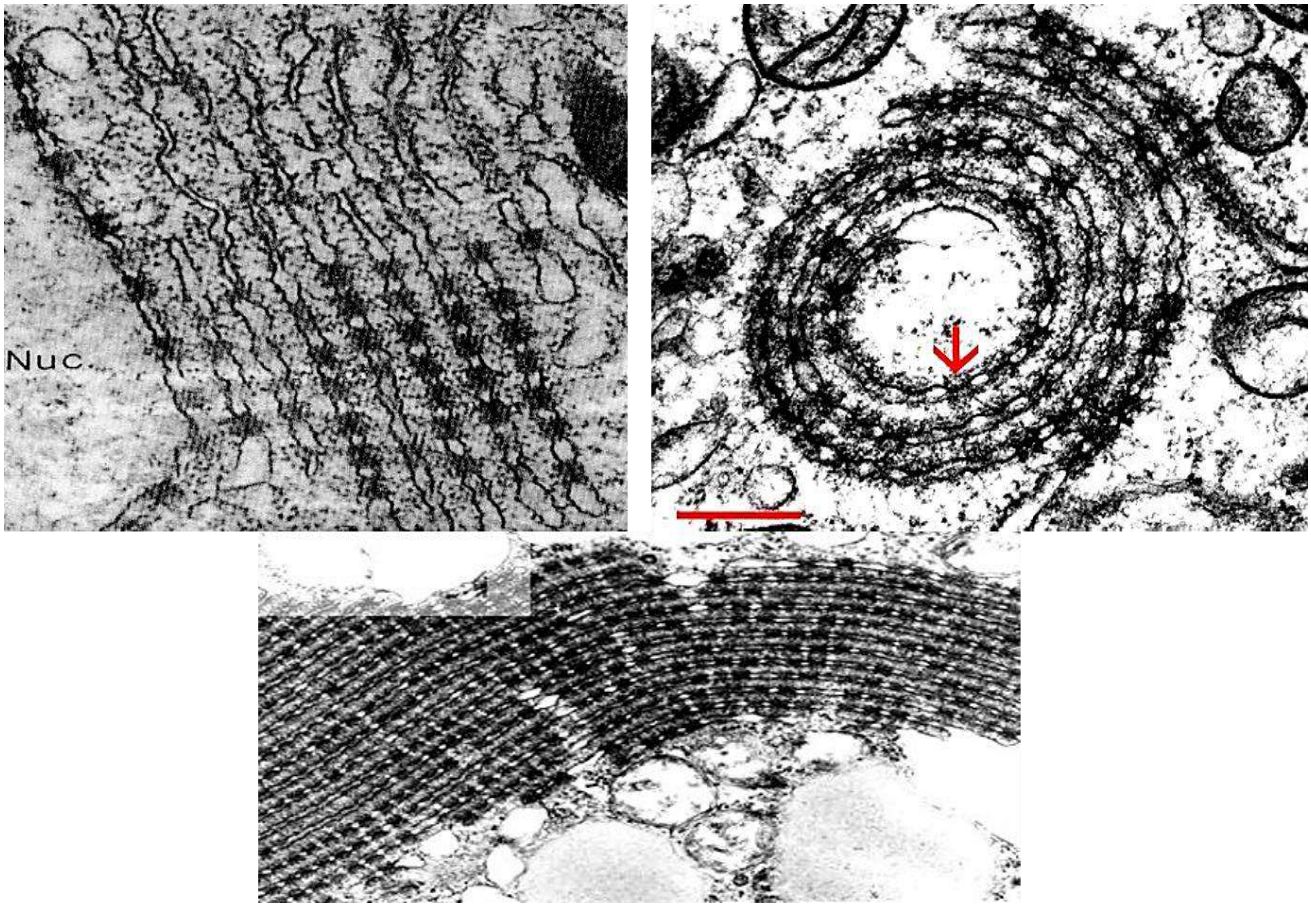
- It appears as birefringent elements in the cytoplasm.

E/M:

- Appears as single or stacks of pairs of parallel membranes enclosing a space which is traversed with circular pore complexes or annuli at regular intervals.

Functions

- May be a messenger function, carrying information from the nucleus to the other cell regions.



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1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

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3- Wheater, P. R., Burkitt, H. G., & Daniels, V. G. (2006). *Wheater's functional histology: A text and colour atlas* (5th ed.). Churchill Livingstone/Elsevier.

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Lecture 3

Non-Membranous Organelles

Learning objectives:

***After this lecture, students should be able to:**

- Describe the structure and types of ribosomes
- Discriminate the structure and function of microtubules and centrosomes
- Define cytoskeleton and enumerate its components.
- Illustrate the structure and function of microfilaments and intermediate filaments.

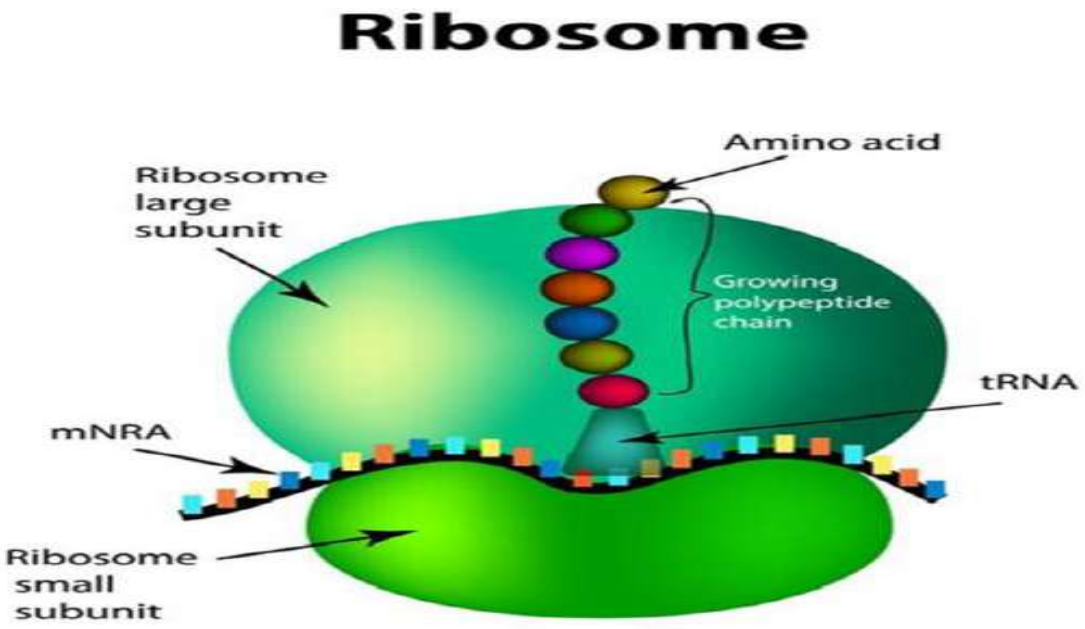
1-Ribosomes

***Ribosomes:** are small, non- membranous organelles about 20 × 30 nm in size, present in all animal cells in varying amounts depending on their activity in protein synthesis.

***A functional ribosome** has two subunits of different sizes (small subunit& large subunit) bound to a strand of mRNA.

*** Chemically;** ribosomes are composed of ribosomal RNA (rRNA) and proteins.

*These ribosomal proteins are themselves synthesized in cytoplasmic ribosomes, but are then imported to the nucleus where they associate with newly synthesized rRNA.



LM appearance: Ribosomes cannot be seen with LM; however, they are responsible for the basophilia of the cytoplasm.

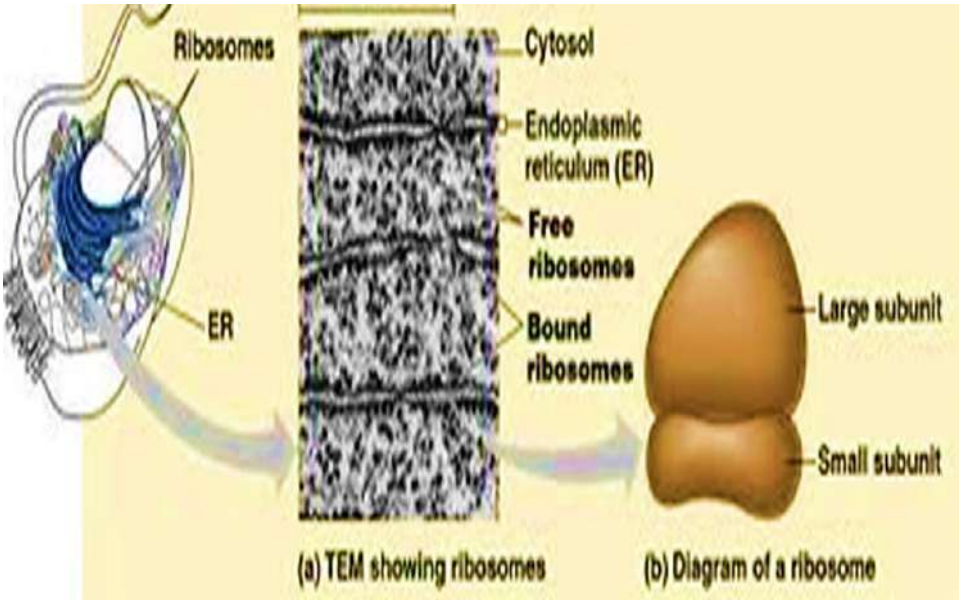
■Cytoplasmic basophilia in cells may be:

a. Diffuse or Localized: in a certain part of the cell e.g. basal basophilia of pancreatic acinar cells.

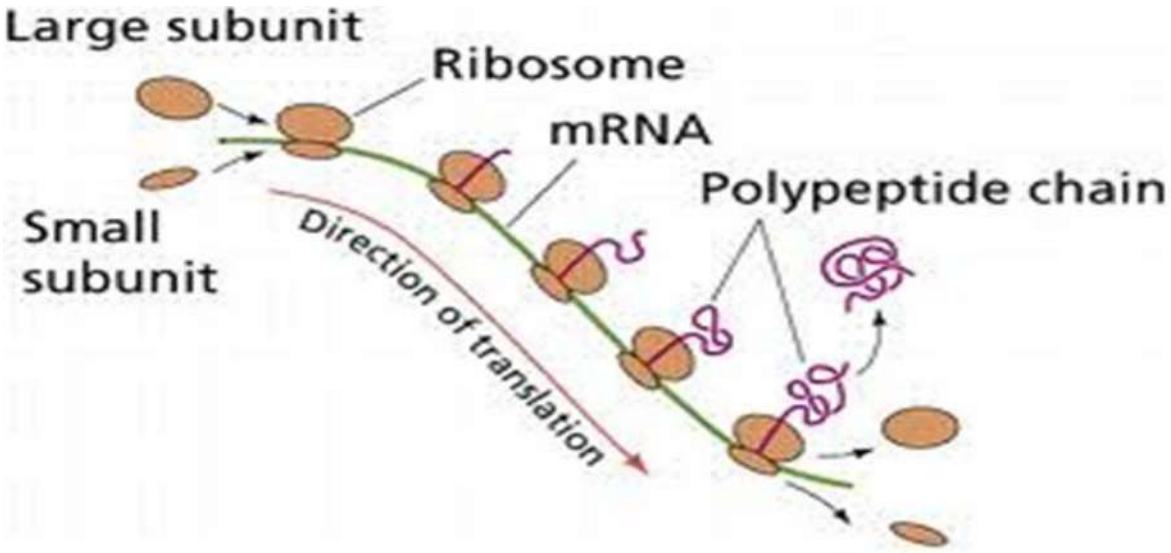
Spotty: in form of basophilic patches such as Nissl's granules of nerve cells

EM appearance:

They appear as small electron dense granules (20-30 nm in diameter)



Single ribosomes are not involved in protein synthesis and have to be organize in the form of polysomes by the aid of messenger RNA



Types of ribosomes:

1. Single ribosome are not involved in protein synthesis

2. polyribosome Group of ribosomes may be linked together by a molecule of mRNA forming a bead-like structures are called polysomes

a. Free polyribosomes:

- They are free particles within the cytoplasm.

- The proteins which are synthesized by the free ribosomes are kept inside the cytoplasm to be used to build up the cell.

b. Attached polyribosomes:

- These are attached to the outer (cytoplasmic) surface of the rER through their large subunits.

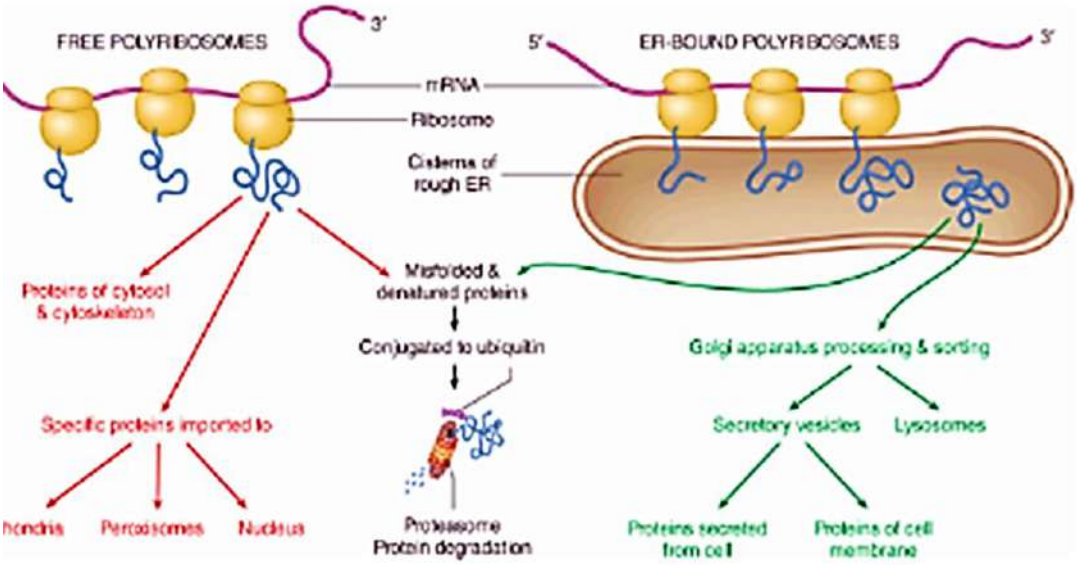
-They synthesize proteins which are packed by the Golgi apparatus into secretory proteins , integral membrane proteins or lysosomal enzymes.

Function of ribosomes:

1-Ribosomes are concerned with protein synthesis. They provide sites for amino acid linking to form polypeptide chains e.g. proteins.

2-The proteins which are synthesized by the free ribosomes are kept inside the cytoplasm to be used to build up the cell.

3-The proteins which are synthesized by the ribosomes attached to the rER are packed by the Golgi apparatus into secretory proteins, integral membrane proteins or lysosomal enzymes.



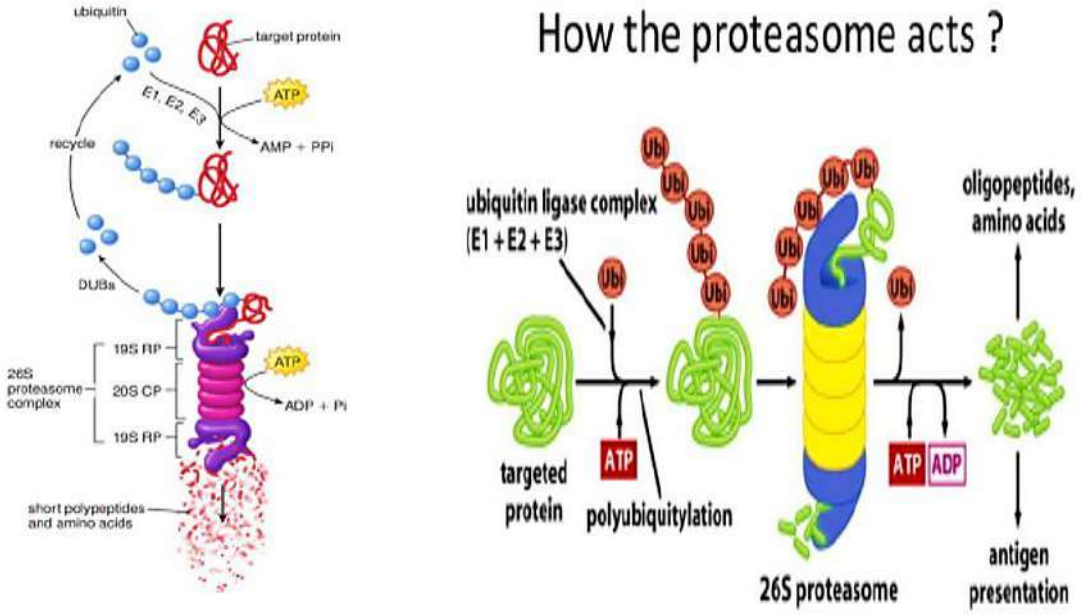
Free and bound ribosomes

2-PROTEASOMES

Definition: Proteasomes are small, non-membranous barrel-shaped (cylindrical) organelles that functioning in degradation of mis-folded or unwanted proteins.

*They are functioning to destroy damaged or unwanted proteins that have been labeled for destruction with ubiquitin.

*Unlike lysosomes, however, which can digest large amounts of cellular debris, the proteasome can only break down one protein at a time.



▲ Proteasome-mediated degradation

Clinical application:

In Parkinson's disease, accumulations of misfolded proteins kill cells in the brain that regulate muscle function, because the proteasome system is not functioning properly.

Function:

1. They function to degrade and digest damaged or unneeded proteins (The denatured or non-functional polypeptides are tagged with the polypeptide ubiquitin to be targeted for destruction).

2. They ensure quality of exported proteins.

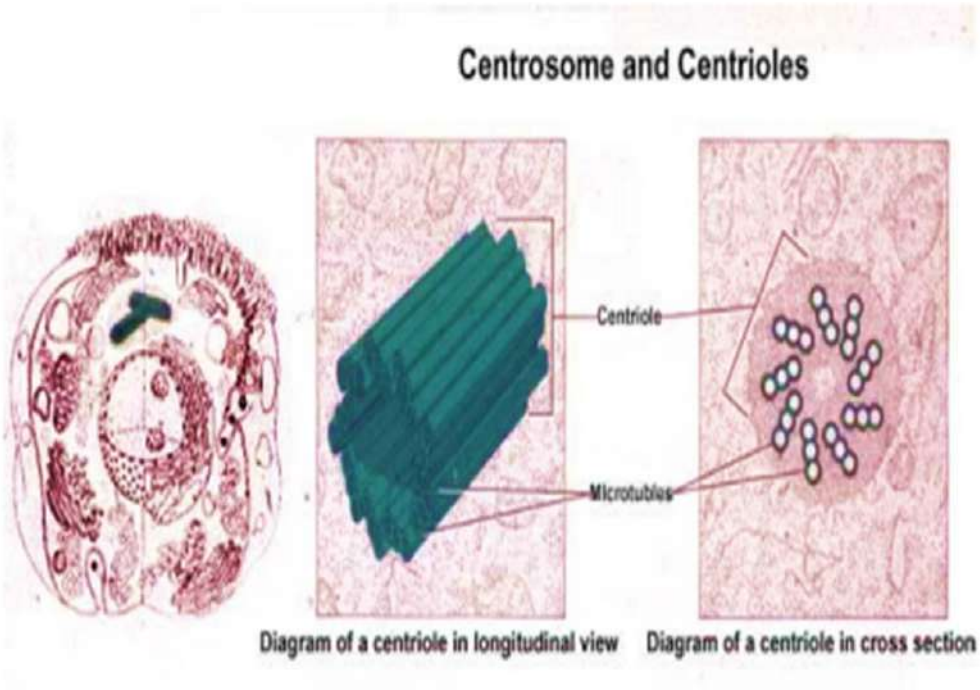
3. They restrict the activity of a specific protein to a certain time.

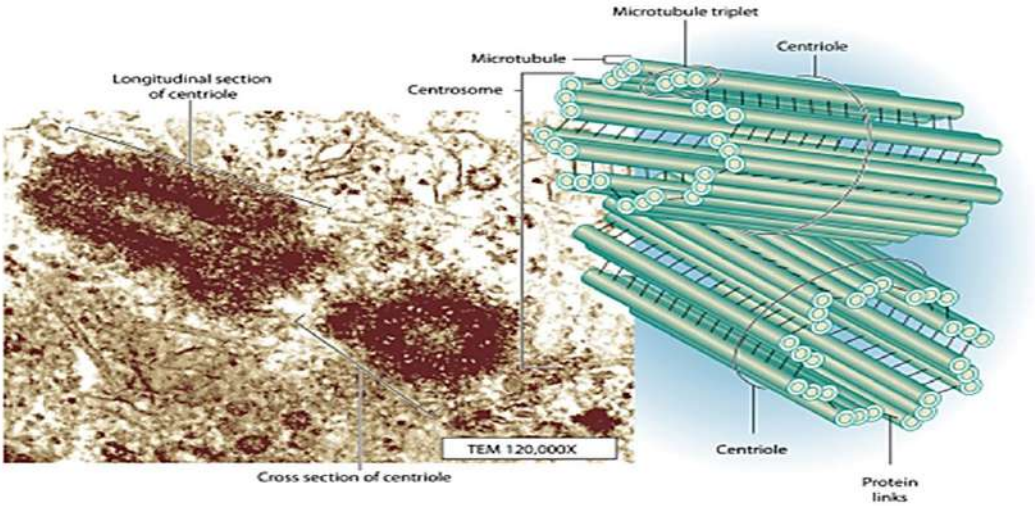
3-Centrosome (Cell centre)

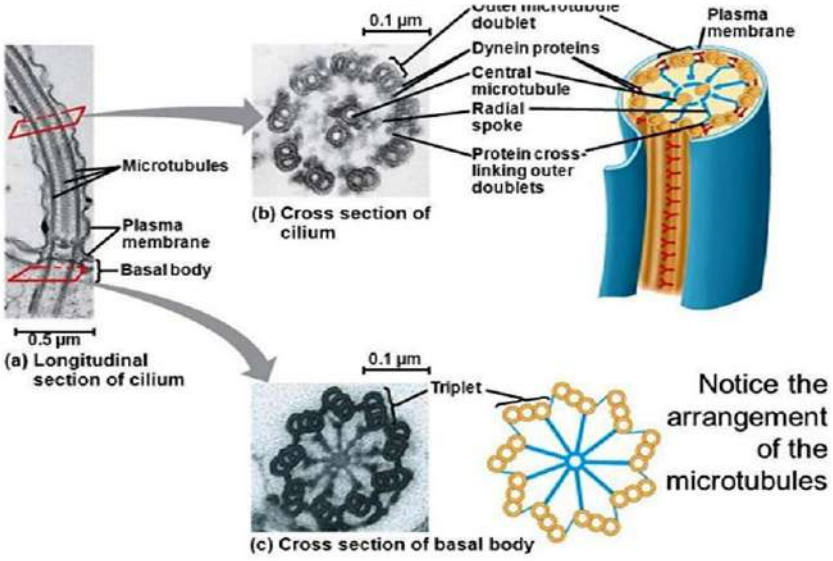
LM: spherical pale area near the nucleus

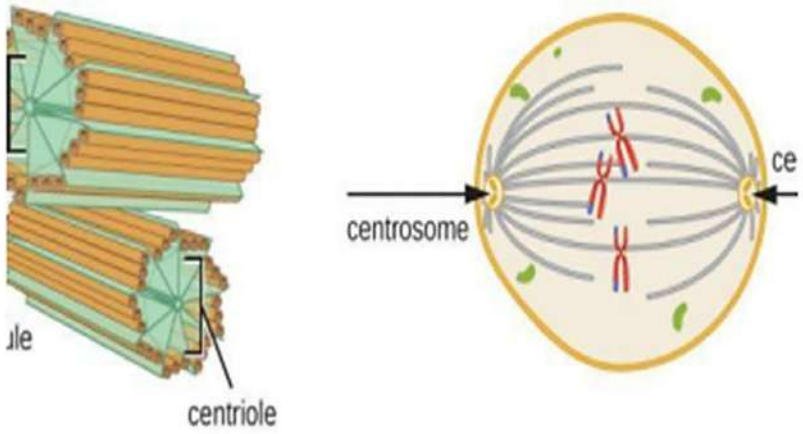
EM: formed of 2 cylinder- like centrioles in the cell center perpendicular to each other surrounded by a granular electron dense area.

▪ The wall of each centriole is made of 9 triplets of longitudinally arranged microtubules, i.e. 27 microtubules.









Function:

It is essential for cell division (It forms the mitotic spindle) .

Formation of:

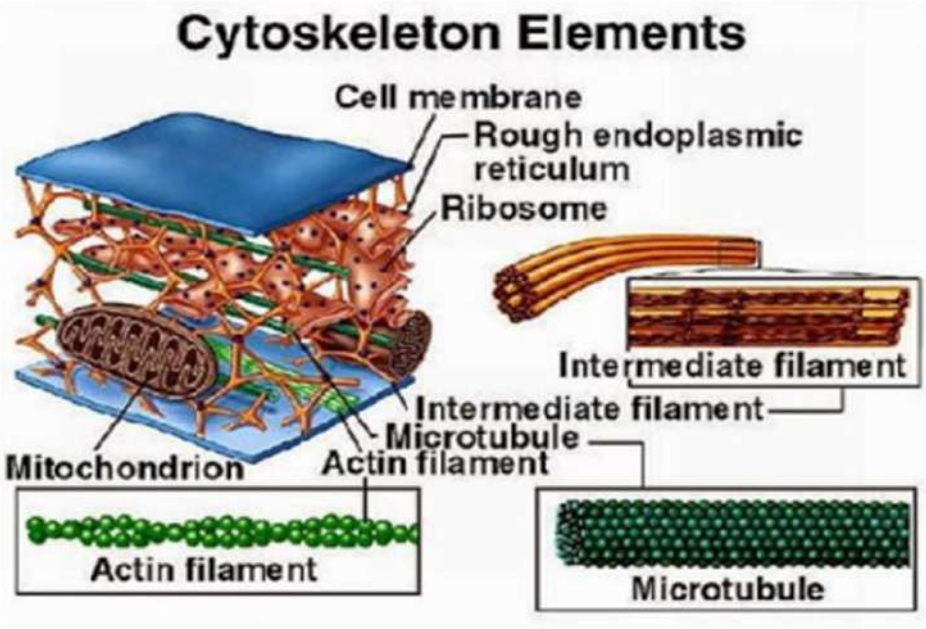
1-Mitotic spindle during cell division (mitosis, meiosis)

2-Cilia and Flagella

3-Basal bodies

4-Microtubules (It considered as the MTOC)

4-Cytoskeleton



Components of cytoskeleton

Definition: It is a dynamic filamentous structure which forms "Skeleton" of the cell.

It provides :

1.structural support for cell components, maintains cell shape,

2.skeleton for movement and stability.

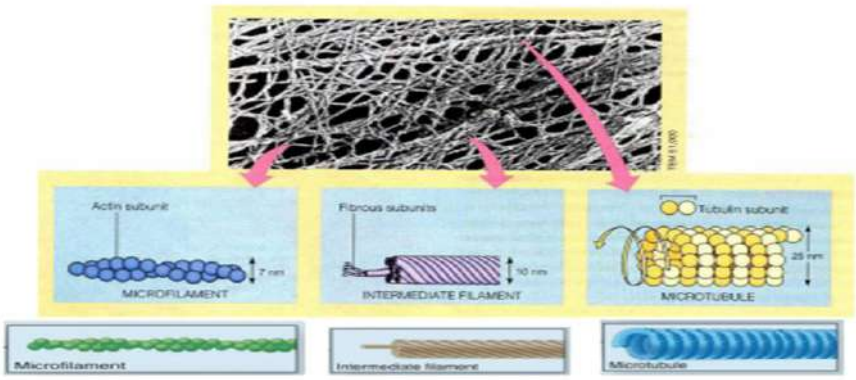
3. a structural framework for the cell, facilitate intra cellular transport, support cell junctions, transmits signals and permits motility

Types

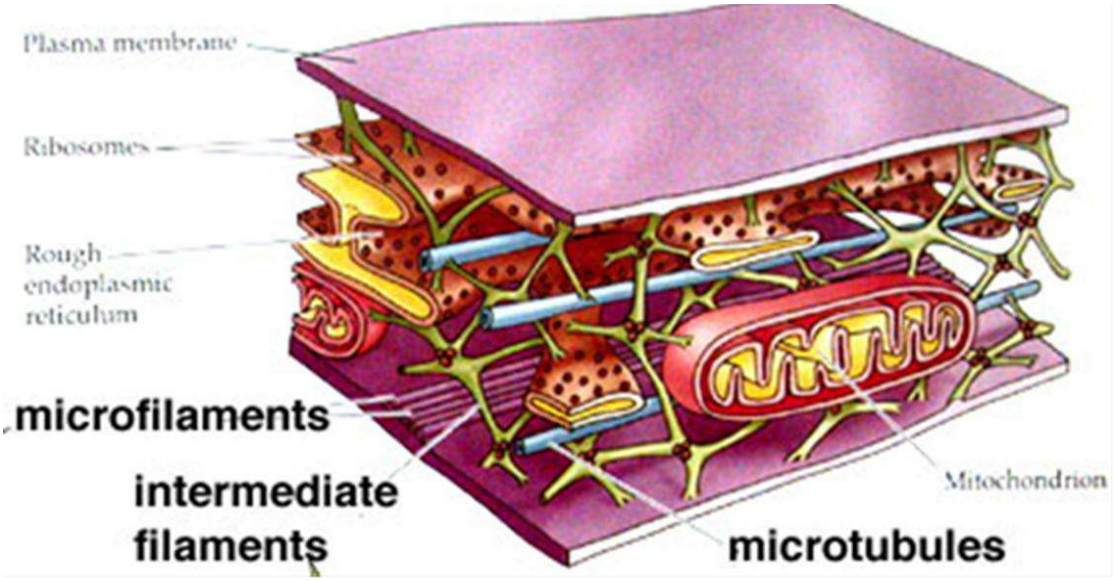
1. Microtubules

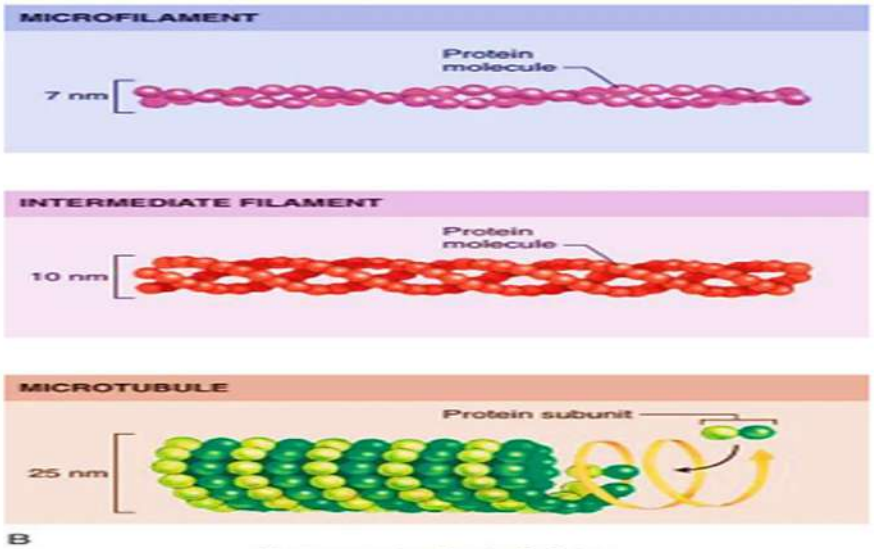
2. Microfilments

3. Intermediate filaments



Components of cytoskeleton





B

Components of cytoskeleton

A- Microfilaments (Actin filaments, thin filaments)

Definition: They are fine fibers 5-7nm in diameter.

globular protein subunits called G actin, which are assembled into a double strand helix.

▪ They are involved in contraction in muscle cells; provide support, maintenance of cell shape, and assist in locomotion in non-muscular cells.

▪ They are present, as a network, under the cell membrane of most cells, in muscle cells and in cores of microvilli.

▪ In cells other than muscle cells, they are dynamic structure (undergoing continuous assembly followed by dissociation).

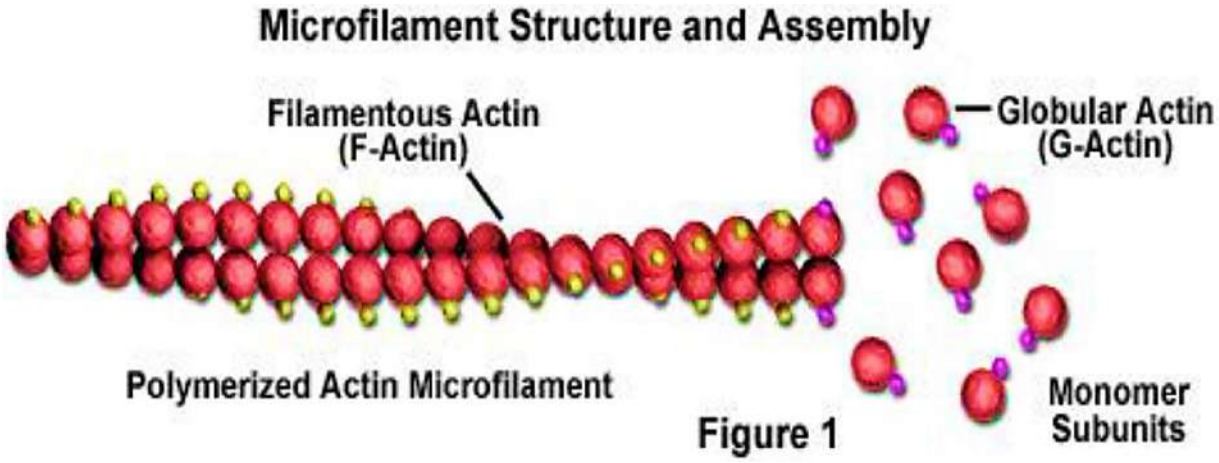


Figure 1

►Function:

1▪ It provide support, and maintains cell shape.

2▪ It helps transportation of material within the cell.

3▪ Its association with myosin in myocytes is responsible for muscle contraction.

4▪ It carries out cellular movements including gliding, contraction, and cytokinesis.

B- Intermediate Filaments

Definition: They are fibrous rope-like protein structures, about 8-10nm in diameter which provide tensile strength for the cell.

*They serve as reinforcing rods for bearing tension, and anchor certain organelles

*They are stable - don't break down (They are the most stable cytoskeletal component, strong mechanical stability to cells)

*They are different in molecular structure in different cells, with variable functions and distribution

Molecular structure: It is

a) Desmin (in striated and smooth muscle).

b) keratins (in epithelial cells).

c) Neurofilaments (in nerve cells).

Glial filaments (in neuroglia cells).

d) Vimentin (in fibroblasts)

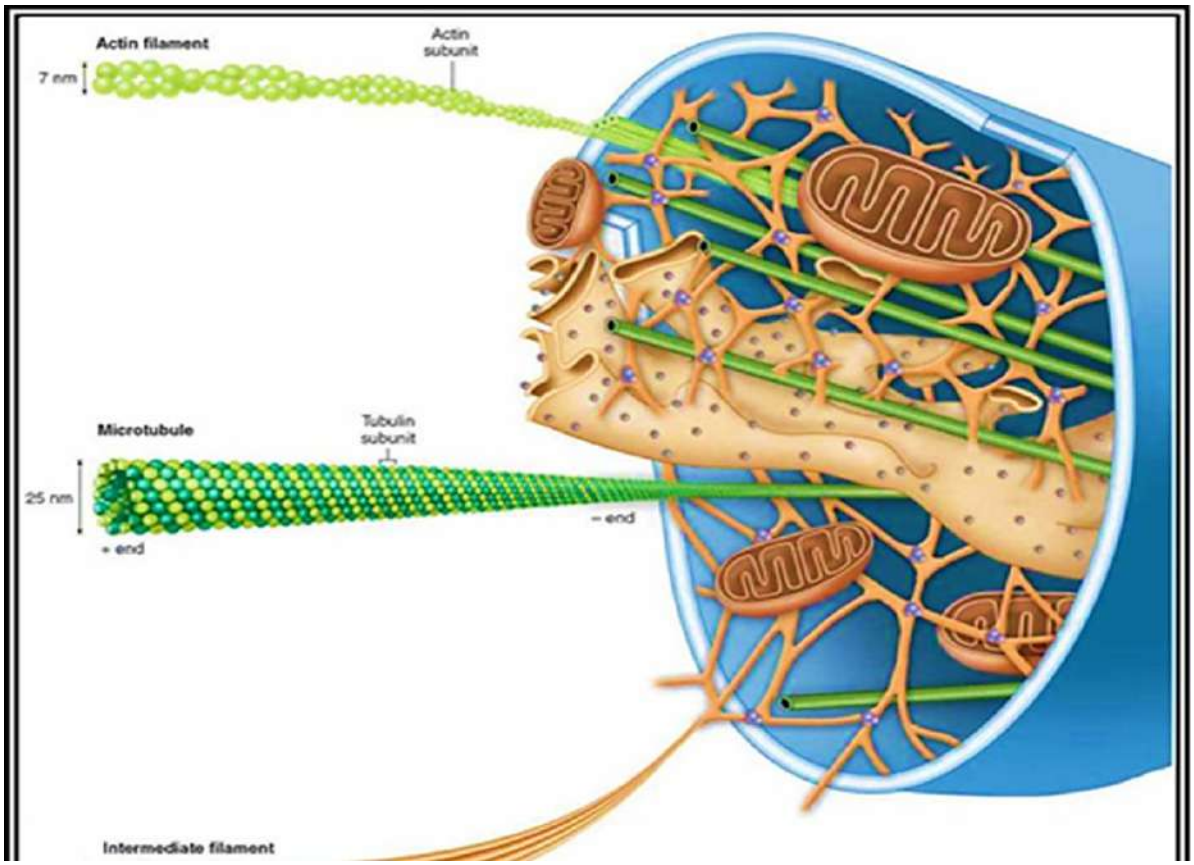
e) Nuclear lamins (associated with the inner membrane of nuclear envelope).

► Function:

1-It is one component of the cytoskeleton which supports the cell, and serves to maintain its shape.

2-Helps to ensure equal distribution of the tension and forces throughout the cells (in smooth muscles and keratinocytes of skin).

3-Involved in muscle contraction.



C.Microtubules

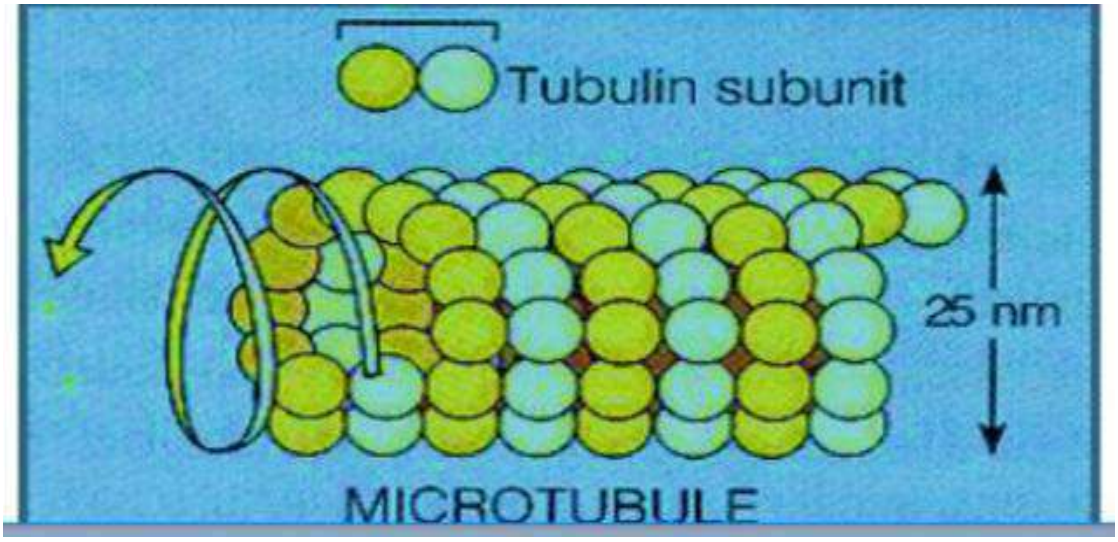
Definition: Are tubular non membranous organelles with variable lengths and measures 20-25nm in diameter.

LM: They appear only after staining with immunofluorescent.

EM: It appears as straight hollow cylindrical tube, of variable lengths, and regular diameter.

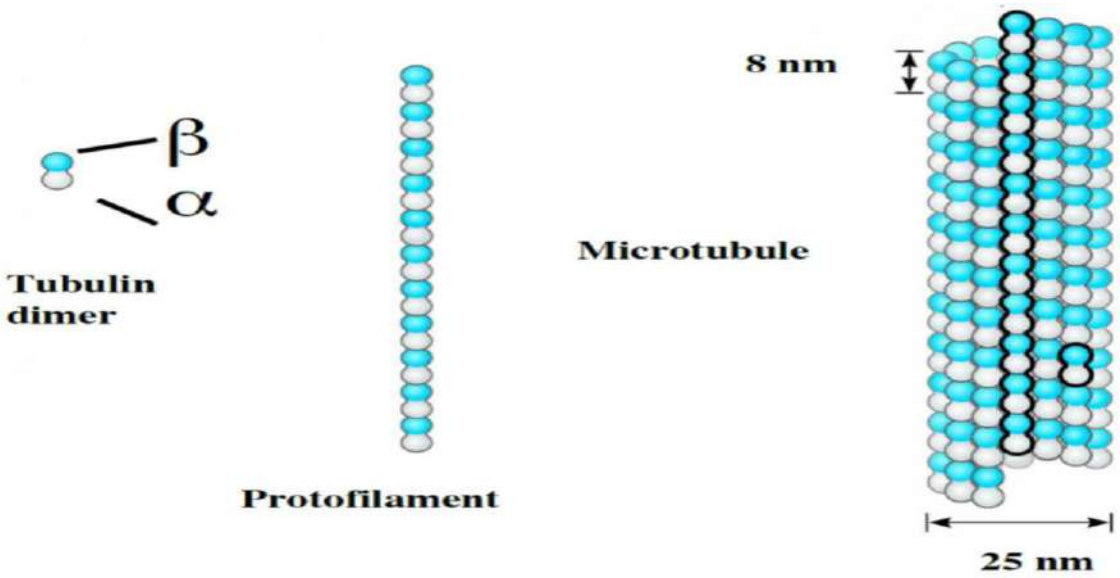
In cross sections they appear as tiny circles.

Molecular structure: It is composed of globular protein subunits called tubulin protein dimers. These subunits are formed of alpha and beta tubulin monomers.



Wall of microtubules is formed of 13 protofilaments (thread like structures).

These protofilaments are consisting of the tubulin protein subunits.



Function:

1-Formation of mitotic spindle (during cell division). They guide movement of chromosomes during cell division

2-They are the structural basis for the cilia, flagella and centerioles.

Medical application:

Anti-mitotic drugs such as colchicines can stop mitosis as they prevent the development of the mitotic spindle, which bind to tubulin and prevent its addition at the plus end. Therefore, the colchicines are used in karyotyping and in treatment of cancer.

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1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

2- Ross, M. H., & Pawlina, W. (2006). *Histology: A text and atlas with correlated cell and molecular biology* (5th ed.). Lippincott Williams & Wilkins.

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Lecture 4

Cytology; Cell inclusion

Learning objectives:

*Recognize types of pigment deposits

*Discuss glycogen granules

*List cell inclusions & enumerate them

*Differentiate between the endogenous & exogenous pigment

Inclusions

Cytoplasmic inclusions: are substances present in the cytoplasm as result of accumulation of cell products ,metabolites and other substances stored in the cytoplasm and have little or no metabolic activity.

*Most inclusions are transitory structure not enclosed by membrane.

There are three types of inclusions:

Stored food, Only glycogen and fat can be stored in animal cells .

Pigments, e.g., melanin and hemoglobin.

Crystals.

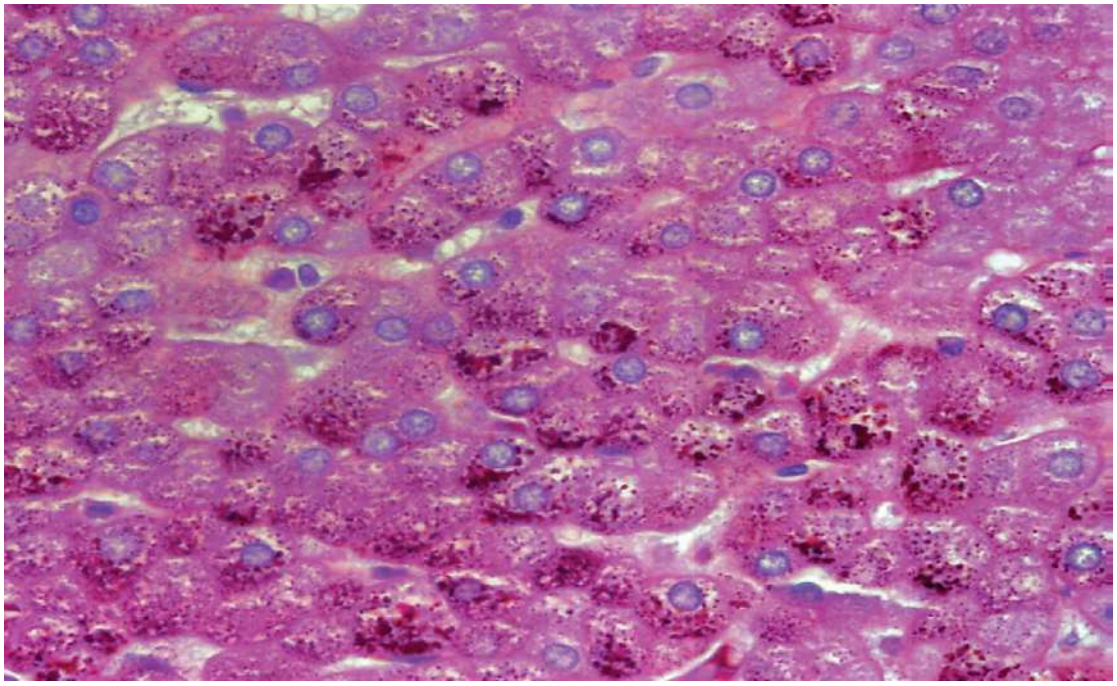
Glycogen

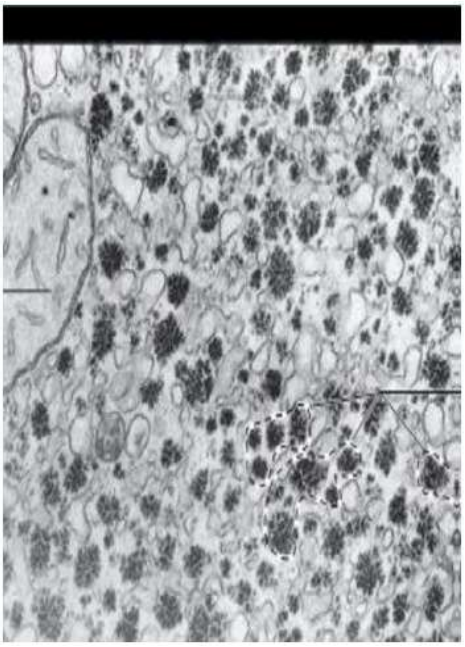
*Glycogen is stored in large amounts only in the liver and muscle cells.

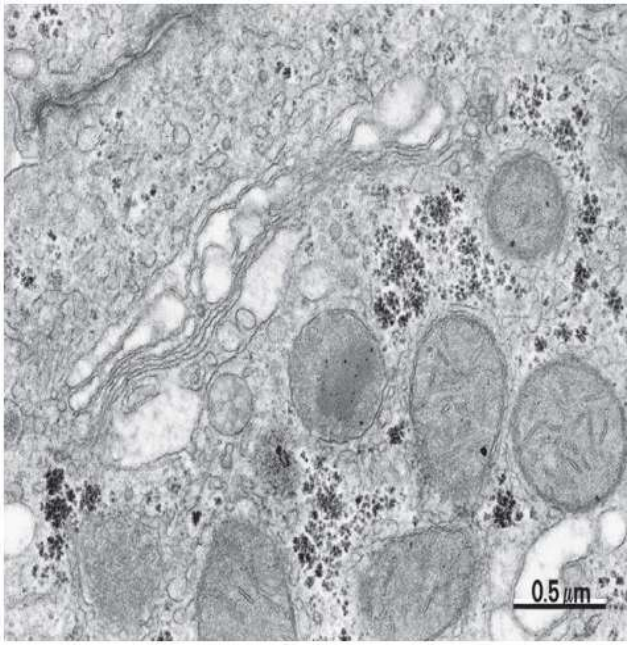
*Glycogen granules aggregates of the carbohydrate in the cytoplasm and the major storage form of glucose and is a ready source of energy.

***With the L.M**. glycogen can be localized in cells after staining with PAS or Best's carmine.

***With E.M.** glycogen appears in the form of clusters of granules usually known as glycogen rosettes. These granules are usually associated with SER.







Fat

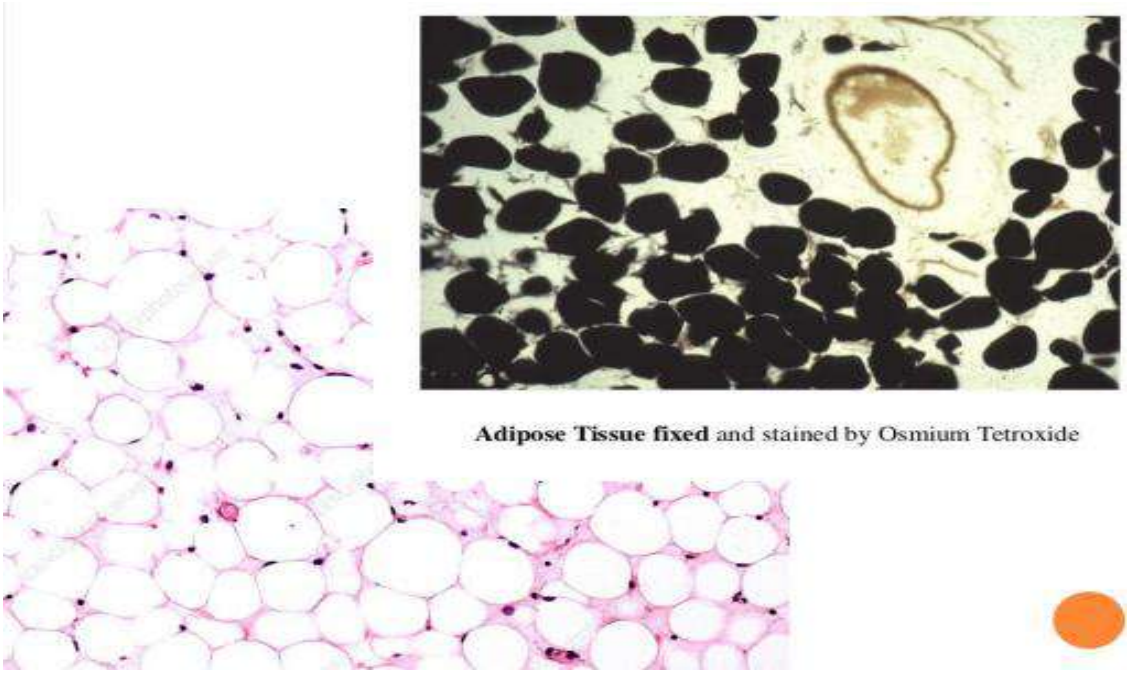
*is stored in special types of cells known as fat cells or adipocyte.

*The latter, when fully engorged with fat, are spherical and have a very thin layer of peripheral cytoplasm which contains the highly flattened nucleus

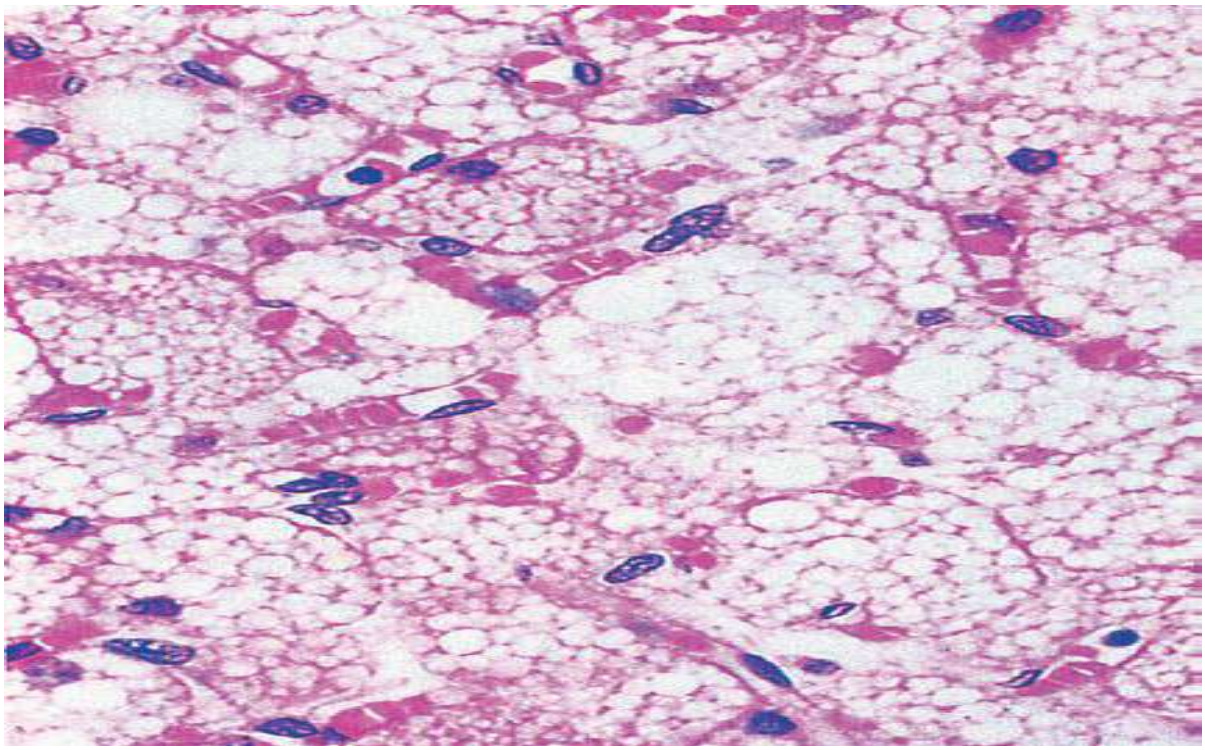
With L.M , in specimens stained with H & E., fat cells appear as signet ring. This appearance results after the fat has been dissolved in alcohol during section preparation, leaving a thin cytoplasmic film with a thickened region containing the nucleus.

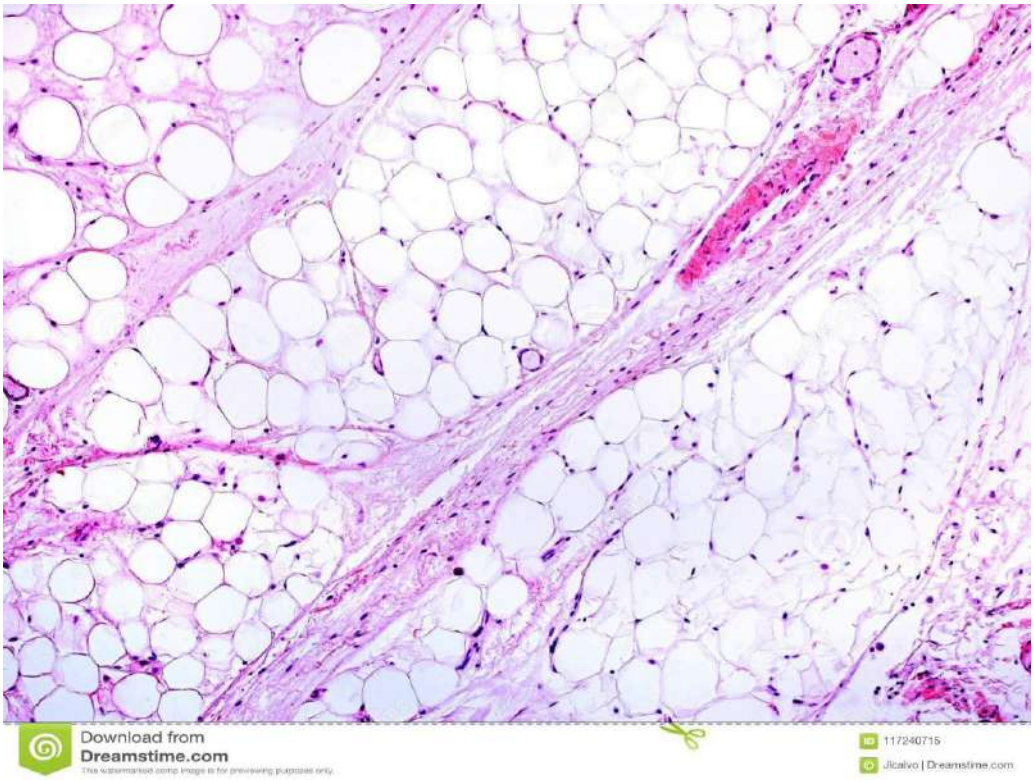
*Also, fat is stained with Sudan black stain.

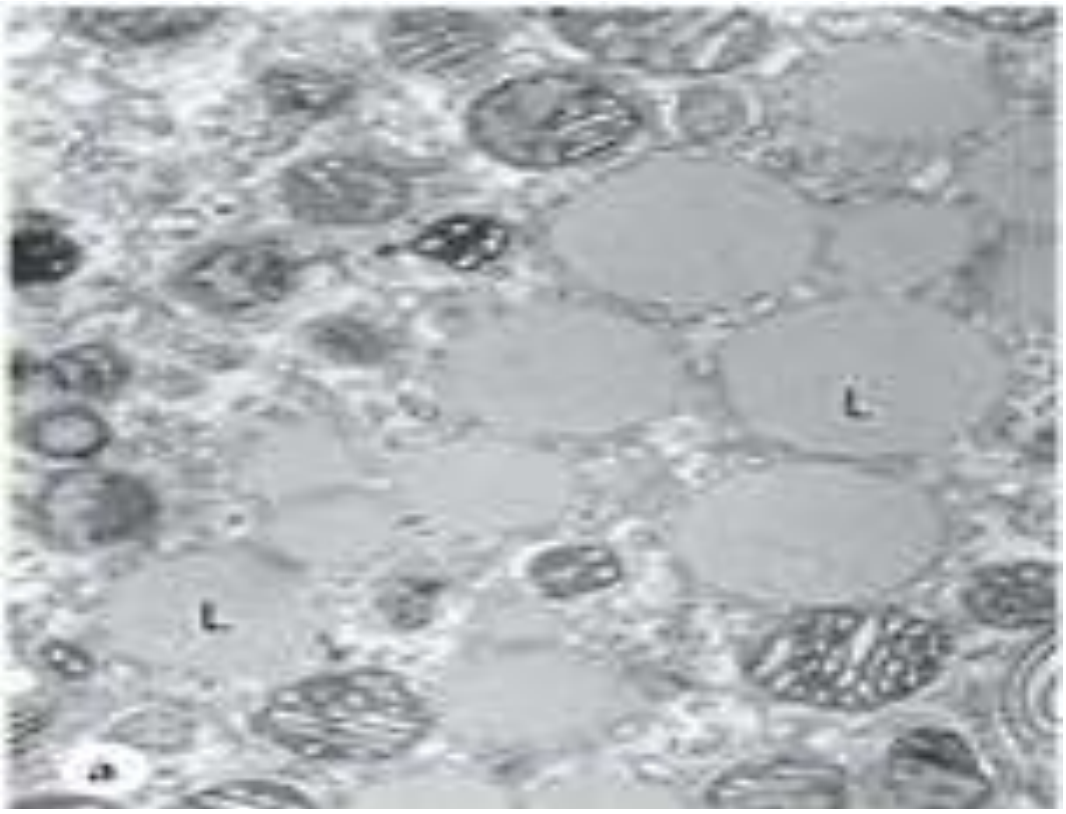
With E.M fat appear as small spherical structures with homogenous matrices.



Adipose Tissue fixed and stained by Osmium Tetroxide




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Pigments

Pigments may get into cells from outside the body and are therefore known as exogenous pigments or may be formed inside cellsand in this case they are known as endogenous pigments.

Exogenous pigments

a-The most important are carotenoid (lipochrome) pigments originating from the carotene present in colored vegetables (e.g., carrots and tomatoes) which enter the body with food.

Some types of carotenes are converted into vitamin A in the skin and serve as important supply of that vitamin.

*There are other exogenous pigments such as dust, metals or tattoo colors .

b- Carbon; Occur as black particles

• Source: Car exhausts and tobacco smoker

• Found in (Lung + lymph node)

• Fail to stain and non reactive

Endogenous pigments

Produced within the tissue cells and serve a physiological function.

They are by-product of normal metabolic process and classified into:

1- Haematogenous

2- Non haematogenous

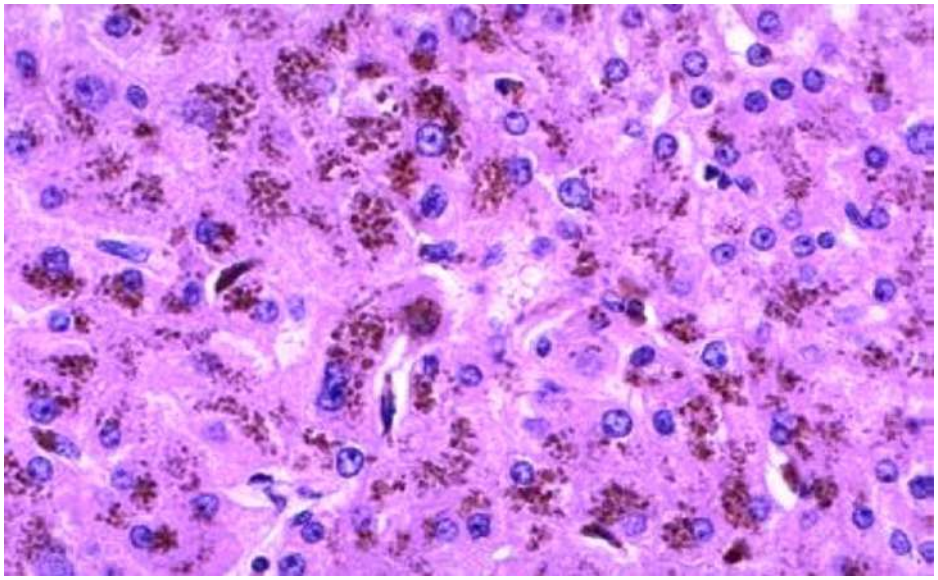
3- Endogenous minerals

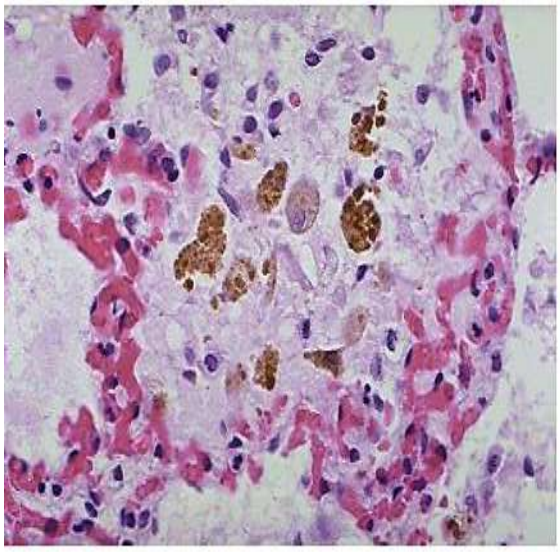
I. Haematogenous Pigments

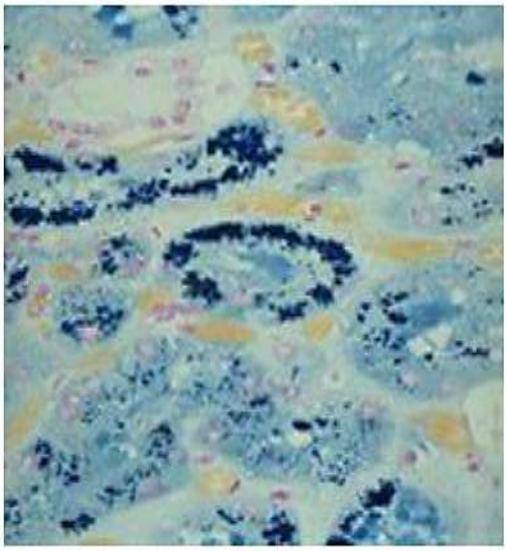
1. Heamoglobin: It is the most important pigment and beside its importance in respiration, it gives the skin its red colour.

haemosiderins, haemoglobin and bile pigment

• Haemosiderins:It results after the breakdown of haemoglobin and represents the iron – containing part are Yellow to brown granules.







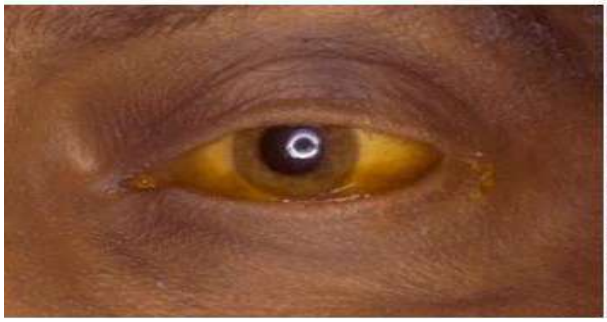
2-Myoglobin is a pigment present in muscle cells**.**

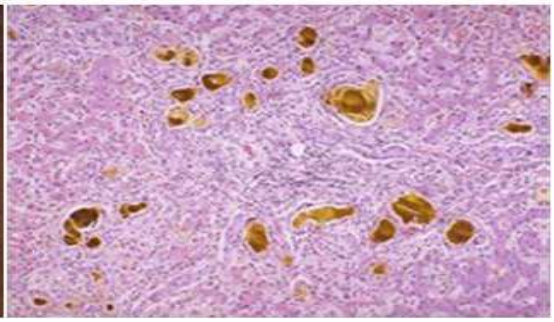
3-Bile Pigments

Yellow to brown pigment result from:

*RBCs breakdown

*Excess bile production





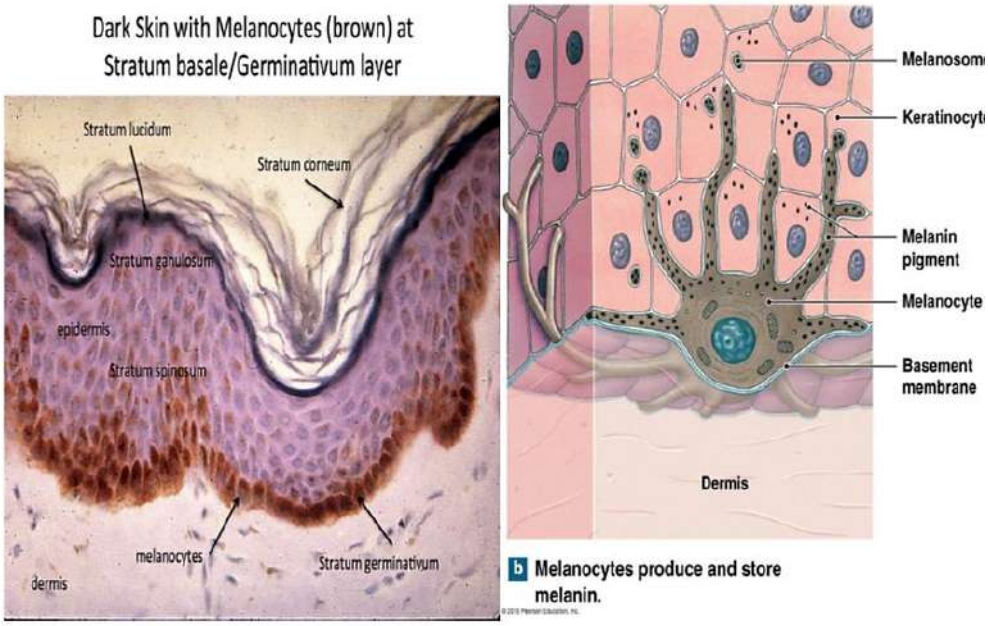
II- Non haematogenous

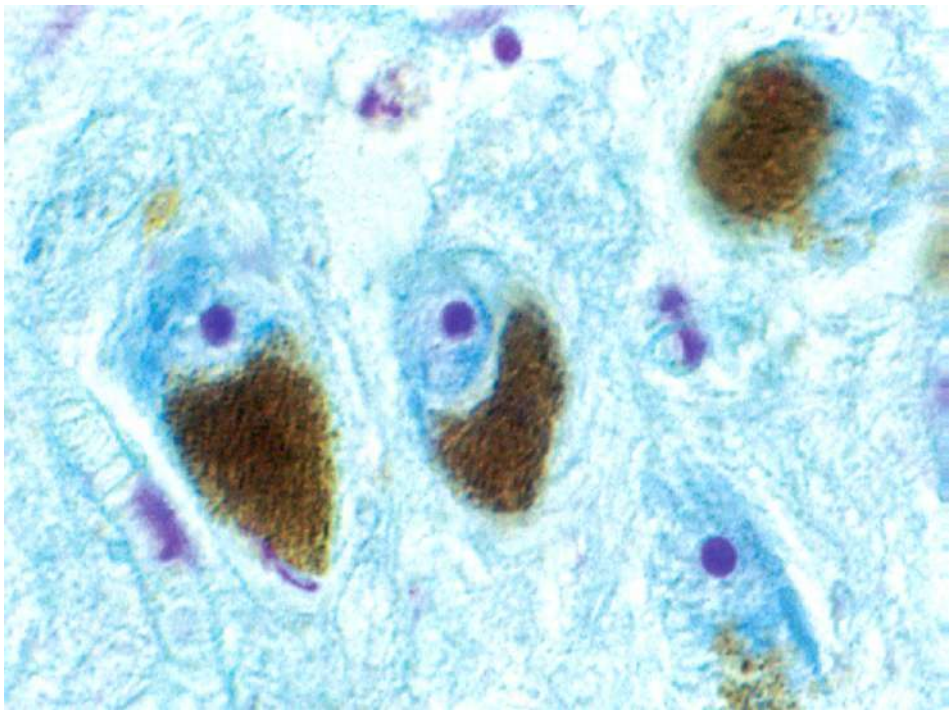
1-melanin pigments:

*This pigment is produced by a special type of cells called melanocytes present in the skin.

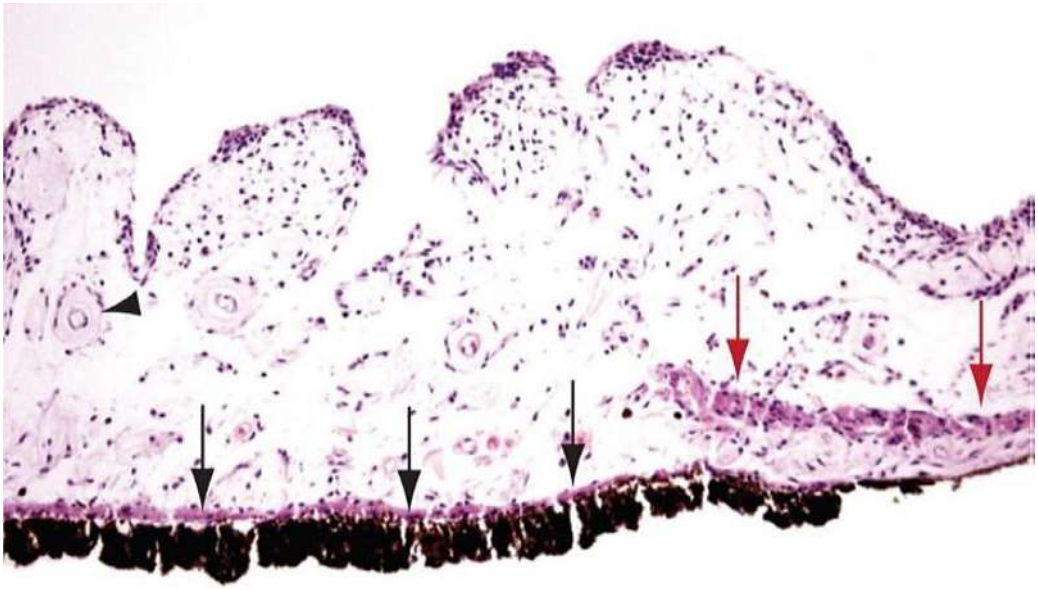
*Melanin is found in cells in the form of brownish granules

*The brown color of melanin together with the yellow colour of carotenoid pigments and the red color of blood haemoglobin constitute the color of the skin.





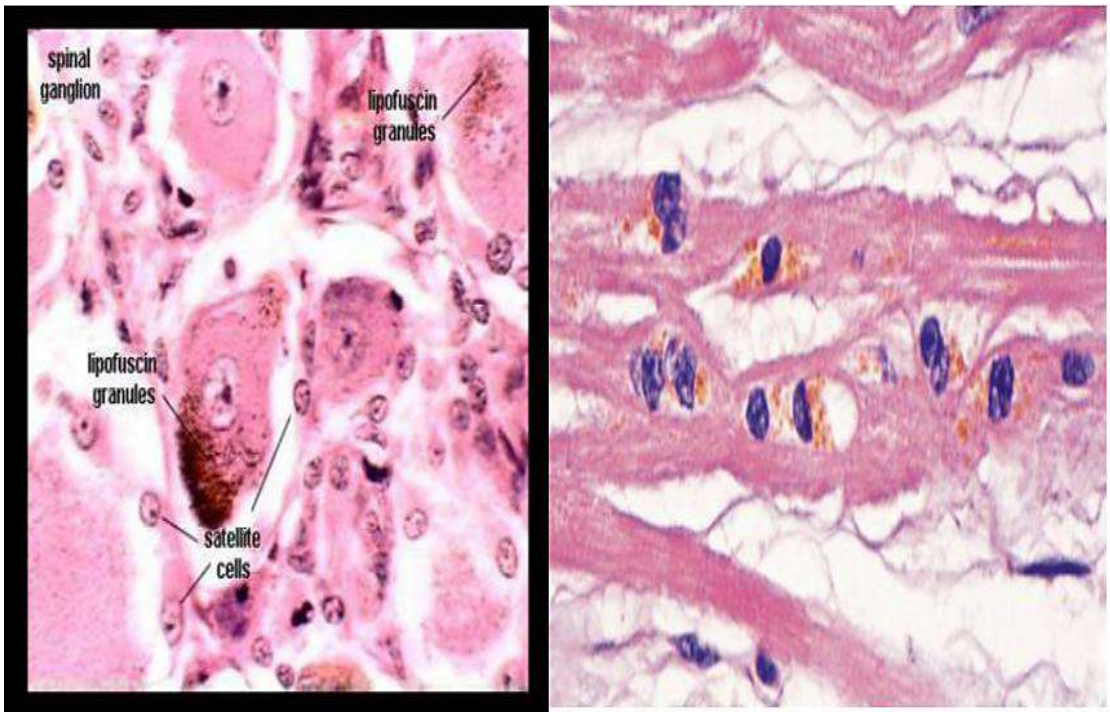
In the eye present in; iris, ciliary body & processes and pigmented epithelial layer of the retinae contain pigments.



III-Endogenous pigments

Lipofuscin: These are brownish pigments contained in membrane –bound granules in the cytoplasm of some cells such as nerve cells, liver cells and muscle cells.

They increase in amount with age and sometimes are called wear and tear pigments they result from secondary lysosomes**.**

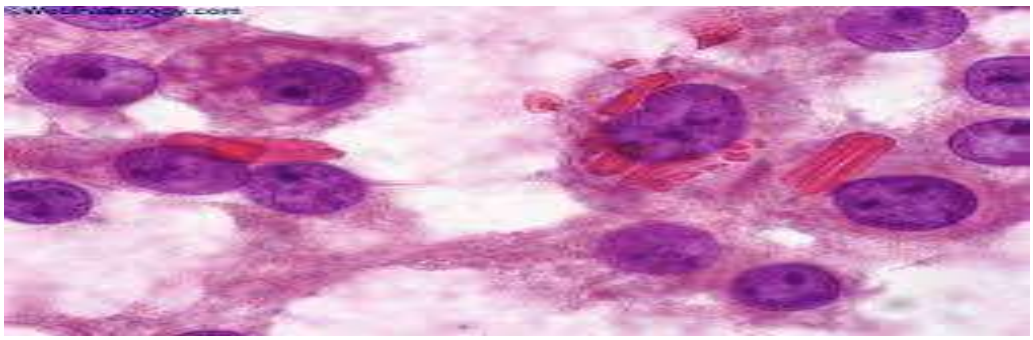


Crystals

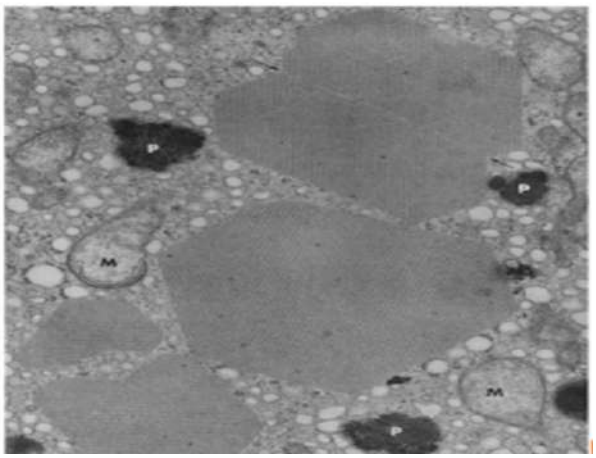
They are organized structures mainly formed of proteins and are present either free in the cytoplasm or inside ER

The interstitial cells of the testis contain a characteristic type of crystals called crystal of Reinke

Sertoli cells contain (crystalloids of charcot bottcher).



CRYSTALS OF LEYDIG CELLS



References:

1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

2- Ross, M. H., & Pawlina, W. (2006). *Histology: A text and atlas with correlated cell and molecular biology* (5th ed.). Lippincott Williams & Wilkins.

3- Wheater, P. R., Burkitt, H. G., & Daniels, V. G. (2006). *Wheater's functional histology: A text and colour atlas* (5th ed.). Churchill Livingstone/Elsevier.

4-Gartner, L. P. (2020). *Textbook of histology* (5th ed.). Elsevier.

5-Kierszenbaum, A. L., & Tres, L. (2025). *Histology and cell biology: An introduction to pathology* (6th ed.). Elsevier.

6-Lowe, J. S., & Anderson, P. G. (2024). *Stevens & Lowe's human histology* (6th ed.). Elsevier.

Lecture 5:

Cytology; Nucleus

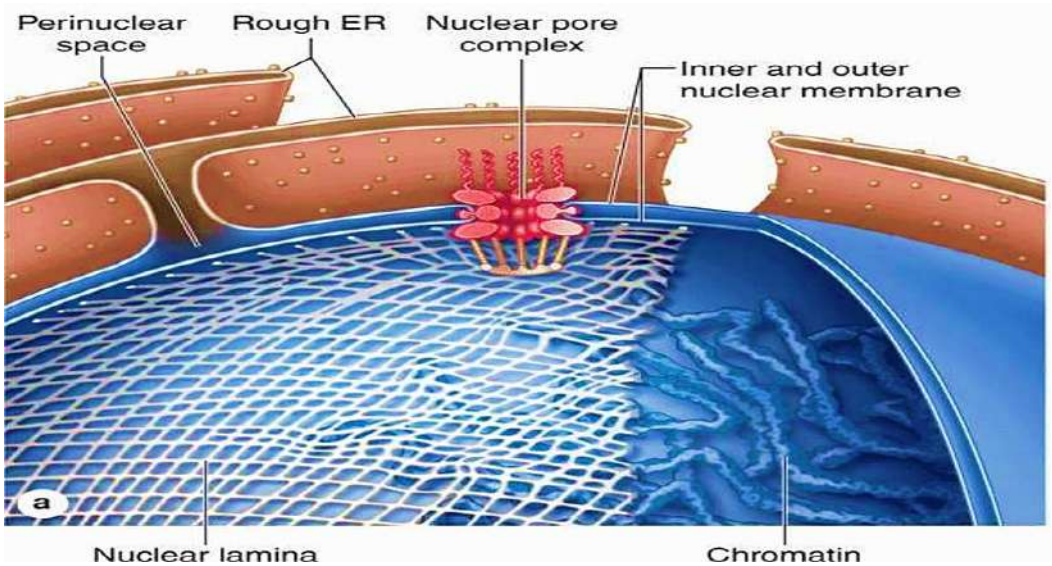
Learning objectives:

*Recognize the structure & function of the nucleolus

*Discuss the structure & function of nuclear envelope

*List the components of the nucleus

* Differentiate between types of chromatin

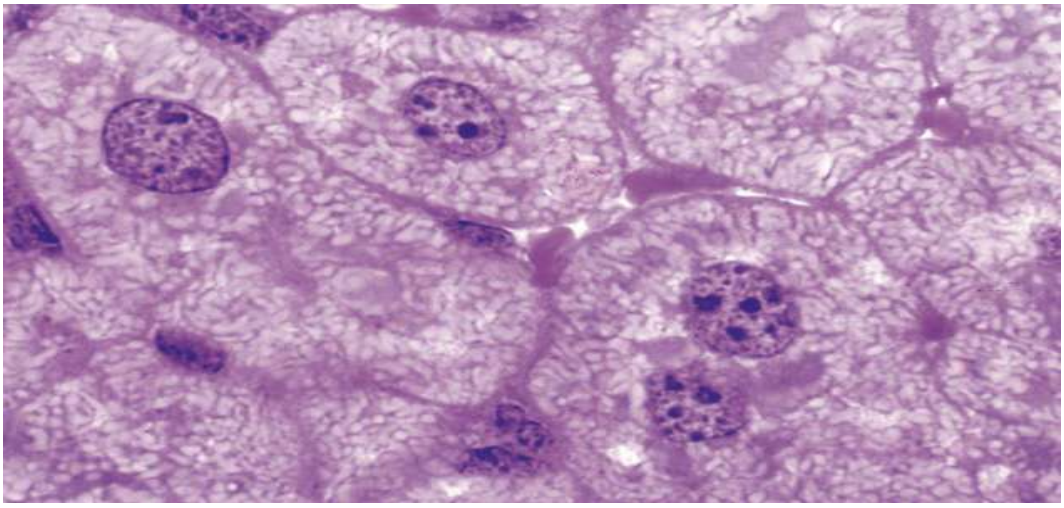


Nucleus

*All cells contain nuclei.

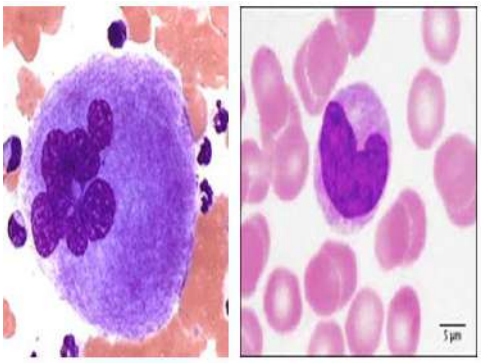
*Only mammalian RBCs are non – nucleated (a nucleated) and they are therefore called corpuscles.

*It is the part of the cell which contains the chromosomes that carry the genetic information.



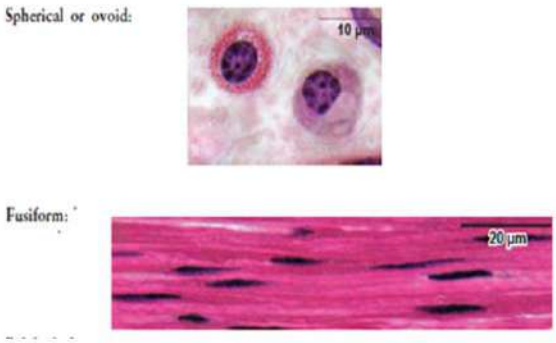
Shape:

Nuclei are of different shapes. They may be spherical, oval, indented (kidney – shaped), spiral, multilobed or lobulated.



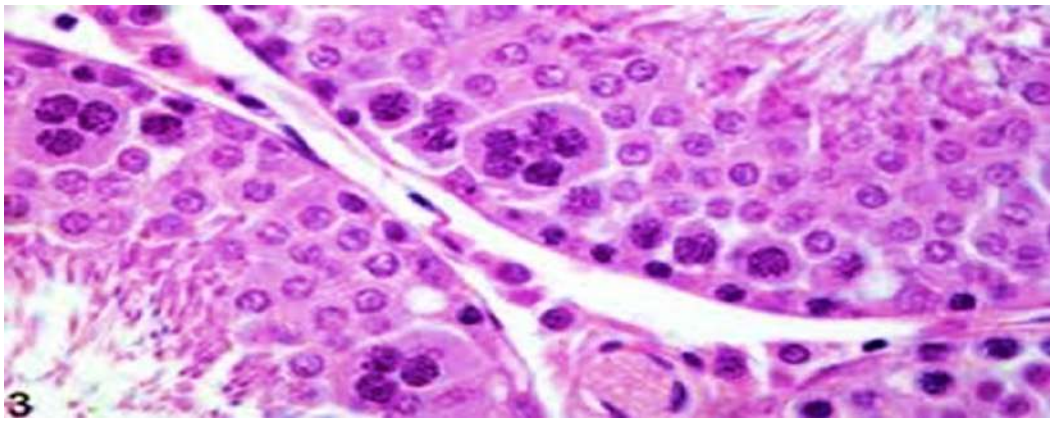
Multilobed

Kidney shaped



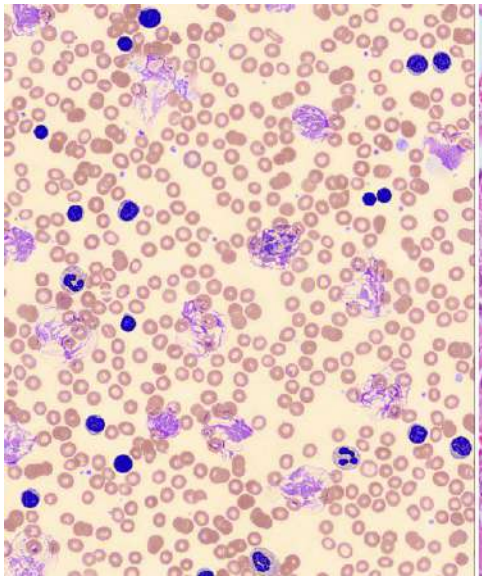
Number

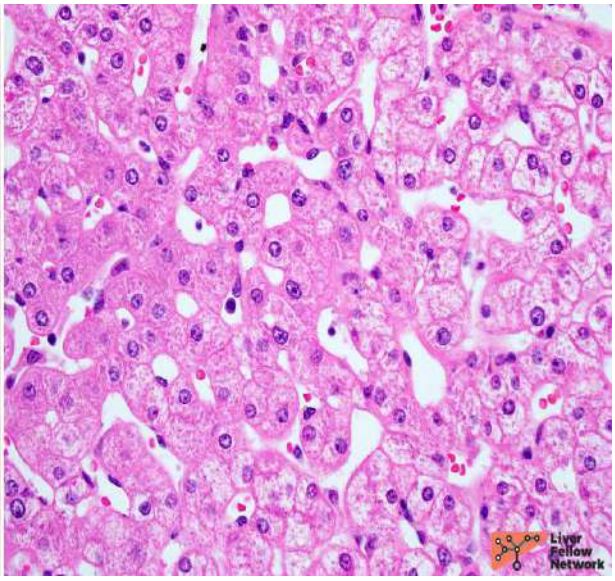
A cell usually contains **one** nucleus but there are **binucleated** cells (liver cells) and **multinucleated** cells (muscle cells). A multinucleated cell may arise either by fusion of a number of cells or by the successive division of the nucleus without the division of the cytoplasm as in the osteoclast (one of the bone cells)

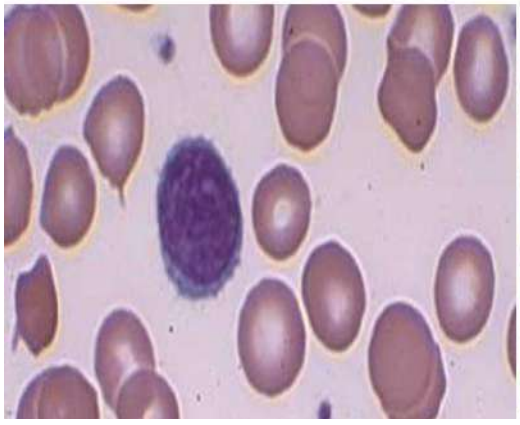


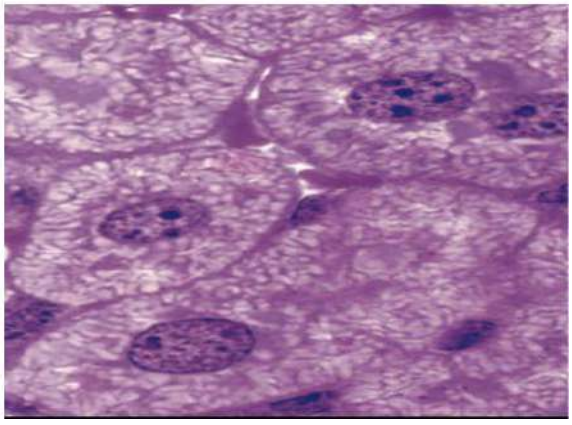
Appearance:

may be deeply stained nucleus or vesicular (open face) nucleus, always deeply basophilic due to abundance of DNA.



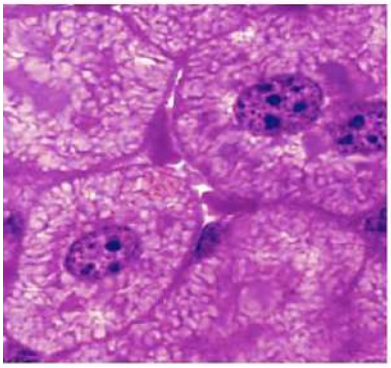


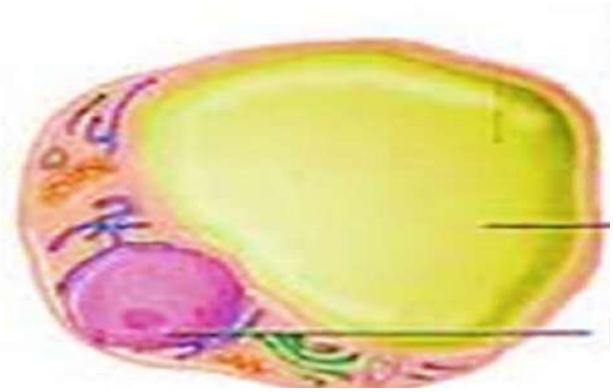




Site:

The nucleus is located in the widest area of the cytoplasm (central) or it is pushed to one side by certain structures or contents of the cell as in fat cells and plasma cells (eccentric).





Structure

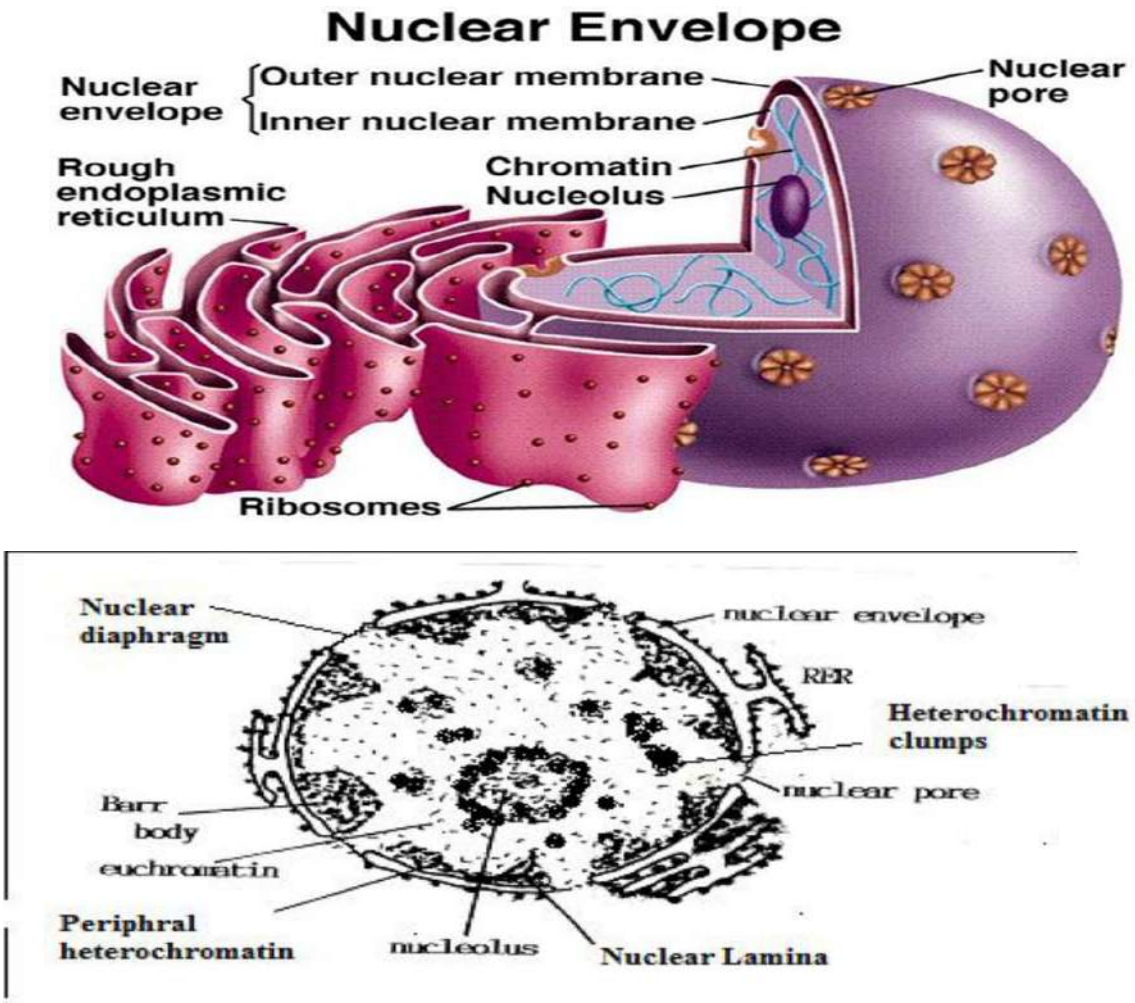
It includes the

1-nuclear envelope

2-chromatin

3-Nucleolus

4-nucleoplasm

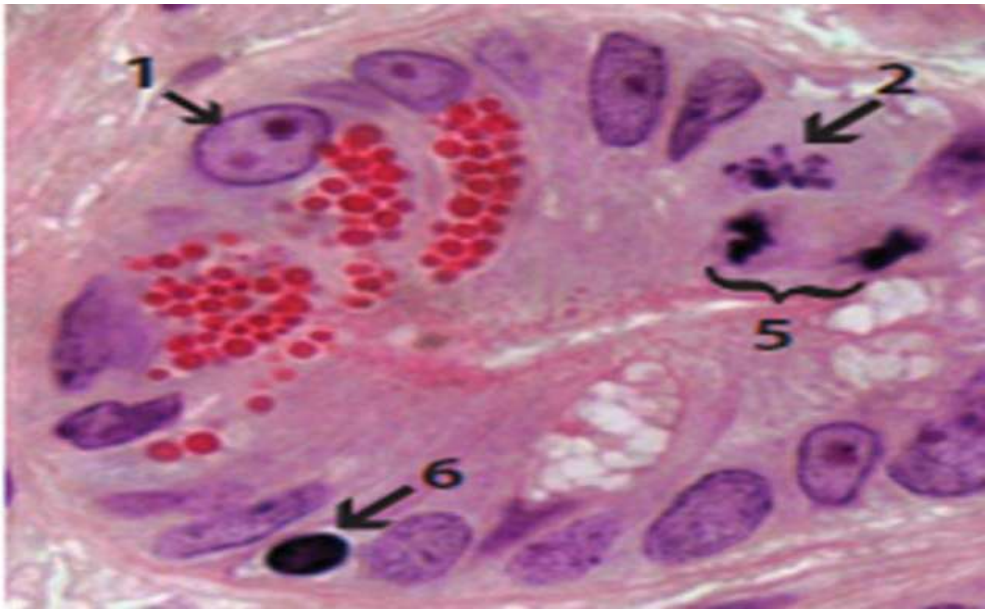


1-Nuclear envelope

Two parallel unit membranes enclosing a narrow(30-50nm) perinuclear space.

It forms a selective barrier between the nuclear and cytoplasmic compartments

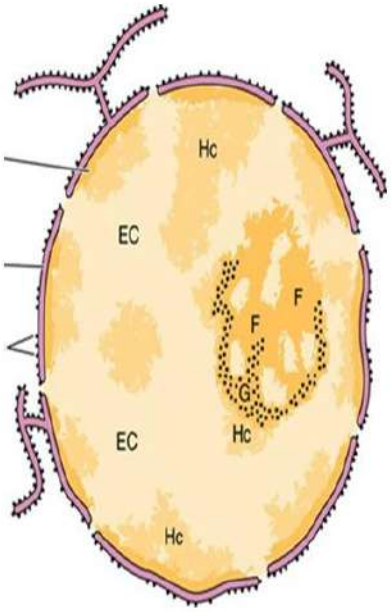
L.M: Dark basophilic membrane which surrounds the nucleus.

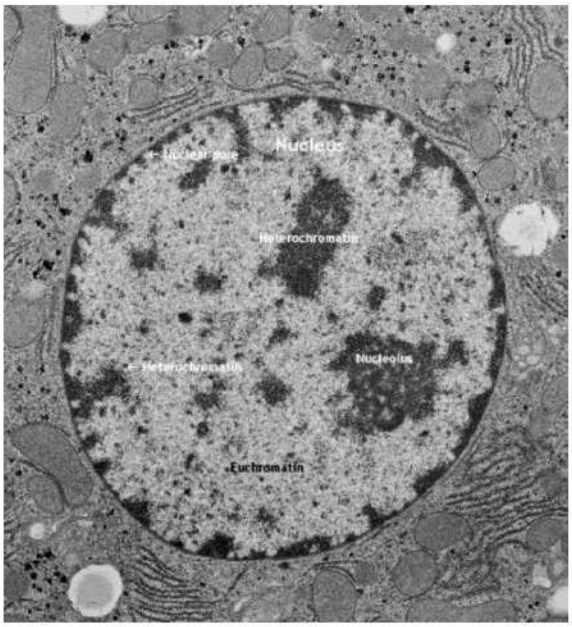


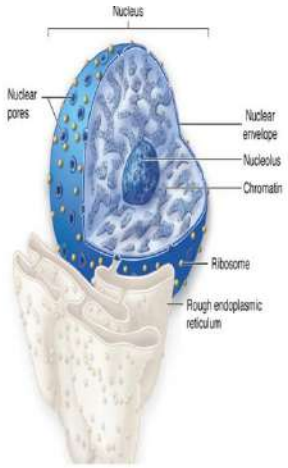
E.M.:

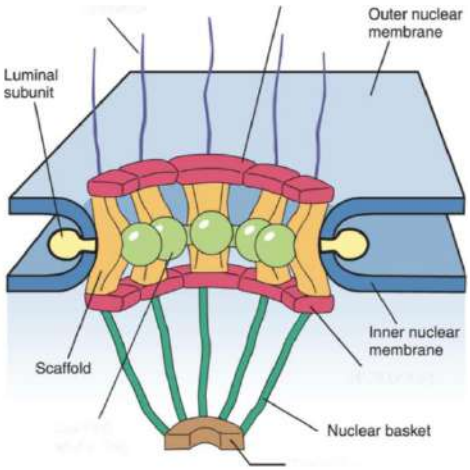
Appears as two concentric membranes separated by a perinuclear space.

Circular pores are present in the nuclear envelope and allow for the exchange of special substances between the nucleus and the cytoplasm. The pore are guarded by pore complex.









2-Chromatin

Chromatin is the substance of chromosomes

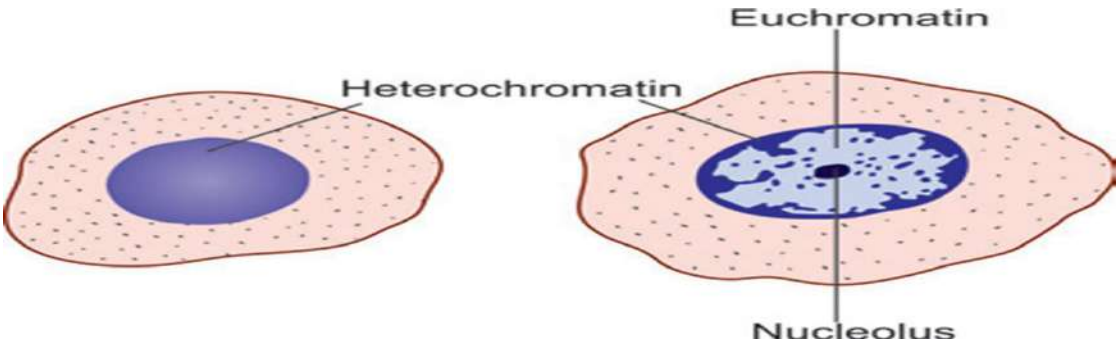
Structure: formed of DNA bound to basic protein called histones

LM: basophilic granules

There are two type of chromatin:

1-Euchromatin(active chromatin)

2-Heterochromatin(Inactive chromatin)



1-Euchromatin

-Very lightly with basic stain

-Most active chromatin

-Finely granulated dispersed chromatin representing uncoiled active parts of chromosomes

-Control protein synthesis

2-Heterochromatin

-Coiled inactive chromatin

-Visible by LM as course granules

-Stain dark with basic stain

-They do not direct any protein synthesis

Types of heterochromatin:

1-Chromatin associated with the nuclear envelope (peripheral chromatin)

2-Chromatin associated with the nucleolus nucleolar organizers

3-Sex chromatin attached to the nuclear envelope (Barr body)

4-Clumps of heterochromatin scattered in nucleoplasm

MEDICAL APPLICATION

*Barr bodies or gender chromatin permit gender to be determined microscopically in patients whose external sex organs do not permit that determination, as in hermaphroditism and pseudohermaphroditism.

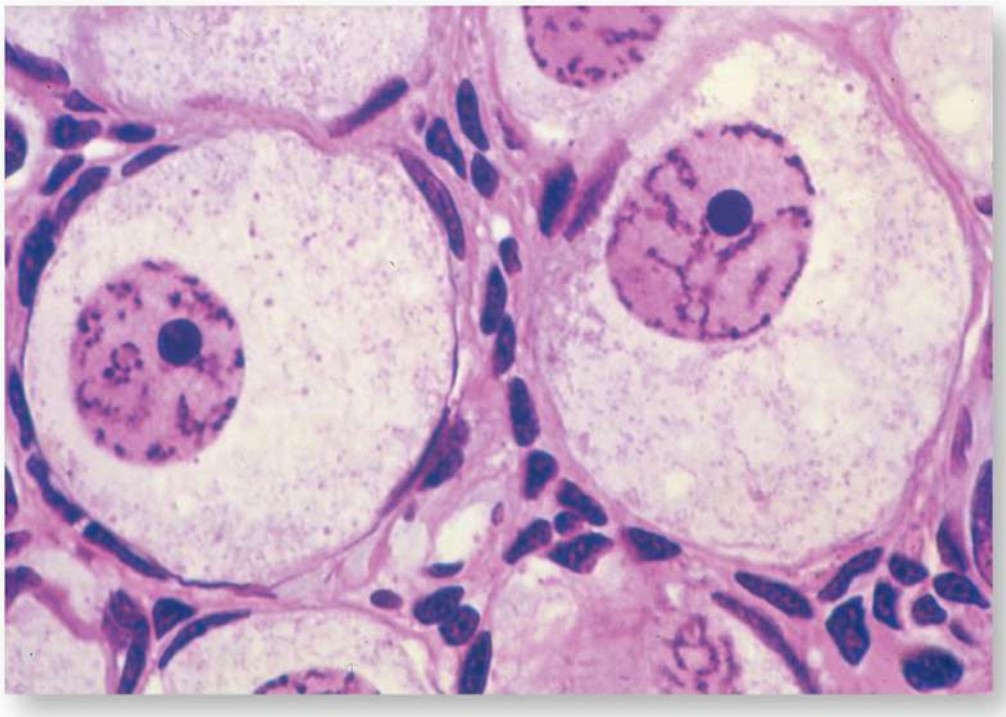
*Sex chromatin analysis also helps reveal other anomalies involving the sex chromosomes, such as the presence of XXY chromosomes (Klinefelter

syndrome), which causes testicular abnormalities and azoospermia (absence of sperm

3-Nucleolus:

*The nucleolus is a spherical basophilic structure present in the nuclei of cells active in protein synthesis.

*The intense basophilia of the nucleoli is due not to heterochromatin but to the presence of densely concentrated rRNA.



EM, the nucleolus is formed of two definite areas:

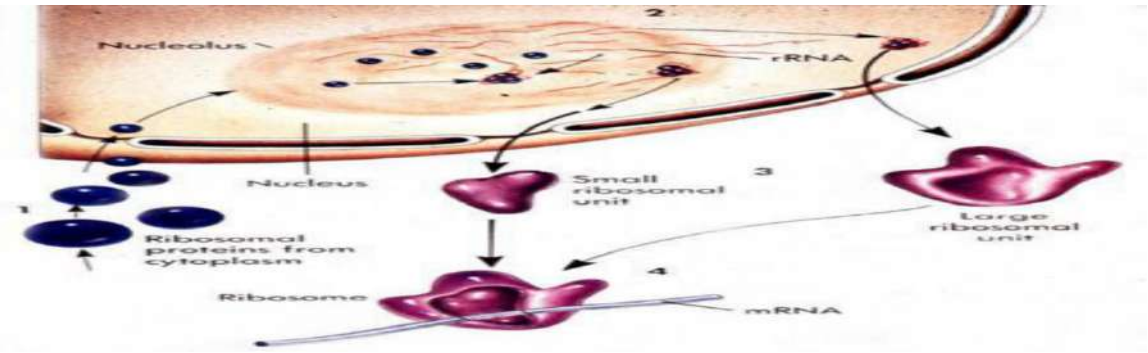
A central fibrous area (pars fibrosa) containing the chromatin filaments of the nucleolar organizer chromosomes as well as filamentous RNA

The outer portion (pars granulose) is granular and is composed of RNA particles quite similar to ribosomal subunits

The filamentous components are **pars fibrosa** which contain chromatin filaments as well as filamentous RNA (rRNA synthesis)

While the granular components **pars granulosa** which contain ribosomal subunits are the sites of ribosomal assembly

The molecules of rRNA synthesized and modified in the nucleolus very quickly associate with the ribosomal proteins coming from the cytoplasm.



4-Nucleoplasm:

Nucleoplasm = nuclear sap = matrix = karyolymph:

A clear colloidal medium for nuclear contents and RNA movement.

It is formed of proteins, carbohydrates, minerals and ions.

According to amount of matrix the nucleus may be:

Open-face(Vesicular) nucleus: with great amount of matrix (as hepatocytes …).

Condensed (Dense) nucleus: with less amount of matrix (as lymphocytes …).

Function of nucleus:

1. The nucleus houses the genes

2. Control all the structural and functional activity of the cell.

3. Cell division

References:

1-Junqueira, L. C., & Carneiro, J. (2005). ***Basic histology: Text & atlas*** **(11th ed.). McGraw-Hill.**

2- Ross, M. H., & Pawlina, W. (2006). ***Histology: A text and atlas with correlated cell and molecular biology*** **(5th ed.). Lippincott Williams & Wilkins.**

Lecture 6

Cytology; Chromosome

Learning objectives:

*Recognize the number of chromosomes (in somatic cells & in gametes

*Discuss the steps of DNA packing to form chromosomes

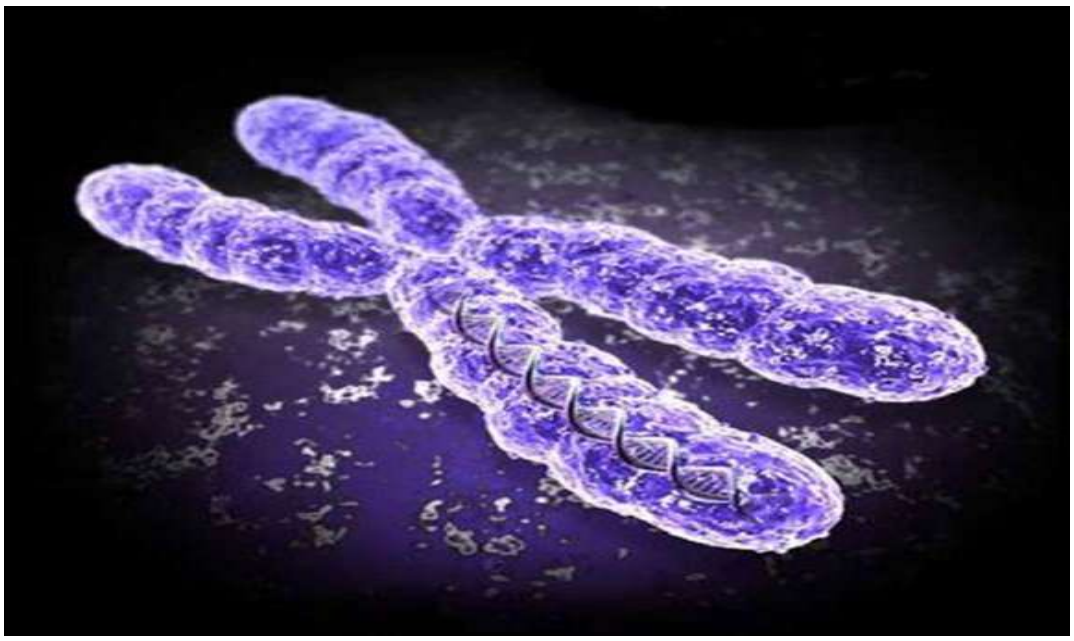
*List the methods of studying chromosomes

*Differentiate between the endocytosis & exocytosis

-Chromosome is a coiled DNA molecule within the cell's nucleus that carries an individual's GENETIC CODE.

- Chromosome's structure is loose and indistinguishable. Only in the stage of cell division immediately before the cell divides (the metaphase)

- It appear as a compact, rod like structure and can be seen under a microscope after applying a special dye to the cell.



Bio NOTES: Chromosomes and Meiosis

There are two types of chromosomes found in cells:

autosomes

sex chromosomes

Your body cells have 23 pairs of chromosomes

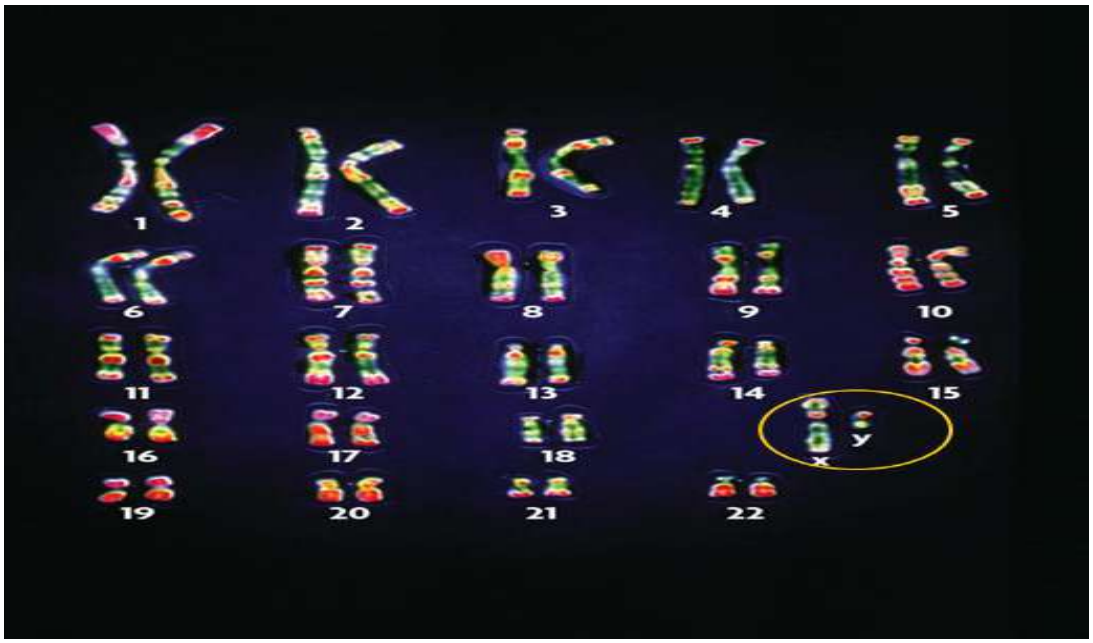
homologous pairs of chromosomes have the same structure- for each homologous pair, one chromosome comes from each parent

Chromosome pairs 1-22 are autosomes.

Sex chromosomes, X and Y, determine gender in mammals

XX = female

XY = male



CHROMOSOMES
in (somatic cells)

*In human cells,there are 46 chromosomes (diploid number) comprising 22 homologous pairs, (the autosomes) and 2 sex chromosomes.

*Homologous means that the members of each pair of autosomes have the same length of DNA and code for the same proteins.

HOMOLOGOUS CHROMOSOMES:

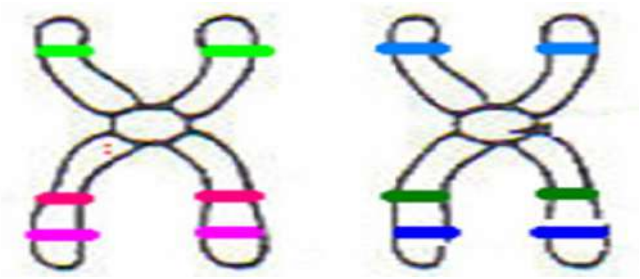
SAME SIZE

SAME SHAPE

CARRY GENES for the SAME TRAITS

BUT _____!

(Don't have to have the SAME CHOICES)



Key Terms:

Diploid refers to complete chromosome sets in a cell (or 2N)

N= the number of chromosomes in a gamete (sex cell)

N chromosomes from the female parent + *N* chromosomes from the male parent= 2N

For humans, 2N = 46

Haploid refers to single chromosomes in a cell (or N)

Gametes (i.e. sperm, egg sex cells) are haploid cells

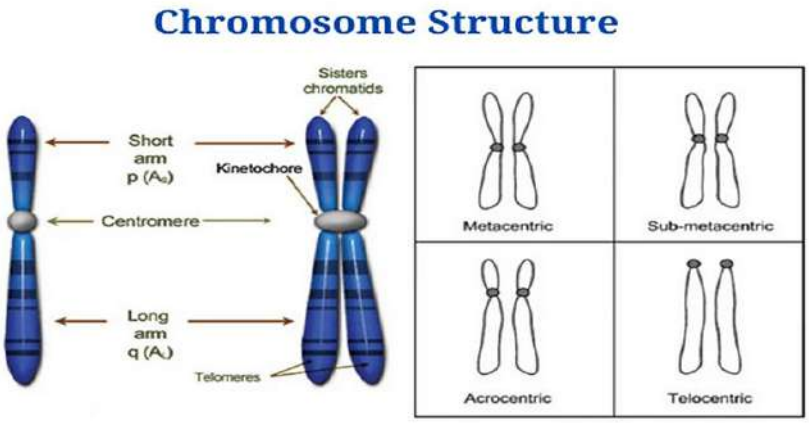
For humans, N = 23

Diploid means there are the full number of chromosomes in cells

Haploid means that there are half the number of chromosomes in cells

CHROMOSOMES
in (Gonadal cells)

Gametes (sex cells) contain haploid number of chromosomes(23),i.e one from each homologous pair. The member of each chromosome pair assigned to a particular gamete is random so, any gamete contains a mix of chromosomes from the mother and father.



*At the point of attachment of the 2 arms of the anaphase chromosome, there is a constriction called primary constriction or the centromere.

*The chromosome is densely stained except the region of the centromere which appears rather transparent, and contains a small granule, the spherule.

*Some chromosomes also have a secondary constriction. It resembles the primary one in having a fixed position on the chromosome, but it does not contain a spherule.

*Secondary constrictions in the nucleolar organizers e.g., chromosomes number 13, 14, 15, 21, and 22 in man, are associated with nucleolar formation.

Types of chromosomes

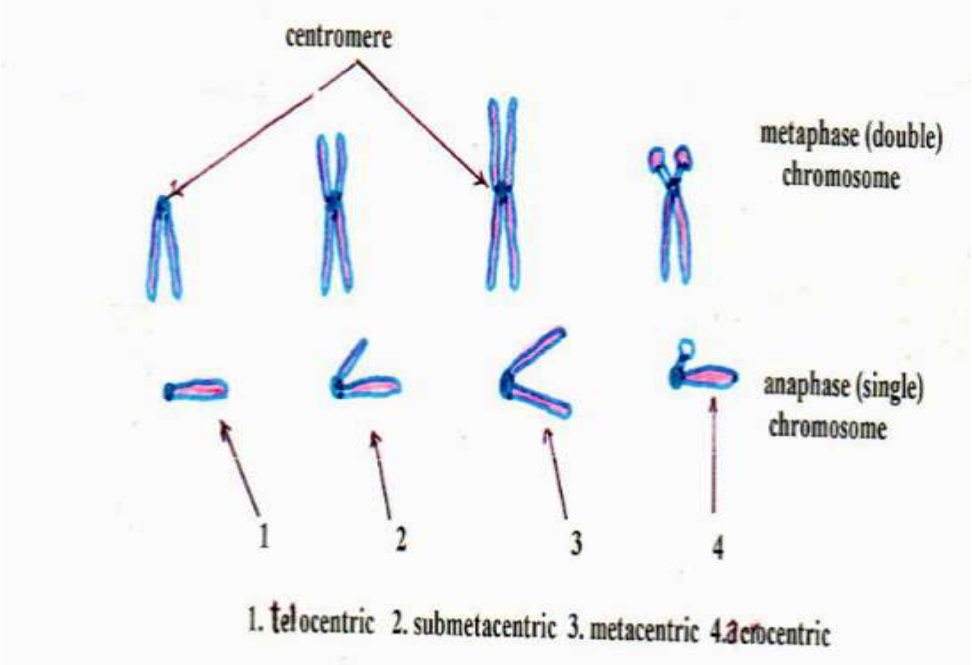
Chromosomes are classified into 4 types according to their shape during metaphase or anaphase depending on the position of the centromere

1-Telocentric chromosomes: having the centromere at the ends

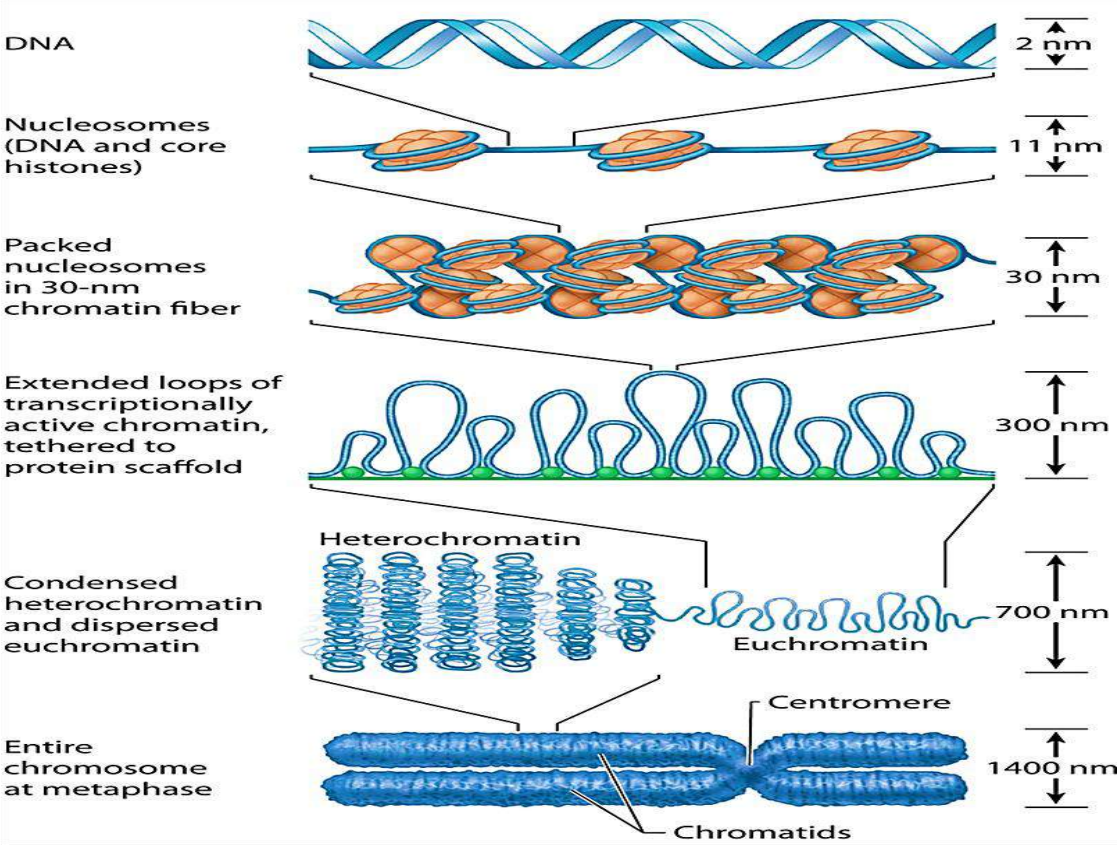
2-Submetacentric chromosomes: Having unequal arms and have (L) shape in the anaphase stage.

3-Metacentric chromosomes: Having arms almost equal in length and are thus V- shaped in anaphase.

4-Straight or acrocentric: These are club – like chromosomes having a very short arm and a much longer one.



DNA packing to form chromosomes

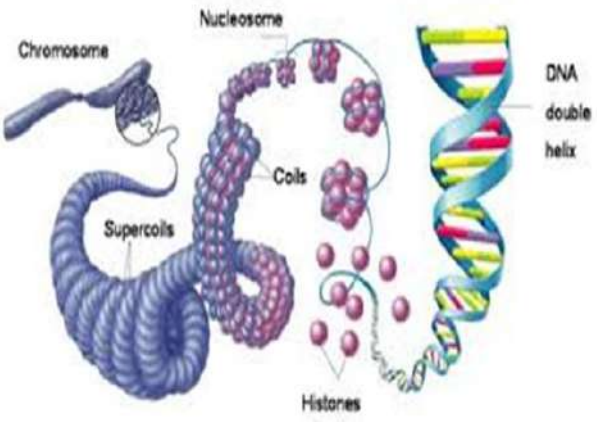


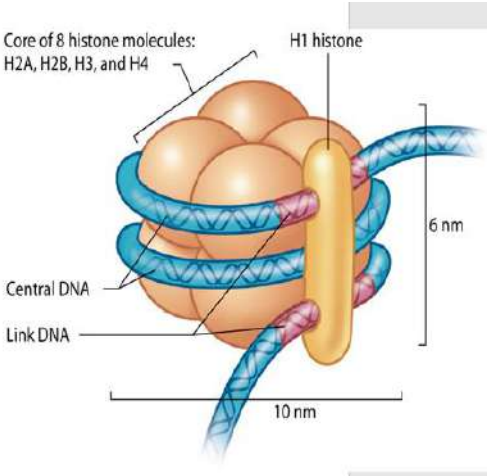
*Several orders of DNA packing occur in chromatin and during chromatin condensation of mitotic prophase.

*It begins with the 2-nm DNA double helix, followed by the association of DNA with histones to form 11-nm filaments of nucleosomes connected by the DNA ("beads on a string").

*Nucleosomes on the DNA then interact in a manner not well understood to form a more compact 30-nm fiber.

*For transcription, DNA forms loops





Karyotyping

► It is the study of the number, type and arrangement of chromosomes of an individual.

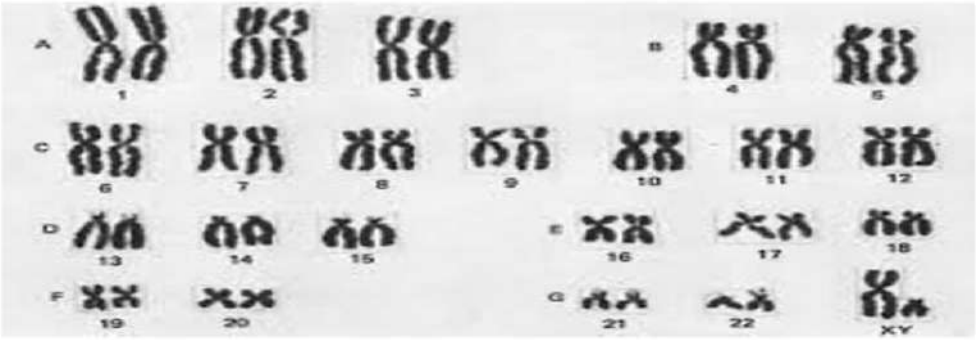
▪ According to the length of chromosome arms and position of the centromere, the chromosomes are numbered serially from l to 22 and arranged into 7 groups (A, B, C, D, E, F and G) from longer to shorter ones.

Method of human karyotyping:

1-The cell is induced to start mitosis.

2-The mitosis is arrested at the metaphase stage using colchicines (an antimitotic drug). At the metaphase stage, the chromosomes are short, condensed, and seen very clearly.

3- Spreading of the chromosomes is done, then they are photographed to make the chromosomal map (karyotyping)



Human karyotypes

Genetic disorders:

1-turner syndrome

2-klinfelter syndrome

3-supermale syndrome

4-superfemale syndrome

References :

1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

2- Ross, M. H., & Pawlina, W. (2006). *Histology: A text and atlas with correlated cell and molecular biology* (5th ed.). Lippincott Williams & Wilkins.

3- Wheater, P. R., Burkitt, H. G., & Daniels, V. G. (2006). *Wheater's functional histology: A text and colour atlas* (5th ed.). Churchill Livingstone/Elsevier.

4-Gartner, L. P. (2020). *Textbook of histology* (5th ed.). Elsevier.

Lecture 7

Cell cycle, development and cell death

Learning objectives:

-Recognize the relation between cell differentiation & cell cycle

-Discuss apoptosis & know its characteristic

- List the features phases of cell cycle & the events occurring in each phase..

- Differentiate between cell death by apoptosis & necrosis..

Cell cycle

The body contains two types of cells:

1. Somatic cells constituting the body of the organism.

2. Gonadal cells (gametes) only in testes (sperms) and ovaries (ova).

*A somatic cell begins as one of the daughter cells of mitotic division. After division, the cell may undergo a period of intense protein synthesis to form granules, filaments and cell organelles and becomes specialized or differentiated.

*On the other hand, the cell may rather go through a number of successive stages of variable durations till divides again. The time from the end of mitosis to the end of the following mitosis is known as cell cycle .

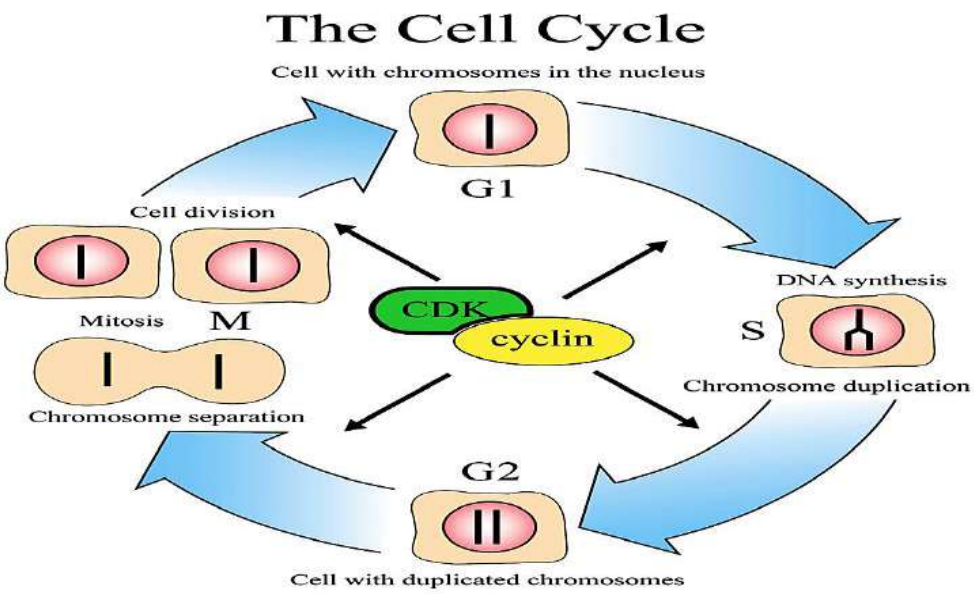
*A fertilized egg (zygote) divides by mitosis to produce 2 identical daughter cells ,,,,and so on.

*Some of these daughter cells progressively specialize and produce the terminally differentiated cells.

*on the other hand, may rather go through a number of successive stages of variable durations till divides again

*The time from the end of mitosis to the end of the following mitosis is known as life cycle of the cell.

*However, most tissues retain some undifferentiated cells, stem cells that are able to divide and replace the differentiated cells.



*Mitosis is the visible manifestation of cell division.

*Another process in which DNA replication occurs is the interphase.

This alternation between mitosis and interphase is known as the cell cycle.

Stages of cell cycle

1. Gap 1 stage (G1): The period starting after the completion of mitosis and during which the nucleus does not show any increase in DNA content. A good amount of protein is formed during this stage.

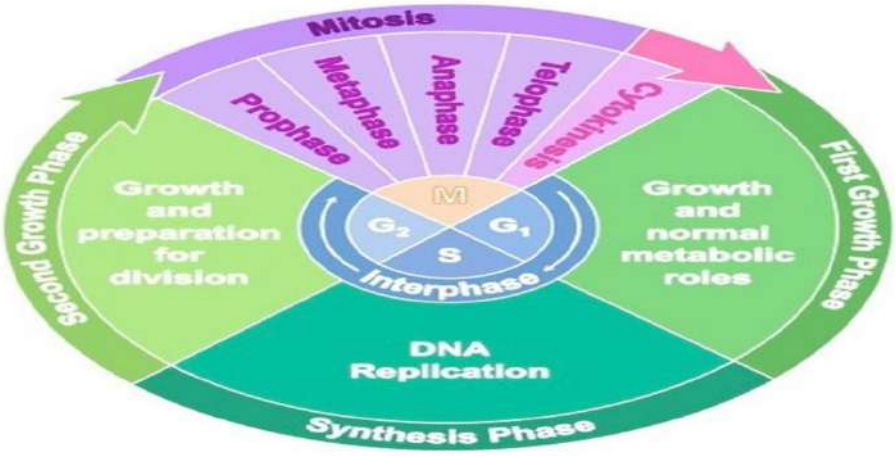
2. Synthesis (S) stage: Starts after G1, and represents the period of DNA and histone (a special nuclear protein) duplication. Any cell entering the S-stage must continue the cycle.

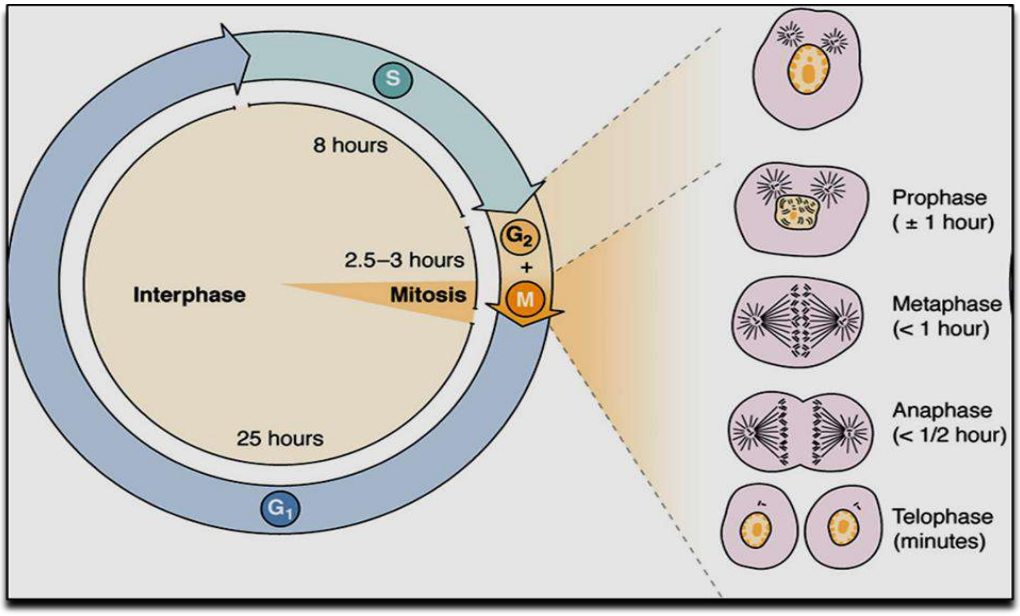
Chromosomes are formed of two filaments or chromatids at the end of this stage and may be called double chromosomes.

3. Gap2 (G2): Is another period of growth. During this stage, the cells synthesize a great amount of cytoplasmic material such as proteins, RNA etc. and the organoids are increased in number.

4. Mitosis (M):

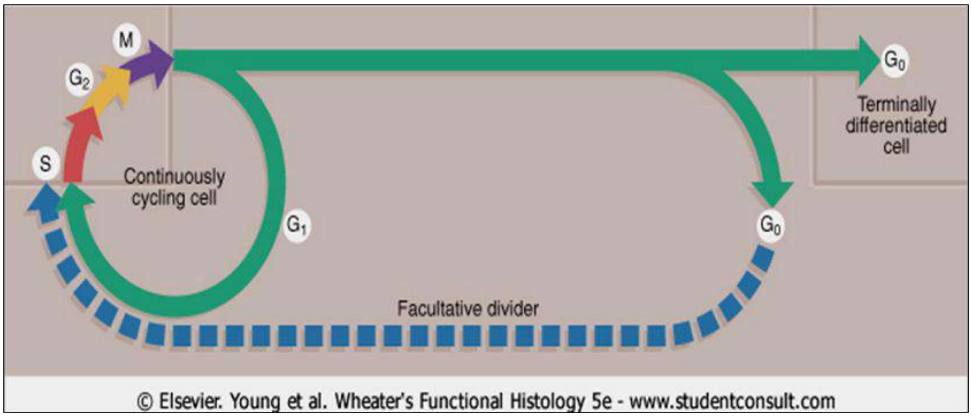
It the period when the chromosomes shorten, align in the middle of the cell and separate into two identical sets each of which forms the chromosomal content of the new daughter cell. The time required for a call cycle to be completed ranges from 10 to 30 hours.



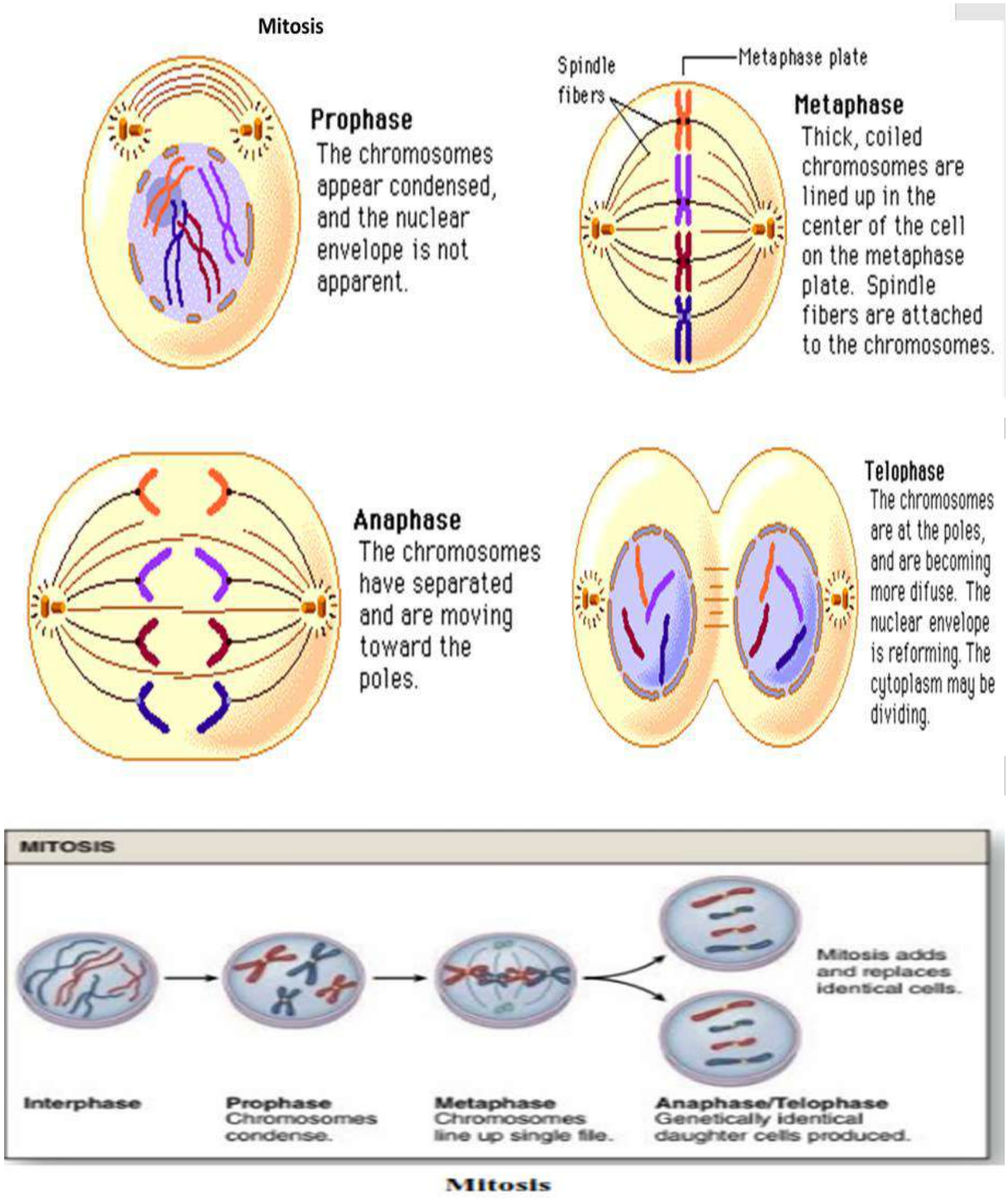


G0 phase. As new postmitotic cells specialize and differentiate, cell cycle activities may be temporarily or permanently suspended, those differentiated cells, as liver cells retain the capacity to re-enter the cell cycle when suitably stimulated.

Others, including muscle and nerve cells, are terminally differentiated



Stages of mitosis



Mitosis

Controls at cell cycle checkpoints

Each phase of the cell cycle has one or more checkpoints where the quality of specific cell activities is checked.

Progression to the next phase of the cycle does not occur until all activities of the preceding phase are completed satisfactorily.

Three important checkpoints are shown here, including:

1-The start or restriction checkpoint just before the start of S

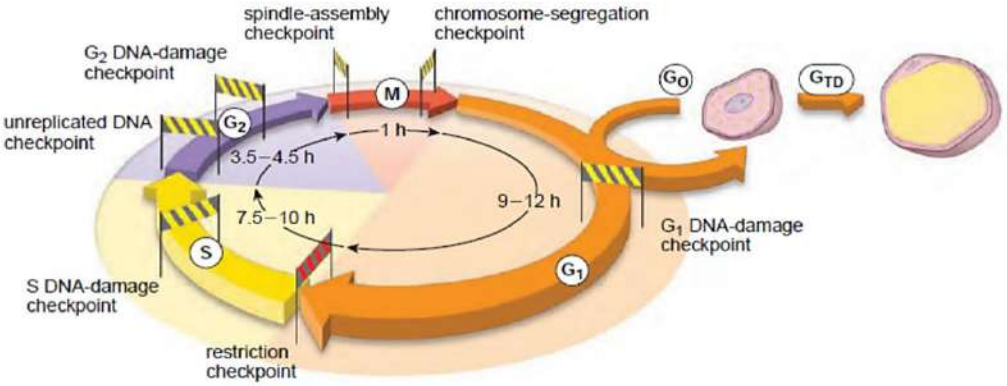
2-The G2 /M checkpoint that ensures that DNA replication is complete

3-The metaphase spindle checkpoint that ensures that all chromosomes will be segregated

Overall progression in the cycle is regulated by proteins called cyclins and cyclin-dependent kinases (CDKs)



Figure 18-12 Essential Cell Biology 3/e (© Garland Science 2010)



Cell death

*Cell death can be defined as "the irreversible cessation of vital activities inside the cell".

*Cells, like individuals, have a life span i.e. they die.

*Death result from the normal process of development where certain cells become functionless and their death is necessary otherwise the function of the individual is altered. Before the cell dies, it goes into a series of changes known as the cell senescence (growing old).

Nuclear changes indicating cell death

1**. Pyknosis:** It is a shrinkage of nuclear material which appears darkly stained.

2. Karyorrhexis: the nucleus disintegrates into tiny fragments.

3. Karyolysis: the nucleus dissolves and disappears.

Types of cell death:

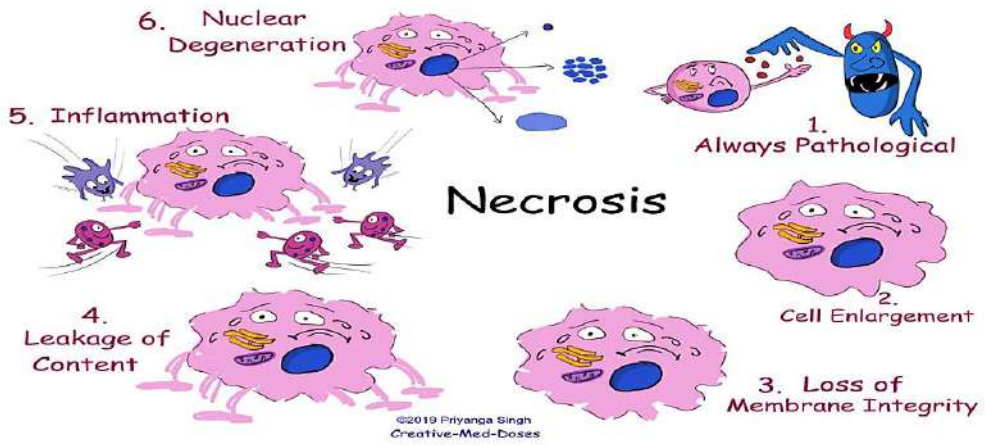
1- Necrosis: Is cell death due to exogenous factors such as injury, toxins or hypoxia.

*The cell swells.

*The chromatin is clumped.

*The cytoplasm becomes weakly stained and the organoids are destroyed.

*The cell bursts and its contents are scattered and then removed by phagocytes.



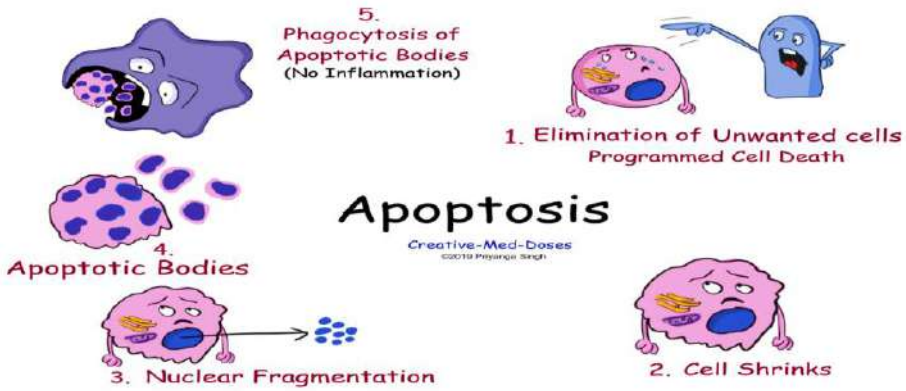
2- Apoptosis: programmed cell death Is the normal cell death due to internal activities within the cell as a result of genetic information.

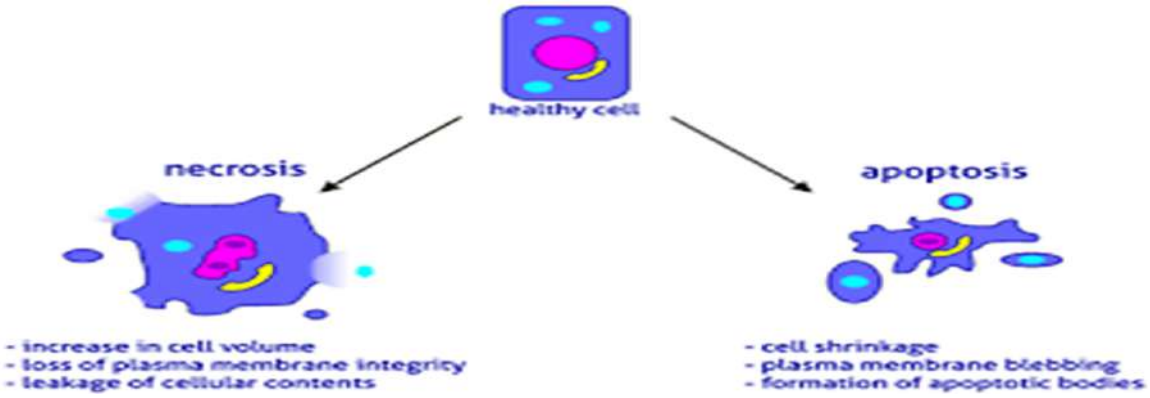
*During apoptosis there is condensation of chromatin and destruction of DNA due to activation of the enzyme endonuclease.

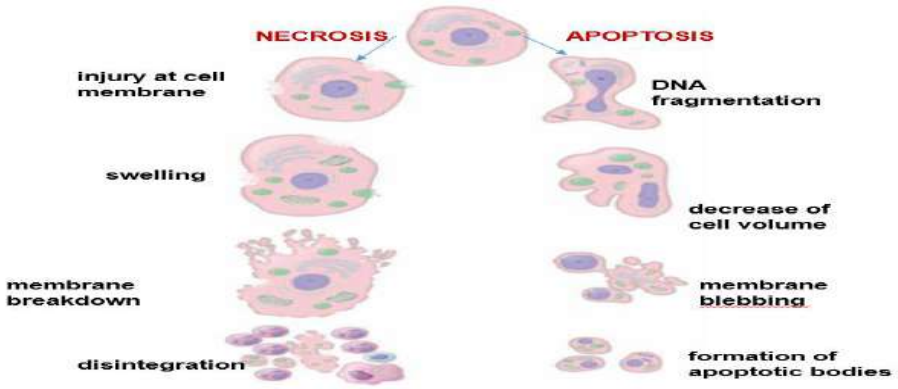
*The cell does not swell. Instead, the cell volume decreases.

*The organoids remain normal.

*At last the cell fragments into a number of membrane-bound globules (apoptotic bodies) which are then phagocytozed by the neighboring cells.







References:

1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

2- Ross, M. H., & Pawlina, W. (2006). *Histology: A text and atlas with correlated cell and molecular biology* (5th ed.). Lippincott Williams & Wilkins.

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Lecture 8

STEM CELLS & TISSUE RENEWAL

Learning objectives:

*Define stem cells

*Recognize the progenitor cells

*Identify types of stem cells

*Determine the tissue's potential to regeneration

*List the tissues with rapidly renewing cell populations

*Identify the tissues with stable cell populations

*Define the terminally differentiated cells

Definition of stem cells

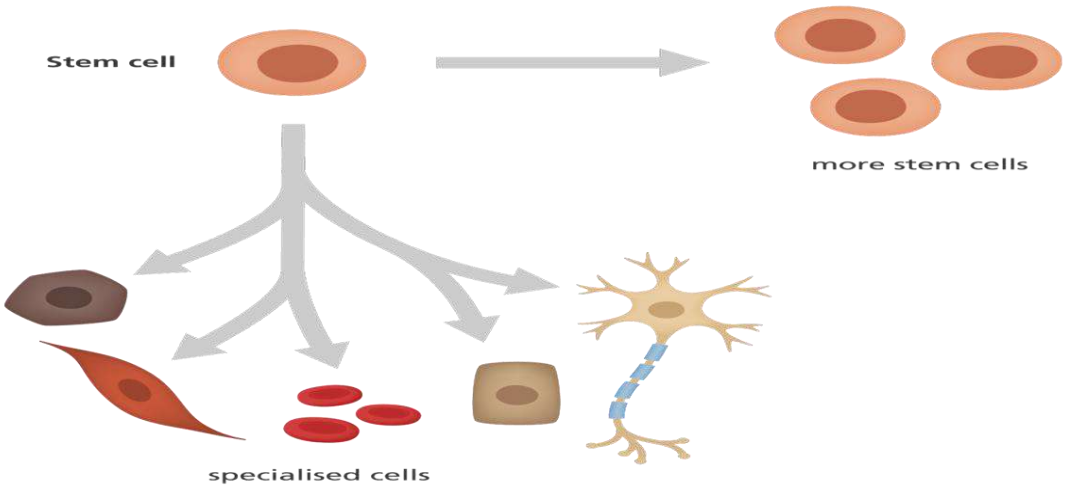
Stem cells are undifferentiated cells that can differentiate into specialized cells and can divide (through mitosis) to produce more stem cells.

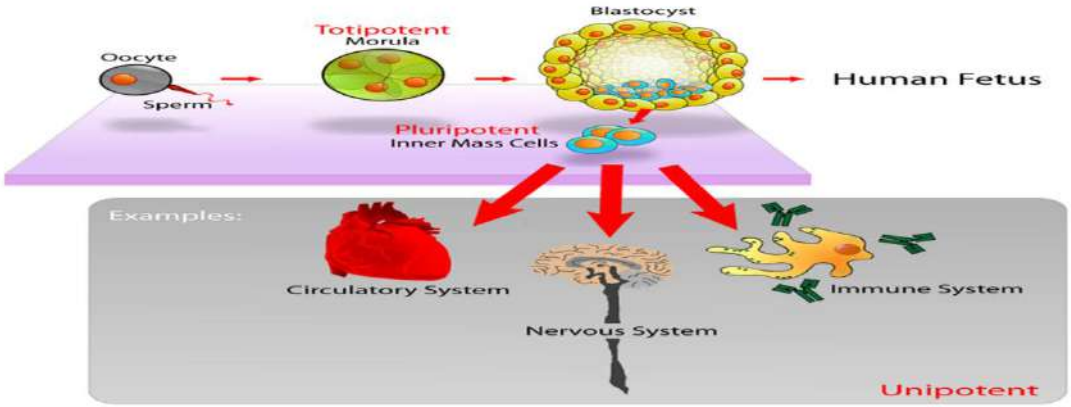
Their progenitors would either remain as stem cells or differentiate to be specialized cells.

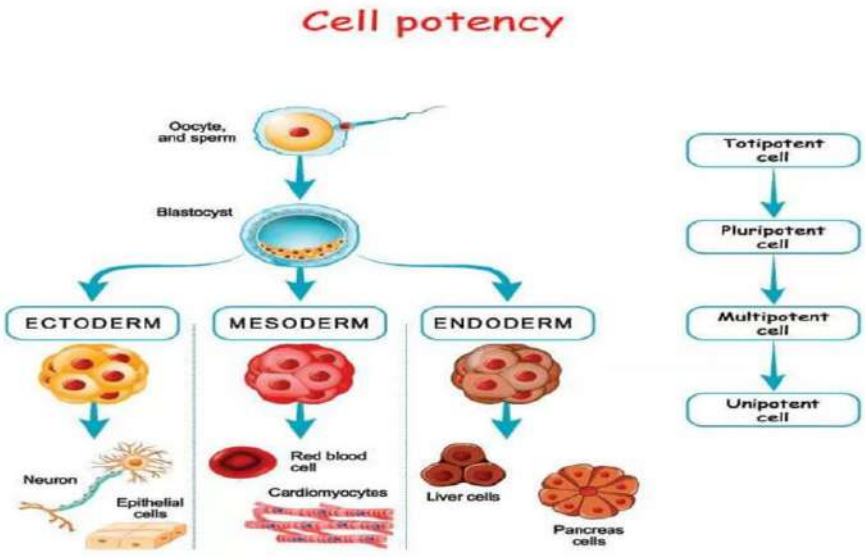
Types of stem cells

1-Embryonic stem cells: which are isolated from the inner cell mass of blastocysts.

They are pluripotent cells (able to differentiate to all derivatives of the three germ layers: (ectoderm, endoderm, and mesoderm).







In a developing embryo:

*Stem cells can differentiate into all the specialized cells—ectoderm, endoderm and mesoderm.

*Their clinical use is limited (due to ethical considerations and their liability to form teratomas).

2- Adult stem cells:

which are found in various tissues.

In adult, stem cells and progenitor cells act as a repair system for the adult body tissues.

▪ They are subdivided into 2 main categories:

A. Hemopoietic stem cells :

They are mainly found in the bone marrow

They are able to differentiate to the different types of blood cells.

B. Mesenchymal stem cells:

➢ They are multipotent cells (can be differentiate into fat cells, muscle cells, cartilage cells, bone cells etc.) according to their surrounding microenvironment.

➢ They are found in the bone marrow, adipose tissue and other sites in the different organs.

➢ Used therapeutically in treating many diseases.

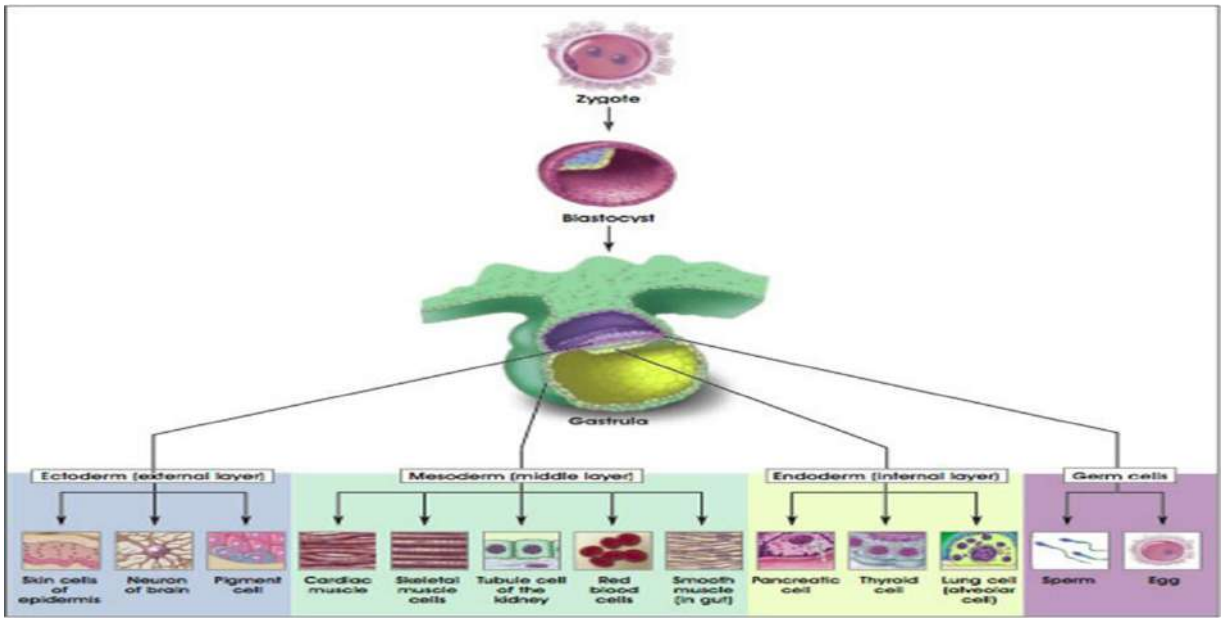
They are able to differentiate to the different types of blood cells.

3- Induced pluripotent stem cells (iPSCs)

These new types of cells are called induced pluripotent stem cells(iPSCs).

They can differentiate into all types of specialized cells in the body.

To create iPSCs, scientists genetically reprogram the adult stem cells, so they behave like embryonic stem cells.



Sources of stem cells

Stem cells of many tissues are found in specific locations where the microenvironment helps maintain their uniquely undifferentiated properties

1-Hematopoietic stem cells: Obtained from umbilical cord blood (at birth).

2-Bone marrow: Obtained by aspiration of marrow tissue (It contains hematopoietic and mesenchymal stem cells).

3-Fetal tissue: Obtained after termination of pregnancy.

4-Adipose connective tissues.

5-Adult tissues: Such as brain, blood vessels, skin, liver and skeletal muscle.

Differentiation of stem cells is affected by:

1- Specific genes.

2- Certain molecules in the microenvironment

surrounding them (i.e., cytokines secreted by other cells).

3- Physical contact with neighboring cells.

Characteristics of stem cells

1- Self-renewal: the ability to go through numerous cycles of cell division while maintaining the undifferentiated state.

2- small population

3- Many stem cells **divide infrequently but the divisions are always asymmetric**, that is, one daughter cell remains as a stem cell while the other becomes committed to a path that leads to differentiation.

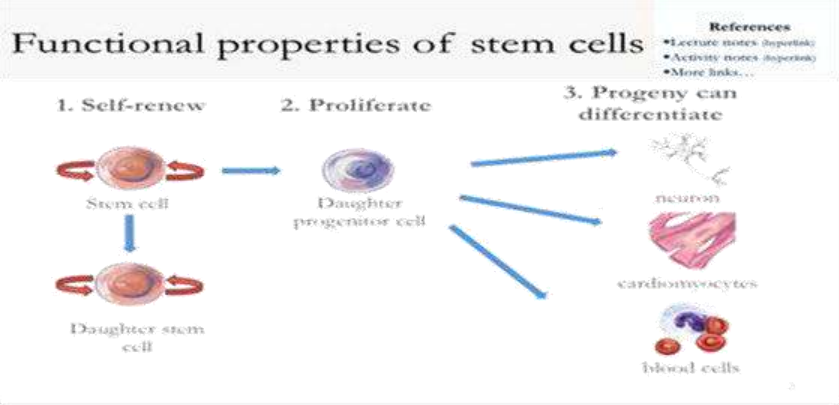
4- Potency: the capacity to differentiate into specialized cell types.

Degrees of Potency:

Totipotent or pluripotent: able to give rise to any mature cell type.

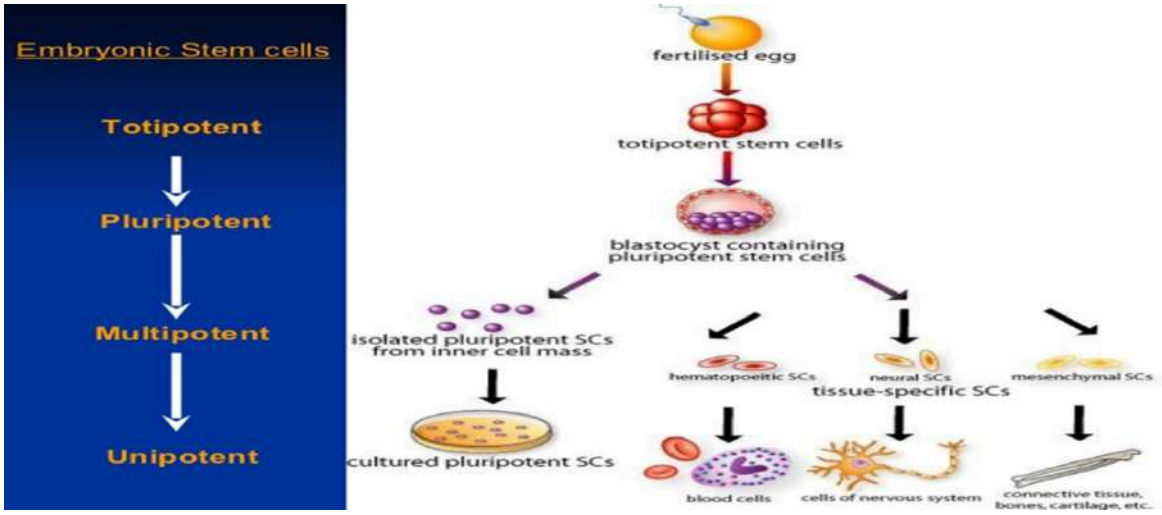
Multipotent: can differentiate into a number of cell types of a closely related family of cells, e.g. hematopoietic stem cells.

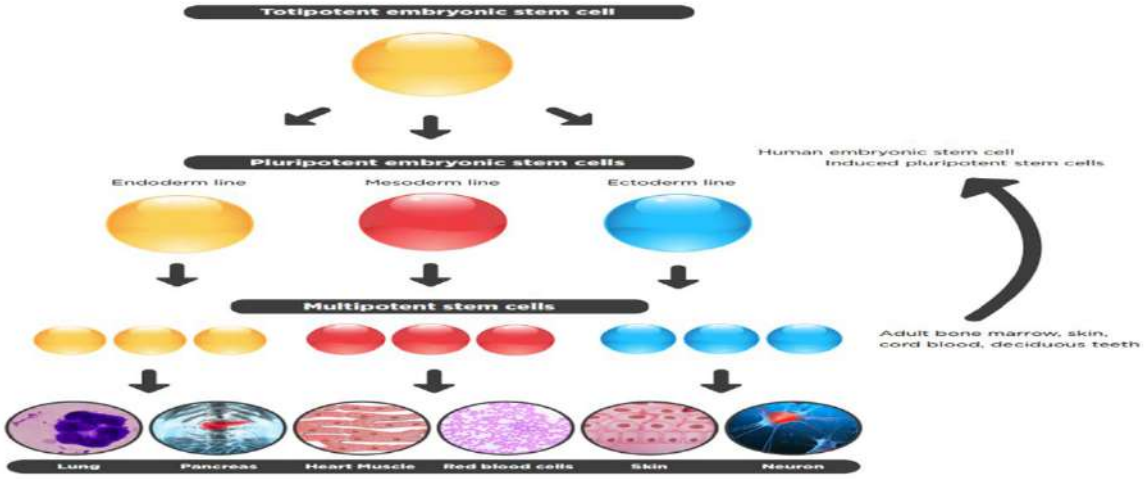
Unipotent: cells can produce only one cell type, their own, but have the property of self-renewal.



5- Plasticity of stem cells:

Plasticity of stem cells means that the stem cells can be shifted from one line of differentiation to another, according to the nature of surrounding microenvironment





Progenitor cells are daughter cells of stem cells

Characters of Progenitor Cells

more rapidly dividing

Have reduced potentiality (committed to a single-cell lineage).

Cells formed by progenitor cells may become terminally differentiated (meaning that renewed cycling cannot occur)

Clinical uses of mesenchymal stem cells

1- Repair of localized tissue defects.

2- Repair of ischemic tissue of heart and lower extremities

3- Repair of non-healed fractures and bone defects.

4- Repair of traumatic cartilage defects and osteoarthritis.

5- Systemic cell transplantation for treatment of degenerative diseases.

Tissue regeneration or renewal

• It is the capacity for mitosis within a tissue, either by differentiated cells or by a reserve of stem cells.

• The cell turnover rate is rapid (rapidly renewing cells) in the epithelium lining the digestive tract, blood cells and skin cells.

• Stem cells are not readily apparent and differentiated cells appear to undergo slow and episodic division to maintain tissue integrity in (Stable cell populations) such as most connective tissues, smooth muscle)

• Some cells cannot divide nor regenerate as nerve cell and cardiac muscle (permenant cells)

References:

1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

2- Ross, M. H., & Pawlina, W. (2006). *Histology: A text and atlas with correlated cell and molecular biology* (5th ed.). Lippincott Williams & Wilkins.

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Lecture 9

Epithelial tissue
Characteristic features & basement membrane

Learning objectives:

After this lecture, students should be able to:

- List the basic types of tissues in the body.
- Discuss the characteristic features of epithelium.
- Define the basement membrane and mention its structure & functions

The organs of the human body are composed of four basic tissue types:

Epithelial, Connective, Muscular and Nervous

Each tissue contains cells as well as extracellular matrix, associate with one another in the variable proportions and morphologies characteristic of each organ

Main characteristics of the four basic types of tissues:

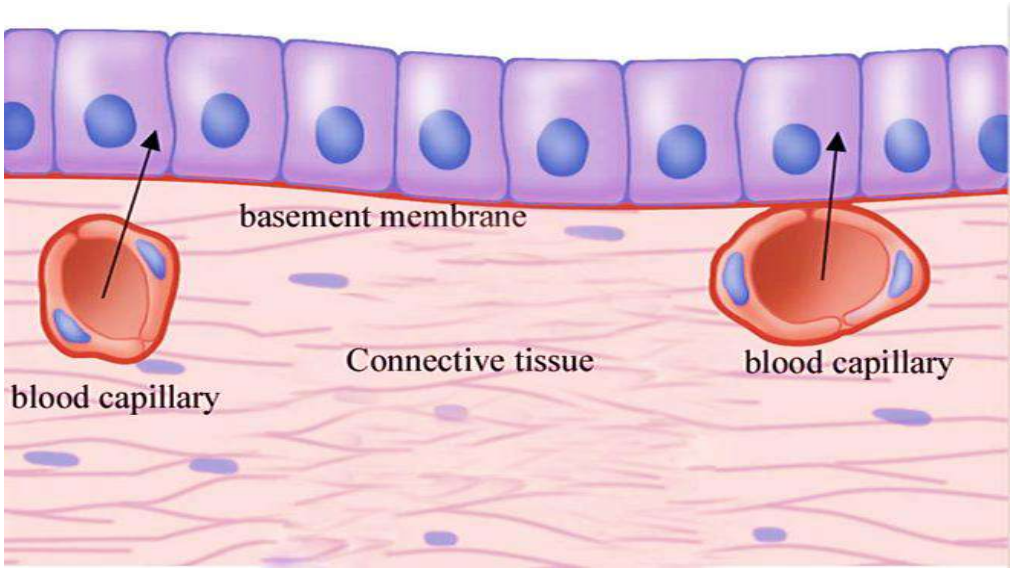
Tissue	Cells	Extracellular matrix
Epithelial	Aggregated cells	Small amount
Connective	Fixed and wandering cells	Abundant amount
Muscle	Elongated contractile cells	Moderate amount
Nervous	Elongated cells with fine processes	Very small amount

Characteristic features of epithelial cells

1-Epithelial tissues are composed of closely aggregated cells, forming cellular sheets that lines the cavities of organs, cover the body surface or form clusters of cells as in glands.

2-The shapes of epithelial cells are variable (squamous, cuboidal, columnar) with nuclear shape corresponding roughly to cell shape.

3-Adjacent epithelial cells are adhering strongly to one another by cellular junctions.



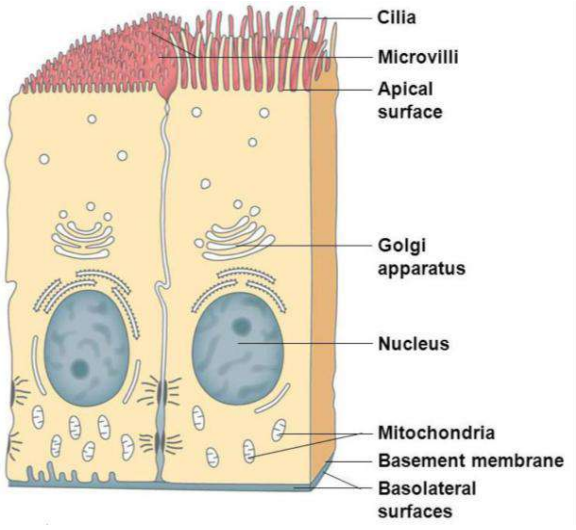
4- The basal surface of all epithelia rests on a thin extracellular sheet "the basement membrane".

5- Epithelia do not themselves normally contain blood vessels "avascular".

6- Most epithelia are adjacent to connective tissue containing blood vessels from which the epithelial cells receive nutrients.

7- The connective tissue that underlies the epithelia lining the organs of the digestive, respiratory & urinary systems is called the "lamina propria".

8- Epithelia which are formed of one cell layer (simple epithelia) generally show “polarity”, with organelles and membrane proteins distributed unevenly within the cell. The region of the cell contacting the connective tissue is called the “basal pole” and the opposite end, is the “apical pole”, with the two poles differ significantly in both structure and function. Regions of cuboidal or columnar cells that adjoin neighboring cells comprise the cells "lateral surfaces"



9- The area of contact between the epithelium and connective tissue may be increased by small invaginations called "papillae" projecting from the connective tissue into the epithelium. Papillae occur most frequently in epithelial tissue subject to friction such as the covering of the skin or tongue.

10-The principal functions of the epithelial tissues include:

- Protection (eg, epidermis).
- Absorption (eg, intestinal lining).
- Secretion (eg, glands).
- Contractile (eg, myoepithelium).
- Sensory (eg, taste buds).

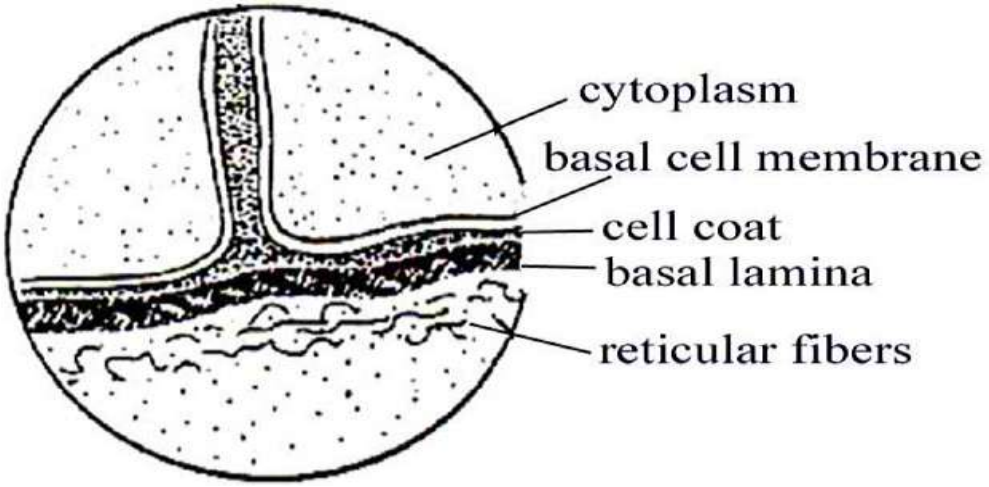
Basement membrane

Definition; Is a thin extracellular sheet of macromolecules between the basal surface of epithelia and the underlying connective tissue.

Structure: the basement membrane appears as a thin layer. It separates the epithelium from underlying tissues. It is secreted from the epithelial cells

➢ **L.M:** The components in this structure can be stained with PAS (glycoproteins in the basal lamina) and silver (reticular fibers in the reticular lamina).

➢ **E.M:** it is composed of two parts. Nearest the epithelial cells is an electron-dense layer, the basal lamina and beneath this layer is a fibrous reticular lamina.



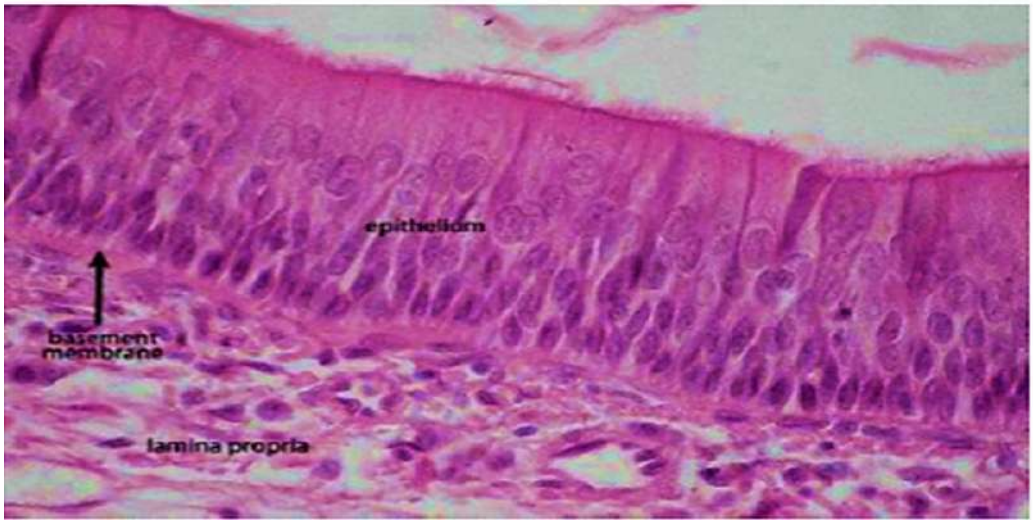
Structure:

1. The cell coat of epithelial cells which appears as a light zone just underneath the outer lamina of the plasma membrane .It is composed of glycoproteins and glycolipids.

2. Basement membrane proper (basal lamina) which is chemically Composed of glycoproteins and collagen material (different from that of the connective tissue fibers). This layer is PAS+ve when seen with the LM.

3. A connective tissue layer highly rich in reticular fibers. This layer is stained by silver. Usually this third layer is not included in the structure of the basement membrane.

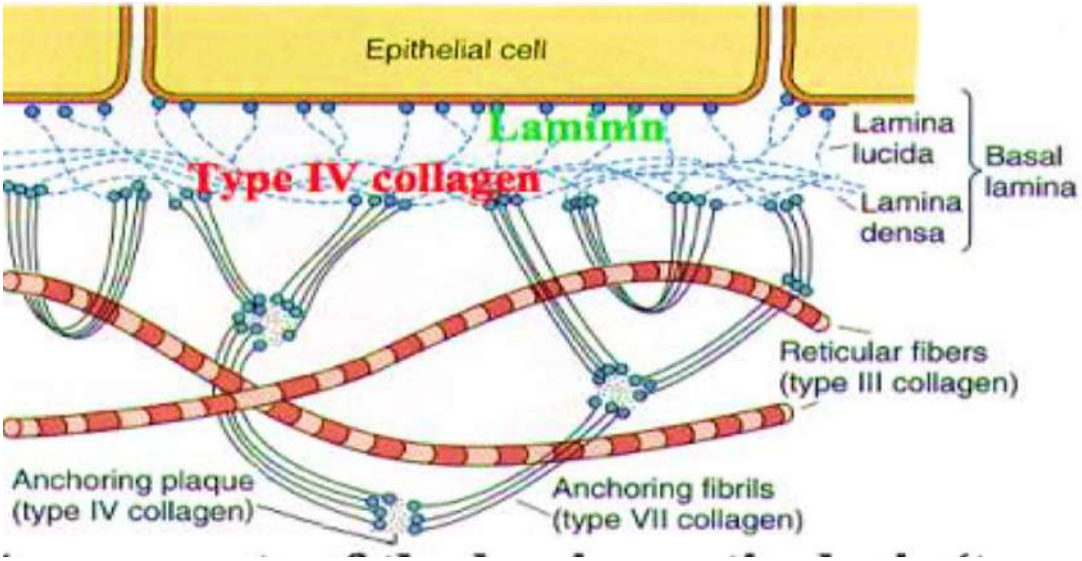
Basement membrane (basal lamina)



*The basal lamina contains glycoproteins and type IV collagen secreted from the basal sides of the epithelial cells.

*The basal surface of the epithelial cells adheres strongly to the basal lamina by "hemidesmosomes".

*The reticular lamina contains reticular fibers (type III collagen) and is bound to the basal lamina by anchoring fibrils of type VII collagen, both of which are produced by cells of the connective tissue.



Functions of basement membrane:

1- Provides structural support for epithelial cells; attach epithelia to underlying connective tissue.

2- Acts as filter of diffusing macromolecules (e.g, in the kidney).

3- Maintains cell polarity.

4- Permits the flow of nutrients, as epithelium is avascular.

5- Controls epithelial growth & differentiation. It forms a barrier to downward epithelial growth. This is only breached in malignancy.

References:

1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

2- Ross, M. H., & Pawlina, W. (2006). *Histology: A text and atlas with correlated cell and molecular biology* (5th ed.). Lippincott Williams & Wilkins.

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Lecture 10

Epithelial tissue

Intercellular junctions & apical cell specializations

Learning objectives:

After this lecture, students should be able to:

▪ List the types of Intercellular junctions.

▪ Discuss the structure of different types of Intercellular junctions.

▪ Define the apical cell specializations and Know the structure & function of microvilli, stereocilia & cilia.

A)Intercellular junctions

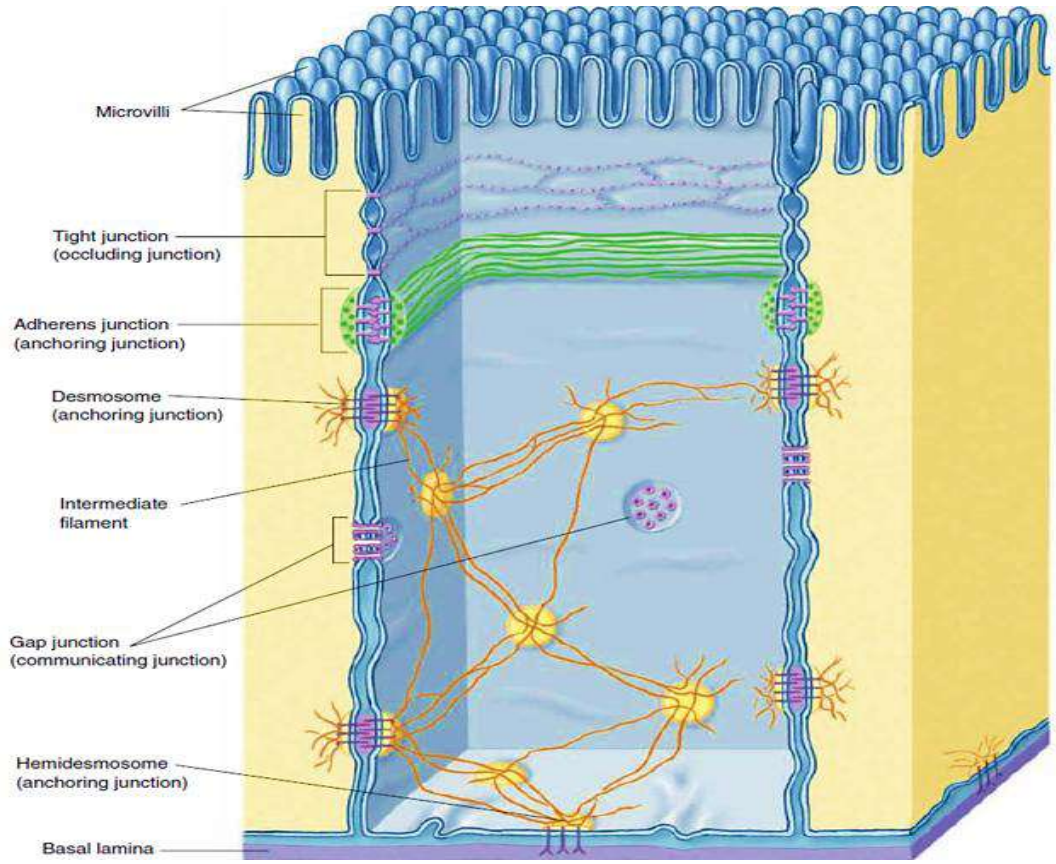
Lateral surfaces of epithelial cells have complexes of several specialized intercellular junctions with different functions.

These junctions include

*Tight or occluding junctions form a seal between adjacent cells

*Anchoring junctions are sites of strong cell adhesion

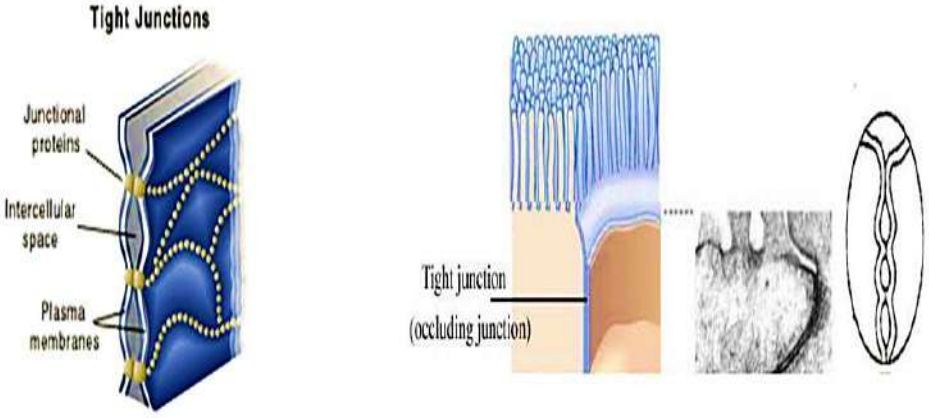
*Gap junctions are channels for communication between adjacent cells



1-Tight or occluding junctions:

Are the most apical of the junctions, present at the apical end of the cells. Tight junctions, also called “zonulae occludens”, The term “zonula” indicates that the junction forms a band completely encircling each cell.

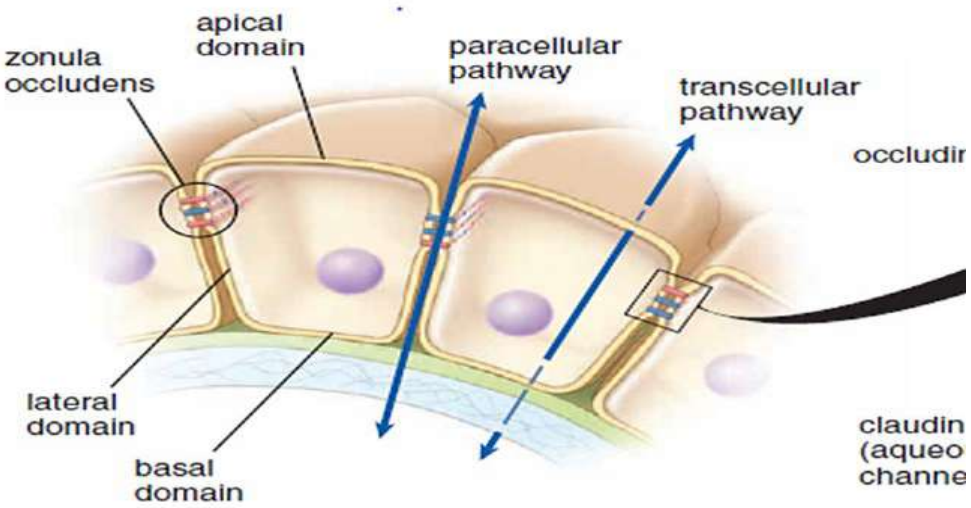
Structure: The adjacent cell membranes appear fused (sealed) or very tightly apposed due to tight interactions between the transmembrane proteins “claudin and occludin".



Function:

1-This intercellular seal ensures that molecules crossing an epithelium by going through the cells (a transcellular pathway) rather than between them (the paracellular pathway), controlling passage of molecules between them (i.e. prevent passive flow of material between the cells).

2- Tight junctions maintain two distinct membrane domains (apical and basolateral).



2-Anchoring junctions:

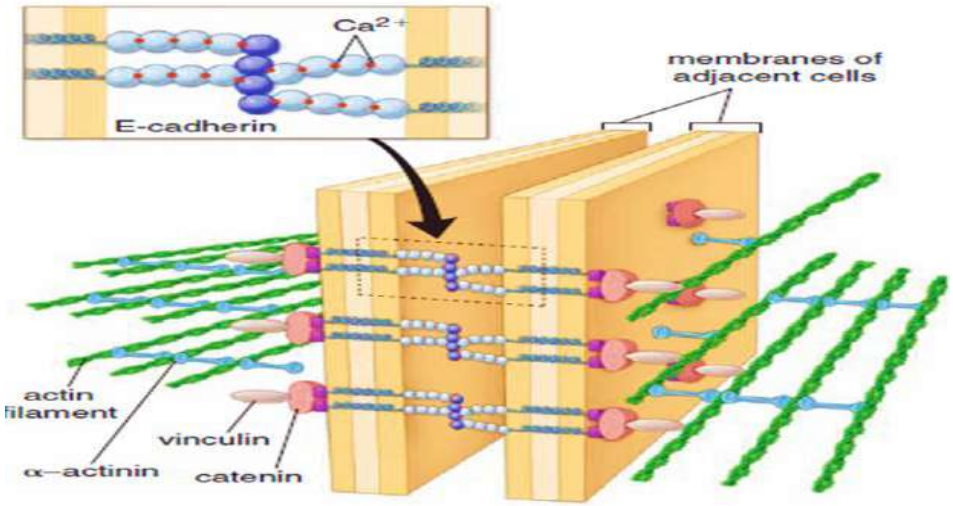
***Zonulae adherens:** encircles the epithelial cells, usually immediately below the tight junction, firmly anchoring a cell to its neighbors

Structure: Cell adhesion is mediated by cadherins transmembrane protein which span the plasma membrane of the cell and bind to identical cadherins on adjacent cells . At their cytoplasmic ends, cadherins bind "catenins" that link to actin. The intercellular component of the zonula adherens appeared as a small electron-dense plaque.

Function:

tight and adherent junctions hold the cells together.

Adherens junctions immediately below the tight junctions serve to stabilize and strenghen the occluding bands

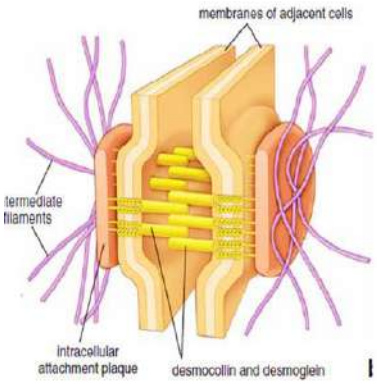


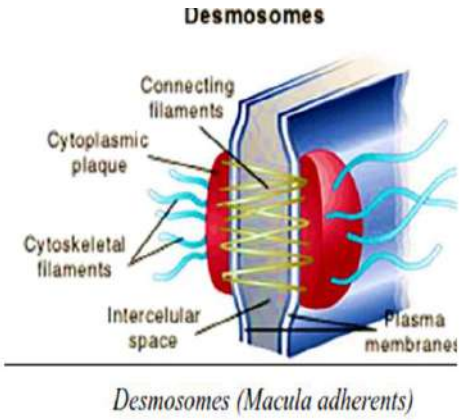
*** Desmosomes (macula adherens):**

-This junction resembles a single" spot-weld" and does not form a belt around the cell.

-Desmosomes are disc-shaped structures. It consists of two cytoplasmic adhesion plaques formed of special proteins and placed on the cytosolic face of the two facing cell membranes.

-The two plaques are connected by transmembrane linker proteins (cadherins) which extend into the intercellular space forming an interlacing network which binds the two cells together and appears as a dense line in the intercellular space





Desmosomes (Macula adherents)

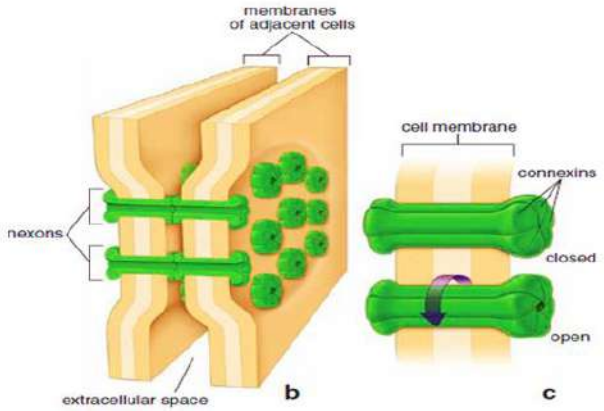
Structure: Desmosomes are disc-shaped structures. They contain cadherin family transmembrane proteins (desmogleins and desmocollins). The cytoplasmic ends of these transmembrane proteins bind plakoglobins, which link to larger proteins called desmoplakins in an electron-dense plaque. Desmoplakins in turn bind intermediate filament proteins "keratin", sometimes referred to "tonofilaments".

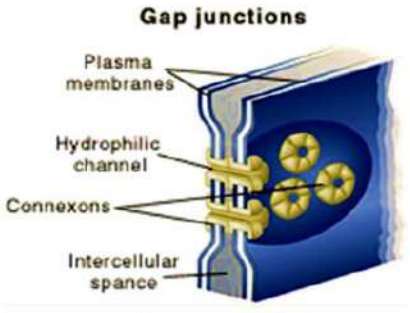
Function: Such intermediate filaments are very strong, and desmosomes provide firm cellular adhesion and strength throughout the epithelium.

3-Gap junctions:

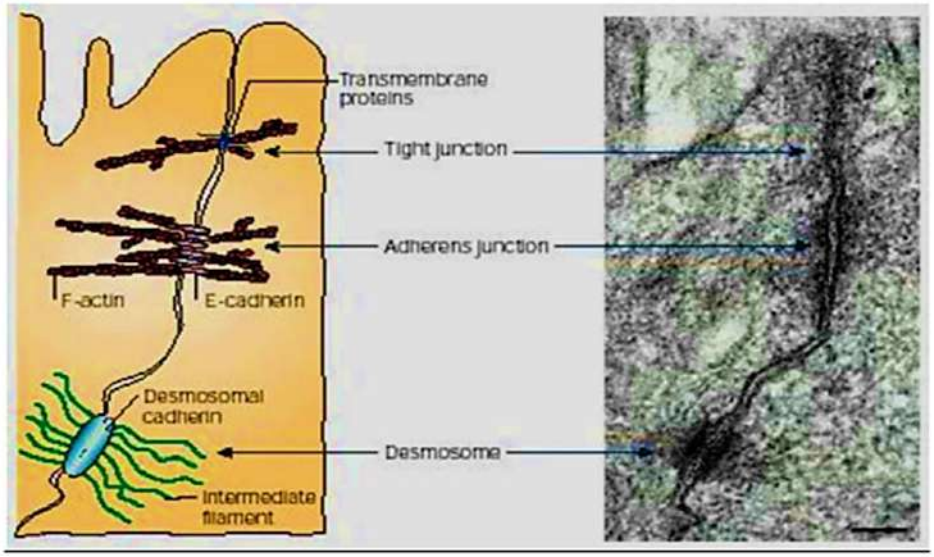
They consist of aggregated transmembrane protein complexes", that form circular patches in the plasma membrane, form hexameric complexes (i.e. six protein subunits) called "connexons", each connexon has a central hydrophilic pore.

Function: permit intercellular exchange of small molecules. In cardiac muscle cells, providing synchronous contraction





Nexus or gap junction



Different types of intercellular junctions

B)Interdigitations

The PM of two adjacent cells may be thrown into a complex interdigitation to give a very wide surface area for intercellular passage of material . The best example is the cells of the of the kidney tubules.

C) Intercellular Canaliculi:

Secretory material from adjacent cells is accumulated in channels known as intercellular canaliculi which carry the secretions to ducts. The intercellular canaliculi between liver cells are examples.

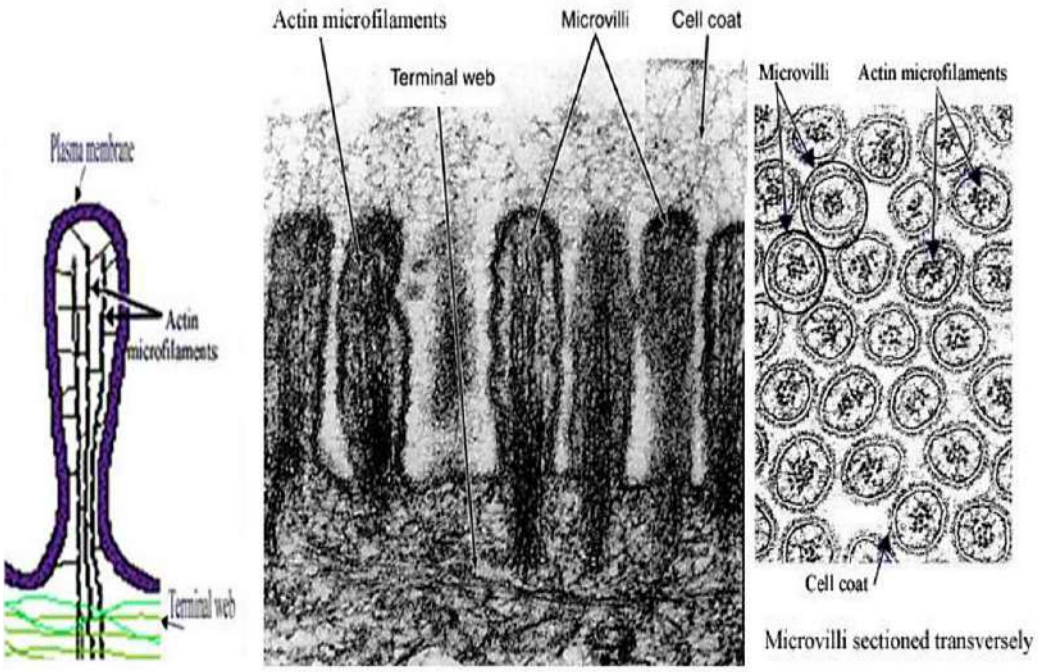
Apical cell specializations

The apical parts of many columnar and cuboidal epithelial cells have specialized structures projecting from the cells.

These include specializations ● Cilia. ● Stereocilia. ●Microvilli.

1-Microvilli

Site: In epithelia specialized for absorption (cells of the small intestine), the apical cell surfaces reveal array of projecting microvilli (L. villus tuft).



Structure: Densely packed microvilli are visible as a "brush or striated border" projecting into the lumen.

The average microvillus is about 1-µm long and 0.1-µm wide.

A thick glycocalyx (cell coat) covering intestinal microvilli contains certain enzymes for certain macromolecule digestion.

Each microvillus contains bundled actin filaments capped by the surrounding plasma membrane. The actin filaments insert into the terminal web at the base of the microvilli

Function: Increase the apical surface area for better absorption.

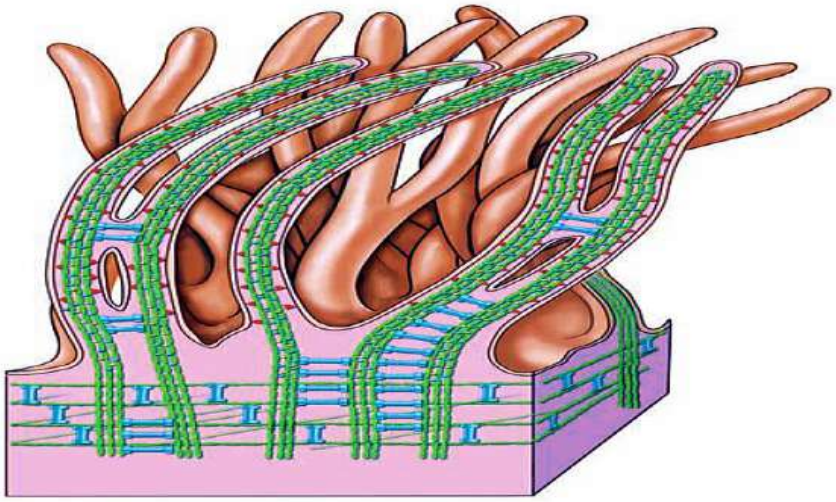
2-Stereocilia:

Stereocilia are a much less common type of apical process on the absorptive epithelial cells lining the male reproductive system.

Structure: Stereocilia resemble microvilli in containing arrays of microfilaments, with similar diameters and with similar connections to the cell's terminal web. However, stereocilia are typically much longer and less motile than microvilli and may show branching distally.

Function: 1- Like microvilli, stereocilia increase the cells' surface area, facilitating absorption.

2-More specialized stereocilia with a motion-detecting function are important components of inner ear sensory cells.

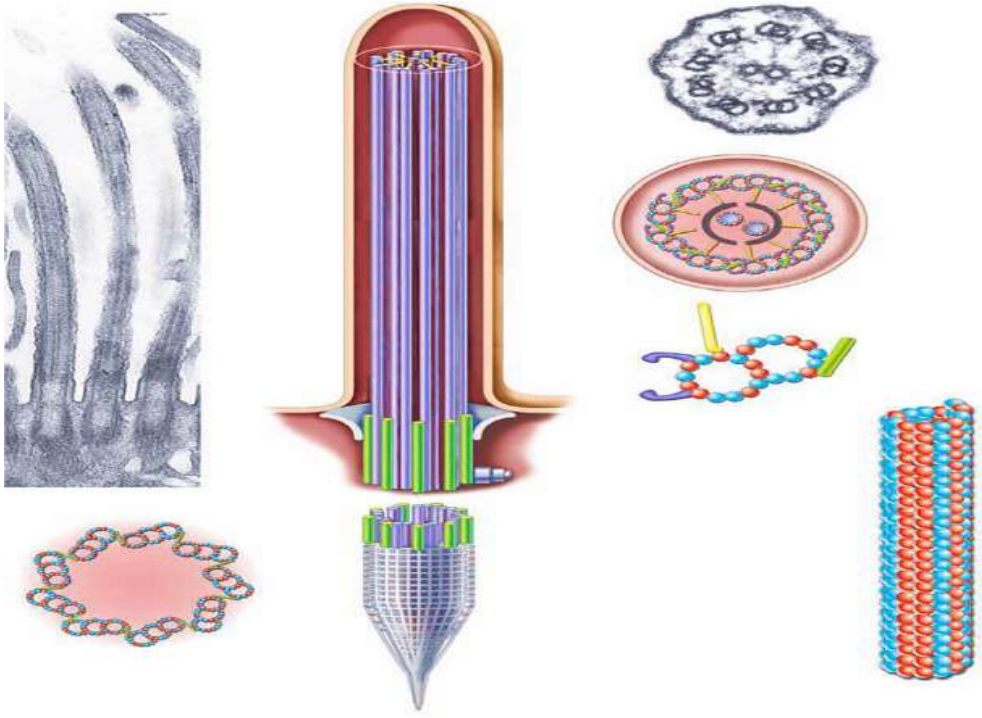


3-Cilia

Cilia are long, highly motile apical structures, larger than microvilli in diameter (typical cilia are 5-10 μm long and 0.2 μm), containing internal arrays of microtubules not microfilament.

Site: At the surface of the cells involved in the transportation of fluids and particles (eg, ciliated cells lining the respiratory tract).

Structure: Each cilium has a core structure consisting of nine peripheral microtubule doublets arrayed around two central microtubules. This 9 + 2 assembly of microtubules is called an "axoneme" (Gr. axon, axis + nema, thread). The peripheral pairs are attached to the central pair by radial spokes. Dynein "arms" bound to one microtubule in each doublet and extend toward a microtubule of the next doublet.

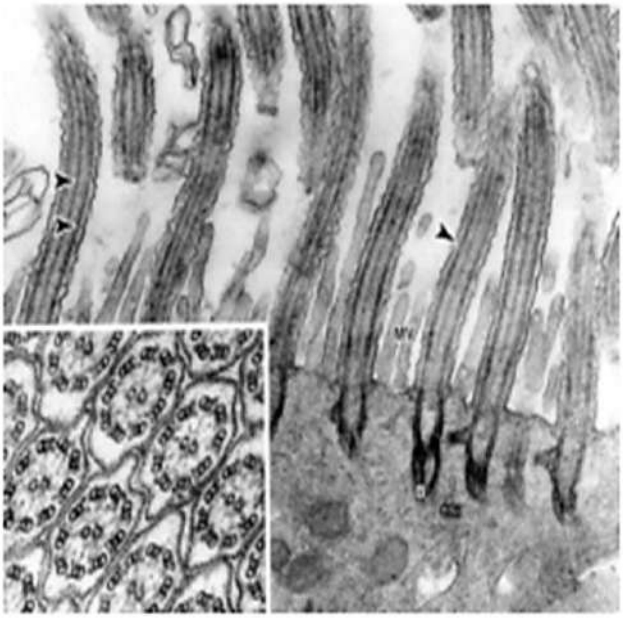


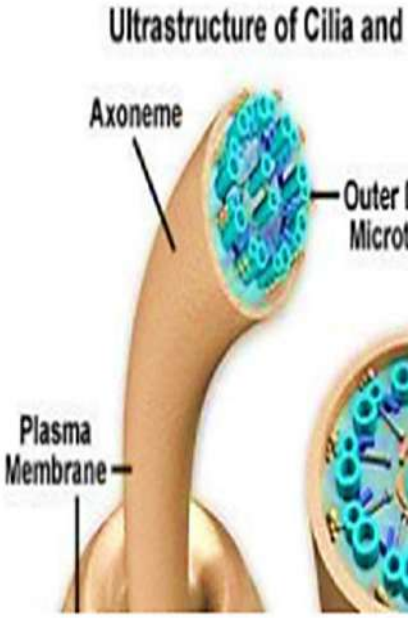
At the base of each cilium is a “basal body”, just below the cell membrane; anchoring the axoneme to the apical cytoplasm. Basal bodies have a structure similar to that of centrioles, with triplets of microtubules. Protofilaments form “rootlets” anchoring the entire structure to the cytoskeleton

Function:

1-Cilia exhibit rapid beating patterns that move a current of fluid and suspended matter in one direction along the epithelium

2-other cell types have at least one short projection called a “primary cilium”, which is not motile but is enriched with receptors for detection of light, odors, motion, and flow of liquid past the cells.





Cilia

Cuticle: The outer surface of certain cells is exposed to harmful substances and therefore needs protection. In this case a substance rich in proteins is deposited

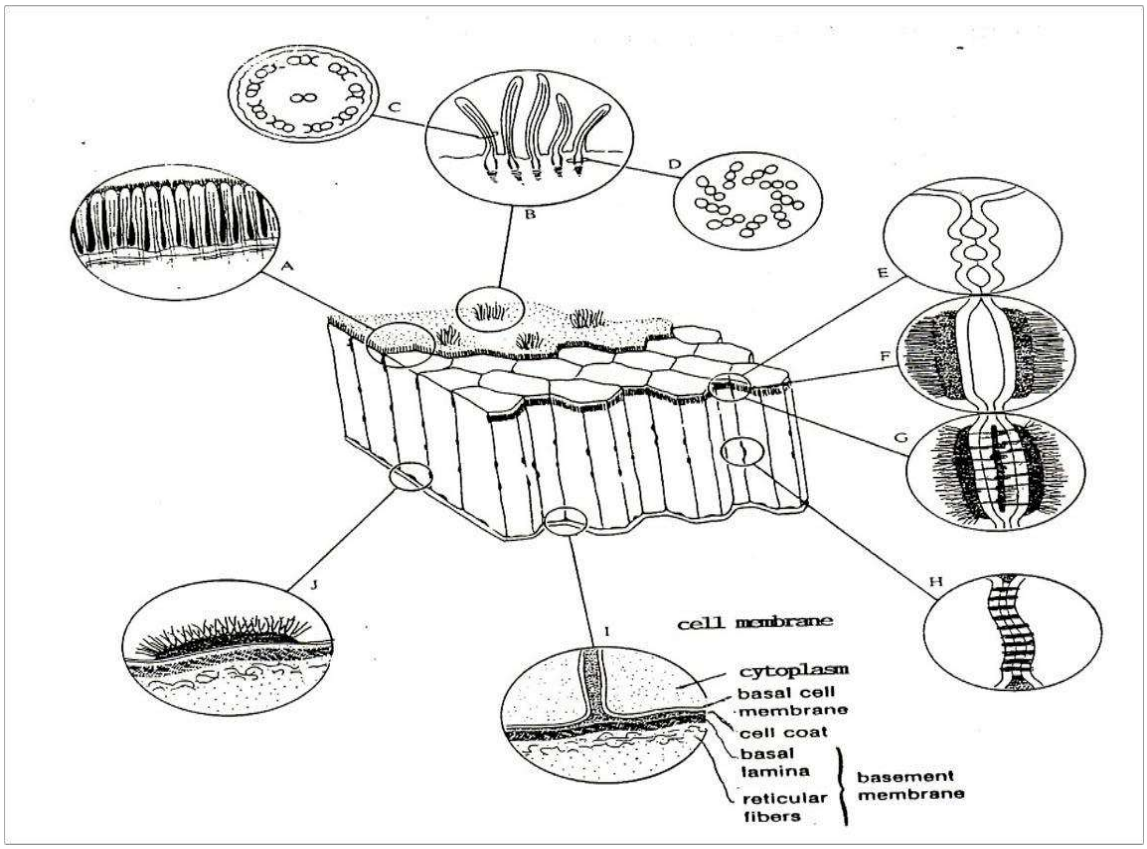
on the outer lamina of the PM and may by called cuticle, e.g. cells lining the urinary bladder or ureter.

Caveoles: The secretory surface of a cell usually shows invagination called caveoles (little caves) which result from the process of exocytosis. Surfaces involved in pinocytosis also show caveoles e.g. endothelial lining of blood capillaries.

Modification (differentiation) of the PM at the cell base:

A) Basal infolding: Membranous infoldings at the base of certain cells are present to help in increasing the surface of the membrane for effective basal flow of material as in the cells of kidney tubules.

B) Hemidesmosomes: are present at the basal membrane of protective cells to hold them firmly to underlying tissues.



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1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

2- Ross, M. H., & Pawlina, W. (2006). *Histology: A text and atlas with correlated cell and molecular biology* (5th ed.). Lippincott Williams & Wilkins.

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6-Lowe, J. S., & Anderson, P. G. (2024). *Stevens & Lowe's human histology* (6th ed.). Elsevier.

Lecture 11

Epithelial tissue

Learning objectives:

After this lecture, students should be able to:

*Identify the types of covering & lining epithelium and their sites.

*Define & classify secretory epithelium.

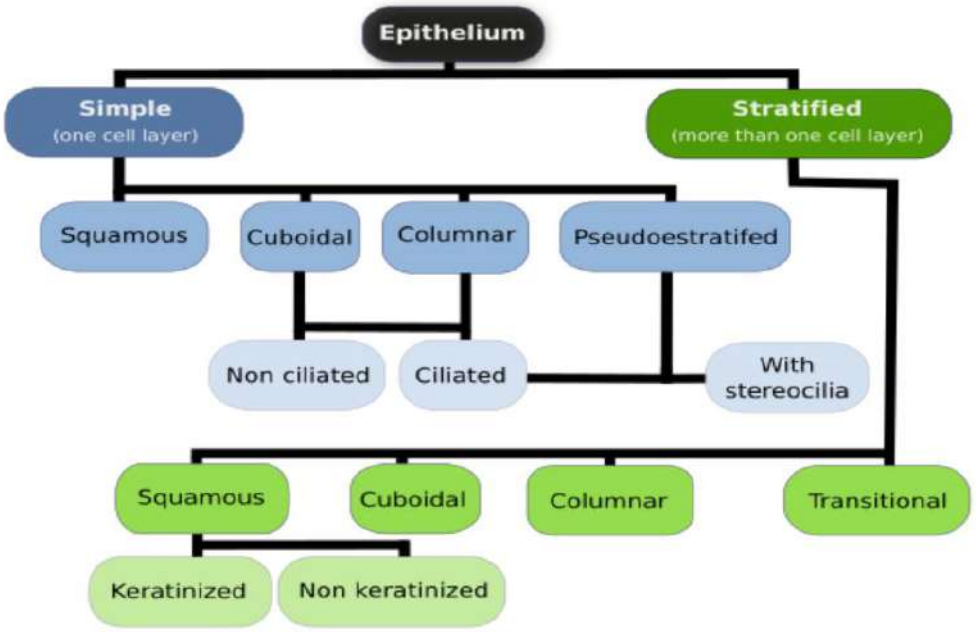
Classification of epithelial tissues:

A)Covering and lining (simple and stratified).

B)Glandular epithelium.

C)Sensory or neuroepithelium.

D)Myoepithelium.



Covering and lining epithelium

*They are the epithelial layers that cover surfaces and line cavities.

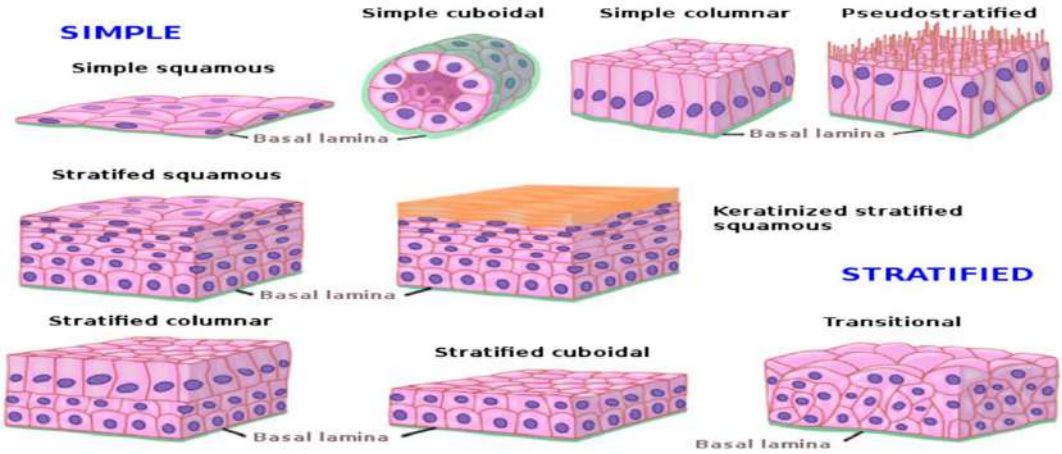
*They are in the form of sheets called membranes.

*All the epithelial membranes in the body are kept wet, except the stratified squamous keratinized epithelium (the epithelial covering of the skin) which is dry to prevent evaporation of water and dehydration of the body.

*The covering and lining epithelium are classified according to the number of cell layers into:

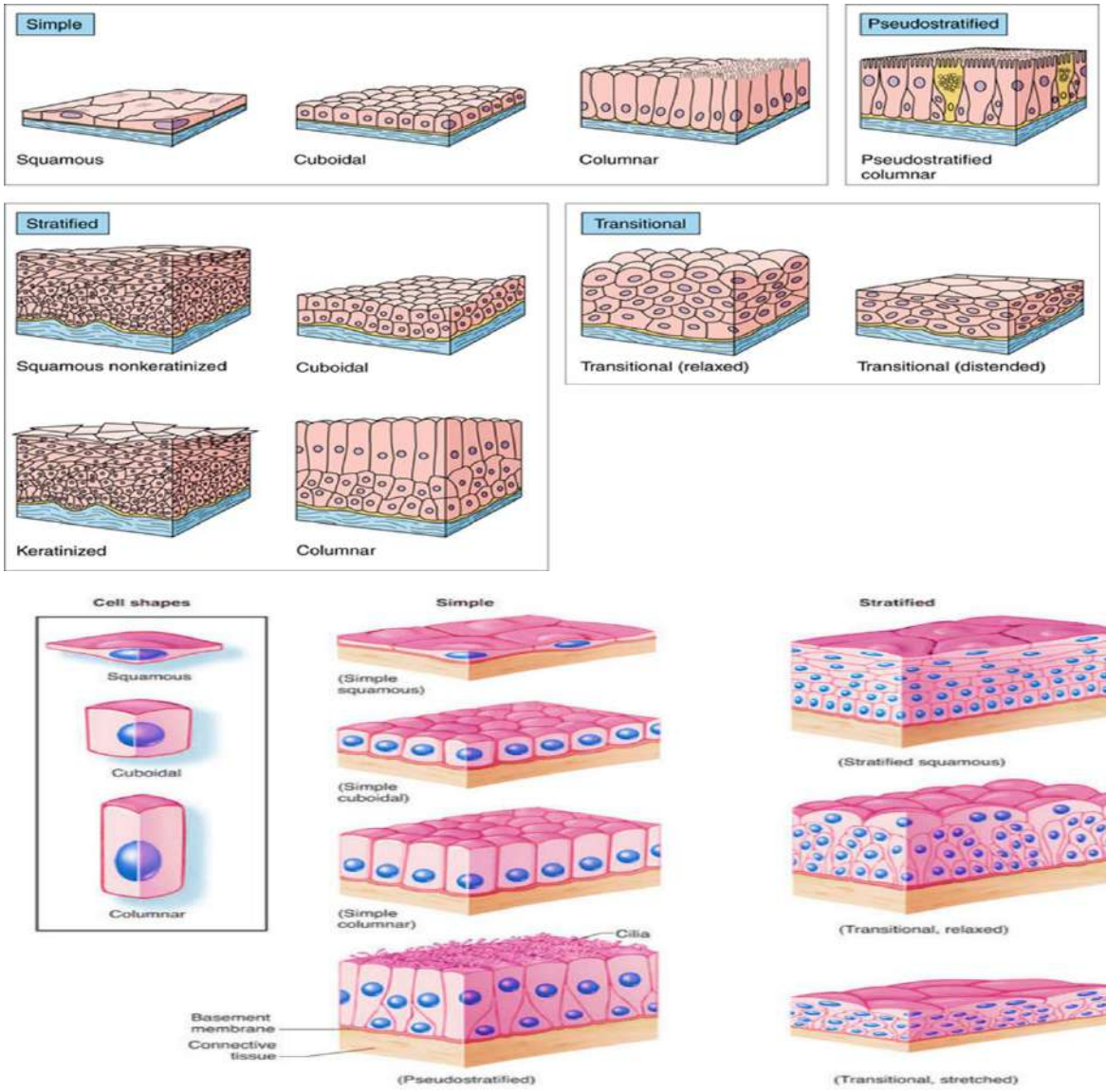
A)simple epithelium : formed of one layer of cells .

B)stratified epithelium : formed of more than one layer of cells .



Cells	Location	Function
Simple squamous epithelium	Air sacs of lungs and the lining of the heart, blood vessels, and lymphatic vessels	Allows materials to pass through by diffusion and filtration, and secretes lubricating substance
Simple cuboidal epithelium	In ducts and secretory portions of small glands and in kidney tubules	Secretes and absorbs
Simple columnar epithelium	Ciliated tissues are in bronchi, uterine tubes, and uterus; smooth (nonciliated tissues) are in the digestive tract, bladder	Absorbs; it also secretes mucous and enzymes
Pseudostratified columnar epithelium	Ciliated tissue lines the trachea and much of the upper respiratory tract	Secretes mucus; ciliated tissue moves mucus
Stratified squamous epithelium	Lines the esophagus, mouth, and vagina	Protects against abrasion
Stratified cuboidal epithelium	Sweat glands, salivary glands, and the mammary glands	Protective tissue
Stratified columnar epithelium	The male urethra and the ducts of some glands	Secretes and protects
Transitional epithelium	Lines the bladder, uretha, and the ureters	Allows the urinary organs to expand and stretch

Types of covering and lining epithelium



I.Simple epithelium

The simple epithelium classified according to the height of cells and shape of their nuclei into:-

Simple squamous epithelium.

Simple cuboidal epithelium.

Simple columnar epithelium.

Pseudostratified columnar epithelium.

A)Simple squamous epithelium:

It is formed of flattened cells which bulge at the area of the nucleus. The nucleus is also flattened and densely stained.

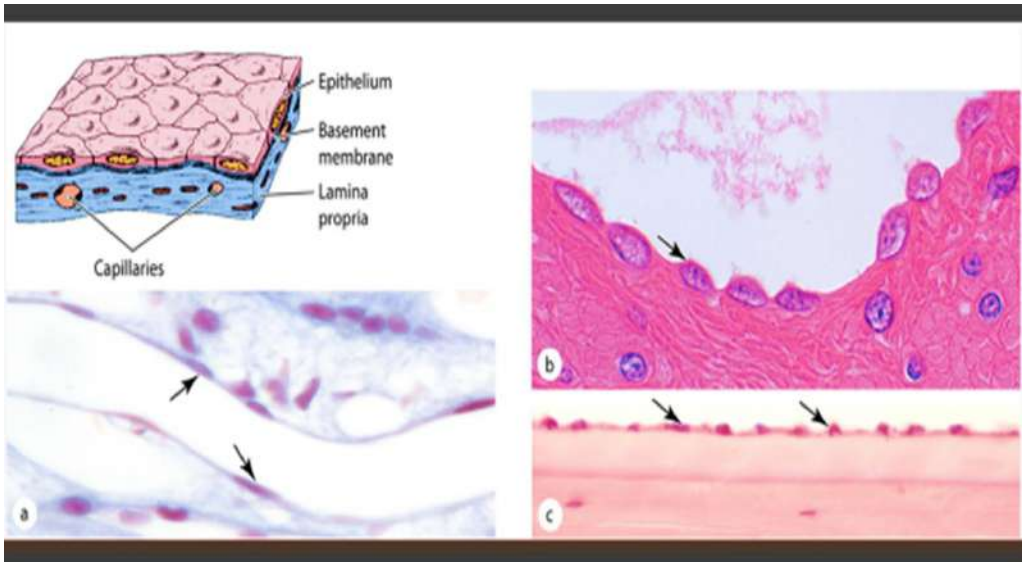
Sites : a) It lines blood vessels (endothelium) and all the body cavities, e.g. pleural, pericardial and peritoneal (mesothelium).

b) It lines the parietal layer of the Bowman's capsule of the kidney

Function:

*Gives smooth surface and prevents friction,

*Secrete biologically active molecules and have an active transport role by pinocytosis.



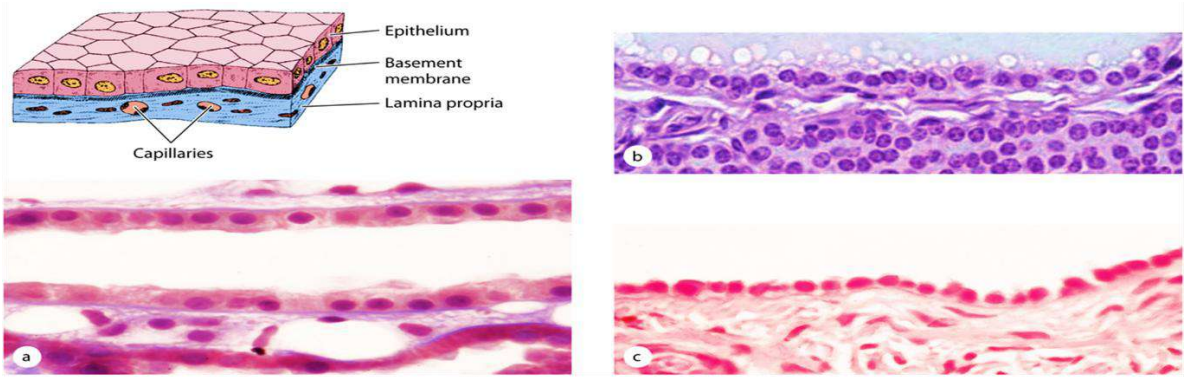
B) Simple cuboidal epithelium:

In a cross section the cells of this epithelium appear as squares, containing spherical relatively large nuclei.

Sites and function:

a) It forms the lining of thyroid follicles, where it provides secretory function.

b) It lines renal tubules where microvilli on the cells serve for reabsorption.



C)Simple columnar epithelium:

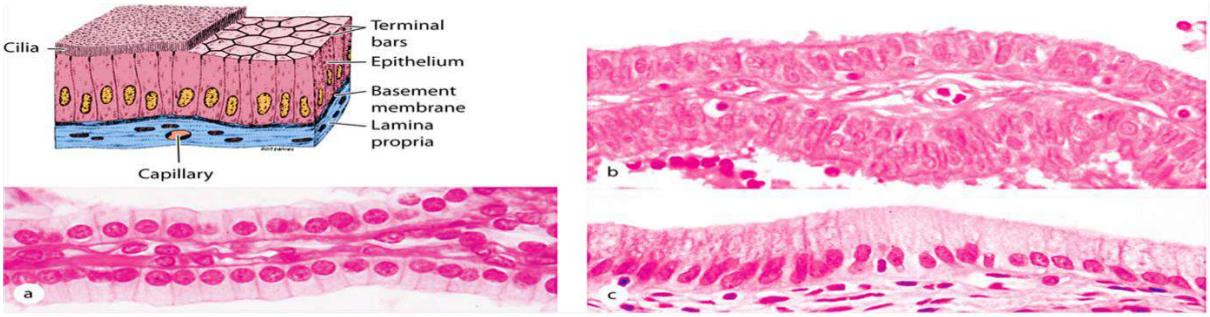
Cells of this type are taller than they are wide. These cells are joined together by junctional complexes.

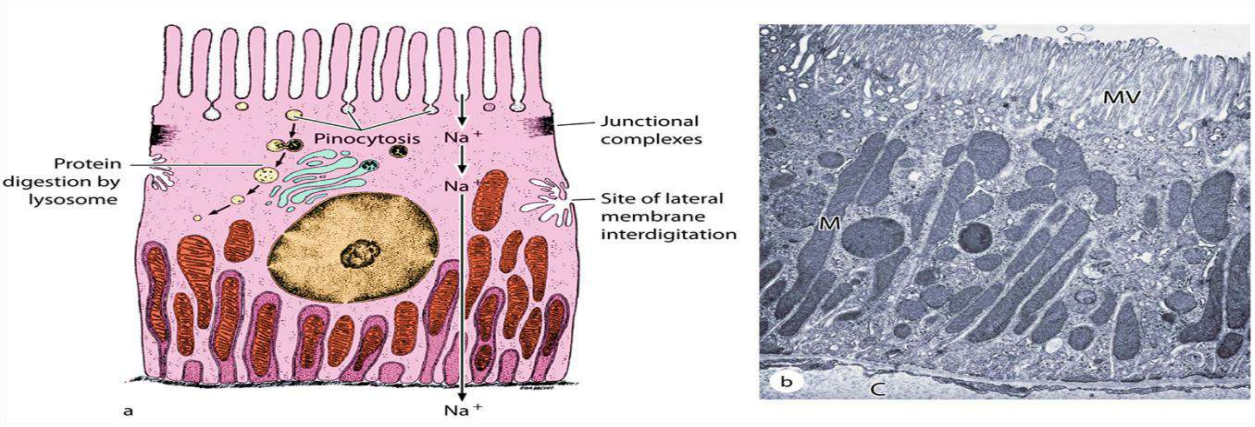
Sites and function:

a-Secretory columnar epithelium as in the lining epithelium of stomach. This epithelium secretes mucus.

b-Secretory-absorptive columnar epithelium as in the lining epithelium of small intestine. The absorptive columnar cells show a striated border (microvilli)

c-Secretory-ciliated columnar epithelium lines the uterus and uterine tubes and is formed of ciliated and secretory columnar cells.





D) Pseudostratified columnar epithelium:

* Some cells are tall and reach the surface of the epithelial membrane and others are too short to reach the surface, however, all the cells rest on the basement membrane.

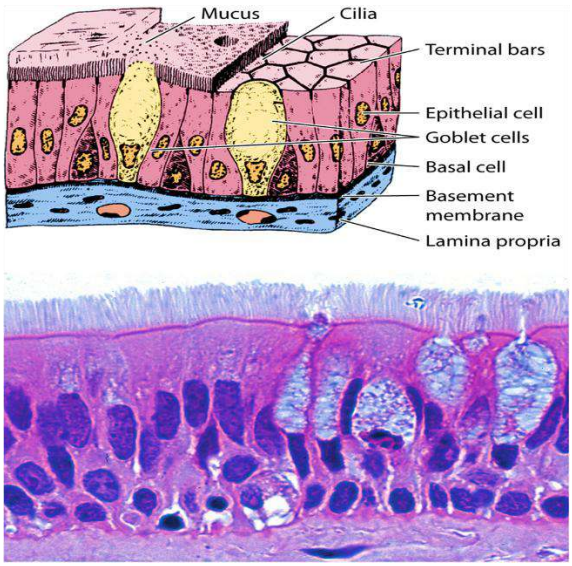
*The tall cells may be secretory goblet cells or ciliated cells. The short basal cells are progenitor cells which differentiate to form cells replacing the tall ones.

Sites:

a) Non- ciliated pseudostratified columnar epithelium is found in vas deferens.

b) Ciliated pseudostratified columnar epithelium with goblet cells lines respiratory passages.

c) Pseudostratified columnar epithelium with immotile cilia (stereocilia) which are actually long microvilli present in the epididymis.



In chronic bronchitis, common among habitual smokers, the number of goblet cells in the lining of airways in the lungs often increases greatly. This leads to excessive mucus production in areas where there are too few ciliated cells for its rapid removal and contributes to obstruction of the airways.

The ciliated pseudostratified epithelium lining the bronchi of smokers can also be transformed into stratified squamous epithelium by metaplasia.

II. Startified epithelium:

Stratified epithelia contain two or more cell layers.

They are classified according to the shape of the superficial cell layer(s) into: squamous, cuboidal, or columnar.

Types:

Stratified squamous epithelium (non–keratinized and keratinized).

2. Stratified cuboidal epithelium

3. Stratified columnar epithelium.

4. Transitional epithelium.

II. Startified epithelium:

Stratified epithelia contain two or more cell layers.

They are classified according to the shape of the superficial cell layer(s) into: squamous, cuboidal, or columnar.

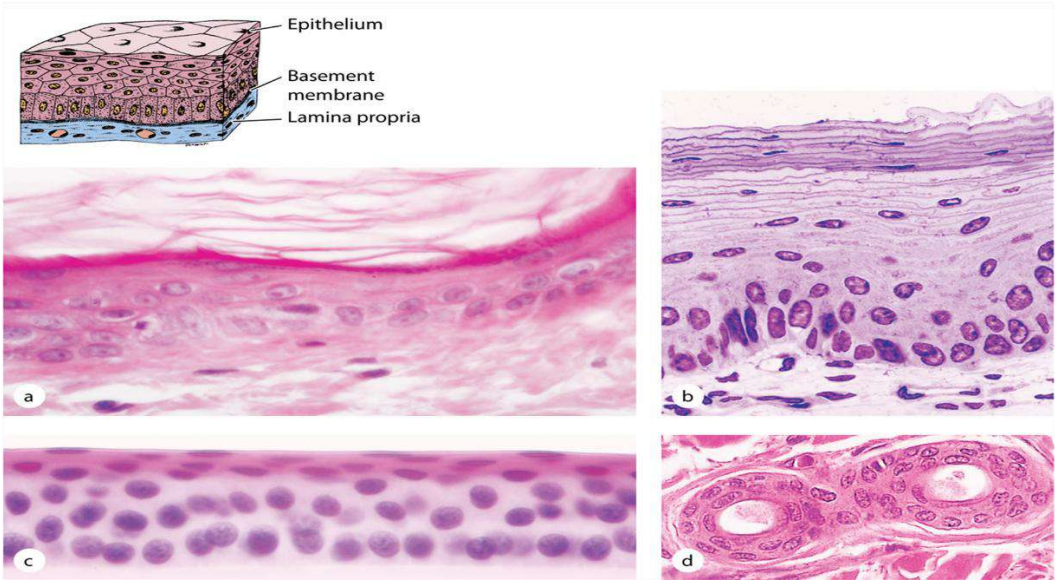
Types:

Stratified squamous epithelium (non–keratinized and keratinized).

2. Stratified cuboidal epithelium

3. Stratified columnar epithelium.

4. Transitional epithelium.



1. Stratified squamous epithelium (non–keratinized):

*Stratified squamous epithelia serve as protective membranes. They are not involved either in absorption or secretion.

*Stratified squamous nonkeratinizing membranes are found on wet surfaces which are subjected to mechanical tearing.

*The basal cells are columnar, followed by polyhedral cells and become gradually squamous at the surface.

Sites: Lining the buccal cavity, esophagus and anal canal and covering the anterior surface of the cornea.

2. Stratified squamous epithelium (keratinized):

It covers dry surfaces as the skin.

The outermost squamous cells undergo certain changes which lead to a layer of keratin, covering the epithelium.

The keratinizing epithelium protects the body from dryness and harmful agents.

3. Stratified cuboidal epithelium:

*Composed of two or three layers of cuboidal cells with the basal layer of cells often appearing nonuniform in distribution.

*It is mainly found lining large ducts of exocrine glands

*It has a protective function.

4. Stratified columnar epithelium:

*The most superficial layer is columnar.

*This type of epithelium is mainly protective.

*It lines very few structures e.g. large ducts of glands.

*Occasionally, the superficial columnar cells may be ciliated as in the fetal esophagus

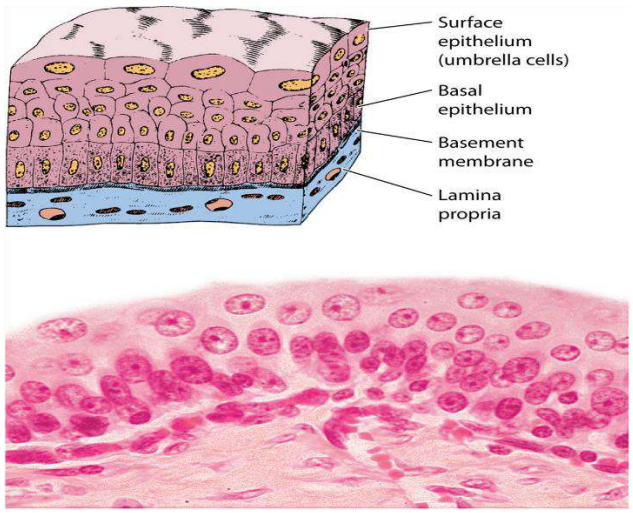
5. Transitional epithelium:

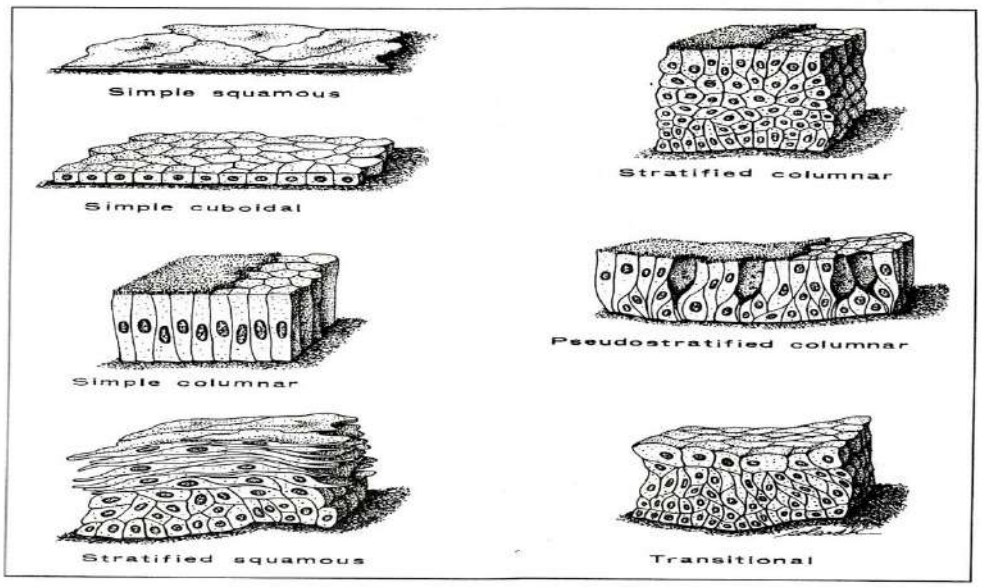
It lines organs which expand greatly like the urinary bladder.

In contracted (empty) state: the superficial layer of cells becomes rounded instead of squamous. The characteristic rounded superficial cells enable the process of stretching without rupture of these cells.

The plasma membrane of the superficial cells facing the lumen with EM its luminal lamella is much thicker than the inner lamina and is formed of dense protein plaques (cuticle) separated by ordinary membranes. The membrane is folded like an accordion. The plaques also serve a protective function against the contents of the urinary passages.

In full (expanded) state: the membrane becomes almost completely extended and the cell becomes squamous.





B) Glandular epithelium :

It is a special type of epithelium characterized by its secretory function.

Glands originate from epithelial cells that leave the surface and penetrate into the underling connective tissue.

Classification of glandular epithelium:

The glandular tissues are classified according to the presence or absence of ducts into:

1. Exocrine glands:

They retain their connection with the surface by having ducts which carry their secretion to the surface of the epithelium e.g. salivary glands.

2. Endocrine glands:

They are ductless glands and lose their connection with the surface. They secrete hormones which pass directly into the blood capillaries e.g. pituitary, thyroid and adrenal glands.

3. Mixed exocrine and endocrine glands:

They consist of the two previous types in the same time e.g. pancreas (exocrine pancreas and Islets of Langerhans).

Exocrine glands are classified into

I. According to the number of cells

1.Unicellular glands formed of one cell e.g. goblet cell

2.Multicellular glands consist of numerous cells e.g. salivary glands

II. According to the shape of the secretory part and the branching of ducts :

***Simple glands:**

Simple tubular glands, e.g. intestinal glands, sweat glands.

Simple acinar glands e.g. mucus secreting glands of penile urethra.

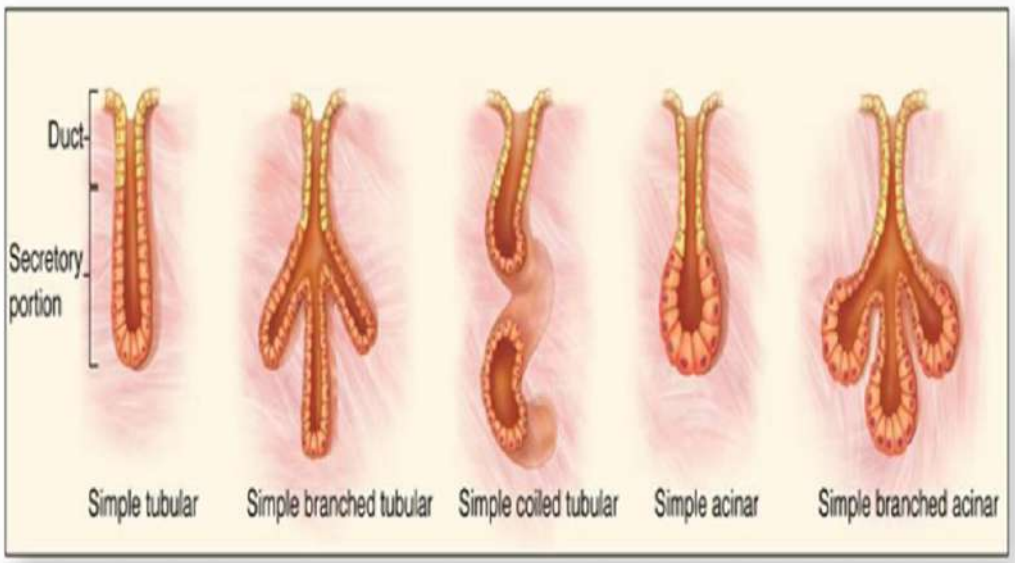
Simple tubulo-acinar glands: They are rare in man.

*** Simple branched glands:**

Simple branched tubular glands e.g. gastric glands.

Simple branched acinar glands e.g. sebaceous glands and tarsal glands.

Simple branched tubulo-acinar glands e.g. minor salivary gland



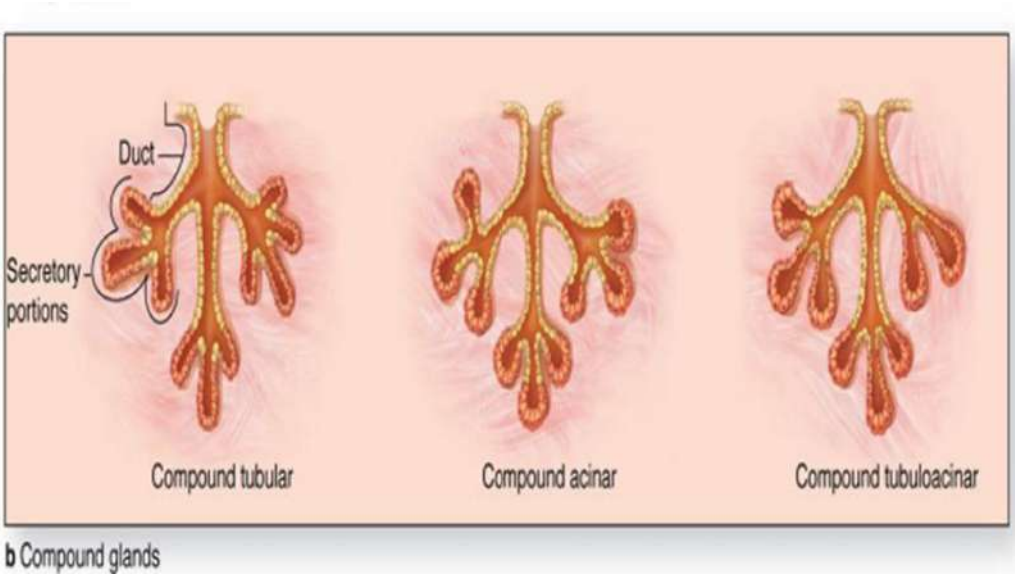
a Simple glands

***Compound glands:**

• Compound tubular glands e.g. lacrimal glands, kidney, and liver.

• Compound acinar glands e.g. mammary glands and pancreas.

• Compound tubulo-acinar glands e.g. major salivary glands.



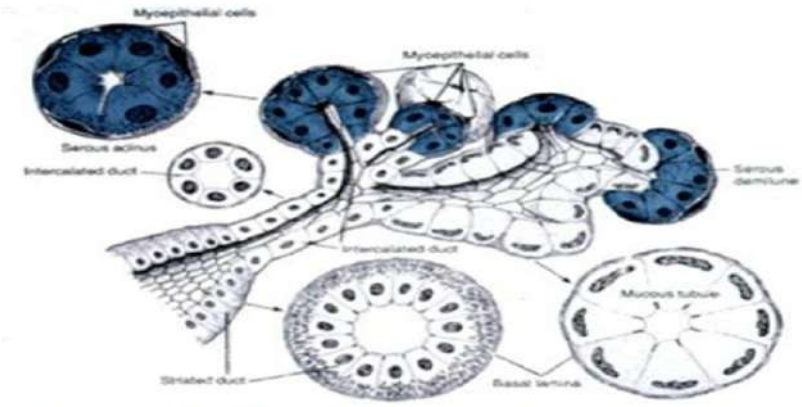
b Compound glands

III. According to the nature of secretion:

*Serous glands secrete serous fluid containing water, albumin and enzymes e.g. pancreas and parotid gland.

*Mucous glands secrete mucin e.g. oesophageal glands and Brunner's glands.

*Mixed seromucous salivary glands secrete serous and mucous secretion e.g. submandibular gland.



Types of Exocrine glands according to nature of secretion

Gland with secretion of special nature

*Sebaceous glands secrete fatty (oily) material.

*Lacrimal glands secrete watery secretion (tears

*Mammary glands secrete milk.

*Waxy (ceruminous) gland

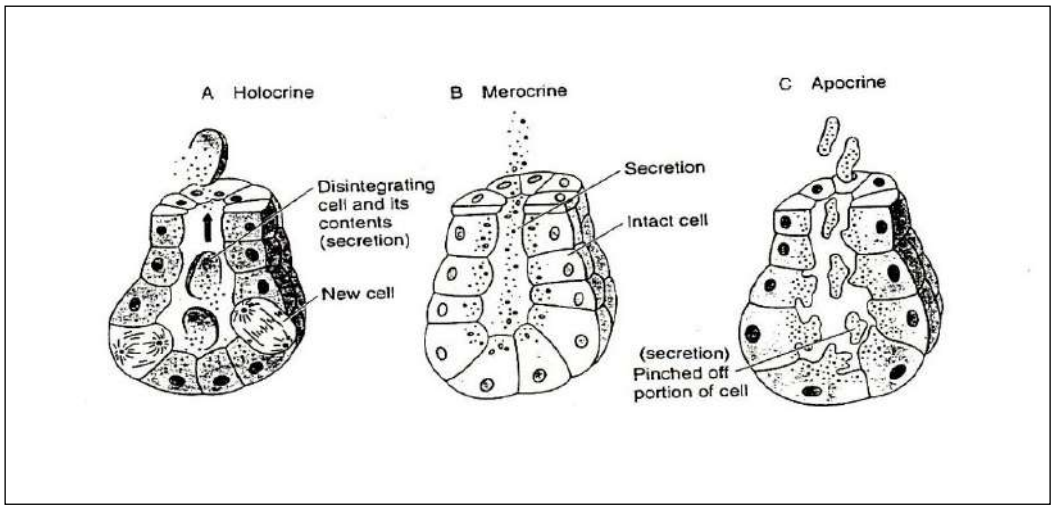
IV. According to the mode of secretion:

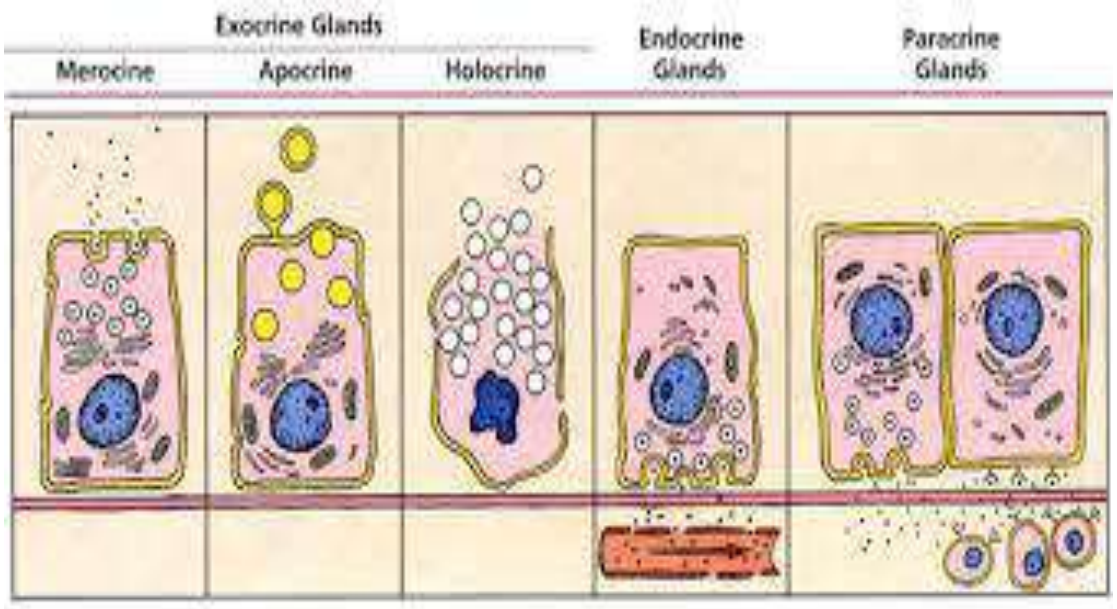
Merocrine glands: The secretory cells are release their contents by exocytosis.

No loss of cell membrane e.g. sweat glands, pancreas and salivary glands.

Apocrine glands: Part of cytoplasm of secretory cells is released along with secretory materials e.g. axillary sweat glands and mammary glands.

Holocrine glands: The cells after being completely full of secretion, they burst and become totally included in the secretory products e.g. sebaceous glands





C) Neuroepithelium (sensory epithelium):

It is a highly specialized type organized during development for perception of external stimuli.

<u>It consists of:</u>

Neuroepithelial (or hair cells).

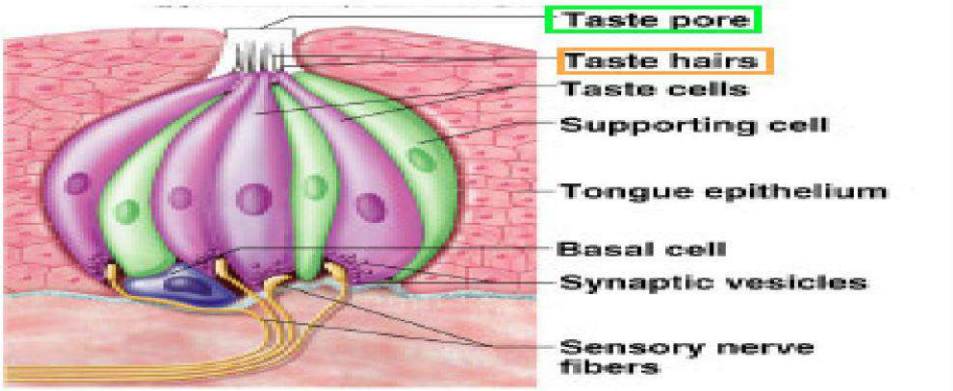
Sustentacular (or supporting cells).

Basal cells (act as stem cells).

Site and function:

1-Taste buds (taste).

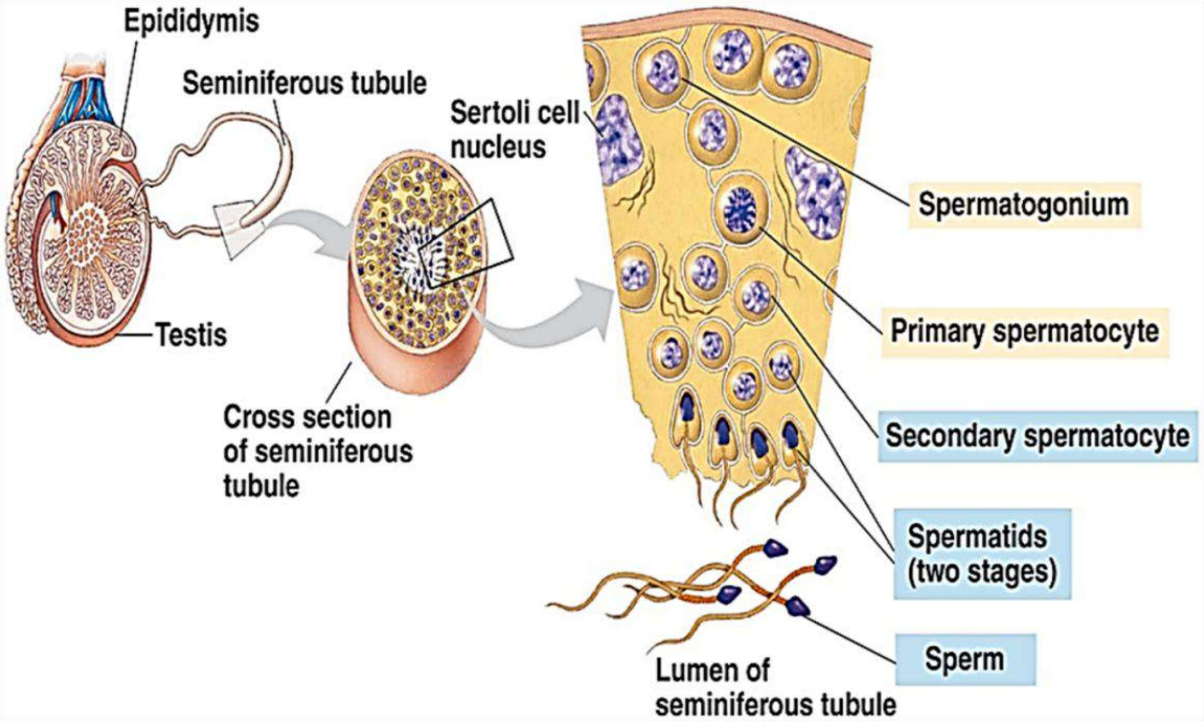
2-Organ of corti in the inner ear (hearing)



D) Germinal epithelium:

*It constitutes an important part in the testis Spermatogenic epithelium becomes active and repeatedly divides (meiotic division) during and after puberty.

*It lines the seminiferous tubules in testis and consists of spermatogenic cells and Sertoli cells.

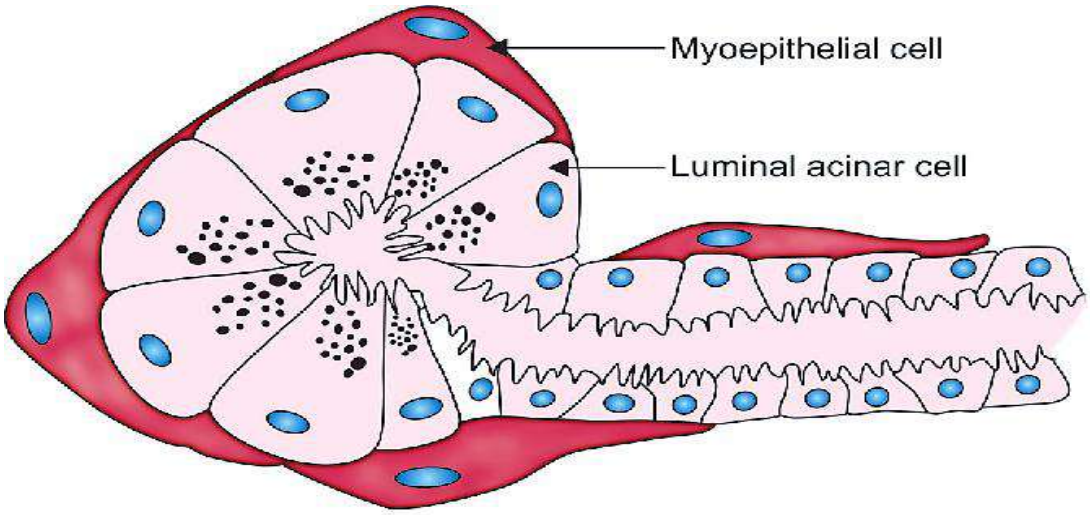


E) Myoepithelium (basket cells) :

Definition: Special type of epithelial cells that are modified to contract.

Structure: Branched cells with multiple processes, which contain contractile filaments.

Site and function: Located around the basal parts of some epithelial secretory cells. They surround the glandular acini and help in squeezing them to discharge their secretion into the ducts, e.g. in sweat glands, salivary glands and mammary glands.



References :

1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

2- Ross, M. H., & Pawlina, W. (2006). *Histology: A text and atlas with correlated cell and molecular biology* (5th ed.). Lippincott Williams & Wilkins.

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Lecture 12

BLOOD

Learning objectives:

After this lecture, students should be able to:

*Describe the cytological characteristics of Blood Elements (RBCs, WBCs, Platelets)

*Differentiate between the stem progenitor and precursor cells of hemopoiesis.

*Differentiate between 2 types of bone marrow, red and yellow.

*Describe the constituents of bone marrow.

*Define the stages of each element development (Haemopoiesis)

Blood is a specialized type of CT formed of blood cells in a fluid matrix (plasma).

Blood cells (45%) are of three types:

Erythrocytes (RBCs): 4-6 million per cubic millimeter.

Leucocytes (WBCs): 4-11 thousand per cubic millimeter.

Thrombocytes (platelets): 150-400 thousand per cubic millimeter.

Plasma (55%) is formed of water, gases, inorganic substances, organic substances, hormones and enzymes. The volume of blood is about 5 liters.

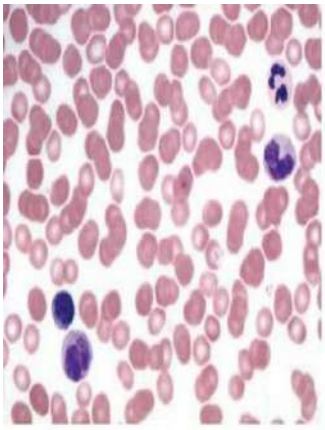
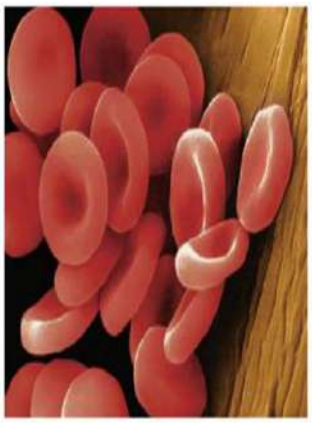
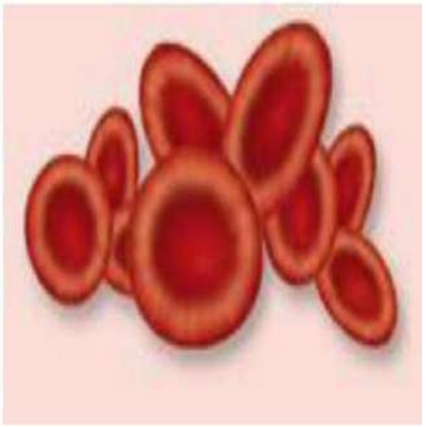
Red blood corpuscles (RBCs) or erythrocytes

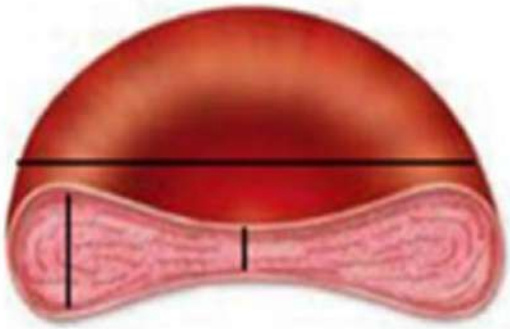
LM: they are rounded biconcave discs (with pale thin center and dark thick periphery).

EM: they are corpuscles and not true cells (as they have neither nuclei nor organelles) so cannot divide.

life span is about 4 months.

Old RBCs are phagocytosed in liver and spleen



Size of RBCs:

Normal size of RBCs: diameter 6-9 (7.5) µm

- central thickness 0.75 µm .

-Peripheral thickness 2.6 µm.

Normal number of RBCs:

-In ♀ 4.5-5 million/3mm (due to menstruation and ♀ hormones).

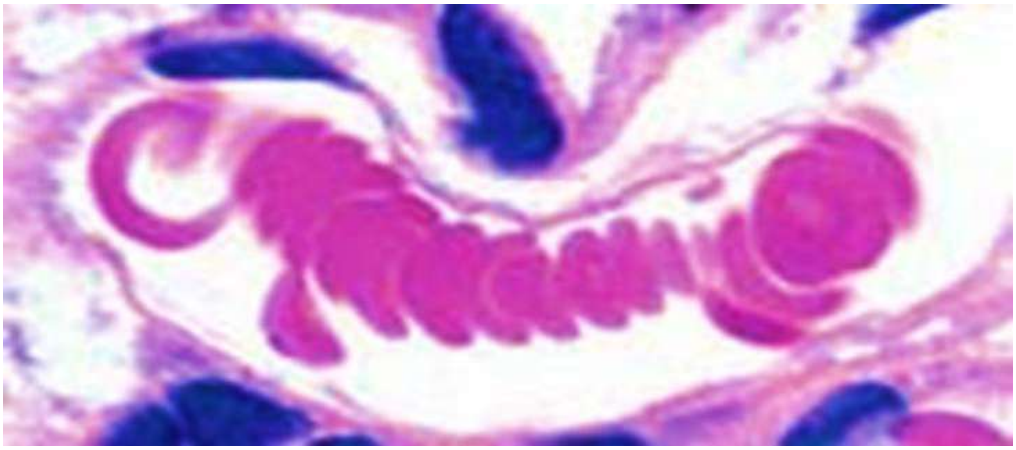
-In ♂ 5-5.5 million/3mm.

*Anemia (RBCs less than 4 million/3mm)

*Polycythemia (RBCs more than 6 million/3mm)

Shape of RBCs:

Normal shape of RBCs: rounded biconcave non-nucleated discs showing rouleaux appearance (as rows of coins).



How the structure of the R.B. Cs is adapted to their function?

1. The biconcave disc appearance increases the surface area by about 25% that increase the respiratory efficiency.

2. Erythrocytes are normally quite flexible, which permits them to bend and adapt to the small diameters and irregular turns of capillaries.

Several cytoskeleta l proteins stabilizes the membrane, maintains the RBC's shape, and provides its elasticity

3. The glycosylated extracellular domains of integral proteins in the Erythrocyte cell membrane include antigenic sites for the ABO blood typing System and Rh antigen

White blood cells (WBCs) or leucocytes

*WBCs are true cells (as they have nuclei, organelles and inclusions).

*They are colorless (devoid of hemoglobin) but appear white when packed together.

*They have amoeboid movement to penetrate capillaries and perform their phagocytic function in CT

Number of WBCs:

*Normal number of WBCs:

WBCs count 4-11 thousand/3mm and around 16 thousand/3mm at birth.

*Abnormal number of WBCs:

Leucopenia (WBCs less than 4 thousand/3mm):

Leukocytosis (WBCs more than 11 thousand/3mm):

Types of WBCs:

according to types of granules leucocytes are classified to:

Granular leucocytes (with specific and non-specific granules):

Basophils,

Eosinophils and

Neutrophils.

Non-granular leucocytes (with only non-specific granules):

Monocytes

lymphocytes (T and B).

	count (/ ^{3}mm)	**diameter (µm)**	**life span**	**nucleus**	**specific granules**
RBCs	4-6 million	7.5 µ	4 months	absent	absent
platelets	150-400 th.	2-4 µ	5-10 days	absent	present
WBCs	4-11 th.	6-18 µ	day-years	present	may be

Differential leucocytic count (percentage of each type of leucocytes relative to total leucocytic count):

	basophils	00 -0.75 %	12-15 µ	months	S-shape	histamine
	eosinophils	01-03 %	12-16 µ	1-2 weeks	bi-lobed	histaminase
	neutrophils	57-67 %	12-15 µ	1-4 days	2-5 segments	collagenase
	monocytes	3-7 %	14-18 µ	days-years	kidney-shape	non
	lymphocytes	25-33 %	06-18 µ	hours-ys	1 segmnt	non
	▪ small	▪ 15-20 %	▪ 06-09 µ		▪ dark	
	▪ large	▪ 05-10 %	▪ 09-18 µ		▪ pale	

T-lymphocytes count 60-80 % of all lymphocytes and can life 2 years while ○ B-lymphocytes count 25-30 % of all lymphocytes and can life 3 months

Basophils (0-0.75 % of leucocytes): concerned with allergy production.

The diameter of basophils is 10-12µm and their life span is months.

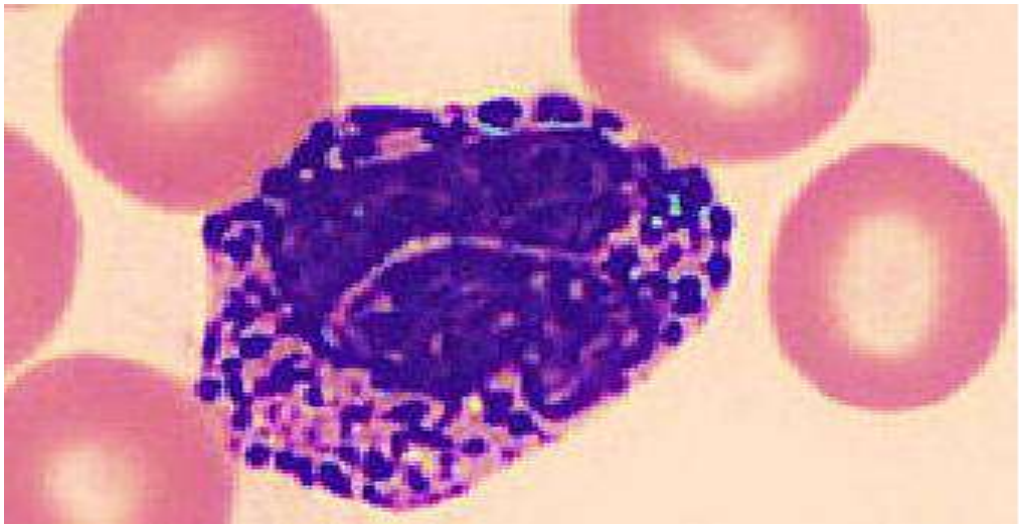
LM: Nucleus of basophil is single, irregular and S-shaped.

EM: Granules:

2ry specific basophilic large granules as that of mast cells (histamine, heparin and eosinophil chemotactic factor "ECF") - these granules mask nucleus and can stained by Gimsa.

Functions: production, storage and secretion of histamine (capillary vasodilator released during allergy), heparin (anticoagulant) and eosinophil chemotactic factor "ECF" (that attract eosinophils to terminate allergy).

Basophilia (increase number of basophils): It occurs in allergic diseases,



Eosinophils (1-3 % of leucocytes):

The diameter of eosinophils is 12-16 µm and their life span is 1-2 weeks.

LM: Nucleus of eosinophil is single, horseshoe and bilobed.

EM: Granules:

1ry non-specific azurophillic granules (primary lysosomes).

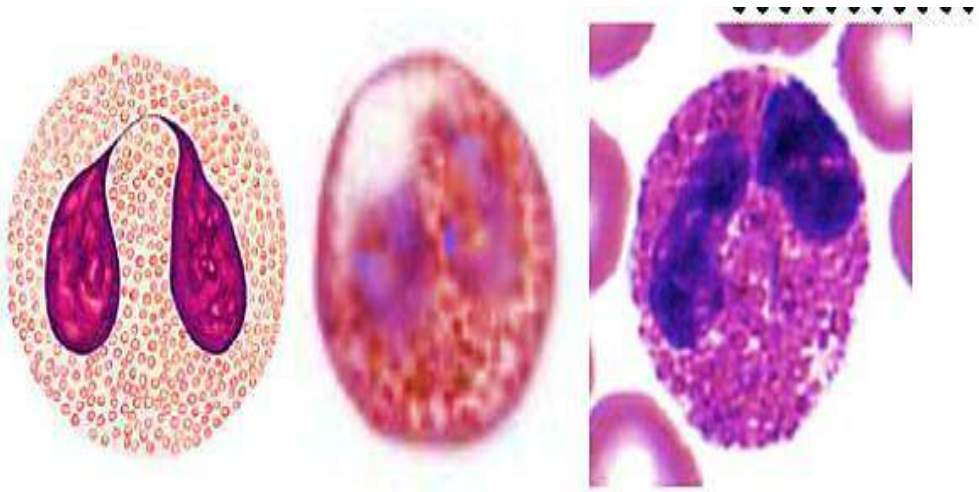
2ry specific acidophilic large granules (histaminase and sulphatase).

Functions: 1-Attracted to allergic sites to terminate allergy by destruction of histamine and phagocytosis of allergic antigen-antibody complexes.

2-Inactivate and kill parasitic larvae.

Eosinophilia (increase number of eosinophils):It occurs in allergic diseases and parasitic infestations.

Eosinopenia (decrease number of eosinophils): It occurs with cortisone therapy that inhibits bone marrow.



Neutrophils (57-67 % of leucocytes):

concerned with non-specific immunity.

The diameter of neutrophils is 12-15 µm and their life span is 1-4 days.

LM: Nucleus of neutrophil is single and formed of connected 2-5 segments (so neutrophils are called "polymorph-nuclear leucocytes").

EM: Granules:

1ry non-specific azurophillic granules (1ry lysosomes).

2ry specific pale large granules (bactericidal phagocytin, bacteriostatic lactoferrin, collagenase and alkaline phosphatase).

Functions: non-specific immunity (which is the first line of defense mechanism)

1-Thy destruct bacteria by proteolytic enzymes present in their granules

2-They secrete trephine substance (that perform healing).

Neutrophilia (increase number of neutrophils): It occurs in appendicitis, tonsillitis, abscess ...

Neutropenia (decrease number of neutrophils): It occurs in typhoid fever, drugs, radiation ...



Monocytes (3-7 % of leucocytes): concerned with non-specific immunity.

The diameter of monocytes is 14-18 µm and their life span is 3 days in blood, 3 months in CT proper and years outside them.

LM: they have large pale kidney-shaped nuclei with pseudopodia.

EM: few organelles with well-developed Golgi apparatus and many azurophilic granules (lysosomes).

Functions: Change to macrophage (and other mononuclear phagocytic cells) to phagocytose small foreign bodies.

Fuse together to form giant cells (as osteoclasts) to phagocytose large foreign bodies.

Monocytosis (increase number of monocytes): It occurs in chronic infections as TB ...



Lymphocytes (25-33 % of leucocytes): concerned with specific immunity. According to size lymphocytes are classified into small & large lymphocytes:

Small inactive lymphocytes (15-20 % of leucocytes):

The diameter of small lymphocytes is 6-9 µm (like RBCs).

LM: they have dark rounded nuclei with pale scanty cytoplasm (forming a thin rim around nucleus).

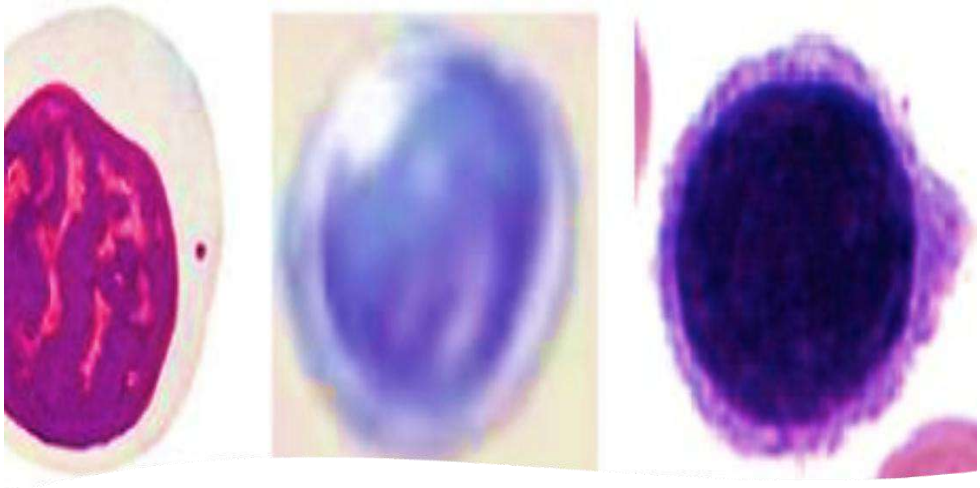
EM: few organelles with many free ribosomes and microvilli.

Large active lymphocytes (5-10 % of leucocytes):

The diameter of large lymphocytes is 9-18 µm.

LM: they have pale indented round nuclei with clear nucleolus and dark abundant cytoplasm.

EM: many mitochondria, many rER and well developed Golgi apparatus.



Function of lymphocytes: specific immunity (which is the second line of defense mechanism).

According to function lymphocytes are classified into B, T & NK lymphocytes:

1)B-lymphocytes (25-30% of circulating small lymphocytes):

*They can life for 3 months.

*They settle in lymphoid tissue, and re-circulate between blood and lymphoid tissue.

*They are bursa dependent as they develop in bursa of fabricius (of birds) and in bone marrow (of human).

*They originate from B-type colony forming cells of bone marrow that differentiate (to B-lymphoblasts then B-lymphocytes) and acquire surface receptors then migrate from bone marrow to peripheral lymphoid organs (lymph nodes, spleen, tonsils ...) as small inactive B-lymphocytes.

*Antigens are picked by T-helper lymphocytes then delivered to small B-lymphocytes their activation to:

Subtypes of B lymphocytes:

1- Plasma cells in which the activated B – lymphocyte changes to plasmablast, then plasma cell which secretes the specific antibody (primary immune response).

2 -B Memory cells : Some activated B – cells remain in lymphatic tissue, to respond rapidly to the second exposure for the same antigen, (secondary immune response).

Functions of B – Lymphocytes: Humoral Immunity

2) T-lymphocytes (60-80% of circulating small lymphocytes):

They can life for 2 years.

They are thymus dependent (as they require thymus gland for their development and maturation).

They originate from T-type colony forming cells of bone marrow that differentiate (to T-lymphoblasts) then migrate from bone marrow to the thymus gland.

o In early life: small inactive T-lymphocytes migrate from the cortex of thymus gland).

o In adult life :lymph nodes and spleen give small inactive T-lymphocytes throughout life.

Antigens or foreign cells (viruses, fungi or tumor cells) activation of small T-lymphocytes to:

1-T-cytotoxic cells (T-killer cells): that performs primary cellular immunity (by secretion of perforins which produce pores in the cell membrane of foreign cells leading to their lyses)

2-T-memory cells: that performs secondary cellular immunity which defends body .against secondary infection (further exposure to the same antigen)

3-T-helper cells: that helps in humoral immunity of B-lymphocytes

4-T-suppressor cells: that suppresses both humoral and cellular immunity

3)Natural killer lymphocytes (5% of circulating lymphocytes):

They can life for 3 months.

They have non-B non-T receptors and can kill some tumor cells and infected cells.

Lymphocytosis (increase number of lymphocytes):

It occurs in chronic diseases (as TB...) and in leukemias.

	RBCs	WBCs
Types:	- one type	- 5 types
Number:	- 4-6 million/^{3}mm	- 4-11 thousand/^{3}mm
Size:	- 6-9 (7.5) µm	- 6 to 18 µm
Shape:	- biconcave discs with rouleaux appearance	- spherical without rouleaux appearance
Color:	- greenish yellow	- colorless
Osmotic pressure:	- liable to hemolysis	- resist hemolysis
LM:	- corpuscles: — no nuclei — no cell organelles — contain hemoglobin	- true cells: — contain nuclei — contain cell organelles — not contain hemoglobin
Function:	- carrying O_2 and CO_2 inside blood vessels (so they have no amoeboid movement)	- phagocytosis outside blood vessels (so they have amoeboid movement)
Development:	- in red bone marrow	- in red bone marrow and in lymphatic tissues
Site:	- in blood	- in blood, in CT and in lymphatic tissues
Life span:	- 4 months	- few dyes to many years

References

1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

2- Ross, M. H., & Pawlina, W. (2006). *Histology: A text and atlas with correlated cell and molecular biology* (5th ed.). Lippincott Williams & Wilkins.

3- Wheater, P. R., Burkitt, H. G., & Daniels, V. G. (2006). *Wheater's functional histology: A text and colour atlas* (5th ed.). Churchill Livingstone/Elsevier.

4-Gartner, L. P. (2020). *Textbook of histology* (5th ed.). Elsevier.

5-Kierszenbaum, A. L., & Tres, L. (2025). *Histology and cell biology: An introduction to pathology* (6th ed.). Elsevier.

Lecture 13

CONNECCTIVE TISSUE

Learning objectives

After this lecture, students should be able to

-Describe their general organization, major functions

-List the types of connective tissue cells

-Enumerate the types of .Describe the structure and ultrastructure of CT cells connective tissue fibers

General characteristics of connective tissue:

It is called connective because it supports, binds, and connects various tissues and organs.

Functions of connective tissue:-

1-It supports and connects various tissues and organs.

2-Its matrix supplies blood vessels, lymphatics and nerves to the organs.

3-Its cells form a defense mechanism.

4-Specific functions for adipose connective tissue.

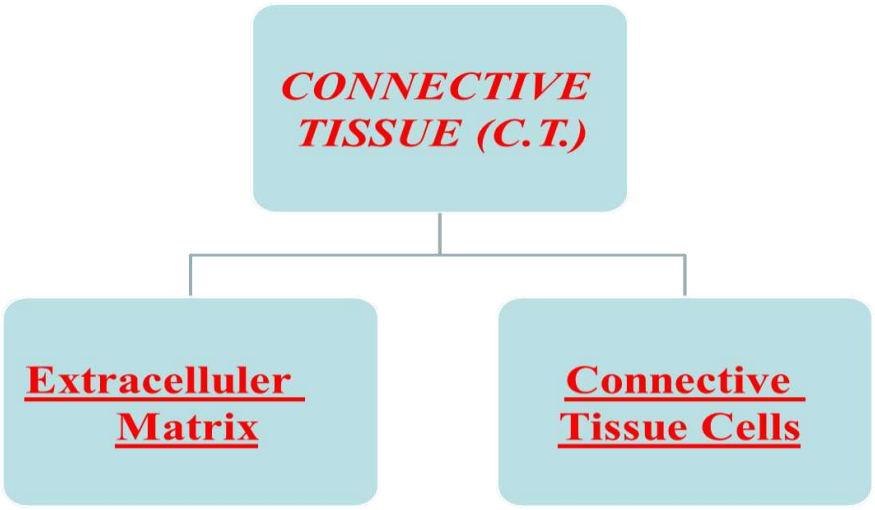
*Storage of fat

*Form supporting pads of fat around organs (e.g kidney)

*It gives the body its natural contour

*Acts as heat insulator.

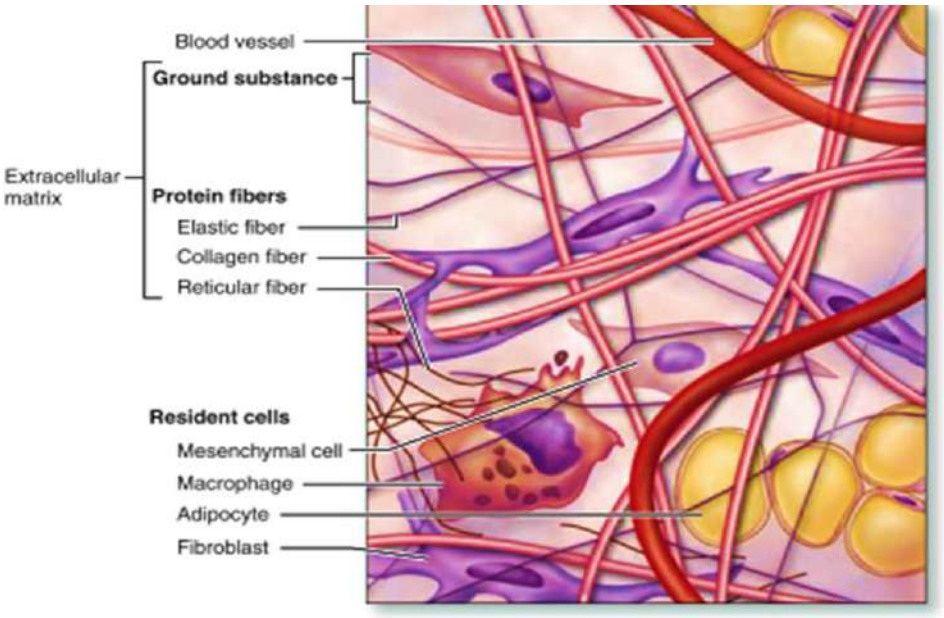
Structure of connective tissue

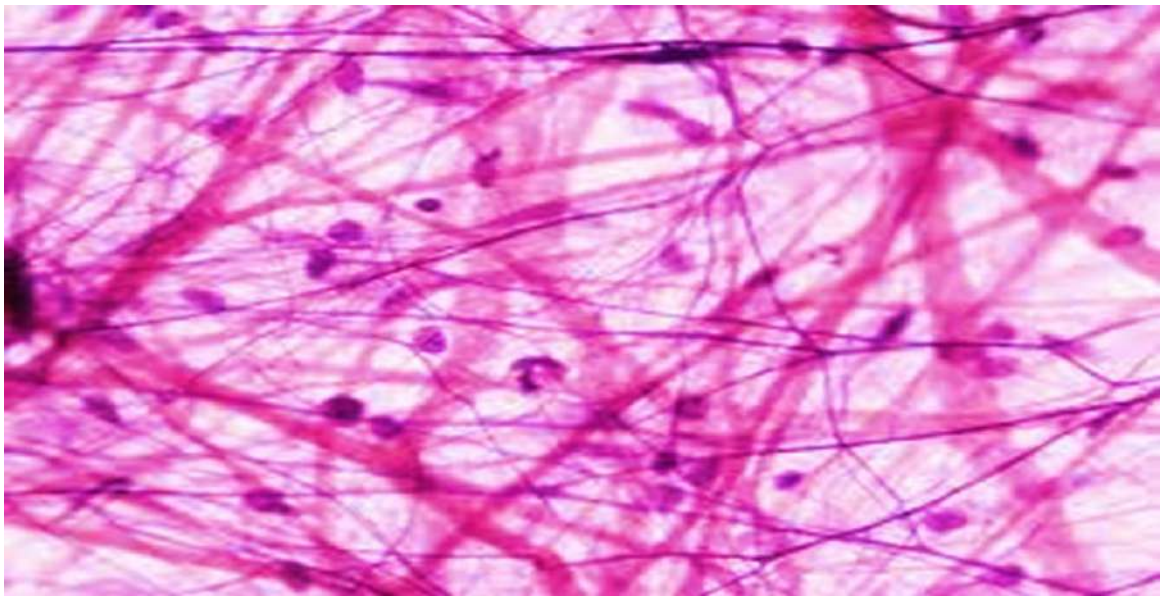


The Extracellular matrix

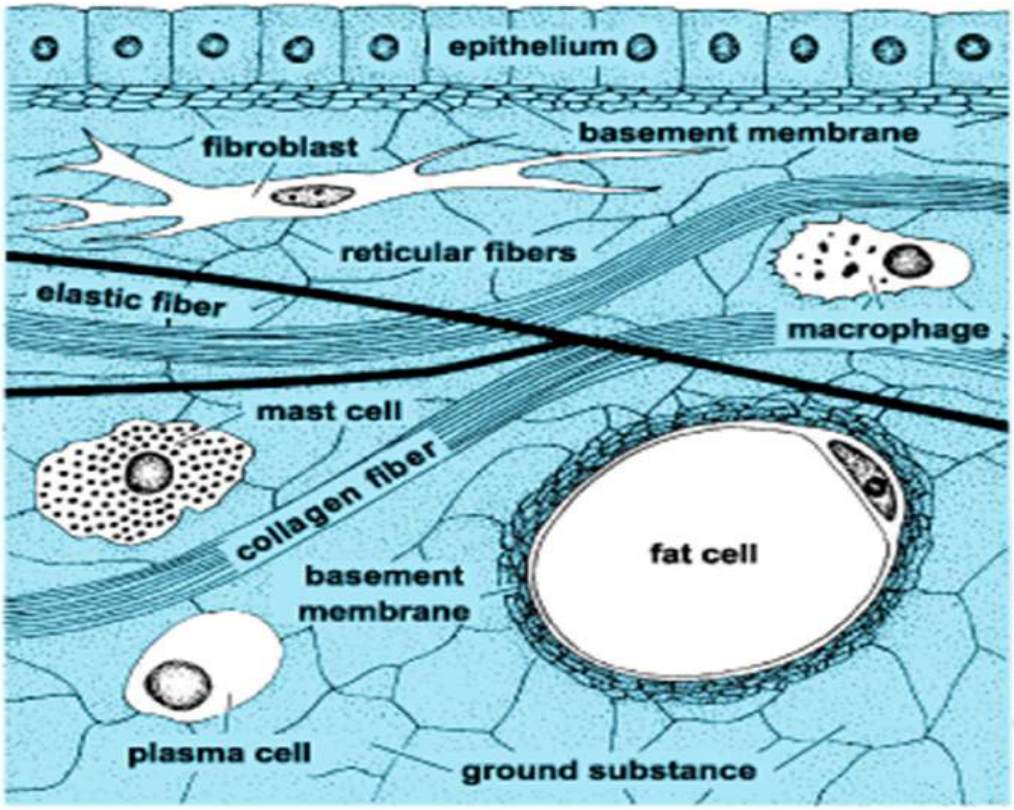
Ground substance

Fibers





CONNECTIVE TISSUE CELLS

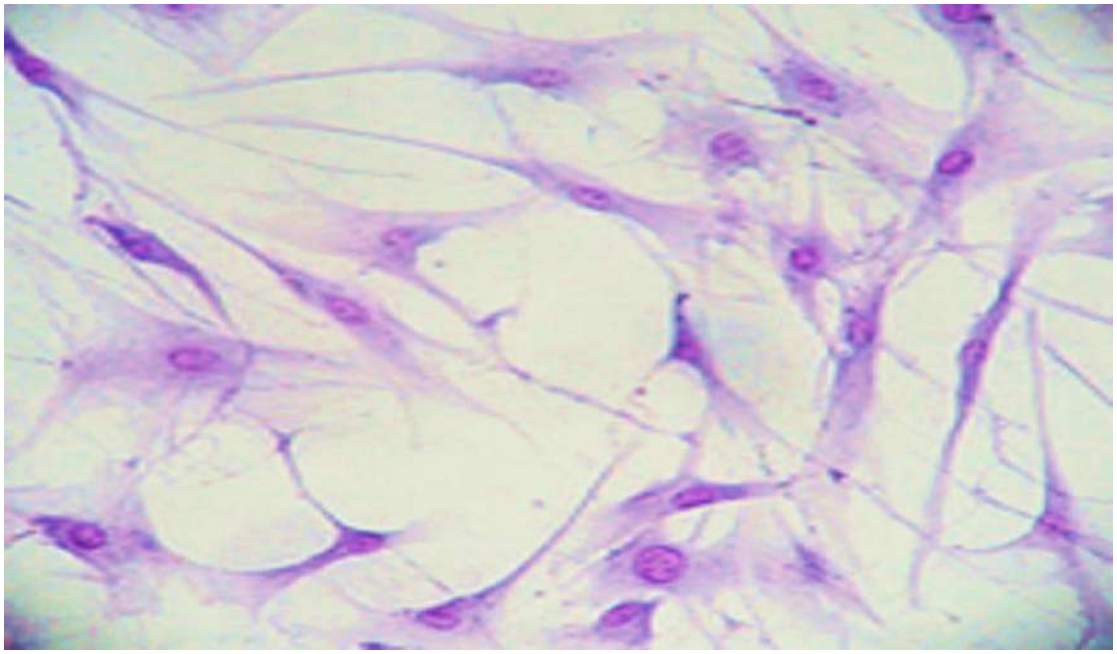


Fixed Cells	Free Cells
-Fibroblast -Fat cell -Mesenchymal cell - Stable long – lived cells	-Macrophage - Mast cell - Plasma cell -Leucocytes -Foreign body giant cells -short – lived cells that are changing population. These are motile cells migrating from blood circulation to the connective tissue.

Undifferentiated Mesenchymal cells:

these are branched cells with pale basophilic cytoplasm

Function: give rise to many other types of C.T cells.



Fibroblast:

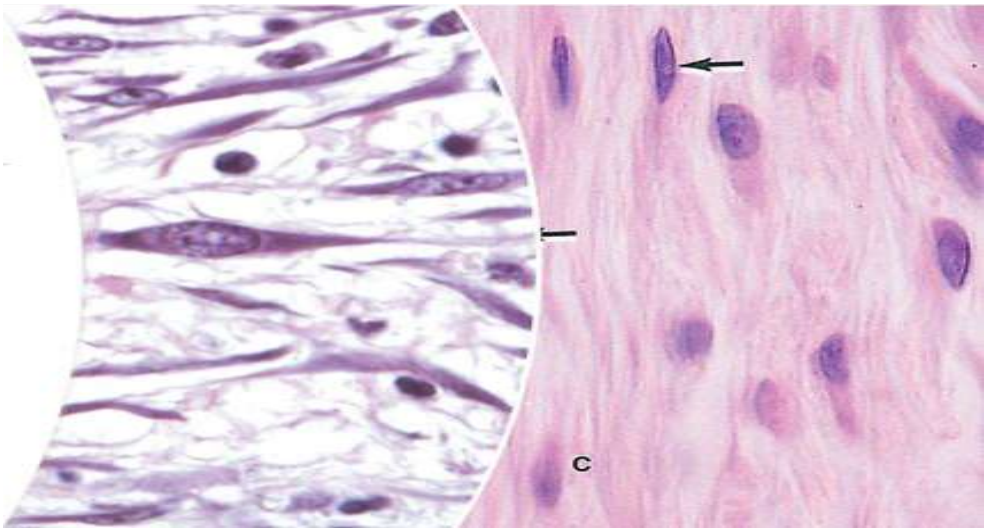
Origin: Mesenchymal cells divide and differentiate into fibroblasts which are active young cells. After getting old, fibroblasts are called fibrocytes.
Fibrocytes or old fibroblasts are flat cells with very little pale acidophilic cytoplasm around flattened or oval nuclei with prominent nucleoli.



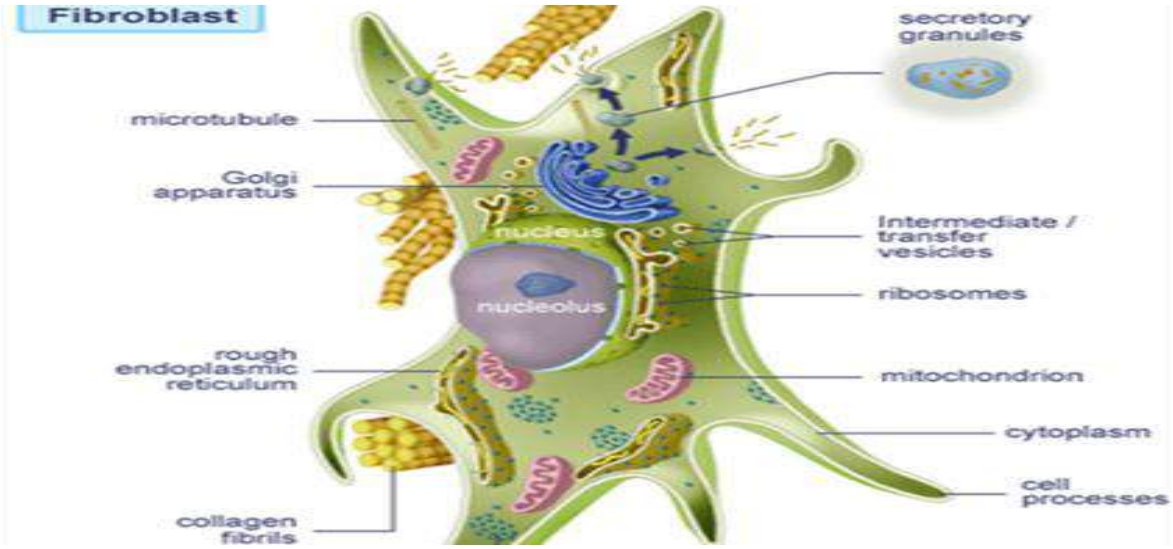
Structure

L/M:

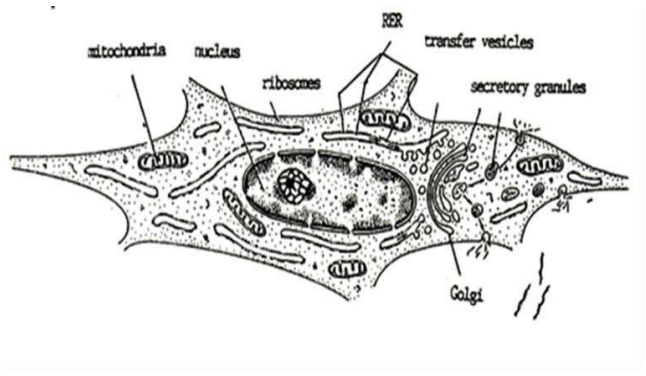
The active young cells appear as branched cells with abundant basophilic cytoplasm and oval pale nucleus with prominent nucleolus.



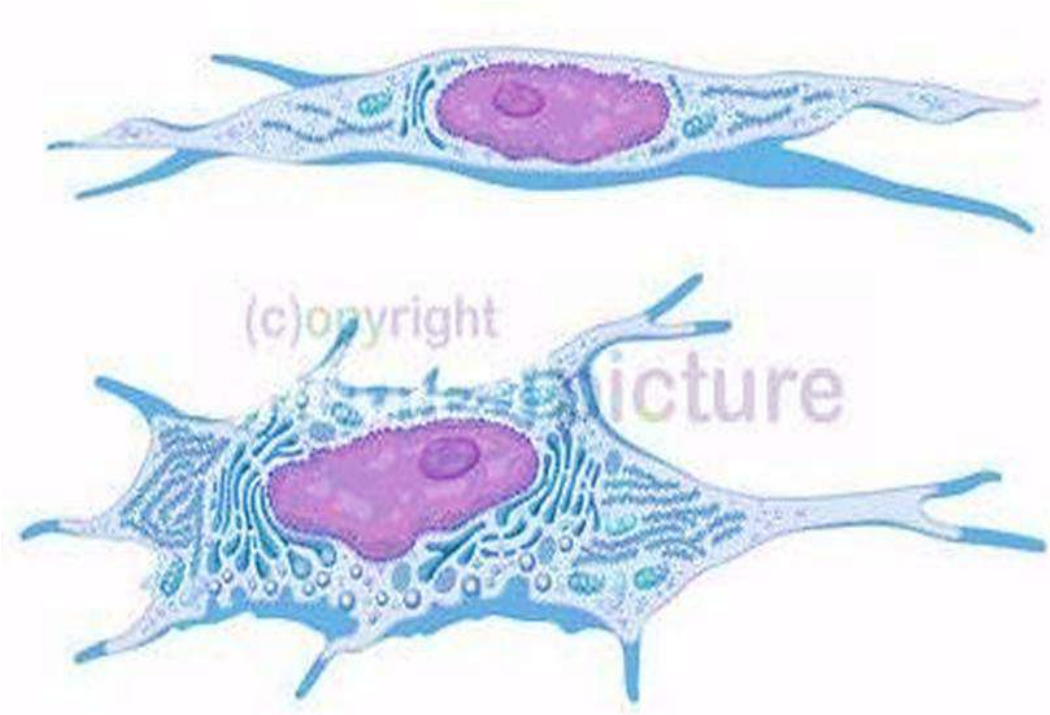
It contains numerous rough endoplasmic reticulum, several well developed Golgi stacks, ribosomes, mitochondria and transfer vesicles present between the rough endoplasmic reticulum and Golgi stacks The secretory vesicles are present between Golgi and the cell surface



Electron microscopic picture of fibroblast



	Fibroblasts	Fibrocytes
	Young active cells	Old inactive cells
By light microscope	a. Larger in size b. Branched cells with abundant basophilic cytoplasm. c. The cells contain large, oval, vesicular nuclei with distinct nucleoli.	a. Smaller in size b. Spindle cells with little lightly acidophilic cytoplasm c. The cells contain spindle deeply stained nuclei with little chromatin.
By electron microscope	The cytoplasm contain: • Well-developed Golgi complex • Well-developed rough E.R. • Abundant Mitochondria	The cytoplasm contain:- • Less-developed Golgi complex • Less-developed rough E.R. • Few Mitochondria
Functions	1- Fibroblasts secrete tropocollagen to form collagenous fibers 2- Fibroblasts secrete elastin to form elastic fibers. 3- Fibroblasts secrete mucopolysaccharide to form connective tissue matrix 4- They play an important role in repair of wounds and formation of scar tissue.	



© medicalpicture no: 35700

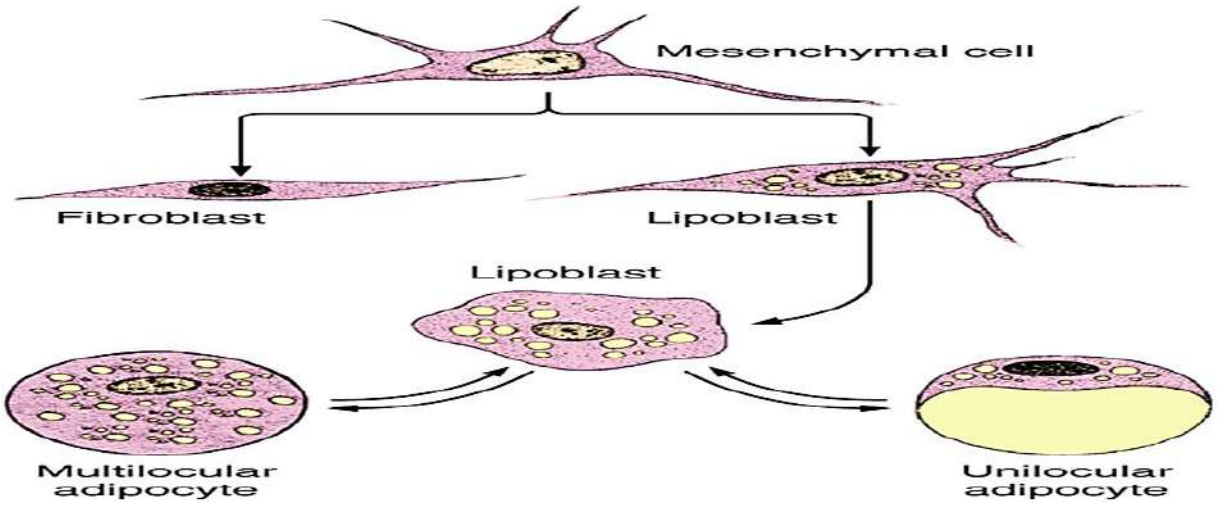
Fat (adipose) cell or adipocyte

They are present singly or in groups

Origin:

Fat cells develop from fibroblasts at sites which accumulate fat, whereas in developing sites they are formed from mesenchymal cells.

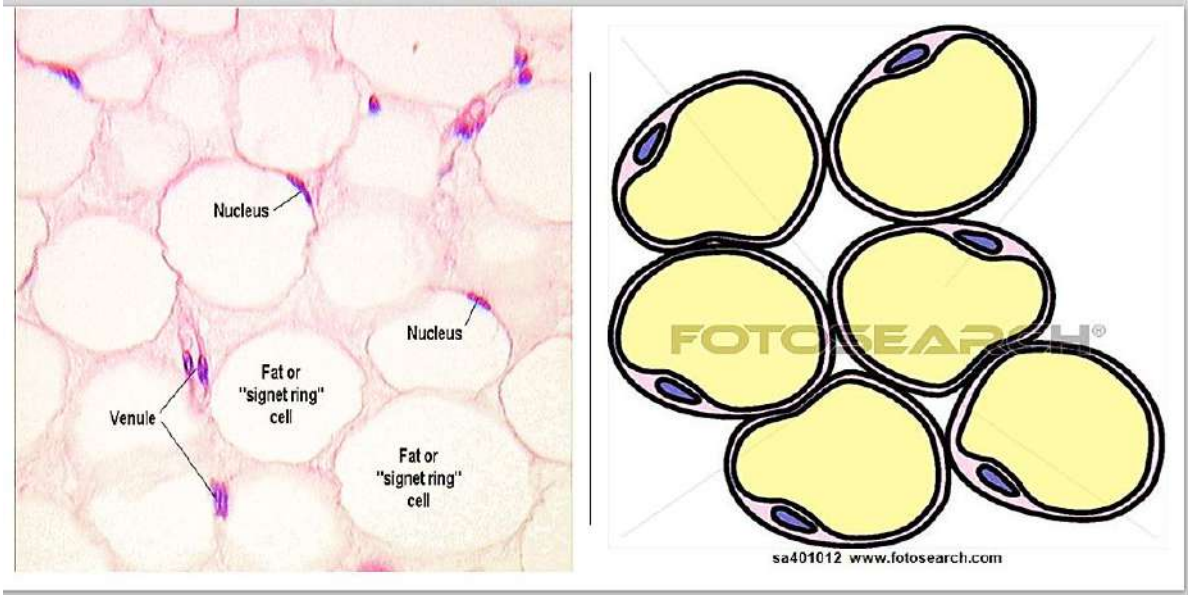
The mother cell (a fibroblast or a mesenchymal cell) accumulates small fat droplets in its two poles then the small droplets of fat unite and form a single large droplet which pushes the nucleus to one side.



Structure: Using H&E stain, fat cells appear in two types:

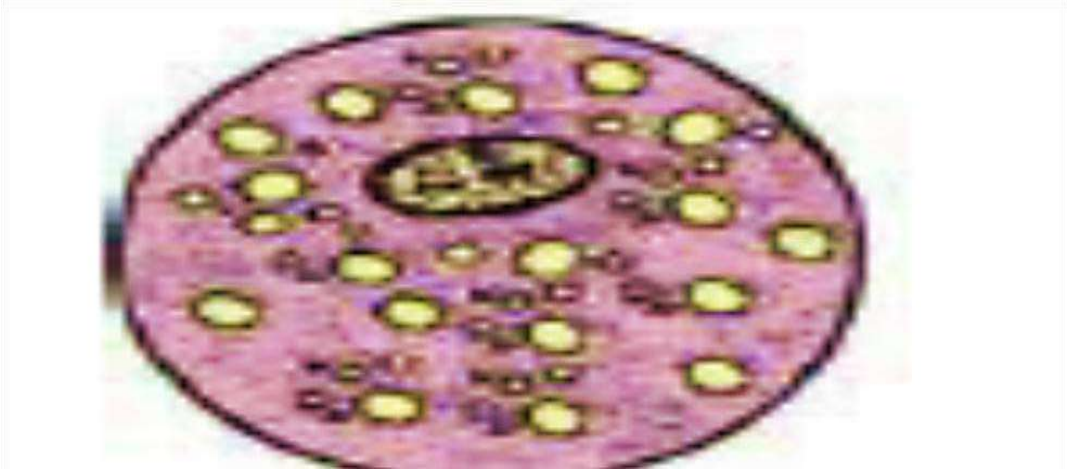
Unilocular fat cells(White fat):

These cells have single large lipid droplet and flat peripheral nucleus. cells appear as signet ring due to the dissolved fat droplet and the peripheral nucleus and thin rim of cytoplasm .



2-Multilocular fat cells (Brown fat):

These cells have multiple small lipid droplets. They have numerous mitochondria and blood vessels.



Plasma cells

Origin:-They arise from B-lymphocytes that are activated by specific antigen.

By light microscope:-

They are rounded or oval cells with strongly basophilic cytoplasm.

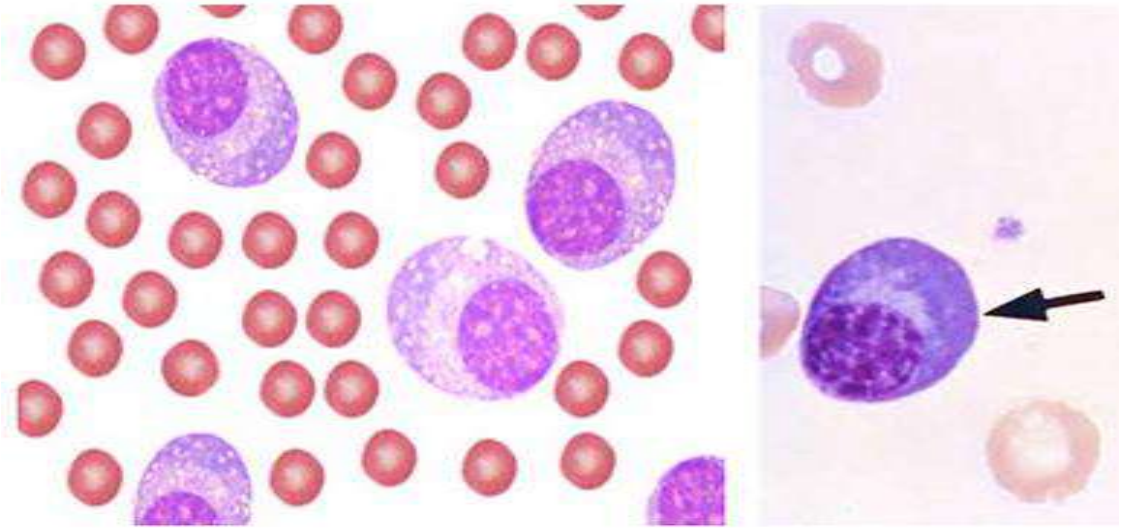
The nucleus is rounded, eccentric with a characteristic cart-wheel or clockface appearance of chromatin.

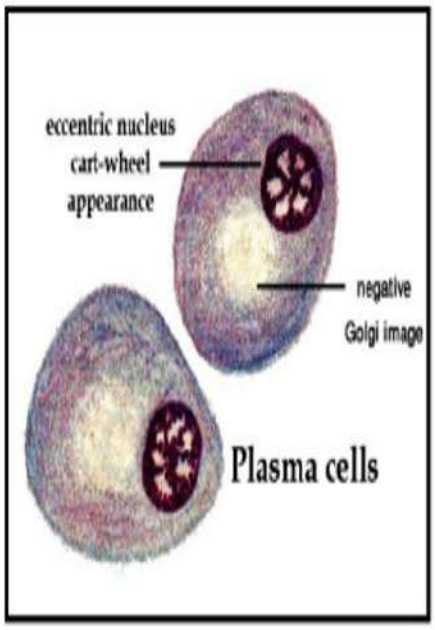
By electron microscope:-

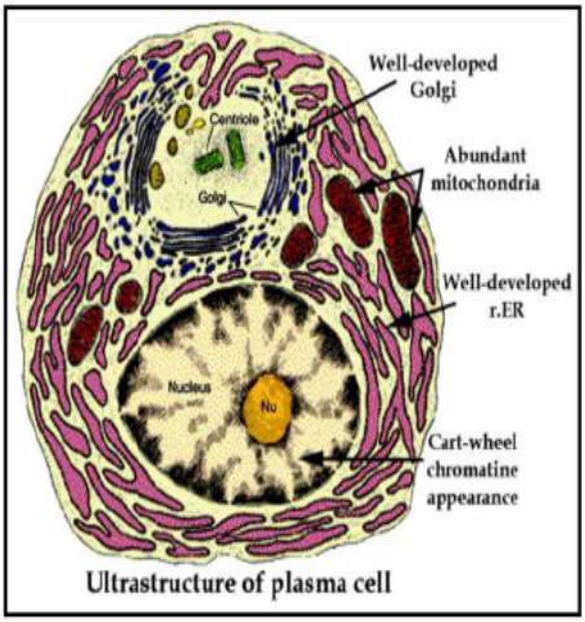
well-developed Golgi apparatus,

well-developed rough ER and

abundant mitochondria.

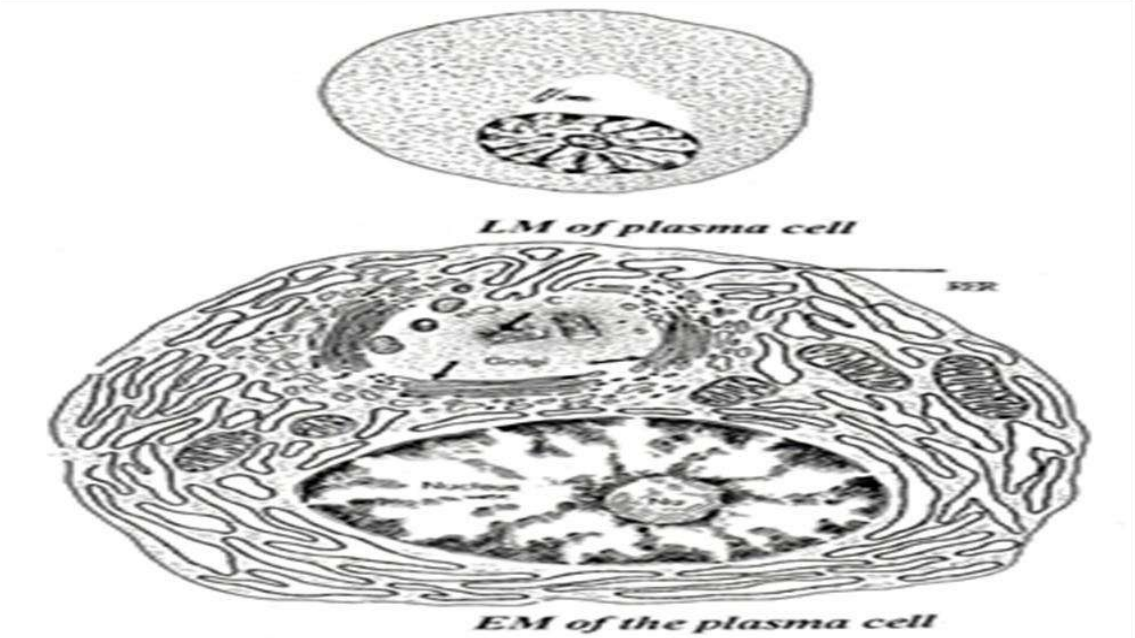




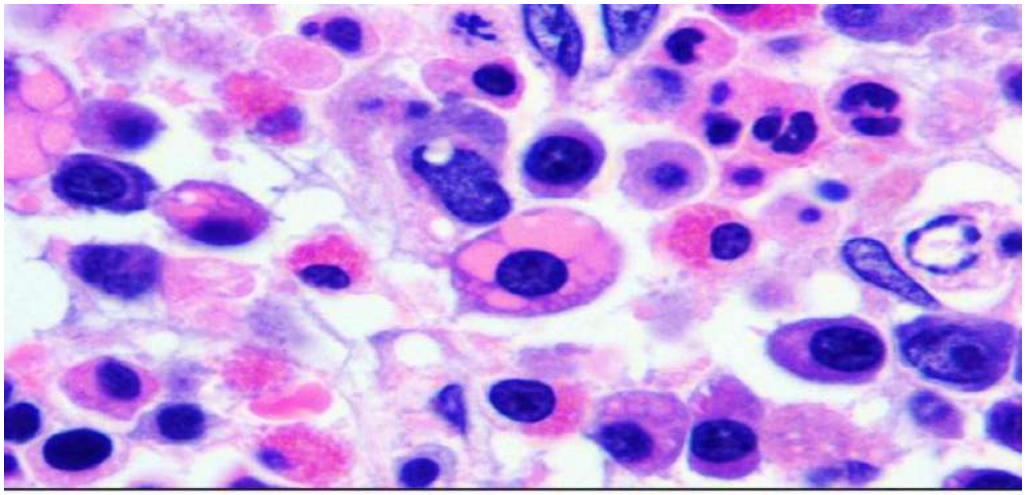


Functions:-

They are capable of synthesizing and secreting immunoglobulin or antibodies



The cytoplasm contains rounded acidophilic bodies, (Russell bodies). These bodies are larger than secretory vesicles and represent accumulations of secretions. They signify the beginning of degeneration in plasma cells.



Russell bodies

Mast cells

They are present in groups along the blood vessels

By light microscope:-

They appear as oval or rounded cells distended with basophilic , metachromatic granules.

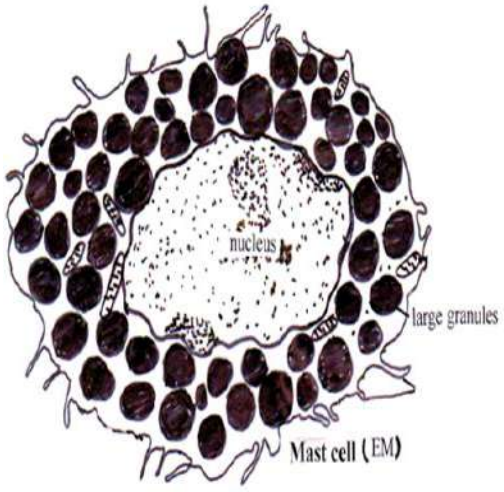
The nucleus is rounded and centrally located.

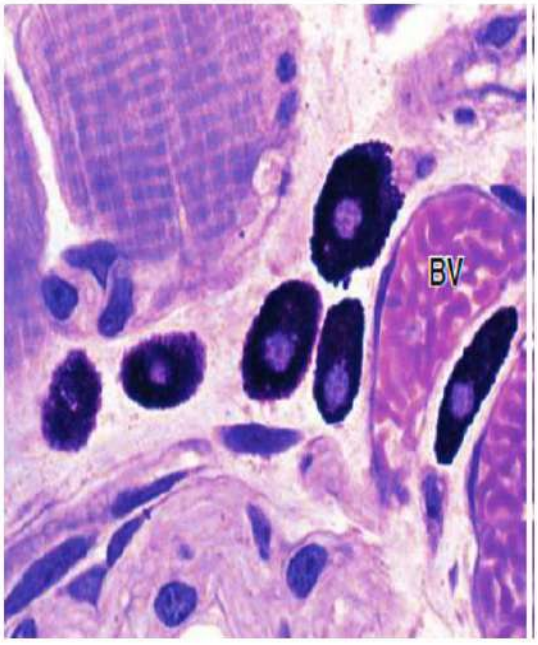
Functions:

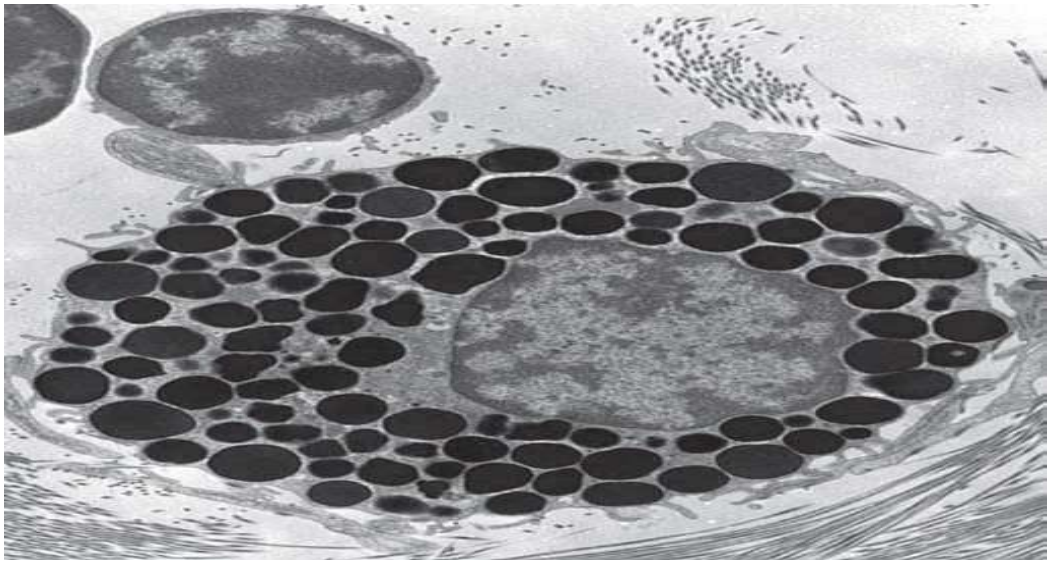
secretion of

Heparin : acts as an anticoagulant,

Histamine : plays a role in allergic conditions.







Macrophages (Histocytes)

Origin:-

They develop from monocytes migrating from the blood to the tissue.

light microscope:-

Macrophages appear oval, rounded and branched cells .

The nucleus is small, kidney shaped, deeply stained and eccentric.

Macrophage can be stained vitally with Trypan blue.

By electron microscope:-

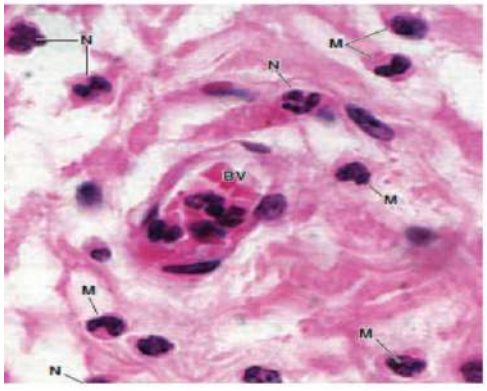
The cellular outlines are irregular.

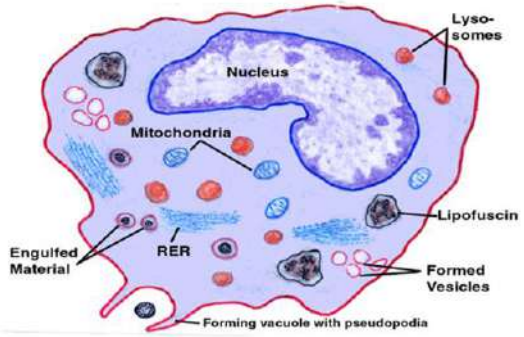
The cytoplasm contains Golgi and numerous phagocytosed particles (Phagosomes), residual bodies and plenty of lysosomes .

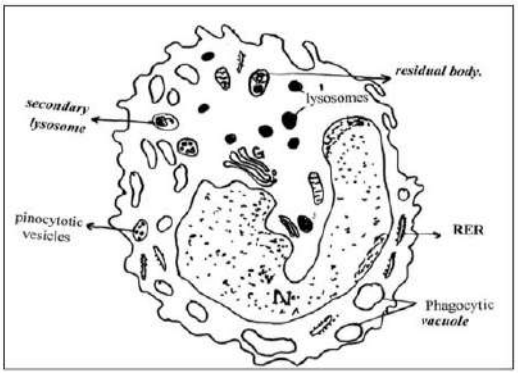
Functions:-

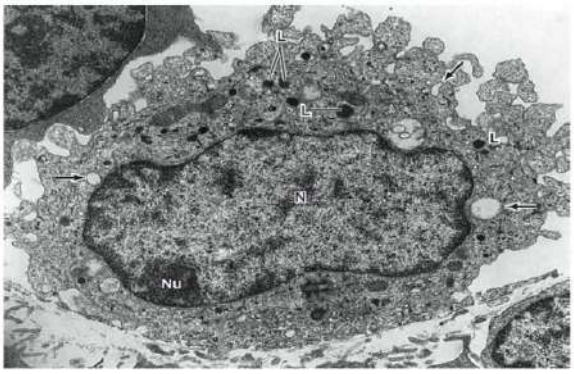
*They are phagocytic cells.

*Also, it has antiviral effect .They secrete interferon which is a compound protein that has antiviral antiproliferative effect









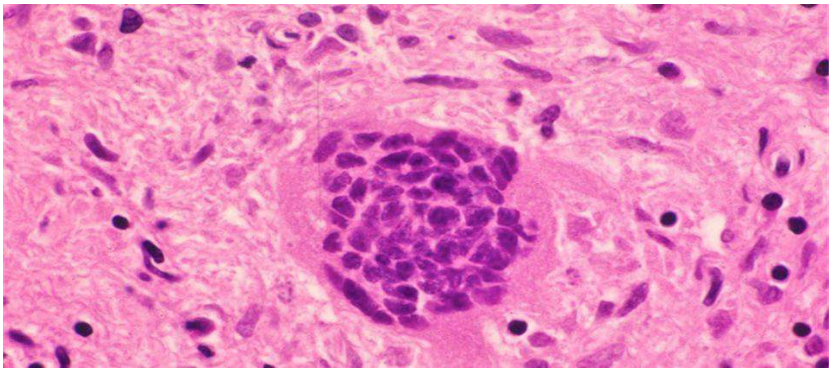
Foreign body giant cells:

Origin:

Originate when a foreign material is too large to be phagocytosed by the macrophages.

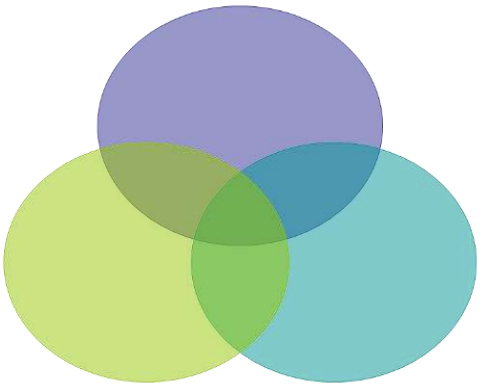
The fusion of two or more macrophages or monocytes form a large cell with two or more nuclei which can engulf the large foreign particles.

Giant cells are common in tuberculosis because tubercle bacilli is large.



The Connective tissue fibers

Collagenic (white) fibers



Elastic (yellow) fibers

Reticular fibers

1. Collagen (white) fibers

Are colorless elongated or wavy bundles which are composed of non- branching small fibrils

longitudinal arranged

It is composed of the protein collagen

They are acidophilic thus staining pink with H&E stain

There are six major types of collagen:

Type I: in connective tissue proper, bone dentin and cementum.

***Type II*:** in hyaline and elastic cartilages.

***Type III*:** reticular fibers.

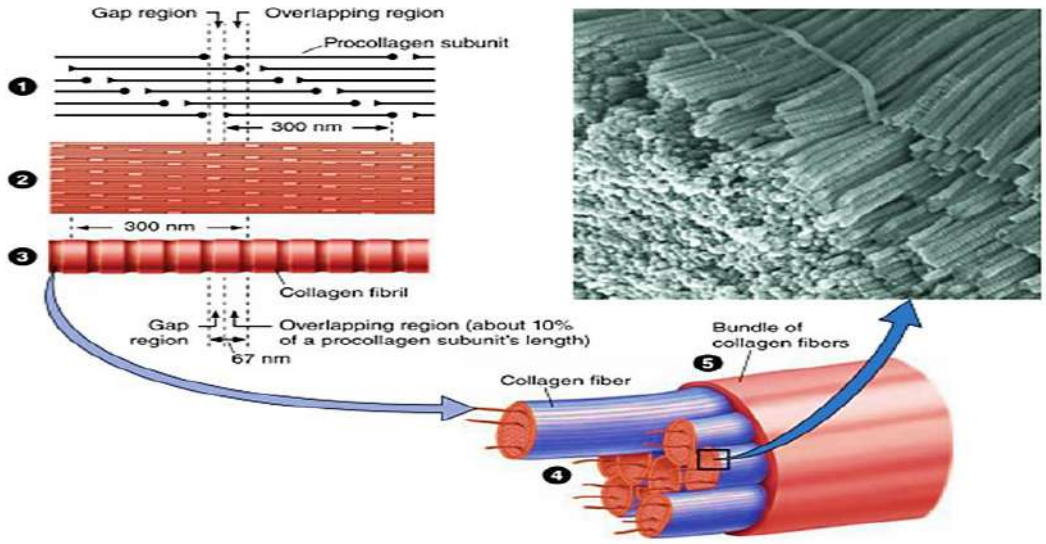
***Type IV*:** lamina densa of the basal lamina

***Type V*:** associated with type I collagen and in the placenta.

***Type VII*:** attaching the basal lamina to the lamina reticularis

Types of collagen fibers

Type of collagen	Location	origin
Type 1	Tendon – Bone -Capsule of organs- white fibrocartilage	Fibroblasts-Osteoblasts
Type 2	Hyaline Cartilage – Elastic Cartilage	Chondroblast – chondrocyte
Type3	Form reticular fibers of organs (Liver – spleen – kidney)	Smooth muscle cells and Fibroblasts
Type 4	Basal lamina of epithelium	Epithelial cells-endothelium

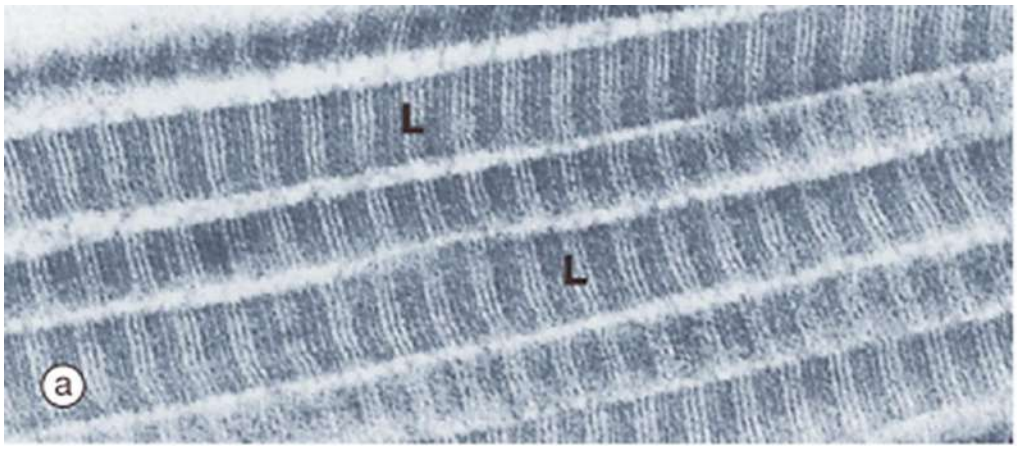


E.M appearance

*Tropocollagen molecules aggregate and become packed together to form fibrils.

*Collagen fibrils are thin, elongated structures having transverse striations (transverse banding periodicity)due to alternative dark and light bands.

*The periodicity (striations) are caused by the regular, overlapping arrangement of the collagen molecule subunits



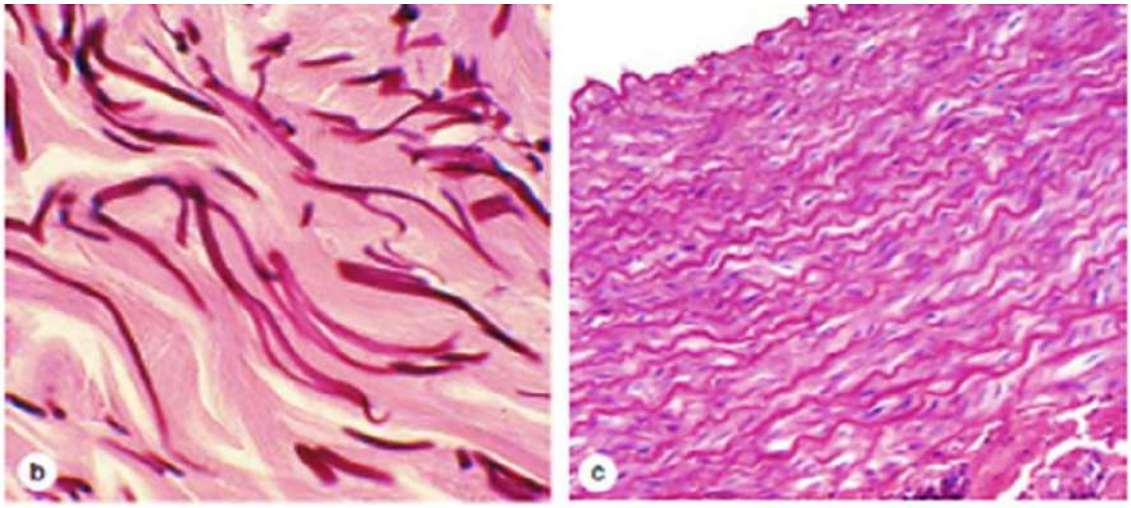
2. Elastic (yellow)fibers

*They are long, thin fibers which have They are mainly composed of the protein elastin

*They branch and may unite and form irregular networks Elastic fibers are weakly acidophilic.

*They are well demonstrated with orcein stain where they take a dark brown colour.

*With E.M., there is no periodicity in elastic fibers (no corss strations).

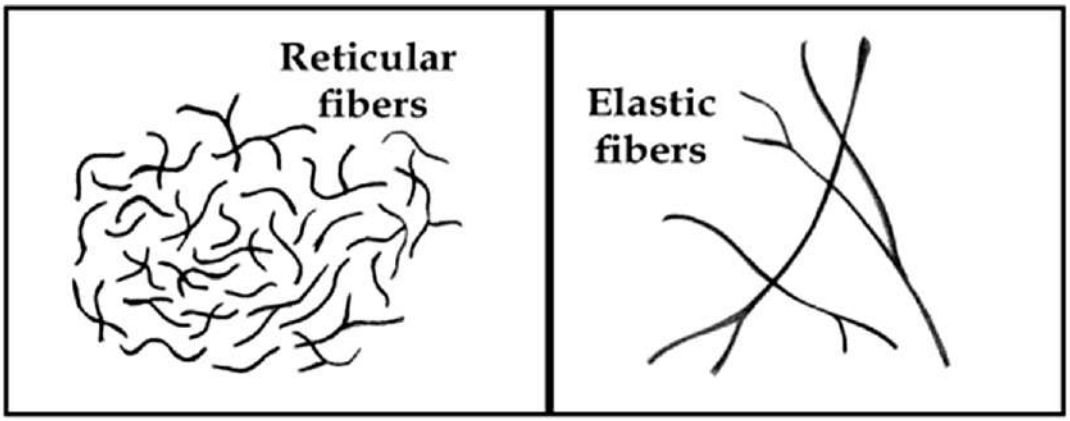


Elastic fibers: b. elastic fibers (dark) seen among collagen fibers. **c.** showing elastic lamellae in the aorta stained with H&E.

Elastic fibers are mostly formed from smooth muscle cells particularly in the wall of blood vessels

Function of elastic fibers:

They provide elasticity and strength to the tissue.



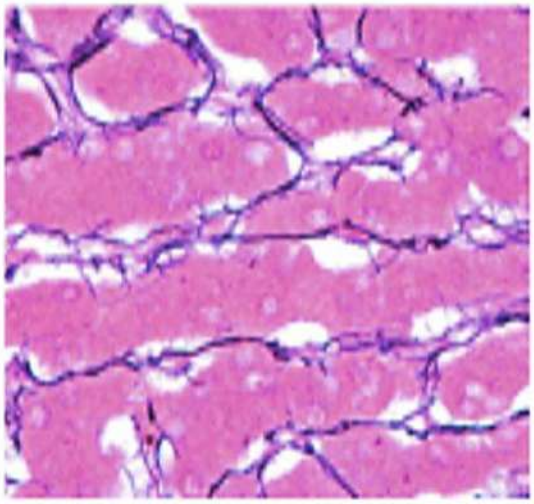
3. Reticular Fibers:

They are Fine fibers arranged in the form of networks

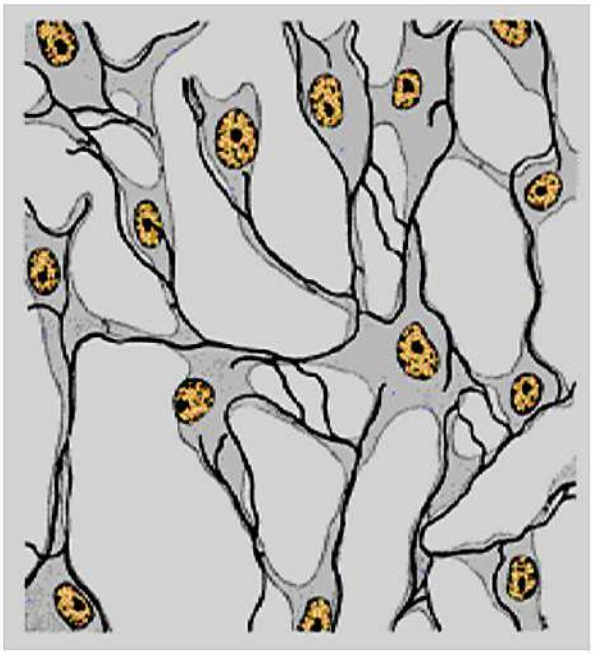
They are not visible in H&E preparations but they can be demonstrated well with silver stains where they take a black colour.

Function of reticular fibers:

They provide a network around the cells of the organs and form the framework of the organs.



Reticular fibers by silver impregnation



	White collagenous fibers	Yellow elastic fibers	Reticular fibers
Characters	They are called white because in fresh state, their color is white	They are called yellow because in fresh state, their color is yellow	they branch and anastomose to form a network
Light microscope	The fibers are acidophilic. Fibers run in wavy bundles.	The Fibers are weakly acidophilic, thin, long and highly refractile The fibers branch	Fibers are very thin and delicate Form a network
ining finity	They are stained: Blue with Mallory stain.	They are stained:- Dark brown with Orcein.	They are stained: Black with silver

References :

1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

2- Ross, M. H., & Pawlina, W. (2006). *Histology: A text and atlas with correlated cell and molecular biology* (5th ed.). Lippincott Williams & Wilkins.

3- Wheater, P. R., Burkitt, H. G., & Daniels, V. G. (2006). *Wheater's functional histology: A text and colour atlas* (5th ed.). Churchill Livingstone/Elsevier.

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Lecture 14

CARTILAGE

Learning objectives:

1-Recognize the general features of cartilage

2-Mention the different histological types of cartilage, their main features & sites

3-Describe detailed structure of chondrocytes

4-Explain the structure of perichondrium & cartilage matrix

5-Discriminate methods of cartilage growth & repair

CARTILAGE

It is a tough supporting form of connective tissue

that develops from mesenchyme and consists chiefly of intercellular substance (matrix & fibers) and cells.

General characters of cartilage:

*It is a special type of C.T characterized by extensive extracellular matrix (ECM)with high concentrations of GAGs, proteoglycans, collagen, elastic fibers, and cells.

*Its intercellular substance is not calcified.

*cartilage matrix lacks blood vessels (avascular); it is nourished by diffusion from blood vessels in the perichondrium.

*cartilage cells can divide

Function of cartilage:-

Form skeleton of the fetus.

Support soft tissue (trachea – contain hyaline cartilage)

Essential for the growth of the long bone.(epiphyseal cartilaginous plate)

Facilitate bone movement

Types of cartilage:

Hyaline cartilage

Elastic cartilage

Fibrocartilage

I-Hyaline Cartilage.

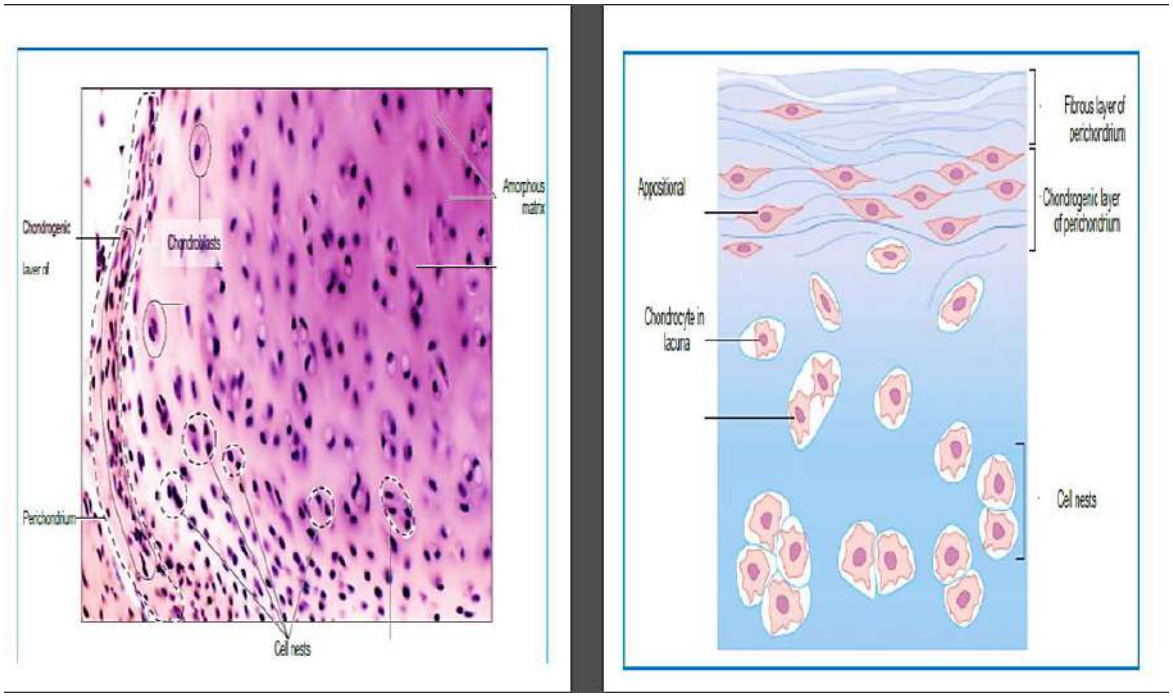
Sites of hyaline cartilage:

In the prenatal life: hyaline cartilage forms the temporary skeleton that is gradually replaced by bone.

In postnatal life : hyaline cartilage is located in the articular surfaces of movable joints, epiphyseal plate of long bones (until its growth is completed), in the respiratory passages (nose, larynx, trachea, bronchi), in the ribs.

Microscopic structure:

Hyaline cartilage is composed of cartilage cells embeded in an extensive extracellular matrix (ECM) and is covered with perichondrium.



Perichondrium:

*Except the articular cartilage of joints, all hyaline cartilage is covered by a layer of dense connective tissue, the perichondrium, which is essential for the growth and maintenance of cartilage.

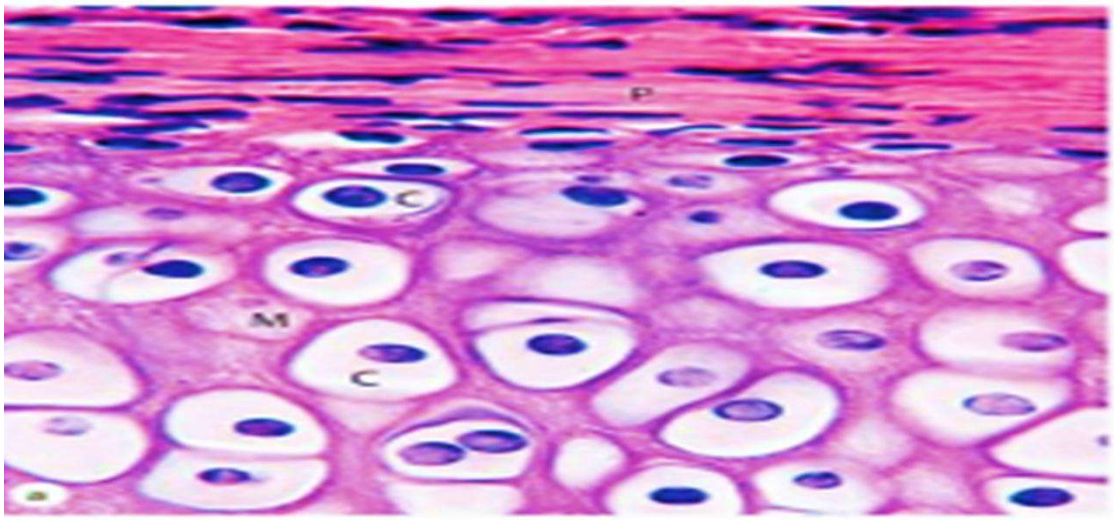
*Articular cartilages lack perichondrium and are nourished by diffusion of oxygen and nutrients from the synovial fluid.

The perichondrium consists of

1- The outer layer consists of collagen type I fibers and fibroblasts

2-The inner layer adjoining the cartilage matrix contain mesenchymal stem cells which provide a source of chondroblasts .

*The chondroblasts are very active ; they divide and differentiate into chondrocytes to form newly formed cartilage by appositional growth.



Cartilage cells

A. Differentiated Chondroblasts (Young Cells):

They are flattened cells with oval nuclei and basophilic cytoplasm, present singly in their lacunae, at the periphery of the cartilage, parallel to the perichondrium.

B. Mature Chondrocytes

are located in lacunae(matrix cavities) in the extensive extracellular matrix(ECM).

With L/M:

• They have rounded nuclei and vacuolated cytoplasm when stained with H & E due to the presence of fat and glycogen.

*The cells are present inside spaces called lacunae.

*Deeper in the cartilage, they are round and may appear in groups of up to eight cells that originate from mitotic divisions of a single chondroblast (cell nest)

With E/M :

The outline of the cell is irregular.

It demonstrates well developed RER and well developed Golgi body.

They synthesize and maintain ECM components .

Function; Chondrocytes synthesize and secrete tropocollagen , sulphated GAGs and proteoglycans of intercellular substance which are accelerated by many hormones and growth factors.



Ultrastructure of the chondrocyte.

The extracellular matrix (ECM)

-Has high concentrations of GAGs, proteoglycans & collagen fibers.

-The dry weight of hyaline cartilage is 40% collagen ,mostly type II embedded in a firm, hydrated gel of proteoglycans and structural glycoproteins.

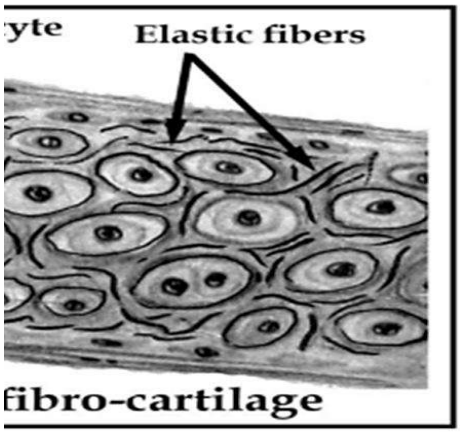
- Water bound to these complex molecules constitutes up 60%-80% of the weight of fresh hyaline cartilage. The high content of bound water allows cartilage to serve as a shock absorber, a role of major functional importance.

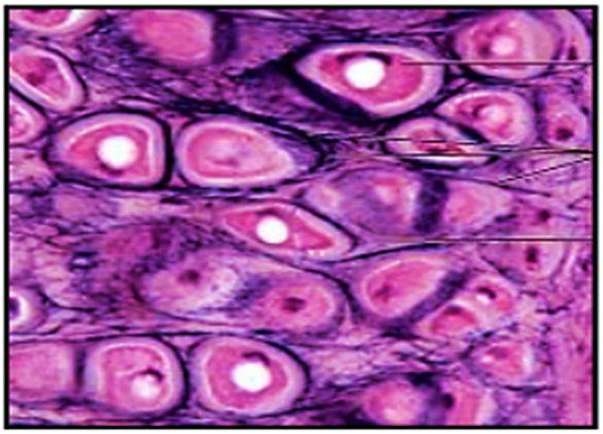
II- Elastic cartilage

With Light Microscopy: similar to the hyaline cartilage but contains abundant network of elastic fibers in the matrix ,so it is more flexible than hyaline cartilage.

- Elastic fibers are demonstrated by orcein stain.

Sites: It is present in tissues that need stiffness and also elasticity as auricles of the ear, wall of the external auditory canals, Eustachian tube, epiglottis and larynx.



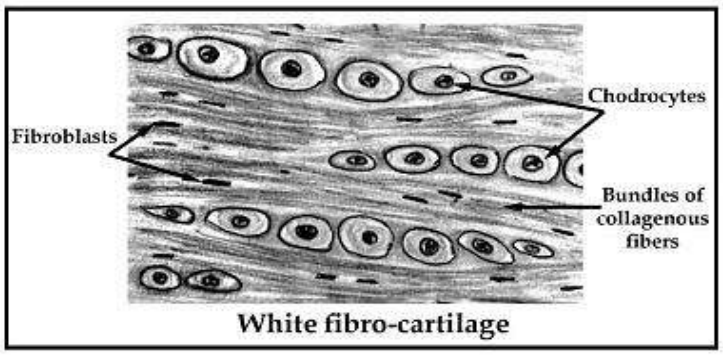


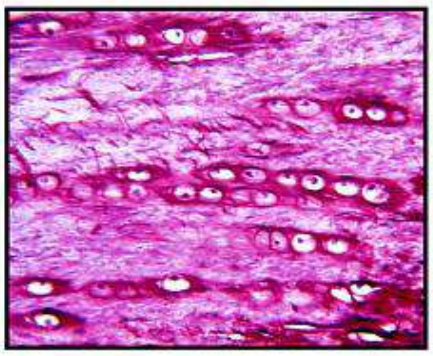
III-Fibrocartilage

With Light Microscopy: groups of chondrocytes surrounded by little hyaline matrix and are separated by bundles of type I collagen with scattered fibroblasts, adding extra strength to this tissue.

-No perichondrium in fibrocartilage.

Sites: Found in intervertebral discs , symphysis pubis, semilunar cartilage of knee joint, lips of glenoid cavity of the hip bone and some tendon insertions.





Cartilage growth and repair

Once formed, the cartilage growth (as in the articular cartilage , epiphyseal plate and during embryogenesis) takes place by:

a- Interstitial growth: Occurs within the substance of the cartilage by mitotic division of pre-existing chondrocytes and lay down matrix around them. By this method cartilage increases in size. This occurs in moderately young cartilage in which the intercellular substance is not too stiff.

b- **Appositional growth**: It occurs elsewhere in the body.

A new cartilage is laid down at one of its surfaces by differentiation of mesenchymal stem cells in the perichondrium into chondroblasts/ chondrocytes which form ECM.

Cartilage repair: Regeneration of cartilage occurs in young cartilage only. In adults, cartilage damage results in connective tissue scar.

References :

1-Junqueira, L. C., & Carneiro, J. (2005). *Basic histology: Text & atlas* (11th ed.). McGraw-Hill.

2- Ross, M. H., & Pawlina, W. (2006). *Histology: A text and atlas with correlated cell and molecular biology* (5th ed.). Lippincott Williams & Wilkins.

3- Wheater, P. R., Burkitt, H. G., & Daniels, V. G. (2006). *Wheater's functional histology: A text and colour atlas* (5th ed.). Churchill Livingstone/Elsevier.

4-Gartner, L. P. (2020). *Textbook of histology* (5th ed.). Elsevier.

Lecture 15

BONE

Learning objectives

1-After this lecture, students should be able to

2-Recognize the general features & functions of bone

3-Describe the detailed structure of bone cells and their functions

4-Identify the structure of bone matrix, periosteum &endosteum

5-Recognize the types of bone

Definition:

Bone is a specialized connective tissue composed of calcified intercellular substance; the bone matrix and bone cells

Function of bone

1-Support

2-Protection (protect internal organs)

3-Movement (provide leverage system for skeletal muscles, tendons, ligaments and joints)

4-Mineral homeostasis (bones act as reserves of minerals important for the body like calcium or phosphorus)

5-Hematopoiesis: blood cell formation

6-Storage of adipose tissue: yellow marrow

Structure of Bone:

It is formed of: Bone matrix, Bone cells, Periosteum and Endosteum.

Bone matrix

It is composed of inorganic and organic materials.

1. The inorganic materials:

- Form about 75% of the dry weight of bone

- Consist mainly of calcium in the form of hydroxyapetite crystals

- Other minerals including Mg+, Na+ and K+ bicarbonate and citrate.

Decalcification of bone will result in flexible bone like a tendon.

2. The organic materials:

- Form about 25% of the dry weight.

- About 90% of the organic material is collagen .

- The remaining are sulphated GAG and glycoprotein together with other materials.

Removal of the organic bone matrix will result in fragile bone on handling(egg-shell consistency).

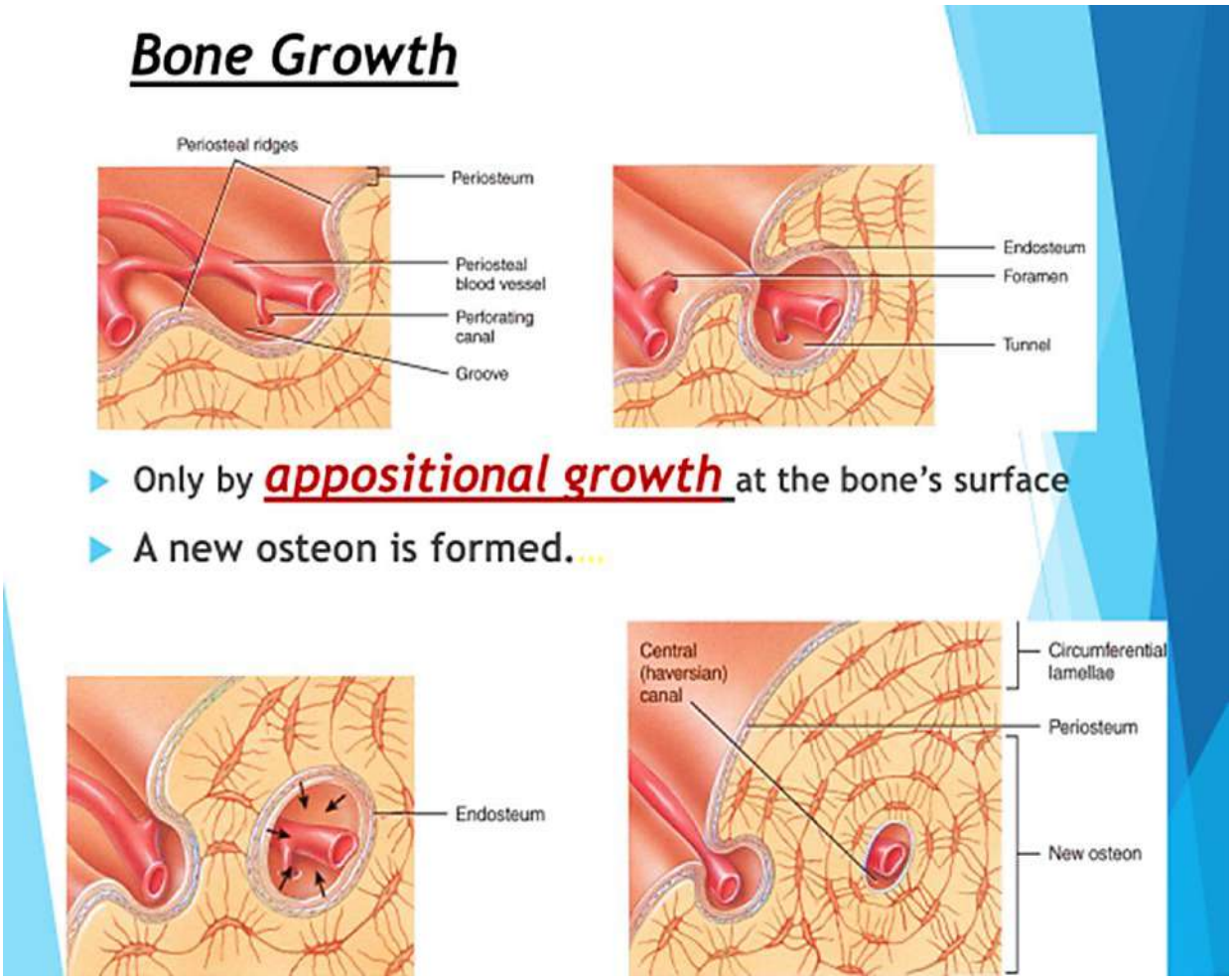
Some differences between cartilage and bone:

1. It grows only by appositional growth.

2. Its intercellular substance is calcified

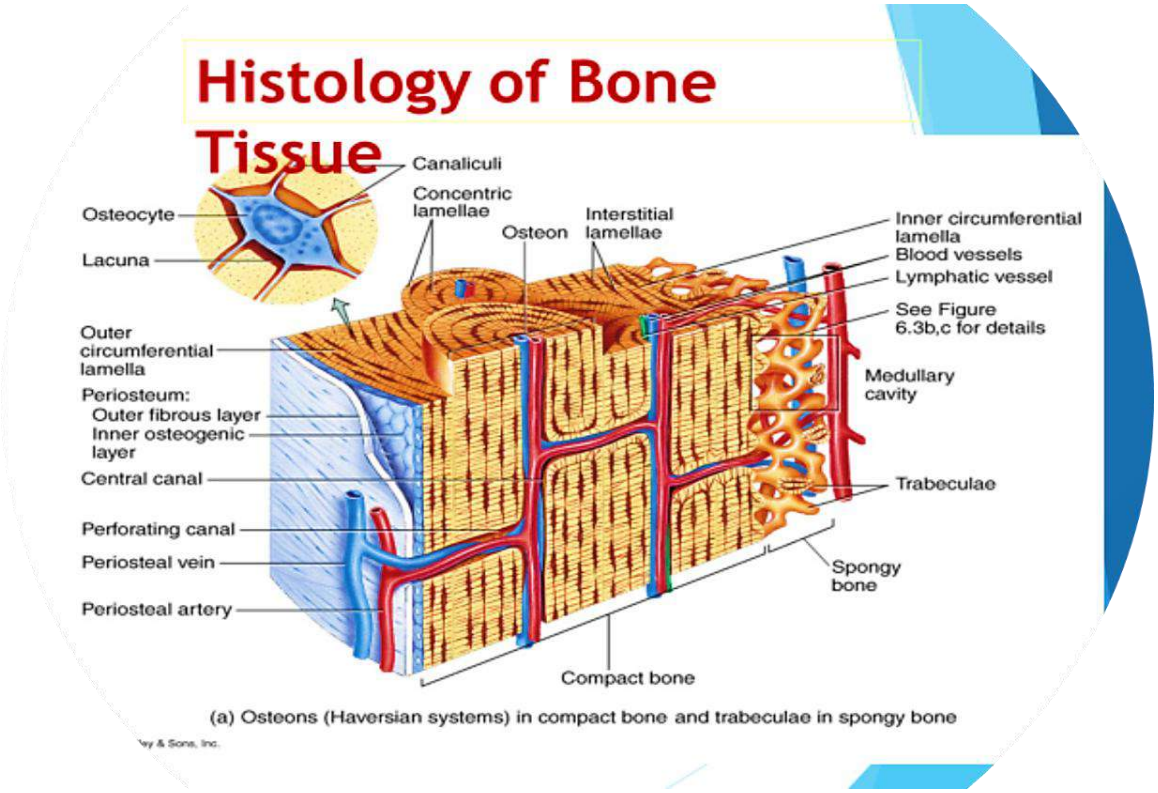
3. Bone matrix is vascular and osteocytes still survive in the calcified matrix.

4. Bone cells cannot divide, not even the osteoblasts, thus there is no interstitial growth.



Differences between bone and Cartilage

Cartilage	Bone	
Cartilage is rigid due to presence of chondritin sulphuric acid in the matrix	Bone is solid due to deposition of calcium salts in the matrix	1
Cartilage cells (chondrocytes) present singly or groups inside lacunae	Bone cells (oestocytes) are present singly inside lacunae	2
Chondrocytes do not communicate by cellular processes	Oestocytes intercommunicate by their thin cellular processes	3
Cartilage is non-vascular but chondrocytes receive nutrients by diffusion from blood vessels in the perichondrium	Bone is vascular: Haversian and Volkmann's canal and bone cancalucli carry blood vessels from periostium to oestocytes	4
Cartilage is three types: Hyaline Elastic White fibrocartilage	Bone is two type: Compact bone Spongy bone	5
Chondrocytes can divide	Oestocytes can not divide	6



Periosteum

It is the C.T that covers the bone.

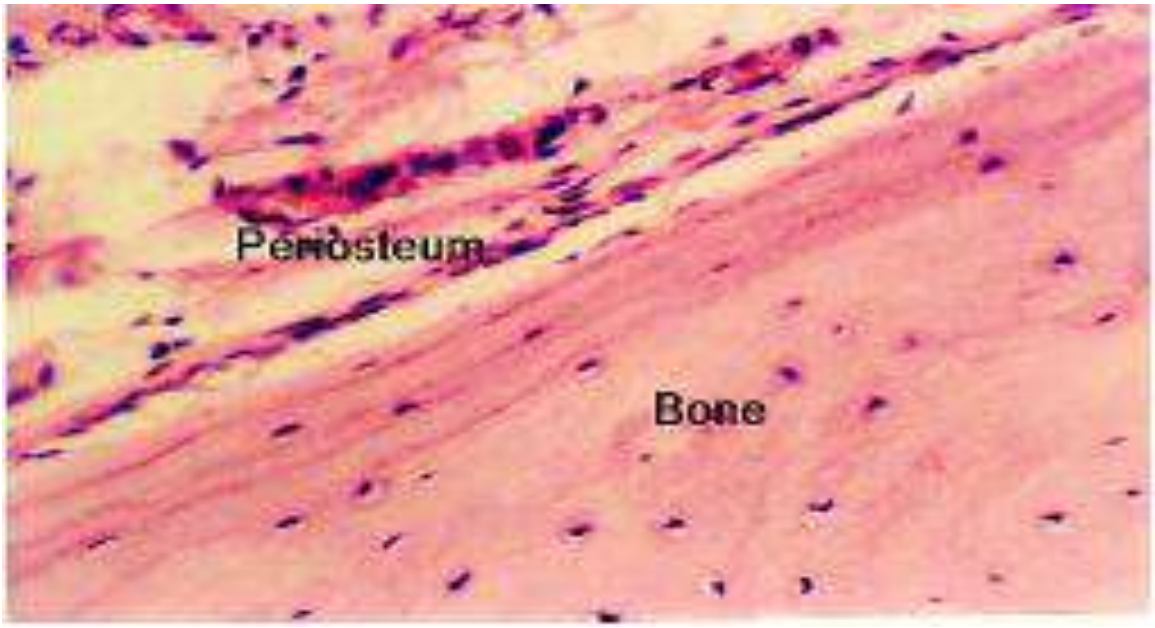
It consists of an outer vascularized dense fibrous layer and an inner osteogenic layer.

Endosteum:

It is a very thin C.T. layer containing flattened osteogenic cells and osteoblasts and lines the bone marrow cavity

Bundles of periosteal collagen fibers, called perforating (Sharpey's) fibers, penetrate the bone matrix, binding the periosteum to bone at sites of tendons &ligament insertion on to the bone surface.

*Osteogenic activity of periosteum is lacked in certain areas which are of clinical significance in cases of bone fracture (surgical neck of femur, Patella of knee ,sites of tendon &ligament insertion on to the bone surface)

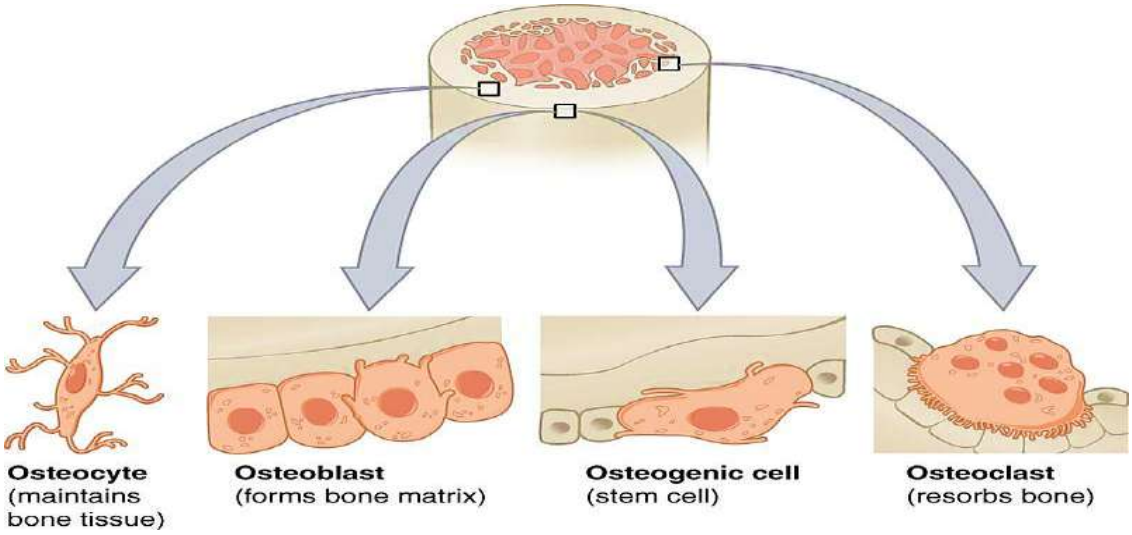


<u>Cells of Bone Tissue</u>

In actively growing bone, four types of cells are distinguishable:

Osteogenic cells, osteoblasts, osteocytes and osteoclasts.

The first 3 types could be considered as different functional phases of the same cell type.



Osteogenic cells:

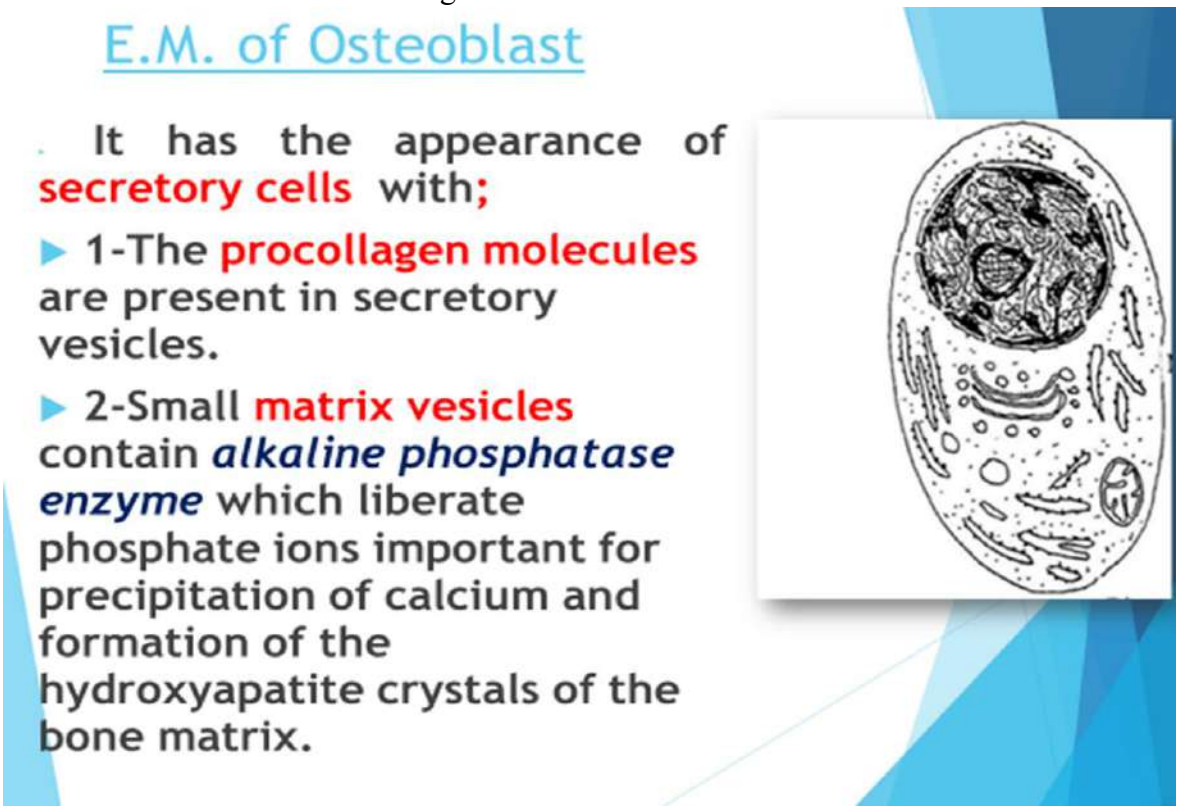
These cells are able to proliferate and differentiate into osteoblasts.

They have pale oval nuclei and scanty, faintly basophilic cytoplasm.

Osteogenic cells are active during the growth of bone and repair of bone after its injury.

Osteoblasts:

They are bone forming cells. They are responsible for synthesis and secretion of macromolecular constituents of organic bone matrix



Osteocytes :

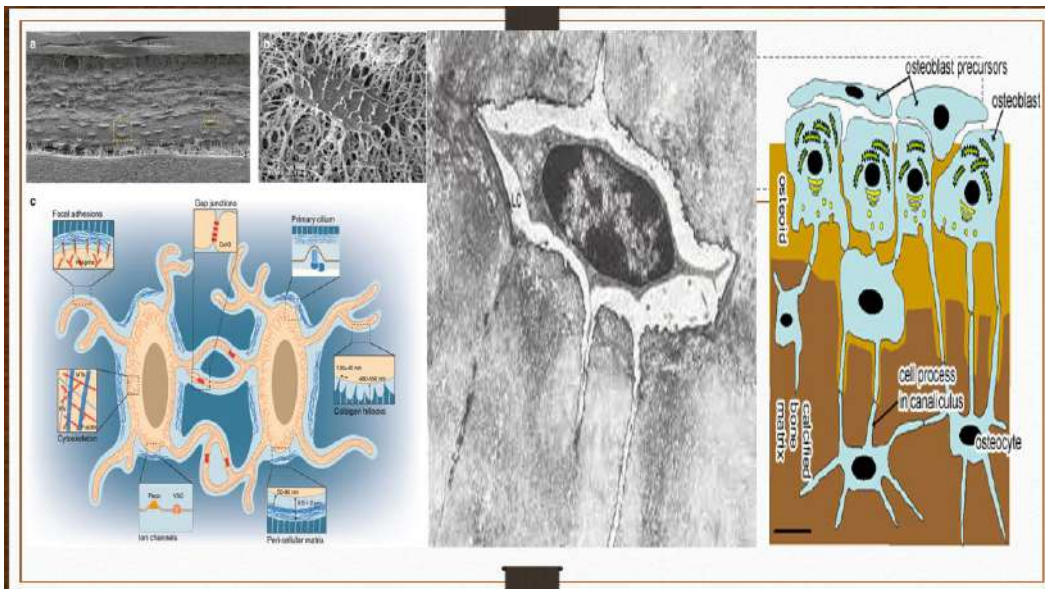
*Once the osteoblast becomes surrounded with intercellular substance it is called osteocyte .

*Osteocytes extend cytoplasmic processes that traverse the matrix through canaliculi.

*Processes of adjacent cells make contact through gap junctions. This coupling provides nourishment and allows the intercellular flow of ions and small molecules (e.g. hormones controlling bone growth and development)

*These cells are actively involved with the maintenance of the bony matrix.

*Death of the osteocytes is followed by resorption of this matrix.



Osteoclasts:

They are large multinucleated cells containing 6-12 nuclei (up to50 nuclei) .

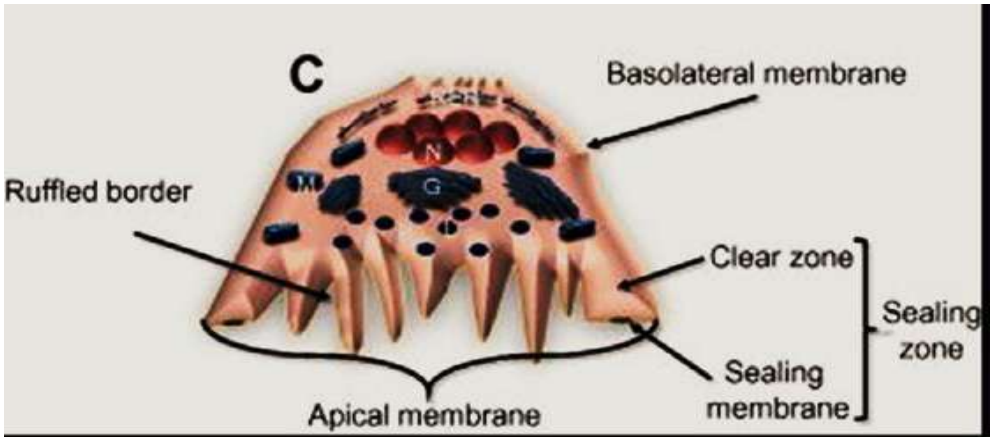
Osteoclasts occupy pits in the surface of the bone called Howship's lacunae.

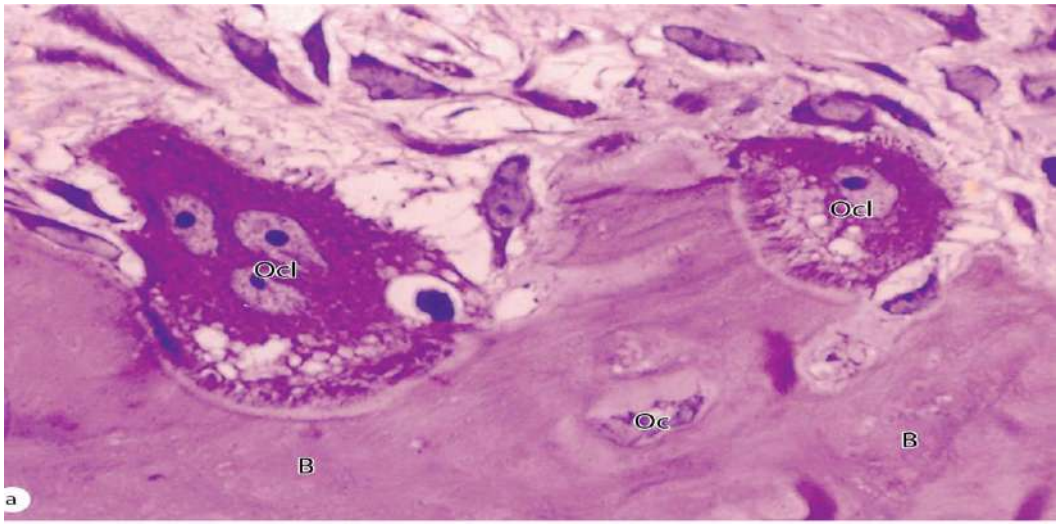
Function of osteoclasts: Bone resorption and remodeling.

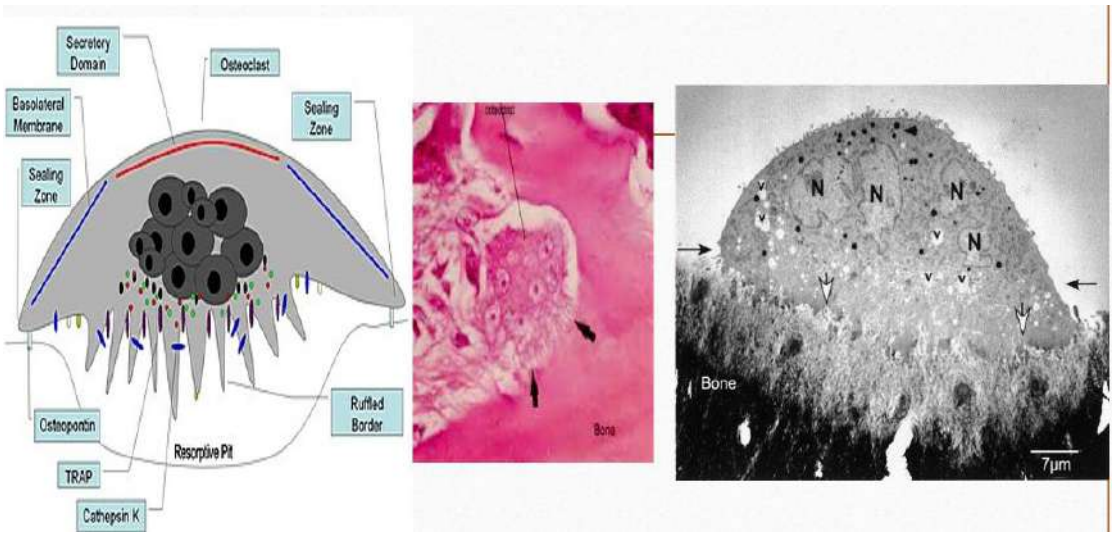
Remodeling: means the maintenance of the shape of the bone by adding bone at new surface and removing or resorbing it at the other surface during the growth of bone.

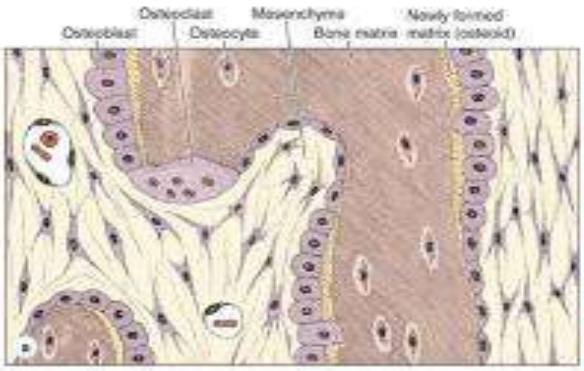
So it is lifelong process (mature bone tissue removed from skeleton (process called bone resorption) and new bone tissue formed (ossification).

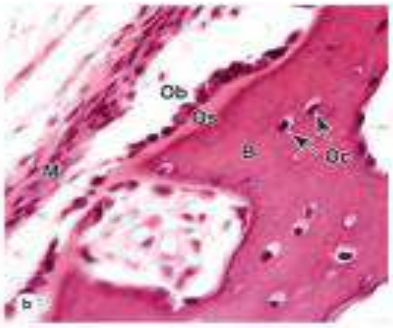
Dynamic process. :Osteocytes & Osteoclasts


C
Basolateral membrane
Ruffled border
Clear zone
Sealing zone
Sealing membrane
Apical membrane


Ocl
Ocl
Oc
B
B
a


Secretory Domain
Osteoclast
Basolateral Membrane
Sealing Zone
Sealing Zone
Ruffled Border
Osteopontin
Resorptive Pit
TRAP
Cathepsin K
Bone
N
N
N
N
Bone
7µm





Source: Anthony L. Mescher: Junqueira's Basic Histology, 14th Edition
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Osteoclast	Osteocyte Actual bone cell	Osteoblast Bone-forming cell	Osteogenic cell	
Develop from monocytes	Develop from osteoblasts	Develop from osteogenic cells	Develop from pericytes	Origin
Present in bone marrow spaces medullary cavity.	Present inside the lacunae in bone matrix.	Present in the periosteum and endosteum.	Present in the periosteum and in the endosteum.	Location
Large cell contains 6 or more nuclei (multinucleated). Its cytoplasm is foamy.	Branched, ovoid cell, with oval nucleus. Its cytoplasm is basophilic.	Small, branched cell with oval, eccentric nucleus. Its cytoplasm is basophilic and shows a negative Golgi image.	Primitive, less differentiated cell, with pale cytoplasm and flat nucleus.	Morphology
Essential in bone resorption during ossification	It can release calcium ions from the bone matrix to the blood in times of need.	It secretes bone collagen It transform into osteocyte.	It divides and differentiates into osteoblast (during growth or at bone fracture)	Functions

Types of bone

Compact bone:

Sites: Shaft of long bones, the outer part of the epiphysis and also the outer part of the short bones.

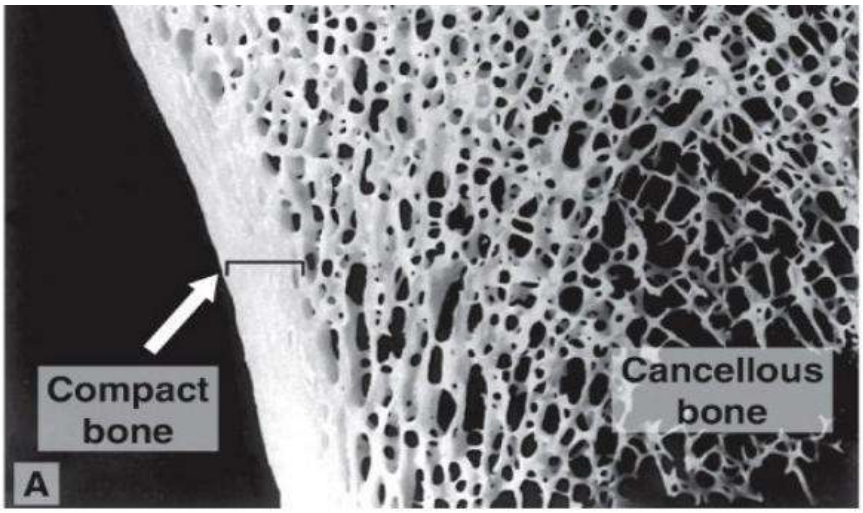
Structure of compact bone:

*It consists of regularly arranged lamellae of bone cells inside lacunae and intercellular matrix which are permeated with the Haversian canals and Volkmen's canals that connect Haversian canals together.

*Compact bone is covered with periosteum and lined with endosteum.

*compact bone is characterized by the presence of Haversian systems.

*Each Haversian system is composed of a Haversian canal surrounded by concentric lamellae of bone (osteons).



Haversian system (osteon)

1) Haversian canal: It runs parallel to the long axis of the bone. It contains loose connective tissue rich in blood vessels. It is lined with osteogenic cells.

2) Concentric bone lamellae: Calcified bone collagen arranged in concentric manner around Haversian canal.

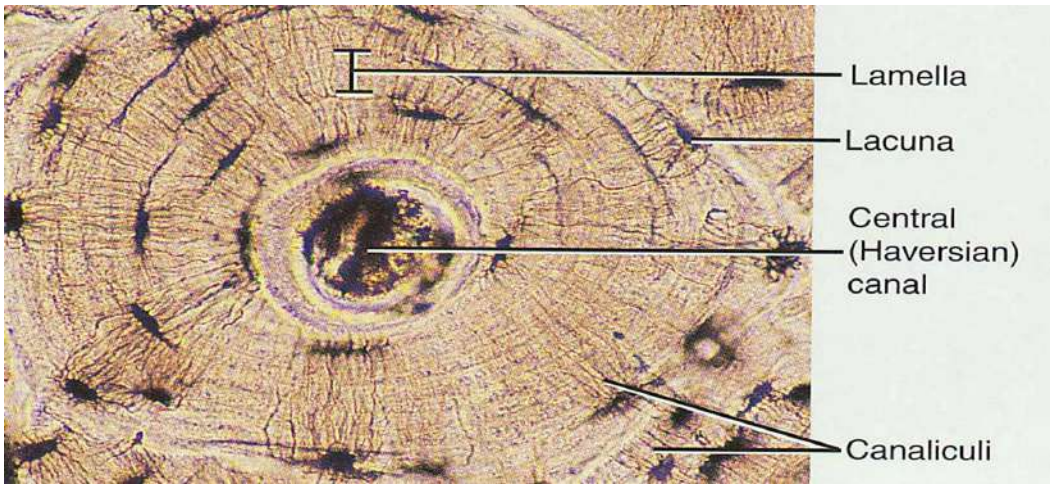
3) Osteocytes: Present inside lacunae. Osteocytes connected with each others by bone canaliculi.

4) Outer circumferential bone lamellae: Parallel to outer bone surface

5) Inner circumferential bone lamellae: Parallel to inner bone surface

6) Interstitial bone lamellae: Present in-between Haversian system.

7) Volkmann's canals: Oblique and transverse canals which connect Haversian canals together.

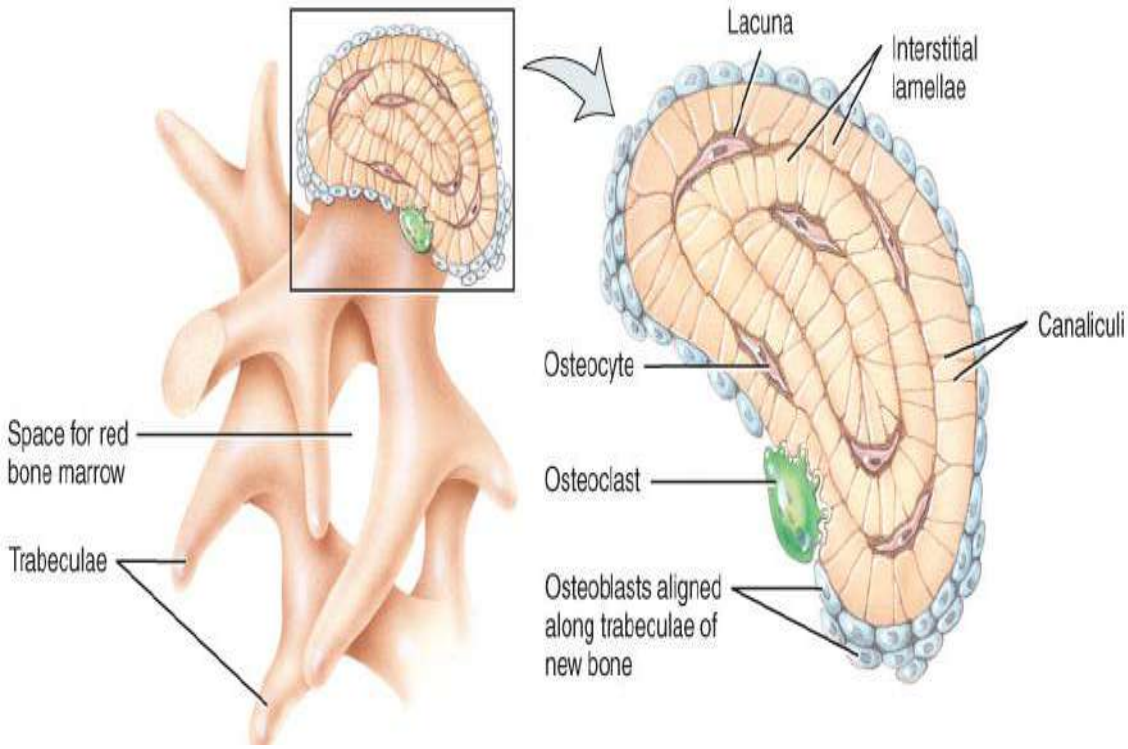


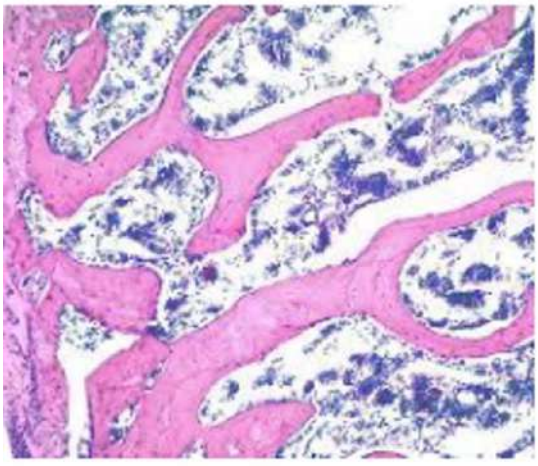
Cancellous bone:

* Cancellous bone is composed of irregular trabeculae of bone separated by bone marrow spaces which are lined with osteogenic cells.

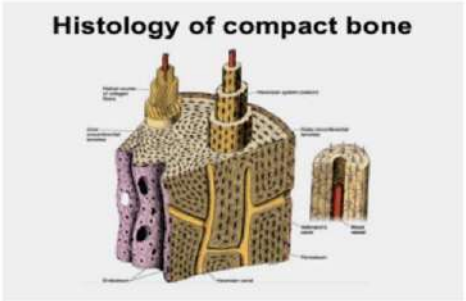
*The bone trabeculae are composed of osteocytes and intercellular substance and covered with periosteum.

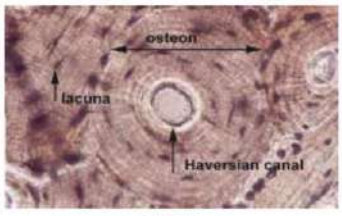
*There are no Haversian systems.



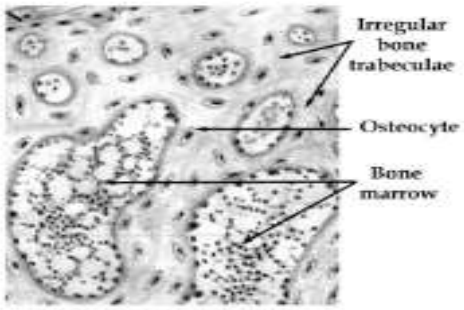
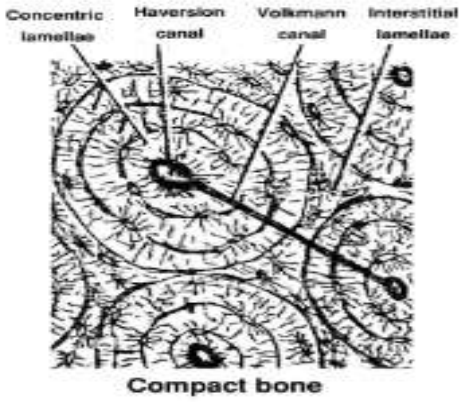


Cancellous bone





Osteon

Cancellous (spongy) bone	Compact bone	
• It looks spongy with many holes separated by irregular rabeculae.	• It looks solid with no apparent holes	
• Epiphysises of long bones. • Ribs and stemum.	• Shafts of long bones. • Flat bones of skull.	Occurrence
• It is formed of irregular bone trabeculae which branch and anastomose. • The bone trabeculae are separated by bone marrow spaces of irregular shape and size  Cancellous bone	• The bone lamellae are deposited in a regular pattern. • The Haversian system or osteon is the structural unit of compact bone.  Compact bone	Structure

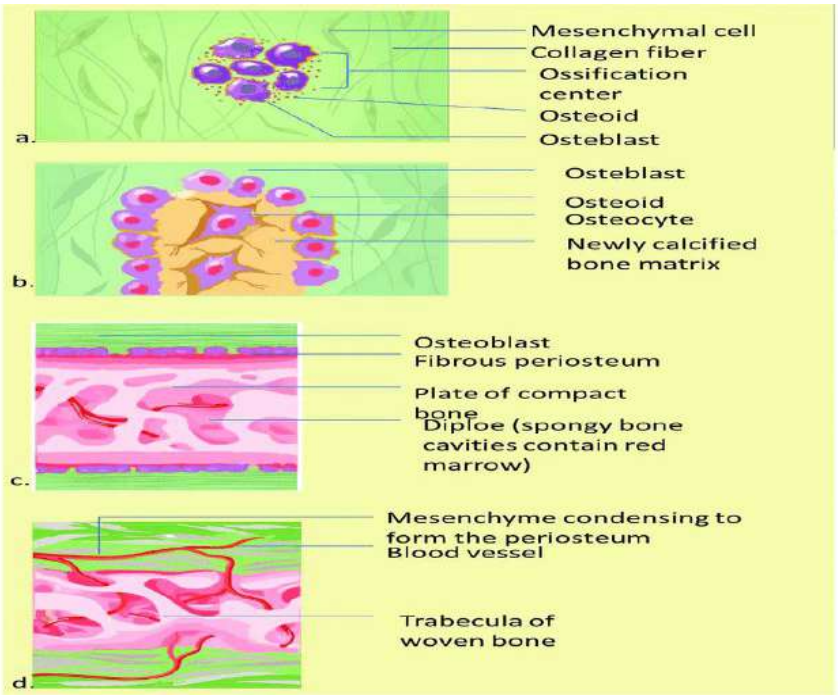
Histogenesis of bone (bone formation, osteogenesis):

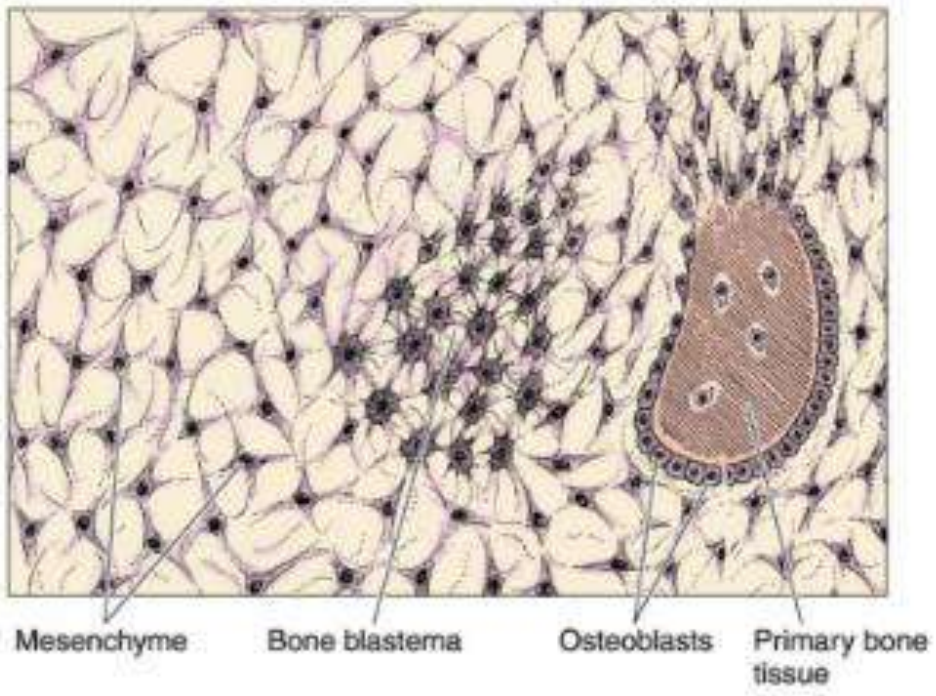
There are two methods by which bone may be formed:

Intramembranous bone formation:

*Bone is formed directly from a primitive C.T. (mesenchynal C. T.). It occurs within a membrane of C. T. during the formation of flat bones of the skull and the clavicle.

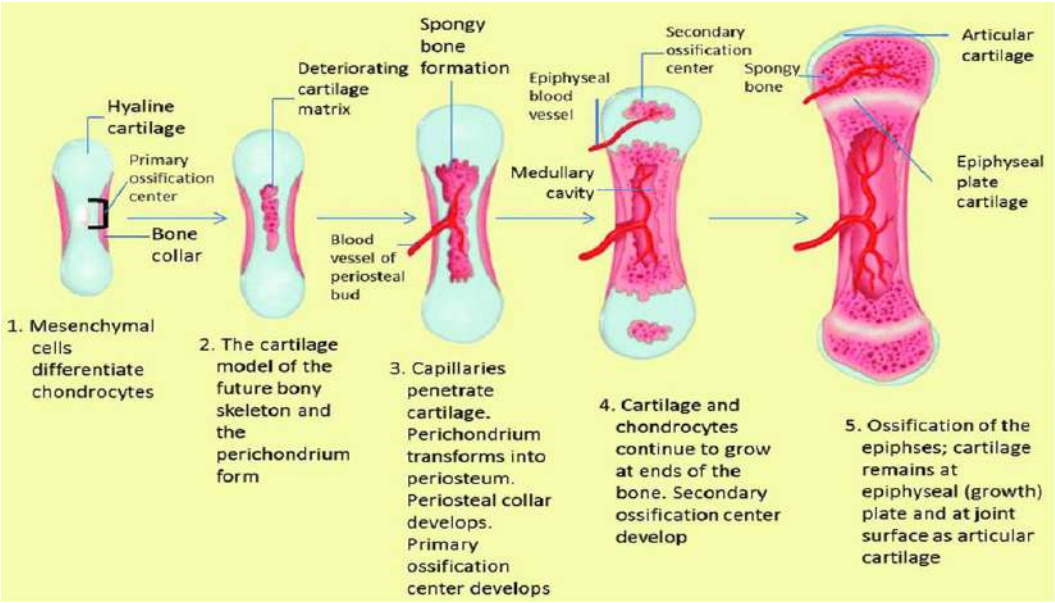
*It contributes and participates in the formation of short bones and epiphyses of long bones.





Intracartilagenous bone formation:

Occurs in cartilage which resembles in shape the future bone. It is mainly responsible for the formation of short and long bones.



Growth of bone at the epiphyseal plate:

It occurs in 4 stages:

a) Proliferation: The chondrocytes of the resting cartilage divide several times giving rise to a large number of cells which are arranged in rows parallel to the longitudinal axis of the long bone.

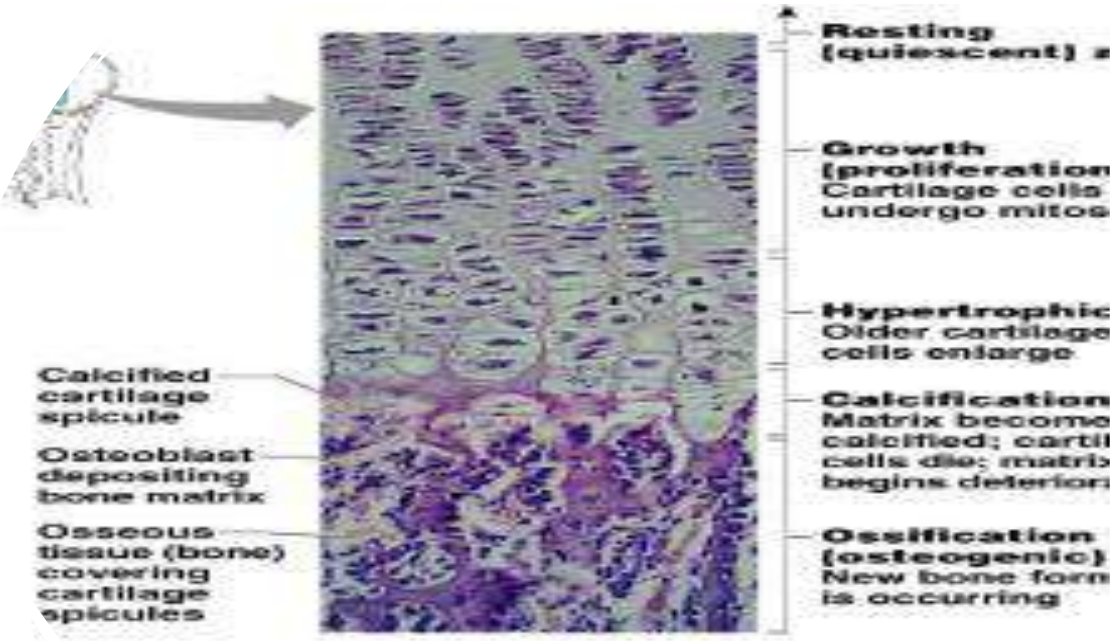
b) Hypertrophy: The chondrocytes increase in size and become rich in phosphatase enzyme.

c) Calcification: Calcium salts are deposited (in the matrix) around the chondrocytes by the activity of phosphatase enzyme. The calcified matrix cuts off the nourishment to the chondrocytes which die and degenerate leaving empty spaces (or canals)

d) Ossification: Blood capillaries and mesenchymal cells penetrate the spaces left after the destruction of chondrocytes. The undifferentiated mesenchymal cells give rise to osteoblasts which deposit bone matrix.

Notice that the bone tissue is formed at the site of cartilage.

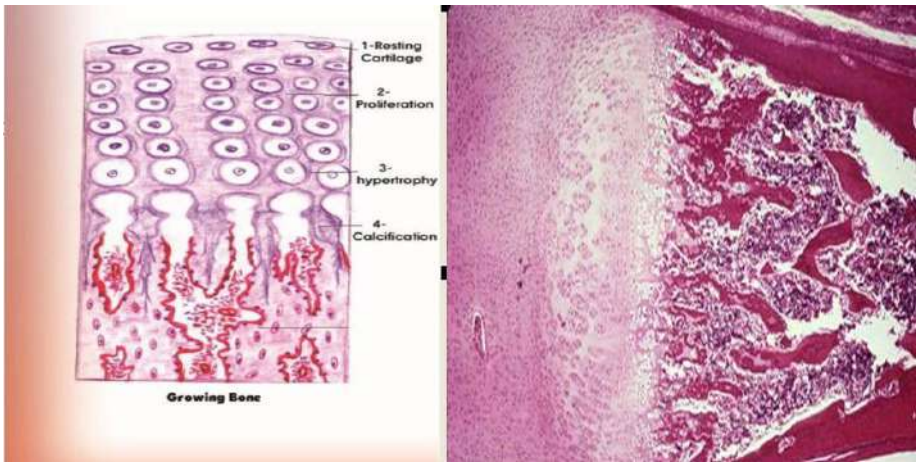
Thus cartilage is not transformed into bone but is rather replaced by bone



Growth of bone at epiphyseal plates

Zones of epiphyseal plates

- Zone of Resting Cartilage
- Zone of Proliferating Cartilage cells
- Zone of Hypertrophic Cartilage cells
- Zone of Calcified Cartilage matrix
- Zone of ossification



REFERENCES:

1-INTEGRATED SYSTEMS (PP. 55 &56), THE NERVOUS SYSTEM (PP. 32-36), BASIC HISTOLOGY (PP. 161- 170).ELSEVIER'S INTEGRATED HISTOLOGY (2007) 1ST EDITION.